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This publication implements Air Force Policy Directive (AFPD) 21-1, Maintenance of Military Materiel. It is the basic Department of Air Force Instruction (DAFI) for all weapon system and support equipment maintenance management guidance. It provides the minimum essential guidance and procedures to safely and effectively maintain, service, and repair weapon systems and support equipment. This publication applies to all military and civilian members of the United States Air Force (USAF), United States Space Force (USSF), Air Force Reserve (AFR), and Air

National Guard (ANG) as well as those with contractual obligation to comply with Air Force publications. Supplements and addendums are written in accordance with (IAW) Department of the Air Force Manual (DAFMAN) 90-161, *Publishing Processes and Procedures*. Supplements must identify and document major command (MAJCOM), AFR, and ANG required deviations (applicability, variance, exception, and differences in organizational placement of responsibilities/processes) in their supplement and addendums with the abbreviation (DEV). Place the "DEV" entry after the paragraph number and directly preceding the affected text, such as "(AMC) (DEV) Use the..." or "(ADDED-AMC) (DEV) Use the...)." MAJCOMs supplements are submitted to the Air Force Maintenance Division (AF/A4LM) @ **AF.A4LM.Maintenance.Policy@us.af.mil** for approval. Supplements and addendums are to be published in the e-Publishing website. The authorities to waive wing and unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, or T-3") number following the compliance statement. See DAFMAN 90-161 for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately, to the requestor's commander for non-tiered compliance items. All approved waivers must be sent to AF/A4LM Maintenance Policy org box: **AF.A4LM.Maintenance.Policy@us.af.mil**, within 30 days of approval, for situational awareness and process improvement considerations. For questions on interpreting this instruction, first contact your MAJCOM maintenance functional activity. Refer recommended changes and questions about this publication through your assigned MAJCOM, AFR or ANG, to the Office of Primary Responsibility (OPR) using the Department of the Air Force (DAF) Form 847, Recommendation for Change of Publication; route DAF Forms 847 from the field through the appropriate functional chain of command. Ensure all records created as a result of processes prescribed in this publication adhere to AFI 33-322, Records Management and Information Governance Program, and disposed IAW the Air Force Records Disposition Schedule located in the Air Force Records Information Management System. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the AF.

(AFMC) DAFI 21-101, *Aircraft and Equipment Maintenance Management*, is supplemented as follows: This supplement identifies specific processes and procedures that are unique to AFMC. This supplement also implements reorganization direction establishing AFMC as servicing MAJCOM for United States Space Force (USSF) and provides DAFI 21-101 MAJCOM/FLDCOM supplemental guidance for USSF in DAFI 21-101 AFMCSUP Addendum D, *United States Space Force (USSF) Non Standard Organization (NSO) Aircraft and Equipment Maintenance Management*. This supplement applies to Air Force Materiel Command organizations and personnel that develop, test, and maintain aerospace platforms and associated equipment. Supplement applicability: Air Force Test Center (AFTC) Test Wing (TW) organizations (96 TW and 412 TW) and the USAF Aircraft Battle Damage Repair Office Air Force Sustainment Center (AFSC) (AFSC/LZD). AFSC Air Logistics Complexes (ALCs) do not follow DAFI 21-101 or this AFMC Supplement, for Depot Maintenance Management guidance see AFMCI 21-100 Volume 1, *Depot Maintenance Principles;* Volume 2, *Depot Maintenance Production*, and Volume 3, *Depot Maintenance Production Support*. This publication applies to Air Force Reserve Command (AFRC) units when assigned or associated with AFMC where AFMC is lead. This publication does not apply to the Air National Guard. Numerous assigned missions within AFMC are not organized under the standard Wing Organization concept for

Aircraft Maintenance Management IAW AFI 38-101, *Manpower and Organization*. Non-Standard Organizations (NSOs) require deviations (applicability, variance, and/or differences in organizational placement of responsibilities/processes) from DAFI 21-101 and this supplement. DEV – Deviation to provide/scope applicability for NSO logistics/maintenance in an addendum(s) to the MAJCOM supplement is approved by Air Force Maintenance Division (AF/A4LM) @ **AF.A4LM.Maintenance.Policy@us.af.mil** IAW DAFI 21-101. Therefore, NSOs will follow the addendum(s) only, not the parent DAFI nor this AFMC Supplement; **Exception:** USSF NSO(s) will follow the USSF addendum to this supplement and the parent DAFI, but not the AFMC Supplement. The respective Addendum(s) will scope NSO requirements in meeting the intent of the DAFI and this AFMC Supplement based on the respective organizational mission(s). The following organizations are identified as NSOs (In the event of a conflict with identification of NSOs in this guidance, AFI21-101_AFMCSUP_Addendum A, *Non Standard Organization (NSO) Logistics Maintenance Management* or DAFI 21-101_AFMCSUP_Addendum C, *Air Force Research Laboratory (AFRL) Logistics Maintenance Management*, will take precedence): Air Force Life Cycle Management Center (AFLCMC) units: 66th Air Base Group (ABG), 88th Air Base Wing (ABW), Cryptologic and Cyber Systems Division (HNC) (specifically the Technical Application Products and Space COMSEC Maintenance mission areas), 645th Aeronautical Systems Group (AESG), Armament Sustainment Division (ASD) (AFLCMC/EB) AFLCMC/EBRM GACP Missiles and Weapons and AFLCMC/EBWWE Weapons Engineering Laboratories (Labs), AFLCMC/USAF Gunsmith Shop (GSS) specifically the geographically separated maintenance function in support of Small Arms/Light Weapons (SA/LW) located at Joint Base Lackland San Antonio, Munitions Sustainment Division (MSD) Laboratory (EBH), and Global Ammunitions Control Point (GACP) Afloat Preposition Fleet (APF) Flight/Cell (AFLCMC/EBHMA); AFTC units: Arnold Engineering and Development Complex (AEDC), 704th Test Group (TG) and Plant 42, and Specialized Management Office (AFTC_SMO); and AFSC units: 72nd Air Base Wing, 72nd Logistics Readiness Squadron Munitions Section (72 LRS/LGRM), 75th Air Base Wing, 649th Munitions Squadron, 78th Air Base Wing, 78 LRS/LGRW Ammo. Throughout this supplement, the civilian equivalent (unless no civilian may hold the position) may be applied to any reference to a military rank, grade or position. All references to aircraft and equipment maintenance in DAFI21-101 include support equipment, test equipment, special projects equipment, etc. The authorities to waive wing/unit level requirements in this publication are identified with a Tier ("T-2, T-3") number following the compliance statement. See DAFMAN 90-161, *Publishing Processes and Procedures*, for a description of the authorities associated with the tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority. For not-tiered AFMC Supplemented items only, submit requests for waiver through chain of command to the AFMC Supplement OPR. Submit recommendations for change or improvement to this instruction on DAF Form 847, *Recommendation for Change of Publication*. For questions on interpreting this instruction, first contact your Quality Assurance Office/activity. Send questions to the OPR at AFMC Maintenance Division (AFMC/A4M) Workflow (**AFMC.A4M.Workflow@us.af.mil**), 4375 Chidlaw Rd, Bldg 262 Room C117 WPAFB OH 45433-5006. Sustainment Operating Instructions (SOIs) are F-35 joint program instructions provided by the Joint Program Office (PO). They are developed with Service/Partner participation and provide source documentation for Air Force policies/instructions specific to the F-35 where legacy instructions may not be adequate for the given topic. AFMC units may use SOI source documentation/information that is relevant and complements DAFI21-101 and AFMC Sup. SOIs requiring a deviation from DAFI21-101 and

DAFI21-101_AFMCSUP requires the unit to submit a waiver. SOIs may be accessed at the following web site: **https://usaf.dps.mil/teams/12097/sustainment/af%20soi%20library/forms/allitems.aspx**. Air Force Research Laboratory applicability is also identified as a NSO, in the event of a conflict with identification of NSO in this guidance, DAFI21-101_AFMCSUP_Addendum C, *Air Force Research Laboratory (AFRL) Logistics Maintenance Management*, will take precedence. This publication must be supplemented by the implementing activities. Ensure all records created as a result of processes prescribed in this publication are maintained IAW Air Force Instruction 33-322, *Records Management and Information Governance Program*, and disposed of IAW the Air Force Records Disposition Schedule located in the Air Force Records Information Management System. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the AFMC.

SUMMARY OF CHANGES

This corrective action revises the following paragraphs: 10.3.1.; 10.3.3.3.3.; 10.11.1.; 10.11.2.1.1.; 10.12.; 10.13.1.; 10.13.1.1., 10.13.6.1 and 11.8.3.15. Additionally, this corrective action deletes **para 6.10.1.3.3.1**. Significant changes include the addition of Maintenance Cyber Discipline requirements, Decentralized Materiel Support, incorporation of four AFIs superseded above and establishes a Wing Avionics Manager Position requirement. Additionally, eTool and Worldwide Identification (WWID) management procedures were expanded to provide standardized enterprise requirements. MAJCOMs/ANG designated to establish Special Certification Roster (SCR) prerequisites to optimize workforce alignment to mission requirements.

SUMMARY OF CHANGES

This publication has been substantially revised and must be completely reviewed in its entirety. Significant changes include the addition of Maintenance Cyber Discipline requirements, Decentralized Materiel Support, incorporation of four AFIs superseded above and establishes a Wing Avionics Manager Position requirement. Additionally, eTool and Worldwide Identification (WWID) management procedures were expanded to provide standardized enterprise requirements. MAJCOMs/ANG designated to establish Special Certification Roster (SCR) prerequisites to optimize workforce alignment to mission requirements.

This publication has been revised and includes published guidance implemented under DAFI 21-101_DAFGM2022-01; Chapter 2, Roles and Responsibilities to support organizational structures where an MXG is not assigned (i.e., AFFORGEN deployed construct); waiver authorities to the appropriate level; accident investigation recommendations; audit corrective actions; recommended changes from the field; and updated references.

(AFMC) This publication has been substantially reorganized to realign the AFMC Supplement with DAFI 21-101_IC date 20 December 2023, and to incorporates DAFI21-101_AFMCSUP_AFMCGM 2022-01, dated 5 July 2022, that implemented the reorganization and transfer of United States Space Force responsibility to publish MAJCOM/FLDCOM level supplemental guidance to DAFI 21-101, *Aircraft and Equipment Maintenance Management*, to AFMC as the identified servicing MAJCOM. Additionally, added a requirement under Maintenance Group Commander responsibilities to establish local procedures for application

intervals of HydroSkip on F-16 transparencies, MAJCOM guidance for Oil Analysis Program (OAP) and Condition Based Maintenance Plus (CBM+). It also incorporates recommendations for change; Quality Assurance (QA) is responsible to ensure modifications or improvements for Aviation Support Equipment (AvSE) (i.e., Common Aviation Support Equipment (CAvSE) and Peculiar AvSE) are submitted to the appropriate authority.

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Chapter 1

MANAGEMENT OVERVIEW, SUPPORTING CONCEPTS AND REQUIREMENTS.

1.1. Introduction. This instruction prescribes basic aircraft and equipment maintenance management policy implementation and procedures used throughout the United States Air Force (USAF) to perform Mission Generation (MG) functions.

1.2. Organization. AF organizations are structured according to AFI 38-101, *Air Force Organization*, or as authorized by the Director of Manpower, Organization & Resources (AF/A1M). Contracted maintenance functions are not required to organize IAW AFI 38-101 but will implement the organization as outlined in their proposal as accepted by the government. For the definition of "Lead Command" see AFPD 10-9, *Lead Command Designation and Responsibilities for Weapon Systems.*

1.2. (AFMC) [DEV] Organization. Completed a High Performing Organization pilot program for DoD and is authorized the following deviations to continue taking advantage of the efficiencies achieved during that program:

1.2.1. **(Added-AFMC)** [DEV] The 412th Maintenance Squadron, Component Repair Flight (412 MXS/MXMA), consolidates Avionics, Armament Flight, and the Electro-Environmental, Battery Shop and Egress functions of the Accessory Flight in addition to Tow Targets sections.

1.2.2. **(Added-AFMC)** [DEV] The 412 MXS, Maintenance Flight (412 MXS/MXMX), consolidates the functions of the 412 MXS Maintenance Support Flight (412 MXS/MXMT) and the pneudraulic and fuels functions of the Accessories Flight (412 MXS/MXMC).

1.2.3. **(Added-AFMC)** [DEV] The 412 MXS, Munitions Flight (MXMW) is responsible for the Armament (412 MXG/MXMR) function for Bomber aircraft only due to the geographic location/separation at Edwards AFB, South Base.

1.3. Maintenance Concept. Per AFPD 21-1, organizational, intermediate, and depot maintenance capabilities for operational readiness will be maintained to ensure effective and timely response to peacetime operations, mobilizations, national defense contingencies, and other emergencies. **Note:** Guidance for the use of Additive Manufacturing to build replacement parts is prescribed in AFI 63-101/20-101, *Integrated Life Cycle Management*.

1.3.1. As a minimum each capability will be able to:

1.3.1.1. Organizational: launch and recover sorties, maintain, and repair materiel coded for organizational level repair.

1.3.1.2. Intermediate: repair materiel coded for organizational and intermediate level repair in back shops, centralized repair facilities, or both.

1.3.1.3. Depot-level Maintenance: Provides the capability to maintain materiel coded for organizational, intermediate and depot levels of maintenance. Includes maintenance requiring the overhaul, upgrading, or rebuilding of parts, assemblies, or subassemblies, and the testing and reclamation of equipment as necessary.

1.3.2. Organizational and intermediate-level maintenance is organized into two mutually exclusive networks, the Mission Generation Network (MGN) and the Repair Network (RN). The MGN is optimized for mission generation at the wing level and consists of authorized "on-

equipment" and "off-equipment" maintenance capabilities required to launch, recover, configure, inspect, and repair AF systems and equipment. The RN supports the MGN by providing the maintenance required to fulfill operational needs outside the capability or capacity of MGN activities. The interface between the two networks takes place when the MGN activity relinquishes control of reparable assets to the RN activity (such as, supply counter turn-in) or changes an end item Purpose Identifier Code (PIC) from an operational activity to a repair network activity (such as, depot maintenance).

1.3.2.1. Most MGN units possess a complement of equipment and supplies necessary to perform on-equipment and off-equipment maintenance.

1.3.2.2. RN units may reside at bases that perform mission generation. RN requirements and processes are identified in DAFI 20-117, *Repair Network Management*.

1.3.3. MAJCOMs assigned combat coded fighter aircraft will coordinate with Mission Design Series (MDS) lead and using commands and the MAJCOM Operations Directorate (MAJCOM/A3) to develop and document standardized MDS and Primary Aerospace Vehicle (Aircraft) Authorized (PAA) specific utilization rate standards in their supplements to this instruction. At a minimum the rates will:

1.3.3.1. Consider aggregated and analyzed unit generation capability data to identify and document standard MDS turn patterns by PAA in their supplements to **Chapter 14**.

1.3.3.2. Account for standard avionics and weapons training configurations.

1.3.3.3. Account for standard Technical Order (TO) driven turn time inspections and Average Sortie Duration.

1.3.3.4. Include a process for supporting units in assessing shortfalls and developing action plans.

1.3.4. Requests for Assistance. If a maintenance activity requires assistance for evaluation, repair, or both, beyond unit capability, requests are made IAW DAFI 21-103, *Equipment Inventory, Status and Utilization Reporting*; TO 00-25-107, *Maintenance Assistance*; and TO 00-20-14, *AF Metrology and Calibration Program*, or automated process as approved by the MDS Program Manager (PM) (for example, C-130 Automated Inspection, Repair, Corrosion and Aircraft Tracking (AIRCAT), F-16 Technical Assistance Request). All requests for assistance must be coordinated through the originating MAJCOM and Lead Command as applicable.

1.4. Aircraft Maintenance Tactics, Techniques and Procedures (TTP). TTPs are developed from lessons learned and best practices that provide valuable reference documents to improve maintenance processes and procedures. Maintenance leaders should utilize the maintenance fundamentals TTP volumes (Aircraft, Munitions and Missile) to support mission generation effectively and efficiently. Maintainers who attend the USAF Advanced Maintenance and Munitions Operations School are trained in advanced operational, expeditionary, and tactical maintenance management concepts stemming from the alumni's development and formalization of TTPs. Maintenance Group Commanders (MXG/CC) should identify their Advanced Maintenance and Munitions Operations School graduates and utilize them as advisors and instructors to enhance mission capability. AFTTP 3-4.21V1, *Aircraft Maintenance*, can be found at: **https://www.e-publishing.af.mil** or **https://usaf.dps.mil/teams/TTP/SitePages/Home.aspx**.

For additional information on Advanced Maintenance and Munitions Operations School and TTP development see Air Force Manual (AFMAN) 21-111, *Advanced Maintenance and Munitions Operations School*.

1.5. Aircraft and Equipment Readiness. Aircraft and equipment readiness is the maintenance mission. The maintenance function ensures assigned aircraft and equipment are safe, serviceable, and properly configured to meet mission needs. Maintenance actions include, but are not limited to, inspection, repair, overhaul, modification, preservation, refurbishment, troubleshooting, testing, analyzing condition, performance, and maintenance documentation. All levels of supervision need to place emphasis on safety, quality, and timeliness in the performance of maintenance. The concept of quality maintenance must be fostered by each supervisor and technician to ensure the integrity and skill of each maintainer is not degraded. To the greatest extent possible, maintenance is accomplished on a preplanned scheduled basis. Planning provides the most effective and efficient use of people, facilities, and equipment; reduces unscheduled maintenance; and allows for progressive actions toward maintaining and returning aircraft and equipment to safe operating condition. Exploiting repair network capability and maintaining visibility of repair cycle assets throughout the maintenance cycle are also critical elements of the equipment maintenance program.

1.5.1. The Air Force has established strategic-level aviation support equipment management within the 440 SCOS: Aviation Support Equipment Management Flight (ASEMF). The ASEMF has the responsibility to align with the integration of Aviation Support Equipment Maintenance Information System (AvSE MIS), also referred to as Defense Property Accountability System (DPAS) Maintenance and Utilization (M&U) Module capabilities. As AvSE MIS (DPAS M&U) comes online and units migrate assets into the new system, ASEMF will start to stand up and data mine available metrics on AvSE from AvSE MIS. These efforts combined (AvSE MIS and ASEMF) give the AF the greatest return to increasing one element of overall readiness in support of the National Defense Strategy (NDS).

1.5.2. Preventive Maintenance. AF units implement and manage the tasks specified in the scheduled recurring maintenance program for their assigned aircraft and associated support equipment (SE). Preventive maintenance is achieved through the inspection requirement concepts described in TO 00-20-1, *Aerospace Equipment Maintenance Inspection, Documentation, Policy, and Procedures*, and applicable weapon system -6 TO.

1.5.3. MAJCOMs that conduct Mission Generation Assessments or similar weapon system logistic evaluations in order to validate unit readiness will:

1.5.3.1. Provide units a standardized assessment report containing, at a minimum, positive, negative, and areas for improvement feedback to facilitate cross tell to like units.

1.5.3.1.1. Analyze unit generation processes to capture and communicate best practices.

1.5.3.1.2. Analyze unit generation performance to identify and communicate noteworthy trends.

1.5.3.2. Directorates of Logistics, Engineering & Force Protection (MAJCOMs A4s) will semi-annually report their top three trends and causal factors to the Logistics Board to facilitate debate to identify and mitigate potential limiting factors. Submit MAJCOM top

three trends to the AF/A4LX workflow no later than 30 days prior to each Logistics Board meeting for consolidation at: **AF.A4LX.Workflow@us.af.mil**.

1.6. Maintenance Discipline. It is the responsibility of all maintenance personnel to comply with all written guidance to ensure required repairs, inspections, and documentation are completed in a compliant, safe, timely, and effective manner. Supervisors are responsible for enforcing and establishing a climate that promotes maintenance and supply discipline. Unless expressly stated otherwise in a particular instruction, waiver, or deviation in this DAFI granted by the appropriate authority, all Airmen must follow AFIs. AFIs do not provide optional guidance, and failure to comply with DAFIs can result in disciplinary action as described in AFI 1-1, *Air Force Standards*. Civilian personnel who violate punitive publications may also be subject to disciplinary action. See DAFMAN 90-161 for more detailed information on the use of punitive language in publications.

1.6.1. Compliance Terminology. For the purposes of this instruction, the following definitions apply:

1.6.1.1. Must, Will - Indicates mandatory requirements. **Note:** "Will" is also used to express a declaration of purpose for a future event.

1.6.1.2. Should - Indicates a preferred method of accomplishment.

1.6.1.3. May - Indicates an acceptable or suggested means of accomplishment.

1.6.2. Use of TOs and TO Supplements. All personnel will enforce compliance with technical data. **(T-2)** Use of prescribed technical data to maintain aircraft and equipment is mandatory and will be conducted and managed IAW TO 00-5-1, *Air Force Technical Order System*.

1.6.3. AFTO Form 492, *Maintenance Warning Tag*. The Air Force Technical Order (AFTO) Form 492, *Maintenance Warning Tag*, is used as prescribed in technical data, local procedures, or both, to flag a condition that could cause damage or injury if ignored. Refer to TO 00-20-1 for additional guidance. **Note:** The AFTO Form 492 is replacing the AF Form 1492, *Warning Tag* referenced in DAFMAN 91-203, Air *Force Occupational Safety, Fire, and Health Standards*. Use of the AF Form 1492 is authorized until supplies are exhausted.

1.7. Communications Security (COMSEC)/Controlled Cryptographic Item (CCI). The Air Force CCI Central Authority is AFLCMC/HNC, Cryptologic and Cyber Systems Division (CCSD), Joint Base San Antonio-Lackland, Texas.

1.7.1. COMSEC/CCI will be managed, and personnel trained IAW AFMAN 17-1302-O, *Communications Security (COMSEC) Operations* and AFI 23-101, *Air Force Material Management*. Ensure all serially controlled and serially tracked COMSEC/CCI information is entered into the Maintenance Information System (MIS) IAW TO 00-20-2, *Maintenance Data Documentation*. **Note:** AFMAN 17-1302-O is accessible through the AF e-Pubs Warehouse Management System (WMS) at **https://wmsweb.afncr.af.mil/wms**.

1.7.1.1. Maintain serial number inventory accountability for all COMSEC/CCI issued or removed to Facilitate Other Maintenance (FOM).

1.7.1.2. Questions concerning COMSEC/CCI can be directed to the AF CCI Central Authority (**aflcmc.hncls.cciworkflow@us.af.mil**).

1.7.2. Maintenance Cybersecurity Discipline.

1.7.2.1. Maintaining positive maintenance cyber discipline practices of Department of Defense (DoD) Information Technology (IT) is critical to sustaining the mission. Department of Defense Instruction (DoDI) 8500.01, *Cybersecurity*, defines both hardware and software that is physically part of, dedicated to, or essential in real-time to the mission assurance of special purpose weapon systems. DoD IT is the most common IT encountered in flightline environments and includes (but is not limited to) electronic tools, aircraft, and associated support equipment. The culture of positive cybersecurity awareness and actions necessary to sustain cyber resiliency is required by all maintenance personnel to mitigate allusive cyber threats and optimize enduring mission generation capabilities.

1.7.2.1.1. All users perform an integral role in prevention, detection, and reporting suspected corrupted software of DoD IT which includes Information Systems (IS) and Platform Information Technology (PIT), which is an electronic platform with information technology for a specific function. See **Table 1.1**.

Table 1.1. Tiered Interface Examples.

TIER	Type of Interface	Examples	Applicable TOs
1	On-Board	EC-130, XX-135, E-3, E-8	Device TO
2	Directly Connected	F-22 PMA, F-35 PMA, Viper MLV, CAPRE, CETS, DTADS	Device TO
3	Indirectly Connected	Test Eqpt., Support Eqpt., ATS, ATE, AIS, VDATS	TO 33-1-38
4	Not Connected	eTools	TO 00-5-1

1.7.2.1.1.1. All users must consult airframe Security Classification Guides, TO 33-1-38, *Cybersecurity for Automatic Test Equipment and Support Test Equipment (ATE/STE)*. TO 33-1-38 provides guidance for Cybersecurity Incident Reporting and refers users to applicable technical manuals, instructions, and publications when determining the classification of cybersecurity incidents and vulnerability documents.

1.7.2.1.2. Authorized and unauthorized uses of IT and PIT. All users must have the ability to distinguish between authorized and unauthorized uses.

1.7.2.1.2.1. Authorized uses must be vetted through a formal cybersecurity assessment process and be directed in specific TO guidance. **(T-2)** The governing TOs or equivalent publications specifically define authorized uses.

1.7.2.1.2.2. Unauthorized uses include connecting any hardware, uploading, or downloading software, or media not explicitly defined by TOs. This includes but is not limited to personal devices, phones, tablets, computers, Universal Serial Bus drives, and similar devices.

1.7.2.1.2.3. DoD IT, derivative AF Publications, and TOs provide users guidance on Automated Computer Program Identification Number System devices acquired from local Communications Squadrons and media obtained from DoD contractors.

1.7.2.1.2.4. All users will follow the applicable TO when directly or indirectly

connecting computers and equipment to the aircraft or support equipment and when uploading or downloading data. **(T-2)**

1.7.2.1.2.5. All users will immediately discontinue use, report, and turn into the appropriate functional authority IT and PIT (WAM, Wing Cybersecurity Office, and the Mission Defense Teams/Cyber Squadron, if assigned) that are suspect for containing malicious software, malicious code, software bugs or unauthorized use. **(T-2)**

1.7.2.1.2.6. All users will complete Maintenance Cyber Discipline Training annually in MyLearning. **(T-2)**

1.7.2.1.3. MDS Lead Commands in coordination with the applicable PM will develop MDS and Support Equipment (SE) cyber threat mitigation methods and procedures for Organizational and Intermediate level maintenance activities. The methods and procedures must detect malicious code, report cyber incidence and issues, and remediate the incidence and issue affecting the MDS or SE. **Note:** Mitigation plan should be developed per DoDI 8500.01; TO 33-1-38, DoD 8570.01, *Information Assurance Workforce Improvement Program*; Military Standard (MIL-STD)-38784A, *General Style and Format Requirements For Technical Manuals*; 17- series AFIs.

1.7.2.1.3.1. The cyber threat mitigation methods must include MDS and SE specific training requirements. **(T-2) Note:** Training requirements could include training aids, for example, computer-based training on how the flightline maintainer should scan support equipment for malicious software.

1.7.2.1.3.2. Lead Commands must ensure current MDS and SE specific malicious code definitions are available to ensure positive cyber threat mitigation management support is available. **(T-2)**

1.7.2.1.3.2.1. MAJCOM and ANG will report system-specific cyber incidents to the applicable MDS Lead Command.

1.7.2.1.3.2.2. MAJCOM and ANG will follow airframe Security Classification Guides, TOs, and applicable technical manuals when providing “cross tell” to inform their subordinate units about system-specific cyber incidents, threats, and issues.

1.7.3. eTools.

1.7.3.1. eTools are portable electronic devices (such as laptop computer, handheld device) that operate in a disconnected mode and are certified to inter-operate on AF networks. eTools are mission critical: the primary purpose is for viewing electronic technical publications and in some cases are used to exchange maintenance data with approved MIS at the point of maintenance. eTools are procured IAW DAFMAN 17-1203, *Information Technology (IT) Asset Management (ITAM)*. **Note:** eTools do not include electronic devices and test equipment issued and configuration managed by a system PM (aircraft test and support equipment).

1.7.3.2. MAJCOM/ANG A4s will develop and implement standardized guidance on the management, use, storage, configuration, content update, security, and cyber hygiene processes necessary to support the approved use of all assigned eTools consistent and IAW

the weapon system MDS specific technical orders and threat specific Air Force and DoD cyber publications.

1.7.3.3. The MAJCOM/ANG A4 guidance will include any assigned command-wide cyber threat awareness and mitigation strategies with reference to supporting publications, technical orders, and MAJCOM eTool OPR contact information in their supplement or addendum to this DAFI.

1.8. Environmental Compliance. It is the responsibility of all maintenance personnel to comply with all written guidance to ensure compliance with hazardous material, hazardous waste management and air emissions record keeping as required for environmental compliance IAW AFI 90-821, *Hazard Communication (HAZCOM) Program*, installation Environment, Safety, and Occupational Health Management System/Environment Management System (ESOHMS/EMS) policy/guidance and applicable environmental requirements and guidance.

1.9. Publications. Units may tailor procedures to the unique aspects of their own maintenance operation and publish directives, instructions, supplements, addendums, and, for functional areas, Operating Instructions (OI) IAW DAFMAN 90-161.

1.9.1. Develop, control, and maintain functional and emergency action checklists. At a minimum, each checklist is titled, dated, and coordinated with the wing safety office. Functional checklists are not to be used in place of or to circumvent technical data for operation, servicing, inspection or maintenance of aircraft, aircraft systems, munitions, and all other equipment supporting aircraft and munitions maintenance.

1.9.2. Methods and Procedures Technical Orders (MPTOs): Due to the close relationship between MPTOs and this DAFI, all changes and revisions to the MPTOs cited in **Attachment 1,** References of this DAFI will be routed from Air Force Material Command (AFMC) to AF/A4LM for content review for conflicts and policy gaps identification and mitigation prior to submission for publication.

1.10. Maintenance Training. Maintenance training provides initial, recurring, and advanced proficiency, qualification, or certification skills needed by a technician to perform assigned duties. Maintenance training includes combat and sortie generation skills not normally integrated into peacetime operations (such as, Multi-Capable Airmen, munitions handling, and external fuel tank build-up, hot refueling). Maintenance training carries an equal priority with the operational training mission. For maintenance training policy and guidance, refer to AFI 36-2650, *Maintenance Training* and MAJCOM supplements.

1.11. Modification Management. A modification proposal is a recommendation to change the operation, use, or appearance of AF equipment. Modifications (temporary, permanent, or safety) to AF aircraft or equipment are expressly prohibited without PM approval. **Note:** PM is used in this publication as defined in AFPD 63-1, *Integrated Life Cycle Management*. Refer to AFI 63-101/20-101, for modification management procedures.

1.11.1. Modifications to Munitions. All proposed modifications to aircraft-carried munitions include AFI 63-101/20-101 and SEEK EAGLE certification IAW AFI 63-101/20-101 and Air Force Pamphlet (AFPAM) 63-129, *Air System Development Process and Procedures*. All modifications to AF nuclear munitions or their associated support and training equipment are nuclear certified IAW AFI 63-125, *Nuclear Certification Program*. All modifications to AF

conventional munitions or their associated support and training equipment are certified IAW DAFI 91-205, *Non-Nuclear Munitions Safety Board*.

1.12. Maintenance Information System (MIS). MIS refers to automated maintenance information systems that support and enable maintenance business processes. The MIS is used to document maintenance actions and track fleet health. The information entered into the MIS is accomplished IAW TO 00-20-2 and matches the content of the aircraft forms. MIS data entries do not have to be accomplished by the same individual who documented the aircraft forms, but employee numbers, man numbers, and User IDs of individuals accomplishing the actual work are entered into the MIS. Red Ball maintenance is documented IAW **Chapter 11**. Data integrity is the responsibility of every member of the unit. All personnel are responsible for ensuring accuracy and completeness.

1.12.1. Units use the approved MIS for their assigned weapon system.

1.12.2. Serial numbers will be documented in the MIS for all serially controlled and tracked assemblies that are identified by an asterisk in the Work Unit Code (WUC) or Logistics Control Number (LCN) manual IAW TO 00-20-2.

1.13. General Safety Guidance. Maintenance personnel are exposed to a large variety of hazardous situations, machinery, equipment, and chemicals. Most hazardous situations can be avoided by following approved procedures, asking for assistance when needed, and using all required personal protective equipment (PPE).

1.13.1. Safety "Knock It Off" and Risk Management. Due to the inherent danger to life, limb, and property associated with maintenance operations, personnel are empowered to terminate an operation or situation which they perceive is unsafe or too dangerous. When supervisors or crew leaders become task-focused, junior personnel are often better able to assess the danger; however, deferring to the experience and judgment of the supervisor or crew leader, they may choose to remain silent, missing an opportunity to break the mishap sequence chain. Maintenance commanders and supervisors are responsible for fostering a culture in their units so that a simple, but recognizable "audible" from anyone can prevent a potential mishap. **Note:** See AFI 90-802, *Risk Management*, and DAFPAM 90-803, *Risk Management (RM) Guidelines and Tools* for additional information.

1.13.2. Visitors. Unit Commanders will not permit visitors to operate any AF equipment unless they are qualified to operate such equipment and are doing so in the performance of their assigned official duties. **(T-2)** Visitors will not be allowed in the flightline area if munitions operations are present IAW DESR6055.09_AFMAN 91-201, *Explosive Safety Standards*.

1.14. Duty Shifts and Rest Periods. MXG/CC or equivalent will establish minimum requirements that ensure units maintain an equitable distribution of supervision (Officer and Senior Non-Commissioned Officer (SNCO)) across all on-duty shifts in their supplement to this DAFI. **(T-3)**

1.14.1. During normal operations, maintenance personnel are scheduled for duty based on a 40-hour work week. Maintenance personnel duty hours are aligned to provide optimal mission support.

1.14.2. Personnel will not be scheduled for more than 12 hours of continuous duty time. **(T-3)** Duty time begins when personnel report for duty and ends when their supervisor releases them. Time spent in exercise or contingency deployment processing lines and in-transit counts toward the total duty day. SQ/CCs are the final approval authority for duty time extensions exceeding the 12-hour limit up to a maximum of 16 hours. **(T-3) Note:** Aircraft or detachment commanders assume this responsibility in Temporary Duty (TDY) or travel status.

1.14.3. Commanders and supervisors will provide a rest period after each shift. **(T-3)** A rest period is a block of time that gives a person the opportunity for 8 hours of uninterrupted sleep in a 24-hour period.

1.14.4. When handling, loading, or performing maintenance on nuclear weapons, conventional munitions, or egress explosives, personnel will follow requirements IAW AFMAN 21-200.

1.14.5. In alert force or standby duty situations where facilities are available for resting, established norms may be exceeded. Adjust rest periods to allow for 8 hours of uninterrupted sleep.

1.14.6. Commanders and supervisors will ensure individuals are afforded adequate duty rest periods and breaks to prevent fatigue or thermal injury. Stop anyone if fatigue may jeopardize safety. In all cases, aircraft commanders or supervisors ensure aircraft maintenance personnel are not required to perform duty when they have reached the point of physical or mental fatigue rendering them incapable of performing their assigned duties safely and reliably.

1.14.7. MAJCOM Commanders will assume the risk for any Flying Crew Chief (FCC) deviations from normal duty shifts, rest periods, and exceedance of the 16-hour maximum duty day. MAJCOM Commanders will publish risk-mitigation procedures in order to meet critical mission requirements.

1.15. Communications. Commanders will develop communication plans according to AF mission requirements. See **Chapter 11** for detailed communication requirements.

1.15.1. Effective maintenance accomplishment requires the ability to communicate across all facets of the maintenance operation efficiently and effectively. Communication technology (government issued Portable Electronic Devices (PED), Portable Maintenance Aid (PMA), radios, cell phones, computers, wireless internet) must be available to expedite personnel, equipment, material, and maintenance data throughout the maintenance complex. **(T-2)**

1.15.2. MAJCOMs will publish guidance for the use and control of personal communications devices (personal cell phones, tablets, and computers) on the flightline, in munitions areas, hangars, and other industrial work areas as required to mitigate cyber risk and ensure compliance with cyber and communication AFI requirements relevant to their operational environments.

1.15.2.1. **(Added-AFMC)** Cell Phones/Cameras in Industrial Areas. Cell phones/cameras are prohibited when directed by safety/classified/special access requirements or TOs. Note these areas will be clearly marked with signs so it is obvious to all that it is an electronic device/cell phone restricted area. **(T-2)**

1.15.2.2. **(Added-AFMC)** Except as noted in **paragraph 1.15.2.1**, use of a privately owned cell phone or communication devices are authorized in the industrial/production

areas except while performing hands-on maintenance activities. Employees will remove themselves from any maintenance activity, aircraft, or high traffic areas prior to using a communication device. **(T-2)**

1.15.2.3. **(Added-AFMC)** Use of portable music/video players, electronic games or other personal electronic hearing impairing devices are authorized in industrial production areas except where audio warning cues are used. Prohibited items do not include, stereos and other sizable devices**.**

1.15.3. Unit Commanders will enforce procedures that prohibit the introduction of government or personal cellular, personal communications system, Radio Frequency (RF), Infrared (IR) wireless devices, and other devices such as cell phones, tablets, and devices that have photographic or audio recording capabilities into areas (e.g., rooms, offices) where classified information is stored, processed, or discussed IAW AFMAN 17-1301, *Computer Security (COMPUSEC)*. Coordinate waiver requests with the applicable Approving Official (AO), and ensure adherence to Certified TEMPEST Technical Authority (CTTA) requirements IAW Department of Defense Directive (DoDD) 8100.02, *Use of Commercial Wireless Devices, Services, and Technologies in the DoD Global Information Grid (GIG)* written approval by the AF CTTA IAW NIST SP 800-53A Revision 4, *Assessing Security and Privacy Controls in Federal Information Systems and Organizations: Building Effective Security Assessment Plans*, and the Enterprise Authorizing Official (AO) IAW DAFI 31-101, *Integrated Defense (ID)* and AFMAN 17-1301.

1.16. Maintenance Repair Priorities. Maintenance repair priorities are listed in **Table 1.2**. This does not prohibit the Production Superintendent (Pro Super), in coordination with the Maintenance Operations Center (MOC), from changing the maintenance repair priority when warranted. During tasked Operational Plan (OPLAN) or operational exercise, the pre-planned maintenance flow determines the job sequence. The maintenance repair priority and the Logistics Readiness Squadron (LRS) delivery priorities (listed in Air Force Handbook (AFH 23-123V, *Materiel Management Reference Information*) are normally identical. Raising or lowering maintenance repair priorities does not necessarily require a corresponding change in the LRS delivery priority. However, the Pro Super may authorize the use of a less responsive LRS delivery priority.

Table 1.2. Maintenance Repair Priority Designators.

PRIORITY	APPLICATION
1	Aircraft on alert status, war plan or national emergency missions, including related Aerospace Ground Equipment (AGE), munitions and Munitions Support Equipment.
2	Primary mission aircraft, related AGE, munitions, and munitions support equipment, for the first 8 work hours after landing or start of recovery or within 6 work hours of a scheduled launch, alert or test flight and during simulated generation or Operational Readiness Exercises (ORE). Air evacuation, rescue, weather (WX) mission aircraft, related AGE, munitions, and munitions support equipment. All transient support, and FAA aircraft. Flight or missile crew training simulator, other training equipment or related AGE required repair, which is impacting the mission by preventing or delaying student training.
3	Primary mission aircraft, engines, air launched missiles and related AGE, munitions and munitions equipment, and equipment undergoing scheduled or unscheduled maintenance, if not performed or repaired will prevent or delay mission accomplishment. Transient air vehicles not otherwise listed. Administrative aircraft within 8 hours of scheduled flight or on alert status with standby crews. Time change requirements for nuclear weapons. Repair cycle assets to satisfy a Mission Impaired Capability Awaiting Parts (MICAP) condition. Spares not available in supply. Critical end items and spares not available in supply. Routine maintenance of aircrew or missile-training simulator, or other training devices or related AGE or sites and aircraft or equipment used for maintenance training. Avionics shop electronic AGE and automated test stations.

4	Routine or extensive repair of primary air mission and related AGE and repair cycle assets. Administrative aircraft undergoing scheduled or unscheduled maintenance. Routine maintenance of AGE not otherwise listed above. War Reserve Materiel (WRM) items due maintenance or inspection. Inspection, maintenance, and Time Compliance Technical Order (TCTO) compliance of Mission Support Kit or Mobility Readiness Spares Package (MRSP) materiel. Extensive repair of aircrew or missile training simulators, other training devices, or related AGE. Inspection, maintenance, and TCTO compliance of munitions and munitions equipment, excluding spares excess to base requirements not listed above. Scheduled calibration and unscheduled repairs on Precision Measurement Equipment (PME) not listed above. Scheduled maintenance to include periodic inspections, routine TCTO, Master Configuration Lists (MCL) Grounding, and Time Change Items (TCIs). Primary mission Comprehensive Engine Management System (CEMS) or equipment including associated AGE undergoing extensive repair or modification.
5	Non-tactical or non-primary-mission aircraft undergoing extensive repair. Fabrication and repair of aeronautical items not carrying a higher priority. Bench stock requirements. Extensive repair of aircrew training devices. Time change requirements not listed above. Routine repair of AGE and repair cycle assets. Alternate and other CEMS or equipment, including associated AGE undergoing extensive repair or modification. Clearing routine delayed discrepancies on training equipment or AGE, and routine maintenance which will not impair or affect mission accomplishment. Equipment requirements.

6	Fabrication and repair of non-aeronautical items. Repair cycle asset shortages required to fill a peacetime operating stock authorization
7	Spares/repair cycle assets excess to base requirements.

1.17. Associate Unit Program/Total Force Integration (TFI). The USAF employs the Associate Unit/TFI program in some locations where RegAF and Air Reserve Component (ARC) units are collocated and share aircraft, equipment, facilities, and other resources IAW AFI 90-1001, *Planning Total Force Associations (TFAS)*, and MAJCOM supplements. For the purpose of this instruction, in an Active Association, the ANG or AFR owns the aircraft, and RegAF personnel will follow ANG or AFR maintenance policy. In an ARC association, AFR owns the aircraft, and ANG personnel will follow AFR guidance, or vice versa. In a classic association, RegAF owns the aircraft, and ANG or AFR personnel will follow RegAF maintenance policy. Type of association is determined by the Program of Record for the associated unit.

1.18. Performance-Based Activities. MAJCOMs may publish the basic responsibilities for managing performance-based activities.

1.18.1. If published MAJCOMs will:

1.18.1. **(AFMC)** Performance-based activities Air Force Materiel Commend (AFMC) Manpower Requirements Branch (AFMC/A1MR) are coordinated/collaborated with the appropriate HQ Directorate of Logistics, Civil Engineering, Force Protection, and Nuclear Integration (AFMC/A4/10) MAJCOM Functional Manager (MFM)/Functional Area Manager (FAM) and the Manpower and Personnel representatives from within the affected AFMC Center.

1.18.1.1. Designate focal points for organizational, functional, and technical questions pertaining to each performance-based activity program.

1.18.1.2. Specify measurement areas and performance levels required for aircraft, systems, and equipment operated or maintained by performance-based activities.

1.18.1.3. Specify the forms, methods of documentation, and frequency of reporting used to assess performance-based activities and ensures these requirements are included in the Quality Assurance Surveillance Plan (QASP).

1.18.1.4. Ensure units with assigned Contracting Officer Representative personnel meet requirements in AFI 63-138, *Acquisition of Services*.

1.18.1.5. Ensure aircraft depot maintenance contracts, Statements of Work (SOW), and Performance Work Statements (PWS) are coordinated with the applicable MAJCOM Functional, to include Munitions.

1.18.2. Unit Commanders will:

1.18.2.1. Designate a focal point for all functional, technical, and contracting officer representative matters pertaining to performance-based activities. **(T-3)**

1.18.2.2. In coordination with the contracting officer and the Program Manager of Functional Services Manager, provide specific guidance to the performance-based activity to ensure proper maintenance discipline and flight worthiness of aircraft and subsystems. **(T-3)**

1.18.2.3. Develop and publish contingency procedures for support of continuing operations in the event of disruption, termination, or default of contracts. **(T-3)**

1.19. Changes to Technical Orders. Official TO updates are the only valid authority for maintaining TO accuracy and currency. To maintain this currency, the AF has instituted an enterprise electronic TO Recommended Change (RC) process.

1.19.1. The Enhanced Technical Information Management System (ETIMS) is the required system for Recommended Change management and this process replaces the AFTO Form 22, *Technical Manual (TM) Change Recommendation and Reply*, AFTO Form 252, *Technical Order Publications Change Request* and the DAF Form 847.

1.19.2. There are specific organizations that may require the deviations or exceptions for those who do not have access to ETIMS. Refer to TO 00-5-1 for specific guidance on the RC process.

Chapter 2

ROLES AND RESPONSIBILITIES

2.1. General. This chapter outlines responsibilities for commanders and key leaders involved in maintenance activities. Decisions made that increase risk to safety of flight will be communicated with operations. In units where there is not a military commander responsible for maintenance, the applicable civilian Director of Maintenance (DOM) will ensure compliance with all responsibilities in this instruction. For organizations without all commanders and key leaders assigned, MAJCOMs will identify equivalent positions of authority commensurate with the responsibilities of the leadership positions identified in this **Chapter** in a MAJCOM supplement to this instruction.

2.1. (AFMC) General. The civilian equivalent (unless no civilian may hold the position) may be applied to any reference to a military rank, grade or position.

2.1.1. For the purpose of this instruction, contractor equivalents are as follows: A1C—aircraft servicer or apprentice/journeyman; SrA (1-year time-in-grade)—aircraft worker or field maintenance worker or higher; SSgt—aircraft mechanic or field maintenance mechanic or higher; TSgt—senior mechanic or craftsman; MSgt—lead mechanic; SMSgt/CMSgt/maintenance officer—foreman, branch chief or higher. MAJCOMs may determine grade and skill level equivalents for civilians.

2.1.2. The functional authority to determine the need and design for civilian uniforms, under the purview of maintenance is delegated to the MAJCOMs/ANG. As such, all functional authority requirements set forth in AFI 36-128, *Pay Settings and Allowances* will also reside at the respective MAJCOM. The routing requirements for final approval of uniform requests and the establishment of uniform allowances will be IAW AFI 36-128.

2.1.3. **(Added-AFMC)** MAJCOM Functional Managers will support the applicable Program Managers Post-Milestone C (the point at which a program is reviewed for entrance into the production and deployment phase) sustainment and product support teams, (e.g., life cycle logistics, systems engineering; Research, Development, Test, and Evaluation; and maintenance management). IAW DAFI 63-101/20-101, *Integrated Life Cycle Management,* paragraph **5.4.21.4.**

2.2. Wing Commander (WG/CC) Responsibilities. The WG/CC allocates resources to meet all mission requirements. The WG/CC will:

2.2.1. Ensure that maintenance organizations are not overtasked with augmentation duties outside maintenance functional areas. **(T-2)**

2.2.2. Conduct a "Wing Standup" meeting. **(T-3)** The MAJCOM/ANG will establish “Wing Standup” meeting frequency that synchronizes communications necessary to optimize fleet readiness and mission generation capability in their supplement to this DAFI.

2.2.2. **(AFMC)** [DEV] Daily “Wing Standup” meeting may be delegated to the MXG/CC or OG/CC. At a minimum, the WG/CC will chair this meeting once a week. **(T-2)**

2.2.3. Ensure a coordinated wing or base instruction is developed that implements procedures to control tools, equipment, electronic devices, and establishes cyber discipline and reporting

requirements that provide operational guidance across all wing or base agencies dispatching to aircraft runway, taxiways, parking, and maintenance areas. **(T-2)**

2.2.3.1. Ensure home station Base Support Plans (BSP) include contingency bare base, comm-out, and cyber-out operating procedures. Periodically exercise operations (as part of scheduled local exercises) in this simulated deployed environment to validate equipment, personnel, and processes provide the required mission generation capability. **(T-2)**

2.2.3.1.1. Develop and maintain local cyber abatement and status reporting procedures that optimize cyber resiliency and the ability to mitigate and recover from cyber threats affecting assigned mission or sortie generation operations. **(T-2)**

2.2.3.1.2. Ensure Wing Cybersecurity Office directs Wing cybersecurity compliance per AFI 17-101, Risk Management Framework (RMF) for Air Force Information Technology (IT), specifically to address cybersecurity assurance of Wing aircraft and support equipment categorized as Operational Technology (OT). **(T-2) Note:** Currently known as Platform Information Technology (PIT).

2.2.4. Ensure maintenance and operations develop a joint annual maintenance and Flying Hour Program (FHP) that establishes a balance between the requirement for sorties and maintenance capability. **(T-2)**

2.2.4.1. Establish a joint MXG and OG planning and scheduling cycle to ensure the best use of aircraft, equipment, and personnel to accomplish short-term sortie production and long-term fleet health. **(T-3)**

2.2.4.2. Approve the weekly; monthly, quarterly, and annual flying or test schedules IAW **Chapter 14**. **(T-3)**

2.2.4.3. DELETED.

2.2.5. Vector future leaders to attend the Senior Leader Mission Generation (SLMG) Course following the course description and criteria listed in **Chapter 11**. **(T-2)**

2.2.6. Ensure a Crashed, Damaged, or Disabled Aircraft Recovery (CDDAR) capability exists for assigned active airfields and runways IAW **Chapter 11** and develop a wing publication IAW DAFMAN 90-161 containing specific responsibilities for all applicable base support agencies. **(T-2)**

2.2.6.1. Collaborate with Fire Emergency Services, Wing Safety, CES, LRS, SFS, MDS, OSS, and other on-/off-base agencies to develop a publication IAW DAFMAN 90-161, that assigns specific responsibilities and procedures to implement a sustainable CDDAR program IAW TO 00-80C-1. Ensure wings with GSU/auxiliary fields outline support requirements in their publication.

2.2.7. Identify specific responsibilities and outline unique materiel management support requirements necessary to optimize wing level maintenance and mission generation operations. Establish processes and responsibilities for maintenance units and work centers without material management support in a local publication. **(T-2)**

2.2.8. Approve hot refueling site certification(s). **(T-3)**

2.3. Wing Vice Commander (WG/CD) Responsibilities. The WG/CD (or equivalent) will:

2.3.1. Manage the Foreign Object Damage (FOD) and Dropped Object Prevention (DOP) Programs. **(T-2)** The WG/CD is the FOD/DOP Prevention Program Manager and will appoint a FOD/DOP Prevention Monitor(s) IAW **Chapter 11**.

2.3.2. Ensure the wing safety office, base operations, and emergency services actively participates in development and coordination of base functional and emergency action checklists. **(T-2)** Ensure content includes and complies with all required safety standards, evacuation distances and MDS specific hazards (such as, egress systems, Hydrazine). **(T-2)** Ensure checklists address all known hazards associated in responding to mishaps, adverse weather, natural disaster, and other emergency responses. **(T-2) Note:** Review, update, and coordinate functional and emergency action checklist with using units every two years.

2.3.3. Chair the FOD/DOP meeting, if required, and will determine minimum required attendees. **(T-3)** The purpose of this meeting is to identify negative trends and develop and execute action plans to resolve them.

2.4. Maintenance Group Commander (MXG/CC) Responsibilities. In addition to the responsibilities listed below, the MXG/CC or equivalent must ensure compliance with the maintenance requirements and programs of this instruction. **(T-3)** Approved variations in the MXG organization does not relieve the MXG/CC of compliance with all the requirements of this DAFI. In organizational structures where an MXG/CC is not assigned (i.e., AFFORGEN deployed construct), all 2.4.1. paragraphs will move up to the WG/CC and all 2.4.2. paragraphs will move down to the Mission Generation Force Element (MGFE) leader, delegable to the senior maintenance leader. Additionally, all 2.4.3. paragraphs' responsibilities will remain with the home unit's MXG/CC, or WG/CC in the absence of an MXG/CC, and any guidance or documentation necessary to support the mission will be brought with the unit.

2.4.1. The MXG/CC will:

2.4.1.1. Ensure standardization of maintenance discipline, procedures, organizational structures, compliance, and management philosophy. **(T-3)**

2.4.1.2. Establish a radiation protection program IAW AFI 48-109, *Electromagnetic Field Radiation Occupational & Environmental Health Program,* when applicable.

2.4.1.3. Designate a focal point for all functional, technical, and COR matters pertaining to performance-based activities. **(T-3)** Refer to **Chapter 1**.

2.4.1.4. Ensure munitions are accounted for IAW DAFMAN 21-201, *Munitions Management*, AFMAN 21-200, *Munitions and Missile Maintenance Management*, AFMAN 21-203, *Nuclear Accountability Procedures*, and AFI 20-110, *Nuclear Weapons-Related Materiel Management*.

2.4.1.5. Ensure a nuclear surety program is implemented (if applicable) IAW DAFI 91-101, *Air Force Nuclear Weapons Surety Program*, and nuclear munitions are maintained, handled, and accounted for IAW AFMAN 21-204, *Nuclear Weapons Maintenance*, AFMAN 21-203.

2.4.1.5.1. For units possessing Nuclear Certified Equipment (NCE), the MXG/CC will ensure personnel are trained in the proper use of nuclear flagwords, mishap, and deficiency reporting instructions IAW AFMAN 91-221, *Weapons Safety Investigations and Reports* and DAFI 91-204, *Safety Investigation and Reports*.

2.4.1.6. Establish CDDAR capability IAW **Chapter 11** and applicable MDS technical data. **(T-2)** The MXG/CC will ensure resources and trained personnel are available to perform responsibilities of the CDDAR Program. **(T-3)**

2.4.1.7. Sustain a Transient Alert (TA) function (if required). **(T-3)** The MXG/CC will establish procedures and furnish necessary personnel and facilities for handling transient aerospace vehicles to ensure that servicing, inspection, and maintenance are consistent with the mission of each transient aerospace vehicle. **(T-3)** Special consideration should be given to medical or air evacuation aerospace vehicle, emergency missions, and special missions.

2.4.1.7.1. Approve FCF requirements, submitted by the TA Section NCOIC/Chief to off-station TA and off-station QA sections. **(T-3)**

2.4.1.7.2. For transient alert, the MXG/CC may direct the WS Personnel to arm, de-arm, and unload an aircraft on which they are not certified and/or qualified if appropriate technical data and support equipment is available. **(T-3)**

2.4.1.7.2.1. The host MXG/CC or authorized representative may delegate this responsibility to the transient aircraft commander/pilot for new or experimental aircraft. **(T-3)**

2.4.1.7.3. **(Added-AFMC)** When issuing or renewing a TA contract, owning/requiring activity will use the Standardized TA Services Performance Work Statement. POC is 763d Enterprise Sourcing Squadron, Contracting (763 ESS/PKB). **(T-2)**

2.4.1.8. Direct QA to develop/manage local OTIs IAW TO 00-20-1 and this instruction for Aircraft Grounding. **(T-3)**

2.4.1.9. Endorse TCTO notification messages to MAJCOM when anticipated problems will prevent compliance within prescribed time limits. **(T-3)**

2.4.1.10. Ensure a read file or equivalent exists for distributing maintenance crosstalk messages, QA newsletters, Higher Headquarters (HHQ) and local policy announcements, technical notifications, and other important maintenance information to all assigned airman. **(T-3)**

2.4.1.11. Ensure direct mission generations support focuses on readiness and the unit's ability to execute daily and deployed flying operations simultaneously. **(T-3)**

2.4.1.12. Chair the FOD meeting in the absence of the WG/CD. **(T-3)**

2.4.1.13. Ensure maintenance requirements (such as, aircraft turnaround, alternate fuel cell, hot refueling, end-of-runway (EOR) check area, engine run spots, explosive load (cargo) areas) are included in the base parking plan. **(T-3)**

2.4.1.14. Communicate constraints with OG to decide what portion of the mission to support and to what degree if conflicts arise between operational requirements and maintenance capability. **(T-3)**

2.4.2. The MXG/CC will:

2.4.2.1. Appoint an MXG Environmental Coordinator IAW AFI 32-7001, *Environmental Management.* Refer to AFI 90-821, installation ESOHMS/EMS policy, guidance, and applicable environmental requirements and guidance, DAFMAN 91-203, and AFMAN 32-7002, *Environmental Compliance and Pollution Prevention*, for additional guidance.

2.4.2.2. Establish and support a Data Integrity Team (DIT). **(T-3)**

2.4.2.3. Ensure the Maintenance Standardization and Evaluation Program (MSEP) requirements are implemented IAW **Chapter 6**. **(T-3)**

2.4.2.4. Establish measures that ensure all maintenance personnel are assigned IAW the Duty Title Tool, are available, and utilized to accomplish critical maintenance tasks necessary to integrate maintenance capabilities that optimize Aircraft Availability (AA). **(T-3)** Maximize utilization of 7-skill level maintenance personnel in the grade of E-5 to E-7 in direct mission generation roles and minimize their use in staff positions or non-maintenance duties. **(T-3)** Note: Consider utilization of Civil Service MXG/SQ Unit Program Coordinators to consolidate programs to maximize availability of sortie generation maintainers on the flightline to maximize AA.

2.4.2.5. Implement an effective Corrosion Prevention and Control Program IAW TO 1-1-8, Application and Removal of Organic Coatings, Aerospace and Non-Aerospace Equipment; TO 35-1-3, Corrosion Prevention and Control, Cleaning, Painting, and Marking of USAF Support Equipment; TO 1-1-691 Cleaning and Corrosion Prevention and Control, Aerospace and Non-Aerospace Equipment.

2.4.2.6. Ensure effective use of the assigned AF Engineering and Technical Services (AFETS), Contractor Engineering and Technical Services (CETS), and contracted Field Service Representatives (FSRs) IAW **Chapter 11**.

2.4.2.7. Ensure the applicable section “safes” all static display aircraft/systems IAW the applicable 00-80-series and weapon system-specific TOs.

2.4.2.8. Ensure the MXG Oil Analysis Program (OAP) complies with **Chapter 11**. **(T-3)**

2.4.2.9. Ensure Engine Health Management Plus (EHM+) duties are performed IAW AFMAN 20-116 for EHM products managed by Air Force Life Cycle Management Center AFLCMC/LP. If EHM data is not managed by AFLCMC/LP, ensure Contractor Logistics Support (CLS) provided EHM products are managed according to respective contract. **(T-3)**

2.4.2.10. Appoint a hot refueling/hot defueling OPR for the Wing (WG) and designate an OPR for hot refuel training (if applicable). **(T-3)**

2.4.2.11. Ensure unit personnel collect and report Aircraft Structural Integrity Program (ASIP) data IAW DAFI 63-140, *Aircraft Structural Integrity Program and Air and Space Equipment Structural Management*, and **Chapter 11**. **(T-3)**

2.4.2.12. Ensure aircraft and equipment sunshades are maintained IAW **Chapter 15**. **(T-3)**

2.4.2.13. Ensure management of the Weight and Balance (W&B) program IAW **Chapter 6**. **(T-3)**

2.4.2.13. **(AFMC)** MXG is the Weight and Balance (W&B) authority. The W&B authority is not required to be W&B certified if they are not performing weight and balance tasks IAW TO 1-1B-50, *Aircraft Weight and Balance.*

2.4.2.14. Coordinate with the Operations Group (OG) and establish Functional Check Flight (FCF), Operational Check Flight (OCF), and High-Speed Taxi Check programs. **(T-3)**

2.4.2.15. Implement the Hangar Queen Program IAW **Chapter 11**. **(T-3)**

2.4.2.16. Ensure compliance with Identification Friend or Foe Program or equivalent IAW **Chapter 11** (if equipped). **(T-3)**

2.4.2.16. **(AFMC)** [DEV] The 412th Electronic Warfare Group (412 EWG) will appoint a Radar Warning Receiver/Radar Threat Warning managers for the 412 TW. **(T-2)**

2.4.2.17. Provide Subject Matter Expertise (SME) support for the development of the wing or installation instruction to control tools, equipment, and electronic devices from all wing agencies dispatching to aircraft parking, runway, taxi areas, and aircraft maintenance areas IAW **Chapter 2** and **Chapter 8**. **(T-3)**

2.4.2.18. Ensure Air Force Repair and Enhancement Program (AFREP) is managed IAW **Chapter 11**. **(T-3)**

2.4.2.19. Establish procedures to ensure assigned units have sufficient eTools available for technical order viewing. **(T-3)**

2.4.2.20. Develop procedures and assign responsibilities to ensure aircraft, aircraft system forms, equipment forms, and MIS documentation are complete, accurate, and are thoroughly reviewed each shift. **(T-3)** Documented procedures as a minimum will include:

2.4.2.20.1. The process for recovering aircraft, aircraft systems from extensive maintenance events, and down time ((cannibalization (CANN), local depot maintenance (MX)) include independent screening and validation that all maintenance actions (IPIs, operational checks, configuration management, W&B, serial number (S/N) tracking (COMSEC/CCI and other significant items as designated by an asterisk in the applicable WUC/LCN Manual)) and AFTO Form 95, *Significant Historical Data*, have been accurately documented in the forms, MIS, or both before being scheduled for a sortie or mission. **(T-3)**

2.4.2.20.2. The process for determining if an OCF or FCF is required. **(T-3)**

2.4.2.21. Ensure that when no 2W1X1 weapons AFSCs are assigned and units are required to install/remove chaff/flare on unique mission aircraft, personnel will be trained and qualified to perform these tasks IAW procedures outlined in AFMAN 21-206, *Aircraft Armament Systems Management,* and **Chapter 11** and approved by the WWM.

2.4.2.22. Ensure storage, physical security, corrupt systems quarantine procedures, and cyber threat mitigation management of MXG assigned PIT and eTools is supported and sustained. **(T-3)** If deployed without an MXG/CC, corrupted systems and cybersecurity issues will be coordinated with the WG/CC.

2.4.2.23. Ensure personnel are trained and appointed as Aircraft Battle Damage Evaluators IAW **Chapter 11**. **(T-3)**

2.4.2.24. Establish minimum requirements that ensure units maintain an equitable distribution of supervision (Officer and Senior Non-Commissioned Officer (SNCO)) across all on-duty shifts. **(T-3)**

2.4.2.25. Ensure compliance with the local Impoundment Program. **(T-3)**

2.4.2.26. Be the Impoundment Release Authority, along with the MXG/CD. **(T-3)**

2.4.2.27. Designate Impoundment Authorities and track them on the SCR. **(T-3)**

2.4.2.28. Determine the need for an FCF for an uncommanded flight control movement. **(T-3)**

2.4.2.29. Coordinate with all wing organizations that work in or enter any maintenance area under his or her responsibility to ensure they have documented tool/equipment control procedures. **(T-3)**

2.4.2.30. Approve all locally manufactured, developed, or modified tools and equipment. **(T-3)**

2.4.2.31. Coordinate to ensure direct mission generations support focuses on readiness and the unit's ability to execute daily and deployed flying operations simultaneously. **(T-3)**

2.4.2.32. Approve items identified in **Table 11.1., Note 1**. **(T-2)** The MXG/CC at their discretion, may delegate approval authority to MXG Squadron Commanders. **(T-3)**

2.4.2.33. Coordinate with the OG/CC or WG/CC to develop procedures to ensure pilots and aircrew members account for all equipment and personal items after each flight and ensure any items that become lost during flight are documented in the aircraft AFTO Form 781A. **(T-3)**

2.4.2.34. Approve CANN Authorities (CA). **(T-3)**

2.4.2.35. Designate qualified TSgts or higher, civilian equivalent, fully qualified/certified contractors, or AFETS/CETS personnel to serve as engine run certifying officials IAW criteria established in **Table 11.1**. **(T-3)**

2.4.2.36. Select maintenance, TD, or AFETS/CETS personnel as instructors, and ensure instructors maintain blade blending proficiency and certification requirements. **(T-3)**

2.4.2.37. Approve and track Hot Pit certifying officials on the SCR. **(T-3)**

2.4.2.38. Ensure CDDAR mobility UTC equipment requirements are available to deploy and accounted for on an AS (if applicable). **(T-3)**

2.4.2.38.1. Coordinate functional and emergency action checklist with Fire Emergency Services, Wing Safety, and the Airfield Operations Flight in developing adverse weather procedures for protecting aircraft and equipment IAW DAFMAN 91-203, DAFI 10-2501, *Air Force Emergency Management Program*, and MDS-specific technical data. **(T-3)**

2.4.2.38.2. In coordination with the MSG/CC, determine unit vehicle/equipment requirements beyond those authorized in the AS(s) to provide 24/7 CDDAR response/runway clearing capability. **(T-3)** Units must identify vehicles and SE

designated to support CDDAR recovery in a local publication to ensure 24-hour availability. **(T-3)**

2.4.2.38.3. Ensure as a minimum, units with a CDDAR requirement possess sufficient equipment to accomplish a recovery of the assigned MDS aircraft. **(T-3)**

2.4.2.38.4. Establish an IFE response capability. **(T-3)**

2.4.2.38.5. Participate in CDDAR training exercises. **(T-3)**

2.4.2.38.6. Manage CDDAR equipment to minimize duplication of resources. **(T-3)**

2.4.2.38.7. Ensure an annual CDDAR equipment inventory is completed and an inventory report containing DPAS Property Accountability (PA) inventory of CDDAR equipment that indicates excess, or shortage items is completed, signed by MXG/CC or equivalent, and sent to the MAJCOM CDDAR OPR, NLT 30 Sep. **(T-3)**

2.4.2.39. Ensure EM is the wing focal point for both the ET&D, Engine Health Management (EHM+) and Condition Based Maintenance Plus (CBM+) program when applicable. **(T-3)**

2.4.2.40. Review and approve the assessment of the wing's ability to execute the FHP, in coordination with the OG/CC, prior to submission to the WG/CC. **(T-3)**

2.4.2.41. Review first look packages before being presented to the WG/CC. **(T-3)**

2.4.2.42. Formalize the next month's flying and maintenance plan prior to presenting it to the WG/CC for approval NLT the third scheduling meeting of the preceding month. **(T-3)**

2.4.2.43. Review the weekly, monthly, quarterly, annual flying or test schedules IAW **Chapter 5** and **Chapter 14**. **(T-3)**

2.4.2.44. Ensure effective management of the Engine Trending and Diagnostic (ET&D) program IAW AFMAN 20-116, *Propulsion Life Cycle Management for Aerial Vehicles*.

2.4.2.45. Focus unit MSEPs on problem areas where improvements are needed. **(T-3)**

2.4.2.46. Ensure maintenance actions for Ground Instructional Training Aircraft (GITA)/Training Aircraft Aids (TAA) are documented IAW TO 00-20-1.

2.4.2.47. Ensure repair cost evaluations are performed and appropriate levels of review and repair authorization are established in squadrons, flights, and repair sections IAW TO 00-20-3, *Maintenance Processing of Repairable Property and The Repair Cycle Asset Control System*, TO 00-25-240, *Uniform Repair/Replacement Criteria for Selected USAF Support Equipment* (SE), and TO 35-1-24, *Air Force Economic Repair/Replacement Criteria For Selected Warner Robins Logistics Complex* (ALC) *Managed Support Equipment* (SE).

2.4.2.48. Collaborate with the WG/CC and forward a recommendation to the MAJCOM if an initial OTI sampling indicates the discrepancy is widespread and has the potential for personal injury and/or further equipment damage. **(T-3)**

2.4.2.49. Approve requests for assistance IAW **Chapter 1** after they are coordinated with Plans, Scheduling, and Documentation (PS&D), Quality Assurance (QA), and all applicable maintenance organizations. **(T-3)**

2.4.2.50. Appoint a Wing Corrosion Manager (2A7X3, 2A7X5) to implement local requirements, ensure implementation of MAJCOM directed requirements and act as the focal point for communicating with external stakeholders. **(T-3)**

2.4.2.51. Be the approval authority for cannibalization of egress explosive components and/or seats to ensure system integrity and validation of explosive CAD/PAD listing. **(T-3)**

2.4.2.52. Establish a Memorandum of Understanding (MOU) for organizational level Maintenance Interoperability or Aircraft Cross-servicing with joint forces or partner nations, if requested. See DAFI 25-301, *Aircraft Cross-Servicing Agreements*, for further guidance.

2.4.2.53. If applicable, establish a Primary, Alternate, Contingency, and Emergency (PACE) plan(s) to conduct operations in a contested, degraded, or operationally limited environment. (T-3) Note: Use AFTTP 3-4.21V1, *Aircraft Maintenance*, Attachment 10, *Contested, Degraded, and Operationally Limited Operations*, as a guide for development of PACE plan and AFI 90-802, *Risk Management*.

2.4.3. The MXG/CC will:

2.4.3.1. Approve and publish In Process Inspection (IPI) listings every two years IAW **Chapter 6. (T-3)**

2.4.3.2. Approve MXG Key Task List (KTL) and Routine Inspection Lists (RIL). **(T-3)**

2.4.3.3. Implement MAJCOM Lead the Fleet (Pacer) Program for engine type IAW AFMAN 20-116. **(T-2)**

2.4.3.4. Ensure aircraft possessed by AFMC for depot maintenance are not cannibalized without coordination through the MAJCOM functional manager who will then request approval from the applicable Air Logistics Complex (ALC) Maintenance Group Commander/Director and Program Manager. **(T-2)**

2.4.3.5. Establish local procedures for management and maintenance of assigned GITA and TAA to ensure they remain useful and safe within guidelines stated in **Chapter 11**, AFI 84-103, *Department of the Air Force Heritage Program*, DAFI 21-103, and 23-series publications. **(T-3)**

2.4.3.6. Appoint a Wing Avionics Manager (WAM) or designated representative to act as the maintenance focal point for wing avionics related programs. **(T-3) Note:** See **Chapter 11** for WAM responsibilities.

2.4.3.7. Determine the Routine Inspection List (RIL) frequency if not mandated by the MAJCOM. **(T-3)**

2.4.3.7.1. Approve Maintenance Supervision inputs and suggested changes to the RIL submitted from QA. **(T-3)**

2.4.3.7.2. Approve the Evaluation and Inspection (E&I) Plan. **(T-3)**

2.4.3.7.3. Establish AQLs for tasks/inspections not included on the MAJCOM AQL listing. **(T-3)**

2.4.3.7.4. Establish quarterly meetings to review a summary of MSEP data. **(T-3)**

2.4.3.8. Oversee GITA/TAA IAW **Chapter 11**. **(T-2)**

2.4.3.8.1. Develop an installation publication or supplement to define the scope of training functions for GITA/TAA use, functional responsibility for funding, operations, maintenance, and records management. **(T-3)**

2.4.3.8.2. Ensure maintenance support of GITA/TAA used for training. **(T-3)** Units that do not have organic maintenance capability will establish a Support Agreement or MOA assigning maintenance responsibility for GITA/TAA training use. **(T-3)**

2.4.3.8.3. Ensure explosive components are removed that are not required to support training requirements.

2.4.3.8.4. Place retained systems and subsystems not currently being used for training into extended storage IAW applicable technical data.

2.4.3.8.5. Ensure standard maintenance practices regarding inspection appearance, cleanliness, ground safety, and prevention of corrosion are met. Corrosion control procedures are outlined in TO 1-1-691.

2.4.3.8.6. Develop and prepare inspection technical data check lists for use in inspecting the condition and safety of equipment before use and ensure inspections are performed. **(T-3)**

2.4.3.8.7. Prepare a separate memorandum for each GITA/TAA, addressed to the appropriate PM for the aircraft and inform them of the systems and subsystems that will be maintained in operational configuration. **(T-3)**

2.4.3.8.8. Report air and space vehicle inventory IAW DAFI 21-103 as required for ground trainers. Aircraft used for ground trainers are exempt from status and utilization reporting.

2.4.3.8.9. Ensure proper coordination and documentation of parts removed from training aircraft are accomplished.

2.4.3.8.10. Designate a GITA/TAA Manager as an additional duty. **(T-3)**

2.4.3.9. Ensure all items being repaired by AFREP have been approved by the appropriate repair authority. **(T-3)**

2.4.3.9.1. Assign an AFREP Manager. **(T-3)**

2.4.3.10. Chair a maintenance and FHP planning meeting quarterly, NLT 14 days before the next quarter. **(T-3)**

2.4.3.11. Establish Minimum Equipment Levels (MELs) for essential maintenance assets to include engines, pods, AGE, vehicles, advocate and reconcile authorized shortfalls and overages. **(T-2)** Coordinate with the applicable MAJCOM functional to advocate with the respective Program Manager to address any requests to change authorized quantities. **(T-3)**

2.4.3.12. Establish the group maintenance awards and recognition program to meet AF and MAJCOM requirements refer to **https://access.afpc.af.mil/Trophies/searchtrophies.aspx**. **(T-3)**

2.4.3.13. Ensure an orientation program is developed and conducted for all personnel newly assigned to MXG maintenance or equivalent maintenance activities IAW AFI 36-2650.

2.4.3.14. Develop a MXG Impoundment Program and ensure compliance with the procedures IAW **Chapter 7**. **(T-3)**

2.4.3.15. Establish written procedures to review and clear "repeat", "recur", and "cannot duplicate" (CND) discrepancies. **(T-3)**

2.4.3.16. Approve the Dedicated Crew Chief (DCC) program, when implemented.

2.4.3.17. Review and coordinate the AGE MEL annually with applicable Maintenance Supervision. **(T-3)** The MXG/CC approves the identified types and quantities of AGE for the MEL.

2.4.3.18. Develop and implement a standardized Impound Official training course and Impoundment Program. **(T-3)**

2.4.3.19. Appoint an Identification Friend or Foe Program Manager for Identification Friend or Foe systems cryptographically keyed by MXG personnel. **(T-3)**

2.4.3.20. Appoint a Radar Warning Receiver/Radar Threat Warning Manager (if equipped as required). **(T-3)**

2.4.3.21. Develop a 10-year facility plan specifying maintenance, upgrade, and replacement projections for the group's facilities. **(T-3)**

2.4.3.22. Ensure aircraft shelters at bases are maintained, unless otherwise stipulated in contracting arrangements, IAW **Chapter 11**. **(T-2) Note:** If an aircraft shelter is used for other than its designed purpose, the using activity will maintain it.

2.4.3.23. Establish and document MXG local manufacture procedures and controls in a supplement to this instruction. **(T-3)**

2.4.3.24. Appoint a Stock Record Account Number (SRAN) Engine Manager (EM) or a Unit Engine Manager (UEM) to accomplish duties outlined in TO 00-25-254-1, *Comprehensive Engine Management System* (CEMS) (D042) *Engine Status, Configuration, and TCTO Reporting Procedures*.

2.4.3.25. Establish written guidance on individual responsibilities and specific procedures for Cannibalization (CANN) actions IAW **Chapter 11**. **(T-3)**

2.4.3.26. Publish directives outlining procedures covering the manufacture of items source coded local manufacture. **(T-3)**

2.4.3.27. Ensure the MT develops and manages an effective engine run certification program. **(T-3)**

2.4.3.28. For calibration limitation Approval Certification Program, appoint maintenance, TD, or PMEL personnel as instructors, and ensure requisite certification and proficiency requirements are tracked in the MIS by course code.

2.4.3.29. For engine management, appoint a qualified 2A6X1, minimum 7-skill level, (or civilian equivalent) technician to manage the EHM+ and CBM+ program IAW AFMAN

20-116 for EHM products managed by AFLCMC/LP. If EHM data is not managed by AFLCMC/LP, ensure Contractor Logistics Support (CLS) provided EHM products are managed according to respective contract.

2.4.3.30. Ensure the Aero Club is operated IAW DAFI 34-101, *Department of the Air Force Morale, Welfare, and Recreation (MWR) Programs and Use Eligibility*, when assigned.

2.4.3.31. **(Added-AFMC)** F-16 (only) Establish local procedures for application intervals of HydroSkip on F-16 transparencies (not to include LUNA transparencies, if installed).

2.4.4. - 2.4.58. DELETED.

2.5. Deputy Maintenance Group Commander (MXG/CD). In organizational structures where an MXG/CC or MXG/CD is not assigned, (i.e., AFORGEN deployed construct), **paragraph 2.5.1** will move up to the WG/CC commander to ensure effective communication across all units.

2.5.1. Establish maintenance production/scheduling meeting. **(T-3)** This meeting is to verify aircraft and equipment utilization, scheduled maintenance requirements, evaluate predictive maintenance forecast, establish work priorities, and coordinate schedule changes. It should synchronize communications necessary to optimize fleet readiness and mission generation capability.

2.5.1.1. -2.5.13. DELETED.

2.6. MXG Senior Enlisted Leader (SEL) Responsibilities. In addition to the Group SEL responsibilities outlined in Brown Book, *The Enlisted Force Structure*, the MXG SEL is responsible to the MXG/CC and will advise and assist the MXG/CC on their responsibilities as outlined in this **Chapter**. In organizational structures where an MXG is not assigned (i.e., AFORGEN deployed construct), the following responsibilities will remain with the home unit's SEL and any guidance/documentation necessary to support the deployed operation will be brought with the unit. The MXG SEL will:

2.6.1. Conduct a quarterly manning meeting with Squadron SELs and Wing Weapons Manager (WWM) to review MXG manning status, Duty Title Tool usage, and ensure manning resources are strategically distributed to provide the greatest possibility for mission success. **(T-3)** The MXG SEL is the focal point for ensuring squadron SELs receive adequate manpower management training.

2.6.1.1. Meeting will consist of a review and evaluation of the impact on the MXG of personnel actions such as: work center/organizational manpower Authorization Change Requests (ACR), AFSC changes, re-training, special duty requests, special assignment actions (SWAP, Palace Chase), SEI balance, overseas Date Eligible for Return from Overseas extensions/In Place Consecutive Overseas Tour (IPCOT) requests, physical profile changes and personnel rotation plans as applicable to enhance mission effectiveness. **(T-3)**

2.6.1.2. MXG SEL will provide the MXG/CC coordinated manning recommendations that develop enlisted individual experience and knowledge for consideration. **(T-3)**

2.6.2. Ensure all AFSC 2A and 2W maintenance personnel are only assigned authorized duty titles. **(T-3)** Reference the Duty Title Tool:

https://usaf.dps.mil/teams/10820/medac/2A_DTT/SitePages/Home.aspx, and the AFSC 2W1 Standardized Duty Titles located at https://usaf.dps.mil/teams/11262/HAF/HAF-A4LW/Standardized%20Duty%20Titles for all assigned AFSC 2W1X1 and 2W100 positions.

2.7. Wing Weapons Manager. See AFMAN 21-206 for duties and responsibilities.

2.7.1. - 2.7.28. DELETED.

2.8. Squadron Commander (SQ/CC) Responsibilities. The SQ/CC will:

2.8.1. Ensure compliance with AFI 90-821, DAFI 91-202, *The US Air Force Mishap Prevention Program*, DAFMAN 91-203 and other publications necessary to perform the commander functions assigned to the squadron.

2.8.2. Establish and administer squadron training programs IAW AFI 36-2650 and DAFMAN 36-2689, *Training Program*; monitor upgrade training, Personnel Reliability Program (PRP) status, and qualifications of assigned work center personnel; and ensure MAJCOM Mandatory Course List requirements are met (if applicable). **(T-3)**

2.8.3. Ensure upgrade training and maintenance qualification programs emphasize quality and are not primarily focused on meeting minimum upgrade time frames. **(T-3)**

2.8.3.1. Ensure all maintenance personnel who utilize DoD Information Technology have received appropriate Maintenance Cyber Discipline Training. **(T-3)**

2.8.4. Monitor all personnel working outside of their primary AFSC to ensure that it does not degrade mission accomplishment. **(T-3)**

2.8.5. Establish a squadron Vehicle Control Program IAW AFI 24-302, *Vehicle Management*.

2.8.6. Establish and manage squadron FCC program IAW **Chapter 11** (if applicable).

2.8.7. Protect and secure munitions as outlined in DAFI 31-101. The SQ/CC will ensure Intrusion Detection Systems requirements are identified when required to store munitions IAW DoDM 5100.76 and DAFI 31-101.

2.8.8. Appoint equipment custodians to manage the Custodian Authorization/Custody Receipt Listing (CA/CRL) (R14) of assigned equipment IAW AFI 23-101 and DAFMAN 23-122, *Materiel Management Procedures*.

2.8.9. Ensure personnel and equipment are identified and prepared to deploy for taskings IAW AFI 23-101, AFI 10-403, *Deployment Planning and Execution*, DAFI 36-3802, *Force Support Readiness Programs*, and AFMAN 10-409-O, *Support to Adaptive Planning*.

2.8.10. Recommend personnel for QA duty positions. **(T-3)**

2.8.11. Designate Flight CC/Chiefs. **(T-3)**

2.8.12. Ensure the UMD is consistent with the approved organizational structure. **(T-3)**

2.8.13. Coordinate support from the local communication squadron or equivalent functional entity to ensure proper eTools configuration (operating system, virus checkers) are maintained. **(T-3)** The SQ/CC will coordinate with lead TODO/Functional System Administrator to resolve TO requirements that are not being satisfied. **(T-3)**

2.8.13.1. Ensure licenses, certification, maintenance, and security of eTools (hardware and software) is conducted IAW 33/17-series AFIs, TO 31S5-4-ETOOL-1, and **Chapter 8**.

2.8.14. Ensure members assigned to the DIT are qualified to accurately assess the Maintenance Data Documentation. **(T-3)**

2.8.15. Ensure maintenance is only performed by personnel who are trained, qualified, and certified, unless under the direct supervision of a trainer or certifier. **(T-3)**

2.8.16. Ensure procedures are followed to properly turn in recoverable and consumable items IAW AFI 23-101. **(T-3)**

2.9. Maintenance Supervision Responsibilities. Maintenance Supervision consists of the Director of Operations (DO) and Maintenance Superintendent (MX SUPT). Maintenance Supervision advises the SQ/CC on technical matters, leads a mission-focused maintenance effort, and manages resources necessary to accomplish the mission. They provide necessary administration to manage assigned responsibilities and control maintenance through Pro Supers, Flight CC/SUPT, and Section NCOICs/Chiefs. The MX SUPT is responsible to the DO. Maintenance Supervision will:

2.9.1. Ensure adequate levels of supervision and manning are balanced across all shifts to accomplish the mission safely and efficiently. **(T-3)**

2.9.2. Ensure timely and accurate engine data is provided to the EM element for all engines IAW **Chapter 14**. **(T-3)**

2.9.3. Enforce procedures to prevent FOD and dropped objects IAW **Chapter 11**. **(T-3)**

2.9.4. Monitor and recommend updates to local IPI requirements and recommendations to QA IAW **Chapter 6**. **(T-3)**

2.9.5. Ensure a sufficient number of personnel are qualified to perform mission critical tasks listed on the SCR **Table 11.1** in **Chapter 11**. **(T-3)** Review and/or recommend individuals for addition to the SCR. **(T-3)** Approve individuals for addition to the SCR. **(T-3)** Review and approve individuals for addition to the SCR. **(T-3)**

2.9.6. Ensure aircraft systems and equipment are available to support unit training objectives. **(T-3)**

2.9.7. Ensure distribution of maintenance cross-tell messages, QA newsletters, policy announcements, technical notifications, and other important maintenance information to all members of the organization. **(T-3)**

2.9.8. Review and evaluate management and production effectiveness. **(T-3)** Maintenance Supervision will analyze personnel and equipment performance history. **(T-3)** Initiate management actions to meet new workloads or correct reported/perceived deficiencies. **(T-3)**

2.9.9. Ensure an annual maintenance plan is developed and reconciled with the flying schedule and flying requirements to ensure maintenance can support the annual flying hour/test program. **(T-3)**

2.9.9.1. Participate in the maintenance planning cycle. **(T-3)**

2.9.9.2. Utilize a sortie production model or equivalent for the assigned MDS (if available). **(T-2)**

2.9.10. Ensure a squadron SERENE BYTE or PACER WARE response capability is available to support reprogramming requirements IAW DAFMAN 10-703, *Electromagnetic Warfare Integrated Reprogramming* (if applicable).

2.9.11. Ensure a squadron Corrosion Control Program is implemented and managed IAW TO 1-1-8, TO 35-1-3, TO 1-1-691, MDS-specific TOs and MAJCOM instructions.

2.9.12. Ensure squadron ASIP responsibilities are accomplished IAW **Chapter 11** and DAFI 63-140.

2.9.13. Develop written procedures in coordination with the WWM, Weapons Safety Manager, and Airfield Management to establish EOR inspection procedures as required by aircraft specific -6 TO, MAJCOM directed requirements and **Chapter 11**.

2.9.13.1. Ensures sufficient personnel, equipment, and facilities are assigned, maintained, and available to properly perform EOR inspections IAW **Chapter 11**.

2.9.13.2. Ensure EOR procedures for transient aircraft are developed IAW TO 00-20-1 and MAJCOM directives.

2.9.14. Review and support the monthly Weapons Load Training (WLT) schedule. **(T-3)**

2.9.15. Ensure deferred maintenance, Pilot Reported Discrepancy, and back-ordered parts are properly managed. **(T-3)**

2.9.16. Review supply products to monitor supply discipline. **(T-3)**

2.9.16.1. Maintenance Supervision will manage DIFMs IAW AFI 23-101. **(T-3)**

2.9.16.2. Monitor and reconcile changes in base-level repair capabilities under their supervision as they occur with the LRS/Material Management activities IAW AFI 23-101.

2.9.17. Ensure lost, damaged, destroyed, or stolen government assets are dispositioned IAW DoD 7000.14-R, *DoD Financial Management Regulation*, Vol 12, **Chapter 7**, Financial Liability for Government Property Lost, Damaged or Destroyed (Formerly Report of Survey).

2.9.18. Ensure Special Purpose Recoverable Authorized Maintenance (SPRAM) accounts are established IAW DAFI 21-103, **Chapter 9**, and maintained IAW AFI 23-101.

2.9.19. Ensure reporting of materiel deficiencies IAW TO 00-35D-54, *USAF-Deficiency Reporting, Investigating, and Resolution*.

2.9.20. Monitor requirements for CTK, special tools, and SE and take necessary action to ensure availability, as required IAW **Chapter 8**.

2.10. Flight Supervision. Flight Supervision consists of the Aircraft Maintenance Unit (AMU) Officer in Charge (OIC)/Superintendent and Flight Commander/Flight Superintendent (Flight CC/Chief). Flight Supervision will:

2.10.1. Provide management and oversight and ensure each section is adequately resourced to efficiently execute their mission. **(T-3)**

2.10.2. Manage, distribute, and adjust the flight's manning to support the maintenance plan across all shifts. **(T-3)**

2.10.2.1. Distribute all levels of supervision based on manning and workload to supervise all duty periods. **(T-3)**

2.10.2.2. Identify imbalances between authorizations and the number of personnel assigned, or between authorized and assigned skill levels or grades to SQ/CC and Maintenance Supervision. **(T-3)**

2.10.2.2.1. Review unit DOC Statements, OPLANs, unit-tasked UTC requirements for personnel and equipment then compare requirements to UMD to identify existing shortfalls. Scrutinize critical AFSCs qualifications and equipment based on their impact on mission generation. Document and report personal and equipment deficiencies through the chain of command and monitor until resolved. **(T-3)**

2.10.2.3. Manage additional duties, leaves, ancillary training, and rotate/assign manning across shifts to balance the workload and minimize negative impacts on the workforce. **(T-3)**

2.10.3. Execute the squadron's Mishap Prevention Program for the flight/work center IAW **Chapter 1** and DAFI 91-202.

2.10.3.1. Ensure all personnel obtain the required safety training, and document it on the AF Form 55, Employee Safety and Health Record, or equivalent IAW DAFI 91-202.

2.10.4. Coordinate occupational and environmental health risk assessments with Bioenvironmental Engineering IAW AFMAN 48-146, *Occupational & Environmental Health Program Management*, to identify, assess and evaluate process hazards in the workplace and identify controls. **(T-3)**

2.10.4.1. Monitor and ensure environmental and applicable health requirements, physicals and respirator training, initial and recurring requirements are accomplished when required for assigned personnel IAW AFMAN 48-146; AFI 48-137, *Respiratory Protection Program*; and AFI 48-127, *Occupational Noise and Hearing Conservation Program.*

2.10.5. Ensure organizational compliance IAW the installation ESOHMS/EMS Program. **(T-3)**

2.10.6. Advocate use of the TO improvement program and ensure work center TO files are maintained IAW TO 00-5-1.

2.10.7. Ensure Materiel Potentially Presenting an Explosive Hazard requirements in DAFMAN 21-201 and TO 11A-1-60, *General Instructions Inspection of Reusable Munitions Containers and Scrap Material* Generated from Items Exposed to or Containing Explosives, are complied with when certifying items associated with explosives such as: Multiple Ejector Rack, Triple Ejector Rack, pylons, launchers, rafts, bomb racks, ejection seats, fire suppression bottles, and gun systems and components.

2.10.7.1. Ensure associated items are explosive free prior to being turned in to LRS or the Defense Logistics Agency Disposition Services (DLADS).

2.10.8. Review deferred maintenance in the MIS and coordinate with the Pro Super to schedule and/or validate task accomplishment. **(T-3)**

2.10.9. Ensure operator inspections and user servicing requirements are accomplished on all assigned support equipment IAW TO 00-20-1.

2.10.10. Ensure records of inspection, lubrication, and maintenance of industrial equipment are maintained IAW TO 00-20-1, TO 34-1-3, *Machinery and Shop Equipment*, to include documentation of records maintained in a MIS.

2.10.10. **(AFMC)** Maintenance records in electronic (other than those required in the MIS) or paper form, will be filed in the file plan per the Records Disposition Schedule (RDS). **(T-2)**

2.10.11. Ensure proper calibration, use, care, handling and transportation of Test Measurement and Diagnostic Equipment (TMDE) IAW TO 00-20-14 and AFMAN 21-113, *Air Force Metrology and Calibration (AFMETCAL) Program Management*, and applicable Calibration Measurement Summaries.

2.10.12. Evaluate maintenance quality, personnel qualifications, and training of assigned personnel. **(T-3)**

2.10.13. Review/update flight IPI requirements listing every two years and route through Maintenance Supervision. **(T-3)**

2.10.14. Ensure only designated personnel identified in the MIS verify MICAPs/Urgency of Need Designator 1A and JA requirements. **(T-3)**

2.10.15. Select personnel to perform special certification tasks IAW **Chapter 11** and **Table 11.1** and forward names to Maintenance Supervision for approval and addition to the SCR. **(T-3)**

2.10.16. Ensure training requirements are executed to support established training plan and individual AFSC Career Field Education and Training Plans (CFETP) IAW DAFMAN 36-2689and AFI 36-2650.

2.10.17. Ensure Cross Utilization Training requirements are identified as required by the unit mission IAW AFI 36-2650.

2.10.17.1. Ensure Cross Utilization Training does not interfere with upgrade/qualification training. **(T-3)**

2.10.18. Review Maintenance Management Analysis (MMA), QA, and other management reports to determine appropriate management actions to meet new workloads, target deficiencies, and identify and correct root causes. **(T-3)**

2.10.18. **(AFMC)** Develops and implements a corrective/preventive action process. (e.g., AF Continuous Process Improvement (CPI), Practical Problem Solving Method (PPSM), 5 Whys, etc.).

2.10.19. Develops maintenance and flying schedules and execute scheduled maintenance plans. **(T-3)**

2.10.20. Establish flight/AMU-specific emergency action procedures to respond to disaster control and severe weather and forward to MOC. **(T-3)**

2.10.20.1. Review unit responsibilities pertaining to aircraft/SE movement and personnel evacuation IAW DAFI 10-2501. **(T-3)**

2.10.21. Manage the flight/AMU's participation in the FOD and DOP program IAW **Chapter 11**.

2.10.22. Oversee the flight/AMU's FCC/Dedicated Crew Chief (DCC) Program (if applicable). **(T-3)**

2.10.23. Establish and enforce a flight/AMU Precious Metals Recovery Program, as applicable, IAW AFI 23-101 and TO 00-25-113, *Conservation and Segregation of Critical Alloy and Precious Metal Bearing Parts and Scrap*.

2.10.24. Assign section supervisors IAW this instruction and the UMD. **(T-3)**

2.10.25. Ensure proper asset management by reviewing MIS data records, the Repair Cycle Asset Management Listing (D23) and other pertinent products to minimize shortfalls. **(T-3)**

2.10.25.1. When applicable, ensure warranty items are loaded in MIS according to applicable directives. **(T-3)**

2.10.25.2. Ensure Deficiency Reports (DR) are accomplished IAW TO 00-35D-54.

2.10.26. Ensure repairable/non-repairable parts are promptly processed through repair channels within the required time frame IAW AFI 23-101.

2.10.26.1. Team with Decentralized Materiel Support (DMS) Flight Service Center to conduct a quarterly reconciliation of all DIFM assets and follow up on delinquent DIFMs and document action taken to correct identified discrepancies. **(T-3)**

2.10.26.2. Immediately identify lost, damaged, destroyed, or stolen assets that require a financial liability investigation IAW DoD 7000.14-R, Vol 12, **Chapter 7**, *Financial Liability for Government Property Lost, Damaged or Destroyed* (Formerly Report of Survey) and forward to Maintenance Supervision for review and processing. **(T-3)**

2.10.27. Approve requirements for bench stocks IAW qualification criteria in DAFMAN 23-122. **(T-3)**

2.10.28. Consolidate lists of items received in supply requiring functional check, operational programming, user calibration or corrosion control/painting. **(T-3)**

2.10.28.1. Submit listing to the LRS Materiel Management Flight IAW TO 00-20-3. **Note:** Does not include TMDE IAW TO 00-20-14.

2.10.29. Coordinate all AGE requirements through the AGE Flight Chief to ensure support capability and eliminate unnecessary duplication of equipment. **(T-3)**

2.10.30. Report cyber threats, incidents, and issues per Wing cyber status reporting procedures. **(T-3)**

2.10.30.1. Consult airframe Security Classification Guides, TOs, and/or applicable technical manuals, instruction and publication when addressing or reporting cybersecurity threats, incidents, and issues. **(T-3)**

2.10.31. Ensure Nuclear Weapons-Related Materiel (NWRM) is controlled IAW AFI 20-110.

2.11. Production Superintendent (Pro Super). Senior NCO responsible for squadron maintenance production. The Pro Super directs the overall maintenance effort of their unit. The Pro Super will be a SNCO or civilian equivalent. **(T-3)** Squadron specific Pro Super responsibilities are outlined in their respective chapters of this DAFI.

2.12. Section NCOIC/Chief. The Section NCOIC/Chief is responsible to the Flight CC/SUPT or AMU OIC/SUPT for the leadership, supervision, and training of assigned personnel. The Section NCOIC/Chief is a first-line manager and supervisor of maintenance production and is the technical authority and advisor in that area. When sections are subdivided, element leaders perform the appropriate functional responsibilities. The Section NCOIC/Chief will:

2.12.1. Establish a Work Center Safety Program IAW DAFI 91-202, DAFMAN 91-203, and include any locally prescribed safety requirements (if applicable).

2.12.2. Monitor, track, and ensure occupational safety, fire prevention, and occupational and environmental health requirements are accomplished for assigned personnel IAW DAFMAN 91-203.

2.12.2.1. Ensure Job Safety Training is documented IAW DAFI 91-202, (AF Form 55 or equivalent) for each assigned individual.

2.12.3. Ensure maintenance is performed by personnel who are trained, qualified, and certified, unless under the direct supervision of a trainer or certifier. **(T-3)**

2.12.4. Advocate use of the TO improvement program and ensure work center TO files are maintained according to TO 00-5-1. **(T-3)**

2.12.5. Establish procedures and ensure configuration control for all applicable software required for the sections assigned systems. **(T-3)** Section NCOIC/Chief will:

2.12.5.1. Access Computer Program Identification Number System (CPINS) in ETIMS or equivalent system. **(T-3)**

2.12.5.2. Ensure technicians check ETIMS/equivalent system for software updates for assigned systems. **(T-3)**

2.12.5.3. Ensure software configuration control is maintained IAW TO 00-5-16, *Management of Computer Program Identification Number System (CPINS)* and equivalent systems are maintained by use of AF approved and authorized publications.

2.12.6. Perform production and supervisory inspections. **(T-3)**

2.12.7. Validate classified parts/materiel are managed IAW AFI 23-101. **(T-3)**

2.12.8. On a daily basis review, monitor, and correct, as needed, the work center's scheduled and deferred events in the MIS. **(T-3)**

2.12.8.1. Close, reschedule, or defer all events beyond their scheduled start date and time (Integrated Maintenance Data System (IMDS)-CDB screen #100/380 and Field Maintenance Command and Control (FMxC2) Screen #8035/8069/67150). **(T-3) Note:** G081 was renamed to FMxC2 August 2022.

2.12.9. Review transcribed AFTO Form 781-series forms, work center MIS data entries for the previous day, and all preceding non-duty days, for job accuracy and completeness (IMDS-CDB Screen #100 and FMxC2 Screen #9154). **(T-3)**

2.12.10. Validate scheduled aircraft document reviews using applicable MIS/records check package and automated aircraft forms IAW **Chapter 14**.

2.12.11. Develop and manage the Work Center Training Program. **(T-3)**

2.12.11.1. Evaluate the quality of maintenance, training, and personnel qualifications, track training requirements and ensure training documentation is complete and accurate. **(T-3)**

2.12.11.2. Conduct On-the-Job training (OJT)/certifying as required. **(T-3)**

2.12.12. Review and recommend changes for maintenance tasks requiring IPIs to the Flight Supervision. **(T-3)**

2.12.13. Review, evaluate, and take corrective action based on QA and other inspection reports. **(T-3)**

2.12.14. Ensure all required work center publications necessary for the work center to meet its functional requirements are current and available for use. **(T-3)**

2.12.15. Ensure section personnel coordinate all flightline maintenance with the Flightline Expediter. **(T-3)**

2.12.16. Manage CTK and supply programs (such as, bench stocks, and operating stocks) IAW **Chapter 8** and **Chapter 9**. **(T-3)** Section NCOIC/Chief will:

2.12.16.1. Ensure sections are organized with tools, equipment, and materiel as close to the Point of Maintenance as possible, as approved by the Flight Supervision, without jeopardizing accountability and control procedures. **(T-3)**

2.12.16.2. Ensure the Bench Stock Review Listing (M04) is reviewed monthly, and all recommendations are adjudicated to meet mission needs most efficiently. **(T-3)**

2.12.16.3. Ensure work centers team with Decentralized Materiel Support or LRS/materiel management activity, to properly identify and dispose of all unserviceable or excess consumable items IAW AFMAN 23-122. **(T-3)**

2.12.17. Ensure custodial responsibilities are accomplished on all assigned equipment IAW AFI 23-101, and DAFMAN 23-122. **(T-3)**

2.12.18. Manage the section's Repair Cycle Program. **(T-3)** The Section NCOIC/Chief will review the D23 and other pertinent supply products to ensure proper supply discipline daily. **(T-3)**

2.12.19. Establish procedures to control, store, and manage Alternate Mission Equipment (AME); Maintenance, Safety, and Protective Equipment; and -21 equipment IAW DAFI 21-103. **(T-3)**

2.12.20. Identify items requiring calibration (does not include TMDE calibrated by the Precision Measurement Equipment Laboratory (PMEL)) or operational check before installation and provide a list of these items to the Flight Supervision. **(T-3)**

2.12.21. Recommend individuals for addition to the SCR to the Flight Supervision. **(T-3)**

2.12.22. Participate in and enforce the Bad Actor Program IAW TO 00-35D-54.

2.12.23. Manage Hazardous Materiel (HAZMAT) and Environment Safety and Occupational Health (ESOH) items IAW AFI 32-70XX-series instructions, and AFI 90-821.

2.12.23.1. Ensure HAZMATs are used IAW TOs and conform to indicated Military Specifications (MIL-Spec) and monitor the Qualified Products List/Qualified Product Database for changes to specified HAZMAT.

2.12.24. Ensure assigned NCE (applies to both nuclear and non-nuclear tasked units) comply with requirements outlined in AFI 63-125 and associated MAJCOM supplements.

2.12.25. Ensure Dull Sword reports are submitted for nuclear deficiencies IAW AFMAN 91-221 and DAFI 91-204.

2.12.26. Ensure aircraft -6 TO system, inspections, TCTOs and aircraft functional checks (except Isochronal (ISO), Phase (PH)/ Hourly Post-flight (HPO)) are accomplished as required to prevent overdue or over flight of equipment. **(T-3)**

2.12.27. Comply with TCTO performing work center requirements below:

2.12.27.1. Report all deficiencies in technical instructions and applicability to the TCTO managing agency and QA. **(T-3)**

2.12.27.2. Attend TCTO planning meetings. **(T-3)** Review the TCTO prior to the meeting and request clarification of any requirements from QA and the appropriate TCTO managing agency during the meeting.

2.12.27.3. Inventory TCTO kits for completeness prior to starting work. **(T-3)** If a discrepancy exists, contact the TCTO managing agency to resolve shortages.

2.12.27.4. Perform the inspection or modification procedures outlined in the TCTO and document results or findings in the MIS. **(T-3)**

2.12.27.5. If an inspection TCTO generates a requirement for parts, the performing workcenter will create a new Job Control Number (JCN) and enter the discrepancy in the AFTO Form 781A, Maintenance Discrepancy and Work Document, or applicable equipment record and order the required parts. **(T-3)** Inspection TCTOs are complete when the inspection is finished.

2.12.27.6. Order and maintain all HAZMAT required to comply with TCTOs and provide document numbers to the TCTO managing agency and supply TCTO monitor. **(T-3)**

2.12.27.7. Validate technical instructions and data on AFTO Form 82, *TCTO Verification Certificate*, when performing TCTO kit proofing IAW TO 00-5-15, *Air Force Time Compliance Technical Order Process*.

Chapter 3

AIRCRAFT MAINTENANCE SQUADRON (AMXS)/GENERATION SQUADRON.

3.1. General. The AMXS/Generation Squadron provides direct MGN support by consolidating and executing on-equipment activities necessary to produce properly configured, mission ready weapon systems to meet contingency or training mission requirements. AMXS personnel service, inspect, maintain, launch, and recover assigned/transient aircraft (if applicable).

3.2. Maintenance Supervision Responsibilities. In addition to common responsibilities outlined in **Chapter 2**, Maintenance Supervision will:

3.2.1. Ensure standardized procedures and organizations among units as applicable to optimize effectiveness. **(T-3)**

3.2.2. Establish hot brake response procedures in coordination with base support agencies, for example, Fire Emergency Services and CDDAR Team. **(T-2)**

3.2.3. Monitor the squadron FCC program, if applicable. **(T-3)**

3.2.4. Ensure personnel use and understand the purpose of the AF Form 2408, *Generation Maintenance Plan*, and the AF Form 2409, *Generation Sequence Action Schedule*, or electronic equivalent. **(T-3)**

3.2.5. Ensure an explosive safety and chaff/flare academics and loading program is established for units without a 2W1 AFSC assigned (when applicable) IAW DESR6055.09_AFMAN91-201.

3.2.6. Publish procedures covering the storage, control, and handling of starter cartridges (if applicable). **(T-3)**

3.2.7. Provide input to MMA for the monthly metrics report to MAJCOM. **(T-2)**

3.2.8. Develops the annual maintenance plan IAW **Chapter 14**.

3.2.9. **(Added-AFMC)** Ensure all personnel who are dispatched for any task requiring landing gear handle movement are authorized and qualified to perform landing gear handle movement and are listed on the Special Certification Roster (SCR), by aircraft type and personnel duties; gear retraction supervisor, 7-level team supervisor, gear handle operator (B-1, B-2, B-52, 135 Series, C-17, C-5, 747, CV-22). **(T-2)**

3.2.9.1. **(Added-AFMC)** Specific duties of the gear retraction supervisor are as follows:

3.2.9.1.1. **(Added-AFMC)** During aircraft and equipment maintenance, gear retraction supervisor will ensure no movement of the gear handle from the full down position without 7-level or civilian equivalent authorization and verification that the handle needs to be moved for the task being performed. **(T-2)**

3.3. Aircraft Maintenance Unit (AMU). AMUs may include the following sections: Aircraft, Specialist, Weapons, Debrief, Supply and Support. MAJCOMs may approve additional sections and AFSC make up within existing sections to efficiently meet unique weapon system maintenance support requirements. **Note:** Organization modifications must be approved IAW AFI 38-101. In a Generation Squadron, AMUs are obsolete, but responsibilities of the unit and duty positions remain.

3.4. AMU OIC/SUPT Responsibilities. Allocates personnel and resources to the production effort. In addition to the common responsibilities in **Chapter 2**, the AMU OIC/SUPT will:

3.4.1. Review Pilot Reported Discrepancies daily and ensure proper maintenance actions are taken. **(T-3)**

3.4.2. Review all aborts and ensure proper maintenance actions are taken. **(T-3)**

3.4.3. Monitor aircraft PH/ISO/Periodic (PE)/Home Station Check (HSC) flow. **(T-3)**

3.4.4. Ensure a sufficient number of personnel are engine run qualified to meet maintenance requirements IAW **Chapter 11**.

3.5. Production Superintendent (Pro Super). In squadrons with eight or fewer assigned aircraft, Pro Super and Flightline Expediter duties may be combined. The Pro Super has responsibilities for:

3.5.1. Make the final determination on aircraft status after reviewing aircraft forms. **(T-3)** Additionally, the Pro Super will review the forms after aircrew debrief. **(T-3)**

3.5.1. **(AFMC)** Make the final determination on aircraft status after reviewing aircraft forms and MIS applicable to specific Mission Design Series (MDS) (e.g. Autonomic Logistics Information System (ALIS), Integrated Maintenance Data System (IMDS), Integrated Maintenance Information System (IMIS), etc.)

3.5.2. Sign the Exceptional Release (ER) IAW TO 00-20-1 when authorized by the MXG/CC IAW **Chapter 11** and **Table 11.1**.

3.5.3. Participate in developing and executing the monthly and weekly flying and maintenance schedules/plans. **(T-3)**

3.5.4. Manage the maintenance production effort by assigning priorities to meet the flying and maintenance schedules. **(T-3)**

3.5.5. Fully understand actions required by the squadron under OPLAN 8010 or contingency plans. **(T-3)**

3.5.5.1. Develop, ensure currency of, and direct the aircraft generation sequence. **(T-3)**

3.5.6. Fully understand and be prepared to implement specific disaster control duties and squadron responsibilities pertaining to aircraft/SE movement and personnel evacuation procedures developed IAW DAFI 10-2501 and emergency response guidebook. **(T-3)**

3.5.6.1. Pro Super will maintain a current copy of the on-base disaster map with cordon overlay and appropriate functional checklists outlining duties during disaster scenarios. **(T-2)**

3.5.7. Determine, track, and report aircraft/systems status IAW DAFI 21-103.

3.5.8. Establish and track Estimated Time In Commission (ETIC). **(T-3)**

3.5.9. Monitor unit CDDAR Program activities and local procedures designed to protect personnel and prevent further damage to aircraft, equipment, and other resources. **(T-3)**

3.5.10. Inform MOC of the maintenance effort and coordinate with MOC, Flightline Expediter, and other squadrons for support. **(T-3)**

3.5.10.1. Pro Super will provide MOC with aircraft/systems status updates as required. **(T-3)**

3.5.11. Verify aircraft/system is in an authorized status IAW MDS-specific Minimum Essential Subsystem List (MESL) or MDS equivalent and DAFI 21-103 prior to verifying MICAP conditions. **(T-3)**

3.5.12. Verify aircraft weapons/load configurations are authorized IAW AFI 63-101/20-101. **(T-3)**

3.5.13. Review Condition Based Maintenance (CBM+) component forecast to ensure proactive maintenance practices are accomplished.

3.6. Flightline Expediter. The Flightline Expediter ensures maintenance is accomplished and coordinates on all aircraft maintenance actions. Flightline Expediters manage, control and direct resources to accomplish maintenance. Flightline Expediters or equivalent will:

3.6.1. Remain on the flightline when maintenance personnel are performing flightline maintenance and launching/recovering aircraft. **(T-3)** Flightline Expediters engage in direct sortie generation activities and work directly for the production superintendent.

3.6.1.1. Not perform production inspections (such as, sign off "Red Xs" and perform IPIs) unless authorized by the SQ/CC. **(T-3)**

3.6.2. Coordinate with the Weapons Expediter, ensure requirements in DAFMAN 21-201 for flightline munitions accountability are strictly followed.

3.6.3. Develop and implement disaster control duties and squadron responsibilities pertaining to aircraft/SE movement and personnel evacuation IAW DAFI 10-2501.

3.6.4. Maintain and have available for immediate use copies of the following as a minimum: flying schedule, emergency action and functional checklists, base grid map with cordon overlay, IPI listings, MESL, Quick Reference List (QRL) (if developed), WUC manual, and tracking device for aircraft status. **(T-3) Note:** Base grid maps and cordon overlays may not always be available during contingency operations at austere locations. Effort should be made to obtain maps and overlays but may not be available.

3.6.4.1. Track, as a minimum, the following aircraft status information: aircraft serial number, location, priority, status and ETIC, configuration, OAP condition codes, fuel load, munitions load, and remarks. Show all limitations against the Full Systems List (FSL) and Basic System List (BSL) column as itemized on the MESL or MDS equivalent. **(T-3)** Ensure devices depicting aircraft status comply with program security requirements. **(T-3)**

3.6.5. Follow established CANN procedures and ensure all CANNs are accurately documented in the aircraft/system forms and MIS as described in **Chapter 11**.

3.6.6. Ensure aircraft OAP sampling is completed IAW **Chapter 11** and applicable technical data. **(T-3)**

3.6.7. Ensure parts are ordered with appropriate priorities and relay document numbers to the Pro Super, MOC, and appropriate technicians. **(T-3)**

3.6.7.1. Ensure timely turn in of DIFM items to DMS/supply IAW AFI 23-101.

3.6.8. Request support beyond unit capability to the MOC. **(T-3)**

3.6.9. Direct AGE drivers to position AGE as required and notify the driver of AGE on the flightline or sub-pools that require maintenance. **(T-3)**

3.6.10. Ensure timely and accurate aircraft status (for example, discrepancies, WUC/LCN, ETIC, job completion) and configuration status is reported IAW DAFI 21-103 to the Pro Super and MOC. **(T-3)**

3.6.11. Ensure completed aircraft forms are provided to the debrief function by the end of the flying day if debriefs have been suspended due to surges. **(T-3)**

3.7. Aircrew and Maintenance Debrief Section. Debrief is conducted at the termination of each sortie/mission or when a sortie/mission is aborted. Aircraft scheduled for turn-around sorties/missions need not be debriefed if returned in landing status Code 1 or 2. However, debriefing is required, regardless of landing status, after the last flight of the day for each aircrew. MAJCOMs operating RPAs will develop and publish debrief procedures for Remote Split Operations in their supplements or addendum for both aircraft and ground control stations to adequately capture all maintenance discrepancies. The Debrief Section will:

3.7.1. Use aircraft fault reporting manuals and include fault codes when documenting discrepancies in the aircraft forms. **(T-3)** Debrief Section will use automated debrief tools such as the Computerized Fault Reporting System. **(T-3)**

3.7.1.1. Debrief Section will develop local aircrew debriefing guides when not provided and managed by the Weapon System PM. **(T-3)** QA will review and approve local aircrew debriefing guides every two years. **(T-3)**

3.7.2. Implement procedures for reporting dropped objects, aborts, code 3 flight control malfunctions and engine malfunctions. **(T-3)**

3.7.3. Use operational utilization update screens in MIS to enter flying time information. **(T-2)** Debrief Section will ensure flying times and installed engine Event History Recorder (EHR) readings, for both home station and deployed sorties/missions, are updated no later than the next duty day after occurrence. **(T-3)**

3.7.4. Check AFTO Form 781H, *Aerospace Vehicle Flight Status and Maintenance*, to ensure updates to airframe time and applicable servicing data (in-flight/hot pit refueling) are entered on the AFTO Form 781H or equivalent and/or applicable debrief system during the pilot/aircrew debrief. **(T-3)**

3.7.5. Input discrepancy verbatim and deviation information, utilization, and applicable flight data (to include landing status, system capability IAW DAFI 21-103 and other applicable cause codes) into the MIS. **(T-3)** Unless using an automated 781 process, do not send AFTO Form 781-series forms to Operations Squadron(s) or to Aviation Resource Management before MIS updates. Use local backup procedures for recording data when the MIS is unavailable.

3.7.6. Utilize MIS to identify and research discrepancies for repeat/recur trends and document them accordingly on the AFTO Form 781A. **(T-3)** Debrief Section will ensure previously documented discrepancies are reviewed and identified as repeat/recurs. **(T-3)**

3.7.6.1. Debrief Section will identify repeat/recurs on automated debriefing sortie recaps and on the AFTO Form 781A. **(T-3)**

3.7.7. Use the appropriate landing status code (**Table 3.1**) and the appropriate system capability code (**Table 3.2**) for the completion of a sortie/mission. **(T-3)**

3.7.8. Provide the MOC with aircraft identification numbers and system WUCs for each aircraft debriefed with a landing status Code-3 IAW **Table 3.1** using the approved MESL or MDS equivalent IAW DAFI 21-103.

3.7.9. Enter one of the deviation cause codes (**Table 3.3**) into the MIS. **(T-3)** Indicate the reason for the deviation and the agency that caused a deviation as referenced in applicable MIS guidance.

3.7.10. Collect and submit ASIP aircraft usage data IAW the MDS specific TOs, DAFI 63-140, and **Chapter 11** of this instruction.

3.7.11. If MIS is not available, use blank printouts as manual documentation method. **(T-3)** If deployed, send documents to home station for data transcribing by the most expeditious means available. Debrief Section will turn in, validate, and reconcile all documents with the MIS when it becomes available. **(T-2)**

Table 3.1. Landing Status Codes.

CODE	STATUS
Code 0	Ground Abort
Code 1	Aircraft Mission Capable (MC) with no additional discrepancies
Code 2	Aircraft or system has minor discrepancies but is capable of further mission assignment.
Code 3	Aircraft or system has major discrepancies in mission essential equipment that may require extensive repair or replacement prior to further mission assignment. The discrepancy may not affect safety-of-flight and the aircraft may be Non-Mission Capable (NMC) flyable.
Code 4	Aircraft or system has suspected or known radiological, chemical, or biological contamination.
Code 5	Aircraft or system has suspected or known battle damage.
Note: Debrief will enter code “8” in MIS for aircraft debriefed as code “4” or “5”. MESL or MDS equivalent requirements determine if aircraft status is NMC or Partially Mission Capable (PMC).	

Table 3.2. System Capability Codes.

CODE	STATUS
Code 0	System flown with a known discrepancy; no additional discrepancies noted. System can be used.
Code 1	System used and performed satisfactorily. No maintenance required.
Code 2	System used and performed satisfactorily. A minor malfunction exists, but system is capable of further mission assignment.
Code 3	System performance was unsatisfactory. This system did not cause an abort.
Code 4	System performance was unsatisfactory. This system caused or contributed to an abort.
Code 5	System out-of-commission prior to takeoff.
Code 6	System installed but not used.
Code 7	System not installed.
Code 8	Aircraft or system has suspected or known radiological/biological contamination.

Table 3.3. Deviation Cause Codes.

CODE	DEVIATION REASON
ATx	Air Traffic
GAA	Ground Abort, before engine start, maintenance
GAB	Ground Abort, after engine start, before taxi, maintenance
GAC	Ground Abort, after taxi, maintenance
HQT	Higher Headquarters
HQN	Higher Headquarters, NAF
HQP	Higher Headquarters, other
MTx	Maintenance
OPx	Operations
SUx	Supply
SYx	Sympathy
WXx	Weather
OTx	Other
Xxx	MAJCOM/local use
Note: Use x for any character for MAJCOM/local use.	

3.8. Aircraft Section. The Aircraft Section is the primary work center responsible for maintaining assigned aircraft. This section performs tasks to include servicing, scheduled and unscheduled maintenance, pre-flights, thru-flights, basic post-flights, home station checks, special inspections, corrosion control, cleaning, ground handling, launch/recovery of aircraft, troubleshooting and adjustment, on-equipment repairs and component removal/replacement, documenting maintenance actions, and managing aircraft forms. AMUs with 18 or more Primary Aerospace Vehicle (Aircraft) Inventory (PAI) aircraft may have two Aircraft Sections. The Aircraft Section consists of Aircraft Technicians. Refer to **Chapter 11** for FCC responsibilities.

3.8.1. Aircraft Technician:

3.8.1.1. Perform ground handling, servicing, -6 inspections, alert duties, maintenance ground tests, corrosion control, lubrication and maintenance and modification preparations, as applicable, on the assigned aircraft/system.

3.8.1.2. Inventory on-aircraft -21 equipment when this responsibility is not assigned to another function.

3.8.1.3. Perform engine operation when qualified and certified.

3.8.2. Dedicated Crew Chief (DCC) Program. The objective of a DCC program is to directly assign a maintenance person to each aircraft to provide continuity/accuracy of aircraft forms, aircraft status, scheduled maintenance, and improve aircraft appearance. DCCs manage and supervise maintenance on their aircraft. DCCs are selected on the basis of initiative, management and leadership ability, and technical knowledge. When authorized, ensure the DCC's and Assistant Dedicated Crew Chief name and rank is stenciled or painted on their assigned aircraft. Use only authorized wing paint scheme and marking procedures in TO 1-1-8. In addition to Aircraft Technician responsibilities, DCCs, if assigned:

3.8.2.1. Accompany their aircraft through scheduled inspections and assist the Inspection Section NCOIC/Chief as needed.

3.8.2.1.1. Attend pre- and post-dock meetings.

3.8.2.1.2. Assist the Inspection Section NCOIC/Chief with completing the required document review and validation at the end of the inspection.

3.8.2.2. Coordinate with Pro Supers and expediters for downtime to accomplish scheduled and unscheduled maintenance.

3.8.2.3. Manage deferred discrepancies.

3.9. Specialist Section. The Specialist Section is responsible for aircraft systems troubleshooting, on-equipment repairs, component removal and replacement, aircraft avionics systems, classified item management, aircraft ground handling, servicing, and cleaning. The section may include avionics, propulsion, hydraulic, and electro/environmental technicians, and other specialties approved through higher headquarters. When used, the Specialist Section Expediter coordinates maintenance priorities with the Pro Supers and Flightline Expediters.

3.9.1. In addition to the common responsibilities in **Chapter 2**, the Specialist Section Chief:

3.9.1.1. Ensures accurate and timely pod and support equipment status is updated or verified daily in Reliability, Availability, Maintainability, for Pods IAW DAFI 21-103 for pods under the control of the Aircraft Maintenance Squadron.

3.9.2. Avionics Specialists:

3.9.2.1. Perform PACER WARE, SERENE BYTE message, or TCTO reprogramming of avionics systems. **(T-3)**

3.9.3. Electronic Warfare (EW) specialist functions may be combined with the avionics specialists. EW Specialists:

3.9.3.1. Maintain inventory control of all installed Electronic Counter Measure (ECM) AME and ECM pods. **(T-3)**

3.9.3.2. Perform reprogramming of avionics/electronic warfare systems (to include electronic attack pods) IAW applicable mission directives, PACER WARE/SERENE BYTE messages, or TCTO requirements. **(T-3)**

3.9.3.3. Load contingency and training configuration settings in ECM pods, infrared countermeasures systems, and Radar Warning Receiver/Radar Threat Warning systems, unless the equipment is assigned to another section. **(T-3)**

3.9.4. Propulsion Specialists:

3.9.4.1. Troubleshoot, repair, and replace aircraft propulsion systems and components. **(T-3)**

3.9.4.2. Perform engine flightline blade blending. **(T-3)**

3.9.4.3. Perform flightline engine borescope inspections. **(T-3)**

3.9.5. Electrical & Environmental (E&E) Specialists:

3.9.5.1. Troubleshoot, repair, and replace aircraft E&E system components including aircraft environmental control, bleed air, vacuum, pneumatic, installed fire extinguishing and suppressant systems, Liquid Oxygen (LOX) and Gaseous Oxygen (GOX) systems, and On-Board Oxygen Generating Systems (OBOGS), On-Board Inert Gas Generation Systems (OBIGGS) and components. **(T-3)**

3.9.5.1. **(AFMC)** [e.g.] Molecular Sieve Oxygen Generating System (MSOGS).

3.9.5.2. Remove and install In Flight Refueling (IFR) carts and fire bottle squibs. **(T-3)** **Note:** Ensure only approved temporary storage locations are used for these components.

3.9.6. Hydraulic Specialists will maintain authorized on-equipment/off-equipment pneumatic and hydraulic systems and components. **(T-3)**

3.10. Weapons Section/Expediter/ Loading Element/ Maintenance Element/ Non-Standard Weapons Sections. See AFMAN 21-206 for duties and responsibilities.

3.10.1. - 3.10.5.8. DELETED.

3.11. Support Section. The Support Section may include the following elements/functions to support unit flightline maintenance activities; support (CTKs/special tools, eTools, test equipment, TOs, bench stock), -21 equipment, AME, mobility equipment and DMS. Personnel will be assigned to the Support Section for a minimum of 12 months. **(T-3)** Personnel may be required to maintain task qualifications/certifications to ensure units can meet readiness and Core Mission Essential Task List (METL) requirements in Defense Readiness Reporting System-Strategic (DRRS-S) IAW AFI 10-201. Support Sections must standardize procedures across the squadron for security, control, and accountability of equipment. **(T-2)** Materiel support procedures in this section do not apply to aircraft supported by Contractor Operated and Maintained Base Supply. The Support Section:

3.11.1. Maintains TOs IAW TO 00-5-1.

3.11.2. Maintains bench, shop, and operating stocks IAW AFI 23-101, and **Chapter 9** of this instruction.

3.11.3. Ensures maintenance, control, and storage of assigned AME, -21 equipment, and Maintenance, Safety, and Protective Equipment IAW DAFI 21-103.

3.11.3.1. Develops local procedures to control and store other equipment not identified as -21 equipment (such as, aircraft galley items, U-2 pod panels, aircraft pylon attachment cover panels, aircraft covers/plugs) using DAFI 21-103 guidelines.

3.11.4. Ensures proper calibration, use, care, handling, and transportation of TMDE IAW TO 00-20-14, AFMAN 21-113, and applicable Calibration Measurement Summaries.

3.11.5. Maintains and manages squadron Land Mobile Radio (LMR) IAW **Chapter 11** (as applicable).

3.11.6. Monitors the status of critical support equipment and testers for serviceability, accountability, and status of TCTO modifications. **(T-3)** Support Section will provide monthly critical support equipment status update to Maintenance Supervision. **(T-3)**

3.11.7. Maintains tools/CTKs IAW **Chapter 8**.

3.12. Unit Decentralized Materiel Support (DMS). In addition to the responsibilities in **Chapter 9** for DMS procedures, DMS personnel:

3.12.1. Requisition parts and use supply management products. Initiate follow-up action when necessary. **(T-3)**

3.12.2. Notifies Flightline Expediter of all back-ordered parts. **(T-3)**

3.12.3. Develops and maintains a QRL as needed and provide it to technicians. **(T-3)**

3.12.4. Tracks and processes DIFM assets, to include warranty parts IAW AFI 23-101. **(T-3)**

3.12.4.1. Notifies unit leadership when DIFM asset turn-in times exceed requirements outlined in AFI 23-101. **(T-3)**

3.12.5. Manages reusable containers IAW DAFI 24-602V2, *Cargo Movement*, and TO 00-20-3. **(T-3)**

3.12.6. Controls and manages aircraft TNB if stored within the Support Section. **(T-3)** When FOM assets are collocated with TNB, the items must be similarly controlled and managed. **(T- 3)**

3.12.7. Coordinates with the Pro Super and Flightline Expediter(s) for “mark for” changes. **(T- 3)**

3.12.8. Manages the unit’s CANN program supply transactions and the associated documentation. **(T-3)**

3.13. (Added-AFMC) Global Hawk (GH) In-Flight Maintenance Procedures and Documentation.

3.13.1. **(Added-AFMC)** GH Remotely Piloted Vehicle (RPV) Documentation. GH RPV reporting will use standard Air Force landing status codes. Mission Crew Element (MCE) and Launch and Recovery Element (LRE) status will be reported separately from aircraft status (i.e., Aircraft Code 1, MCE Code 1, LRE Code 1). Aircraft, MCE and LRE will retain separate aircraft forms binders. Mission capable status will also be tracked separately. Discrepancies discovered during flight will not be documented on the AFTO Form 781As or entered into IMDS for any asset directly involved with the mission while the aircraft is airborne. Units will develop a manual debrief data sheet which is maintained and controlled by the debrief section

by mission number. This debrief data sheet will be used to annotate in-flight maintenance and pilot reported discrepancies. Continuation sheets will be annotated as #2, #3, etc., of XX pages. Once the mission ends and the post-mission debrief is completed, the 781 forms, IMDS and mission capable status will be annotated. **(T-2)**

3.13.2. **(Added-AFMC)** GH RPV Ground Station In-flight Maintenance procedures. The pilot in control of the system during the mission will always be the final authority as to what, when and for how long maintenance actions will be performed during flight. UAS Ground Station In-flight Maintenance is intended to provide repair of maintenance discrepancies while allowing the mission to continue. Discrepancies discovered during flight will not be documented on the AFTO Form 781As or entered into IMDS until after the mission is ended. Upon notification from the pilot, maintenance technicians will be dispatched. Maintainers will respond with the debrief data sheet, provided by the debrief section. The pilot will document the discrepancy and provide written release for maintenance on the debrief data sheet. Maintenance will not be performed until the pilot releases for maintenance. The maintainer will return the debrief data sheet (with maintenance actions annotated) to the debrief section for inclusion into the final mission debrief. If additional maintenance actions are responded to during the same mission, the same debrief data sheet will be used. Use continuation sheets as required. **(T-2)**

Chapter 4

MAINTENANCE SQUADRON (MXS)/GENERATION SQUADRON.

4.1. General. The MXS/Generation Squadron supports MGN operations by providing centralized back shop support to perform on and off equipment maintenance tasks that are assigned to a specific back shop function. The MXS provides both organizational and intermediate level maintenance described in the "Maintenance Concept" section in **Chapter 1** of this DAFI. Bases with permanently assigned Centralized Repair Facilities (CRF), which support enterprise RN functions, will develop, and document the division of responsibilities between the MGN and RN, as outlined in **Chapter 13** of this DAFI, to ensure both local and enterprise mission requirements are met. IAW AFI 38-101, the MXS may consist of personnel from various AFSCs organized into flights: Propulsion Flight, Avionics Flight, TMDE Flight, Accessories Flight, AGE Flight, Fabrication Flight, Armament Flight, Maintenance Flight, and Munitions Flight. The MXS maintains AGE, munitions, off-equipment aircraft, and support equipment components; performs on-equipment maintenance of aircraft and fabrication of parts; and provides repair and calibration of TMDE. **Note:** For purpose of this instruction, MXS represents MXS, Equipment Maintenance Squadron (EMS), and Component Maintenance Squadron (CMS).

4.1. (AFMC) [DEV] General. The 412 MXS is organized into seven flights versus nine.

4.1.1. **(Added-AFMC)** [DEV] The 412 MXS/MXMA, Component Repair Flight, consolidates Avionics Flight, Armament Flight, and the Electro- Environmental, Battery Shop and Egress functions of the Accessory Flight in addition to the Tow Targets Section.

4.1.2. **(Added-AFMC)** [DEV] The 412 MXS/MXMX, Maintenance Flight, consolidates the functions of the Maintenance Flight and the pneudraulic and fuels functions of the Accessories Flight.

4.1.3. **(Added-AFMC)** [DEV] The 412 MXS/MXMW, Munitions Flight is responsible for the Armament (412 MXS/MXMR) function for Bomber aircraft only due to the geographic location/separation at Edwards AFB, South Base.

4.2. Maintenance Supervision Responsibilities. Maintenance Supervision manages the resources to accomplish the workload. **Note:** In a Generation Squadron, MXSs are obsolete but responsibilities of the unit and duty positions remain. In addition to general responsibilities in **Chapter 2**, Maintenance Supervision:

4.2.1. Reviews and consolidates monthly maintenance plan inputs from flights/sections and forward to Maintenance Operations PS&D. **(T-3)**

4.2.2. Optimizes local repair capability by ensuring base level repair constraints (for example, lack of equipment, manpower, parts) are elevated to the Repair Network Manager (RNM) and applicable stakeholder(s) IAW DAFI 20-117, TO 00-20-3, and **Paragraph 1.3.2** of this DAFI, and MAJCOM supplements to keep repair at the lowest level. **(T-2)**

4.2.3. Ensures EOR procedures for transient aircraft are developed IAW TO 00-20-1 and MAJCOM supplements. **(T-3)**

4.2.4. Ensures the MXS and WS develop procedures for required weapons loading actions on transient aircraft, storage of transient aircraft impulse cartridges, and requisition and maintenance of weapons safing equipment for common transient types of aircraft. **(T-3)**

4.2.5. Ensures local manufacture capability and fabrication process is controlled IAW this instruction. **(T-3)**

4.2.6. Ensures MXS personnel utilize Engineering Technical Service (ETS) personnel and the Joint Engineering Data Management Information and Control System (JEDMICS) **https://jedmics.af.mil/webjedmics/index.jsp** to obtain information and specifications when the information in TOs does not provide enough detail. **(T-3) Note:** For drawings not available electronically, contact the appropriate JEDMICS help desk.

4.2.7. DELETED.

4.3. MXS Production Superintendent (Pro Super). Monitor backshop production and flightline operations and coordinate support and priority with other squadron Pro Supers and MOC. **(T-3)** Focuses overall maintenance efforts toward MXG maintenance priorities. Identifies production requirements and shortfalls to Maintenance Supervision. **(T-3)**

4.3.1. & 4.3.2. DELETED.

4.4. Accessories Flight. The Accessories Flight normally consists of four sections; Electrical and Environmental (E&E), Egress, Fuels, and Hydraulics and is responsible for performing on/off-equipment maintenance of systems and equipment.

4.4.1. Accessories Flight CC/Chief Responsibilities. In addition to the common responsibilities in **Chapter 2**, the Accessories Flight CC/Chief:

4.4.1.1. Ensures an egress training program is established IAW this instruction. **(T-3)**

4.4.1.2. Coordinates with squadron superintendents or civilian equivalent to ensure assigned personnel rotation plans are established to comply with core task upgrade requirements. **(T-3)**

4.4.1.3. Ensures explosives are controlled and stored in approved storage areas/containers IAW DESR6055.09_AFMAN91-201.

4.4.2. Electrical and Environmental (E&E) Section. The E&E Section performs authorized local manufacture, repair, overhaul, testing, modification, and inspection of aircraft and SE electrical components, wiring harnesses, batteries, and charging units. The E&E Section:

4.4.2.1. Ensures battery disposal procedures meet applicable environmental standards and batteries are controlled for accountability purposes. **(T-0)**

4.4.2.2. Performs on/off-equipment maintenance on aircraft electrical and environmental systems and components. **(T-3)**

4.4.2.3. Repairs LOX/GOX/Liquid Nitrogen servicing units/carts. **(T-3) Note:** AGE performs chassis, enclosure, and trailer maintenance on gaseous and cryogenic servicing units and all maintenance on Self-Generating Nitrogen Servicing Carts.

4.4.2.4. Performs off-equipment maintenance for aircraft and aircrew Carbon Dioxide cylinders. **(T-3)**

4.4.2.5. Performs off-equipment maintenance on type MA-1 portable breathing oxygen cylinders (portable walk around bottles) and regulators, to include removing/replacing the regulator and purging the bottle. **(T-3) Note:** Ownership and storage of these cylinders remain with the appropriate support section.

4.4.3. Egress Section. The Egress Section maintains aircraft egress systems, components, and trainers (such as, aircraft ejection seats, extraction and escape systems, egress components of jettisonable canopies, explosive components of escape hatches/doors) and stores egress explosive components that are removed to FOM. Wings will identify the base level organization responsible for locating inadvertent beacon activations on the flightline and configuring survival kit personnel locator beacons (on-aircraft). **(T-3)**

4.4.3.1. The Egress Section:

4.4.3.1.1. Performs all off-equipment ejection seat maintenance in the egress maintenance facility compliant with Defense Explosives Safety Regulation 6055.09_DESR6055.09_AFMAN 91-201and TO requirements.

4.4.3.1.2. Ensures all personnel use the Demand Response Team during any task requiring the removal/installation of explosive components, and during egress final inspections. **(T-3)=3**

4.4.3.1.2.1. Demand Response Teams will be comprised of individuals who are certified to perform egress maintenance. **(T-3)** At least one team member must be a certified egress journeyman. **(T-3)**

4.4.3.1.2.1.1. **(Added-AFMC)** Egress augmentees may be utilized as part of an egress maintenance team to perform on-equipment maintenance, such as ejection seat and canopy removals or installations and egress final inspections. Do not use augmentees to perform egress off-equipment maintenance actions. **(T-2) Note:** Ejection seats undergoing periodic maintenance are exempt from the demand-response concept once explosive components have been removed.

4.4.3.1.3. Coordinate with PS&D and monitor the weekly maintenance schedule to identify egress items requiring removal for scheduled time changes/maintenance. **(T-3)**

4.4.3.1.4. Utilize a facility that meets the requirements of DAFMAN 32-1084, *Facility Requirements*. Locations are established IAW DESR6055.09_AFMAN 91-201 to store explosive components and ensure they are properly licensed.

4.4.3.1.4. **(AFMC)** Egress facilities will have limited access to ensure system integrity. **(T-2)**

4.4.3.1.4.1. Egress Section will ensure licensed explosive area will not exceed the licensed Net Explosive Weight capacity for each Hazard Division without approval from Wing Safety. IAW DESR6055.09_AFMAN 91-201.

4.4.3.1.4.1. **(AFMC)** Only egress section personnel will be authorized unescorted entrance to the egress licensed explosive location. **(T-2)**

4.4.3.2. The Egress Section NCOIC/Chief will:

4.4.3.2.1. Ensure ejection systems are “safed” IAW with 00-80G-series technical orders and DESR6055.09_AFMAN 91-201 prior to an aircraft being placed on static display. **(T-3)**

4.4.3.2.2. Ensure egress systems on training aircraft are de-armed/“safed” IAW MDS specific TOs when an aircraft is used for Fire Emergency Services and/or aircrew extraction training. **(T-3)**

4.4.3.2.3. Ensure aircraft (to include GITA) are "safed" IAW 00-80-series TOs.

4.4.3.2.4. Ensure all permanently decommissioned static display aircraft explosive devices are removed and turned in to munitions inspections IAW DAFMAN 21-201. Egress Section will sign the appropriate block on the AF Form 3580, *USAF Heritage Program Aerospace Vehicle Static Display Egress and Safety Certificate*, which is retained by the Historical Property Custodian(s). **(T-3)**

4.4.3.2.5. Request assistance from Explosive Ordnance Disposal (EOD) when egress explosive devices are damaged or suspected to be unsafe. **(T-3)**

4.4.3.2.5.1. **(Added-AFMC)** When requesting changes or updates to the need-date of any Cartridge Actuated Device/Propellant Actuated Device (CAD/PAD) item, coordinate with Plans Scheduling & Documentation (PS&D) and Munitions Accountable System Officer (MASO) in sufficient time to preclude emergency issue requests. Refer to DAFMAN 21-201, *Munitions Management*, for emergency issue and contingency issue requests.

4.4.3.2.6. Establish egress training program requirements and conduct reviews IAW AFI 36-2650. **(T-2)**

4.4.3.2.6. **(AFMC)** Classification training of Egress personnel will meet mandatory training requirements contained in AFI 36-2650, *Maintenance Training*, DAFI 91-202, *The US Air Force Mishap Prevention Program*, this AFI, and the CFETP. **(T-2)**

4.4.3.2.6.1. As a minimum, the program will include: a master training plan, explosive safety certification requirements, and MIS time change documentation qualification minimums. **(T-3)**

4.4.3.2.6.1.1. **(Added-AFMC)** Civil service and contractor egress maintenance personnel who possess, as a minimum, one year of experience within the last three years performing egress intermediate- and organizational-level maintenance, repair, inspections, etc., may be considered for a waiver of classification training requirements. Waiver requests will be submitted to the applicable 2A6X3 MAJCOM functional manager for review then forwarded to the 2A6X3 Career Field Manager for final approval/disapproval. If waiver is disapproved, individuals must complete classification training. **(T-2)**

4.4.3.2.6.2. Certification requirements.

4.4.3.2.6.2.1. Trained and certified egress 5-levels and above reassigned to another base or unit are not required to complete the Organizational Maintenance (on-equipment) egress technician course unless required by the Egress Section NCOIC/Chief.

4.4.3.2.6.2.1.1. **(Added-AFMC)** Personnel are certified to perform egress systems maintenance by demonstrating adequate proficiency to a designated certifying official in the egress systems work center. Certification pass/fail criteria will be established by the egress work center supervisor. **(T-2)** Document certification in accordance with AFI 36-2650. Retrain any individual who fails the practical evaluation until the individual demonstrates adequate proficiency or withdraw the individual from training. If egress

personnel are withdrawn from training, retrain in accordance with AFI 36-2650.

4.4.3.2.6.2.1.2. **(Added-AFMC)** Initial Certification of Non-Egress who augment egress technicians or perform Quality Assurance Evaluations of Egress Systems: Non-egress personnel must successfully complete an AETC egress technician course for the specific aircraft to be maintained. **(T-2) Exception:** AFE personnel do not have to complete the technician course unless they are to be employed as egress augmentees.

4.4.3.2.6.2.1.3. **(Added-AFMC)** After completing the AETC technician course, non-egress personnel will be initially certified in accordance with **paragraph 4.4.3.2.6**. Non-egress personnel will also be recertified in accordance with **paragraph 4.4.3.2.6.4.2.2 (T-2)**

4.4.3.2.6.3. Decertification requirements:

4.4.3.2.6.3.1. Decertify egress personnel after not having performed egress maintenance for more than 18 months. **(T-3)** Instructing and inspecting egress maintenance is not considered performing maintenance.

4.4.3.2.6.3.2. Document decertification IAW DAFMAN 36-2689.

4.4.3.2.6.4. Recertification requirements:

4.4.3.2.6.4.1. Recertify egress personnel who have not performed egress maintenance for 18 months. **(T-3)**

4.4.3.2.6.4.2. Recertification must be accomplished by a 2A673 trainer and certifier. **(T-3)**

4.4.3.2.6.4.2.1. **(Added-AFMC)** The purpose of recertification is to ensure personnel still maintain the required knowledge and skills to safely maintain and/or inspect egress systems.

4.4.3.2.6.4.2.2. **(Added-AFMC)** Recertify non-egress personnel at least every 180 days. **(T-2)**

4.4.3.2.6.4.2.3. **(Added-AFMC)** Contracting Officer Representative (COR)/ QA evaluator personnel exclusively performing contract surveillance do not have to be recertified. If these personnel are performing egress maintenance, they must be recertified. **(T-2)**

4.4.3.2.6.4.2.4. **(Added-AFMC)** Recertification procedures are identical to initial certification procedures and will be accomplished in accordance with **paragraph 4.4.3.2.6.2.1.1**. **(T-2)** Document recertification in accordance with AFI 36-2650.

4.4.3.2.7. Review and validate all egress familiarization training documents at least every 24 months. **(T-3)**

4.4.3.2.8. Ensure the egress Time Change Item (TCI) data in the MIS is accurate. **(T-3)**

4.4.3.2.8.1. Ensure automated data products will be updated whenever an egress item is replaced to ensure the annual TCI forecast is correct. **(T-3)**

4.4.3.2.8.2. Ensure separate databases are not used to manage the egress TCI program. **(T-3)**

4.4.3.2.8.3. Ensure component background information is provided to PS&D to include a list of all components having multiple part numbers with a different service life. **(T-3)**

4.4.3.2.9. Reconcile and verify each aircraft's egress data annually with PS&D. **(T-3)**

4.4.3.2.9.1. Document the annual verification on the AF Form 2411, *Inspection Document* maintained in the aircraft jacket file. **(T-3)**

4.4.3.2.10. Establish local egress systems documentation requirements: **(T-3)**

4.4.3.2.10.1. Egress Section will maintain an egress tail number binder for each assigned aircraft. **(T-3) Note:** This binder should not be confused with the aircraft jacket file of historical records maintained by PS&D but serves to standardize the format for maintaining egress documentation requirements across AF egress shops. As a minimum the binder will include:

4.4.3.2.10.1.1. Binder Spine - Aircraft assigned Serial Number. **(T-3)**

4.4.3.2.10.1.2. Front Cover - Delayed Discrepancies. **(T-3)**

4.4.3.2.10.1.3. Tab A – PRA or 5th Gen equivalent (Cross reference sheet stating digital location of data). **(T-3)**

4.4.3.2.10.1.4. Tab B – Completed In-Shop Maintenance Flow Sheet(s). **(T-3)**

4.4.3.2.10.1.5. Tab C – Completed Major Inspection Flow sheet(s) (for example, 36 month), or equivalent (if Mission Design Series applicable). **(T-3)**

4.4.3.2.10.1.5.1. Annotate pull check/inspection results on the flow sheet. **(T-3)**

4.4.3.2.10.1.6. Tab D – Egress Configuration Screen (IMDS 257 or 5th Generation equivalent), current manual Cartridge Actuated Device/Propellant Actuated Device (CAD/PAD) collection sheets. **(T-3)**

4.4.3.2.10.1.7. Tab E – Completed Time Change Item Replacement Sheets, for example, locally generated or equivalent. **(T-3)**

4.4.3.2.10.1.8. Tab F – Significant Historical Data (AFTO 95 or equivalent (cross reference to MIS maybe used), Depot/Program Depot Maintenance (PDM) package, -107 TAR or ETARs). **(T-3)**

4.4.3.2.10.1.9. Tab G – Misc. (example, PAIR, Extension Letters). **(T-3)**

4.4.3.2.10.1.10. Tab H AF Form 2411, *Inspection Document*. **(T-3)**

4.4.3.2.11. A certified egress inspector (for example, Red X, IPI certified and tracked on the SCR IAW **Table 11.1**) will inspect any integral part of the egress system when any maintenance other than a visual inspection is performed. **(T-3)**

4.4.3.2.11. **(AFMC)** Newly assigned uncertified egress personnel who have completed Classification Training may assist in performing egress systems

maintenance. These personnel will never clear (sign off) AFTO Form 781-series entries, MIS or condition tags until certified. **(T-2)**

4.4.3.2.11.1. The inspection must be an egress final inspection unless another inspection is prescribed by technical data. **(T-3)**

4.4.3.2.11.2. Egress personnel will conduct an egress final every 30 days on ejection seats that have integrated personnel/recovery parachutes and/or survival kits as part of the seat if prescribed by technical data. **(T-3)**

4.4.3.2.12. Egress Sections may store spare parachutes and survival kits for use in responding to Red Ball and unscheduled maintenance events.

4.4.4. Fuel Systems Section. The Fuel Systems Section repairs, functionally checks, and inspects aircraft fuel systems, fuel tanks, hydrazine systems, in-flight refueling receptacle systems, and related components. This section also performs maintenance on AME external fuel tanks, Conformal Fuel Tanks, and Weapons Bay Tanks and provides temporary storage for Conformal Fuel Tanks, and Weapons Bay Tanks.

4.4.4.1. The Fuels Systems Section will:

4.4.4.1.1. Maintain serial number inventory accountability for all removable external fuel tanks IAW DAFI 21-103. **(T-3)**

4.4.4.1.2. Purge and preserve fuel tanks for storage and shipment. **(T-3) Note:** LRS is responsible for the storage, delivery, and shipment of fuel tanks in their possession.

4.4.4.1.3. Establish a local Memorandum of Agreement (MOA) or Memorandum of Understanding (MOU) (MXS with squadron or equivalents) governing the storage, issue, receipt, and inventory control of in-use removable external fuel tanks. **(T-3)**

4.4.4.1.3. **(AFMC)** Within the Memorandum of Agreement (MOA)/Memorandum of Understanding (MOU), local units with approval from MXG/CC (or equivalent) will designate which squadron(s) is responsible for management of external fuel tanks and conformal fuel tanks. **(T-2)**

4.4.4.1.4. Perform all maintenance and inspections on WRM fuel tanks. **(T-3)**

4.4.4.1.4.1. Meet quarterly with installation War Reserve Materiel Officer/WRM Non-Commissioned Officer (NCO) and LRS representatives to review inspection criteria for stored WRM tanks, schedule tank inspections and maintenance, and report discrepancies identified during WRM monthly walk-through inspections. **(T-3)**

4.4.4.2. In addition to the common responsibilities outlined in **Chapter 2**, the Fuel Systems Section NCOIC/Chief will:

4.4.4.2.1. Establish controls to prevent unauthorized entry into fuel cell and hydrazine repair areas. **(T-3)**

4.4.4.2.2. Provide required qualification training to all personnel who enter aircraft fuel tanks and/or open fuel tank areas to perform maintenance or to provide assistance. **(T-3)**

4.4.4.2.3. When required, ensure Hydrazine Response Teams are formed. The HRT supervisors must possess 2A6X4 AFSC. Team members can be any qualified personnel provided they are certified IAW with TO 00-25-172, *Ground Servicing of Aircraft and Static Grounding/Bonding*, TO 00-105E-9, *Aerospace Emergency Rescue and Mishap Response Information*, TO 42B1-1-18, *General Procedures for Handling of H-70*, and review MDS-specific TOs and MAJCOM/Lead Command directives for additional information on hydrazine hazards and management. **(T-2)**

4.4.4.2.3.1. Ensure initial and refresher hydrazine safety training is completed for all hydrazine response team members IAW TO 42B1-1-18.

4.4.4.2.3.2. Integrate Hydrazine Response Team responsibilities into the CDDAR Program and local In-Flight Emergency (IFE) functional checklists (as applicable). **(T-3)**

4.4.4.2.4. Perform safety inspections on facilities to ensure open tank repair areas, and equipment used for open fuel tank or hydrazine maintenance meet MDS-specific TOs, TO 42B1-1-18 and TO 1-1-3, *Inspection and Repair of Aircraft Integral Tanks and Fuel Cells* requirements.

4.4.4.2.5. Establish a Confined Space Entry Program IAW TO 1-1-3 and AFMAN 91-203.

4.4.4.2.6. Establish a Respiratory Protection Program IAW AFI 48-137.

4.4.4.2.6.1. All respiratory training requirements are documented on AF Form 55 or equivalent IAW DAFI 91-202.

4.4.4.2.7. **(Added-AFMC)** Monthly review IMDS to schedule external fuel tanks for inspection or Time Compliance Technical Order (TCTOs). **(T-3)**

4.4.4.2.8. **(Added-AFMC)** Coordinate with Aircraft Maintenance Squadron (AMXS) or equivalents to schedule aircraft with non-grounding fuel leaks through the fuel systems repair facility to prevent further deterioration. **(T-3)**

4.4.4.3. Ensure assigned personnel receive periodic physical examinations as established by the base medical service as required IAW TO 1-1-3. Track periodic physical examinations in the MIS. **(T-3)**

4.4.4.3. **(AFMC)** Nestable Fuel Tank Build Up (NFTBU) Team Members and Training. The fuel systems section NCOIC identifies 2A6X4 personnel as NFTBU cadre members and establishes a training program. **(T-2)** The NFTBU team will be augmented by non-2A6X4 personnel in the wing during build up operations. **(T-3)** NFTBU Cadre team members will:

4.4.4.3.1. **(Added-AFMC)** Attend initial NFTBU training at an Air Education and Training Command (AETC) Training Detachment (TD). **(T-2)**

4.4.4.3.2. **(Added-AFMC)** Conduct annual refresher NFTBU training for all fuel systems section personnel tasked for any UTC and document completed training in the MIS. **(T-2)**

4.4.4.3.3. **(Added-AFMC)** Reviews UTC MISCAP statement as it applies to the unit's tasking and ensures availability of trained fuel systems personnel and serviceable

equipment/tools to support requirements. **(T-2)** (Units are no longer required to maintain ready-trained augmentees. Commanders must provide NFTBU augmentees to fill UTC requirements at the time of tasking). **(T-2)**

4.4.5. Hydraulic Section. The Hydraulic Section performs on- and off-equipment maintenance on pneumatic and hydraulic systems, components (except environmental and egress systems) and provides maintenance support for SE and test equipment. The Hydraulic Section also maintains hydraulic test stands, pumping units, and associated components.

4.4.5.1. The Hydraulic Section will:

4.4.5.1.1. Perform maintenance on munitions loading and handling equipment with discrepancies that exceed the munitions flight repair capabilities.

4.4.5.1.2. Maintain and inspect refueling drogues, booms, and refueling receptacle systems for large aircraft.

4.4.5.1.2. **(AFMC)** [DEV] Aircraft Maintenance Units (AMUs) will inspect refueling drogues, booms, and refueling receptacle systems for 412 TW aircraft.

4.4.5.1.3. Repairs, overhauls, and bench checks flight control, landing gear, and hydraulic power system components (such as, brakes, struts, accumulators, reservoirs, actuators).

4.5. Aerospace Ground Equipment (AGE) Flight. The AGE Flight is normally organized as a consolidated maintenance unit (repair, inspection, and servicing sections). The AGE Flight is responsible for providing powered and Non-Powered AGE (NPA) as defined in TO 00-20-1 to support both aircraft and non-aircraft weapon systems.

4.5. (AFMC) Aerospace Ground Equipment (AGE) Flight. The AGE Flight may be organized in any manner, deemed necessary by the MXS commander or equivalent, to best support and execute local aircraft sortie generation and aircraft maintenance.

4.5.1. The AGE Flight will:

4.5.1.1. Maintain and inspect AGE, IAW TO 00-20-1, and equipment specific TOs in support of sortie production and back shop maintenance activities.

4.5.1.2. Pick up, service, deliver, repair, and perform approved modifications, TCTOs, inspect AGE and perform corrosion control tasks.

4.5.1.3. Utilize AF Form 864, *Daily Requirement and Dispatch Record*, or MAJCOM-approved electronic product to record all equipment pickup and delivery.

4.5.1.3. **(AFMC)** Electronic form equivalents may be used.

4.5.1.4. Perform chassis, enclosure, and trailer maintenance on gaseous and cryogenic servicing units.

4.5.1.5. Manage maintenance and inspection scheduling activities for flight-maintained equipment. **Note:** Maintain oversight of additional requirements for assigned Nuclear Certified Equipment IAW AFI 63-125.

4.5.1.6. Safeguard any Item Unique Identification (IUID) marks during maintenance activities to the extent possible. **(T-3)** In the event the Unique Item Identifier (UII) is

damaged during maintenance activities, the AGE Flight notifies the responsible Equipment Custodian.

4.5.2. AGE Flight Chief Responsibilities. In addition to the applicable Flight CC/Chief responsibilities in **Chapter 2**, the AGE Flight Chief will:

4.5.2.1. Review and coordinate the AGE MEL annually with applicable Maintenance Supervision. **(T-3)**

4.5.2.1.1. Provides copies of the approved MEL to the MOC. **(T-3)**

4.5.2.2. Track AGE status/scheduling daily using the MIS. **(T-3)** Once available and training is complete, DPAS M&U will be the MIS for tracking AGE Status and Scheduling. **(T-2)** Additional guidance for stand up and integration of DPAS M&U is available at: **https://usaf.dps.mil/teams/afdpassemis**.

4.5.2.2.1. Provide status and ETIC information to the MOC when it falls below MEL. **(T-3)**

4.5.2.3. Ensure newly assigned AGE receives acceptance inspections IAW TO 00-20-1. **(T-3)**

4.5.2.4. Control fuel dispensed from issue tanks IAW DAFI 23-201, *Fuels Management*.

4.5.2.5. Ensure the Uniform Repair and Replacement Criteria Program is implemented IAW TO 00-25-240 and TO 35-1-24.

4.5.2.6. Coordinate welding requirements with the Fabrication Flight Chief. **(T-3)**

4.5.2.7. Manage AGE CANN actions IAW **Chapter 9** and **Chapter 11**. **(T-3)**

4.5.2.8. Establish and monitor the AGE Operator Training Program and assist in the development of course control documents in conjunction with Maintenance Training (MT). **(T-3)**

4.5.2.9. Ensure an AGE Corrosion Control and Prevention Program is maintained and a field number system is established IAW TO 35-1-3, TO 1-1-8, TO 1-1-691, MAJCOM instructions, and equipment specific TOs.

4.5.2.10. Develop and implement a tracking system to prioritize complete repainting for AGE equipment based on a “worst is first” principle.

4.5.2.10.1. Coordinate with Fabrication Flight Chief for work beyond the AGE work center capability.

4.5.2.11. Ensure equipment is prepared for storage or shipment IAW TO 35-1-4, *Processing and Inspection of Support Equipment for Storage and Shipment*, and applicable end item TOs.

4.5.2.12. Submits annual transient aircraft landing data to the respective MAJCOM AGE functional manager by 1 February. Data will reflect previous year's TA aircraft landings by aircraft MDS and is obtained from local Transient Alert managing office.

4.5.2.13. Establish AGE sub-pools, as needed, in coordination with OSS’s Airfield Operations Flight.

4.5.2.14. Ensure AGE tow vehicles are two-way radio equipped, permanent or hand-held, to expedite AGE deliveries. **(T-3)** Ensure any permanent installation of radios are accomplished IAW AFI 24-302. **(T-3)**

4.5.3. AGE Pro Super Responsibilities (if not assigned, the AGE Flight Chief will fulfill these responsibilities). The AGE Pro Super will:

4.5.3.1. Monitor the production of AGE Flight and recommend equipment/personnel adjustments to the AGE Flight Chief as required. **(T-3)**

4.5.3.2. Monitor adherence to AGE Flight's safety, training, and CTK programs. **(T-3)**

4.5.3.3. Monitor serviceability status of equipment parked in sub-pools. **(T-3)**

4.5.3.4. Monitor distribution, control, and condition of AGE Flight's assigned vehicles.

4.5.3.5. Monitor shop equipment for condition and documentation. **(T-3)**

4.5.3.6. **(Added-AFMC)** AGE Pro Super will coordinate with Fabrication Pro Super for work beyond the AGE work center capability. **(T-3)**

4.5.4. AGE Production Support Section. The AGE Production Support Section provides administration and ancillary services for TO file maintenance, supply support, and fuels management. A full-time Materiel Management Journeyman/craftsman (AFSC 2S0X1) should be assigned to the AGE Production Support Section when the workload warrants. In addition to the applicable Section NCOIC/Chief responsibilities outlined in **Chapter 2**, the AGE Production Support Section NCOIC/Chief will:

4.5.4.1. Manage the AGE Flight's TO libraries IAW TO 00-5-1.

4.5.4.2. Manage the AGE Flight's tool storage and issue areas IAW **Chapter 8**. **(T-3)**

4.5.4.3. Manage the AGE Flight's TMDE program IAW TO 00-20-14 and AFMAN 21-113. **(T-3)**

4.5.4.4. Manage the AGE Flight's materiel management function IAW **Chapter 9** and AFI 23-101.

4.5.4.4.1. Pre-assembled part kits are authorized; if required, assemble from bench stock in minimum quantities necessary to support workload requirements.

4.5.4.5. Coordinate the AGE Flight's scheduling function with PS&D. **(T-3)** The AGE Production Support Section NCOIC/Chief will:

4.5.4.5.1. Maintain AGE historical records. **(T-3)**

4.5.4.5.2. Prepare an AGE scheduled maintenance plan and maintain a current equipment scheduling report for all assigned equipment. **(T-3)**

4.5.4.5.3. **(Added-AFMC)** In coordination with PS&D, ensure all inspections, TCTOs, time changes, maintenance and repair actions are scheduled and performed. **(T-2)**

4.5.4.6. Manage the AGE Flight's organizational fuel tank(s) IAW DAFI 23-201.

4.5.4.7. Manage the AGE Flight's HAZMAT/ESOH programs IAW AFI 90-8XX series ESOH instructions and the AFI 32-70XX series environmental instructions.

4.6. Armament Flight. See AFMAN 21-206 for duties and responsibilities.

4.6.1. - 4.6.4.2. DELETED.

4.7. Avionics Flight. The Avionics Flight normally consists of some combination of; a Communication-Navigation Section, an Radio Frequency (RF) Multiplexing Section, an Instrument and Flight Control Systems (IFCS) section, a Weapons Control System Section, a Sensors Section, an Electronic Warfare System (EWS) Section, an Avionics Intermediate Section, a Computer Section, a Surveillance Radar Section, a Combat Systems Section, a Cryptographic Section, an Offensive Avionics Section, and a Cyber/Information Security Section. **Note:** Do not authorize additional manpower positions to form sections resulting from local management decisions.

4.7. (AFMC) [DEV] Avionics Flight. Within the 412 MXS, the Component Repair Flight will perform this function and will provide oversight to all responsibilities and duties called out for these sections in DAFI 21-101. There are no additional mission support requirements for AFMC.

4.7.1. Sections within the Flight are responsible for maintaining avionics systems and components and the associated test/support equipment. They perform authorized equipment repairs, TCTOs, component programming/reprogramming, troubleshooting, CND/BCS screening of line replaceable units (LRUs), sub-component removal and replacement, management, programming, and status reporting for assigned pods and SE, and in-work classified avionics systems component management. Repairs above and beyond those listed require approval from the appropriate approval authority (Lead Command, Depot).

4.7.1. **(AFMC)** See **paragraph 4.7** [DEV] Avionics Flight.

4.7.2. Avionics Flight CC/Chief Responsibilities. In addition to the applicable Flight CC/Chief responsibilities listed in **Chapter 2**, the Avionics Flight CC/Chief will:

4.7.2.1. Support Wing EW system programming. **(T-3)**

4.7.2.2. Ensure control and storage of assigned AME IAW DAFI 21-103. Develop local procedures for control and storage of items not specified in -21 TOs. **(T-3)**

4.7.2.3. Ensure avionics POD status is updated in RAMPOD or approved MIS IAW DAFI 21-103. **(T-3)**

4.7.2.4. Ensure personnel do not make unauthorized or false transmissions on international distress frequencies IAW TO 31R2-1-251, *General Instructions-Transmission of False Distress Signals on Emergency Frequencies*. **(T-3)**

4.7.2.5. Ensure cryptography components are controlled and maintained IAW National Security Agency publications and directives for the functions performed. **(T-3)**

4.7.2.5. **(AFMC) Note:** Due to the increased prevalence of aircraft COMSEC/Controlled Cryptographic Item line Replaceable Units (LRU) that contain Cryptographic Ignition Keys (CIK), especially true with new and future Mission Design Series (MDS), consult with local COMSEC Responsible Officer (CRO) or COMSEC account users to ensure LRUs turned into supply comply with zeroization requirements, i.e., CIK removal.

4.7.2.6. When applicable, determine maintenance responsibility for aircraft adapter group equipment. **(T-3)**

4.7.3. Section NCOICs will maintain AFTO Form 95, or equivalent, on selected, repairable, serialized components for which historical failure data will enhance repair. **(T-3)** Historical records are mandatory for SPRAM LRUs, and items asterisked in weapons system -6 TOs. Historical records will be maintained IAW TO 00-20-1.

4.7.3.1. The record will remain with the component anytime it is undergoing maintenance. **(T-3) Note:** Data is provided from these records, upon request, to the analysis function to aid in defining avionics maintenance problems and recommended solutions.

4.7.4. Avionics Flight's within established Repair Networks will collaborate with the Repair Network Manager (RNM) for maintenance constraint resolution. If the SMR code in the 4th position is "F" (I-Level Repair) will require RNM and all applicable stakeholders to determine next step or appropriate Action Taken Code to use when resolving the maintenance constraint.

4.7.5. Implement the “Bad Actor” program IAW TO 00-35D-54. The purpose of the Air Force Bad Actor Program is to identify serial-numbered items that enter the repair cycle at an abnormally high rate when compared to the total population of like assets and to repair them or remove them from the exhibit holding activity.

4.7.6. Repair Monitor Responsibilities. Monitors the status of items processed into the section for repair. Each shift may have a repair monitor assigned. Maintain records used by the repair monitor according to AFI 33-322. Each Repair Monitor will:

4.7.6.1. Process items into and out of the section, ensuring all documentation is accurate and complete. **(T-3)**

4.7.6.2. Advise the section NCOICs and Pro Supers of item status. **(T-3)**

4.7.6.3. Assist the section NCOICs in managing the DIFM program by complying with MAJCOM instructions to ensure ordered and received parts are documented; and uses, maintains, and files, management and computer records. **(T-3)** Repair Monitors will maintain and update a working copy of the D-23, *Repair Cycle Asset Management Listing*, sorted by location and detail number. **(T-3)**

4.7.6.4. Designate and maintain an AWP area, ensure accurate documentation, and submit supply assistance requests, as required. **(T-3)**

4.7.6.5. Track and monitor MICAP status for all assigned DIFM and parts affecting section repair capabilities using automated Integrated Logistics System -Supply (ILS-S) reports. **(T-3)**

4.7.6.6. Ensure the MIS is updated with current supply data, location changes and DIFM status changes. **(T-3)**

4.8. Fabrication Flight. The Fabrication Flight may consist of four sections; Aircraft Structural Maintenance (ASM), Metals Technology, Nondestructive Inspection (NDI), and Low Observable Aircraft Structural Maintenance (LOASM) and is responsible for performing on/off-equipment maintenance of systems and equipment.

4.8.1. Fabrication Flight CC/Chief Responsibilities. In addition to the applicable Flight CC/Chief responsibilities outlined in **Chapter 2**, the Fabrication Flight CC/Chief will:

4.8.1.1. Provide local manufacture capability to meet mission requirements and monitor all local manufacture work order requests. **(T-3)**

4.8.1.2. Coordinate AGE welding requirements with the AGE Flight Chief. **(T-3)**

4.8.1.3. *Assist WG Corrosion Manager in efforts to ensure effective corrosion prevention and control requirements, wash rack procedures, and established paint schemes are accomplished IAW TO 1-1-691, TO 1-1-8, TO 35-1-3, MAJCOM/Lead Command instructions, and MDS-specific TOs.*

4.8.1.4. Coordinate with squadron SUPTs or civilian equivalent to ensure assigned personnel rotation plans are established to comply with core task upgrade requirements. **(T-3)**

4.8.2. Aircraft Structural Maintenance (ASM) Section. Manages structural repair, corrosion control, Low Observable, inspection, damage evaluation, repair, manufacture, and/or modification of metallic, composite, fiberglass, plastic components, and related hardware associated with aircraft and SE. In addition to applicable Section NCOIC/Chief responsibilities in **Chapter 2**, the ASM Section NCOIC/Chief will:

4.8.2.1. Ensure appropriate resources are available to all personnel to chemically or mechanically inspect, remove, and treat corrosion on aircraft, engines, AGE, and components. **(T-3)**

4.8.2.2. Monitor the aircraft wash and corrosion inspection schedule in the weekly and monthly maintenance plans. **(T-3)**

4.8.2.3. Provide training and assistance to sections managing their own corrosion programs to include cleaning operations, corrosion prevention, inspection, removal, and treatment techniques. **(T-3)**

4.8.2.4. Develop maintenance procedures IAW **Chapter 11** of this instruction, DAFMAN 91-203, and ensure assigned ASM personnel are trained and qualified on aircraft intake maintenance. **(T-3)**

4.8.2.5. Review the Qualified Product List/Qualified Product Database for changes to cleaners that must conform to a MIL-Spec as specified in applicable TOs for aircraft wash rack. **(T-3)**

4.8.2.6. Stock supplies and equipment necessary to support aircraft and equipment washing, inspection, and treatment. **(T-3)**

4.8.3. Metals Technology Section. Inspects, repairs, services, manufactures, fabricates, performs heat treating, cleans, welds, and tests aircraft and equipment, components, and tools. In addition to responsibilities outlined in **Chapter 2**, the metals technology section NCOIC will:

4.8.3.1. Ensure assigned welders are certified in all base metal groups prescribed by the MAJCOM Fabrication functional manager (or equivalent) IAW TO 00-25-252, *Aeronautical Equipment Welding*. **(T-3)**

4.8.3.1.1. Ensure assigned welders conducting Gas Tungsten Arc Welding, Gas Metal Arc Welding, or Shielded Metal Arc Welding repairs on support equipment are certified IAW TO 00-25-252.

4.8.3.1.2. Ensure welding proficiency is documented IAW TO 00-25-252.

4.8.3.2. Provide safety briefings stressing Arc radiation hazards. **(T-3)**

4.8.3.3. Ensure special tools, jigs, and fixtures are designed, fabricated, protected, and properly stored. **(T-3)**

4.8.4. Nondestructive Inspection (NDI) Section. Performs NDI of aircraft, engines, AGE, other equipment and manages the Oil Analysis Program (OAP). Inspection findings are limited to a description of the size, location, and type of any defect discovered. NDI personnel do not make serviceability determinations except for "inspect only" TCTOs and if NDI actions constitute a completed maintenance action. In addition to the applicable Section NCOIC/Chief responsibilities in **Chapter 2**, the NDI Section NCOIC/Chief will:

4.8.4.1. Ensure OAP requirements are accomplished (if applicable to assigned MDS) IAW AFI 21-131, Joint Oil Analysis Program and **Chapter 11**. **(T-3)**

4.8.4.1.1. If the NDI laboratory providing OAP support is not located on the same base as the supported unit, or the supported unit does not have NDI/OAP personnel assigned, assign the OAP responsibilities to the owning organization IAW TO 33-1-37-1, *Joint Oil Analysis Program Volume II* and TO 33-1-37-3, *Joint Oil Analysis Program Laboratory Manual, Volume III*, TO 33-1-37-4, *Joint Analysis Program Manual, Volume IV*. The owning organization provides samples in an expeditious manner to the supporting OAP laboratory.

4.8.4.1.1.1. The owning organization will establish collection points and procedures to receive and forward OAP samples to the supporting laboratory, monitor sample collection, assign control numbers, and provide blocks of sample control numbers for use in other squadrons. **(T-3)**

4.8.4.2. Advise Maintenance Supervision, MOC and the owning work center of abnormal OAP trends. **(T-3)**

4.8.4.3. Ensure capability exists to perform optical, dye-penetrant, magnetic particle, ultrasonic, eddy current, radiographic, and special inspections as required. **(T-3)**

4.8.4.4. Ensure process control procedures IAW TO 33B-1-2, *Nondestructive Inspection General Procedures and Process Controls* are completed at the required or established frequency. **(T-3)**

4.8.4.5. Establish technique files using AFTO Form 242, *Nondestructive Inspection Data*, and TO 33B-1-1, *Nondestructive Inspection Methods Basic Theory*. **(T-3) Note:** Locally developed inspection techniques for use on aircraft and their components will be approved by the responsible ALC NDI manager prior to use. All other non-aircraft related AFTO Form 242 established techniques may be approved by the lab Chief.

4.8.4.6. Maintain coordination with the base medical service that provides occupational physicals, emergency treatments, film badge services, and acts as radiographic advisors IAW DAFMAN 48-125, *Personnel Ionizing Radiation Dosimetry*, and TO 33B-1-1.

4.8.4.7. Ensure a Radiation Safety Program is established IAW TO 33B-1-1.

4.8.4.8. Control and dispose of radiographic silver-bearing materiel IAW AFI 23-101.

4.8.4.9. Ensure radiographic film files and computed radiography files contain, as a minimum:

4.8.4.9.1. The last complete set of radiographs taken by owning organization, for each assigned aircraft and engine by serial number or identification number. **(T-3)**

4.8.4.9.2. The name of the person who interpreted the radiography. **(T-3) Note:** Radiography identification procedures will be followed IAW TO 33B-1-1.

4.8.4.9.2.1. Ensure the person interpreting the film also initials the set of radiographs or a locally developed interpretation worksheet, as applicable. **(T-3)**

4.8.4.9.3. All NDI radiographic film exposures, to include paper, will be filed and maintained for all One Time Inspection (OTI), TCTO, -6 TO, -9 TO, and -36 TO inspection requirements. **(T-3)** The NDI Section NCOIC/Chief will ensure disposition of radiographic film IAW Air Force Records Disposition Schedule located at **https://afrims.cce.af.mil. (T-3)**

4.8.4.10. Ensure all NDI technicians are certified IAW TO 33B-1-1.

4.8.5. Low Observable (LO) Aircraft Structural Maintenance Section. LO ASM Section manages structural repair, corrosion control, composite repair, LO coatings. **Note:** The Fabrication Flight CC/Chief will determine which tasks listed in **Paragraph 4.8.2** in this instruction (ASM Section) will be applicable to this section based on flight configuration. In addition to applicable Section NCOIC/Chief responsibilities in **Chapter 2**, the LO ASM Section NCOIC/Chief will:

4.8.5.1. Provide inspection, damage evaluation, repair, manufacture, and/or modification of LO components, and related hardware associated with aircraft. **(T-3)**

4.8.5.2. Ensure appropriate resources are available to perform all LO related tasks. **(T-3)**

4.8.5.3. Stock supplies and equipment necessary to support aircraft inspection, and treatment. **(T-3)**

4.8.5.4. Monitor the inspection schedule in the weekly and monthly maintenance plans. **(T-3)**

4.8.5.5. Ensure protective/LO coatings are applied to aircraft, AGE, applicable munitions, and components IAW applicable TOs. Ensure protective/LO coatings are applied IAW local, state, and federal environmental directives. **(T-0)**

4.8.5.6. Provide training and assistance to sections managing their own LO programs. **(T-3)**

4.9. Maintenance Flight. May consist of Repair and Reclamation, Wheel and Tire, Inspection, and TA Sections.

4.9. (AFMC) [DEV] Maintenance Flight. Within the 412 MXS, the Maintenance Flight will include Hydraulic and Aircraft Fuels System Sections and will provide oversight to all responsibilities and duties called out for these sections in DAFI 21-101.

4.9.1. The Maintenance Flight CC/Chief will comply with the common Flight CC/Chief responsibilities in **Chapter 2** and locally established management requirements. **(T-3)**

4.9.2. Repair and Reclamation Section. When established, removes, replaces, and rigs flight control surfaces/systems on assigned aircraft. Troubleshoots, rigs, and replaces landing gears, actuated doors, canopies, and associated equipment requiring component maintenance beyond

the capability of other activities. MAJCOM or MXG/CC may identify delineation of complex tasks to optimize maintenance capability in supplements or addendums to this DAFI as required.

4.9.2.1. Repair and Reclamation Section, when established, will remove, install, and repair towed targets and airborne reel pods. **(T-3)**

4.9.2.1. **(AFMC)** [DEV] The 412 MXS Component Repair Flight/AIS will remove, install, and repair towed-targets and airborne reel pods.

4.9.3. Wheel and Tire Section. Manages the build-up, repair, test, and storage of wheel and tire assemblies and components. Wheel and Tire Section will:

4.9.3.1. Degrease and disassemble wheel components for NDI inspection IAW TO 4W-1-61, *Maintenance and Overhaul Instruction - All Types Aircraft Wheels*, prior to processing through the ASM and NDI Sections.

4.9.3.2. Clean, inspect, and properly store (do not co-mingle) wheel bearings. **(T-3)**

4.9.4. Aircraft Inspection Section. Performs aircraft PH, PE, ISO, or letter check inspections. **Note:** Section may be divided into separate elements for each type aircraft maintained. In addition to the applicable Section NCOIC/Chief responsibilities in **Chapter 2**, the Inspection Section NCOIC/Chief will:

4.9.4.1. Ensure assigned non-powered SE (such as, dock stands) is maintained. **(T-3)**

4.9.4.2. Review inspection schedules and ensure dock teams are available to meet inspection needs. **(T-3)**

4.9.4.3. Develop standardized inspection flow plan to aid in managing the inspection progress and to control dock personnel and support specialists. **(T-3)**

4.9.4.3.1. DELETED.

4.9.4.3.2. Inspection Section NCOIC/Chief will ensure flow plan data remains current with -6 TO requirements. **(T-3)**

4.9.4.4. Inform the MOC and owning agency of all MICAP parts. **(T-3)**

4.9.4.5. Provide PS&D with an inspection document record upon completion of the inspection. **(T-3)**

4.9.4.6. Ensure components are tagged with an AFTO Form 350, *Reparable Item Processing Tag*, IAW TO 00-20-2.

4.9.4.6.1. Ensure serially controlled components are reinstalled on the same aircraft and position from which they were removed. **(T-3) Exception:** If it is absolutely necessary to install serially controlled components in a different position, the Inspection Section NCOIC/Chief will notify PS&D to update the records.

4.9.5. Transient Aircraft (TA) Section. Recovers, services, inspects, maintains, and launches transient aircraft. Transient aircraft are those aircraft not assigned to a base that are enroute from one location to another that may require routine servicing. Aircraft are not considered transient aircraft when deploying to, staging from, or departing from any location for the purpose of flying sorties or conducting training, either with or without the necessary

maintenance support from the aircraft's home base. MOC coordinates specialist support for transient aircraft through appropriate squadrons. For off-station recovery procedures refer to owning MAJCOM instructions and command-to-command agreements. In addition to the applicable Section NCOIC/Chief responsibilities outlined in **Chapter 2**, the TA Section NCOIC/Chief will:

4.9.5.1. Recover and deliver all deceleration chutes for assigned, transient, and tenant aircraft to the AFE. **(T-3)**

4.9.5.2. Complete reimbursement documentation. **(T-3)**

4.9.5.2.1. AFTO Form 726, T*ransient Aircraft Service Record*, may be used for documenting maintenance servicing requirements and necessary billing information and is prescribe in TO 00-20-1.

4.9.5.3. Record arrivals and departures of transient aircraft on AF Form 861, *Base/Transient Job Control Number* Register or locally approved form if it captures all AF Form 861 fields. **(T-3)** TA Section NCOIC/Chief (or equivalent) will:

4.9.5.3.1. Assign each aircraft a single Event Identification Description (EID) for all support general work performed by TA. **(T-3)**

4.9.5.3.2. Enter, as a minimum, "P" for park, "I" for inspect, "S" for service, "L" for launch, and "E" for EOR in the job description/remarks block. **(T-3)**

4.9.5.3.3. Forward completed AF Form 861 for contracted TA activities to the COR monthly. **(T-2)** The COR forwards completed forms to the applicable contracting officer managing the TA contract for inclusion in the contract file.

4.9.5.3.4. Route the AF Form 861 for non-contracted TA activities to the Maintenance Flight CC/Chief for review. **(T-3)**

4.9.5.3.4.1. After review, the TA Section NCOIC/Chief will file AF Form 861 for a minimum of 1 year. **(T-3)**

4.9.5.3.4.2. AF Form 861 may be used as a reference to quantify tasked performed to validate manpower and equipment requirements against current AF standards.

4.9.5.4. Close out support general EIDs daily. **(T-3)**

4.9.5.4.1. Use the same last four digits on subsequent days for the same aircraft.

4.9.5.4.2. Use a separate EID for each discrepancy that is not support general.

4.9.5.5. Ensure that when a FCF is required on transient aircraft, QA at the transient base serves as the focal point and ensures all FCF requirements are completed. **(T-3)**

4.9.5.5.1. DELETED.

4.9.5.5.2. If no off-station agencies exist, owning MXG/CC and owning OG/CC will issue guidance directly to the aircraft commander and off-station maintenance personnel. **(T-3)**

4.9.5.6. Supervise maintenance performed by assigned personnel on transient aircraft. **(T-3)**

4.9.5.7. Maintain the appropriate TOs for aircraft that can be expected to transit the function on a regular basis. **(T-3)**

4.9.5.8. Ensure personnel are trained and strictly adhere to oil sample requirements specified in the respective -6 TO. **(T-3)**

4.9.5.9. Ensure personnel authorized to run engines are qualified IAW **Chapter 11**.

4.9.5.9.1. Request the aircrew to run engines if TA or maintenance personnel are not authorized.

4.9.5.9.2. If qualified aircrew members are not available, contact MOC to request assistance from the home station.

4.9.5.10. Ensure transient aircraft status changes are reported to MOC. **(T-3)** If support is required, the MOC notifies the home station for support.

4.9.5.11. Ensure EOR procedures for transient aircraft are developed IAW TO 00-20-1.

4.9.5.12. Ensure procedures exist for required weapons loading actions on transient aircraft, transient aircraft impulse cartridge tracking and storage, and weapons "safing" equipment requisition and maintenance for frequently transiting aircraft. **(T-3)**

4.9.5.12.1. Arming, de-arming and munitions unloading/loading operations on transient aircraft will be performed by a weapons load crew certified/qualified on the munitions and aircraft. **(T-3)**

4.9.5.12.2. DELETED.

4.9.5.12.2.1. In such cases, the aircrew will be available for consultation on aircraft peculiarities. **(T-3)**

4.9.5.12.2.2. If these criteria cannot be met, request assistance from higher headquarters.

4.9.5.13. Ensure checklists exist to ask pilots about explosive egress systems pertaining to unfamiliar aircraft that do not normally transit their base. **(T-3)**

4.9.5.13.1. Aircrew members remove and install flight status safety pins on aircraft when transient maintenance personnel are not qualified.

4.9.5.13.1.1. DELETED.

4.9.5.13.1.2. If TA cannot accomplish the required inspections, servicing, or repairs because of a lack of qualified personnel, facilities, or materiel (or there is no TA support available), and the transient aircraft commander does not wish to continue the flight without accomplishment of these items, the transient aircraft commander is responsible for requesting assistance through the appropriate external organizations.

4.10. Munitions Flight. Controls, accounts for, stores, ships, receives, inspects, maintains, assembles, and delivers conventional, precision guided and nuclear munitions. Manages and maintains all assigned tools, test and munitions handling equipment. Refer to AFI 21-2XX series instructions for specific guidance. **Note:** Munitions may be part of the MXS or established in a Munitions Squadron IAW AFMAN 21-200.

4.10. (AFMC) [DEV] Munitions Flight. The 412 MXS/MXMW, Munitions Flight is also responsible for the Armament function for Bomber aircraft due to the geographic location/separation at Edwards AFB, South Base.

4.11. Propulsion Flight. Maintains aircraft engine propulsion units, propulsion components, and propellers. Performs engine/module/accessory disassembly, inspection, assembly, test, and repair. Responsible for Jet Engine Intermediate Maintenance (JEIM); Engine Test Stands (ETS) and Noise Suppression Systems (NSS); accessory and Quick Engine Change (QEC) repair; small gas turbine; module/accessory repair section; support equipment; and turboprop/turbo-shaft repair, engine PH/ISO inspections, as required. Sections may be combined or grouped at the discretion of the squadron commander. When an engine CRF is co-located with an operational wing, a MOA or MOU may be developed to clarify mutual support responsibilities. In addition, the flight will be the focal point for common propulsion support equipment, for example, flexible borescopes, engine trailers and download equipment.

4.11. (AFMC) [DEV] Propulsion Flight. In addition to the normal propulsion flight responsibilities, the 412 MXS Propulsion Flight Business Unit will perform duties and responsibilities of Engine Management (EM) Section listed in **Chapter 5** and **Chapter 14**.

4.11.1. In addition to the applicable Flight CC/Chief responsibilities in **Chapter 2**, the Propulsion Flight CC/Chief will:

4.11.1.1. Perform as the wing focal point for propulsion maintenance programs, focusing on continuity, compliance, and standardization, provide advice to wing leadership on propulsion issues and monitor all aspects of wing propulsion maintenance program. **(T-3)**

4.11.1.2. Act as the wing 2A6X1 AFSC functional manager and provide technical guidance to maintain propulsion systems to support the wing mission. **(T-3)**

4.11.1.3. Coordinate with Engine Manager (EM) and organization leadership to support War Readiness Engine (WRE) requirements. **(T-3)**

4.11.1.3.1. Track the status of ready spare engines using a visual display or automated product showing serial number, configuration (type and position, if applicable), time remaining until next scheduled engine removal, overhaul or reconditioning, preservation date, type accomplished, re-preservation due date, OAP code (if applicable), and remarks. **(T-3)**

4.11.1.4. Review production data to ensure propulsion units and components processed through the flight are repaired and functionally checked IAW TO 2-1-18, *Aircraft Engine Operating Limits and Factors*, including QEC configuration when applicable. **(T-3)**

4.11.1.5. Coordinate with the EM to ensure accurate engine and equipment status reporting IAW **Chapter 14**, DAFI 21-103, AFMAN 20-116, AFPAM 63-129, and TO 00-25-254-1.

4.11.1.6. Develop guidelines to comply with AF and wing OAP requirements IAW 33-series TOs and **Chapter 11**. **(T-3)**

4.11.1.7. Review/analyze all unscheduled engine or module removals and ETS rejects. **(T-3)**

4.11.1.7.1. Review/analyze major component failure trends. **(T-3)**

4.11.1.8. Ensure in-shop CANN actions are accomplished IAW local procedures, **Chapter 9**, **Chapter 11**, and TO 00-20-2.

4.11.1.8.1. Ensure local procedures are coordinated with Engine Management (EM) to ensure sufficient time remains on TCIs prior to CANN action approval. **(T-3)**

4.11.1.9. Coordinate with base civil engineering to provide maintenance on NSS and ETS supporting structures that are categorized as real property. **(T-3)** If the wing or squadron is a tenant, incorporate this maintenance requirement into the host-tenant support agreement. **(T-3)**

4.11.1.9.1. Ensure NSS and/or ETS repair discrepancies that exceed the base repair capability are reported in Web Applications Software Product (WASP). **(T-3) Note:** Entering repair requirements into WASP establishes official repair request and ensures visibility to MAJCOM and SE Product Group Manager at Robins AFB (AFLCMC/WN).

4.11.1.9.1. **(AFMC)** Edward's brick and mortar facility does not fall under the Warner Robins Air Logistics Complex Support Equipment PGM and therefore WASP tracking is not required for the facility.

4.11.1.10. Ensure an uninstalled engine run qualification and certification program is established IAW **Chapter 11**.

4.11.1.11. Ensure specialized and long-life shipping devices and containers are accounted for and maintained in a serviceable condition IAW AFI 23-101 and TO 00-85-20, *Engine Shipping Instructions*.

4.11.1.12. Ensure engines and engine components removed from crash damaged aircraft are correctly dispositioned for termination IAW DAFI 21-103 and disposed of IAW AFI 23-101.

4.11.1.13. Ensure an engine flexible borescope certification and blade-blending certification program, for each Type, Model, Series (TMS) possessed, is established IAW **Chapter 11**.

4.11.1.14. Monitor scheduled and unscheduled engine removals to balance Propulsion Flight workload with production capability and coordinate with EM section to program engine removals for the weekly and monthly maintenance plans. **(T-3)**

4.11.1.14.1. Coordinate with EM to develop a 6-month plan to smooth surges in the engine maintenance workload. **(T-3)**

4.11.1.14.1.1. Use automated methods to develop the 6-month plan and include scheduled engine removals for TCIs, PE, TCTOs and a projected unscheduled removals factor.

4.11.1.14.1.2. Ensure Reliability-Centered Maintenance principles IAW AFMAN 20-116 are followed. **(T-3)**

4.11.1.15. Ensure Engine Automated Work Package (EAWP) user permissions mirror current training and certification authorizations. **(T-3)**

4.11.1.15.1. Ensure EAWP users use the EAWP program in lieu of a work folder to meet minimum requirements of this DAFI.

4.11.1.16. Coordinate with the OAP laboratory to obtain maximum benefits from OAP data when abnormal wear-metal trends are indicated. **(T-3)**

4.11.1.16.1. Ensure all OAP responsibilities are performed IAW **Chapter 11**.

4.11.1.16.2. Establish procedures to monitor OAP trends. **(T-3)**

4.11.1.16.3. Ensure personnel are trained to identify and respond to wear metal limits for assigned and maintained engines and are trained to perform sampling procedures IAW TO 33-1-37-2.

4.11.1.16.4. Ensure oil samples taken at the ETS are promptly delivered to the OAP laboratory. **(T-3)**

4.11.1.16.5. Act as a central point-of-contact for all abnormal OAP laboratory results. **(T-3)**

4.11.1.16.6. Forward information to the OAP laboratory concerning actions taken as a result of OAP recommendations. **(T-3)**

4.11.1.16.7. Review OAP response time (from sampling to receipt at the laboratory and return to the unit) to ensure processing time meets mission needs. **(T-3)**

4.11.1.17. **(Added-AFMC)** [DEV] The 412 MXS Propulsion Flight Chief is responsible for management of the Engine Blade Blending Training and Certification Program and will ensure all related responsibilities contained in DAFI 21-101 are adhered to.

4.11.2. Support Section. The Support Section manages the flight's HAZMAT program and operates tool storage areas. DMS or designated personnel process supply requests to facilitate the issue request, tracks MICAP due-outs, monitors bench stock, conducts bench stock/adjusted stock level reviews IAW DAFMAN 23-122. **(T-3)** In addition to the applicable Section NCOIC/Chief responsibilities outlined in **Chapter 2**, the Support Section NCOIC/Chief will:

4.11.2.1. Ensure a flight due-out release point and holding bins are established, and Urgency of Need Designator "A" and Urgency Justification Code BQ requirements are verified. **(T-3)**

4.11.3. Jet, Turboprop, Turbo-shaft Engine Intermediate Maintenance (JEIM) section. Stores, builds up, tears down, inspects, modifies, and repairs engines, QEC kits, and tests components. In addition to the applicable Section NCOIC/Chief responsibilities in **Chapter 2**, the JEIM Section NCOIC/Chief will:

4.11.3.1. Plan and monitor the progress of propulsion system maintenance production, ensuring maintenance schedules are met by anticipating materiel required and managing delays to prevent schedule disruptions to support operational requirements and maintain required WRE levels. **(T-3)**

4.11.3.1.1. Report production to Propulsion Flight CC/Chief and immediately inform EM of engine status changes.

4.11.3.2. Ensure personnel prepare propulsion units and components for shipment and properly identify units to be returned to depot. **(T-3)**

4.11.3.2.1. Attach CEMS and/or MIS paper products to life-limited components IAW 00-20-series TOs if required by the source of repair.

4.11.3.3. Ensure documentation of TCTO compliance IAW 00-20-series TOs.

4.11.3.4. Ensure CEMS and/or MIS products obtained from EM are used for all assigned engines. **(T-3)**

4.11.3.4.1. CEMS and/or MIS products will list all parts and serial numbers installed on the engine. **(T-3)**

4.11.3.5. Establish procedures to ensure all parts and serial numbers are inventoried when an engine is received or released by the section. **(T-3)**

4.11.3.5.1. The JEIM Section NCOIC will notify EM when a different serial numbered part is installed or changed so the automated record is updated. **(T-3)**

4.11.3.5.2. EAE is the change correction authority on Part Number/Serial Number Record updates in EAWP. **(T-3)**

4.11.3.6. Ensure an engine work folder is established for each engine during PE, reconditioning, or other maintenance. **(T-3)**

4.11.3.6.1. One work order is initiated in MIS for an entire job.

4.11.3.6.1.1. MIS work orders are completed during inspection, reconditioning or maintenance.

4.11.3.6.1.2. Separate JCN/Work Center Event (WCE)/Work Event Separator are initiated for discrepancies found during the look phase of an inspection, subsequent to repair, or when maintenance is required beyond the scope of the JEIM induction JCN.

4.11.3.6.2. Establish engine work folders on all possessed engines and EM or JEIM will maintain the folders until the engine is transferred. **(T-3)** As a minimum, engine work folders will contain the following:

4.11.3.6.2.1. List of all parts, TCTOs and TCI requirements for the engine. **(T-3)**

4.11.3.6.2.2. Engine/Module/Accessories Information Worksheet. **(T-3)** This document is used to provide a quick synopsis of maintenance accomplished. Minimum requirements will include engine serial number, type, position (if applicable), engine operating time, date started work, date turned serviceable, job control number, maintenance required, reason for removal, list of time change and TCTO requirements. **(T-3)**

4.11.3.6.2.2.1. A supervisory review of signature blocks (Crew Chief, Support Section, EM Section) to verify all repair requirements have been accomplished and are correctly documented in the work folder. **(T-3)**

4.11.3.6.2.2.2. Validation that a JCN was created by the JEIM/Module/Accessories Section or EM section and used to account for

maintenance events completed in the process of repairing the engine and modules. **(T-3)** This procedure ensures all maintenance data is documented against one JCN and engine failure information is connected to the in-shop action.

4.11.3.6.2.3. Receiving Inspection Worksheet. **(T-3)** The worksheet is used for documenting items to be accomplished by JEIM prior to engine repair. Minimum requirements will include: FOD check of engine inlet and exhaust, inspection of engine for general condition and fluid leakage, EHR/Turbine Engine Monitoring System data (if applicable), ET&D (if applicable), borescope inspection (if applicable), a check with OAP lab for possible problems, and a list of unique or problem areas to be checked prior to engine disassembly or maintenance. **(T-3)**

4.11.3.6.2.4. Serially Controlled/Time-Tracked Item Replacement Record. **(T-3)** This document shows a list of components replaced by nomenclature, old and new part number (if applicable), and serial number.

4.11.3.6.2.5. Daily Summary Record. **(T-3)** This document provides a synopsis of maintenance performed during each shift.

4.11.3.6.2.5.1. Each entry in the Daily Summary Record includes the Employee Number of the person who accomplished the maintenance action. For EAWP users, this process may be automated.

4.11.3.6.2.5.2. Include a sufficient reference in the summary block (such as, work package, TO) used to perform the task or determine the work performed (subordinate work packages are not required to be listed if the work package for the primary task identifies all required work packages for the task).

4.11.3.6.2.5.3. At the end of each shift, the crew chief who verified the entries listed in the Daily Summary Record will annotate their shift, rank, last name, and employee number. **(T-3)** Units may use a general purpose or MAJCOM/locally approved form.

4.11.3.6.2.6. IPI Worksheet. **(T-3)** This form includes the WUC, nomenclature, specific step required for the IPI, and space for employee numbers and signatures of technicians and inspectors performing maintenance. **Note:** Organizations using digital systems may file a printed report in lieu of signatures (such as, Interactive Electronic Technical Manual (IETM)).

4.11.3.6.2.7. Parts Requisition Record. **(T-3)** This document is used to list all parts (including TCIs) on order. As a minimum, this document will include the following headings: Engine/ Module/Accessory, TMS, Engine/Module/Accessory Serial Number, Nomenclature, Part Number, National Stock Number (NSN), Requisition Number, Priority, Status, and DIFM Clear with “Yes” and “No” sections. **(T-3)**

4.11.3.6.2.8. JEIM ETS Preparation Worksheet. **(T-3)** This worksheet contains a list of items/tasks to be accomplished by JEIM prior to sending an engine to the ETS. As a minimum, document the following:

4.11.3.6.2.8.1. Inlet and exhaust FOD inspection; any pre-run servicing required (such as, cap open lines, cannon plugs, engine intake and exhaust

inspection). **(T-3)**

4.11.3.6.2.8.2. A thorough tool inventory and an inspection for loose hardware. **(T-3)**

4.11.3.6.2.8.3. The section supervisor will document a review of the work folder to ensure maintenance performed or required actions are documented. **(T-3)**

4.11.3.6.2.9. ETS Pre-run Worksheet. **(T-3)** ETS personnel will complete this document prior to an engine run. **(T-3)**

4.11.3.6.2.9.1. As a minimum, this document will include the following headings: Engine TMS; Engine Serial Number; Engine Operating Time (EOT)/Cycles; JCN; Remarks; Pre-run Emergency Briefing Accomplished with run Supervisor's Name, Signature and Date sections; and Inspection with Area, Employee Number, and Date sections. **(T-3)**

4.11.3.6.2.9.2. As a minimum, area inspections will include Inlet FOD/Foreign Object (FO); Exhaust FOD/FO; Engine Exterior and FO; General Engine Serviceability; Test Stand/Thrust Bed/Test Equipment for FO; CTK Inventory Complied With (C/W); Engine Servicing Check; all preliminary engine installation and run requirements C/W; and serviceable fire extinguisher on hand. **(T-3)**

4.11.3.6.2.9.2.1. Each area inspection will have the performing technician's employee number and date accomplished annotated. **(T-3)**

4.11.3.6.2.10. ETS Post Run Worksheet. **(T-3)** This document is used to document items/tasks accomplished by ETS personnel after engine run.

4.11.3.6.2.10.1. As a minimum, this document will include the following headings: Engine TMS; Engine Serial Number; EOT/Cycles; JCN; Maintenance Actions Performed; Engine Monitoring System (EMS) operational/performance data collected and provide to EM; ETS Supervisors Post-run Review with Name, Signature and Date; and Area Inspections, Employee Number and Date. **(T-3)**

4.11.3.6.2.10.2. As a minimum, area inspection will include: Inlet FOD/FO; Exhaust FOD/ FO; CTK Inventory C/W; Post-Run OAP Samples C/W (if applicable); AFTO Form 350 or AFTO Form 20, *Caution and Inspection Record*, annotate: Engine Preservation Type and Date; Throttle Secured to Off Position (if applicable) and Tagged; Cap Open Lines/Cannon Plugs; Install Intake/Exhaust Covers; Servicing Amount; ETS Discrepancies Cleared; 7-Level Inspection of Components Replaced or Disconnected; and Final Leak Check. **(T-3)**

4.11.3.6.2.10.3. ETS personnel will leak-check items not accessible with the engine installed in or on the aircraft prior to leaving ETS. **(T-3)**

4.11.3.6.2.10.4. Each area inspection will have the performing technician's employee number and date accomplished annotated. **(T-3)**

4.11.3.6.2.11. Final Inspection Worksheet. **(T-3)** This document is used to document JEIM requirements after repair or testing has been completed.

4.11.3.6.2.11.1. As a minimum, this worksheet will include: FOD inspection of intake, exhaust, and external engine; borescope engine (if applicable); ensuring throttle is secured and tagged to "off" position (if applicable); capping, plugging and covering fittings and lines; attaching AFTO Form 350 to lines, fittings or plugs that require "leak check" when installed in aircraft (items not accessible in aircraft must be leak checked on ETS); attaching AFTO Form 350 and/or serviceable tag to engine, ensure supply accounts and MIS entries have been cleared. **(T-3)**

4.11.3.6.2.12. Borescope Worksheets. **(T-3)** Borescope inspection worksheets will be used for engines requiring borescope documentation. **(T-3)**

4.11.3.6.2.13. Uninstalled Engine/Module Blade Blending/FOD Damage Worksheet. **(T-3)** This worksheet is used to document blade blending/FOD damage for uninstalled engines/modules. As a minimum, this worksheet will include Engine/Module Serial Number, Date, Discrepancy, Stage and Corrective Action including number of blades blended, depth of damage before and after blend, area of damage and Employee Number. **(T-3)**

4.11.3.6.2.14. Reliability Centered Maintenance Worksheets (if applicable). **(T-3)**

4.11.3.6.2.14.1. For JEIM engine builds, a copy of the "Reliability Centered Maintenance Build Options" and "Reliability Centered Maintenance Calculator Summary" worksheets are maintained in the engine work folder or EAWP for documenting life-limited component engine build recommendations.

4.11.3.6.2.14.1.1. Utilize Reliability Centered Maintenance calculator software accessed through CEMS prior to engine build.

4.11.3.6.2.14.1.2. The sheets are required only if life-limited components (excluding LRU) are removed and replaced during the JEIM engine build and the TMS engine is available in the Reliability Centered Maintenance Calculator. The calculator is not used for engines which do not have the calculator developed.

4.11.3.6.2.15. Worksheets that document engine historical information, critical maintenance management stages, and employee numbers of technicians and supervisors completing maintenance and inspections.

4.11.3.6.2.15.1. Supplement work folders and worksheets to fit unit needs.

4.11.3.6.2.15.2. Flights may use computer-generated products, provided they include all required information. If TMS has an established EAWP, it will be utilized. **(T-3)**

4.11.3.6.2.15.2.1. EAWP users will ensure all maintenance discrepancies are documented in the system's appropriate discrepancy block. **(T-3)**

4.11.3.7. Ensure MICAPs are processed in Enterprise Solution-Supply; ensure all pertinent data is included. **(T-3)**

4.11.3.7. **(AFMC)** Ensure Mission Impaired Capability Awaiting Parts (MICAP) are processed in Integrated Logistics Systems-Supply; ensure all pertinent data is included.

4.11.3.8. Upgrade, downgrade and cancel MICAP requirements. **(T-3)**

4.11.4. Noise Suppression Systems (NSS) and Engine Test Stands (ETS) Section. Tests engines to evaluate the quality of maintenance, engine performance, and accomplish engine preservation including engines installed on aircraft in coordination with owning squadron. In addition to the applicable Section NCOIC/Chief responsibilities outlined in **Chapter 2**, the NSS and ETS Section NCOIC/Chief will:

4.11.4.1. Assign primary and alternate AutoTAR or Air Force Lifecycle Management (AF-PLM) custodians to perform -107 ETAR and status updates in AutoTAR/AF-PLM for AFLCMC/WN-managed NSSs and ETSs. **(T-2)**

4.11.4.2. Monitor repair activity and ensure reporting and status updates are timely, accurate, and kept current in AutoTAR/AF-PLM. **(T-3)**

4.11.4.3. Ensure NSS and ETS personnel accomplish minor maintenance, make adjustments to engines, document engine condition, and collect EMS operational/performance data and deliver to EM. **(T-3)**

4.11.4.3.1. Ensures ETS components are calibrated on site, if practical. **(T-3)**

4.11.4.4. Brief maintenance personnel on NSS/ETS operating/emergency procedures. **(T-3)**

4.11.4.5. Handle and report halon releases IAW AFMAN 32-7002.

4.11.5. Module/Accessory Repair Section. Repairs, stores, and maintains fuel nozzles, fuel manifolds, oil pumps, accessory housings, afterburners, thrust reversers, augmentors, engine components, time change modules, and shop replaceable units. Operates and maintains the bearing room IAW TO 44B-1-15, *General Instructions - Jet Engine Anti-friction Bearing Handling, Removal, Cleaning, Inspecting, and Installation at Jet Engine Base Maintenance Facilities* (if applicable).

4.11.6. Small Gas Turbine Engine Section. Repairs and maintains small gas turbines used in aircraft. In addition to applicable Section NCOIC/Chief responsibilities outlined in **Chapter 2**, the Small Gas Turbine Engine Section NCOIC/Chief will ensure personnel are qualified to operate small gas turbine engines and test stands. **(T-3)**

4.11.7. Engine Equipment Maintenance Section. Maintains, manages, and stores engine support and removal/installation/transportation equipment and trailers. The Engine Equipment Maintenance Section NCOIC/Chief will:

4.11.7.1. Ensure engine removal/installation/transportation trailers and adapters status is properly reported IAW DAFI 21-103 and MAJCOM supplements. **(T-3)**

4.11.7.2. Track and schedule all inspections and maintenance, removal, installation, transportation trailers and adapters in the MIS. **(T-3)**

4.11.7.3. Ensure equipment forms and MIS documentation are complete, accurate, and accomplished for all maintenance and scheduled inspections. **(T-3)**

4.11.7.4. Ensure status is accurately reflected in both the maintenance forms and the MIS. **(T-3)**

4.11.8. Propeller Section. Repairs, builds up, tears down, inspects, tests, and modifies propellers, valve housings, pump housings, and associated components.

4.11.9. Quick Engine Change (QEC) Kit Management. QEC kit removals and installations are coordinated with the SRAN EM and loaded in MIS as a part number-serial number item, reflecting where the kit is installed or spared.

4.11.9.1. In addition to repair cycle procedures outlined in **Chapter 9** in this instruction, the technician removing a QEC kit item from an engine will complete an AFTO Form 350, enters the reason for removal in Block 14, and annotates the QEC kit inventory for each repairable item. **(T-3)**

4.11.9.1. **(AFMC)** [DEV] QEC kits removed to facilitate other engine maintenance, but will be reinstalled on the original engine and remain with that engine during shipment are required only to be tagged by an AFTO Form 350, *Reparable Item Processing Tag*, or similar method when removed, and no inventory action is required.

4.11.9.2. Technicians will complete the AF Form 596, *Quick Engine Change Kit Inventory* for on repair cycle items and QEC kit unique items, when an engine enters the section for tear down. **(T-3)**

4.11.9.2. **(AFMC)** [DEV] QEC kits removed to facilitate other engine maintenance, to include teardown, but will remain with the original engine require no inventory to be accomplished.

4.11.9.2.1. If TO requirements restrict reuse of items, the technician will mark the AF Form 596 with an asterisk to show a demand has been placed on supply. **(T-3)**

4.12. Test, Measurement, and Diagnostic Equipment (TMDE) Flight. Maintains, calibrates, and certifies TMDE, traceable through the AF Primary Standards Laboratory (AFPSL) to the National Institute of Standards and Technology (NIST), or other AF Metrology and Calibration (AFMETCAL)-approved source. Provides base-level support of aircraft, precision-guided munitions, ground systems, and other equipment assigned to the base or GSU. TMDE Flight ensures Calibration, certification, and maintenance of TMDE is accomplished IAW AFMAN 21-113, TO 00-20-14, TO 33K-1-100-1, *Calibration Procedure for Maintenance Data Collection Codes and Calibration Measurement Summaries*. A Rapid Assistance Support for Calibration may also be assigned.

4.12.1. In addition to applicable Flight CC/Chief responsibilities in **Chapter 2**, the TMDE Flight Chief will:

4.12.1.1. Establish a PMEL Quality Assurance Section IAW AFMAN 21-113 and TO 00-20-14. **(T-3)**

4.12.1.2. Publish a monthly PMEL Activity Summary and route it through the Maintenance Supervision to the SQ/CC (or organizational equivalent). **(T-3)**

4.12.1.2.1. The report format will comply with TO 00-20-14 and meet local requirements. **(T-3)**

4.12.1.3. Ensure all Groups with TMDE Flights assigned include the TMDE Quality Program Activity Summary in the QA monthly summary IAW **Chapter 6**. **(T-3)**

4.12.2. The NCOIC, PMEL Logistics will:

4.12.2.1. Ensure TMDE monitors are properly trained and maintain a database or log to track training events (dates, names, organizations). **(T-3)**

4.12.2.2. TMDE will be scheduled with input priorities IAW TO 00-20-14, Air Force Metrology and Calibration Program.

4.12.2.2.1. - 4.12.2.2.3. DELETED.

Chapter 5

MAINTENANCE OPERATIONS (MXO).

5.1. General. MXO is directly responsible to the MXG/CC for the administration, analysis, training management of assigned personnel, and programs and resources necessary to support the group's production effort. MXO is comprised of the following sections: EM, PS&D, MMA, MOC, MT, and Programs and Resources (P&R). In missile organizations, MXO will be organized as a Maintenance Operations Squadron, as applicable. For the purposes of this instruction, the term MXO is equivalent with Maintenance Operations Flight for ANG units.

5.1. (AFMC) [DEV] General. Within the 412 MXG, the 412 MXS Propulsion Flight Business Unit will perform duties and responsibilities of EM section listed in DAFI 21-101.

5.2. Maintenance Operations. MXO is the central agency for monitoring and developing long-range strategies of fleet management to sustain the health of the fleet. Fleet management is defined as the effective utilization of available resources to accomplish the aircraft support cycle from planned maintenance events to operations schedule execution. It is a disciplined and prioritized scheduling effort that optimizes support to aircraft requirements such as flying/operational events, ground training events, scheduled maintenance inspections, aircraft/system configuration control, aircraft/system modification schedules and aircraft/system recovery maintenance. Effective fleet management results in consistent availability of quality aircraft/systems to meet operational requirements. The Maintenance Operations Superintendent position will be filled by SNCO 2RXXX personnel and act as the 2R Base Functional Manager for their component. **(T-2)**

5.2. (AFMC) Maintenance Operations. Maintenance Operations Superintendent position can be filled by civilian equivalent.

5.2.1. Base 2R Functional Manager will be the senior ranking enlisted member, possessing a 7-skill level or higher, assigned to the host MAJCOM of the installation. **(T-2)** They are responsible for both AFSC 2R0X1 and 2R1X1 personnel development, training, and assignment for their component. HQ MAJCOM personnel may not fill this position.

5.2.2. Develop a 2R0X1 and 2R1X1 training plan for all assigned personnel, to include tenant units. Determine utilization and assignment of installation AFSC 2R0X1 and 2R1X1 personnel and a rotation plan for their respective 2R0X1 and 2R1X1 wing resources based on mission requirements. **(T-3)**

5.2.3. Ensure 2R0X1 and 2R1X1 are trained and qualified by establishing a structured training program. **(T-3)** Ensure training sessions for 2R0X1 and 2R1X1 are conducted and documented monthly. Maintain training session documentation to include attendance logs for at least one year.

5.2.4. Maintenance Operations, Director of Operations /Superintendent (DO/SUPT). In addition to the applicable Maintenance Supervision responsibilities in **Chapter 2**, the MXO OIC/SUPT will:

5.2.4.1. Develop and publish the wing operations/maintenance schedule in coordination with other squadrons and submit to both the OG/CC and MXG/CC for approval. **(T-3)**

5.2.4.2. Determine long-range fleet health maintenance priorities. **(T-3)**

5.2.4.3. Manage the data collection process, review data, and verify analysis for maintenance data collection requirements. **(T-3)**

5.2.4.4. Evaluate and provide trend analysis information to the MXG/CC and SQ/CCs. **(T-3)**

5.2.4.5. Ensure aircraft status is properly reported and maintained IAW DAFI 21-103.

5.2.4.5.1. Coordinate with the PS&D Aerospace Vehicle Distribution Officer (AVDO) to ensure all assignment and possession changes are accurately reported to the MAJCOM AVDO IAW DAFI 21-103 and AFI 16-402, *Aerospace Vehicle Programming, Assignment, Distribution, Accounting, and Termination*.

5.2.4.5.1. **(AFMC)** Include the AFMC AVDO (AFMC/A4M) and the AFMC Aerospace Vehicle Inventory and Fleet Manager (AFMC/A3F) in coordination for all proposed assignment/possession changes. This is regardless of the scale of the aerospace vehicle and regardless of the MIS or Accountable Property System of Record (APSR) being used. **(T-2)**

5.2.4.6. Initiate, review, and validate special analysis studies. **(T-3)**

5.2.4.6.1. Determine planning factors for the next year's flying hour program. **(T-3)**

5.2.4.6.2. Utilize a sortie production model or equivalent for the assigned MDS, if available. **(T-3)**

5.2.4.6.3. **(Added-AFMC)** Ensure data is valid and submitted to meet MAJCOM reporting suspense requirements. Before submission, reconciliation and coordination with squadron and group leadership is required to ensure comments fully explain causes of the data. Comments should focus on mission requirements, improving equipment performance, identifying emerging support problems, and projecting current trends. Also, when operational requirements are not achieved, determine and explain the root causes behind it. **(T-2)**

5.2.4.7. Develop procedures to update Geographical Location (GEOLOC) codes for all on and off-station possessed aircraft and ensure GEOLOC codes are updated/correct in the MIS "Location Subsystem" (FMxC2 units are exempt as long as a HHQ agency accomplishes this requirement). **(T-3)**

5.2.4.7. **(AFMC)** Ensure processes and procedures are in place and personnel are designated to support accomplishment of Flying Hour Program tracking and reconciling IAW intent of DAFI 21-103, *Equipment Inventory, Status and Utilization Reporting* **(T-2)**

5.2.4.8. IMDS units will use code "XXXX," and FMxC2 units will use "CCCC" for classified locations. **(T-3)**

5.2.1.8.1. DELETED.

5.2.4.9. Host DFTs/CFTs, provide in-briefs on unit-specific maintenance and tool-control requirements, review plans, coordinate/monitor status of aircraft and progress of repair work. **(T-3)**

5.2.4.10. Participate in the review of base level repair capability IAW TO 00-20-3, AFREP in **Chapter 11** of this DAFI, and supplements.

5.2.4.11. Publish wing notification requirements for munitions-loaded or unloaded aircraft. **(T-3)**

5.2.4.12. Establish/sustain local radio call signs for maintenance LMR networks IAW AFI 17-210, Radio Management. **(T-3)**

5.2.4.13. Develop a training plan for individuals assigned to QA that will inspect MXO functions. Minimum training requirements will include MIS (FMxC2/IMDS/or equivalent) online and background products for inspections, time changes, TCTOs, and aircraft configuration management. **(T-3)** For units with IMDS, the QA inspector will be trained on the use of DS Maintenance Scheduling Module (MSM) to provide the capability for a qualified QA inspectors to evaluate and report PS&D compliance with functional requirements. **(T-3)** The senior 2R in the MXO will provide assistance to the inspector as required. **(T-3)**

5.2.4.14. Ensure MAJCOM Master Course Listing includes 2R weapon system familiarization courses requirements. **(T-3)**

5.2.4.14.1. As a minimum, the course will include weapon system/communications electronics familiarization, flightline and shop operations, organizational structure and roles of each group, squadron, and flight. **(T-3)**

5.2.4.14.2. Analysts will attend the course within 6 months of assignment to the unit. **(T-3)**

5.2.4.14.2. **(AFMC)** [DEV] Not required if local familiarization course does not exist for civilian personnel previously qualified.

5.2.4.14.3. For remote assignments, analysts will attend within 1 month of assignment. **(T-3)**

5.2.4.14.4. DELETED.

5.2.5. Maintenance Operations Center (MOC). The MOC monitors and coordinates sortie generation, maintenance production, and execution of the operations and maintenance schedules while maintaining visibility of fleet health indicators. Through coordination with maintenance units, the MOC communicates priorities for competing limited resources (such as, fuel or calibration docks, wash racks, and dispatched specialists from the maintenance squadron(s) (for example, egress) based on daily operations schedule and maintenance priorities. The exchange of information between squadrons and the MOC must be in sufficient detail to allow the MOC to comply with reporting requirements and to identify potential problems.

5.2.5.1. The MOC will:

5.2.5.1.1. Monitor the status of aircraft/systems, as directed, (through the use of electronic or manual visual aids) including ETIC, progress of FCFs, and location of each aircraft on station. **(T-3)**

5.2.5.1.2. Track contingency and exercise aircraft generation activities. **(T-3)**

5.2.5.1.2.1. Maintain and update aircraft generation line up and display aircraft status using AF Form 2408, *Generation Maintenance Plan* and AF Form 2409, *Generation Sequence Action Schedule* or locally computer-generated equivalents.

(T-3) Note: If the tail number, mission number or specifically tasked no-later-than times are linked, this form becomes classified. See DODM 5200.01V1_AFMAN 16-1404, *Information Security Program: Overview, Classification and Declassification*.

5.2.5.1.2.2. Monitor and report aircraft generation progress with a minimum of the following information: ETIC, location of each aircraft, status of generation actions, progress against timeline necessary to meet mission requirements. **(T-3) Note:** The display format should be compatible with OPLANS and command post displays.

5.2.5.1.3. Utilize the Enhanced Maintenance Operations Center. **(T-3) Note:** BLADE will be used to track classified aircraft status.

5.2.5.1.4. Track aircraft maintained or supported by the unit but not on station. (Aircraft cross-country). **(T-3)**

5.2.5.1.5. Coordinate maintenance on the alert force, if applicable. **(T-3)**

5.2.5.1.6. Ensure status boards depict aircraft status and location comply with Security Program guidelines. **(T-3)**

5.2.5.1.7. Monitor the status and ETIC of MEL-designated AGE if it falls below critical levels. **(T-3)**

5.2.5.1.8. Monitor the status of ECM and sensor pods IAW AFI 10-201, *Force Readiness Reporting*. **(T-3)**

5.2.5.1.8.1. When MC pod availability falls below requirements per the DOC or OPLAN, the MOC will track/monitor the following information: pod serial number, status (AWP/Awaiting Maintenance (AWM)), MICAP NSN, off-base requisition numbers, and ETIC. **(T-3)**

5.2.5.1.9. Classify information IAW AFI17-130, *Cybersecurity Program Management.*

5.2.5.1.10. Verify aircraft status and ETICs with the Pro Super(s) and ensure they are properly documented in the MIS IAW DAFI 21-103**.** Reference AFCSM, 21-564, V2, *Status and Inventory Reporting Software User Manual* or equivalent for MIS guidance.

5.2.5.1.10.1. When the Production Superintendent (Pro-Super) or equivalent notifies the Maintenance Operations Center (MOC) that an aircraft is "Crew Ready" the MOC will review the Maintenance Information Systems (MIS) for each Crew Ready aircraft to ensure there are no open Red Xs. **(T-3)** If open Red X(s) are present in the MIS, the MOC will notify the Pro-super or equivalent for action. **(T-3)**

5.2.5.1.11. The MOC will verify aircraft status using the MIS and ETIC before reporting it. **(T-3)**

5.2.5.1.12. Inform affected activities of changes in priorities, plans, and schedules. **(T-3)**

5.2.5.1.13. Coordinate on changes to the operations schedule with applicable agencies by use of AF Form 2407. **(T-3)**

5.2.5.1.14. Ensure all deviations to the daily operations schedule are reviewed and accurately reported. **(T-3)**

5.2.5.1.14.1. Forward a copy of each AF Form 2407 and the daily flying schedule, with all annotated deviations, to MMA. **(T-3)**

5.2.5.1.15. Request support services outside the scope of the MXG (such as, standby firefighting capability, aircraft water, snow removal, fueling and defueling service, civil engineer support, or control tower clearances for ground movement of aircraft and equipment). **(T-3)**

5.2.5.1.15.1. Coordinate on all aircraft engine runs and all aircraft ground movements conducted by maintenance personnel prior to execution. **(T-3)**

5.2.5.1.16. Develop, coordinate, implement, and maintain functional and emergency action checklists. **(T-3)** Checklists contain those actions required to be taken by functional area(s).

5.2.5.1.16.1. Functional checklists are required for use during actions such as nuclear mass loads, Broken Arrow, Dull Swords, Bent Spear, aircraft crash/mishap/incident, aircraft FOD, aircraft damage, flightline fire, severe weather warning or evacuation, runway closure, hazardous chemical release (example, Hydrazine, Broken Pod Glass release of Thorium Fluoride or Americium), Quick Reaction Checklists, injuries resulting from aircraft maintenance and any other unusual circumstances deemed necessary.

5.2.5.1.16.2. For OPLAN 8010 notification, use the plan implementation checklists.

5.2.5.1.16.3. Use unit OPLANs as a guide in developing these checklists.

5.2.5.1.16.4. DELETED.

5.2.5.1.16.5. The MOC will maintain checklists that implement all approved MAJCOM and local requirements. **(T-3)**

5.2.5.1.16.6. Establish a command/contingency focal point to coordinate ABDR and or Joint Combat Assessment Team (JCAT) response requests with AFMC. **(T-3) Note:** If data collection forms are required forms will be forwarded to the Aircraft Battle Damage Repair (ABDR) Technical Support Office (TSO) and JCAT. **(T-3)** For ABDR TSO, CLASSIFIED messages must be sent to: **usaf.wright-patt.afsc-lg.mbx.afsc-lgpm-abdr-tso@mail.smil.mil** and UNCLASSIFIED messages must be sent to NIPR: **afsc.lgpm.abdrtso@us.af.mil** for filing in the historical archives. For JCAT, CLASSIFIED data must be submitted via the Intellipedia Secret Internet Protocol Router Network (SIPRNET) link **https://www.intellipedia.intelink.sgov.gov/wiki/Portal:Joint_Combat_Assessment_Team**. UNCLASSIFIED data can be sent to: JCAT (**JCAT@us.af.mil**).

5.2.5.1.17. Coordinate munitions delivery priorities with flying units and munitions maintenance activities, and control when tasked. **(T-3)**

5.2.5.1.17.1. Maintain a contact list and notify the base Fire Emergency Services

and all applicable agencies that require notification of munitions-loaded or unloaded aircraft. **(T-3)**

5.2.5.1.17.1.1. The MOC will provide agencies with the aircraft type, tail number, location, type of explosives, and arming status. **(T-3)**

5.2.5.1.18. Upon notification of deployments, ensure all deploying equipment is identified and loaded into the IMDS-CDB, Aerospace Expeditionary Force subsystem or designated MIS equivalent for the duration of the deployment. **(T-3)**

5.2.5.1.19. Monitor and manage reporting of Hangar Queen aircraft/systems IAW **Chapter 11**.

5.2.5.1.19. **(AFMC)** Ensure aircraft boards are visibly marked to show Hangar Queen Status, date of last flight, ETIC, expected fly date and identify the Hangar Queen Manager. **(T-2)**

5.2.5.1.20. Notify Flightline Expediters of all OAP code conditions. **(T-3)**

5.2.5.1.20. **(AFMC)** Notify Flightline Expediters of all OAP code conditions through Maintenance Operations Center (MOC).

5.2.5.1.21. Ensure facilities and visual aids meet the following minimum standards:

5.2.5.1.21.1. A completely enclosed room with air conditioning and heating. **(T-3)** An observation room is permitted.

5.2.5.1.21.1.1. Doors to the MOC and the observation room will be either mechanically or electrically locked to control access. **(T-3)**

5.2.5.1.21.2. Isolate MOC electrical power circuits and provide a standby power source and emergency lighting. **(T-3)**

5.2.5.1.21.2.1. The MOC will establish procedures to operate standby power sources. **(T-3)**

5.2.5.1.22. Maintain the status and location of all transient aircraft. **(T-3)**

5.2.5.1.22.1. Post the priority of each transient aircraft on the status board, based on the maintenance priorities listed in **Table 1.2** **(T-3)**

5.2.5.1.22.2. Coordinate with the appropriate agency for aircraft maintenance support. **(T-3)**

5.2.5.1.22.3. Contact WS for arming or de-arming of transient aircraft IAW **Chapter 11**. **(T-3)**

5.2.5.2. MOC Maintenance Communications. Reliable, redundant, and effective communications systems are essential for efficient operation. Communications equipment will be operated and managed IAW AFI 17-210, DAFI 17-220, *Spectrum Management*, and AFI 17-130. The MOC NCOIC/SUPT will:

5.2.5.2.1. Establish a procedure to process requests for specific radio equipment to support MXG maintenance activities IAW DAFMAN 23-122.

5.2.5.2.1.1. Specific radio allowances are stated in Allowance Standard 660 at **https://usaf.dps.mil/teams/12599/AFKN_Docs/ASC%20Prefaces/ASC%2066**

0%20Ground%20Communications%20-%20Operations%20Support.pdf.

5.2.5.2.2. Ensure a Very High Frequency (VHF) and/or Ultra High Frequency (UHF) and/or High Frequency (HF) radio is authorized and available to provide communications between aircraft and maintenance.

5.2.5.2.3. Ensure the MOC has a hotline on the secondary crash phone net. **(T-3)**

5.2.5.2.3.1. When required, direct communications lines will be provided to QA, Munitions Control, EOD, airfield operations, base fire department, NDI, control tower and the central security control. **(T-3)**

5.2.5.2.4. Develop and exercise comm-out procedures to include loss of radios, Local Area Network (LAN) and phone. **(T-3)**

5.2.5.2.5. Ensure MOC personnel receive initial radio operating training before assuming duties involving radio operations IAW AFI 17-210 and **Chapter 11**. **(T-3)**

5.2.6. Engine Management. EM manages unit efforts to maintain adequate engine support for mission requirements by monitoring engine removals and replacements, component tracking, engine TCTOs and TCIs, engine records in the MIS and CEMS; and perform Engine Manager duties. Functions supporting EM will be combined within the wing and physically co-locate with the Propulsion Flight. **(T-3)** The SRAN Engine Manager works and is co-located with the EM section. **(T-3)**

5.2.6. **(AFMC)** [DEV] Engine Management. Within the 412 MXG, the 412 MXS Propulsion Flight Business Unit will perform duties and responsibilities of Engine Management (EM) Section listed in DAFI 21-101.

5.2.6.1. Specific EM responsibilities are detailed AFMAN 20-116, AFPAM 63-129, TO 00-25-254-1, -2 and **Chapter 14**.

5.2.7. Plans, Scheduling, and Documentation (PS&D). PS&D is responsible for coordinating aircraft maintenance requirements and utilization scheduling between maintenance, operations, and external agencies. PS&D oversees the entire maintenance scheduling effort throughout the wing and notifies applicable senior managers of scheduling process discrepancies and recommended courses of action.

5.2.7.1. Specific PS&D responsibilities are detailed in **Chapter 14**.

5.2.8. Maintenance Management Analysis (MMA). MMA tracks, analyzes, and presents information to help senior leadership assess the health of the units' weapon systems and equipment. MMA acts as the group POC for MIS issues and performs analyses to assess and improve unit performance (such as, effectiveness and efficiency of unit resources and logistical support processes). The MIS provides the main source of information used by analysts to assess unit performance and capability.

5.2.8. **(AFMC)** Maintenance Management Analysis (MMA). The senior 2R2 or civilian equivalent member is the unit functional manager and monitors and controls analysis resources for the unit. All issues relating to MIS database management, and training should be directed to the unit functional manager first. The unit functional manager will elevate issues beyond the unit control to the wing’s analysis functional manager. **(T-2)** This includes allocations of AETC formal school quotas.

5.2.8.1. MMA will:

5.2.8.1.1. Be centrally organized but may locate analysts in the squadron to enable maximum responsiveness and effectiveness. **(T-3)**

5.2.8.1.1.1. When analysts are located in the squadron, they will still work directly for the MMA Section NCOIC/Chief who will provide their training and monitor the quality/relevancy of their workload. **(T-3)**

5.2.8.1.2. Provide information on analysis services, capabilities and training to units and maintenance supervision. **(T-3)** At a minimum, the following will be covered; MIS changes, on-going special studies, trend analysis, performance indicators/metrics TTP. **(T-3)** For units where MMA is centralized, perform monthly visits to maintenance work centers directly associated with aircraft generation. **(T-3)**

5.2.8.1.2.1. Work with MTS and/or FTD for opportunities to provide training on analysis services and capabilities (example, Mx Orientation, DCC Course). **(T-3)**

5.2.8.1.2.2. DELETED.

5.2.8.1.3. Calculate maintenance metrics and compare unit performance against MAJCOM and locally developed goals (if applicable). **(T-3)**

5.2.8.1.4. Develop products to track, monitor and identify seasonal and cyclical trends at the group, squadron, and unit level for MAJCOM reportable leading and lagging indicators. **(T-3)**

5.2.5.1.4.1. & 5.2.5.1.4.2. DELETED**.**

5.2.8.1.5. Review data for anomalies, variations, and trends to identify areas requiring further study. **(T-3)**

5.2.8.1.5.1. When significant seasonal or cyclical trends are identified in leading or lagging indicators, accomplish and document further analysis. **(T-3)**

5.2.8.1.5.1.1. Provide presentations, reports, studies/analyses, and briefings as requested or deemed appropriate. **(T-3)**

5.2.8.1.5.2. Units with like MDSs will compare and contrast metrics to identify significant variation. **(T-3)**

5.2.8.1.5.2.1. When significant variation is identified, accomplish and document further analysis. **(T-3)**

5.2.8.1.5.2.2. Provide presentations, reports, studies, analyses, and briefings as requested or deemed appropriate. **(T-3)**

5.2.8.1.6. Monitor wing, group, and squadron utilization rates. **(T-3)**

5.2.8.1.6.1. When operational requirements are not achieved, determine if assignable causes are present. Recommend corrective actions or measures when necessary. **(T-3)**

5.2.8.1.7. Assist unit leaders with the application and interpretation of maintenance data. **(T-3)**

5.2.8.1.8. Coordinate with PS&D and unit's Maintenance Supervision to provide monthly airframe, facility, and personnel capabilities (as required), attrition, and spare factors for use in planning the annual FHP. **(T-3)**

5.2.8.1.8.1. MAJCOMs will publish attrition and spare factors computations in a supplement to this instruction reference TO 00-20-2.

5.2.8.1.8.1. **(AFMC)** Instructions are noted in **Attachment 8**.

5.2.8.1.8.1.1. MMA will use MAJCOM supplement guidance to calculate attrition and spare factor computations. **(T-3)**

5.2.5.1.8.1.2. DELETED.

5.2.8.1.9. Analyze equipment performance trends to identify problems affecting the unit mission and, whenever possible, provide predictive analytical information with recommendations to unit's Maintenance Supervision. **(T-3)**

5.2.8.1.10. Verify accuracy of Job Data Documentation (JDD), flying schedule deviations, aircraft status and utilization within the MIS. **(T-3)**

5.2.8.1.10.1. Validate data entered into the MIS as part of daily analysis duties and inform affected agencies of discrepancies. **(T-3)**

5.2.8.1.10.2. Identify erroneous or missing data to the responsible agency for correction or completion. **(T-3)**

5.2.8.1.11. Control the assignment of unit work center and mnemonic codes. **(T-3)**

5.2.8.1.11.1. Coordinate with P&R, MTS, or responsible agency on the assignment of alphanumeric and work center codes. **(T-3)**

5.2.8.1.11.2. Publish written guidance to control these codes when not provided by higher headquarters. Multiple mnemonic codes may be used within a work center code to accommodate different AFSCs assigned. **(T-3)**

5.2.8.1.11.3. Coordinate new or revised mnemonic codes with affected activities for planning purposes. **(T-3)**

5.2.8.1.12. Be responsible for MIS database management. **(T-3)**

5.2.8.1.13. Assists MIS users in developing procedures for collecting information from deployments and exercises where the MIS is not available. **(T-3)**

5.2.8.2. In addition to the applicable Section NCOIC/Chief responsibilities outlined in **Chapter 2**, the MMA Section NCOIC/Supervisor will:

5.2.8.2.1. Ensure growth of analysis personnel by developing and maintaining a plan to rotate personnel through various duty positions within MMA.

5.2.8.2.1.1. Establish and coordinate plans for rotating 2R0XX personnel to increase field knowledge and experience every 24 months, not to exceed 36 months. **(T-3)**

5.2.8.2.1.2. This rotation plan applies to TSgts and below as well as 3- or 5-skill level personnel of any rank. **(T-3)**

5.2.8.2.2. Define the daily, weekly, monthly, and annual roles and responsibilities for each duty position within MMA. **(T-3)**

5.2.8.2.3. Ensure the CFM approved AFSC duty titles are utilized for all MMA personnel. **(T-3)**

5.2.8.2.4. **(Added-AFMC)** IMDS-CDB/FMxC2 Users Group. Establish an IMDS-CDB/FMxC2 Users Group to identify user problems, provide on the spot training to correct user documentation problems, and to discuss other issues relating to operation of the system. MMA will conduct meetings in conjunction with the monthly DIT meeting. **(T-2)** An agenda (pre-meeting) and meeting minutes (post-meeting) will be published and sent to all work centers. The use of electronic means (telecon, Defense Collaboration Services, etc.) for the meeting is authorized.

5.2.8.2.5. **(Added-AFMC)** Set priorities and resolve conflicts on all automated maintenance systems priorities.

5.2.8.2.6. **(Added-AFMC)** Coordinate with wing plans personnel to develop planning data and inputs.

5.2.8.2.7. **(Added-AFMC)** Ensure timely submission of data to meet MAJCOM reporting suspense and validity of data submissions. **(T-2)** Work with the senior maintainers on all comments written to explain the meaning of the data presented.

5.2.8.2.8. **(Added-AFMC)** Ensure database managers (DBMs) can be reached during off duty hours to support processing requirements for IMDS-CDB. **(T-3)**

5.2.8.2.9. **(Added-AFMC)** MMA Team Concept. MMA teams identify problem areas for additional study using MIS data gathering, research, and investigation. Deficiency Analysis/QA assists by providing in-depth technical expertise. The MMA team will present MXG leadership with completed studies focusing on the cause and effect of problem areas, and include recommendations for course(s) of action as appropriate. **(T-2)**

5.2.8.3. Maintenance Information Systems (MIS). For management of IMDS-CDB, FMxC2, and Reliability and Maintainability Information System (REMIS), follow AFCSM 21-556, Vol 2, *Intro to IMDS CDB*, MAJCOM/Lead Command guidance, unit procedures, and REMIS user manuals. Personal computers and software used as "stand-alone" systems are not considered MIS.

5.2.8.3. **(AFMC)** The command functional manager at HQ AFMC Maintenance Management Branch (AFMC/A4MM) (**AFMC.A4MM@us.af.mil**) is responsible for all issues related to MIS. They will provide technical advice to units on problems beyond their expertise.

5.2.8.3.1. Request to modify/create new functionality within IMDS-CDB IAW AFCSM 21-556V2. FMxC2 units will submit a System Change Request for any new requirements or corrections to existing features. **(T-3)**

5.2.8.3.2. Database Manager (DBM) will identify functions that require subsystem monitors and provide applicable training to those responsible for ensuring the accuracy/sustainment of their subsystem. **(T-3)**

5.2.8.3.2.1. Subsystem monitors will be appointed by the Section Chief of subsystem functions (such as, PS&D Section Chief appoints and signs appointment letter for configuration management). **(T-3)**

5.2.8.3.2.2. MMA is responsible for the overall management of the JDD subsystem. **(T-3)**

5.2.8.3.3. MAJCOMs will provide guidance describing the management of the MIS listed in **paragraph 5.2.8.3** within their command.

5.2.8.3.3. **(AFMC)** No further guidance required. All requirements listed meet MAJCOM requirements.

5.2.8.3.4. At a minimum, the MMA MIS/Host DBM will ensure:

5.2.8.3.4.1. MAJCOM coordination occurs when problems exist that are beyond the scope of responsibilities of Host DBMs. **(T-3)**

5.2.8.3.4.2. IMDS-CDB/FMxC2 security is maintained IAW AFI 17-130, *Cybersecurity Program Management*.

5.2.8.3.4.2.1. MMA personnel coordinate MIS access permission requirements to enable MDD on non-possessed aircraft. **(T-3)**

5.2.8.3.4.3. MMA personnel provide expertise on IMDS-CDB/FMxC2 for resolution of problems beyond the work center and sub-system monitors' control. **(T-3)**

5.2.8.3.4.4. Support is provided to tenant organizations and users. **(T-3)**

5.2.8.3.4.5. Coordination with the Defense Enterprise Computing Center or AF Network Control Center on all matters concerning IMDS-CDB. **(T-3)**

5.2.8.3.4.6. The Defense Enterprise Computing Center supports all requirements concerning the operation and maintenance of IMDS-CDB. **(T-3)**

5.2.8.3.4.7. Scheduled MIS downtime is published for users. **(T-3)**

5.2.8.3.4.8. MMA personnel control and monitor submissions of IMDS-CDB Difficulty Report(s). **(T-3)**

5.2.8.3.4.9. Coordination occurs on matters pertaining to the interface of other automated systems with IMDS-CDB. **(T-3)**

5.2.8.3.4.10. A functional checklist is developed to establish timelines and MIS data capture requirements for use in the event of a weapon system mishap. **(T-3)**

5.2.8.3.4.10.1. The checklist must require immediate capture and isolation of the historical data for the mishap weapon system regardless of the time or day of week. **(T-3)** Contact the Host DBM to immediately put the IMDS-CDB in File Update Mode (when required) until the functional checklist can be completed. FMxC2 equipment records will be locked using screen 9012 (Lock/Unlock Aircraft/Data Records). **(T-3)**

5.2.8.3.4.11. Support of the Communications-Electronics maintenance community referring to DAFI 21-103 and TO 00-33A-1001, *Methods and Procedures, General*

Cyber Defense Operations Activities Management Procedures and Practice Requirements, for maintenance analysis and host DBM responsibilities.

5.2.8.3.4.12. Control of access to specific IMDS-CDB programs and subsystems by utilizing Transaction Identification Codes (TRICs) security profiles or screen 9057 (program access) for FMxC2. **(T-3)**

5.2.8.3.4.12.1. Audit permissions to IMDS security profiles and FMxC2 access keys annually. Take appropriate measures when a compromise is suspected or reported. **(T-3)**

5.2.8.3.4.12.2. Semi-annually audit IMDS ELC access. **(T-3)**

5.2.8.3.4.13. IMDS-CDB subsystem managers are informed of the status of applicable TRICs prior to turning the TRIC on or off. **(T-3)**

5.2.8.3.5. MMA is the focal point for codes used on the maintenance data documentation process. This includes maintenance deferred codes, delayed status codes, flying scheduling deviation and cause codes listing. **(T-3)**

5.2.8.3.5.1. Use the MAJCOMs approved codes and narrative for official use. Code listings will be adjudicated yearly by all commands. **(T-2)**

5.2.8.3.5.2. Recommended changes, additions, and deletions to these codes and/or data elements will be adjudicated and approved by all MAJCOMs. MMA will submit request to their respective MAJCOM to initiate and coordinate any updates to the code listings. **(T-2)**

5.2.8.3.6. Data Integrity. MMA is the OPR for the Data Integrity Team (DIT). All maintenance units will participate in the DIT program. **(T-3)**

5.2.8.3.6.1. The purposes of the DIT include: (1) ensuring the unit has complete and accurate data in the MIS and aircraft forms, (2) identifying and quantifying problems within the unit preventing complete and accurate documentation, and (3) identifying and correcting the root causes for poor data integrity. The DIT is established to evaluate/isolate/eliminate documentation problems in IMDS-CDB/FMxC2/DPAS M&U Module. MMA will ensure that all assigned DIT members are trained in the use of MIS applicable programs for the data integrity review/correction process. **(T-3)** Errors identified by the DIT team will be reconciled IAW **Paragraph 5.2.5.3.6.5**. **(T-3) Note:** DIT should be aware of Logistics Interference (LI) and data errors that may be the result of purposeful efforts. DIT will note possible LI and monitor for trends and report unusual behavior IAW AFI 17-130.

5.2.8.3.6.2. The DIT will include, at a minimum, one representative from each squadron under the MXG. It will include participation from PS&D, MOC, DMS, EM, Debrief Section, and QA as determined by MMA. **(T-3)**

5.2.8.3.6.3. MAJCOMs will determine the frequency of DIT meetings.

5.2.8.3.6.3. **(AFMC)** DIT meetings will be held monthly. **(T-2)**

5.2.8.3.6.4. Representatives will be at least a SrA that possesses a 5-skill level and is familiar with the unit's assigned weapon system(s). **(T-3)**

5.2.8.3.6.5. As a minimum, the following functions will be performed by the DIT:

5.2.8.3.6.5.1. Educate group, squadron, and AMU leadership on the importance of data integrity including the impacts of erroneous data.

5.2.8.3.6.5.2. Ensure MIS accurately reflects AFTO Form 781-series forms entries.

5.2.8.3.6.5.3. Run maintenance action review background reports for all work accomplished by squadrons and work centers.

5.2.8.3.6.5.3.1. Audit the report by JCN/WCE (Work Event Separator for FMxC2) to identify suspected errors.

5.2.8.3.6.5.3.2. Responsibility for correcting errors belongs to the performing work center.

5.2.8.3.6.5.3.2.1. The use of general WUCs/LCNs when a component level WUC/LCN exists will be counted as an error. **(T-3)**

5.2.8.3.6.5.3.3. Audit for Weapons System status, flying schedule deviation and utilization errors. **(T-3)**

5.2.8.3.6.5.3.4. Use of automated processes is authorized.

5.2.5.3.6.5.4.1. DELETED.

5.2.8.3.6.5.3.5. Develop a system to track, measure and report data integrity errors by work center, AMU, and squadron. **(T-3)**

5.2.8.3.6.5.3.5. **(AFMC)** Do not limit scope of DIT to JDD. **(T-2)**

5.2.8.3.6.5.3.6. Implement training programs that are geared toward correcting data integrity trends. **(T-3)**

5.2.8.3.6.5.3.6. **(AFMC)** Provide feedback to users via users' meetings, periodic memorandums, letters, etc. **(T-2)**

5.2.8.3.6.5.3.7. **(Added-AFMC)** When the operational requirements are not achieved, MMA will perform an investigation to determine the cause. **(T-2)** As a minimum the following areas will be considered:

5.2.8.3.6.5.3.7.1. **(Added-AFMC)** Are operational requirements realistically based on availability of equipment? **(T-2)**

5.2.8.3.6.5.3.7.2. **(Added-AFMC)** What are the causes for flying schedule deviations (cancellations, aborts, additions or early takeoff (ET)/late takeoffs (LT). **(T-2)**

5.2.8.3.6.5.3.7.3. **(Added-AFMC)** Are specific aircraft, equipment, systems, or subsystems contributing to a disproportionate share of deviations/turbulence? **(T-2)**

5.2.8.3.6.5.3.7.4. **(Added-AFMC)** Is specific equipment failing to perform as scheduled? Does this equipment require more or less maintenance than others? **(T-2)**

5.2.8.3.6.5.3.7.5. **(Added-AFMC)** Are there enough people to meet mission needs? Are certain work centers documenting significant overtime or show consistently high utilization rates? **(T-2)**

5.2.8.3.6.5.3.7.6. **(Added-AFMC)** Is there a good balance of skills within AFSCs and between the units? **(T-2)**

5.2.8.3.6.5.3.7.7. **(Added-AFMC)** Do higher rates for repeat/recur discrepancies indicate training/experience shortfalls? **(T-2)**

5.2.8.3.6.5.3.7.8. **(Added-AFMC)** Is there sufficient time to schedule and work maintenance problems? **(T-2)**

5.2.8.3.6.5.3.7.9. **(Added-AFMC)** Are trends significant? Are the trends short term (6 months or less) or long term? Where will the unit likely be in 6-12 months? **(T-2)**

5.2.8.3.6.5.3.7.10. **(Added-AFMC)** Are there seasonal or cyclical variations? Are current variations outliers? **(T-2)**

5.2.8.4. **(Added-AFMC)** Analytical Process. The analytical process consists of identifying contributory factors, manipulating raw data into meaningful formats, computing management indicators, performing statistical measurements, and creating accurate, complete, and easy to understand presentations.

5.2.8.4.1. **(Added-AFMC)** Management Contributions to the Analytical Process. Maintenance managers have a significant impact on the usefulness of the MMA to the unit. Managers should constantly review how information is being organized and presented.

5.2.8.4.2. **(Added-AFMC)** Analytical Studies. MMA will provide results of investigations, analyses, or studies to work centers. **(T-2)** The most effective study is one that goes beyond superficial conclusions. It helps solve a problem relative to mission performance. Specific studies are provided to the requester, and a file copy is retained for future reference. Disseminate the study by electronic means or include it in a monthly maintenance summary. The study should state assumptions up front, and should be summarized to state how the significance is measured.

5.2.8.4.3. **(Added-AFMC)** Maintenance Analysis Referrals. A referral is a procedure used to identify, investigate, and propose corrective action for management problems. These are highly effective tools to make agencies aware of common problems.

5.2.8.4.4. **(Added-AFMC)** Dedicated AMU Analysis (where applicable). The AMU analysis function is intended to provide dedicated analytical support for the AMU. The MMA section NCOIC may dedicate an analyst to each AMU. The dedicated AMU analyst works for the MMA section NCOIC. Accomplishing AMU tasks are the dedicated analyst's primary responsibility. The MMA section NCOIC is responsible for the overall effectiveness of the AMU analysis program. As a minimum, the AMU analyst will: **(T-2)**

5.2.8.4.4.1. **(Added-AFMC)** Review maintenance debriefing data to track in-flight discrepancies and deviations on each aircraft. **(T-2)** Review aircraft status inputs from the MOC forWork Unit Code (WUC) accuracy. **(T-2)** Closely monitor

fix time on Code 3 discrepancies and report results to AMU supervision daily. **(T-2)** Brief problem aircraft and systems to AMU supervision daily. **(T-2)**

5.2.8.4.4.2. **(Added-AFMC)** Brief comparative AMU data weekly and monthly as required by the MXG/CC. **(T-2)**

5.2.8.4.4.3. **(Added-AFMC)** Validate cannibalization (CANN) documentation in IMDS-CDB at least weekly, with AMU supply and inform AMU supervision of its accuracy. If errors exist, a more frequent validation may be required. Advise AMU supervision of recurring problems. **(T-2)**

5.2.8.4.4.4. **(Added-AFMC)** Monitor the Unit Type Code (UTE) rate for the AMU. **(T-2)**

5.2.8.4.4.5. **(Added-AFMC)** Analyze programmed and actual attrition factors. **(T-2)**

5.2.8.4.4.6. **(Added-AFMC)** Attend AMU scheduling/production meetings at least once per week. **(T-2)**

5.2.8.4.4.7. **(Added-AFMC)** Present Maintenance Analysis. Present data by using summaries, charts, graphs, tabular displays, and narratives. These data presentations should show the relationships among various factors. Data presentations should be displayed or presented in time to be useful in plans or reports. A printed monthly maintenance summary, tailored to the needs of the unit, is an excellent method of presenting data. **(T-2)**

5.3. Maintenance Training (MT). MT consists of the Training Management Element and the Development and Instructor Element. Maintenance Training assists SQ/CCs by providing Unit Training Managers (UTM) to manage the enlisted specialty training program. MT will:

5.3.1. Provide initial, recurring, and advanced proficiency, qualification, or certification training needed by a technician to perform duties in their primary AFSC and manage course codes to track training IAW AFI 36-2650 and DAFMAN 36-2689.

5.3.2. Serve as the single point of contact for all training matters affecting maintenance. **(T-3)**

5.3.3. The MT Superintendent/NCOIC will maintain administrative responsibility for UTMs whether UTMs are centralized or decentralized. **(T-3)**

5.3.4. Develop and administer appropriate Maintenance Cyber Discipline training. Training will be tailored to DoD Information Technology used locally and will emphasize authorized, and unauthorized uses, prevention, detection, remediation, and provide an overview of recent negative trends and effective mitigation techniques. **(T-3)**

5.3.4.1. Coordinate with Quality Assurance, MXG/CC designated responders, and facilitators for action when discovery that DoD Information Technology lacks the capability to perform a mission function. **(T-3)** Example: Enhanced Technical Information Management System containing software for a TO task that does not have a hardware TO approving its installation or use.

5.4. Programs and Resources (P&R). P&R serves as the MXG focal point for interaction with external functional support activities to ensure critical mission generations support, infrastructure personnel management and resources are configured to maximize mission capability/AA. P&R

program configurations may vary based on mission but generally P&R coordinates with functional base OPRs to interact on Facility Management (FM), Vehicle Control (VCO), Support Agreements, Unit Safety, Security, Resources, Manpower, Environment, Deployment and Maintenance Information System programs as applicable. Units will maximize consolidation of personnel assigned in P&R as described in **Paragraph 2.4.14**. P&R will:

5.4. (AFMC) Programs and Resources (P&R). Developmental testing is conducted throughout the acquisition and sustainment processes to assist engineering design and development, and verify that critical technical parameters have been achieved. Developmental Test & Evaluation (DT&E) supports the development and demonstration of new materiel solutions or operational capabilities as early as possible in the acquisition life cycle. After full- rate production or fielding, DT&E supports the sustainment and modernization of systems. To support integrated testing, as many test activities as practical are conducted in operationally relevant environments without compromising engineering integrity, safety, or security. Developmental testing leads to and supports a certification that the system is ready for dedicated operational testing IAW DoDI 5000.02, *Operation of the Adaptive Acquisition Framework*, and DAFMAN 63-119, *Mission-Oriented Test Readiness Certification*. P&R will:

5.4.1. Develop, maintain, and coordinate all applicable AFI-directed programs and plans affecting maintenance. **(T-3)**

5.4.1.1. **(Added-AFMC)** Coordinate with the Maintenance Group (MXG)/Squadron SUPTs to manage Test and Evaluation (T&E) resource requirements for MXG.

5.4.1.2. **(Added-AFMC)** 412 Logistics Programs (412 MXG/MXOPL).

5.4.1.2.1. **(Added-AFMC)** Programs. The Life Cycle Logistics Management Specialist (LCLMS) support DT&E, and Operational Test and Evaluation (OT&E) activities IAW DoDI 5000.02 and DoDD 5000.01, *The Defense Acquisition System.* Provide continued emphasis and awareness by early involvement in DT&E and OT&E activities. The Life Cycle Logistics Specialist will establish standards and procedures for accomplishment of DT&E and OT&E logistics functions. Ensure compliance with statutory directives; Air Force environmental management, safety, security instructions; and TOs. Develop and provide training and tools to improve the DT&E and OT&E process within the Acquisition Community. Will assist in development of logistics support test plans, data analysis plans, and technical reports. The Local Checklists (LCL) will assist local management in test execution during DT&E and OT&E program involvement as defined IAW DAFI 99-103, *Capabilities-Based Test and Evaluation*. The LCLMS is the Logistics Subject Matter Expert (SME) and focal point for Engineering and Program Management personnel during all DT&E and OT&E activities. They will plan, direct, and manage the development and acquisition of logistics support plans for System Program Offices at the Group level. **(T-2)** The process of Integrated Product Support (IPS) Elements for testing, evaluating and, supporting a weapon system from cradle to grave is defined in the Defense Acquisition Guidebook. LCLMS' will determine whether logistics support and system readiness objectives are met. LCLMS' apply the use of the 12 IPS Elements (DAFI 63-101/20-101) during DT&E, OT&E and Sustainment. Programs conduct DT&E throughout the systems life cycle, from program initiation through system sustainment, to reduce

design and programmatic risks and provide assessments. DT&E can occur as either contractor testing, government testing or a mix of both.

5.4.1.2.2. **(Added-AFMC)** LCLMSs provide complete, logistics planning and support to a broad system, program, or functional area of assignment.

5.4.1.2.2.1. **(Added-AFMC)** Lead logistics planning efforts and secure support for ground and/or flight testing of a broad system, program or functional area through the efforts of self and other team members.

5.4.1.2.2.2. **(Added-AFMC)** Direct and coordinate the efforts of TW MXG managers and other team members in developing capability estimates and logistics support planning for assigned area of responsibility.

5.4.1.2.2.3. **(Added-AFMC)** Lead and balance test requirements for the various programs within the assigned area against existing or available support capabilities and capacities.

5.4.1.2.2.4. **(Added-AFMC)** Prepare and present maintenance briefing material to inform and advise the MXG staff, visitors and higher authorities on logistics support capabilities and status of programs and projects.

5.4.1.2.2.5. **(Added-AFMC)** Develop cross-coordination staffing and the preparation of correspondence for the MXG on matters involving multiple maintenance functional elements; plans for modernization; develop and submit budgetary estimates through resource or manpower managers.

5.4.1.2.3. **(Added-AFMC)** Represent the MXG at initial test and evaluation planning meetings, in development of test and evaluation master plans, program management plans, program test plans, program introduction documents, integrated logistics support plans, and logistics support plans.

5.4.1.2.3.1. **(Added-AFMC)** Perform Program Analyst and Project Manager duties for programs unique to the maintenance complex; such as maintenance back shop and jet engine development or sustainment work.

5.4.1.2.3.2. **(Added-AFMC)** Manage collaboration and attain assistance from qualified specialists or SMEs to develop and implement logistics plans for a variety of projects.

5.4.1.2.3.3. **(Added-AFMC)** Organize and delegate work within the team, provide guidance and training to assigned specialists, reviews work, and provides input to performance evaluations.

5.4.1.2.4. **(Added-AFMC)** Develop and implement plans for logistics support of the TW mission.

5.4.1.2.4.1. **(Added-AFMC)** Review planning documents for assigned area and develop detailed support plans including estimates of manpower requirements by skill, career field and special training requirements; funding requirements; facility management and formulation or review of plans and specifications for facilities modification or for new facility requirements.

5.4.1.2.4.2. **(Added-AFMC)** Prepares and reviews services and support

agreements between the AFTC, TW, MXG, contractors, NASA, AFMC and other MAJCOMS, Air Force, other services, and other DOD agencies. Ensures all support agreements requiring MAJCOM involvement are properly routed into the MAJCOM for needed reviews, coordinations, signatures, etc. Ensures any support agreements involving aerospace vehicle transfers or possessions are submitted to AFTC Plans in preparation for subsequesnt HQ AFMC review (AFMC/A4M, AFMC/A3F, and others as required by content) and final AFMC/A3F approval.

5.4.1.2.4.3. **(Added-AFMC)** Review and participate in negotiation of host-tenant and interagency support agreements.

5.4.1.2.4.4. **(Added-AFMC)** Perform functional area chief duties for MXG service contracts, including the development of statements of work, performance work statements, and other contractual documents; serve as liaison with the AFTC Contracting Office.

5.4.1.2.4.5. **(Added-AFMC)** Perform a variety of special assignments to include conducting special one-time and feasibility studies to determine support capability, assess new requirements and develop more effective ways to do business.

5.4.1.2.5. **(Added-AFMC)** Manage equipment, support equipment and, material requirements to support assigned programs.

5.4.1.2.5.1. **(Added-AFMC)** Manage deployment of MXG resources (aircraft, manpower, equipment, spare parts, etc.) to meet test requirements or war/contingency taskings; loan of government property as Government Furnished Equipment, transportation plans, OPSEC/COMSEC plans, reception of other forces deploying to Edwards AFB to meet test requirements or war/contingency taskings. Conduct studies and analysis in support of bed down activities for new test programs and manage commitment of logistics resources for support of assigned programs/projects and logistics test plans.

5.4.1.2.5.2. **(Added-AFMC)** Manage acquisition logistics support for production weapons systems and for systems in pre-production phases by conducting continual follow-up on changes in test schedules, acquisition of materials and support equipment, availability of funds, status of maintenance manpower, training, and support capabilities, adjusts logistics resources as required to ensure test milestones can be achieved.

5.4.1.2.5.3. **(Added-AFMC)** Formulate innovative solutions to diverse logistics problems encountered with unique test assets.

5.4.1.3. **(Added-AFMC)** 412 MXG Organizational Design and Improvement (412 MXG/MXOPO).

5.4.1.3.1. **(Added-AFMC)** Develop workforce resource requirement inputs for T&E activities associated with assigned MXG Research, Development, Test and Evaluation programs IAW AFI 38-101, DAFI 99-103, DoDD 3200.11, *Major Range and Test Facility Base (MRTFB)*, DoDI 7000.14, *Department of Defense Financial Management Policy*, DoDD 5010.42, *DoD-Wide Continuous Process Improvement (CPI)/Lean Six Sigma (LSS) Program*, and DoDI 5010.43, *Implementation and*

Management of the DoD-Wide Continuous Process Improvement/Lean Six Sigma (CPI/LSS) Program.

5.4.1.3.1.1. **(Added-AFMC)** Serve as the MXG and its sub-organizations' focal point for MXG manpower requirements determination for planning and resourcing current, future or potential T&E programs. Perform studies to assess requirements, disconnects, offsets, initiatives, workload and/or adjustments related to military, civilian, or overhire authorizations. Develop decision level data using appropriate resourcing tools, past practices, standards or functional or SME technical estimates. Develops project inputs for statement of capabilities (SOC), Program Objective Memorandum (POM), manpower estimate report, briefs or strawman Unit Manning Document (UMD) to support T&E requirement programing and coordinate efforts with external organizations.

5.4.1.3.1.2. **(Added-AFMC)** Serve as the MXG and its subordinate units' action officer for organization change requests. Provide recommendations to meet capability and capacity requirements based on current and/or future mission objectives. Develop, manage, and coordinate supporting document with external organizations, and track progression IAW AFI 38-101.

5.4.1.3.1.3. **(Added-AFMC)** Develop decision level data for organizational or process solutions to resolve complex T&E mission capacity and capability issues. Support strategic planning with alternate course of actions (COA), and/or recommendations on maintenance capabilities and resources needed to meet T&E mission requirements effectively and efficiently.

5.4.1.3.1.4. **(Added-AFMC)** Develop management tools as required to aid leadership in making data driven decisions related to mission, resource priorities, and strategic action plans. Tools include support for, but not limited to data gathering, metrics reporting, requirements forecasting, Logistics Test and Evaluation (LT&E) data management, resource availability, and Major Graded Area focus areas IAW DAFI 99-103, AFI 1-2, *Commander's Responsibilities*, and DAFI 90-302, *The Inspection System of the Department of the Air Force*.

5.4.1.3.2. **(Added-AFMC)** Serve as the MXG commander's focal point for AF CPI to provide visible leadership and promote AF CPI culture of innovation and teamwork. Trained and or certified Green Belt (GB) and Black Belt (BB) practitioners will provide employment of AF CPI methodologies to fulfill AFI 1-2 and DAFI 90-302 requirements to correct unit deficiencies using Root Cause Analysis (RCA) IAW AFI 38 Series.

5.4.1.3.2.1. **(Added-AFMC)** GB and BB practitioner will use AF CPI policy guidance, related efficiency methodologies, processes, procedures, tools, and training curriculums to implement and manage CPI throughout the MXG and its sub-organizations.

5.4.1.3.2.2. **(Added-AFMC)** Establish AF CPI program governance via the MXG Corporate Board and implement strategic alignment across the MXG. Develop mission, vision, goals, objectives, and action items and track progression toward meeting goals.

5.4.2. Act as the resource advisor to the MXG/CC. **(T-3)**

5.4.2. **(AFMC)** [DEV] Within the 412 TW, 412 MXG/CC is not required to appoint a Programs and Resource Advisor. This responsibility will be managed by personnel assigned to the 412 TW.

5.4.3. Coordinate with the MXG/Squadron SUPTs to manage manpower authorizations for the MXG. **(T-3)**

5.4.3. **(AFMC)** 412 MXG, Manpower (412 MXG/MXOPM).

5.4.3.1. **(Added-AFMC)** Act as the MXG and sub organization Commander's and supervisor's manpower adviser for personnel staffing actions, strategic human capital planning, workforce management programs, policies, accountability, reporting requirements, and issue resolution.

5.4.3.2. **(Added-AFMC)** Assist commanders and/or supervisors in management of their civilian and military manpower authorizations to support T&E manpower requirements identified in DAFI 99-103, Program SOCs, and DoDD 3200.11 IAW AFI 38-101, and DAFPD 36-1, *Appropriated Funds Civilian Management and Administration.*

5.4.3.3. **(Added-AFMC)** Manage position control numbers and request for personal actions (RPA) staffing actions from "cradle to grave" in support of T&E. Validate requester data and request required information when documentation is incomplete. Coordinate with wing, Center and AFPC agencies as necessary to ensure effective actions. Manage RPA fill action hiring certifications issues and close out actions. Provide periodic status updates.

5.4.3.4. **(Added-AFMC)** Develop COAs to resolve complex management issues and applies critical thinking to resolve organizational staffing challenges.

5.4.3.5. **(Added-AFMC)** Provide reporting metrics, periodic reports, and tools as directed by MXG leadership to assist in the determination of organizational priorities in the areas of training, staffing, recruitment, and workforce shortfalls and/or surpluses to enable data-driven decisions.

5.4.3.6. **(Added-AFMC)** Develop and manage metrics and reports to higher echelons as requested. Metrics include, but are not limited to; data calls, taskers, and/or guidance from higher headquarters, manpower and occupational demographics, staffing levels, activities necessary to manage POM plus up or reduction actions, reasonable accommodations, medical retirements, training, classifications, capacity, and capability etc. Chair periodic meetings as required. Meetings will include leadership and other personnel necessary to review metrics, address and correct staffing concerns and establish staffing priorities.

5.4.3.7. **(Added-AFMC)** Ensure authorized and assigned personnel are adequate to support the unit mission and tasking plans. Coordinate with wing or Air Logistics Complex manpower office representatives for assistance in preparing requests to MAJCOM for UMD adjustments.

5.4.3.8. **(Added-AFMC)** Maintain current Unit Personnel Manpower Roster. Work with MXG Chief, Enlisted Advisor, WAM and unit superintendents to allocate projected gains appropriate billets. Maintain record of approved actions and verify that personnel data subsystems are updated.

5.4.3.9. **(Added-AFMC)** Distribute projected gain/loss lists and maintenance manpower requests to appropriate work centers and establish suspense for updates. Ensure approved actions have been received for update and submit consolidated updates to the appropriate program or system.

5.4.3.10. **(Added-AFMC)** Develop inputs for mass realignment request. Provide research, and validation of UMD. Request Table 30 updates when new Office Symbol Code and/or Personnel Accounting Symbol code is received. Build and submit spreadsheet for mass realignment to AFPC.

5.4.3.11. **(Added-AFMC)** Focal point for T&E reimbursable workforce requirements. Review SOCs for MAJCOM, Special Program Office and/or higher echelon directed workload. Forwards inputs to TW Manpower, Financial Management, and Civilian Employment Cost Management Committee for TW review, validation, coordination, and approval. Monitor overhires and notify units of pending expiration and coordinates revalidation of SOC requirements and funding availability as required. Advises management of available options and coordinates with AFPC to ensure RPA action submitted to extend and/or convert appointment.

5.4.4. Serve as the focal point within the MXG for management of facilities. **(T-3)**

5.4.4.1. Ensure fire detection and foam suppression training is included in unit facility manager/ occupant training briefings. **(T-3)**

5.4.4.2. **(Added-AFMC)** 412 MXG, Logistics Site Support (412 MXG/MXOL).

5.4.4.2.1. **(Added-AFMC)** Serve as MXG/CC facility advisor for management of T&E facility support to include renovations and major facility modifications IAW AFI 32-1001, *Civil Engineer Operations.*

5.4.4.2.1.1. **(Added-AFMC)** Assign Site Support Specialist as necessary to support these functions.

5.4.4.2.1.1.1. **(Added-AFMC)** Manage planned addition, revision, or alteration to any facilities specialized utility system - such systems are facility access control security systems, complex mission critical computer room cooling systems Non-Real Property Installed Equipment (NRPIE), specialized aircraft electrical systems NRPIE, specialized aircraft specific cooling and activation systems i.e. polyalphaolefin, avionics, and hydraulics unique NRPIE test support systems.

5.4.4.2.1.1.2. **(Added-AFMC)** Develop and coordinate commercial contracts IAW AFI 64-102, *Installation Contracting*.

5.4.4.2.1.1.3. **(Added-AFMC)** Focal point for functional, technical, and Contract Officer Representative matters pertaining to performance-based activities IAW DoDI 5000.72, *DoD Standard for Contracting Officer's Representative (COR) Certification*.

5.4.4.2.1.1.4. **(Added-AFMC)** Prepare project statements of work, manage contract budget, and service contracts - such as custodial, trash removal, shop towel, coverall, and industrial gases., cable TV, NRPIE camera systems, elevator services, and NRPIE maintenance and any facility-related awarded

unfunded requirements IAW AFI 64-102.

5.4.4.2.1.1.5. **(Added-AFMC)** Develop and manage program specific Letters of Technical Direction (LOTD) in support of mission critical NRPIE such as access control, computer room air conditioning, and aircraft unique hangar aircraft support equipment.

5.4.4.2.1.1.6. **(Added-AFMC)** Manage preparation of real-time program requirement Rough-Order-Magnitude (ROM) cost estimates and Initial Government Estimates for modification or renovation of logistics and T&E facilities for support of current and future test requirements.

5.4.4.2.1.1.7. **(Added-AFMC)** Responsible for the installation, inspection, and maintenance of aircraft Sun Shades IAW **Chapter 15**, Aircraft Sun Shade Sustainment.

5.4.4.2.1.1.8. **(Added-AFMC)** Perform building inspection, maintenance, and repair on real property, and NRPIE , fire, safety, security, etc.).

5.4.4.2.1.1.9. **(Added-AFMC)** Develop and prioritize maintenance facility work orders.

5.4.4.2.1.1.10. **(Added-AFMC)** Ensure identified deficiencies and corrective actions are completed. Activities include but are not limited to repair, modification, and facility maintenance.

5.4.4.2.1.1.11. **(Added-AFMC)** Develop MXG Emergency Management and Continuity of Operations Plan and coordinate training and exercise requirements across the MXG.

5.4.4.2.1.1.12. **(Added-AFMC)** Perform facility drills to include fire drills and man down exercises.

5.4.4.3. **(Added-AFMC)** Act as the facility resource planning advisor to the MXG/CC IAW AFI 32-1015, *Integrated Installation Planning*.

5.4.4.3.1. **(Added-AFMC)** Develop and manage 10-year facility plan specifying maintenance, upgrade, and replacement projections for T&E logistics facilities, and coordinate plan with Civil Engineer (CE).

5.4.4.3.2. **(Added-AFMC)** Develop TW long range requirements for base site plan.

5.4.4.3.3. **(Added-AFMC)** Provide plan updates; review, and coordinate with CE annually.

5.4.4.3.4. **(Added-AFMC)** Develop and conduct comprehensive plans and studies to prevent and correct problems and ensure TW capacity for current and future test programs.

5.4.4.3.5. **(Added-AFMC)** Manage preparation of long term and potential program requirement ROM cost estimates and IGE for modification or renovation of logistics and T&E facilities for support of current and future test requirements IAW UFC 3-740-05.

5.4.4.3.6. **(Added-AFMC)** Develop and manage space planning and allocation database for test program support.

5.4.4.3.7. **(Added-AFMC)** Focal point for Military Construction Projects to support TW activities.

5.4.4.3.8. **(Added-AFMC)** Process T&E program resource allocation requests through TW functional areas.

5.4.4.3.9. **(Added-AFMC)** Provide management input to the facility board working group to ensure proper prioritization of logistics resource allocation.

5.4.4.3.10. **(Added-AFMC)** Develop and manage TW logistics facility integrated project list with CE.

5.4.4.3.11. **(Added-AFMC)** Manage MXG energy conservation plan.

5.4.4.4. **(Added-AFMC)** Partner with appropriate agencies to procure services and resources for maintenance, repairs, and modifications of facilities supporting logistics and T&E programs.

5.4.4.4.1. **(Added-AFMC)** Schedule and chair facility construction and upgrade project meetings for T&E programs.

5.4.4.4.2. **(Added-AFMC)** Develop and manage facility design and prepare cost estimates for T&E programs.

5.4.4.4.3. **(Added-AFMC)** Maintain the TW T&E programs space activity database.

5.4.4.4.4. **(Added-AFMC)** Manage project meetings for new construction and modification of existing facilities per program requirements.

5.4.4.4.5. **(Added-AFMC)** Plan, program, budget, and monitor projects IAW the Planning Programming Budgeting and Execution process for MAJCOM T&E resources.

5.4.4.4.6. **(Added-AFMC)** Plan T&E resource investment and modernization efforts for test program support.

5.4.4.5. **(Added-AFMC)** Assist with coordination of site surveys for deployment locations and maintain copies of the Expeditionary Site Plan Part I for deployment locations IAW AFI 10-404, Base Support and Expeditionary Site Planning.

5.4.4.5.1. **(Added-AFMC)** Focal point for MXG and Operations Group (OG) Combined Test Force (CTF) site planning.

5.4.4.5.2. **(Added-AFMC)** Provide Test Program Support IAW DAFI 99-103 and DAFI 99-103_AFMCSUP, *Capabilities-Based Test and Evaluation*.

5.4.4.5.3. **(Added-AFMC)** Review program introduction and test concept documents for test program needs and provide program support.

5.4.4.5.4. **(Added-AFMC)** Prepare Cost and Resource Estimates. Assess MXG capacity and capability to support including identification of constraints and concerns.

5.4.4.5.5. **(Added-AFMC)** Focal point for MXG final SOC inputs for test program support.

5.4.4.5.6. **(Added-AFMC)** Manage infrastructure for T&E facilities and resources for system life cycle activities.

5.4.5. Serve as the Environmental Coordinator focal point within the MXG. **(T-3)**

5.4.6. Serve as the focal point for MXG deployment planning and execution actions. **(T-3)** If designated as a UTC pilot unit IAW DAFI 10-401, *Operations Planning and Execution*, P&R will:

5.4.6.1. Coordinate with other UTC tasked units on cargo and equipment authorizations/requirements to develop and maintain a standardized package to meet specific mission capability requirements. **(T-3)**

5.4.6.2. Coordinate with the unit equipment custodian(s) to review equipment changes and new equipment requirements driven by changes to UTCs and/or Allowance Standards (AS). **(T-3)**

5.4.6.3. Assist with coordination of site surveys for deployment locations and maintain copies of the Expeditionary Site Plan (ESP) **Part I** for deployment locations IAW AFI 10-404, *Base Support and Expeditionary (BAS&E) Site Planning*.

5.4.6.4. Coordinate with QA biennially to verify aircraft MDS Hot Pit refueling capabilities are current and accurate in Base Support and Expeditionary and ensure applicable ESP/BSP **Parts I** and **II** accurately reflect unit capabilities IAW AFI 10-404, if applicable. **(T-3)**

5.4.7. Oversee local, functional or host country unique support agreements applicable to the MXG IAW AFI 25-201, *Intra-Service, Intra-Agency, and Inter-Agency Support Agreements Procedures*.

5.4.8. Develop and coordinate MXG commercial contracts as directed by the MXG/CC. **(T-3)**

5.4.9. Manage readiness reporting for the MXG IAW AFI 10-201.

5.4.10. Coordinate with LRS Deployment & Distribution Flight to obtain unit assistance in interpreting guidance for marking, packing, and marshaling of tasked equipment IAW DAFMAN 24-604, *Preparing Hazardous Materials for Military Air Shipments*, DAFI 10-401 and DESR6055.09_AFMAN 91-201.

Chapter 6

QUALITY ASSURANCE (QA)

6.1. General. Maintenance quality and equipment reliability is the responsibility of all maintenance personnel. The combined efforts of QA personnel, maintenance leaders, and technicians are necessary to ensure high quality maintenance production and equipment reliability. The QA staff evaluates the quality of maintenance accomplished and performs necessary functions to manage the MSEP. Personnel assigned to QA are not an extension of the work force and will not be tasked to perform sortie production inspections unless authorized by the MXG/CC or MGFE/CC on a case-by-case basis. **(T-2)** QA serves as the primary technical advisory agency in the maintenance organization, assisting maintenance supervision at all levels to identify, validate and/or resolve quality, proficiency and/or compliance issues impacting mission generation. The evaluation and analysis of deficiencies and problem areas identified are key functions of QA that highlight and reveal underlying causes of poor quality in the maintenance production effort. Aircraft and equipment condition and personnel proficiency are validated through the MSEP and recorded in the Logistics Evaluation Assurance Program (LEAP) QA database. Civil service and contracted personnel are to follow requirements established in their respective civilian position description/contract and accepted quality assessment system.

6.2. Responsibilities. QA is responsible to the MXG/CC or equivalent to perform as the primary technical advisory agency for maintenance actions and to assist work center supervisors in reviewing tasks involved in supporting the maintenance effort. MXG QA Inspectors have the authority to observe, correct and document applicable maintenance activities performed within the MXG. QA will:

6.2.1. Implement and administer the MSEP and other programs as applicable to include:

6.2.1.1. Product Improvement Program (PIP). **(T-2)**

6.2.1.1.1. DR. **(T-2)**

6.2.1.1.2. Product improvement inputs. **(T-2)**

6.2.1.2. Aircraft and equipment impoundment procedures IAW **Chapter 7**.

6.2.1.3. Functional Check Flight (FCF) program IAW this **Chapter**. **(T-2)**

6.2.1.4. W&B Program IAW this **Chapter**.

6.2.1.5. Hot Refuel/Defuel and Aircraft-to-Aircraft Refuel Programs if applicable. **(T-2)**

6.2.2. Review and analyze aircraft aborts, IFEs, and incidents involving damage to equipment or injury of personnel to determine if trend analysis, cross-tell or MSEP focus is warranted. **(T-2)**

6.2.3. Comply with the configuration management program requirements IAW **Chapter 14**. **(T-2)**

6.2.4. In coordination with PS&D, comply with TCTO Program requirements IAW **Chapter 14**, TO 00-5-1 and TO 00-5-15.

6.2.5. In conjunction with PS&D, develop a local Job Standard (JST) for both gaining and losing aircraft and equipment transfer inspection IAW **Chapter 14**. **Note:** For the purpose of

this instruction JST is an alpha code identifying the type of job represented in a job standard (defined in IMDS User's Manual AFSM 21-566, Volume 2).

6.2.6. Coordinate with PS&D on all AFTO Form 103s. **(T-2)**

6.2.7. Manage OTIs. **(T-2)**

6.2.8. Augment evaluations at the request of the WS. **(T-2)** Flightline weapons loading inspections/evaluations are the responsibility of WS evaluators.

6.2.9. Evaluate unit maintenance management procedures, including locally developed forms, publications, OIs, checklists, for accuracy, intent, and necessity as referenced in this DAFI.

6.2.10. Evaluate cyber hygiene, discipline practices, and cyber incident reporting per applicable TO, Wing directive, and this instruction for compliance. **(T-2)**

6.2.10.1. Report weapons system and support equipment cyber related deficiencies to the applicable Program Manager IAW T.O. 00-35D-54.

6.2.10.1.1. Elevate resolution of deficient cyber-requirements or issue via an AF Form 1067, Modification Proposal IAW AFI 63-101/20-101, and/or maintenance assistance request (such as, Technical Assistance Request) IAW TO 00-25-107 *Maintenance Assistance*.

6.2.11. **(Added-AFMC)** Publish and maintain a list of approved profile JSTs. **(T-2)**

6.3. QA Superintendent (QA SUPT) Responsibilities. In addition to the applicable Flight CC/SUPT duties in **Chapter 2** the QA SUPT will:

6.3.1. Develop and maintain a master training plan to train all QA Inspectors, and include augmentees, if applicable. **(T-2) Note:** See **Paragraph 5.2.1.13** for minimum MAJCOM training requirements for inspectors inspecting MXO function.

6.3.2. Develop and monitor the MSEP using the LEAP QA database and provide supervisors access to MSEP data. **(T-2)**

6.3.3. Notify the appropriate agencies when deficiencies are found in (AF, MAJCOM/Lead Command, WG, Group (GP)) instructions. **(T-2)**

6.3.4. Review maintenance-related instructions, supplements, operating instructions, forms, and local/functional and emergency action checklists every two years or when source data changes for accuracy, intent, and necessity. **(T-2)**

6.3.4. **(AFMC)** Units will develop a local tracker to ensure publications are reviewed for currency and accuracy every two years per HAF direction. Units are also required to maintain currency/review procedures for publications outlined in DAFMAN 90-161. **(T-2)**

6.3.4.1. The QA SUPT will document the review once complete. **(T-2)**

6.3.5. Review JSTs annually or when source data changes for accuracy, intent, and necessity. **(T-2)**

6.3.5.1. The QA SUPT will document the review once complete. **(T-2)**

6.3.6. Ensure management and special inspections are performed (when required). **(T-2)**

6.3.7. Ensure the GP portion of the FOD Prevention Program is conducted IAW **Chapter 11**. **(T-2)**

6.3.8. Oversee and implement the GP Impoundment Procedures IAW **Chapter 7**. **(T-2)**

6.3.9. Coordinate on all requests for locally manufactured, developed, and modified tools and equipment, and maintain records for approved requests. **(T-2)**

6.3.9.1. This includes pictures or drawings and a description of the use for each item.

6.3.9.1.1. If applicable technical data contains the option for use of these tools and equipment, QA does not need to coordinate or maintain the records on these items.

6.3.9.1.2. Locally manufactured, developed, or modified equipment for weapons loading, maintenance and the armament systems flight must be coordinated through the WWM before routing to QA. **(T-2)**

6.3.10. Verify IPI requirements from MAJCOM and sources outlined in TO 00-20-1 and publish combined MXG IPI listing every 2 years as a minimum or when source data changes. **(T-2)**

6.3.11. Develop KTL/RIL to supplement MAJCOM listings (if required). **(T-2)**

6.3.11.1. Provide copies of approved KTL/RIL to all affected organizations. **(T-2)**

6.3.12. Ensure Acceptable Quality Level (AQL) Standards are developed for all tasks including key tasks and routine inspections not included on the MAJCOM AQL. **(T-2)**

6.3.13. Ensure agendas and presentations are compiled for the MSEP summary. **(T-2)**

6.3.14. Review wing requests for assistance IAW **Chapter 1** and **Chapter 14**.

6.3.15. Designate a Chief Inspector. **(T-2)**

6.3.16. Designate individuals to be the Technical Order Distribution Office (TODO) and Product Improvement Manager (PIM). **(T-2)**

6.3.17. Designate individuals to be the W&B and FCF Program managers. **(T-2)**

6.3.18. Monitor the ASIP IAW **Chapter 11**.

6.3.19. Maintain DOP program oversight IAW **Chapter 11**.

6.3.20. When hot refueling is performed by maintenance personnel, ensure Hot Refueling Program is accomplished IAW TO 00-25-172 and this DAFI.

6.3.21. Ensure non-resident organizations hot refueling aircraft at an AF certified hot pit coordinate site/personnel certification, utilization and documentation requirements with the supporting QA and Airfield Management/Operations. **(T-2)**

6.3.21.1. Using units are responsible for maintaining currency/reporting requirements. **(T-2)**

6.3.22. Ensure designated Responders/facilitators respond to DoD IT incidents that lack the capability to perform a mission function. **(T-2)**

6.3.22.1. Consolidate and report incidents in conjunction with appropriate POC (e.g., WAM, TODO, or designated official). **(T-2)** Example: ETIMS containing software for a TO task that does not have a hardware TO approving its installation or use.

6.3.23. **(Added-AFMC)** As the selecting/hiring official for QA inspectors, ensure the individual is a qualified SME in the functional area (i.e., avionics, Aerospace Ground Equipment (AGE), aircraft maintenance, fabrication, munitions, etc.) and meets position requirements. **(T-2)**

6.4. Chief Inspector Responsibilities. The Chief Inspector is responsible to the QA SUPT for ensuring functions listed below are performed and is responsible for applicable Section NCOIC/Chief duties in **Chapter 2**. The Chief Inspector will:

6.4.1. Use assigned inspectors/augmentees to provide on-the-spot assistance to correct problems. **(T-2)**

6.4.2. Spot-check TOs, inspection work cards, checklists, job guides and WUC manuals during evaluations and inspections for currency and serviceability. **(T-2)**

6.4.3. Assist MMA with investigations and studies. **(T-2)**

6.4.4. Review QA LEAP MSEP inspection summary inputs for accuracy and content. **(T-2)**

6.4.5. Initiate actions when additional attention is required to resolve adverse maintenance trends or training problems. **(T-2)**

6.4.5.1. Actions include preparing cross-tell information bulletins and messages for MXG/CC release to other similarly equipped units and higher headquarters.

6.4.6. Review and compile inputs for updating the IPI listing. **(T-2)**

6.4.6.1. Maintain a copy of the MXG/CC or equivalent approved IPI listing with the signature and date of review and certification. **(T-2)**

6.4.7. Review Category II major discrepancies for trends quarterly. **(T-2)**

6.4.7.1. If frequency or severity of identified discrepancies warrant inclusion of that item into a specific TO governing an action or inspection, the QA Chief Inspector must submit an RC or develop a local work card, local page supplement or checklist IAW TO 00-5-1.

6.4.8. Establish procedures for QA Inspectors to document completed inspections. **(T-2)**

6.4.9. Perform inspections on GITA IAW **Chapter 11**. **(T-2)**

6.4.10. Construct and maintain a master standardized AFTO Form 781-series forms binder IAW TO 00-20-1.

6.4.11. Develop an aircrew briefing checklist specifically for high-speed taxi checks (see **Paragraph 6.14**). **(T-2)**

6.4.12. Review MSEP data monthly to identify high-missed carded items from Personnel Evaluations and Quality Verification Inspections (QVI). **(T-2)** A high-missed carded item is defined as any work card item missed at least three times during a one-month period.

6.4.12.1. Coordinate with MMA to identify any relationships with repeat, recur and CND trends. **(T-2)**

6.4.12.2. Include this data in the monthly MSEP summary. **(T-2)**

6.4.13. Conduct Evaluator Proficiency Evaluations on each inspector. **(T-2)**

6.4.13.1. Evaluators Proficiency Evaluation (EPEs) will be conducted while the Chief Inspector assesses one Personnel Evaluation (PE) and one technical inspection (QVI/Special Inspection (SI)). **(T-2)**

6.4.13.2. Each QA Inspector, permanent or augmentee, must pass both EPEs prior to performing unsupervised evaluations and inspections. **(T-2)**

6.4.13.2. **(AFMC)** QA augmentees require an annual Evaluators Proficiency Evaluation on either a PE or technical inspection. **(T-2)**

6.4.14. Document QA Inspector training in the MyTraining. **(T-2)**

6.4.15. Ensure each discrepancy/finding within rated inspections, evaluation, and observation reports have been assigned both a Category (CAT I or CAT II) and a Finding Severity (Maj or Min).

6.5. Quality Assurance Inspector Responsibilities. QA Inspectors will:

6.5.1. Evaluate flightline and back shop maintenance tasks/inspections and MXO functions to include items identified by the KTL/RIL. **(T-2)**

6.5.2. Enter inspection and evaluation reports into the LEAP QA database. **(T-2)**

6.5.3. Perform QA review of Dull Swords, TCTOs, OTIs, modification proposals, DRs, RCs, instructions, and supplements. **(T-2)**

6.5.4. Provide training/instruction as applicable to address deficiencies identified during evaluations/inspections. **(T-2)**

6.5.5. Evaluate forms and MIS documentation to evaluate compliance IAW MXG written procedures described in **Chapter 2**. **(T-2)**

6.5.6. Evaluate maintenance TO files that are kept on the aircraft (G files). **(T-2)**

6.6. Quality Assurance Inspector Training. As a minimum, the local QA Inspector Training Plan will include the applicable items listed below to ensure QA program standardization. **(T-2)**

6.6.1. Training must cover inspection and evaluation techniques, documenting inspection worksheets and actions to prevent injury to personnel or property/equipment damage. **(T-2)**

6.6.1.1. **(Added-AFMC)** Training must cover the evaluation and inspections of functional area(s) (e.g., Avionics, AGE, PMEL, Munitions, etc.) or programs (e.g., weight and balance, Functional Check Flight (FCF)) and be documented in the individuals applicable training records. **(T-2)**

6.6.2. All EPEs must be tracked in the MIS and/or the LEAP QA database. **(T-2) Note:** Additional requirements for Nuclear Weapons Certifying Officials are located in AFI 21-204.

6.6.3. QA Inspectors inspecting outside of their AFSC will be task qualified on a Work Center Job Qualification Standard (WJQS) in MyTraining for the KTL requirements they evaluate. **(T-2)**

6.6.3.1. Chief Inspectors will identify other critical tasks requiring DAF Form 797, *Job Qualification Standard Continuation/Command JQS*, qualification (QA WJQS) within MyTraining as required. **(T-2)**

6.6.3.2. For all other tasks, QA Inspectors must be familiar with the requirements/procedures of tasks they evaluate. **(T-2)**

6.6.3.2.1. Cross-utilization for QA Inspectors is not allowed for 2W1, 2W0, 2A7X2, or 2P0X1 career field specific tasks. Inspectors will hold the respective AFSC, have accomplished any required ancillary/safety training, and evaluate IAW their respective AFIs/TOs. **(T-2)**

6.6.3.2.1.1. QA inspectors evaluating Scheduling and Analysis functions must be trained and qualified. **(T-2) Note:** See **Paragraph 5.2.1.13** for additional requirements.

6.6.4. & 6.6.5. DELETED.

6.6.6. MAJCOMs will determine if QA personnel who conduct engine run evaluations are required to maintain the engine run proficiency requirements outlined in **Chapter 11** and document requirements in their supplement to this DAFI. **(T-2)**

6.6.6. **(AFMC)** QA personnel who conduct engine run evaluations are not required to maintain the engine run proficiency requirements outlined in **Chapter 11**. **(T-2)**

6.6.7. - 6.6.10.2. DELETED.

6.7. Maintenance Standardization and Evaluation Program (MSEP). The purpose of the MSEP is to provide units with a method of evaluating technical compliance and measure how well they comply with established standards.

6.7.1. Units will develop a MSEP and conduct local inspections to ensure their programs, processes, maintenance technician proficiency, equipment condition and other focus areas are in compliance with AF, MAJCOM, and local directives. **(T-2)** The unit level MSEP is not applicable to contract maintenance activities unless required by the contract SOW or PWS.

6.7.1.1. The MSEP will be developed in conjunction with inputs from assigned squadron Maintenance Supervision and Group Leadership and will be executed by QA. **(T-2)**

6.7.1.2. DELETED.

6.7.1.3. The following types of evaluations, inspections and observations support the MSEP: PEs, QVIs, SIs, Management Inspection (MI)s, Detected Safety Violation (DSV)s, Technical Data Violation (TDV)s, Unsatisfactory Condition Report (UCR)s, and when directed, other inspections.

6.7.1.3.1. These inspection terms may differ in the LEAP QA database however, MAJCOMs must provide non-standard terms and definitions in their supplement to this DAFI when used.

6.7.1.3.2. Develop the monthly MSEP on the minimum personnel evaluations, inspections or observation requirements as depicted in Air Force Manpower Standard 21QX Quality Assurance Application Tool, located on the Air Force Manpower Analysis Agency SharePoint site:

https://cs2.eis.af.mil/sites/11190/AFMDandAFMS/AFMS/Forms/FAC1stFAC2nd.aspx.

6.7.2. Unit MSEP Focus Areas. QA will assess how units are meeting compliance goals and will identify areas of opportunity for improvement. **(T-2)** A unit's MSEP will focus on:

6.7.2.1. Compliance with and currency of TOs and directives. **(T-2)**

6.7.2.1.1. Ensure personnel at all levels are responsible and accountable for enforcing mandatory standards and ensuring all applicable TOs and directives are complete, current, and used.

6.7.2.2. Aircraft, systems, and equipment forms documentation. **(T-2)**

6.7.2.2.1. Ensure forms used to document any maintenance related action for aircraft, systems or equipment are documented IAW 00-20 series TOs, specific equipment TO requirements, and other applicable directives and supplements. **(T-2)**

6.7.2.2.2. MSEP will validate compliance with the MXG's or equivalents' written procedures to ensure aircraft/system forms, equipment forms and MIS documentation are complete, accurate, and accomplished for each shift as referenced in **Paragraph 2.4.53**. **(T-2)**

6.7.2.2.2. **(AFMC)** Referenced in **paragraph 2.4.2.20**.

6.7.2.3. Aircraft, Systems and Equipment Inspections. **(T-2)**

6.7.2.3.1. Ensure aircraft and equipment, including munitions, are inspected IAW TOs and directives. **(T-2)**

6.7.2.4. Compliance and Management of Occupational Safety and Health, Environmental, Bioenvironmental, Housekeeping, and FOD Programs. **(T-2)**

6.7.2.4.1. Personnel at all levels are responsible for minimizing risk to equipment, personnel, and the environment. **(T-2)**

6.7.2.5. Training. **(T-2)**

6.7.2.5.1. Verify training is correctly documented and ensure individuals are qualified/certified to perform evaluated tasks. **(T-2)**

6.7.2.6. Unit-Directed Programs. **(T-2)**

6.7.2.6.1. Verify units' programs are in compliance with local directives. **(T-2)**

6.7.2.7. Key Task List (KTL). The KTL is an AF, MAJCOM or unit developed list of required inspections that cover tasks that are complex and tasks affecting safety of flight.

6.7.2.7.1. MAJCOMs will identify minimum KTLs for each MDS. **(T-2)**

6.7.2.7.1. **(AFMC)** Units will use Lead Command KTLs for the specific MDS assigned. **(T-2)** The Center/Wing/Group may add additional tasks to the KTL. Lead Command KTLs can be found at the following link: **https://usaf.dps.mil/teams/40368/Comand%20Quality%20Assurance/Forms/AllItems.aspx?id=%2Fteams%2F40368%2FComand%20Quality%20Assurance%2FTest%20Wing%20%26%20Non%2DStd%20Orgs%2FTest%20Wing%20QA**

%2FMAJCOM%20RIL%5FKTL%5FAQL%20Files&viewid=96c81608%2D2a90%2D4e7

6.7.2.7.2. All maintenance actions/functions listed on the KTL require mandatory call-in to QA each time the maintenance action/function is accomplished. **(T-2)**

6.7.2.7.2.1. QA evaluators will respond and perform an evaluation. **(T-2)**

6.7.2.7.2.1.1. The MXG/CC may waive the inspection or designate authorized a representative(s) to waive the inspection.

6.7.2.7.2.1.2. QA will track all KTLs called in, waived, or completed and maintain a list of MXG-designated KTL waiver authorities. **(T-2)**

6.7.2.7.3. QA will review and update the KTL list at least every 2 years to ensure it encompasses those maintenance actions/functions directly affecting quality of maintenance. **(T-2)**

6.7.2.8. Routine Inspection List (RIL). The RIL is an AF, MAJCOM, or unit developed list of routine inspections that must be performed.

6.7.2.8.1. QA will consolidate Maintenance Supervision inputs and suggested changes for the RIL. **(T-2)**

6.7.2.8.2. Additional RIL requirements for nuclear capable units are located in AFMAN 21-200.

6.7.2.8.3. Tasks will not be removed from the RIL without issuing authorities' approval (such as, AF, MAJCOM, MXG/CC). **(T-2)**

6.7.2.8.4. The RIL must contain the following if applicable to the unit:

6.7.2.8.4.1. Pre-flight. **(T-2)**

6.7.2.8.4.2. Thru-flight. **(T-2)**

6.7.2.8.4.3. Basic post-flight. **(T-2)**

6.7.2.8.4.4. HSC/HPO inspections. **(T-2)**

6.7.2.8.4.5. Aircraft forms/MIS documentation. **(T-2)**

6.7.2.8.4.6. Equipment forms/MIS documentation. **(T-2)**

6.7.2.8.4.7. Aircraft and munitions flightline accountability/accountable property system of record (APSR). **(T-2)**

6.7.2.8.4.8. Aircraft ground handling. **(T-2)**

6.7.2.8.4.9. Launch and recovery. **(T-2)**

6.7.2.8.4.10. Servicing tasks. **(T-2)**

6.7.2.8.4.11. Technical data. **(T-2)**

6.7.2.8.4.12. CTK Program. **(T-2)**

6.7.2.8.4.13. TMDE calibrations when the performing work center is not a PMEL IAW TO 00-20-14. **(T-2)**

6.7.2.8.4.14. AGE maintenance. **(T-2)**

6.7.2.8.4.15. AGE flightline use. **(T-2)**

6.7.2.8.4.16. Housekeeping. **(T-2)**

6.7.2.8.4.17. Vehicles. **(T-2)**

6.7.2.8.4.18. Aircraft washes/aircraft corrosion inspections. **(T-2)**

6.7.2.8.4.19. Supply discipline (example, TNB, DIFM management & coding). **(T-2)**

6.7.2.8.4.20. Equipment washes/ equipment corrosion inspections. **(T-2)**

6.7.2.8.4.21. Environmental compliance. **(T-2)**

6.7.2.8.4.22. NWRM accountability and forms documentation. **(T-2)**

6.7.2.8.4.23. TCTO Program. **(T-2)**

6.7.2.8.4.24. Time-Change Program. **(T-2)**

6.7.2.8.4.25. FHP management. **(T-2)**

6.7.2.8.4.26. Maintenance Cyber Discipline. **(T-2)**

6.7.2.8.4.26.1. Upload and download of software and data. **(T-2)**

6.7.2.8.4.26.2. Malicious code detection and reporting. **(T-2)**

6.7.2.8.4.26.3. Verify personnel upload required CPINS updates prior to use. **(T-2)**

6.7.2.8.4.26.4. Verify personnel validate antivirus scans of external media are complete prior to use. **(T-2)**

6.7.2.8.4.26.5. Account for ATS/ATE in the appropriate APSR. **(T-2)**

6.7.2.8.4.26.6. Cyber hygiene. **(T-2)**

6.7.2.8.4.27. Repair Network Node Management. **(T-2)**

6.7.2.8.4.28. **(Added-AFMC)** Plans Scheduling & Documentation (PS&D).

6.7.2.8.4.29. **(Added-AFMC)** MAJCOM recommended QA routine inspections. Refer to the following link and review the current Lead Command Routine Inspection List (RIL) for additional focus recommendations to tailor unit Evaluation and Inspection plan as needed**:** HQ AFMC A4M Functions - Test Wing QA - All Documents (dps.mil)

6.7.2.9. QA will coordinate with the Munitions Activity (Munitions Supervision with in the MUNS) and will develop quarterly standards (such as, number of inspections and frequency) for the following areas:

6.7.2.9.1. Munitions accountability. **(T-2)**

6.7.2.9.2. Munitions storage practices, security, and safety. **(T-2)**

6.7.2.9.3. Munitions inspections. **(T-2)**

6.7.2.9.4. Munitions materiel handling and test equipment. **(T-2)**

6.7.2.9.5. Munitions stockpile management. **(T-2)**

6.7.2.9.6. Tactical missile reporting system. **(T-2)**

6.7.2.9.7. Munitions infrastructure (such as, adequacy of lightning protection and grounding systems, bonding of facility doors, adequate power conversion equipment). **(T-2)**

6.7.2.9.8. Munitions training programs. **(T-2)**

6.7.2.9.9. Maintenance Cyber Discipline. **(T-2)**

6.7.3. Unit MSEP Evaluation and Inspection (E&I) Plan. QA will develop an E&I Plan specifying numbers of approved areas and types (PE, QVI, SI, MI) of inspections and evaluations to complete during the month. Types of inspections must be separated and will not be combined (such as, PE/QVI). **(T-2)**

6.7.3.1. The E&I Plan, and changes to it, will be coordinated through each squadron Maintenance Supervision. **(T-2)**

6.7.3.2. The E&I Plan will be reviewed and updated monthly based on trends in the maintenance complex and will be adjusted to meet the MXG/CC's focus areas. **(T-2)**

6.7.3.3. When developing the E&I Plan, the QA SUPT will:

6.7.3.3.1. Address areas of concern identified by maintenance managers and the WWM. **(T-2)**

6.7.3.3.1.1. **(Added-AFMC)** Consider historical PEs, Quality Verification Inspections (QVI), and other inspections data e.g., items with pass rates below 80 percent; Not Mission Capable (NMC) causes; aborts and trends; In-Flight Emergencies (IFE) and trends; high component or system failure rates; repeat/recurring/CND discrepancies trends; Deficiency Report (DR)/ Service Report (SR) trends; suspected training deficiencies; Technical Data Violations (TDVs), Detected Safety Violations (DSVs), and tasks outlined in aircraft -6 TOs.

6.7.3.3.2. Tailor the plan for each squadron, flight, and section. **(T-2)**

6.7.3.3.2.1. **(Added-AFMC)** Identify the assessment type (i.e., task, specific item, procedure or process) and minimum number (determined by a documented methodology (e.g., ANSI- Z1.4 or data analytics tool) for Personnel Evaluations (PE), QVI, and RIL and assessment areas to be conducted monthly. **(T-2)**

6.7.3.3.3. Coordinate and distribute the E&I Plan. **(T-2)**

6.7.4. Evaluation Criteria.

6.7.4.1. Acceptable Quality Levels (AQL). AQLs denotes the maximum allowable number of minor findings that a process or product may be charged for the task to be rated "Pass" and are used to minimize subjectivity in assessing tasks identified by the MSEP.

6.7.4.1.1. MAJCOMs may develop standardized AQLs by weapon system and establish procedures to review at least annually.

6.7.4.1.2. MXG/CCs establish AQLs for local tasks/inspections.

6.7.4.1.2.1. AQLs need to be derived/revised from QA performance-based data.

6.7.4.1.3. AQLs/baselines for nuclear maintenance, cruise missile maintenance and nuclear weapons handling tasks are defined in AFMAN 21-200.

6.7.4.1.4. All findings will be rated Cat I or Cat II with a severity of Major or Minor. **(T-2)**

6.7.4.2. Discrepancy Categories.

6.7.4.2.1. Category I (CAT I). A required inspection/TO/AFI procedural item missed or improperly completed. This category is a specific AFI requirement, work card item or TO step, warning, caution, or note for a specific condition or action. Use findings of major or minor to indicate the discrepancy's relative severity.

6.7.4.2.2. Category II (CAT II). An obvious defect, which could have been readily detected by a technician or supervisor, but is not a specific AFI requirement, work card item or TO step, warning, caution, or note for that specific evaluated task. Use findings of major or minor to indicate the discrepancy's relative severity.

6.7.4.3. Findings.

6.7.4.3.1. A major finding is defined as a condition that may endanger personnel, jeopardize equipment or system reliability, impact safety of flight or warrant discontinuing the process or equipment operation.

6.7.4.3.2. Any major discrepancy will result in an automatic inspection failure. **(T-2)**

6.7.4.3.3. The QA Inspector will intercede and declare a major finding when one additional action will result in one of the following: endanger personnel, jeopardize equipment or system reliability, impact safety of flight or warrant discontinuing the process or equipment operation. **(T-2)**

6.7.4.3.3.1. The QA Inspector will write up the major finding even though the jeopardizing action was never taken due to their intercession. **(T-2)**

6.7.4.4. A minor finding is defined as an unsatisfactory condition that requires repair or correction, but does not endanger personnel, impact safety of flight, jeopardize equipment reliability or warrant discontinuing a process or equipment operation.

6.7.4.4.1. CAT II minors will be documented for trends but must not be counted against the AQL. **(T-2)**

6.7.4.4.2. Soft FO contained in tool kits or found in cargo areas of aircraft which pose no FOD threat are classified as a minor finding since it will require more than one additional action to meet the definition of a major finding.

6.7.5. Observations. This category represents observed events or conditions with safety implications or technical violations not related to an evaluation or inspection, are considered unsafe, in violation of established procedures, or in the case of equipment, unfit for operations. Observations include: DSVs, TDVs and UCRs. The LEAP QA database is used to document any of the following conditions:

6.7.5.1. DSV. An observed unsafe act by an individual.

6.7.5.1.1. The QA Inspector must stop the unsafe act immediately. **(T-2)**

6.7.5.1.2. The QA Inspector will not document a separate DSV on an individual undergoing a PE since the unsafe act automatically results in a "Fail" rating on the PE. **(T-2)**

6.7.5.1.2.1. The QA Inspector will use DSV verbiage in the PE summary when a safety violation is committed during a PE. **(T-2)**

6.7.5.2. TDV. An observation of any person performing maintenance without the proper technical data available, available but not in use or not following the correct sequence of steps (if directed).

6.7.5.2.1. The technician must have knowledge of all general directives associated with the job prior to performing the task. **(T-2)** However, those general directives need not be present at the job site.

6.7.5.2.2. Do not document a separate TDV on an individual undergoing a PE but use TDV verbiage in the PE summary since failure to use technical data automatically results in a "Fail" rating.

6.7.5.3. UCR. An unsafe or unsatisfactory condition, other than a DSV, chargeable to the work center supervisor.

6.7.5.3.1. UCRs will be documented even when it is not possible to determine who created the condition. **(T-2)**

6.7.6. Evaluations. An evaluation represents the direct evaluation of a logistics action, inspection, or training conducted/performed by an individual or team. Evaluations are used to evaluate job proficiency, degree of training, and compliance with technical data or instructions.

6.7.6.1. PE. A PE is an over-the-shoulder (direct) evaluation of a maintenance action, inspection, or internal MXG support process (such as, PS&D, Analysis, MOC). Individuals performing, supervising, or evaluating tasks are subject to a PE. PEs may be performed on individuals working alone or as part of a team.

6.7.6.1.1. Rate PEs “Pass or Fail” based on established AQLs/standards. **(T-2)**

6.7.6.1.2. Document the PE in the LEAP QA database. **(T-2)**

6.7.6.1.3. PEs will be accomplished on all technicians who perform maintenance. **(T-2)** MAJCOMs will “establish” the frequency. **(T-2)**

6.7.6.1.3.1. **(Added-AFMC)** Ensure personnel performing maintenance on aircraft/equipment receive a PE every 24 months. For failed PE, conduct a follow-up evaluation on the same task within 30 days (extensions may be given by exception). **(T-2)**

6.7.6.1.4. Personnel in any AFSC certified to perform nuclear maintenance or logistics operations (example, limited general maintenance, transfer, transport) will also comply with applicable PE requirements in AFMAN 21-200.

6.7.6.1.5. Types of PEs.

6.7.6.1.5.1. Individual Evaluations. This is a QA over-the-shoulder (direct)

evaluation of a technician or supervisor performing a job.

6.7.6.1.5.2. Team Evaluations. This is a QA over-the-shoulder (direct) evaluation of technicians and supervisors performing a team task.

6.7.6.1.5.2.1. A team task is one requiring more than one person to complete the task (for example, refueling, ECM pod up/downloading, bomb build-up, towing, weapons maintenance, pylon installation).

6.7.6.1.5.2.2. Team evaluations must accurately assess the proficiency of each individual under evaluation.

6.7.6.1.5.2.3. Refer to AFMAN 21-200 for nuclear weapons maintenance and handling evaluations.

6.7.6.1.6. QA Inspectors will conduct on each NDI technician, for each NDI method annually (every 2 years for the ARC) to ensure effective trending on NDI methods. **(T-2)**

6.7.6.2. Performing a PE. When performing a PE, the QA Inspector will brief the individual or team on the evaluation and how it will be rated. **(T-2)**

6.7.6.2.1. The QA inspector will determine what task will be evaluated. **(T-3)**

6.7.6.2.2. The PE will include an evaluation of the individual's training records, SCR (if task requires), tools, equipment, TMDE (use/impact of certification label limitation on maintenance being performed), and TOs used to perform the task. **(T-2)**

6.7.6.2.2.1. **(Added-AFMC)** Additionally, as part of the PE, sample MIS for proper aircraft/equipment documentation of the completed maintenance action.

6.7.6.2.3. The evaluation starts when the individual or team begins the task, or portion of the task to be evaluated, and is completed when the task or previously determined portion of the task is finished. **(T-2)**

6.7.6.2.4. Provide feedback to the individual or team and supervision upon completion. **(T-2)**

6.7.6.3. Rating PEs. QA Inspectors rate each evaluation based on AQLs/standards. The rating applies only to the specific task evaluated and not to other tasks that a technician or supervisor is qualified to perform. Upon completion of a failed evaluation, the QA Inspector must provide on-the-spot feedback. **(T-2)** Determine ratings as follows:

6.7.6.3.1. Pass: Number of discrepancies does not exceed AQL/standards.

6.7.6.3.2. Fail: An evaluation that results in any of the following:

6.7.6.3.2.1. Number of discrepancies exceeds the established AQL/standards.

6.7.6.3.2.2. A technician fails to detect a major discrepancy while complying with an inspection or TO requirement.

6.7.6.3.2.3. A technician fails to comply with a technical data step.

6.7.6.3.2.3.1. QA Inspectors will notify individuals immediately during the PE that a TDV was committed. **(T-2)**

6.7.6.3.2.3.2. Do not document a separate TDV on an individual undergoing a PE, since failure to use technical data automatically results in a "Fail" rating.

6.7.6.3.2.4. A technician demonstrates a lack of technical proficiency, system knowledge or demonstrated knowledge commensurate with skill grade.

6.7.6.3.2.5. Training and certification not documented.

6.7.6.3.2.6. A technician commits a safety violation.

6.7.6.3.2.6.1. Use the word "Safety" when a safety violation is committed during a PE.

6.7.6.3.2.6.2. Do not document a separate DSV on an individual undergoing a PE since the unsafe act automatically results in a "Fail" rating on the PE.

6.7.6.3.2.7. A technician fails to document maintenance actions in appropriate equipment records.

6.7.6.3.2.8. For nuclear weapons maintenance, an unsatisfactory rating must be given when any deficiencies or applicable unsatisfactory conditions exist IAW AFMAN 21-200. **(T-2)**

6.7.7. Inspections: An inspection represents inspections of equipment, programs, and processes to ensure compliance with established standards. Inspections are rated as "Pass" or "Fail". Inspections include:

6.7.7.1. QVIs. A QVI is an inspection of equipment condition, or a maintenance process, an assessment following a maintenance inspection, servicing or repair action, or verification that a technician or supervisor properly completed an inspection or repair action.

6.7.7.1.1. QVIs will not be conducted after equipment operation when such operation could invalidate indications of proper job accomplishment.

6.7.7.1.2. Limit QVIs to the same inspection card deck or technical data required for the job. This inspection does not require disassembling parts, removing stress panels or like actions.

6.7.7.1.3. A QVI required for -6 TO inspections may be accomplished by checking a portion of the required card or area.

6.7.7.1.4. The QVI report should reflect deficiencies by the individual who accomplished the task and identify specific discrepancies.

6.7.7.1.5. Document these discrepancies in active equipment records and forms, for example AFTO Form 781A, AFTO Form 244, *Industrial/Support Equipment Record*.

6.7.7.1.6. Rate QVIs "Pass" or "Fail" by comparing the number of discrepancies with the established AQLs/standards.

6.7.7.1.6.1. Pass: Number of discrepancies does not exceed established AQL/standard.

6.7.7.1.6.2. Fail: An inspection that results in any of the following:

6.7.7.1.6.2.1. A technician fails to detect a major discrepancy while complying

with an inspection or TO requirement.

6.7.7.1.6.2.2. Number of CAT I minor discrepancies exceeds the established AQL/standard.

6.7.7.1.6.2.3. A technician is not signed off in training records as task qualified.

6.7.7.1.6.3. Document the QVI in the LEAP QA database.

6.7.7.1.6.3.1. Each QVI is chargeable to the technician or supervisor who signed off/cleared the "corrected by" block or "inspected by" block of the applicable maintenance form or equipment record.

6.7.7.1.6.3.2. When evaluating the technician who signed off the "inspected by" block, evaluate only the items normally verified by signing off the "Red-X".

6.7.7.1.6.3.3. Only one evaluation is scored for each inspection.

6.7.7.2. Management Inspection (MI). Perform these inspections to follow-up on trends, conduct investigations or conduct research to get to the root cause of problems. MXG/CCs, SQ/CCs or work center supervisors may request MIs. MIs may encompass PE/QVI trends and other inspection data, NMC causes, aborts and trends, IFE trends, high component or system failure rates, suspected training deficiencies, and tasks outlined in aircraft -6 TOs. MI results are reported to the requester. MIs can be non-rated and may be counted in QA trends. The LEAP QA database will be used to document MIs. **(T-2)**

6.7.7.3. Special Inspections (SIs) are inspections not covered by QVIs, PEs or MIs. SIs may include, but are not limited to, aircraft and equipment forms inspections, document file inspections, CTKs, TO files, vehicle inspections, housekeeping, safety practices, FOD Program, SIs may be condition, procedural, or compliance oriented.

6.7.7.3.1. The LEAP QA database will be used to document special inspections. **(T-2)**

6.7.7.3.2. SIs will be rated as "Pass" or "Fail" based on established AQLs/standards.

6.7.8. Discrepancy Reporting. Report all discrepancies to the applicable work centers. **(T-2)**

6.7.8.1. QA will provide an authoritative, validated source reference for all reported discrepancies (such as, work cards, job guides, WUC manuals, checklists, occupational safety requirements, TOs, and other applicable references). **(T-2)**

6.7.9. Units will grade their MSEP evaluations monthly using objective ratings based on the following five-tier rating system:

6.7.9.1. Outstanding: 95-100% **(T-2)**

6.7.9.2. Excellent: 90-94.99% **(T-2)**

6.7.9.3. Satisfactory: 80-89.99% **(T-2)**

6.7.9.4. Marginal: 70-79.99% **(T-2)**

6.7.9.5. Unsatisfactory: 0-69.99% **(T-2)**

6.7.9.6. Inspections and evaluations performed (such as, PE, SI, QVI) are rated "Pass/Fail". Exception: Unless otherwise directed by AFMAN 21-200 and AFI 21-204 for Nuclear Weapons PEs and Certification Program.

6.7.9.7. Ratings are calculated by dividing the total number of inspections passed by total completed. For example, QA inspects 10 aircraft preflights with the following results: 9 "passes" and 1 "failure". Divide the total "passes" by the total inspections (9/10=0.90) 90 percent for an "Excellent" rating.

6.7.9.7.1. Deduct 0.5 percentage points from overall percentage grade for each TDV, DSV, and UCR monthly. For example, a squadron earns an overall rating of 90 percent, "Excellent", however, QA observed 4 TDVs and 3 DSVs. Multiply the sum (7) by 0.5 and subtract the product (3.5) from the original 90 percent. The adjusted total is 86.5 percent; therefore, the squadron is rated "Satisfactory".

6.7.10. A cumulative MXG score will be determined by dividing the Group's total number of inspections and evaluations passed by the total inspections and evaluations completed. **(T-2)**

6.7.10.1. Deduct 0.5 percentage points for each TDV, DSV, and UCR from the overall percentage grade, using same formula in **Paragraph 6.7.9.7.1**.

6.7.11. Monthly Summary. QA will publish and distribute the monthly summary to the MXG/CC or equivalent and inspected organizations. **(T-2)**

6.7.11.1. For security purposes, classified portions of the MSEP will be published separately from the main summary. **(T-2)**

6.7.11.2. QA will compile the summary from inspection data and attach the load crew evaluation statistics provided by WS. **(T-2)**

6.7.11.3. The MSEP summary should include visual information, graphs, narratives, quality trends identified through inspections and evaluations, discussion of common problem areas and descriptions of successful programs or initiatives.

6.7.11.3. **(AFMC)** MSEP Summary. The MSEP summary shall be distributed to HQ AFMC/A4MM. **(T-2)**

6.7.11.4. The following areas must be addressed in the summary:

6.7.11.4.1. Compliance with and currency of TOs and directives to include unit. **(T-2)**

6.7.11.4.2. Aircraft and equipment forms documentation. **(T-2)**

6.7.11.4.3. Compliance and management of Safety, Environmental, Housekeeping, cyber discipline/Hygiene and FOD Programs. **(T-2)**

6.7.11.4.4. Training Program. **(T-2)**

6.7.11.4.5. Key Task List (KTL). **(T-2)**

6.7.11.4.6. Routine Inspection List (RIL). **(T-2)**

6.7.11.4.7. TDVs, DSVs, and UCRs. **(T-2)**

6.7.11.4.8. Munitions Program. **(T-2)**

6.7.11.4.9. TMDE PMEL Activity Summary (if TMDE lab assigned). **(T-2)**

6.7.11.4.10. High-missed carded items. **(T-2)**

6.7.11.4.10.1. A high-missed carded item is defined as any work card item missed at least three times during a one-month period.

6.7.11.4.10.2. Units should use the high-missed carded items to enhance maintenance training programs, detect trends and improve the quality of maintenance.

6.7.11.4.10.3. MMA will review items to identify any relationships with repeat, recur and CND trends. **(T-2)**

6.7.11.4.11. Narrative Report: The monthly narrative report must contain an analysis of the MSEP results, a summary of significant CAT I and II discrepancies, technical inspections, and recommendations for improvement.

6.7.11.4.11.1. Prior to preparing the narrative report, QA will conduct a study of trends. **(T-2)**

6.7.11.4.12. Trend Analysis. QA will review previous reports to determine if inspected areas have improved or declined. **(T-2)**

6.7.11.4.12.1. Consistent high scores in any category may indicate the programs emphasis is not focused on the unit's actual problem areas. Low scoring areas may require a reassessment of the corrective actions taken by management. Continuous communication between MMA, unit leadership, maintenance supervision, and QA personnel is essential.

6.7.11.4.12.2. Highlight trends and root causes in the summary.

6.7.11.4.12.2.1. **(Added-AFMC)** Establish and document procedures to follow-up highlighted trends and/or root causes to closure e.g.,Logistics Evaluation Assurance Program (LEAP) RCA/Corrective Action Plan (CAP). **(T-2)**

6.7.12. - 6.7.12.2. DELETED.

6.7.13. **(Added-AFMC)** HQ AFMC/A4MM Command QA Functional Manager shall:

6.7.13.1. **(Added-AFMC)** Oversee the AFMC MSEP, and serve as POC for AFMC QA functions.

6.7.13.2. **(Added-AFMC)** Ensure development and review of Commanders Inspection Program Self-Assessment Checklists (SACs) IAW DAFI 90-302_AFMCSUP, *The Inspection System of the Department of the Air Force*.

6.7.13.3. **(Added-AFMC)** Review results of HQ AFMC Inspector General (AFMC/IG) inspections and unit MSEP reports for needed policy actions.

6.7.13.4. **(Added-AFMC)** Host and chair the Quality Assurance Working Group (QAWG).

6.7.13.5. **(Added-AFMC)** QAWG members include unit QA Chiefs/reps. Additionally, HQ AFMC/IG, HQ AFMC MFMs/SME, Contracting Officer Representative (COR) and designated union representatives may attend as required.

6.7.13.6. **(Added-AFMC)** QAWG will conduct a technical interchange meeting every 2 years in person (funding permitting) and participate in teleconferences at least bi-monthly. **(T-2)**

6.7.13.7. **(Added-AFMC)** Manage MAJCOM responsibilities for LEAP database and perform as LEAP Functional Administrator duties as stated in **paragraph 6.8**.

6.8. LEAP QA Database. Units will use the LEAP QA database to capture MSEP data. **(T-1)** The LEAP User's Manual provides information on registration, site management, and evaluation/inspection input and can be found at: **https://leap.disa.mil/mi/LEAPQA/Account/Login.aspx?ReturnUrl=%2fmi%2fLEAPQA%2fDefault.aspx**.

6.8. (AFMC) LEAP QA Database. Logistics Evaluation Assurance Program (LEAP) - Logistics Evaluation Assurance Program (LEAP) (disa.mil)

6.8.1. LEAP QA Database Roles and Responsibilities. Roles in LEAP are assigned based on each user's authorized level of control needed. The roles themselves operate in a hierarchical manner with each successively higher role possessing all of the rights of the subordinate roles. Only one role will be assigned to a LEAP user at any given time.

6.8.1.1. Application Administrator. This role is limited to Defense Information System Administration (DISA) programming staff and Program Management Office (PMO) personnel only. They perform Department of the AF-wide database management and modification. LEAP Application Administrator will:

6.8.1.1.1. Host a quarterly (virtual) Functional Review Board (FRB) and an annual (Physical or Virtual) Functional Review Board. **(T-2)**

6.8.1.1.1.1. Attendees will include: DISA Programming staff, PMO, AF/A4LM LEAP representative, Functional Administrators, and when possible, Site Managers. **(T-2)**

6.8.1.2. Provide initial training to Functional Administrators and Site Managers as necessary. **(T-2)**

6.8.1.3. Update the LEAP User's Manual as necessary. **(T-2)**

6.8.1.4. Provide a monthly status of all System Change Requests (SCRs). **(T-2)**

6.8.2. Functional Administrator. LEAP Functional Administrators provide overall database management and typically operate at the MAJCOM level. Usually assigned to a MAJCOM's policy section, they ensure proper use and alignment of the database with current policy guidelines. They also operate as the primary focal point for all LEAP-related issues within their MAJCOM and coordinate directly with the LEAP PMO. LEAP Functional Administrators will:

6.8.2.1. Manage access to LEAP either directly or through local site offices that have the capability for delegation for Site Managers only, for example, FMxC2 Manager on site. **(T-2)**

6.8.2.2. Develop and assign LEAP Finding Codes IAW MAJCOM directives. **(T-2)**

6.8.2.3. Create, modify, and manage sites within the LEAP Database for their MAJCOM. **(T-2)**

6.8.2.4. Transcribe and manage Command Task Lists (CTL) in LEAP IAW MAJCOM directives (example, RILs).

6.8.2.5. Assist local Site Managers with day-to-day operations in LEAP to include troubleshooting and reporting SCRs to the PMO. **(T-2)**

6.8.3. Site Manager. LEAP Site Managers provide local oversight for their respective Group (or equivalent). Possessing "base level" control, they are able to approve and modify LEAP Users and assign roles up to and including other Site Managers. While there is no limitation to how many Site Managers can be assigned per site, the level of control available should warrant assignment based on appropriate rank/position (typically, QA Superintendent and Chief Inspector). LEAP Site Managers will:

6.8.3.1. Manage access to LEAP by coordinating new user documentation with their assigned FMxC2 Manager and approving accounts in LEAP after user registration. **(T-2)**

6.8.3.2. Modify user accounts according to need. **(T-2)**

6.8.3.3. Ensure LEAP users are deactivated or downgraded in LEAP when out-processing the QA office. **(T-2)**

6.8.3.4. Create Flights and Sections in LEAP to which evaluations will be assigned. **(T-2)**

6.8.3.5. Assign RILs (Command Task Lists/Site Task Lists) within the LEAP application to Organizational Sites in order to facilitate entry of the MXG's Evaluation and Inspection Plan. **(T-2)**

6.8.3.6. Build and manage the E&I Plan (if used) in LEAP and ensure it incorporates all MAJCOM specific requirements. **(T-2)**

6.8.4. Read-Only Guest. LEAP Read-Only Guests are intended to be supervisory personnel and unit leadership who require regular access to LEAP Reports and Evaluation documentation.

6.9. QA Product Improvement Programs (PIP). QA runs PIP for the maintenance complex. Combined with daily maintenance data reporting, the PIP monitors, and reviews maintenance data to identify opportunities to improve aircraft and equipment. PIP includes the following programs:

6.9.1. DR Reporting.

6.9.2. TO RC requests.

6.9.3. SMR code change request(s).

6.9.4. Configuration Management (CM) and Modification Management Program; AF Form 1067, Modification Proposal; and TCTOs.

6.9.4.1. QA is responsible for CM and Modification Management. This includes reviewing, submitting, and tracking unit modification proposals being worked by MAJCOMs/Lead Commands and ensuring proper implementation of approved modification instructions or TCTOs.

6.9.4.1. **(AFMC)** QA is responsible to ensure modifications or improvements for AvSE are submitted to the appropriate authority.

6.9.4.1.1. **(Added-AFMC)** For Common AvSE (CAvSE), submit requests to AFLCMC Technology Modernization Program (AFLCMC/WN) Technology Modernization Program utilizing the following e-mail addresses:

6.9.4.1.1.1. **(Added-AFMC)** AFLCMC/WNZ Support Equipment use: **AFLCMC.WNZ.STMPWorkflow@us.af.mil**;

6.9.4.1.1.2. **(Added-AFMC)** AFLCMC Automatic Test Systems (AFLCMC/WNA), AFLCMC Automatic Test Systems use: **AFLCMC.WNA.ATSTechandMod@us.af.mil**.

6.9.4.1.2. **(Added-AFMC)** For Peculiar AvSE (MDS specific), submit requests through the MAJCOM to the Weapons Sustainment Team.

6.9.4.2. QA will manage/document modifications IAW **Chapter 14**, AFI 63-101/20-101, TO 00-20-2, and TO 00-5-15.

6.9.4.2. **(AFMC)** Additionally, see AFMCI 21-126, Temporary 2 (T-2) *Modification of Aerospace Vehicles*.

6.9.4.3. QA will establish a process for updating the weapon systems MIS that require manual updates for TCTO configuration. **(T-2)**

6.9.4.4. **(Added-AFMC)** In addition to accomplishing duties and responsibilities listed in **paragraph 6.9.5**., The PIM will:

6.9.4.4.1. **(Added-AFMC)** Evaluate the overall unit configuration management program by reviewing technical, managerial, and documentation aspects of the program. Report any deficiencies to appropriate local manager or as directed in TO 00-5-15, *Air Force Time Compliance Technical Order Process*, and AFMC instructions. Immediate action, urgent action, and safety TCTOs require particular attention and emphasis. **(T-2)**

6.9.4.4.2. **(Added-AFMC)** Monitor and document initial compliance on TCTOs and determine depth and frequency of inspection coverage. Coverage is directly related to the complexity of the TCTO as well as to how critical the system or the component is to be modified. **(T-2)**

6.9.4.4.3. **(Added-AFMC)** Support verification, validation and TCTO kit proofing, One Time Inspection (OTI) and command-directed modifications. **(T-2)**

6.9.4.4.4. **(Added-AFMC)** Along with the TCTO monitor, attend all TCTO planning and reconciliation meetings. **(T-2)**

6.9.4.4.5. **(Added-AFMC)** Ensure all test aircraft that do not meet TCTO applicability criteria due to installed modifications are identified and tracked in a comprehensive database system. This non-applicability list will reference the waiver of compliance signed by the MXG/CC. **(T-2)**

6.9.4.4.6. **(Added-AFMC)** Coordinate with the Maintenance Operations Flight (MOF) PS&D and the Combined Flight Test modification manager (when necessary),

the PIM will ensure all aircraft that are de-modified have a conformity check (de-mod reconciliation) complied with to ensure aircraft TCTOs are reinstated (As applicable). **(T-2)**

6.9.5. Product Improvement Manager (PIM). The PIM promotes deficiency reporting and provides a sound PIP based on inputs from maintenance activities. The PIM interacts with assigned AFETS personnel, FSR and MAJCOM/Lead Command as applicable to remain cognizant of ongoing and new improvement initiatives. The PIM emphasizes and promotes product improvement initiatives and ensures maintenance personnel are familiar with them by circulating flyers/newsletters, visiting commander's calls, presenting the program at maintenance orientation briefings, and making routine visits to maintenance areas.

6.9.5.1. Deficiency Reporting. DR is the process of reporting prescribed by TO 00-35D-54. Maintenance processing of warranty items is located in TO 00-20-3. The PIM's will:

6.9.5.1.1. Monitor the DR process to ensure items are properly loaded in the MIS database. **(T-2)**

6.9.5.1.2. Ensure compliance with acceptance inspection reporting requirements on DRs for aircraft returning from depot or contractor maintenance. **(T-2)**

6.9.5.1.3. Ensure DRs are submitted using Joint Deficiency Reporting System (JDRS) at **https://jdrs.mil. (T-2)**

6.9.5.1.4. Review the DR prior to releasing to the ALC or AFMC Maintenance Wings IAW TO 00-35D-54. **(T-2)**

6.9.5.1.5. Verify each report against pertinent publications and assign the appropriate precedence and category. **(T-2)**

6.9.5.1.6. Screen reported deficiencies for possible unit-unique contributing factors and initiate management action on unsatisfactory conditions resulting from local procedures or a lack of technical capability. **(T-2)**

6.9.5.1.7. Perform/coordinate a technical review of DRs returned to the unit without an adequate response to determine whether resubmitting with additional information is warranted. **(T-2)**

6.9.5.1.8. Perform exhibit-processing oversight by coordinating with the ALC and the LRS to ensure proper exhibit control and handling. **(T-2)**

6.9.5.2. RC Process. The PIM will review and route RCs IAW TO 00-5-1.

6.9.5.3. SMR code change request. Submit an SMR code change request IAW TO. 00-25-195, *AF Technical Order System Source, Maintenance, and Recoverability Coding of Air Force Weapons, Systems, and Equipment*. The PIM will:

6.9.5.3.1. Track the status of SMR change requests. **(T-2)**

6.9.5.3.2. Conduct a technical review of SMR change requests returned from depots and item managers with an unsatisfactory answer to determine whether to resubmit with additional information. **(T-2)**

6.9.5.3.3. Coordinate repair evaluation meetings when approved SMR change requests affect several agencies. **(T-2)** Ensure units with active AFREP coordinate SMR code changes through the PIM prior to submission to the MAJCOM AFREP Manager. **(T- 2)**

6.9.5.3.4. Serve as focal point for base-level repair and manufacturing capability. **(T-2)**

6.10. Technical Order Distribution Office (TODO). The TODO ensures TOs and CPINS are managed IAW AFI 63-101/20-101, TO 00-5-1, TO 00-5-15 and TO 00-5-16. TO 00-5-1 provides criteria for establishing levels of TO distribution activities. Additionally, TODOs will control electronic technical data configuration IAW **Chapter 8**. Establish the PMEL TODO under the control of the TMDE Flight. **(T-2)** The TODO will:

6.10.1. Coordinate with QA SME for each incoming TCTO to determine applicability. **(T-2)**

6.10.1.1. All TCTOs received from outside agencies need to be routed through QA for the review process.

6.10.1.2. TCTO applicability is determined by aircraft serial number for aircraft, engine serial number for engines, and by part number or other specific criteria for commodities.

6.10.1.3. TCTOs need to be manually or electronically date stamped to reflect the date the electronic or hard copy is received.

6.10.1.3.1. Date stamping all TCTOs with the date received indicates QA has reviewed the TCTO and that applicability has been determined.

6.10.1.3.2. TCTO electronic date stamping can be accomplished by either (1) utilizing a locally developed spreadsheet containing the minimum following information: TCTO number, MDS, receiving TODO name, applicability determination and the date received, all of which must be associated with the corresponding TCTO or (2) inserting the receipt date on the TCTO utilizing the Adobe Tools feature. For either option, the date received will be entered by the QA TODO responsible for tracking TCTOs. **(T-2)**

6.10.1.3.2.1. If used, the TCTO tracking spreadsheet or Adobe-inserted date stamped TCTOs will be electronically secured and controlled by the receiving QA office. **(T-2)**

6.10.1.3.3. Only date stamped TCTOs are authorized for use. **(T-2)**

6.10.1.4. Post TCTO file copies IAW TO 00-5-1.

6.10.1.4.1. TCTO file copies may be posted and distributed in electronic format provide all requirements of TO 00-5-1 and AFI 33-322 are sustained. Electronic TCTO distribution is automated for ETIMS/IETM.

6.10.1.5. For hard copy TCTOs, provide a file copy of the TCTO to PS&D. **(T-2)**

6.10.1.6. Ensure personnel assigned as TODO/Technical Order Distribution Account (TODA) managers meet requirements set forth in TO 00-5-1 and AFI 63-101/20-101. **(T-2)**

6.10.2. Manage the QA Central TO File. **(T-2)**

6.10.2.1. As a minimum, the QA Central TO File must contain copies of general and procedural TOs and copies of all TCTOs pertaining to the assigned aircraft and equipment

owned, operated, or maintained. (Paper copies for paper TOs or local access for digital TOs) **(T-2)**

6.10.2.2. The file is kept to meet QA requirements, not to duplicate TOs held by maintenance work centers.

6.10.3. Manage TO accounts using ETIMS IAW TO 00-5-1. **(T-2)**

6.10.4. Limit use of Local Work Cards, Local Job Guides, Local Page Supplements or Local Checklists to accomplish maintenance on AF equipment. **(T-2)** Locally prepared technical instructions will not be used to circumvent approved technical data (see TO 00-5-1). **(T-2)**

6.10.4.1. The TODO will review and manage all locally developed products IAW TO 00-5-1 and MAJCOM supplements for safety and adequacy of procedures. **(T-2)**

6.10.4.2. Local Work Cards, Local Job Guides, Local Page Supplements and Local Checklists need to be reviewed for currency when source reference data changes.

6.10.4.3. TODO will develop local publications IAW DAFMAN 90-161 to ensure compliance with these policies. **(T-2)**

6.10.5. Prepare a list of all changes and revisions to indexes, TOs, inspection work cards and checklists. **(T-2)**

6.10.5.1. This list will include TO number and date received. **(T-2)**

6.10.5.2. The TODO will date stamp the cover page of all paper TOs, changes, supplements, Local Work Cards, Local Job Guides, Local Checklists and CPINS to reflect the date the hard copy is received. **(T-2)**

6.10.5.2.1. This list will be sent out electronically weekly to all affected agencies. **(T-2)**

6.10.5.2.2. Supervisors should review the list of changes and ensure all personnel are aware a change or revision has been received.

6.10.5.3. "Immediate" action TCTOs must be implemented upon receipt, and "Urgent Action" TCTOs, safety supplements and interim supplements must be brought to the attention of local maintenance supervision within 24 hours of receipt. **(T-2) Note:** Communication of a grounding, or potential grounding event (TCTO, OTI, PM event) to weapon systems operating in a deployed environment and/or under the authority of an operational command must follow the grounding procedures outlined in DAFMAN 11-401, *Aviation Management*, and AFI 63-101/20-101. Units will not directly communicate home station requirements to its deployed forces but provide communication through their respective MAJCOM. **(T-2)**

6.10.6. Ensure all authorized technical data variances are kept with aircraft and equipment historical records IAW **Chapter 14**. **(T-2)**

6.10.7. If designated as Lead TODO (primary as designated in block 5 of the AFTO Form 43 per TO 00-5-1), will conduct a management inspection on other maintenance TODOs/TODAs in the maintenance complex at least annually along with performing spot checks of TO files. **(T-2)**

6.10.7.1. As part of this inspection, the TODO will confirm TODO/TODA personnel and Library Custodian have completed the mandatory minimum requirements of TO System training. **(T-2)**

6.10.7.2. The Lead TODO(s) will coordinate with other TODOs and TODAs, and local Client Support Administrators, Functional Systems Administrators, and applicable functional OPRs to ensure eTools are configured with current and only authorized software to support TO and maintenance documentation. **(T-2) Note:** Coordination with the local Communications Squadron to verify network configuration requirements for eTools are available and sustained to meet the requirements listed in TO 31S5-4-ETOOL-1-WA-1. Additional user support available through the Air Force Technical Order Functional Support Team, **af.etimstofst@us.af.mil** or DSN 872-9300.

6.10.8. Control the electronic data configuration on applicable eTools IAW **Chapter 8**. **(T-2)**

6.10.8.1. TODO/ Functional Systems Administrators will develop local procedures to quarantine eTools and eTool update history in the event of a mishap. **(T-2)**

6.10.9. Maintain records of ETIMS IAW TO 00-5-1.

6.10.9.1. TODOs will set up software sub-accounts with each appropriate shop/section and ensure each shop/section has the most current software on hand. **(T-2)**

6.10.9.2. TODOs will include ETIMS or equivalent system in the routine and annual checks required by TO 00-5-1. **(T-2)**

6.11. One-Time Inspections (OTI) program. The OTI program is managed by the MXG IAW TO 00-20-1. OTIs are normally look-only actions to verify the existence of suspected equipment conditions or malfunctions.

6.12. Functional Check Flights (FCFs) to include Operational Check Flights (OCFs).

6.12.1. Check Flights are performed to ensure an aircraft is airworthy and/or capable of accomplishing its mission. Additional guidance may be found in DAFMAN 11-401, AFMAN 11-202 V3, *Flight Operations*, DAFMAN 13-201, *AF Airspace Management*; TO 1-1-300, *Maintenance Operational Checks and Check Flights*; TO 00-20-1; and applicable -6 TOs and -1 Flight Manuals.

6.12.1.1. OCF will be kept to a minimum and are not used to replace TO 1-1-300, *Maintenance Operational Checks and Check Flights* or MDS specific -6 TO Functional Check Flight (FCF) requirements. OCFs must be flown by experienced aircrews (not required to be an FCF qualified aircrew), must be briefed by QA for aircraft condition, and accomplished following the same maintenance criteria as FCFs. **(T-2)**

6.12.2. The QA FCF Program Manager will:

6.12.2.1. Establish local FCF procedures IAW TO 1-1-300 and checklists for any specific local aircraft requirements to include configuration, administration, control, and documentation of the FCF Program. **(T-2)**

6.12.2.1.1. Coordinate these procedures with OG Standardization/Evaluation and publish them in a wing publication/supplement IAW DAFMAN 90-161. **(T-2)**

6.12.2.2. Support the appropriate squadron to facilitate FCF.

6.12.2.3. Maintain an information file containing unit directives concerning FCF procedures and an FCF checklist for each MDS assigned for briefing aircrews. **(T-2)**

6.12.2.3.1. DELETED.

6.12.2.3.1.1. **(Added-AFMC)** Information file will also include: **(T-2)**

6.12.2.3.1.1.1. **(Added-AFMC)** Mission profile for each type of assigned aircraft, consisting of checks to be accomplished, presented in consecutive order. **(T-2)**

6.12.2.3.1.1.2. **(Added-AFMC)** TO 1-1-300, *Maintenance Operational Checks and Check Flights*. **(T-2)**

6.12.2.3.1.1.3. **(Added-AFMC)** List of authorized FCF crew members signed by the OG/CC. **(T-2)**

6.12.2.4. Ensure a FCF checklist is used for each FCF. **(T-2)**

6.12.2.5. Ensure all FCFs are debriefed with the appropriate debrief function. **(T-2)**

6.12.2.5.1. During debriefing, the FCF checklist and aircraft forms will be reviewed to determine if all requirements have been accomplished. **(T-2)**

6.12.2.5.2. After completing the review, the FCF checklist will be sent to PS&D for inclusion in the aircraft jacket file. **(T-2)**

6.12.2.6. Maintain a copy of the AF Form 2400, *Functional Check Flight Log*, or equivalent automated product for deficiency and trend analysis. **(T-2)**

6.12.2.6.1. **(Added-AFMC)** Review the log monthly for trends indicating problems requiring further analysis or corrective actions. Identify deficient areas and forward to the MXW/CC and OG/CC for required review or action. **(T-2)**

6.12.3. FCF-qualified QA Inspectors will:

6.12.3.1. Ensure the FCF aircrew is briefed on the purpose and extent of the flight, previous maintenance problems and discrepancies recorded on the aircraft or engines related to the FCF. **(T-2)**

6.12.3.2. Review aircraft W&B documents. **(T-2)**

6.12.3.3. Ensure AF Form 2400 or an equivalent automated product is maintained to provide information for evaluation and analysis. **(T-2)**

6.12.3.3.1. Include the date and time of the FCF, aircraft serial number, reason for FCF, name of debriefer and name of aircraft commander. **(T-2)**

6.12.3.3.2. The AF Form 2400 or equivalent automated product will also indicate if the aircraft was released for flight, reasons for any non-release, action taken, and date completed, and the date maintenance documents were forwarded to PS&D or records section. **(T-2)**

6.12.3.4. Ensure all maintenance actions are completed and all AFTO Form 781-series forms are documented IAW MDS specific -6 TO and TO 00-20-1 or electronic equivalent. **(T-2)**

6.12.3.5. All maintenance actions on transient aircraft requiring FCF must be reviewed by QA prior to FCF. **(T-2)**

6.12.3.5.1. If the aircraft MDS/type is not assigned at the transient base, then the owning unit must provide a qualified FCF pilot/crew and maintenance as required. **(T-2)**

6.12.4. Flight Requirements. The mandatory requirements for FCF are outlined in TO 1-1-300 and the applicable -6 TO. FCF profiles are determined by the maintenance requirement causing the FCF. The FCF profile will be tailored for the discrepancy causing the FCF by applying the following guidance:

6.12.4.1. Require a clean configuration whenever FCFs are flown for flight controls, fuel controls or engine changes. **(T-2)**

6.12.4.2. Do not remove fixed wing pylons, fixed wing tip tanks and fixed external stores unless they interfere with fuel scheduling, aerodynamic reaction, air loading, signal propagation. **(T-2)**

6.12.4.3. DELETED.

6.12.5. FCF Release. An FCF release occurs upon the successful completion of all requirements as determined by the FCF aircrew. The final decision to release rests solely with the aircraft commander. An aircraft may be released for flight if a malfunction occurs during an FCF, which is not related to the condition generating the FCF and the original condition checks good.

6.12.5.1. An FCF conditional release may occur when the aircraft does not successfully complete FCF requirements due to a specific system malfunction. The FCF aircrew, in coordination with maintenance, determines a FCF conditional release if the malfunction may be corrected without generating another FCF. If upon review of the corrective action, the FCF aircrew accepts the maintenance action as a satisfactory repair of the malfunction, they may release the aircraft from FCF.

6.12.6. MAJCOMs will determine the process and level of command that will issue instructions for FCF procedures away from home station in their supplements to this DAFI.

6.12.6.1. **(Added-AFMC)** While performing an FCF away from home station, the applicable requirements for FCFs of the transient/host base must be followed. The transient/host base QA office must be contacted and should be the focal point for all FCF requirements. **(T-2)**

6.13. Inflight Operational Checks. Inflight operational checks (as applicable) will be accomplished IAW TO 1-1-300, TO 00-20-1 and applicable -6 and -1 TOs.

6.13.1. Document inflight operational checks IAW TO 00-20-1.

6.14. High Speed Taxi Checks. The MXG/CC and OG/CC may authorize high speed taxi checks when a maintenance ground operational check requires aircraft movement at higher-than-normal taxi speeds (with qualified FCF aircrews) to operationally check completed maintenance.

6.14.1. High speed taxi checks (as applicable) will be accomplished IAW TO 1-1-300 instead of FCFs. **(T-2)**

6.14.1.1. Process aircraft forms through QA using FCF procedures. **(T-2)**

6.14.1.2. QA will develop an aircrew briefing checklist specifically for high-speed taxi checks, to include the required FCF briefing items and pertinent warnings, cautions. **(T-2)**

6.14.2. Configure aircraft with the minimum -1 operational fuel requirements. **(T-2)**

6.14.3. Ensure aircraft is prepared for flight and the Exceptional/Conditional Release is signed off prior to conducting high speed taxi checks. **(T-2)**

6.15. Weight and Balance (W&B) Program. QA will maintain the W&B Program IAW TO 1-1B-50, *Basic Technical Order for USAF Aircraft Weight and Balance*.

6.15. (AFMC) Weight and Balance (W&B) Program. The W&B program NCOIC/Manager will attend an in-resident W&B course. **(T-2)** MXG/CC is the approval authority to waive the J3AZR2AXXX 0W1A Weight and Balance Practical Course for weight and balance technicians. **(T-2)**

6.15.1. W&B manuals for Class I and II aircraft are maintained in a central file. **(T-2)**

6.15.1.1. The Lead Command will standardize the method of supplemental handbook storage and physical location for like-MDS aircraft. **(T-2)**

6.15.1.1. **(AFMC)** [DEV] Weight and balance handbook storage and physical location will be determined by MXG/CC or equivalent for test/mission unique aircraft. **(T-2)**

6.15.2. QA will manage W&B on commercial derivative aircraft IAW **Chapter 6**. **(T-2) Note:** The contractor is responsible for managing W&B programs on contract logistics supported aircraft.

6.15.3. The W&B Program Manager will ensure:

6.15.3.1. Sufficient personnel are qualified on assigned aircraft IAW TO 1-1B-50. **(T-2)**

6.15.3.2. All assigned aircraft are weighed IAW prescribe MDS specific publications and directives. **(T-2)** The W&B Program Manager will:

6.15.3.2.1. Keep W&B documents required by TO 1-1B-50 for each assigned aircraft. **(T-2)**

6.15.3.2.2. Use the Automated Weight and Balance System and maintain a back-up copy of all W&B documents. **(T-2)**

6.15.3.3. Procedures are established for routing completed TCTO and modification information for W&B changes. **(T-2)**

6.15.3.4. Essential W&B data and changes to the basic weight and moment are available for appropriate mission planning (such as, Standard Configuration Loads, updates to supplemental handbook). **(T-2)**

6.15.3.5. Periodic serviceability inspections are accomplished on unit-stored and maintained W&B equipment (as applicable). **(T-2)**

6.15.3.6. Coordination with Maintenance Supervision in developing a W&B Preparation Checklist if the aircraft -5 TO is not comprehensive enough for the task. **(T- 2)**

6.15.3.7. The SCR reflects W&B certification. **(T-2)**

6.15.4. W&B Qualified QA Inspector Responsibilities. The W&B Qualified QA Inspector will:

6.15.4.1. Verify scale readings and accomplish/oversee the actual computations. **(T-2)**

6.15.4.2. Supervise the preparation, leveling and weighing of the aircraft IAW MDS specific –2 and –5 series TOs and TO 1-1B-50.

6.15.4.3. Inspect W&B documents before flight when locally accomplished modifications affect the basic aircraft weight and moment. **(T-2)**

6.15.4.4. Review computations for accuracy. (T-2)

6.16. (Added-AFMC) Chafing Awareness Program. This program is mandatory for fighter aircraft units and other MDS IAW applicable MDS technical data. QA must monitor and track instances of wire, harness and metal line/tube chafing. A randomly selected 10 percent of assigned aircraft are inspected when notification is received of a potential chafing problem involving like model, lot number or block of aircraft. **(T-2)** Record pertinent information and take pictures as required facilitating entry into the applicable maintenance information database.

6.16.1. **(Added-AFMC)** The Chief Inspector shall recommend initiating an OTI if the sampled aircraft indicates a chafing problem or the detected chafing is an operational safety hazard. The MXG/CC will be the final approval authority on the scope of the OTI. **(T-2)**

6.16.2. **(Added-AFMC)** QA will utilize a database for the purpose of tracking wire and harness chafing problems identified through OTIs and maintenance cross-tell reports. Consult the database before expending man-hours performing inspections. **(T-2)**

Chapter 7

IMPOUNDMENT PROCEDURES

7.1. Aircraft and Equipment Impoundment. Aircraft or equipment is impounded when intensified management is warranted due to system or component malfunction or failure of a serious or chronic nature. Refer to DAFI 91-204 for aircraft and equipment involved in accidents, mishaps, or incidents. Impounding aircraft and equipment enable investigative efforts to systematically proceed with minimal risk relative to intentional/unintentional actions and subsequent loss of evidence.

7.1.1. **(Added-AFMC)** Coordinate with the Wing Safety Office to determine reporting requirement, when property damage to aircraft, equipment, commodities, or resources exceeds the Class D threshold $25,000 or more but less than $60,000.

7.2. Specific Guidance. MXGs will develop and implement a standardized Impound Official training course. The course will include review of applicable AFIs, TOs, impoundment clearing procedures, and leverage FSR and AFETS expertise to ensure Impound Official is prepared to assume all duties and responsibilities of an impoundment official.

7.2.1. Local program procedures, requirements and responsibilities will be captured in a local supplement to this instruction. **(T-3)**

7.2.1.1. QA is the OPR for the Impoundment Program and will develop local impoundment checklists. **(T-3)**

7.2.1.1. **(AFMC)** Unit QA will ensure development of a local operating instruction for local procedures and development of a local impoundment checklists (see **Attachment 9** for sample Impoundment checklist) for use, notification, release officials, specific forms entries, appointment and responsibilities of impoundment investigating officials, special handling or tagging of parts from impounded items sent through the repair cycle and conducting impoundment investigations across organizations. **(T-2)**

7.2.1.2. QA in coordination with MT, will serve as the OPR and focal point for the management of the Impound Official training course. The course completion is mandatory prior to being assigned as an Impound Official. **(T-3)**

7.2.2. - 7.2.2.1. DELETED.

7.2.3. The Impoundment Release Authority determines the need for a one-time flight and will coordinate appropriate authorization IAW TO 00-20-1. **(T-3) Note:** If the aircraft/equipment were impounded for a mishap, coordinate with the safety office prior to releasing the aircraft/equipment from impound. **(T-3)**

7.2.3.1. **(Added-AFMC)** The MXG/CC or CD will appoint an individual in writing as the designated Impoundment release authority at deployed location(s) under their control. The owning MXG/CC, MXG/CD, or equivalent, is the impoundment release authority for aircraft transiting to enroute locations. **(T-2)**

7.2.4. MAJCOMs will determine the amount of time unit QAs will maintain copies of finalized Impoundment reports in their supplement to this DAFI. **Note:** See Air Force Records Information Management System (AFRIMS) Reference.

7.2.4.1. **(Added-AFMC)** QA will maintain copies of finalized Impoundment reports for a minimum of 2 years. Ensure incidents are documented in the applicable component/subcomponent historical records as applicable **(T-2)**

7.2.5. **(Added-AFMC)** Impoundments may be transferred from an aerospace vehicle and/or an equipment item to a sub-component once the sub-component is removed (engine, gun system, etc. determined to be causal for asset impoundment and is removed from the aircraft).

7.2.6. **(Added-AFMC)** Units will ensure procedures are developed to safeguard/secure possible investigative evidence such as data records (e.g. crash survivable flight data recorder, seat data recorder, Cockpit voice recorder) per DAFI 91-204, *Safety Investigation and Hazard Reporting*, upon aircraft impoundment for potential safety related incidents. **(T-2)**

7.3. Impoundment Authorities.

7.3.1. Impoundment Authorities will:

7.3.1.1. Make impoundment determination when extraordinary measures are required to address any degradation of aircraft airworthiness or serious aircraft/equipment anomaly, not listed in section 7.5. **(T-2)**

7.3.1.2. Select the Impoundment Official. **(T-2)**

7.3.1.2.1. - 7.3.1.2.3. DELETED.

7.4. Impoundment Official Responsibilities. The Impoundment Official is designated as the single POC for impounded aircraft or equipment and will hold the minimum rank of MSgt and will be tracked on the SCR. **(T-2)** The Impoundment Official will:

7.4. (AFMC) Impoundment Official Responsibilities. MSgt or Equivalent.

7.4.1. Be responsible for controlling and monitoring the investigation of impounded aircraft or equipment. **(T-2)**

7.4.2. Use established checklists to guide the sequence of actions. **(T-2)**

7.4.3. Control and track access of personnel to impounded aircraft or equipment. **(T-2)**

7.5. Mandatory Impoundments. Aircraft and/or equipment will be impounded:

7.5.1. DELETED.

7.5.2. Following an aircraft mishap as defined in DAFI 91-204 and DAFMAN 91-223, *Aviation Safety Investigations and Reports*. **(T-2)**

7.5.3. When support equipment is known or suspected to have been a factor in a mishap or may have contributed to injuries. **(T-2)**

7.5.4. Following an un-commanded flight control movement. **(T-2)**

7.5.5. Following an inadvertent ordnance release or explosive mishap. **(T-2)**

7.5.6. When authorized procedures are not adequate or the unit is unable to identify/repair loaded nuclear weapons system malfunctions IAW DAFI 91-101. **(T-2)**

7.5.7. Following aircraft engine anomalies to include but not limited to:

7.5.7.1. Unselected propeller reversal. **(T-2)**

7.5.7.2. Flameout/stagnation (for single engine aircraft). **(T-2)**

7.5.7.3. Unselected power reversal. **(T-2)**

7.5.7.4. Engine case penetrations, ruptures, or burn-through from an internal engine component. **(T-2)**

7.5.7.5. When an aircraft experiences a loss of thrust sufficient to prevent maintaining level flight at a safe altitude. **(T-2)** This includes all cases of multiple engine power loss or roll back.

7.5.7.6. Engine damage due to a foreign object and source of FO is determined to be internal to the engine. **(T-2)** For Propeller driven aircraft, both the propeller and engine will be impounded as a single unit when the engine has confirmed internal damage due to a foreign object. **(T-2)**

7.5.7.7. Engine damage which occurs during transport. **(T-2)**

7.5.8. Following an in-flight fire. **(T-2)**

7.5.9. When an aircraft experiences an in-flight loss of all pitot-static system instruments or all gyro stabilized attitude or direction indicators. **(T-2)**

7.5.10. When there is evidence of intentional damage, tampering, or sabotage. **(T-2)** **Exception:** Intentional damage due to combat is not mandatory.

7.5.11. When physiological incidents attributable to aircraft systems or cargo occurs. **(T-2)**

7.6. Impoundment Procedures.

7.6.1. When the Impoundment Authority directs impoundment, a Red X symbol will be placed in the applicable AFTO Form 781A for aircraft, applicable engine work packages for uninstalled engines or AFTO Form 244 for equipment (or electronic form equivalents) with a statement indicating the reason for impoundment and the name of the assigned Impoundment Official. **(T-2)**

7.6.2. The MOC will be notified when an impoundment decision has been made. **(T-2)**

7.6.3. Aircraft or equipment records will be controlled at the discretion of the Impoundment Official. **(T-2)** When required, the Impoundment Official will:

7.6.3.1. Obtain and secure the current aircraft forms and the aircraft jacket file for aircraft, applicable engine work packages for uninstalled engines, or the AFTO Form 244 for equipment (or electronic form equivalents). **(T-2)**

7.6.3.2. Notify the MIS DBM administrator to isolate the aircraft or equipment serial number in order to prevent any changes and maintain the integrity of the historical data until the aircraft or equipment is released. **(T-2)**

7.6.3.3. Request and collect any training records, required to complete the impoundment investigation. **(T-2)**

7.6.3.4. On aircraft impounded for potential safety related incidents, ensure the Cockpit Voice Recorder/Flight Data Recorder circuit breakers are pulled immediately after engine shutdown or before applying external power to safeguard Cockpit Voice Recorder/Flight Data Recorder data, if equipped. **(T-2)**

7.6.3.5. Ensure impounded aircraft/equipment is identified by cordon with cones, ropes or placards indicating impound condition and aircraft location. **(T-2)**

7.6.4. Impoundment Official will limit maintenance actions on impounded aircraft or equipment until the cause is determined. **(T-2)**

7.6.4.1. The Impoundment Official will determine what maintenance can be performed in conjunction with the maintenance required to release the aircraft or equipment from impoundment. **(T-2)**

7.6.4.1.1. Impoundment Official will coordinate with Wing Safety or equivalent for aircraft, equipment, and parts disposition to support any ongoing safety or mishap investigation. **(T-2)**

7.6.4.2. Parts removed from impounded aircraft or equipment will be carefully controlled until impound has been cleared and/or a safety investigation determination has been made. **(T-2)** This is to ensure that parts, once confirmed as the cause for impoundment, are available to be processed as DR exhibits.

7.6.5. The Impoundment Official will select a team of qualified technicians dedicated to determining the cause of the problem that led to the impoundment. **(T-2)**

7.6.5.1. Impoundment team members will be relieved of all other duties until released by the Impoundment Official. **(T-2)**

7.6.6. Once the cause of the malfunction or failure has been positively determined, the Impoundment Official will brief the Impoundment Release Authority on findings, corrective actions, and requests release of the aircraft or equipment from impoundment. **(T-2)**

7.6.7. If the cause of the discrepancy could potentially affect other aircraft or equipment in the fleet, QA will provide cross-tell information for up-channeling to the MAJCOM and the designated Lead Command IAW AFPD 10-9. **(T-2)**

7.6.8. Clear impoundments from forms/MIS IAW TO 00-20-1. **(T-2)**

7.6.9. If the cause of a reported malfunction cannot be determined or a positive corrective action cannot be confirmed, the Impoundment Release Authority will determine if further actions are required (such as, requesting depot assistance, further troubleshooting, FCF/OCF). **(T-2)**

7.6.10. MAJCOMs will publish guidance outlining impoundment and release procedures for transient aircraft.

7.6.10. **(AFMC)** Unit QA will ensure development of a local operating instruction to include, procedures for off-station aircraft impoundment while transient, home station notification, release officials, forms entries, appointment and responsibilities of impoundment investigating officials, special handling or tagging of parts from impounded items sent through the repair cycle and conducting impoundment investigations across organizations. **(T-2)**

7.6.10.1. At locations where no MXG/CC or designated representative is available, the aircraft assigned MXG/CC may temporarily delegate Impoundment and Release Authority.

7.7. Rules of Impoundment Specifically for Explosive-Related Events/Mishaps. When an inadvertent release or an explosive mishap is reported, the following procedures will apply:

7.7.1. In-flight:

7.7.1.1. When the involved aircraft returns to the de-arm or parking area, the aircraft will be impounded. **(T-2)** Limit maintenance actions to those required to make the aircraft safe.

7.7.1.2. The MXG/CC, MOC, Munitions Control, WWM, QA and Wing Safety or equivalent will be notified of the impoundment action. **(T-2)**

7.7.1.3. The aircraft with unsafe munitions will be parked and isolated in an area approved by the weapons safety office or equivalent and airfield management. **(T-2)**

7.7.1.4. Investigate and report the incident IAW DAFI 91-204.

7.7.2. Ground:

7.7.2.1. The senior ground crew member will be in charge of the aircraft or equipment until relieved and will ensure involved persons remain at the scene. **(T-2)**

7.7.2.2. Protect other aircraft or equipment located near the incident if an explosive hazard exists. **(T-2)**

7.7.2.3. Do not change the position of any switches except as needed for safety. **(T-2)**

7.7.2.4. Limit maintenance actions to those actions required to make the aircraft or equipment safe. **(T-2)**

7.7.3. Preserve mishap evidence to the maximum extent possible. **(T-2)** An example would be segregating an aircraft gun versus destroying it if it poses no immediate danger. This allows for evaluation of all the evidence and the ability to recreate the mishap conditions.

7.7.4. If an incident, malfunction, or mishap is suspected to have occurred or caused by in-use, installed, or otherwise configured munition (live or inert), or a 20 or 30MM gun system jam creating a safety condition, notify the Global Ammunition Control Point Air Force Life Cycle Management Center, Munition Division (AFLCMC/EBH) Munitions Rapid Response Team: DSN: 312-777-2666; COMM: (801) 775-2666, and the MAJCOM munitions staff. **(T-2)**

7.7.4.1. Refer to DAFI 91-202 for additional information about the Munitions Rapid Response Team. **Note:** The Munitions Rapid Response Team can also provide units technical assistance in resolving recurring 20 or 30MM gun system jams and malfunction isolation.

7.7.5. For impoundments involving nuclear loaded weapon systems see **Paragraph 7.5.6**. **(T-2)**

Chapter 8

TOOL AND EQUIPMENT MANAGEMENT

8.1. Tool and Equipment Management. The objectives of the Tool and Equipment Management Program are to prevent and eliminate FOD to aircraft, engines, missiles, training, and support equipment, and to reduce costs through strict effective control and accountability of assets. To ensure standardization among maintenance units, commanders and key leaders are responsible for executing an effective tool program. MAJCOMs will identify small unique unit tool and equipment management requirements in a supplement, addendum or deviation as described in the purpose statement of this DAFI. DFT/CFT will adhere to local tool control policies and procedures provided in the MXG/MXO in-brief (see **Paragraph 5.2.1.9**) when working on aerospace equipment possessed by the unit. The AF enterprise Tool Accountability System is TCMax®. **(T-2) Exception:** N/A to aircraft/weapon system programs that provide tool and equipment accountability support as an internal function of the program (e.g., F-35 ALIS).

8.1. (AFMC) Tool and Equipment Management. Small unique units designated as NSOs will follow guidance in DAFI 21-101 AFMC Sup Addendum A or Addendum C for AFRL.

8.2. Guidelines for Program Management. Wings will document procedures for the control and management of all tools/equipment used for aircraft/aerospace equipment maintenance or which enter the flightline or aerospace equipment maintenance industrial areas, to include all wing organization's (Hospital, CE, vehicle Mx, Security Forces), to provide mission support in a wing level publication IAW DAFMAN 90-161. As a minimum, guidance will address the following:

8.2. (AFMC) Guidelines for Program Management. Units will use the Blanket Purchasing Agreement **(**BPA) for acquiring tools. **(T-2)**

8.2.1. Standardized procedures for security, control, and accountability of tools and equipment. **(T-3)**

8.2.1.1. DELETED.

8.2.1.2. **(Added-AFMC)** Include standardized procedures for clecos. **(T-2)**

8.2.2. Inventory requirements. **(T-3)** As a minimum, units will conduct and document an annual inventory of all tools and equipment. **(T-3)**

8.2.3. Procedures for warranted tool management. **(T-3)**

8.2.3. **(AFMC)** Warranty/Quality Tool Program. The purpose of the warranty/quality tool program is to ensure high quality; industrial-strength warranted tools are available for use in aircraft and equipment activities. Warranty tools will be purchased using the BPA or through local contracts with a warranty tool vendor. Units desiring to establish a warranty tool program must coordinate their program through both the supply and contracting squadrons. Detailed management procedures are contained in AFI 23-101, *Air Force Materiel Management*. **(T-2)** The following general guidelines apply:

8.2.3.1. Procedures to tag/segregate unserviceable warranty tools. **(T-3)**

8.2.3.2. **(Added-AFMC)** Procedures for etching/de-etching of tools. Ensure method does not void warranty. **(T-2)**

8.2.3.3. **(Added-AFMC)** The unit purchasing tools will work with the local contracting office to establish a BPA for acquiring tools. **(T-2)**

8.2.4. Procedures for control and management of replacement, expendable and consumable hand tools, HAZMATs, and other items contained in CTKs. **(T-3)**

8.2.5. Procedures for transfer of tools/CTKs at the job site (on-site transfers). **(T-3)**

8.2.5.1. Ensure tool accountability and control is maintained when transfer occurs between the individuals. As a minimum the individuals involved in the transfer will accomplish a joint inventory and document accordingly. **(T-3)**

8.2.6. Procedures for lost or missing tools. **(T-3)**

8.2.6.1. **(Added-AFMC)** Procedures for aircraft that have already taxied or are flying when tools, equipment, TOs, or any object is lost and was used on the aircraft. **(T-2)**

8.2.7. Assignment of Equipment Identification Designators (EID) for CTKs, non-CA/CRL equipment, and assignment of CTK numbers for tools. **(T-3)**

8.2.8. Procedures for issue, marking, and control of PPE, tools, or equipment (such as, hearing protectors, reflective belts, headsets) assigned/issued to individuals. **(T-3)**

8.2.8. **(AFMC)** At a minimum, these markings will identify the individual. **(T-2)**

8.2.8.1. At a minimum, individual issued equipment will be marked with first name initial, last name, and employee number. **(T-3)** For members without employee numbers, organizational items will be marked with first name initial, last name, and organization. **(T-3)**

8.2.9. Procedures to ensure positive accountability and control of rags. **(T-3)**

8.2.9.1. A rag is defined as a remnant of cloth purchased in bulk or a standardized, commercial quality, vendor-supplied shop cloth used in general industrial, shop, and flightline operations.

8.2.9.1.1. Cheesecloth is considered a rag; however, paper products/paper towels are not considered rags.

8.2.9.1.1. **(AFMC)** Procedures if paper product is used, identify if and when they will be controlled. **(T-2)**

8.2.9.2. Rags should be uniform in size and color.

8.2.9.3. Marking or identifying each shop rag with a CTK number is not necessary.

8.2.10. Procedures to limit numbers of personnel authorized to procure tools. **(T-3)**

8.2.11. Procedures for control of locally manufactured or developed tools and equipment. **(T-3)**

8.2.12. Procedures for FSRs/DFTs/CFTs when working on equipment within the unit. **(T-3)**

8.2.13. Standardized procedures and responsibilities for decentralizing CTKs, tools, and equipment outside tool room/support section to meet mission requirements. **(T-3)**

8.2.13.1. Inventory and accountability requirements described in this DAFI apply equally to all decentralized CTKs tools, and equipment. **(T-3)**

8.2.14. Procedures for control of response equipment permanently stored/located in trailers or vehicles. **(T-3)**

8.2.15. Procedures for requiring a second party or on-duty supervisor inspection of CTKs when conditions warrant a single person shift. **(T-3)**

8.2.15.1. The same individual that signs out a CTK cannot sign it back in. **(T-3)**

8.2.16. Procedures for controlled access to tool rooms. **(T-3)**

8.2.17. **(Added-AFMC)** Procedures for long term tool kit (CTK) storage. **(T-2)**

8.2.18. **(Added-AFMC)** Procedures for control of equipment issued/dispatched to work areas. **(T-2)**

8.2.19. **(Added-AFMC)** Procedures for control of shop machine accessories/attachments. **(T-2)**

8.2.20. **(Added-AFMC)** Procedures for control of tooling included in TCTO/Mod Kits. **(T-2)**

8.2.21. **(Added-AFMC)** Procedures for loaned tools to include issue, tracking, duration, and verification. **(T-2)**

8.2.22. **(Added-AFMC)** Procedures for training personnel on tool control and accountability measures. **(T-2)**

8.3. General Program Guidelines.

8.3.1. The Flight CC/SUPT will designate CTK custodians. **(T-3)**

8.3.1.1. CTK custodians are responsible for tool, HAZMAT, and consumable asset accountability and control. **Exception**: A separate person may be designated as the HAZMAT monitor.

8.3.2. Flight CC/SUPT and/or Section NCOICs/Chiefs (or equivalents) will determine the type, size, contents, and number of CTKs required for their work centers. **(T-3)**

8.3.2.1. The WWM will make this determination for load crew CTKs, when assigned. **(T-3)**

8.3.2.2. **(Added-AFMC)** Consider deployment taskings, TDYs, and special test project requirements when making CTK determinations. Ensure user input is considered when determining quantity, content and layout of tool kits. **(T-2)**

8.3.3. Design CTKs to provide for quick inventory and accountability of tools. CTKs and tools will be clearly marked with the Equipment Identification Designator (EID) (follow guidance below). **(T-3)**

8.3.3.1. **(Added-AFMC)** If a tool is to be permanently removed from a CTK/Tool Kit (TK) ensure any previously assigned World Wide Identification (WWID) markings are de-etched (reference warranty procedures if applicable). **(T-2)**

8.3.4. CTK contents will be standardized to the maximum extent possible within functional elements of a squadron that have similar missions (such as, aircraft flights/sections and Combat Armament Support Team). **(T-3)**

8.3.5. Each tool, item of equipment, or consumable contained in a CTK will have an assigned location identified either by inlay cuts in the shape of the item, shadowed layout, label, or silhouette. **(T-3)**

8.3.5.1. No more than one item will be stored in a cutout, shadow, or silhouette. Tools too small to be etched individually may be place in CTKs as a set (such as, drill bits, Allen wrenches, apexes, or paired items like gloves, booties). **(T-3)**

8.3.6. A Master Inventory List (MIL) will be required for each CTK or series of identical CTKs. **(T-3)**

8.3.6. **(AFMC)** Tools will be described on the MIL to accurately depict the actual size or type to ensure positive tool control, i.e. screwdriver, common, 6 inches; socket, 3/8 drive X 5/8. **(T-2*)*** CTK custodian or designated representative must review and sign the MIL. A copy of the MIL will also remain in each CTK/equipment kit at all times for inventory purposes. **(T-2)**

8.3.6.1. The WWM will approve/sign a single MIL to be used as the standard for all Load Crew CTKs on like mission-design-series aircraft; a copy will be maintained in each support section. **(T-3)**

8.3.6.2. When items such as dispatchable support equipment or dispatchable special tools are issued separately (not contained in a CTK) and contain multiple parts that are required for its use (such as, cartridges containing consumables, cables, hoses, adapters), a MIL of all the items will be provided with the support equipment or special tools to facilitate positive accountability of all items during checkout, transfer, and check in. **(T-3)**

8.3.6.3. The MIL resides in the TCMax®, but a hard copy of the signed MIL must reside with each dispatchable CTK to provide the ability to verify the inventory regardless of location. **(T-3)**

8.3.6.4. If items such as identification tags or dust caps are attached to tools/equipment, they will be secured in a manner that will minimize any possibility of FOD. **(T-3)**

8.3.6.4.1. Items not permanently attached, will be marked/etched with the appropriate CTK number. **(T-3)**

8.3.6.4.2. All items will be listed on the MIL. **(T-3)**

8.3.6.5. Consumables may be placed in CTKs. If so, they will be identified on the MIL as consumables. **(T-3)** Examples of consumables include safety wire, adhesive, wire bundle lacing, solder.

8.3.6.5.1. Do not include common hardware items such as bolts, nuts, and/or screws unless they are required for the tool to perform its intended function. Cartridges or equivalents containing consumable items whether disposable or not will be accounted for to mitigate FOD hazards. **(T-3)**

8.3.6.5.2. **(Added-AFMC)** If not using the MIL or TCMax® to document missing/removed tools then use AFMC Form 61, *Missing/Removed Tools and Equipment*, to annotate missing/removed tools. **(T-2)**

8.3.6.6. Tool sets placed within a CTK will be identified on the CTK MIL by total number of items in the set (such as, Allen wrench set - 9 each Allen wrenches + container for a total of 10). **(T-3)**

8.3.6.6.1. Items identified as too small to be marked, etched, or stamped, as approved by QA, will be annotated with a description of the individual items contained within the set on the CTK/MIL (such as, the variation in the size of the items contained within the set, Apex, file, drill bit, size). **(T-3)**

8.3.6.7. Missing, removed and/or broken tools/items will be documented in the TCMax® if they cannot be replaced immediately. **(T-3)**

8.3.6.7.1. In addition, for dispatchable CTKs, dispatchable support equipment, and dispatchable special tools containing multiple parts, missing, removed and/or broken tools/items will be documented on a MAJCOM/locally generated form, or on the hard copy MIL. **(T-3)**

8.3.6.7.1.1. If a MAJCOM/locally generated form is used, the form will be kept with each dispatchable CTK, dispatchable support equipment and dispatchable special tools. **(T-3)**

8.3.6.7.1.2. Pencil/pen may be used for hard copy MIL documentation and erased/lined through when cleared.

8.3.6.7.2. The EID will be removed from any permanently removed item/tool. **(T-3)**

8.3.6.7.3. A permanently removed (without planned replacement) item/tool constitutes a change to the inventory and requires a new MIL.

8.3.6.7.4. The CTK custodian has the authority to interchange "like" (form, fit, function) items.

8.3.7. Equipment and accessories that do not present a FOD potential and will not leave the work center, support section, or tool room, need not be included in a CTK; however, this equipment must have designated storage locations established. **(T-3)**

8.3.7.1. Designated locations may be work areas or stations.

8.3.8. The CTK Custodian will establish designated locations for test equipment and common accessories (such as, waveguides, attenuators, fittings, cables, adapters) that are not part of a CTK. **(T-3)**

8.3.8.1. As a minimum, designated locations will be labeled to identify the contents. **(T-3)**

8.3.8.2. Industrial shop machinery accessories/attachments (example, blades, arbors, chucks, gears) need not be controlled as tools; however, these items will be maintained in designated storage locations for accountability. **(T-3)**

8.3.8.2.1. As a minimum, storage cabinets/drawers will be labeled to identify the contents. **(T-3)**

8.3.9. Tools/expendable items used for titanium engine blade blending or oxygen system maintenance will be kept in special purpose kits separate from other tools. **(T-3)**

8.3.9.1. In addition to normal CTK identification, the titanium engine blade blending kits will be marked “For Titanium Engine Blade Blending Only”. **(T-3)**

8.3.9.2. In addition to normal CTK identification, oxygen system maintenance kits will be marked “For Oxygen System Use Only”. **(T-3)**

8.3.10. Discard removable (slide on) pocket clips and spare parts from tools when possible (flashlights, continuity testers, small screwdrivers) prior to placement in tool kits. **(T-3)**

8.3.10.1. Do not disassemble or damage tools for sole purpose of removing clips (example, tape measures, rubber switch guards).

8.3.11. Tools not controlled through CTK procedures are NOT authorized on the flightline, or in any maintenance area (for example, personal Mini Maglite® flashlights, Leatherman®, Buck Knives®). **(T-3)**

8.3.11.1. Units will mark and control equipment that a workcenter assigns/issues to an individual IAW MAJCOM supplements. **(T-3)**

8.3.11.2. Personally purchased tools are not authorized. **(T-3)**

8.3.12. Flashlights, lanterns, portable lighting devices and light sources will conform to the requirements of TO 00-25-172 when used during servicing operations; TO 1-1-3 when used during fuel cell maintenance; and DESR6055.09_AFMAN 91-201 when used in explosive environments. **Note:** Aircraft and equipment TOs may dictate additional restrictions.

8.3.13. **(Added-AFMC)** Tool Replacement Procedures. A stock of spare tools is authorized to replace broken, worn, or missing tools to prevent unnecessary work delays. Spare and consumable tools are highly pilferable and pose an increased fraud, waste, and abuse potential. **(T-2)** MXG/CC (MXW/CC or equivalent functional authority for maintenance) will establish strict controls to include the following:

8.3.13.1. **(Added-AFMC)** Inventory and document all replacement Spare tools quarterly. **(T-2)**

8.3.13.2. **(Added-AFMC)** Replacement spare tools will not be issued until the receipt of the unserviceable tool or completed AFMC Form 310, *Lost/Found Item Report*, has been received. **(T-2)**

8.3.13.3. **(Added-AFMC)** Replacement spare tools must be marked with the CTK WWID prior to issue. **(T-2)**

8.3.13.4. **(Added-AFMC)** If previously issued serviceable spare tools are to be used as replacement tools, completely de-etch any prior assigned markings before placing in spare tools storage. **(T-2)**

8.4. TMDE Management Guidelines. Support Sections will designate a TMDE Monitor who will act as the focal point with the designated servicing PMEL (whether on-base or off-base) for managing the TMDE calibrations requirements for the owning work center. **(T-3)** The TMDE Monitor will:

8.4.1. Establish procedures for turn-in and pick-up of TMDE requiring calibration. **(T-3)**

8.4.2. Coordinate emergency calibration requirements. **(T-3)**

8.4.3. Review quarterly TMDE schedules and annual master identification (ID) lists within 5 duty days of receipt from servicing PMEL. **(T-3)**

8.4.3.1. Forward any corrections to the servicing PMEL within 3 duty days to have the PMEL Automated Management System/MIS updated. **(T-3)**

8.4.4. Take necessary actions to minimize the late delivery of TMDE for scheduled calibration. **(T-3)** Servicing PMEL will notify OWC of overdue TMDE under established procedures. **(T-3)**

8.4.5. Use PMEL Automated Management System or equivalent MIS (as coordinated with supporting PMEL) to control TMDE processed for maintenance. **(T-3)**

8.4.6. Ensure TMDE submitted for calibration has all required documentation complete, the AFTO Form 350 (as applicable) provides adequate malfunction description and accessories/ items required for calibration accompany the TMDE to include batteries (as applicable). **(T-3)**

8.4.7. Ensure classified TMDE is protected IAW AFI 16-1404.

8.4.8. Ensure TMDE shipped off base for calibration or repair and return is shipped by traceable means and IAW DAFI 24-602V2.

8.4.8.1. The TMDE Monitor will maintain a file consisting of all supporting documentation for each type of shipment. **(T-3)**

8.4.8.2. Safeguard any IUID marks during calibration/TMDE activities to the extent possible. In the event the UII is damaged during calibration activities, the TMDE Monitor will notify the responsible Equipment Custodian and/or EAE to replace the mark with the same UII. **(T-3)**

8.4.9. For deployment purposes, ensure equipment, tools, and HAZMAT items are properly identified, prepared, and documented IAW AFI 10-403.

8.5. Tool Accountability and Serviceability. Flight CC/SUPT and Section NCOICs/Chiefs, through CTK Custodians, are responsible for tool and equipment accountability and control (knowing where tools are and who has responsibility for them). When a person signs for a tool or piece of equipment, they are considered the user and accountable for the item until it is returned to the tool room and accountability transfers back to the CTK Custodian (through a representative or tool room employee).

8.5.1. All units must use TCMax® for accountability and control of tools and equipment. **(T-3)** Contractors and MEOs are not required to use TCMax® unless specified in the Performance Work Statement/Statement of Work. **Note:** Refer to **Paragraph 8.1** for exceptions.

8.5.1.1. Units are required to electronically back up TCMax® at least once a month. **(T-3)**

8.5.1.1.1. This backup must be kept physically and electrically separate from the computer that houses the tool control database. **(T-3)**

8.5.1.2. Units will use TCMax® to:

8.5.1.2.1. Track the issuance and receipt of all assigned tools, equipment, tool kits, HAZMAT items, TOs (does not apply to TOs, equipment and HAZMAT kept in a shop and not dispatched). **(T-3)**

8.5.1.2.1.1. HAZMAT items issued for one time use (oil cans, hydraulic cans, mixing compounds) are supply items and do not have to be tracked in TCMax®. However, HAZMAT and supply procedures will be followed. **(T-3)**

8.5.1.2.2. Track authorizations/restrictions for special tools/equipment (by individual). **(T-3)**

8.5.1.2.3. Track CTK and Support Section inspections. **(T-3)**

8.5.1.2.4. Track spare, lost, damaged, and/or removed tools. **(T-3)**

8.5.1.2.5. Develop and manage tool/equipment inventories. **(T-3)**

8.5.1.2.6. Develop and manage deployment kits (import/export). **(T-3)**

8.5.1.3. If TCMax® is not available (such as at a deployed location), units will use the AF Form 1297, Temporary Issue Receipt, a MAJCOM, or locally approved form for accountability and control of CTKs, equipment, and tools. **(T-3)**

8.5.1.3. **(AFMC)** If an automated system is not available, units may use AFMC Form 62, *CTK Inventory and Control Log*, to sign in and out CTKs/TKs and use AFMC Form 61 to document Missing/Removed tools.

8.5.2. The CTK Shift Supervisor will account for all dispatchable/decentralized CTKs, tools, and equipment at the beginning and end of each shift. **(T-3)**

8.5.2. **(AFMC)** If an automated system is not available, units may use AFMC Form 62 or computer generated equivalent to document separate shift inventories.

8.5.2.1. Shift inventories must be documented by both outgoing and incoming personnel. **(T-3)**

8.5.2.2. CTKs present during tool room shift inventories do not need to be opened for inventory.

8.5.3. At least annually or when the CTK Custodian changes, conduct a comprehensive inventory of all dispatchable/decentralized tools, non-Custodian Authorization/Custody Receipt Listing (CA/CRL) equipment, and CTKs. **(T-3)**

8.5.3.1. The purpose of this inventory is to perform an extensive inspection of all tools and non-CA/CRL equipment, to include condition, identification markings, and accuracy of the MIL/CRL Supplemental Listing.

8.5.3.2. CTK Custodians will ensure all tools are inspected for serviceability IAW TO 32-1-101, *Use and Care of Hand Tools and Measuring Tools*.

8.5.3.3. CTK Custodians will document these inventories and maintain the most current inventory. **(T-3)**

8.5.4. Users will perform a visual inventory of all dispatchable/decentralized CTKs when issued for use, at the completion of each job or tasks, and when returned to the tool storage facility. **(T-3)** Users will refuse unserviceable items and return to CTK custodian. **(T-3)**

8.5.4.1. Users will accomplish a CTK inventory prior to operation of any aircraft or equipment when maintenance actions are performed (such as, engine run, landing gear retraction, flight control operational checks). **(T-3)**

8.5.4.2. Users will perform an immediate and complete inventory of all CTKs when returning to the work area after sheltering for real-world/exercise events. **(T-3)**

8.5.4.3. Users will ensure dispatchable tools, equipment, eTools and CTKs are locked and/or secured when left unattended. **(T-3)**

8.5.4.4. CTK's signed out to deployed ABDR Individual/Assessor are not considered "issued-for use" unless signed open on an AFSC Form 309, *AFSC Tool Control Inventory Record*, or issued from a local deployed support section for the purpose of inventory requirements.

8.5.5. eTools: Units will use the following procedure to maintain positive control of assigned eTools:

8.5.5.1. Manage eTools IAW TO 00-5-1, and this instruction. **(T-3)**

8.5.5.2. Track dispatchable eTools in TCMax®. **(T-3)**

8.5.5.3. Ensure only serviceable eTools with current technical data are available for checkout, and any missing plugs/covers/doors are documented IAW **Paragraph 8.3.6.7**. **(T-3)**

8.5.5.4. Make maximum use of eTool warranties. **(T-3)**

8.5.5.5. Ensure eTools are used for official and authorized purposes IAW TO 31S5-4-etool, 17 & 33 Series instructions, MAJCOM guidance, and this DAFI.

8.5.5.5.1. Not install unauthorized files or software (such as, games, mp3s). **(T-3)**

8.5.5.5.2. Not use unauthorized external media devices to retrieve data from removable hard drives. **(T-3)**

8.5.5.6. Coordinate with the local Cybersecurity Liaison and/or Information System Security Managers to identify publish local guidance on restrictions for the use of eTools/PEDs in classified processing areas. **(T-3)**

8.5.5.7. Establish procedures for shipping TOs, eTools, and required support equipment needed to ensure eTools availability to support mobility and deployed operational requirements. **(T-3)**

8.5.5.8. If applicable, units will update Defense Integration and Management of Nuclear Data Services (DIAMONDS) hardware and status IAW TO 11N-3150-8-1, *USAF DIAMONDS Policy and Procedures*. **(T-3)**

8.5.5.8.1. For accountability, DIAMONDS laptops and hardware must be managed and tracked in TCMax®, but do not require placement on unit equipment account IAW TO 11N-3150-8-1.

8.6. Tool and Equipment Marking and Identification.

8.6.1. To ensure tool rooms have unique identifiers, wings (or equivalent) must ensure other units within the same wing or Personnel Assignment Symbol (PAS) code do not duplicate the WWID. **(T-2)** MAJCOMs/ANG will develop, sustain, and review annually a complete listing of all the WWID utilized within their MAJCOM/ANG. MAJCOMs will update AF/A4LM with WWID changes as they occur. Reference the AF/A4LM SharePoint® site: **https://usaf.dps.mil/sites/haf-a4/A4L/AF_A4LM/Policy/SitePages/Home.aspx**

8.6.1.1. All units must permanently mark their tools and equipment with the standard EID. **(T-3)** GSU may use the parent wing EID. Replacement spare tools stored in the tool room do not need to be etched until placement in a specific CTK.

8.6.1.2. The EID will consist of nine characters (numbers/letters) of which the first four characters will be a unique WWID code. **(T-3)**

8.6.1.2.1. The WWID identifies the base (first and second character), unit (third character), and shop (fourth character). The remaining five characters are available for tool/CTK equipment numbering.

8.6.1.2.1.1. The first two characters of the WWID in the EID are based on the wing/unit PAS base code. Multiple wings (or equivalent) at the same base (example, ANG, AFR, and RegAF) must have different WWID codes. **(T-3)**

8.6.1.2.1.2. The third and fourth characters designate the unit and shop by using unique/distinguishable characters. To ensure tool rooms have unique identifiers, wings (or equivalent) must ensure other units within the same wing or PAS code do not duplicate the first 4 characters of the EID. **(T-3)**

8.6.1.2.1.3. Request additional "base" code information from AF/A4LM at: **AF/A4LM Maintenance Policy, AF.A4LM.Maintenance.Policy@us.af.mil**.

8.6.1.3. The unit will establish the remaining five characters (any combination of numbers/letters) for CTKs, tools, and dispatchable equipment identification. **(T-3)**

8.6.1.4. Units must place the 9-digit EID on all CTKs, tools not assigned to a box, and dispatchable equipment that is of sufficient size. **(T-3)**

8.6.1.4.1. The 9-digit EID must be placed on the outside of dispatchable CTKs. **(T-3)**

8.6.1.4.2. Tools located inside the toolbox may be marked with less than 9-digits but must contain the 4-digit WWID and will have identifying character(s) that ties the tool back to the CTK. **(T-2)** For example, tools inside an assigned dispatchable CTK "U6JG00001" may be marked "U6JG1." Units may affix non-metallic barcode labels on tools to prevent re-etching as long as the use of the tool and its work environment does not normally result in excessive damage to the label making it unreadable.

8.6.1.4.3. Tools will be marked with the most current EID. **(T-3)**

8.6.1.4.4. All previous CTK identifiers will either be removed or marked out (this does not include PMEL markings). **(T-3)**

8.6.1.4.5. Small tool sets and/or items that cannot be marked as described in **Paragraph 8.3.6.6** above (such as drill bits, Allen wrenches in sets, apexes) will be maintained in a container marked with the EID and an identifying character(s) that ties the tool back to the CTK along with the number of tools contained. **(T-3)**

8.6.1.4.5.1. The container is counted as one of the items.

8.6.1.5. MXG/CCs may require use of the EID in addition to AFTO Form 66, *TMDE Bar Codes (Polyester Film)*, for TMDE routinely (example, once per week) dispatched from a work center or use of the AFTO Form 66 alone.

8.6.1.6. For items that physically or mechanically check tolerances that require calibration, do not etch, or stamp an EID in any manner that will affect calibration or the ability to calibrate. **(T-3)**

8.6.1.6.1. If marking is in question consult TO 00-20-14 and/or PMEL to validate applicable marking criteria.

8.6.2. Permanently mark (by etching or other means) grease guns, dispensing cans, spray bottles, pump oilers, and similar containers with the type of grease, fluid, or other liquids and Military Specification (MILSPEC) of the contents. **(T-3)** If the MILSPEC is subdivided into Grades, Classes, or Types, include that info on the permanent marking. **(T-3)**

8.6.2.1. If no MILSPEC exists, the item will be marked with the manufacturer's name, part number/NSN from the applicable Safety Data Sheet. **(T-3)**

8.6.2.2. Keep hoses and fittings separate for each type of grease. **(T-3)**

8.6.2.3. If containers are used to hold or apply substances classified as hazardous materials, ensure labeling requirements are IAW AFI 90-821, *Title 29 Code of Federal Regulations 1910.1200(f), Occupational Safety and Health Standards, Toxic and Hazardous Substances*. **(T-0)**

8.6.3. Prior to etching tools and equipment, consult applicable technical order to ensure no special circumstances apply for the item being etched (such as, fiberglass handled hammers are etched IAW TO 32-1-101 and safety glasses. **(T-3)**

8.6.4. CTKs, tools, and dispatchable equipment that may possess a unique serial/tracking number must be marked with an EID number. **(T-3)**

8.6.4.1. If the item cannot be marked, etched, or stamped, annotate the additional designator on the CTK contents list. **(T-3)**

8.6.5. Items that are assembled and are not intended to be disassembled during use, require only one mark/etch/stamp and one entry in the MIL (such as, scribes, flashlights, grease guns, feeler gauges). **(T-3)**

8.6.6. Remove the EID from unserviceable tools and tools removed from the CTK (with the exception of warranty tools where removal of EID may void the tool warranty) and update TCMax® and the MAJCOM/locally generated form or hard copy MIL accordingly. **(T-3)**

8.6.7. Ensure all tools which are accountable on a CA/CRL must be marked with a UII. **(T-3)** Units need to safeguard any IUID marks. In the event the UII is damaged, notify the responsible Equipment Custodian and/or EAE to replace the mark with the same UII.

8.7. Locally Manufactured, Developed, or Modified Tools and Equipment. All locally manufactured, developed, or modified tools and equipment used on aerospace equipment must be approved by the MXG/CC, their equivalent, or a designated representative and meet the requirements described in **Chapter 9**.

8.7.1. This procedure does not apply to tools and equipment authorized for local manufacture in MDS specific technical data or equivalent engineering approved document. QA will coordinate on all requests for approval and use of locally manufactured, developed, or modified tools or equipment. **(T-3)**

8.7.2. Work centers will review items and requirements every two years for applicability and current configuration. **(T-3)** See **Chapter 6** and **Chapter 9** for additional guidance.

8.7.3. Weapons loading, maintenance and armament systems locally designed equipment must be coordinated through the WWM. **(T-3)** Weapons loading, maintenance and armament systems locally designed equipment must meet the following requirements:

8.7.3.1. In the event munitions/armament LME, is not included in technical data or listed on the MMHE Focal Point website **https://cs2.eis.af.mil/sites/10134/sitepages/home.aspx**, contact the MMHE Focal Point AFLCMC/EBDW, 615 Apalachicola Road, Suite 101, Eglin AFB, FL 32542-6845 to establish/validate LME disposition.

8.7.3.1.1. Units must use MMHE Focal Point-designed munitions/armament LME for new procurements if a design exists and fills the requirement. **(T-2)**

8.7.3.1.1.1. Munitions/armament LME is specialized equipment designed to interface with or support munitions or armament suspension equipment (such as, tools, handling dollies, storage racks, maintenance stands, transport adapters). All munitions/armament LME contained on the MMHE Focal Point web site meets applicable DAFMAN 91-203 and USAF standards and is approved for local manufacture and use at unit level AF-wide. Drawing packages for these items are available to the unit via the MMHE Focal Point website.

8.7.3.1.1.2. Munitions/armament LME specifically designed to interface with or support munitions that are not contained in technical data or on the MMHE Focal Point website (such as, hardened/protective aircraft shelter missile racks, Y-stands, munitions chocks, specialized tools) must be coordinated at unit level and forwarded to the MAJCOM Functional Manager for coordination, evaluation, or both. **(T-2)**

8.7.3.1.1.2.1. If the MAJCOM Functional Manager determines the item has AF utility, the drawings will be forwarded to ACC/A4WC for review and addition to the MMHE Focal Point Master Project List that results in formal development and placement onto the MMHE Focal Point website.

8.7.3.1.1.3. Munitions/armament LME not designed to interface with or support munitions that are not contained in technical data or on the MMHE Focal Point web site must be approved by QA. **(T-2)**

8.7.3.1.1.3.1. Units are encouraged to forward any such approved LME for possible inclusion on MMHE Focal Point web site by sending an approved drawing package to the MAJCOM Functional Manager for coordination/evaluation.

8.7.3.1.2. All LME must meet applicable DAFMAN 91-203 and USAF standards.

8.7.3.1.3. All equipment designated for use with nuclear weapons test and handling must meet requirements in DAFI 91-101.

8.7.3.1.4. All weapons loading, maintenance and armament systems LME must be maintained and inspected for serviceability on a regular basis IAW applicable TO 00-20-series, TO 35D-1-2, *Maintenance Instructions WIPB-Miscellaneous Munitions*

Handling and Support (Munitions Related), and TO 35D-2-2 *Munitions Materiel Handling Equipment Miscellaneous And Locally Manufactured – Armament Related*.

8.7.3.1.4.1. AFTO Form 244, or equivalent, must be maintained for all LME items (racks, stands, adapters) except hand tools. **(T-3)**

8.7.3.1.4.2. Equipment without technical data must, as a minimum, be inspected every 180 days for corrosion, physical defect, and lubrication as required. **(T-3)**

8.8. Tool Room Operations and Tool Security.

8.8.1. Operations. Tool rooms will be set up to ensure accountability. **(T-3)**

8.8.1.1. Procedures will be established to ensure custodial control. **(T-3)**

8.8.1.2. Tools will not be issued individually from dispatchable CTKs. **(T-3)**

8.8.1.2.1. When a recurring need exists for common tools to be issued individually, (such as, hammers, screwdrivers, pliers, drills, wrenches) individual issue bins/drawers may be established as a CTK within the tool room. **(T-3)**

8.8.1.3. Process reports for tools that are lost, damaged, or destroyed, due to neglect IAW AFI 23-101. **(T-3)**

8.8.2. Security.

8.8.2.1. The tool room must be capable of being locked and afford protective measures such as monitoring, 24-hour coverage, or controlled key access. **(T-3)**

8.8.2.1.1. When all CTKs are not capable of being secured in the tool room, the Section NCOIC/Chief will design a process to prevent the unauthorized use or access to tools and equipment. **(T-3)**

8.8.2.1.2. Due to space and facility limitations, it may not be possible to store oversized tool kits in the tool room.

8.8.2.2. Locks will be used on tool kits stored outside the CTK to provide a physical barrier to opening the container lid, drawer or door and prevent the unauthorized removal of tools. **(T-3)**

8.8.2.2.1. Locks are not required on tools and equipment that are stored within secured tool rooms or work centers.

8.8.2.2.2. Tools and equipment will never be secured to the exterior of an aircraft. **(T-3)**

8.8.2.2.3. Tool kits located within high traffic, controlled movement areas or that could limit aircraft movement or be exposed to jet blast are required to be locked when unattended/not in use and moved to un-obstructive/exposed location. **(T-3)**

8.8.2.2.3.1. Alert Aircraft in Protection Level Areas 1, 2, and Alert Status Aircraft in Protection Level Area 3. CTKs in these areas that are directly supporting alert status aircraft do not have to be locked when unattended and not in use as long as they are inventoried at the beginning of each shift, after each maintenance task, and at the end of each shift.

8.9. Lost Item/Tool Procedures.

8.9.1. Authorization to clear Red X's when an item/tool cannot be located should be no lower than Maintenance Supervision IAW **Chapter 2**. **Exception:** For contingency operations with absence of Maintenance Supervision, aircraft commander assumes responsibility.

8.9.2. Supervisors need to ensure all assigned personnel are familiar with lost item/tool procedures. If an item/tool or a portion of a broken item/tool is discovered missing, the following procedures apply:

8.9.2.1. The person identifying the missing item/tool will search the immediate work area for the item/tool. **(T-3)**

8.9.2.1.1. If not found, after completing an initial search the individual will notify the expediter/Pro Super or equivalent. **(T-3)**

8.9.2.2. Place a Red X in the aircraft or equipment forms of all affected aircraft or equipment with a description of the item/tool and a specific, last known, location of the item/tool. **(T-3)**

8.9.2.3. Expediter/Pro Super or equivalent will immediately notify the Flight CC/SUPT, Support Section, MOC, and QA. **(T-3)**

8.9.2.3.1. Initiate a thorough search for the item/tool. **(T-3)**

8.9.2.3.2. Initiate the lost tool report if tool is not located during initial search. **(T-3)**

8.9.2.3.3. **(Added-AFMC)** Initiate AFMC Form 310, if the item is not found within two hours of start of search. Maintain reports in QA section for two years. **(T-2)**

8.9.2.4. If it is suspected that the item/tool has fallen into an inaccessible or unobservable aircraft area, perform an NDI or use borescope equipment to locate the lost item/tool. **(T-3)**

8.9.2.4.1. If the item/tool is in an inaccessible area that poses no FOD threat and the action is to leave the item/tool in place, the x-ray (or equivalent) with the identification of the exact item/tool location and copies of all information concerning the lost item/tool are maintained in the aircraft historical file until the item/tool is recovered. **(T-3)**

8.9.2.5. If at any time during the investigation the item/tool is found, notify the Flight CC/SUPT, Support Section, MOC, QA, expediter, Pro Super or equivalent, and the owning work center. **(T-3)**

8.9.2.5.1. If the item/tool is found, but is inaccessible, Maintenance Supervision may explore other possible actions to include having the unit or a DFT disassemble the aircraft to remove the item/tool.

8.9.2.5.1.1. If the aircraft MDS is one that has a PDM or is scheduled for depot modification, any inaccessible lost item/tool will be listed with location on the AFTO Form 345, Aerospace Vehicle Transfer Inspection Checklist and Certification, for removal by the depot. **(T-3)**

8.9.2.6. Maintenance Supervision will determine when the search for the lost item/tool may be discontinued. **(T-3)** If the item/tool is not found:

8.9.2.6.1. Notify the MOC and the MXG/CC when the search for the lost item/tool has been discontinued. **(T-3)**

8.9.2.6.2. Ensure lost item/tool report is completed IAW locally established procedures. **(T-3)**

8.9.2.6.3. If applicable, ensure the TCMax® is documented IAW **Paragraph 8.3.6.7**. **(T-3)**

Chapter 9

MATERIEL MANAGEMENT SUPPORT

9.1. General. Material management plays a critical role in optimizing mission generation capabilities. Although assigned to LRS, all DMS personnel must be integrated into daily mission generation operations both at home station and deployed. The critical nature of direct interaction between maintenance and materiel management activities at the point of maintenance provides units direct access to materiel management SMEs to accurately identify, communicate, acquire, or disposition materiel management support necessary to maximize combat capability. This **Chapter**, coupled with AFI 23-101, provides the minimum materiel management support requirements necessary to provide the best possible opportunity for success in meeting mission generation requirements. The AF has consolidated materiel management support under the AF Sustainment Center. The AF Sustainment Center provides fleet-wide supply support to all AF weapon systems and leverages consolidated repair facilities and ALCs capability to optimize weapon system availability.

9.2. Decentralized Materiel Support. Decentralized Materiel Support personnel coordinate maintenance and materiel management actions and manage supply transactions for the MXG. The duty location for DMS personnel is the applicable maintenance unit as agreed upon by the MXG and base 2S0 functional manager. Personnel rotations, as determined by the base 2S0 functional manager will be accomplished with consideration of operational, career development, and training requirements. **(T-3)**

9.2.1. In units with Decentralized Materiel Support personnel, the LRS Superintendent exercises control and career-field development opportunities for Decentralized Materiel Support (2S0XX) personnel. **(T-3)**

9.2.2. In units/work centers directly supported by Decentralized Materiel Support personnel, the Logistics Readiness Squadron Materiel Management Flight is responsible for ensuring materiel management support roles and responsibilities listed are completed IAW 23-series publications.

9.2.3. LRS Superintendent will ensure complete DMS coverage of MXG duty periods, to include weekend duty. **(T-3)**

9.2.4. At a minimum, DMS personnel will provide the following support functions to the maintenance unit: monitor and track assets in the repair cycle, resolve supply support problems, and report aircraft parts status and changes in base level repair capability to maintenance supervision. **(T-3)**

9.2.4.1. Decentralized Materiel Support personnel also support maintenance in processing issue requests, researching sources of supply, entering manual requisitions (part number only), updating exception code lists, and resolving other peculiar maintenance supply problems. **(T-3)**

9.2.4.2. DMS personnel will manage processes, in addition to those previously described, such as parts ordering, backorder review and validation, Readiness Spares Packages, and TNB. **(T-3)**

9.3. Supply Discipline. Supply discipline is the responsibility of all military and civilian personnel regardless of grade or position. Personnel at all levels need to ensure the practice of good supply discipline.

9.4. Readiness Spares Package Review. Maintainers play a critical role in the annual Readiness Spares Package review process. This role includes active maintenance participation in the base level validation process conducted by the LRS and their MAJCOM during the annual Readiness Spares Package pre-review process in preparation for the Weapon System Program Manager final review. Close maintenance-materiel management collaboration is essential to ensure RSPs are properly sized to support contingency maintenance requirements. See DAFMAN 23-122 for further details.

9.5. Bench Stock. Flight CCs/SUPT and/or Section NCOICs/Chiefs will determine the contents of their bench stock IAW qualification criteria in DAFMAN 23-122. **(T-3)** Examples of bench stock items include nuts, bolts, cotter keys, washers, resistors, capacitors, light bulbs, sealants, and batteries. Bench stock levels are managed and based predominantly on consumption. Monthly and semi-annual bench stock listings are provided by the LRS/DMS. A thorough review of these listings is extremely important to ensure that bench stock supports the mission efficiently and economically.

9.5.1. Bench stock assets are organizationally purchased and therefore organizationally owned. Appointment of DMS personnel as bench stock monitors must be agreed upon by the applicable Squadron Superintendents.

9.5.2. Mark bins containing 50 percent or less of the authorized quantity to facilitate monthly inventories. **(T-3)** Do not include items coded TCTO, unacceptable for AF use and critical in bench stock. Controlled Item Code S and C may be in bench stock with MAJCOM approval. Other controlled Item Codes are authorized with written approval from Unit Commander. See DAFMAN 23-122 for additional information.

9.5.3. Work center supervisors will:

9.5.3.1. Semi-annually complete a bench stock joint review with the Customer Support Liaison Element, Materiel Control. **(T-3)**

9.5.3.1.1. During these reviews, special emphasis needs to be given to items with no demands in the past year and items with excessive quantities not supported by demands. The continuance of stocking such items is the exception and not the normative process. See DAFMAN 23-122 for further details.

9.5.3.2. Work with DMS or LRS/materiel management activity to ensure demilitarization criteria is identified, and used or excess items are disposed of IAW procedures in DAFMAN 23-122.

9.6. Consumable Readiness Spares Package. The Consumable Readiness Spares Package process provides requirement and asset visibility, has automated transfer and deployment procedures, has the capability to provide the correct priority and project-coded replenishment requisitions, and eliminates redundant requirements. Additionally, Consumable Readiness Spares Package procedures provide MAJCOMs with a standard process to support consumable item requirements during contingency operations. Refer to DAFMAN 23-122 for Consumable Readiness Spares Package procedures and options.

9.7. Shop Stock. Shop stock includes gas cylinders, random length bar stock, sheet metal, plastic, fabric, electrical wire, and similar items not normally included in bench stocks. Maintain shop stock for day-to-day operations. Monitor shop stock to prevent materiel from becoming excessive or outdated. Shop stock should not normally exceed 90-days usage, or the unit of issue or unit pack, whichever is greater. Store shop stock near/adjacent to bench stock items, if practical, but do not mix them together. Clearly identify materiel as "Shop Stock" and label them with noun, national stock number or part number, unit of issue, and shelf-life, if applicable.

9.8. Operating Stock. Operating stock includes connector dust covers, hydraulic line caps/plugs, and similar items that are normally recovered after use and re-used. Store operating stock near/adjacent to bench stock items, if practical, but do not mix them together. Monitor operating stock to prevent it from becoming excessive or outdated. Retain partially used bench stock items in bench stock and not in operating stock. Identify, tag, and turn in items with no forecasted use IAW AFI 23-101. Clearly identify items as "Operating Stock" and label them with noun, national stock number or part number, unit of issue, and shelf-life as applicable.

9.9. Work Order Residue. Work order residue includes expendable bit/piece items left over from maintenance work orders or bench stock deletions. Store work order residue near/adjacent to bench stock items, if practical, but do not mix them together. Ensure excesses are consolidated for turn-in to LRS at least annually. Clearly identify items as "Work Order Residue" and label them with noun, national stock number or part number, unit of issue, and shelf-life as applicable. Control all work order residues used on or around aircraft, uninstalled engines, and AGE.

9.10. Adjusted Stock Levels. Adjusted stock levels are used when the demand level or consumption is inadequate to support the requirement. A single occurrence of a mission limiting status is not sufficient reason to establish an adjusted stock level but should result in an LRS/materiel management activity review of demand data for accuracy. The using work center, with assistance from LRS/materiel management activity, will prepare the request IAW DAFMAN 23-122 and provide adequate justification (such as, seasonal materiel requirements, long lead-time items, infrequent use components that cause an NMC condition and result in a new procurement or excessive lead-time to restock). Route the request through the applicable Squadron Maintenance Supervision for approval prior to submitting to LRS/materiel management activity. Using work centers will maintain a master file of approved adjusted stock level items and follow-up on all requests until completed. **(T-3)**

9.11. Shelf-life Items. Using work centers will control the quantity and inspect (Type I and Type II) shelf-life items kept in unit bench stock, operating/shop stock, and work order residue IAW DAFMAN 23-122. Personnel managing bench, shop, or operating stocks will:

9.11.1. Identify serviceable shelf-life items/locations with a colored and/or highlighted label that clearly states the items expiration date. **(T-3)**

9.11.2. Check expiration dates on issued items and do not accept outdated items. **(T-3)** Refer to DAFMAN 23-122 for outdated and/or unserviceable shelf-life items.

9.11.3. Not open shelf-life containers until needed and use the oldest items first. **(T-3)**

9.11.4. Ensure shelf-life material stored in other than original containers are marked with original shelf-life expiration codes. **(T-3)**

9.11.5. Recycle, reclaim, or turn-in for disposal, shelf-life items which are loose in the bin and expiration dates cannot be determined. **(T-3)**

9.12. Equipment Items. Flight CCs/SUPTs and/or Section NCOICs/Chiefs will review equipment items needed for mission accomplishment IAW AFI 23-101.

9.12.1. Equipment Custodians will contact the EAE for assistance in researching and preparing documents for gaining authorizations and ordering equipment items IAW AFI 23-101. Refer to DAFMAN 23-122, for the required procedures to order and deploy equipment items.

9.13. Special Purpose Recoverable Authorized Maintenance (SPRAM). SPRAM assets are fault isolation spares, shop standard spares, training spares, -21 TO spares (AME), test station spares, and stand-alone spares. These assets are Expendability, Recoverability, Reparability Code (ERRC) XD/XF items, which are controlled and managed as in-use supplies.

9.13.1. Flight CCs/SUPTs and/or Section NCOICs/Chiefs will review all SPRAM authorizations annually and certify as valid IAW AFI 23-101, DAFMAN 23-122, and DAFI 21- 103. **(T-3)**

9.14. Supply Assets Requiring Functional Check, Calibration, or Operational Flight Programming. Maintenance sections must identify items requiring functional checks, calibration, or operational flight programming prior to use. **(T-3)**

9.14.1. Maintenance sections will prepare a list of items, (including the repair section's organization and shop code) for items requiring functional checks, calibration, or operational flight programming. **(T-3)**

9.14.1.1. The list will be routed through Maintenance Supervision to the LRS. **(T-3)**

9.14.1.2. This list will be updated/validated IAW DAFMAN 23-122. **(T-3)**

9.14.2. The LRS/management materiel activity issues the items to repair sections when assets are initially received on station, when functional checks, calibration, or programming is due or when serviceability is doubtful.

9.14.3. If a Part issue requiring a functional check, ensure it is not restricted in the weapon system -6 TO. Refer to TO 00-20-3 for functional check and frequency requirements.

9.15. Time Compliance Technical Order (TCTO) Kit Procedures. TCTO kit management is a shared responsibility between maintenance and supply IAW TO 00-5-15 and AFI 23-101.

9.15.1. Initiate requests for kits, parts, and special tool requirements through LRS as outlined in **Chapter 14**.

9.15.2. Transfer TCTO kits with aircraft or equipment. DAFMAN 23-122, TO 00-5-15, and TO 00-5-1 contain detailed guidance for the transfer of TCTO kits.

9.15.3. Retain TCTO kits for aircraft returning to the unit for TCTO compliance.

9.16. Supply Points. Supply points may be established within individual work centers when time or resources required to move items dictates the need to do so.

9.16.1. Storage space for the supply points is provided by the supported work center.

9.16.2. Management of the supply point processes will be agreed to and documented by participating group commanders. **(T-3)** Participating group commanders will require appointment of supply point monitors to manage and account for supply point assets as part of their agreement. **(T-3)** LRS Materiel Management Activities will maintain overall accountability and control of supply point assets. **(T-3)**

9.16.3. Supply points must be reconciled semi-annually by the Supply Point Monitor. **(T-3)**

9.16.3.1. One of the semiannual reconciliations will be done at the same time as the annual supply point inventory IAW AFI 23-101.

9.17. Local Manufacture. Local manufacturing is an essential part of maintenance unit support. The applicable end-item TOs identify items subject to local manufacture and specific procedures for processing are in DAFMAN 23-122.

9.17.1. DELETED.

9.17.2. MXG directive at a minimum will include:

9.17.2.1. Procedures that specify coordination requirements to include QA, EAE, Wing/Base Safety, and endorsement by the approval authority. **(T-3)**

9.17.2.2. DELETED.

9.17.2.3. Identifying drawing, sample, technical data and DD Form 1348-6, *DoD Single Line Item Requisition System Document*, source requirements as required. **(T-3)**

9.17.2.3.1. Ensure guidance identifies that drawings are obtained from the appropriate repository (such as, Engineering Data Service Center or JEDMICS).

9.17.2.4. Establishing coordination process for all the appropriate fabricating sections to determine the bits and pieces required to manufacture the item. **(T-3)**

9.17.2.4.1. Coordinating bit and piece parts requirements and availability with the LRS/DMS.

9.17.2.5. Identifying all work centers that have action on the AFTO Form 350 for items requiring multiple section processing. **(T-3)**

9.17.3. **(Added-AFMC)** Efforts should be made to purchase items in-lieu of local manufacture if practical when the item has been specifically designed and manufactured for the intended use/purpose.

9.18. DIFM Management.

9.18.1. DIFM inputs are critical to recording and getting credit for proper repair cycle times.

9.18.2. DIFM status codes are broken down into three categories: delayed maintenance time, repair time, and AWP time. Repair time is the only time recorded and used to determine the number of assets that should be stocked. Not using the proper codes when they change reduces the number of assets on base.

9.18.3. The roles and responsibilities for DIFM management are identified in AFI 23-101. The LRS/DMS provides the D23 or equivalent to assist each repair section in DIFM Management. The D23 is provided in both maintenance location and stock number sequence. Repair sections

use the D23 to manage the flow of serviceable and unserviceable DIFM assets in the repair cycle and to ensure the DIFM status and location is updated.

9.18.3.1. If a parts request is backordered and the removal of the unserviceable DIFM item does not further limit or restrict the operational capability of the end item, it will be removed and sent to the applicable support section for either repair, NRTS approval, or condemnation with a subsequent turn-in to LRS/materiel management activity (as a credit DIFM) IAW TO 00-20-3.

9.18.3.1.1. Repair assets to the fullest extent authorized.

9.18.3.2. Repairable components will be processed, repaired, and returned to the FSC within the required time frame IAW AFI 23-101.

9.18.3.3. The D23 will not be used to manage serviceable assets.

9.18.4. Repair Cycle Throughput. Throughput is the average time it takes to move individual items through the repair cycle. Timelines for turn-in are outlined in AFI 23-101.

9.18.5. Units will establish local procedures for the control of repair cycle assets throughout the maintenance repair cycle IAW AFI 23-101 and DAFMAN 23-122.

9.18.5.1. Procedures will include methods of accounting for all components and accessories, procedures for control of assets in AWP or AWM status, and procedures and responsibilities for cross CANN, removal of bits and pieces, and scheduling and control of repair cycle assets. **(T-3)**

9.18.6. AWP and cross-CANN assets will be controlled and managed IAW DAFMAN 23-122.

9.18.6.1. Maintenance activities will closely control reparable assets in AWP status. **(T-3)**

9.18.7. Maintenance Turn-In to Supply. Maintenance is responsible for DIFM items until the item is returned to LRS/DMS.

9.18.7.1. Work centers must properly tag and secure repair cycle assets and place items in a leak–proof containment liner (no leaks/stains/tears/punctures), as required. **(T-3)**

9.18.7.1.1. To prevent spillage, any item containing any type of residual fluid, regardless of hazard classification, will be drained, purged, preserved, capped, plugged, and placed in a leak-proof containment liner before placement into a serviceable reusable container for storage or shipment. **(T-3)**

9.18.7.1.2. The work center must comply with packaging, environmental control, inert certification, purge, and preservation requirements as specified in applicable TOs, DAFI 24-602V2, DAFMAN 24-604, and place sufficient copy(s) of the technical document(s) for handling, storage, shipping, and distribution of copies inside the container. **(T-3)**

9.18.7.2. Include AFTO Form 350, **Parts I** and **II**, and a condition tag or label with all items turned into supply IAW TO 00-20-3. **(T-3) Note:** Some DIFM assets may require additional tags.

9.18.7.2.1. Enter the correct action taken code on AFTO Form 350, **Part II.**

9.18.7.3. Accomplish proper reclamation and demilitarization actions on condemned repair cycle assets IAW DAFMAN 23-122 and AFH 23-123, Vol. 2, **Part I**, **Sec. 6C**.

9.18.7.3. **(AFMC) AFMCI 23-111,** *Reclamation of Air Force Property***, and AFI 23-101.**

9.18.7.3.1. **(Added-AFMC)** Base maintenance organizations must properly dispose of consumable, serviceable and unserviceable, items in coordination with the LRS/Materiel Management Activity. Items will be segregated by demilitarization (DEMIL) code and demilitarized by generating unit as required, prior to being turned into Defense Logistics Agency Disposition Services (DLADS). Reference DoDM4160.28-V2, *Defense Demilitarization Program Administration*, paragraph 3.1., Table 1 for DEMIL code definitions/requirements. The Materiel Management Activity will reissue and redistribute serviceable consumable items. **Note:** DEMIL code A are the only assets that do not have a DEMIL requirement but are still required to be transferred to DLADS and will not be disposed of as natural waste.

9.18.7.4. DIFM items (serviceable or unserviceable) will be processed and turned in to LRS IAW AFI 23-101.

9.19. Tail Number Bins (TNB).

9.19.1. Establishment and management of TNBs is a shared responsibility between maintenance and supply. TNBs are storage locations established and controlled to store issued parts awaiting installation and parts removed to FOM. TNBs are set up by tail number, serial number, or identification number.

9.19.2. TNB items used to satisfy MICAP conditions are not CANNs. If a TNB asset is issued to satisfy a part request, maintenance personnel will:

9.19.2.1. Reorder the item and notify the expediter of the new document number. **(T-3)**

9.19.2.2. Update the aircraft forms and the MIS. **(T-3)**

9.19.2.3. If a due-out is created prior to transfer of these items, notify the LRS/materiel management activity to change the "mark-for" field on the due-out detail. **(T-3)**

9.19.3. Seal and store partially completed TCTO kits and parts in the TNB and mark the container or package with the tail number, serial number, or equipment identification number and TCTO number. **(T-3)**

9.19.4. Maintain security and control of TNB assets. **(T-3)**

9.19.5. Track property placed in the TNB by tail number, serial number, or equipment identification number. Each entry will indicate:

9.19.5.1. Date received. **(T-3)**

9.19.5.2. Noun/nomenclature. **(T-3)**

9.19.5.3. Document number. **(T-3)**

9.19.5.4. Status (FOM, Issue/Due-Out Release (ISU/DOR), TCTO). **(T-3)**

9.19.5.5. Removal information (date, time, signature, and employee number of the person who picked up the property). **(T-3)**

9.19.5.6. Remarks. **(T-3)** Enter "NONE" if no remarks are necessary.

9.19.5.7. Current JCN. **(T-3)**

9.20. CANN Actions. See **Chapter 11** for CANN procedures and responsibilities.

9.21. Bench Check and Repair Policy. Maintenance sections bench check items as part of the off equipment troubleshooting process. When workload requires, the Section NCOIC/Chief determines the priority for bench check actions. Specific procedures for bench check and repair policy are provided in TO 00-20-3. The following general guidelines apply:

9.21.1. Order required parts "fill or kill."

9.21.1.1. If the part is not in stock and a MICAP condition exists, backorder the new request.

9.21.1.2. Determine local repair capability before requisitioning off-base support or going lateral support.

9.21.2. Remove the suspected item, fill out the AFTO Form 350, and annotate it as repair and return. Attach AFTO Form 350 to the item; place the item in the repair cycle; and annotate the name of the repair section on the form.

9.21.3. Bench-check, repair, take NRTS action, or condemn the item.

9.21.3.1. If the item is repaired or otherwise determined to be serviceable, the repair section informs the Support Section the item is available for pick-up so on-equipment maintenance action may resume.

9.21.3.2. If the item cannot be repaired, the repair section informs the Support Section to initiate a backordered request and takes appropriate NRTS and condemnation action on the unserviceable asset.

9.22. Maintenance Turn-Around Record Update Processing. Work centers processing TRNs will coordinate with LRS/DMS and follow requirements outlined in AFI 23-101, DAFMAN 23-122, and AFH 23-123.

9.23. Buildup Items. Maintain items requiring build-up prior to use (such as, wheels and tires) in supply points in a built-up configuration.

9.23.1. Send items to appropriate work centers for build-up and return them to the supply point for re-issue.

9.23.1.1. Use AF Form 1297 or control log to control assets sent for build-up when the supply point is operated by other than maintenance personnel.

9.23.1.2. Validate AF Form 1297 daily if over 10 days old.

9.23.2. Local procedures will be established to control assets when maintenance operates the supply point and assets are sent to another organization for build-up. **(T-3)**

9.24. DR Exhibits. DR exhibit procedures for issue, turn-in, and storage are contained in TO 00-35D-54 and AFI 23-101. DRs will be input into the JDRS at **https://jdrs.mil.**

9.25. Destruction of TOP SECRET Material. Destruction of TOP SECRET material requires a receipt according to DODM 5200.01, Vol 3, DOD Information Security Program: Protection of

Classified Information and AFI 16-1404. A copy of the destruction certificate will be included with the turn-in documentation. **(T-0)**

9.25.1. Provide sensitive instruments interior container protection. **(T-3)**

9.25.2. Ensure a copy of the LRU/SRU historical record accompanies turn-in of all items. **(T-3)**

9.26. Certifying Items Associated With Explosives. Ensure items such as MERS, TERS, pylons, launchers, rafts, bomb racks, ejection seats, fire suppression bottles, gun systems and components are certified explosive-free prior to turn-in to LRS and/or Defense Logistics Agency (DLA) Disposition Services. **(T-3)** Refer to TO 11A-1-60 and DAFMAN 21-201 for specific certification requirements.

9.27. (Added-AFMC) Repair Network Enhancement Program (RNEP) (formerly IREP). MXG/CC is the OPR for the RNEP program. RNEP provides maintenance group senior leadership a forum to evaluate current aircraft weapons systems resource and support status, highlight specific problem areas, focus on local repair initiatives to include the Air Force Repair Enhancement Program (AFREP) processes, and discuss ways to improve the overall repair cycle process. **(T-2)**

9.27.1. **(Added-AFMC)** RNEP Meeting. The meeting will be held at least quarterly; chaired by the MXG/CC. LRS CC/CL and MSG/CC will attend RNEP for awareness of materiel management issues affecting aircraft maintenance and mission generation. Recommended participants (as required) include the following: LRS, FSC, representatives from maintenance units, operations and maintenance (O&M) resource advisors, maintenance analysis, AFREP (if applicable), QA, Aircrew Flight Equipment, and others as determined by the MXG/CC. The responsible asset manager should be the focal point to lead the discussion of the key data about a specific part. MXG/CC will conduct a due in from maintenance (DIFM) status review and determine RNEP meeting content based on local requirements. **(T-2)**

9.27.1.1. **(Added-AFMC)** Subject Matter Review. One of the objectives of the RNEP meeting is to increase overall base self-sufficiency for repair and reduce the overall cost of operations. Topics discussed vary based on local requirements, but should include key elements of asset management and costs associated with each of the maintenance stock fund divisions. The number of items reviewed in each topic is determined locally. **(T-3)** Units can determine the specific format/visual aids used for presentation and may include the following information:

9.27.1.1.1. **(Added-AFMC)** Asset Profile/Top Projected MICAP Situations. An asset profile is an in-depth review of an asset identified as critical to mission accomplishment or that causes frequent MICAP situations. Data in an asset profile may include number authorized and on-hand, number repaired and not repaired, number of MICAPs, MICAP cause codes, average repair cycle days, average AWP days, monthly demand, item cost, and financial value of assets in the repair cycle. The overall health of the assets should include reasons for MICAP situations and solutions to resolve them. **(T-3)**

9.27.1.1.2. **(Added-AFMC)** Test Station Equipment Profile. Test station in-commission time is critical to efficient repair cycle output. Test, Measurement, and Diagnostic Equipment and other shop deficiencies may have a negative effect on the

base repair cycle process. The wing should focus on actions which maximize test station capability. **(T-3)**

9.27.1.1.3. **(Added-AFMC)** Wing Self-Sufficiency Initiatives. Initiatives include discussion of new wing, group and squadron AFREP initiatives and other local self-sufficiency repairs. Discussions must include how initiatives are crossfed to appropriate depot, Lead Command and all other like-MDS bases. Discuss product improvement initiatives and maintenance related to Airmen Power by Innovation (API). **(T-3)**

9.27.1.1.4. **(Added-AFMC)** High Cost Maintenance. Unit funded TCTOs/modifications, high cost work centers, SPRAM back orders, financial value of parts in the repair cycle, etc. **(T-3)**

9.27.1.1.5. **(Added-AFMC)** Top CANN Items. Items with significant CANN histories. Review information which includes the number of times items were CANNed in the last 30 days, average CANN occurrences over the last 6 months, projected get well date, and the time required to CANN the item. **(T-3)**

9.27.1.1.6. **(Added-AFMC)** Unit Aircraft Engine Status Review. A status review summary should include number in work, projected production date, and supply drivers for work stoppage. **(T-3)**

9.27.1.1.7. **(Added-AFMC)** Repair Cycle Bottlenecks. Review any area, which impedes the repair cycle process, such as frozen supply, records, supply, rejects, test station backlogs, personnel deficiencies, manpower shortages etc. Review high-time DIFM. **(T-3)**

9.27.1.1.8. **(Added-AFMC)** AWP Summary. Analyze due-outs causes and back order priorities to determine if supply action is required to correct any deficiencies/problems. **(T-3)**

9.27.1.1.9. **(Added-AFMC)** Repair Cycle Throughput. Throughput is the average time it takes to move individual items through the repair cycle. Review/compare the 12-month average versus the current month repair cycle time to determine if progress is being made. 2LM reparable items must be processed without delay as established in AFI 23-101. Non-2LM repairable items must be processed within 4 days for serviceable assets and 10 days for unserviceable assets. **(T-3)**

9.27.1.1.10. **(Added-AFMC)** Part Store Issue Effectiveness. Percentage of aircraft parts issued from the flightline part store vs. the main warehouse. Disregard this element when LRS does not segregate aircraft components into a separate warehouse. **(T-3)**

Chapter 10

MUNITIONS POLICY AND WEAPONS LOAD CREW PROGRAM

10.1. See AFMAN 21-206 for Munitions Policy and Weapons Load Crew Program.

10.1.1. - Table 10.3.. DELETED.

Chapter 11

ADDITIONAL MAINTENANCE REQUIREMENTS AND PROGRAMS

11.1. Facility Housekeeping and Contamination Control. Units will publish housekeeping and contamination procedures which protect the health of workers and maintain areas as free as practical from surface contamination. **(T-3)** Units will:

11.1.1. Ensure Bioenvironmental Engineering approved workplace-housekeeping procedures are employed to prevent the spread of contamination within a work center. **(T-3)**

11.1.2. Emphasize controlling the source of the contamination and ensure workplace personnel follow proper work procedures, PPE use, and hygiene practices. **(T-3)**

11.1.3. Ensure housekeeping procedures will account for the dangers and hazard exposures found in the work center and will be consistent with mitigation methods outlined in DAFMAN 91-203. **(T-3)**

11.2. Personal Wireless Communications Systems Management.

11.2.1. Maintenance Communications. Reliable, redundant, cyber resilient, and effective communications systems are essential for efficient maintenance operations. These systems will provide accurate, timely, secure, programmable frequency and jam resistant communications needed to securely accomplish the maintenance mission in a fully deployed and isolated mode.

11.2.2. Commanders or designated representative will coordinate base Communication Squadron or equivalent to ensure compliance with Personal Wireless Communications Systems management requirements IAW AFI 17-210, DAFI 17-220, *Spectrum Management*, AFI 17-130, and DAFMAN 17-1203. The following general guidelines apply:

11.2.2.1. Allowance for specific radios is shown in AS 660, *Equipment Allowances for Non-Weapon Systems Communications Requirements*, Repair Cycle Data Listing. Process requests for specific radio equipment to support maintenance activities IAW AFI 23-101, AFI 17-210, DAFI 17-220, AFI 17-130, and AFH 23-123 V3, *Air Force Equipment Management*.

11.2.3. A VHF/UHF radio is authorized for use in maintenance operations to facilitate communications between aircraft and maintenance personnel. Additionally, aircrews may relay advance aircraft status information to maintenance personnel using VHF/UHF channels.

11.2.3.1. Maintenance Operations will coordinate procedures for use of these radio communications with operations and other essential wing organizations. **(T-3)**

11.2.3.2. For effective flightline operations, more than one non-tactical radio nets are authorized when large numbers or different types of weapon systems are assigned or when Allowance Standards specify.

11.3. MAJCOM/ANG Special Certification Rosters (SCR). The SCR is a management tool providing supervisors a clear and concise listing of personnel who have been appointed to perform, evaluate, and/or inspect work of a critical nature. Normally, only maintenance requirements that have a definite potential for personal injury or damage to equipment will be included in the SCR. Other tasks requiring special training or qualifications may be managed on the SCR. The SCR is used to build personnel rosters for deployments, shift schedules, and assess workforce capability.

AF/A4LM establishes mandatory SCR Item Titles in Column A of **Table 11.1**. Based on the evolving complexity of weapon systems and the associated task diversity the process of assigning prerequisite to SCR tasks are assigned to using MAJCOMs/ANG as follows: MAJCOMs/ANG A4s will coordinate with their applicable Lead Commands to develop and document SCR item "Prerequisite" criteria in Column B of **Table 11.1** based on task complexity of their assigned weapons systems. MAJCOMs/ANG may add additional items and remove non-applicable Items on their SCR using the standardized **Table 11.1** format provided. MAJCOM/ANG Supplements must include their complete SCR table when coordinating supplement approval with AF/A4LM as described in the opening **Paragraph** of this DAFI. Special Certification approval authority will be accomplished IAW notes at the bottom of the SCR. The MXG/CC and CD are not required to be on the SCR by virtue of their position as the SCR approval authority.

11.3.1. DELETED.

11.3.1.1. Squadron Maintenance Supervision approves individuals in their primary AFSC based on their experience and technical expertise regardless of their assigned skill or position. **(T-3)**

11.3.1.2. MXG/SEL will review and sign SCR actions for those individuals administratively assigned to MXO (QA, AFREP). **(T-3)**

11.3.1.2.1. MXG/SEL will coordinate with the Field Training Detachment (TD) CC/SUPT to validate currency of Field TD personnel on the SCR. **(T-3)**

11.3.1.3. WWM will review and sign WS SCRs. **(T-3)**

11.3.2. TFI units will establish a process for approving SCR additions in a MOA/MOU to provide visibility across participating organizations. **(T-3)**

11.3.3. The MXG/CC may waive selected 5-skill level personnel, in the rank of SrA or higher, for tasks normally requiring a 7-skill level requirement to facilitate the production effort. Waived 5-skill level personnel should be closely monitored and kept to the minimum required to accomplish mission generation.

11.3.3.1. Maintenance Supervision or equivalent will retain file copies of approved waivers. **(T-3)**

11.3.3.1.1. Approved waiver file copies may be discarded if SCR specifically identifies task as waived in the Maintenance Information System.

11.3.3.2. Certified weapons load crew chiefs (load crew member position number 1) by virtue of their task certification and position, serve as inspectors for weapons loading tasks only and do not require a waiver. **Note:** 2W0XX certified munitions inspectors are exempt from these requirements.

11.3.4. MAJCOM Waiver Policy. If local conditions require assignment of other than MAJCOM approved mandatory SCR grade (to include civilian equivalents) and skill level prerequisite requirements and cannot be fulfilled using the MXG/CC authority stated in **Paragraph 11.3.3** then the MXG/CC (or equivalent) must request a waiver from the MAJCOM. **(T-3)**

11.3.5. MAJCOMs may add additional mandatory critical tasks or inspections they deem necessary.

11.3.5.1. Identify each task on the SCR by a specific course code.

11.3.6. SCR Documentation. Flight CCs/SUPTs and Section NCOICs/Chiefs will review each individual's qualifications prior to recommending approval to perform SCR tasks to the appropriate approval level. **(T-3)**

11.3.6.1. AF Form 2426, *Training Request and Completion* or MAJCOM-approved (ANG locally approved) form is used by the work center supervisor to add or remove an individual to the SCR. Additionally, removal from the SCR may be accomplished by lining through the task on the SCR and notifying the training section to update the MIS.

11.3.6.1. **(AFMC)** AFMC Form 64, *Request for Placement on Special Certification Roster*, may be used in lieu of AF Form 2426, *Training Request and Completion*.

11.3.6.2. The appropriate level of authority approves the individual for addition to the SCR as listed in **Table 11.1**.

11.3.6.3. On approval, the UTM IAW AFI 36-2650, loads the approved name into the Maintenance Information System.

11.3.6.4. Flight CCs/SUPTs and Section NCOICs/Chiefs will retain their copy of AF Form 2426 or MAJCOM-approved form until they verify proper loading. **(T-3)**

11.3.6.5. Appointment letters are not required if loaded in the MIS.

11.3.6.6. Work center supervisor, AMU/Flight supervision, Maintenance Supervision, SQ/ CC, or MXG/CC may decertify individuals at any time and remove them from the SCR.

11.3.7. Units will ensure a current copy of the SCR is taken on all deployments. **(T-3)**

Table 11.1. Mandatory Special Certification Roster and Prerequisites.

	A	B
ITEM	**Mandatory SCR Item Titles**	**MAJCOM/ANG Prerequisites**
1	All Systems “Red-X” (no egress, welding, munitions, fuel cell (in-tank work)) (Note 1)	Refer to MAJCOM Supplement
2	Exceptional Release (ER) (Note 1)	Refer to MAJCOM Supplement
3	“Red-X” Downgrade (Note 1)	Refer to MAJCOM Supplement
4	All Systems In Process Inspection (no egress, welding, munitions, fuel cell in-tank work) (Note 1)	Refer to MAJCOM Supplement
5	Installed Engine Run Certifying Officials (Note 1)	Refer to MAJCOM Supplement
6	Aircraft Inlet/Intake/Exhaust Certifying Officials (Note 1)	Refer to MAJCOM Supplement
7	Flexible Borescope Certifying Officials (Note 1)	Refer to MAJCOM Supplement
8	Engine Blade Blending Certifying Officials (Note 1)	Refer to MAJCOM Supplement
*9	“Red-X” by Mission Design Series (Notes 2 and 6)	Refer to MAJCOM Supplement
*10	IPI by Mission Design Series (Notes 2 and 6)	Refer to MAJCOM Supplement
11	“Red-X” and/or In Process Inspection - Limited (Note 2)	Refer to MAJCOM Supplement
*12	DELETED.	DELETED.
13	NWRM packaging (Notes 4 and 5)	Refer to MAJCOM Supplement
14	Installed Engine Run by Mission Design Series (Note 2)	Refer to MAJCOM Supplement
15	Engine Blade Blending	Refer to MAJCOM Supplement
16	QA hot pit certifying officials and QA hot pit certifier augmentees (squadron certifying officials) (Note 1)	Refer to MAJCOM Supplement
17	Hot Refueling PAD Supervisor/"A" Member (Note 2)	Refer to MAJCOM Supplement
18	Hot Refueling Team Member (“B” or “D” member) (Note 2)	Refer to MAJCOM Supplement
19	Aircraft to Aircraft Refueling Supervisor (Note 2)	Refer to MAJCOM Supplement

20	Uninstalled Engine Operations (Test Stand and ETS) Run by TMS (Note 2)	Refer to MAJCOM Supplement
21	Uninstalled Engine Run Certifying Officials by TMS (Note 1)	Refer to MAJCOM Supplement
22	Aircraft Inlet/Intake/Exhaust Inspection/Certification (Note 2)	Refer to MAJCOM Supplement
23	Engine Flexible Borescope Inspections (Note 2)	Refer to MAJCOM Supplement
24	Chief Servicing Supervisor (Heavy Aircraft/Commercial Derivative Aircraft) (Note 2)	Refer to MAJCOM Supplement
*25	Concurrent Servicing Supervisor/Aircraft Turnaround Supervisor (Fighter Aircraft) (Note 2)	Refer to MAJCOM Supplement
26	W&B Certified/Clear Red X (refer to TO 1-1B-50) (Note 2)	Refer to MAJCOM Supplement
27	Impoundment Official (refer to **Chapter 7**) (Note 2)	Refer to MAJCOM Supplement
*28	Impoundment Authority (refer to **Chapter 7** of this DAFI) (Note 1)	Refer to MAJCOM Supplement
29	CANN Authority (Note 1)	Refer to MAJCOM Supplement
30	Auxiliary Power Unit (APU) Operation (In Cockpit) (Note 2)	Refer to MAJCOM Supplement
*31	Calibration Limitation Approval (refer to TO 00-20-14) (Notes 2)	Refer to MAJCOM Supplement
32	CDDAR Team Chief (Note 1)	Refer to MAJCOM Supplement
33	Weapons Task Qualification Manager (WTQM) (Note 1)	Refer to MAJCOM Supplement
34	Weapons Task Qualification Crew (WTQC) (Note 2)	Refer to MAJCOM Supplement
35	NSS and T-9/T-10/T-11/T-12/T-20 sound suppressor Fire Control Panel (Note 2)	Refer to MAJCOM Supplement
36	Aircraft Rapid/Hot Defueling Supervisor (Note 2)	Refer to MAJCOM Supplement
37	Clear Red X when a lost item/tool cannot be located (refer to **Chapter 8**) (Note 1)	Refer to MAJCOM Supplement
38	Aircraft APU Run Certifying Officials (In Cockpit) (Note 1)	Refer to MAJCOM Supplement

Notes:
1. Approved by MXG/CC or equivalent may be delegated IAW **Paragraph 2.4.2.27.**
*2. Approved by Maintenance Supervision or equivalents. May delegate approval authority to the OIC/SUPT or Flight CC/SUPT.
*3. DELETED.
4. Munitions inspectors who are trained and certified may annotate serviceability tags for munitions items (TO 11A-1-10).
5. Appointed by the unit commander (or equivalent) of units possessing NWRM.
6. MAJCOMs determine the AFSC applicability.

Table 11.1. (AFMC) Mandatory Special Certification Roster and Prerequisites.

	A	**B**
ITEM	**Mandatory SCR Item Titles**	**MAJCOM/ANG Prerequisites**
1	All Systems "Red-X" (no egress, welding, munitions, fuel cell (in-tank work)) (Note 1)	MSgt or higher (or civilian equivalent).
2	Exceptional Release (ER) (Note 1)	
3	"Red-X" Downgrade (Note 1)	
4	All Systems In Process Inspection (no egress, welding, munitions, fuel cell in-tank work) (Note 1)	
5	Installed Engine Run Certifying Officials (Note 1)	MSgt or higher (or civilian equivalent), or a fully qualified/certified contractor or AFETS/CETS representative and possess one of the following AFSCs: 2A671, 2A571/2/4, and 2A373/7/8. One year minimum engine-run experience on applicable MDS. MXG/CC may waive qualified TSgts.
6	Aircraft Inlet/Intake/Exhaust Certifying Officials (Note 1)	Qualified/certified 7 or 9- skill level(or civilian equivalent), or a fully qualified/certified contractor, AFETS or CETS representative and possess one of the following AFSCs: 2A6X1, 2A5X1/2/4, 2A3X3/7/8. One year minimum experience on applicable MDS/TMS.
7	Flexible Borescope Certifying Officials (Note 1)	
8	Engine Blade Blending Certifying Officials (Note 1)	
9	"Red-X" by Primary AFSC (PAFSC) and Mission Design Series (For multiple Mission Design Series, list separately) (Note 2)	SSgt or higher, minimum 7-skill level or civilian equivalent (includes MXG/CC-appointed exceptional SrA per **paragraph 11.3.3.**).
10	IPI by PAFSC and Mission Design Series (For multiple Mission Design Series, list separately) (Note 2)	

11	"Red-X" and/or In Process Inspection - Limited (Note 2)	5-skill level personnel certified on limited tasks as determined by the unit (Note 1); 5-level Certified Weapons Load Crew Chiefs on loading tasks only.
12	"Red-X" and/or In Process Inspection - Cross Utilization Training (For multiple MDSs, list separately), for tasks outside PAFSC (Note 2)	SSgt or higher, minimum 7-skill level (or civilian equivalent), Use for personnel certified on tasks in other AFSCs through CUT training.
13	NWRM packaging (Notes 4 and 5))	Refer to AFI 20-110, Chapter 2.
14	Installed Engine Run by Mission Design Series (Note 2)	SrA or higher, minimum 5-skill level (or civilian equivalent), with at least 6 consecutive months experience on MDS for which engine run training is required. (Experience must have occurred immediately prior to course enrollment). The MXG/CC may waive the weapons system experience. MXG/CCs may waive qualified 5-skill level A1C for critical manpower shortages. The time on weapon system may also be waived by MXG/CC.
15	Engine Blade Blending	Minimum 5-skill level 2A3X3/7/8, 2A5X1/2/4, and 2A6X1 or civilian equivalent. (Note 2).
16	QA hot pit certifying officials and QA hot pit certifier augmentees (squadron certifying officials) (Note 1)	Fully qualified/certified TSgt or higher 2AX7X AFSC (or civilian equivalent) with a minimum of one year hot pit experience on the applicable TMS. MXG/CC may waive time requirement.
17	Hot Refueling PAD Supervisor/"A" Member (Note 2)	Minimum 5-skill level, 2AX5X (or civilian equivalent), with a minimum of 1 year flightline maintenance experience.
18	Hot Refueling Team Member ("B" or "D" member) (Note 2)	Flightline maintenance AFSC, with a minimum of 1 year flightline maintenance experience.
19	Aircraft to Aircraft Refueling Supervisor (Note 2)	Minimum 5-skill level with a minimum of 1 year weapon system experience.
20	Uninstalled Engine Operations (Test Stand and ETS) Run by TMS (Note 2)	SSgt or higher 7-skill level 2A6X1 (or civilian equivalent) with a minimum of 6 months current experience on each applicable TMS,

		unless previously qualified (N/A to short tour assignments). If previously qualified on a different TMS, the 6-month experience requirement may also be waived. MXG/CC may waive 5-skill level SrA with minimum of 6 months' time on applicable TMS.
21	Uninstalled Engine Run Certifying Officials by TMS (Note 1)	Fully qualified/certified TSgt or higher 2A671 AFSC, civilian equivalent, contractor, or AFETS/CETS personnel with a minimum of one year engine run experience on the applicable TMS. (One year run experience not applicable to short tour assignments). The MXG/CC may waive qualified SSgts and may authorize MT uninstalled engine run instructors as certifying officials.
22	Aircraft Inlet/Intake/Exhaust Inspection/Certification (Note 2)	Minimum 5-skill level, 2A3X3/7/8, 2A5X1/2/4, and 2A6X1 (or civilian equivalent).
23	Engine Flexible Borescope Inspections (Note 2)	Minimum 5-skill level 2A3X3/7/8, 2A5X1/2/4, and 2A6X1 (or civilian equivalent).
24	Chief Servicing Supervisor (Heavy Aircraft/Commercial Derivative Aircraft) (Note 2)	Minimum 5-skill level with 1 year weapons system experience.
25	Concurrent Servicing Supervisor (Fighter Aircraft) (Note 2)	For A-10, F-15, F-16, F-22A aircraft, minimum 7-skill level with a minimum of 1 year weapons system experience.
26	W&B Certified/Clear Red X (refer to TO 1-1B-50) (Note 2)	7-Skill level (or civilian equivalent), with a minimum of 1 year time on weapon system. For units that maintain multiple weapon systems, as long as the technician is certified on one airframe they do not require an additional year of experience for every subsequent airframe W&B certification. Time requirement may be waived by MXG/CC.
27	Impoundment Official (refer to **Chapter 7** of this instruction) (Note 2)	

28	Impoundment Authority (refer to **Chapter 7** of this AFI) (Note 1)	
29	CANN Authority (Note 1)	
30	Auxiliary Power Unit (APU) Operation (In Cockpit) (Note 2)	SrA or higher, minimum 5-skill level (or civilian equivalent), with at least 6 consecutive months experience on MDS for which APU Operation training is required. The MXG/CC may waive the weapons system experience if member proficiency/performance supports this action and may waive qualified 5-skill level A1C to fill critical manpower shortages. MXG/CCs may also institute more stringent requirements as necessary (e.g. must be engine run qualified).
31	Calibration Limitation Approval (refer to TO 00-20-14) (Notes 2 and 3)	SSgt or higher, minimum 7-skill level (or civilian equivalent).
32	CDDAR Team Chief (Note 1)	MSgt or higher or civilian equivalent. MXG/CC may waive grade requirement.
33	Weapons Task Qualification Manager (WTQM) (Note 1)	TSgt or higher, minimum 7-skill level AFSC 2A871X or 2AX7X (or civilian equivalent).
34	Weapons Task Qualification Crew (WTQC) (Note 2)	Lead will be SSgt or higher, minimum 7-skill level 2AX7X (or civilian equivalent); other crew member minimum 5-skill level 2AX5X (or civilian equivalent).
35	NSS and T-9/T-10/T-11/T-12/T-20 sound suppressor Fire Control Panel (Note 2)	SrA or higher, (or civilian equivalent) with AFSC 2A6X1 must have a minimum 6 months experience.
36	Aircraft Rapid/Hot Defueling Supervisor (Note 2)	Minimum 5-skill level, 1 year of flightline experience, with 6 months weapon system experience.
37	Clear Red X when a lost item/tool cannot be located (refer to **Chapter 8** of this instruction) (Note 1)	Operations Officer/MX SUPT or above.
38	Aircraft APU Run Certifying Officials (In Cockpit) (Note 1)	7-skill level (or civilian equivalent), or a fully qualified/certified contractor or AFETS/CETS representative. MXG/CCs may also waive qualified 5-skill level SSgts.

39	T-38 Flight Control /System Rig Certifier (Note 1)	Minimum 7-skill level or civilian equivalent, 12 months experience adjusting system rigging. Certifies specialized MX rig teams to accomplish flight control, canopy, landing gear, door and other systems rigging.
40	T-38 Flight Control/System Rig Team Member (Note 1)	Minimum 7-skill level or civilian equivalent, 12 months experience performing system rigging. Performs flight control, canopy, landing gear, door and other system rigging as team member when directed. Member may be CUT trainee.
Note: For tankers, airlift, bomber and passenger aircraft include the following items 41-47 on SCR.		
41	Landing Gear Retraction Supervisor (Note 2)	(Supervisor) SSgt or higher or civilian equivalent, minimum 7-skill level.
42	Landing Gear Retraction Member (Note 2)	(Gear handle operator) minimum 5-skill level, and 1 year flightline experience.
43	Jacking Supervisor (Note 2)	Minimum 7-skill level or civilian equivalent, 6 months system experience.
44	Jacking Manifold Operator (Note 2)	Minimum 5-skill level or civilian equivalent, 6 months system experience.
45	Integral Jacking Member (C-17) (Note 2)	Minimum 6 months C-17 experience (Note 2)
46	Integral Jacking Supervisor (C-17) (Note 2)	Minimum SSgt (or civilian equivalent), 6 months minimum C-17 system experience (Note 2)
47	Integral Jacking Certifying Official (C-17) (Note 2)	Minimum 7-skill level or civilian equivalent, with minimum 1 year C-17 experience
48	MICAP Approval (Note 2)	MSgt or higher (or civilian equivalent)
49	Tow Team Supervisor	SrA or higher, minimum 5-level with a minimum of 6 months weapons system experience.

Note:
1. Approved by MXG/CC or equivalent may be delegated IAW p**aragraph 2.4.2.32.**
2. Approved by Operations Officer/MX SUPT or equivalents.
3. Operations Officer/MX SUPT may delegate approval authority to the AMU OIC/SUPT or Flight CC/SUPT.
4. Munitions inspectors who are trained and certified may annotate serviceability tags for munitions items (TO 11A-1-10).
5. Appointed by the unit commander (or equivalent) of units possessing NWRM.

11.4. Aircraft Grounding.

11.4.1. Aircraft grounding is defined in DAFMAN 11-401, *Aviation Management*.

11.4.1.1. Communication of a grounding, or a potential grounding event (TCTO, OTI, PM event) to weapon systems operating in a deployed environment and/or under the authority of an operational command must follow the grounding procedures outlined in DAFMAN 11-401 and 63-101/20-101.

11.4.1.1.1. Units must not directly communicate home station requirements to its deployed forces but provide communication through their respective MAJCOM. **(T-2)** **Note:** This section does not apply to conditions which are clearly limited to the affected unit/base (such as, lost tool, fluid contamination, aircraft and equipment damage of known origin, or other strictly local event). In these circumstances, the affected unit follows the impoundment procedures specified in **Chapter 7**.

11.4.2. Initial Investigation.

11.4.2.1. The OTI will require an inspection of a representative number of systems or units (aircraft, engines, missiles, or munitions) of the same mission and design to determine if the condition exists on other aerospace equipment within the wing's assigned aircraft/systems or equipment. **(T-2)**

11.4.2.1.1. If initial sampling indicates the discrepancy is widespread and has the potential for personal injury and/or further equipment damage, wings will coordinate with MAJCOM. **(T-2)**

11.4.2.2. If there is no repair or corrective action specified in technical data, QA will also submit a technical assistance request through the MAJCOM to the appropriate weapon system program manager IAW TO 00-25-107 or equivalent process. **(T-2)**

11.4.3. Grounding Authority. The approved procedures for grounding aircraft or stand-down for operational reasons are determined and executed IAW DAFMAN 11-401.

11.4.3.1. Notification and final reporting for grounding and release status will be accomplished IAW AFMAN 10-206, *Operational Reporting (OPREP)*.

11.4.3.2. Annotate aircraft grounding in the aircraft forms IAW TO 00-20-1.

11.5. Ramp Inspection Program. Public Law 99-661, section 2638, National Defense Authorization Act for Fiscal Year 198 requires a pre-flight safety inspection of all internationally scheduled charter missions for the transportation of members of the Armed Forces departing the United States. **(T-0)**

11.5.1. Air Mobility Command (AMC) is lead for the DoD in the management and administration of the Ramp Inspection Program.

11.5.1.1. AMC will publish specific guidance for this Program in a supplement/addendum/or equivalent publication IAW DAFMAN 90-161 to communicate requirements and processes necessary for MAJCOMS to effectively comply with public law requirements in all using commands.

11.5.1.2. AMC will coordinate with other MAJCOMs as required to accomplish ramp inspections to ensure the maximum efficiency and utilization of resources.

11.5.1.3. When requested by AMC, MAJCOMs if able should provide support to reduce the TDY and manpower impact associated with the execution of this program.

11.6. Red Ball Maintenance. The term "Red Ball" is a traditional descriptor, recognized throughout aircraft maintenance, and defines a situation requiring a sense of urgency and priority actions. Red Ball maintenance normally occurs two hours prior to launch and until aircrew have released the aircraft back to maintenance. The Red Ball maintenance concept is intended to prevent late takeoffs and aborts by having qualified maintenance personnel available (such as, in a truck or standby in the shop) during launch and recovery operations to troubleshoot, isolate, and repair system malfunctions. Red Ball maintenance does not authorize technicians to take shortcuts or deviate from TOs, disregard personnel safety requirements or fail to properly document the aircraft forms and the MIS for all completed repair actions.

11.6.1. Units will ensure all maintenance repair actions (does not apply to incorrect switch settings due to operator error) are documented in the aircraft forms and MIS during Red Ball, launch, or EOR operations and cleared from the aircraft forms prior to flight. **(T-3)**

11.6.2. Maintenance repair actions must be cleared in the MIS as soon as possible. **(T-3)** It is imperative that maintenance documentation is performed regardless of the timing of the action in the generation and launching of the aircraft.

11.6.3. All grounding inputs must be cleared from the forms prior to flight. **(T-3)**

11.6.4. If aircraft status changes, an ER must be re-accomplished by a certified individual upon completion of maintenance and before the aircraft is released for flight IAW TO 00-20-1.

11.6.5. Units will develop written procedures to capture, document, and clear Red Ball maintenance actions in the event the MIS is down. **(T-3)**

11.6.5.1. Procedures must require MIS entry of Red Ball maintenance actions as soon as the MIS becomes operable. **(T-3)**

11.7. Maintenance Recovery Team. MAJCOMs will publish standardized procedures to recover assigned aircraft at remote locations.

11.7.1. Procedures at a minimum will identify how resources, including personnel, supplies, and equipment will be made available to support transient aircraft recovery.

11.7.2. If required, establish multiple command MOUs/MOAs/collaboration necessary to achieve efficient aircraft recovery and MIS documentation.

11.7.3. **(Added-AFMC)** The MXG/CC is responsible for deploying Maintenance Recovery Teams and equipment to recover aircraft at remote locations, as directed by this instruction. Units will develop checklists. Unit resources, including personnel, supplies, and equipment, will be made available to support en route aircraft recoveries. **(T-2)** The MXG/CC will:

11.7.3.1. **(Added-AFMC)** Ensure adequate equipment resources, as authorized by applicable AS, are available to perform remote aircraft recoveries. **(T-2)**

11.7.3.2. **(Added-AFMC)** Ensure enough individuals are qualified to perform and complete all anticipated tasks, including Maintenance Information System (MIS) documentation, IPIs, and Red X sign-offs. Consider the experience level of the individuals you select when the need is identified to troubleshoot repeat/recurring discrepancies, especially those that caused in-flight emergencies or involve safety of flight. The MXG/CC has the authority to grant approval for a single person to sign off their own work when only one individual deploys on an Maintenance Recovery Team IAW TO 00-20-1, *Aerospace Equipment Maintenance Inspection, Documentation, Policies, and Procedures*. **(T-2)**

11.8. Foreign Object Damage (FOD) Prevention Program. All personnel (military, civilian, and contractors) working in, on, around, or traveling through areas near aircraft, flightline munitions, AGE, engines, or components thereof will comply with FOD prevention. **(T-2)** FOD prevention training requirements are outlined in AFI 36-2650. This section establishes minimum requirements for a FOD Prevention Program.

11.8.1. The WG/CD is responsible for ensuring an effective FOD prevention program is established.

11.8.2. Definition. FOD: Any damage to an aircraft, engine, aircraft system, component, tire, munitions, or SE caused by a foreign object(s) (FO) which may or may not degrade the required safety and/or operational characteristics of the aforementioned items.

11.8.3. FOD Prevention.

11.8.3.1. On aircraft, uninstalled engines, LRUs and AGE. Openings, ports, lines, hoses, electrical connections, and ducts will be properly plugged or capped to prevent FO from entering the systems. **(T-2) Note:** Do NOT place items (such as, trash bags, rags, cloths) inside open cavities or ducts. When no approved manufactured coverings and/or caps are available for use, securely cover open ducts and/or cavities externally to prevent foreign objects from being introduced. Prior to installation, inspect openings, cavities, and ducts for FO.

11.8.3.1.1. Items that are actively being disconnected, installed, and/or removed will be capped IAW technical data or at completion of the task. **(T-2)**

11.8.3.1.2. At no time will items, (such as, aircraft forms binders, video tape recorder tapes, checklists, tools.), be placed in or on engine intakes. **(T-2) Note:** Does not apply to technicians performing inlet maintenance, inspections and blade blending requiring lights, files, or other tools inside aircraft inlets.

11.8.3.1.3. Inventory all items IAW **Chapter 8**. **(T-2)**

11.8.3.2. MAJCOMs in coordination with Safety, applicable MDS Lead Commands and Weapon System Teams will review FOD, Incident, and Mishap reports to determine if MAJCOM directed IPI or KTL additions are needed to mitigate identified FOD trends.

11.8.3.2.1. Units will establish MDS specific procedures that ensure pre-launch removal and post-recovery installation of intake/inlet plugs and covers (such as, pitot tubes to include ejection seats as required) remain installed on aircraft as close to crew show as possible to prevent FOD, as determined by MXG/CC guidance. **(T-3)**

11.8.3.2.2. Units will establish MDS specific FOD prevention guidance that standardizes mitigation procedures when performing high FOD risk maintenance task (use of plugs/barrier paper, tape, inlet/intake/ECS maintenance or equivalent tasks). **(T-2)**

11.8.3.2.3. Technicians will ensure engine inlet run-up screens and anti-personnel guards are used IAW applicable weapon system TOs. **(T-2)**

11.8.3.3. Covers (such as, engine, pitot tube(s) to include ejection seat) need to remain installed on aircraft as close to crew show as possible to prevent FOD, as determined by MDS/local MXG/CC guidance.

11.8.3.4. Technicians should use a light source of sufficient illumination to inspect the aircraft intakes and exhaust for FO/FOD.

11.8.3.5. Whenever physical entry into an aircraft intake or exhaust is required, technicians will wear a pocket-less, zipper-less, button-less, bunny-suit marked "Intakes Only" and cloth over-booties or stocking feet, (Boots may be worn if not restricted for use by TO/MAJCOM and are authorized by the MXG or equivalent), boots if worn must be inspected and FOD removed from boots prior to installing cloth over-booties. **(T-2)**

11.8.3.5.1. When performing intake inspections while wearing a Chemical Warfare Defense Equipment, pockets will be emptied, and all accessories removed. **(T-2)**

11.8.3.5.1.1. During exercises/inspections, the Chemical Warfare Defense Equipment will be removed, and the bunny-suit will be utilized. **(T-2)**

11.8.3.5.1.2. Chemical Warfare Defense Equipment will only be worn during “real world” situations. **(T-2)** To minimize the potential for FOD and intake damage where Chemical Warfare Defense Equipment zippers are exposed, cover them with any type of tape and account for the tape upon completion of the inspection. **(T-2)**

11.8.3.6. Each base will develop a local flightline clothing policy that addresses wearing of hats, badges, and passes aimed at FOD prevention while considering climate and safety. **(T-2)** As a minimum, it will include the following requirements:

11.8.3.6.1. Restricted area badges will be secured with a subdued non-metallic cord or plastic armband when worn on the flightline. **(T-2)**

11.8.3.6.2. Restricted area badges will be removed when performing intake/inlet/exhaust inspections if personnel physically enter these areas. **(T-2)**

11.8.3.6.3. Metal insignias/badges will not be worn on the flightline. **(T-2)**

11.8.3.6.4. Wigs, hairpieces, metal hair fasteners, earrings, or any other jewelry/loose items that may fall off without notice, are not authorized on the aircraft and industrial work areas. **(T-2)**

11.8.3.6.5. Escorts of visiting personnel will ensure FOD prevention measures are taken. **(T-3)**

11.8.3.7. All maintenance production areas must have FO containers readily accessible. **(T-2)**

11.8.3.7.1. All vehicles driven primarily on the flightline for direct aircraft maintenance support activities must be equipped with secured and lidded FO containers. **(T-2) Note:** Permanently affixed FO containers must be approved by Vehicle Management prior to installation IAW AFI 24-302.

11.8.3.8. Control all work order residue used on or around aircraft, uninstalled engines, and AGE. **(T-2)**

11.8.3.8.1. **(Added-AFMC)** Establish and tailor rivet replacement procedures for local operation of assigned weapons systems. Include the procedures as part of the Foreign Object Damage (FOD) orientation/familiarization for personnel working in these areas. Include work order residue control procedures for all maintenance performed in and around intake areas. **(T-2)**

11.8.3.8.2. **(Added-AFMC)** Aircraft Structural Maintenance shops and Low Observable Aircraft Structural Maintenance shops will develop LCL for intake repairs IAW **Chapter 6** and TO 00-5-1. All parts and pieces installed and removed from the aircraft will be documented and verified by a 7-level. The checklist will be completed on the job site and turned into QA, or equivalent, within 24 hours of repair completion. Aircraft Structural Maintenance and Low Observable Aircraft Structural Maintenance technicians are not required to meet the training requirements of **paragraph 11.16**. **(T-2)**

11.8.3.9. Rags will be controlled and accounted for IAW **Chapter 8**. **(T-2)**

11.8.3.9.1. Rag control applies to all organizations and personnel performing aircraft, missile, munitions, and equipment maintenance.

11.8.3.10. FOD walks are mandatory to remove FO from ramps, runways, taxiways, maintenance areas and access roads.

11.8.3.10.1. In addition, mechanical/vacuum sweepers, magnetic bars or sweeping by hand are highly encouraged to supplement FOD walks.

11.8.3.11. When FOD is discovered on a transient aircraft, depot input/output or CRF engine, the host FOD monitor, or aircrew must notify the owning organization within 24 hours. **(T-2)**

11.8.3.11.1. An informational copy of the FOD report must be provided to the owning organization’s flight safety office and FOD monitor to ensure compliance with DAFI 91-204. **(T-2)**

11.8.3.11.2. For depot input/output or CRF engine. If the FOD is found during the receiving inspection at one of the aforementioned locations, it will be tracked/charged (if necessary) to the owning MAJCOM unit. **(T-2)** If discovered any other time at one of the aforementioned locations, it will be tracked/charged to the ALC or CRF. **(T-2)**

11.8.3.11.3. **(Added-AFMC)** When transient/deployed aircraft incur FOD, the host unit will conduct the investigation and notify the owning organization using command-reporting procedures. **(T-2)**

11.8.3.11.4. **(Added-AFMC)** The owning organization is responsible for FOD incidents and investigations on transient aircraft/engines when one of the following conditions apply:

11.8.3.11.4.1. **(Added-AFMC)** When the owning organization's maintainers are deployed with the aircraft and the FOD is a direct result of transient/deployed unit negligence. **(T-2)**

11.8.3.11.4.2. **(Added-AFMC)** FOD discovered by transient alert or by contractor personnel during acceptance inspections will be charged to the base from which the aircraft last departed if a FOD inspection was not accomplished/documented and there were no intermediate stops. (**T-2)**

11.8.3.12. Ensure local FOD Prevention Program addresses the elimination of FOs to include aircraft cockpits and flight decks before and after flight. **(T-2)**

11.8.3.12.1. When an item is lost on or in the vicinity of aircraft or equipment, lost item/tool procedures in **Chapter 8** will be followed.

11.8.3.12.2. DELETED.

11.8.3.12.3. These procedures will be documented in the wing tool/equipment management publication referenced in **Paragraph 8.2**. **(T-2)**

11.8.3.13. Use extreme care during engine ground runs. Jet blast and helicopter hover power check areas need to be free of debris that could cause FOD.

11.8.3.14. Special emphasis is required for items such as: remove before flight streamer attachment, safing pin condition, hinge pin security, dust and FO prevention cover condition/security, and aircraft forms binder condition. Periodically check these types of items for FO prevention compliance.

11.8.3.14.1. Units will account for -21 equipment and covers IAW DAFI 21-103.

11.8.3.14.2. Weapons Expediters must ensure all mission specific safing gear is controlled and accounted for to preclude loss and potential FOD. **(T-2)**

11.8.3.15. Refer to DAFI 13-213, *Airfield Driving and* DAFMAN 91-203, for vehicle FOD check requirements.

11.8.3.16. Grounding wires/points:

11.8.3.16.1. Two Allen head screws, or equivalent, will be utilized to secure cable to grounding clip. **(T-2)**

11.8.3.16.1.1. Screw heads will be coated with sealant or screws will be staked in order to prevent screws from backing out. **(T-2)**

11.8.3.16.1.2. Unused screws will be removed. **(T-2)**

11.8.3.16.2. All grounding points will be kept clean of debris at all times and will be a high interest item for FOD walks. **(T-2)**

11.8.3.17. Use of magnetic bars on the flightline is optional. If used, the magnetic bars will be towed by, or attached to vehicles primarily used on the flightline and will be inspected and made FOD free daily. **(T-2)**

11.8.3.18. A locally manufactured tool for removing debris from tire treads is authorized for use and will be identified to the vehicle by using the vehicle ID number. **(T-2)**

11.8.3.19. Remove metal identification bands from all tubing (except aircraft installed egress system components) and cables on the aircraft.

11.8.3.19.1. With the exception of factory-installed ID tags attached to cargo chains/devices to identify the type being used, remove metal identification bands from cargo tie-down chains/devices prior to use around aircraft.

11.8.3.19.2. Do not remove manufacturer installed metal identification bands from hydraulic hoses.

11.8.3.19.3. Mark hydraulic lines IAW TO 42E1-1-1, *Organizational, Intermediate and Depot Maintenance; Aviation Hoses and Tube Manual.*

11.8.3.19.4. **(Added-AFMC)** Leave inspection and load tolerance tags attached. **(T-2)**

11.8.3.20. Use X-ray, borescope, and other equipment to locate FO in inaccessible areas.

11.8.3.21. WG/CD may institute real-time FOD detection (automated FOD detectors) based upon risk analysis IAW AFI 90-802 and DAFI 91-202.

11.8.4. FOD Prevention Responsibilities.

11.8.4.1. The WG/CD will be assigned as the FOD Prevention Program Manager and will appoint a qualified maintenance AFSC, civilian equivalent or contractor if designated by SOW or PWS, to the position of FOD Monitor. **(T-2)**

11.8.4.1.1. The appointed individuals name will be posted in a prominent place within the unit on a locally developed visual aid which also provides contact information. **(T-2)**

11.8.4.2. The WG/CD will:

11.8.4.2.1. Ensure all personnel actively support the FOD Prevention Program. **(T-2)**

11.8.4.2.2. Provide local guidance to ensure each FOD mishap is investigated, and action taken to solve any underlying problems. **(T-2)**

11.8.4.2.3. Review all unit FOD mishap reports and analyze the reports and other data for trends identifying areas requiring management action. **(T-2)**

11.8.4.2.4. Coordinate FOD prevention needs with the airfield manager and other agencies when construction is in progress on or near the flightline, or other areas where FOD incidents could occur. **(T-2)**

11.8.4.2.5. Ensure FOD prevention is part of QA inspections. **(T-2)**

11.8.4.2.6. Coordinate with the airfield manager to identify and properly mark FOD checkpoints. **(T-2)**

11.8.4.2.7. **(Added-AFMC)** Develop and implement a FOD awareness and prevention training program. **(T** At a minimum, the training program must address the following:

11.8.4.2.7.1. **(Added-AFMC)** Standardized training documentation requirements. **(T-2)**

11.8.4.2.7.2. **(Added-AFMC)** FOD awareness and prevention practices. **(T-2)**

11.8.4.2.7.3. **(Added-AFMC)** Initial FOD training. Ensure newcomers are trained on work center specific FOD awareness and prevention practices prior to starting work in their assigned work area. **(T-2)**

11.8.4.2.7.4. **(Added-AFMC)** Annual FOD training. Ensure personnel receive annual FOD awareness and prevention training. **(T-2)**

11.8.4.2.7.5. **(Added-AFMC)** Ensure all maintenance, operations, base support, and contractor personnel who work in, around, or drive through maintenance and/or operational areas are trained on FOD prevention. **(T-2)**

11.8.4.3. Tenant Unit FOD Prevention Responsibilities. The host base FOD Prevention Program Manager will incorporate tenant units in the host unit program. **(T-2)**

11.8.4.3.1. Tenant units will establish their own FOD Prevention Program but will still participate in the host program and comply with host program requirements. **(T-2)**

11.8.5. FOD Monitor. The Wing FOD Monitor's office will be located within QA or at the discretion of the WG/CD. **(T-3)** The Wing FOD Monitor, at a minimum, will:

11.8.5.1. Inform all wing agencies of FOD hazards. **(T-2)**

11.8.5.2. Develop wing procedures to document and perform spot checks of "high risk" areas weekly. **(T-2)**

11.8.5.3. Be involved in each FOD investigation and help ensure corrective actions are sound. **(T-2)**

11.8.5.4. Monitor and recommend changes to FOD prevention training. **(T-2)**

11.8.5.4.1. Those units having several types of aircraft assigned will have their FOD prevention training incorporated into one wing/center training program. **(T-2)**

11.8.5.4.2. Units will ensure an initial FOD awareness and responsibilities briefing is given to all newly assigned personnel. **(T-2)**

11.8.5.5. Periodically inspect and report damaged pavement, flightline construction, or other hazards in or near aircraft parking ramps or taxiways to the airfield manager and monitor status to ensure timely repairs. **(T-2)**

11.8.6. FOD Investigation and Reporting.

11.8.6.1. When suspected or confirmed FOD is discovered, the MOC will be notified immediately. **(T-2)**

11.8.6.1.1. Upon notification, the MOC will immediately notify the Wing FOD Monitor, and Wing Safety, IAW DAFMAN 91-223.

11.8.6.1.2. **(Added-AFMC)** MOC will immediately notify Airfield Management when cut or damaged tires are discovered after taxiing or aircraft tows. **(T-2)**

11.8.6.2. Units must make every attempt to determine the root cause of FOD-related mishaps before returning engines and modules to the depot for investigation. **(T-2)**

11.8.6.2.1. If engines/modules are returned to the depot, an information DR will be completed and forwarded IAW procedures outlined in DAFI 91-204 and TO 00-35D-54.

11.8.6.2.2. All FOD-mishap engines and modules returned to the depot must be properly marked on the outside of the packaging as a FOD-mishap asset. **(T-2)**

11.8.6.2.3. Mark container or package in red with the following statement, "FOD mishap investigation required." **(T-2)**

11.8.6.3. FOD incidents are classified as preventable and non-preventable. Only preventable FOD over $60K (parts and labor) are to be chargeable to the FOD rate. FOD is considered preventable except when the damaged can be attributed to the following:

11.8.6.3.1. Caused by natural environment or wildlife. This includes hail, ice, animals, insects, sand, and birds. Report this type of damage IAW DAFI 91-204. Do not include these in the FOD rates.

11.8.6.3.2. From internal engine materiel failure as long as damage is confined to the engine.

11.8.6.3.3. Caused by materiel failure of an aircraft component if the component failure is reported as a DR using the combined mishap DR reporting procedures of DAFI 91-204 and TO 00-35D-54.

11.8.6.3.4. Found during depot overhaul for maximum operating time.

11.8.6.4. Additionally, the following apply:

11.8.6.4.1. Engine damage caused by improper anti-ice/de-ice procedures by either flight or ground crews are considered preventable.

11.8.6.4.2. Engine or airframe damage caused by gunnery or rocket mission ricochets are considered non-preventable provided mission parameters were not exceeded and range cleaning was sufficient.

11.8.6.4.3. Engine and propeller damage caused by rocks, stones, wood, or other objects ingested during low hover operations or unimproved runway landings are considered non-preventable, provided mission parameters were not exceeded.

11.8.6.4.4. MAJCOMs will determine reporting criteria for FOD incidents that result in a blade blending requirement IAW applicable tech-data.

11.8.6.4.4. **(AFMC)** Damage beyond blend repair, where hardware may be suspect for cause, and/or further unserviceable damage exists beyond the 1st stage fan will be considered reportable FOD. FOD found on the 1st stage fan which can be blended IAW with applicable tech-data may be considered non-reportable. **(T-2)**

11.8.6.5. Preventable FOD over $60K incurred at ETS or on trim pad will be chargeable. **(T-2)**

11.8.6.6. Appropriate MAJCOM offices will assist in resolving any questionable FOD issues, preventable or non-preventable.

11.8.6.6. **(AFMC)** HQ AFMC/A4M Propulsion Functional Manager shall assist in resolving questionable FOD issues.

11.8.6.7. The Wing FOD Monitor will provide an initial report of all FOD incidents to the MAJCOM FOD monitor within 24 hours of occurrence. **(T-2)**

11.8.6.7.1. A follow-up report will be required every 45 days until closeout. **(T-2)** Use the FOD report format as listed in **Attachment 6**.

11.8.6.7.2. MAJCOMs will determine FOD standards, MDS specific flying hour source data, period of time for calculation, reporting procedures, and meeting frequency for units that exceed standards in their supplement to this DAFI.

11.8.6.7.2. **(AFMC)** Wing FOD monitor will provide notification of a FOD incident within 24- hrs. of the occurrence via the HQ AFMC/A4M Propulsion Management SharePoint Site **https://usaf.dps.mil/teams/22355/SitePages/FOD-DOP-Program-Management.aspx**. If the 24 hour notification tool is used, an initial report must be created within two weeks of the occurrence.

11.8.6.7.2.1. **(Added-AFMC)** FOD standard for fighter/rescue/bomber aircraft (F-15, HH60, B-1) is 1.0 and airlift/tanker aircraft (C-17, C-130, KC-135) is 3.0. **(T-2)**

11.8.6.7.2.2. **(Added-AFMC)** MDS specific flying hours source data is defined as the flying hours flown as identified in the Maintenance Information System (MIS). **(T-2)**

11.8.6.7.2.3. **(Added-AFMC)** Units will conduct monthly FOD Prevention Committee Meetings, whenever the unit exceed the MAJCOM standard. Meetings will continue until the unit completes three consecutive months without exceeding the standard. The FOD meeting may be combined with other meetings. **(T-2)**

11.8.6.7.2.4. **(Added-AFMC)** Wing FOD monitor shall submit FOD calculations monthly for each MDS using the HQ AFMC/A4M Propulsion Management SharePoint site: **https://usaf.dps.mil/teams/22355/Lists/FOD%20Rate/Test%20Wing.aspx**, no later than the 15th of the month following the reporting month.

11.8.6.7.2.5. **(Added-AFMC)** Wing FOD monitor shall provide a final report the HQ AFMC/A4M Propulsion Management SharePoint site: **https://usaf.dps.mil/teams/22355/Lists/ALC%20FOD%20Incident%20Report/Standard%20View.aspx** upon completion of the FOD investigation. **(T-2)**

11.8.6.7.2.6. **(Added-AFMC)** If the FOD incident is deemed a mishap IAW DAFI 91-204, the Wing FOD monitor will still report the FOD incident then work in tandem with the Wing SE office to properly report the mishap. **(T-2)**

11.8.6.8. FOD rates are computed by MDS as follows: Number of Preventable FODs (damage exceeding $60K) ÷ Aircraft Flying Hours X 10,000 = FOD Rate. **Note:** ALCs compute FOD rates as follows: Number of Preventable FODs (damage exceeding $60K) ÷ Aircraft Flying Hours X 1,000 = FOD Rate. ALCs compute aircraft flying hours by using acceptance flights, functional check flights, ground runs, and the number of un-installed ETS starts.

11.8.7. FOD Prevention Committee Meeting. This meeting is mandatory for units that exceed the MAJCOM-established standard. **(T-2)**

11.8.7. **(AFMC)** Additionally, the following will be considered for minimum attendance at the FOD Prevention Committee Meeting: Center FOD Monitor, the designated Union representative, Center or Wing Maintenance Training, Flight test representative, Contracting (As applicable), Center tool control manager, and the ABW/CV. **(T-2)**

11.8.7.1. - 11.8.7.1.1. DELETED.

11.8.7.2. Meeting agenda items should include issues that resulted in the wing exceeding the FOD standard, such as:

11.8.7.2.1. Total number of airframes, engine, and tire FOD incidents during the reporting period. Indicate quantity and cause. Current status of all other pending incidents will be discussed.

11.8.7.2.2. Mechanical/vacuum sweeper status.

11.8.7.2.3. Review and refinement of the existing FOD prevention program.

11.8.7.2.4. New directives/actions established to minimize FOD.

11.8.7.2.5. Status and condition of engine run-up screens as applicable.

11.8.7.2.6. Results of X-rays for FOs during engine bay inspections, acceptance inspections, and PH inspections. Maintenance trends should be discussed when an increase in FO is discovered during these X-rays.

11.8.7.2.7. Identification of potential FOD sources.

11.8.7.2.8. Lost tools/items.

11.8.7.2.9. Increased potential for FOD within the next 30-60 days.

11.8.7.2.10. Dropped objects. Pay particular attention to those that result in downstream FOD.

11.8.7.2.11. Breakdown of FOD inspections/assessments.

11.8.7.2.12. Cockpit FO incidents.

11.8.7.2.13. Recognition of personnel making significant contributions to FOD prevention (such as, golden bolt program, FOD poster contests, or other FOD recognition programs locally developed at each unit).

11.8.7.2.14. **(Added-AFMC)** In addition to the agenda items, the minutes will include as a minimum a list of attendees and absentees. Attendees list will identify the wing FOD monitor and provide functional address symbol and duty phone number for

all personnel. Meeting minutes will be made available to FOD committee members. **(T-2)**

11.8.8. Bird Strikes. Consult TO 1-1-691 for bird strike clean-up procedures and DAFMAN 91- 223, *Aviation Safety Investigation and Reports* for bird strike reporting procedures.

11.9. Dropped Object Prevention (DOP) Program. A dropped object is any aircraft part, component, surface, or other item lost during aircrew operations (unless intentionally jettisoned) from engine start to engine shutdown. Inadvertently released munitions are not considered dropped objects and will be reported IAW DAFI 91-204. **Note:** Missing Chaff/Flare/Decoy endcaps/LO coating are not reportable dropped objects.

11.9.1. Responsibilities. All units, which fly, service, or maintain aircraft, will develop a DOP Program with the following provisions: **(T-2)**

11.9.1.1. MAJCOM DOP monitors or aircraft functional managers will act as OPR for all dropped object inquiries IAW MAJCOM established standards.

11.9.1.2. The WG/CD serves as the Wing DOP Program Manager and will appoint a Wing DOP Monitor. **(T-2)**

11.9.2. Investigation. The DOP Monitor will investigate each dropped object incident. **(T-2)**

11.9.2.1. Every effort needs to be made to determine the precise cause to ensure positive corrective action is accomplished. Anytime a materiel or design deficiency is the cause, or suspected cause, a DR will be submitted IAW TO 00-35D-54, even when an exhibit is not available.

11.9.2.2. Investigation results will be distributed to each appropriate work center for inclusion in personnel training and education programs. **(T-2)**

11.9.3. Reporting. Units will follow MAJCOM DOP Program reporting procedures. **(T-2)**

11.9.3. **(AFMC)** Wing DOP monitor will provide notification of a DO incident within 24-hrs. of the occurrence via the HQ AFMC/A4M Propulsion Management SharePoint Site: **https://usaf.dps.mil/teams/22355/Lists/24%20hr%20Notification/Allitemsg.aspx** . If the 24-hr. notification tool is used, an initial report must be created within two weeks of the occurrence. Moreover, the final dropped object report will be uploaded to **https://usaf.dps.mil/teams/22355/Lists/FOD%20Incident%20Report/Standard%20View.aspx** after occurrence. In addition, if it involves casualties, property damage, or if adverse publicity is likely, provide data to command post for reporting IAW AFMAN 10-206, *Operational Reporting (OPREP)*. The base/wing safety office shall be notified of all dropped objects within 24 hours of occurrence. **(T-2)**

11.9.3.1. Transient Aircraft. The host Wing DOP Monitor will be responsible to investigate dropped objects from a transient aircraft. **(T-2)**

11.9.3.1.1. The host Wing DOP Monitor will provide the home station Wing DOP Monitor with sufficient data to generate a report for trending and tracking purposes. **(T-2)**

11.10. Aircraft Structural Integrity Program (ASIP). The ASIP includes requirements for collection and evaluation of aircraft usage data to update or confirm the original design or baseline spectrum and to adjust maintenance intervals on an individual aircraft basis. The

Loads/Environment Spectra Survey data is collected via flight data recorders of instrumented aircraft to evaluate the loads spectrum. The Individual Aircraft Tracking data is collected via flight data recorders or manual forms such as "bubble sheets" and the data is used to make maintenance/inspection/force structure decisions. Both the Loads/Environment Spectra Survey and Individual Aircraft Tracking usage data programs are established by applicable MDS-specific TOs and DAFI 63-140 and require coordinated action by a number of base-level maintenance activities to achieve the required data capture rates. An effective ASIP aircraft usage data collection program is essential to establish, assess and support inspections, maintenance activities, repairs and required modification/replacement actions. MAJCOMs will:

11.10.1. Publish ASIP roles and responsibilities for each assigned weapon system IAW DAFI 63-140.

11.10.1. **(AFMC)** The following are the interim roles and responsibilities until AFMC supplement to DAFI 63-140 is published by HQ AFMC/A4/10 Engineering Divion (AFMC/A4/10/A4-EN).

11.10.1.1. **(Added-AFMC)** AFMC Aircraft Structural Integrity Program (ASIP) OPR will coordinate with other command ASIP OPR (as necessary), unit ASIP Program Manager, and PO ASIP Manager to resolve Individual Aircraft Tracking (IAT) data submission problems, aircraft hardware/software discrepancies, and programmatic issues that affect valid data capture rates.

11.10.1.2. **(Added-AFMC)** Maintenance Group Commander will:

11.10.1.2.1. **(Added-AFMC)** Appoint an officer, NCO, or civilian as the MXG ASIP Program Manager. **(T-2)**

11.10.1.2.2. **(Added-AFMC)** Ensure effective measures are in place to capture and report ASIP aircraft usage data (Individual Aircraft Tracking and/or Loads Environment Spectrum Survey) to achieve the required data capture/validity rate.

11.10.1.3. **(Added-AFMC)** MXG ASIP program manager will:

11.10.1.3.1. **(Added-AFMC)** Develop MXG ASIP Program Manager, AMU ASIP monitors, and Debrief section responsibilities and training program (as applicable).

11.10.1.3.2. **(Added-AFMC)** MXG ASIP program manager will track and submit to AFMC/A4M (**AFMC.A4M.Workflow@us.af.mil**) a quarterly IAT status report IAW Attachment X for all assigned manned/unmanned, home station/deployed aircraft, by MDS. **(T-2)**

11.10.1.3.2.1. **(Added-AFMC)** Unit quarterly IAT reports will be submitted to the AFMC ASIP OPR (**AFMC.A4M.Workflow@us.af.mil**) NLT the fifth duty day of the quarter (Jan, Apr, Jul, & Oct).

11.10.1.3.2.2. **(Added-AFMC)** Unit quarterly IAT reports will include amplifying remarks to explain IAT valid data submission rates below 90%, and listed by month in the remarks section of the template, see **Table 11.2**.

Table 11.2. (Added-AFMC) AFMC Individual Aircraft Tracking (IAT) Status Reporting Template.

BASE	UNIT	MDS(s)	Reporting Quarter
Sample Base	96 MXG	F-XX	Jan –Mar 2022

ITEM	VALUE
Number of Recorder or DTC Downloads Required	225
Number of Recorder or DTC Downloads Submitted	72
Recorder Submission Rate	32%

Number of Sorties Flown	635
Number of Forms Submitted	550
Percent Captured	88.60%

COMMENTS
The objective is to submit 100% of required collected data. Search in progress for missing forms. Download hard drive corrupt all files lost, new hard drive on order EDD 30 Oct 2014.

Monthly break down (AFTO Form 239)			
Jan sorties flown:			
Jan downloads required:		Downloads submitted:	
Jan forms submitted:			
Feb sorties flown:			
Feb downloads required:		Downloads submitted:	
Feb forms submitted:			
Mar sorties flown:			

BASE	UNIT	MDS(s)	Reporting Quarter
Mar downloads required:		Downloads submitted:	
Mar forms submitted:		0	

11.10.1.3.3. **(Added-AFMC)** Units possessing manned or unmanned weapon systems that do not have an approved IAT system will contact the AFMC ASIP OPR (**AFMC.A4M.Workflow@us.af.mil**) for reporting guidance.

11.10.1.3.4. **(Added-AFMC)** MXG ASIP program manager will coordinate with the MAJCOM ASIP OPR and MDS ASIP Manager to obtain guidance and resolution when ASIP usage data collection and submission issues are encountered.

11.10.1.3.5. **(Added-AFMC)** MXG ASIP program manager will coordinate with the MAJCOM ASIP OPR and PO ASIP Manager to obtain feedback on data capture rates and to implement corrective actions as needed to achieve the required rates.

11.10.1.3.6. **(Added-AFMC)** MXG ASIP program manager will provide oversight and ensure units are complying with planning and collection of ASIP collection plans and data IAW paragraph **11.10.1.4.1** and **11.10.2.1** of this instruction.

11.10.1.4. **(Added-AFMC)** AMU ASIP monitors will:

11.10.1.4.1. **(Added-AFMC)** Develop and publish procedures to collect and submit ASIP aircraft usage data at home station and deployed locations. **(T-3)**

11.10.1.4.2. **(Added-AFMC)** Ensure of collection of IAT ASIP data at the unit level, IAW **paragraph 11.10.2.1** of this instruction.

11.10.1.4.3. **(Added-AFMC)** When valid ASIP submission rates fall below 90%, devise plan in coordination with the MXG ASIP program manager to improve valid capture rates.

11.10.2. Ensure operational units continuously meet authorized reporting requirements established by Program Offices.

11.10.2. **(AFMC)** Operational Requirements.

11.10.2.1. **(Added-AFMC)** AMUs/Mx Units will collect, track, and submit IAT data (no later than the end of the next duty day) IAW specific aircraft technical orders and/or ASIP Master Plans. **(T-2)**

11.10.3. Document causes and corrective actions for units that fail to meet reporting requirements and retain until resolved or relief of the reporting requirement is granted from the Program Office in writing.

11.11. Identification Friend or Foe (IFF) Program.

11.11.1. MAJCOMs will establish an Identification Friend or Foe Program for aircraft in their command (if equipped).

11.11.1.1. MAJCOM programs will identify additional requirements necessary to ensure status of Identification Friend or Foe systems meets mission requirements.

11.11.2. DELETED.

11.11.3. Equipped aircraft will be checked prior to its first sortie of the day during contingency operations. **(T-2)**

11.11.4. **(Added-AFMC)** AFMC Units will ensure fully operational IFF Mode V systems by performing checks on all aircraft departing the Continental United States (CONUS) to overseas locations. **(T-2)**

11.12. Radar Warning Receiver/Radar Threat Warning Testing.

11.12.1. MAJCOMs will identify weapon systems with enhanced on-board diagnostics and internal testing capabilities which do not require external testing in their supplement to this instruction (if equipped).

11.12.1. **(AFMC)** B-1s, B-2s, F-22s, and F-35s have on-board diagnostic and internal testing capabilities. Therefore, no external test equipment is required for these MDSs.

11.12.1.1. MAJCOMs will determine non-contingency system functional check requirements necessary to ensure Radar Warning Receiver/Radar Threat Warning systems are maintained operationally ready to meet mission requirements IAW the MESL or MDS equivalent.

11.12.1.1. **(AFMC)** MXG/CC will determine non-contingency system functional check requirements.

11.12.2. DELETED.

11.12.2.1. The Radar Warning Receiver/Radar Threat Warning Manager if assigned, will coordinate test procedures with the Wing Electronic Warfare Officer and the MXS, if applicable. **(T-2)**

11.12.2.2. The Radar Warning Receiver/Radar Threat Warning Manager if assigned will ensure each unit accomplishes the required minimum number of checks as defined below. **(T-2)**

11.12.2.2.1. For contingency missions, the Radar Warning Receiver/Radar Threat Warning Manager will coordinate with the Electronic Warfare Officer/Electronic Combat Officer who will determine system check requirements and specific threats to be simulated. **(T-3)**

11.12.3. When an aircraft is found to have a malfunctioning Radar Warning Receiver/Radar Threat Warning system prior to flight, the Aircraft Commander (AC) determines the course of action based on operational needs and requirements.

11.13. Cannibalization Program.

11.13.1. General. CANN actions may be necessary when a condition prevents the accomplishment of a mission, and the required assets are not immediately available from supply. Prior to performing a CANN action, verify the required component cannot be sourced from LRS, TNB or back shop. When authorizing a CANN, the expenditure of man-hours and potential damage to equipment need to be weighed against the expected benefit. High risk

CANNs should not be performed unless priority aircraft are involved, or lack of ready equipment will impede mission accomplishment. See **Table 1.2** (Reference AFTTP 3-4.21V1).

11.13.2. Definition. CANN is the authorized removal of a specific assembly, subassembly, or part from one weapon system, system, support system, or equipment end item for installation on another end item to satisfy an existing supply requisition and to meet priority mission requirements with an obligation to replace the removed item. Weapon systems, support systems, or equipment include aircraft, missiles, drones, RPA, uninstalled engines, uninstalled engine modules, aircrew and/or launch crew training devices, Communications-Electronics equipment, AGE, TMDE, serviceable uninstalled pods, and guns.

11.13.3. Responsibilities. CANN Authorities (CA) will be tracked in the MIS and SCR (see **Table 11.1**). **(T-2)**

11.13.3.1. CA will be SNCOs, officers or civilian equivalents. **(T-2)** These personnel are typically Pro Supers.

11.13.3.2. Those who are authorized to approve CANNs will not further delegate their responsibility. **(T-2)**

11.13.4. If an assembly is cannibalized to satisfy a condition caused by lack of bits and pieces (such as, washers, nuts, and bolts), the assembly is counted as a CANN, and the bits and pieces are considered transfer actions. Bits and pieces removed from an end item (without removing the assembly) for installation on another end item are considered individual CANN actions.

11.13.5. When a required part cannot be delivered and installed on time, the CA may approve the CANN of parts before the initiation of CANN documentation (such as, Red Ball maintenance). The CA will give this approval only after confirming the part is not readily available in LRS, TNB, forward supply points, or back shops. **(T-2)**

11.13.5.1. The CA will notify the appropriate supply activity to change the “mark-for” components in the document number. **(T-2)**

11.13.5.2. The CA will also ensure complete documentation is accomplished for each CANN action. **(T-2)**

11.13.6. When TCIs, serially controlled items, items affecting compliance of a TCTO, or other components with inspection requirements that align to specific hourly, calendar, or event limits are considered for CANN, the CA will coordinate with PS&D or EM to ensure adequate time remains on the item to justify the CANN and to ensure appropriate records are updated. **(T-2)**

11.13.6.1. If the CANN action takes place, the performing work center will update the MIS and notify PS&D or EM. **(T-2)**

11.13.7. Installed engines are not end items; installed engines are considered a LRU (similar to a radar component, gun, seat, canopy, radio, multifunction display unit).

11.13.7.1. If a functional LRU is removed from one end item to put on another end item to fill a “hole” which was caused by a supply requisition, (the requisition could be against the LRU), then this is considered a CANN.

11.13.8. Restrictions.

11.13.8.1. Egress system component CAD/PAD cannibalization actions are considered "High-Risk" and should not be performed unless priority aircraft are involved (example, higher headquarters/alert status), or lack of ready equipment will impede mission accomplishment.

11.13.8.1.1. To ensure system integrity and validation of the explosive CAD/PAD listing, cannibalization of egress explosive and seat components require MXG approval. **(T-3)**

11.13.8.1.2. After cannibalization actions, Egress Red X discrepancies in the aircraft AFTO Form 781As will not be cleared until verification that CAD/PAD S/N content matches the S/N content entries in the MIS. **(T-2)**

11.13.8.1.3. Only (2A6X3) Egress personnel will accomplish this action. **(T-2)**

11.13.8.2. CANN actions involving parts from ABDR training aircraft, AF Museum Aircraft, Maintenance Training Devices, GITA, TAA, or DLADS will not be accomplished without authorization from the Program Office. **(T-2)**

11.13.8.2.1. Parts will not be removed from static display/AF Museum Aircraft except as authorized by AFI 84-103. **(T-2)**

11.13.8.2.2. If the part is approved for CANN, it must not be put into service until all necessary inspections (such as, NDI, pressure checks, operational checks, TCTOs) have been accomplished using specific guidance from the item manager to ensure proper serviceability. **(T-2)**

11.13.8.3. Units will not CANN parts from aircraft possessed by AFMC (B or D PIC) without first coordinating through the MAJCOM functional manager who will request approval from the applicable PM. **(T-2)**

11.13.8.4. An aircraft that has been extensively cannibalized will not be launched on an overseas or cross-country sortie/mission on the first flight following CANN rebuild without the owning MXG/CC or equivalent approval. **(T-2)**

11.14. Hangar Queen Aircraft.

11.14.1. General. The objective of this program is to ensure the entire fleet remains healthy and all possible management actions are carried out to ensure aircraft do not remain inoperative for extended periods. MAJCOMs will establish a Hangar Queen Management Program.

11.14.1. **(AFMC)** See AFMC Sup paragraph **11.14.3** through **11.14.4.2** for Hangar Queen Management Program.

11.14.2. Definitions. A “Hangar Queen” is a unit-possessed aircraft that has not flown for at least 30 calendar days. Aircraft are exempt from accruing Hangar Queen time for up to 10 days immediately following DFT/CFT repair or maintenance; however, if an aircraft is not flown after the 10th day, the 10 days are included in the total number of days since last fly date to determine the Hangar Queen category computation. Hangar Queen aircraft will be further defined by the following three categories:

11.14.2.1. Category 1: Aircraft that have not flown for 30 to 59 calendar days. **(T-2)**

11.14.2.2. Category 2: Aircraft that have not flown for 60 to 89 calendar days. **(T-2)**

11.14.2.3. Category 3: Aircraft that have not flown for 90 or more calendar days. **(T-2)**

11.14.3. All aircraft placed on higher HHQ alert status are exempt from the Hangar Queen Management Program and reporting throughout the duration of alert status/posturing.

11.14.3. **(AFMC)** AF Test Center aircraft undergoing Temporary Modifications or De-Modifications for the purpose of Operational or Developmental testing are exempt from Hangar Queen Reporting for the duration of the modification efforts.

11.14.3.1. **(Added-AFMC)** When an aircraft becomes a Category 1 Hangar Queen, establish a maintenance recovery plan. Brief aircraft maintenance and supply status once weekly to the wing standup meeting. Category 1 aircraft are managed locally. **(T-2)**

11.14.3.2. **(Added-AFMC)** When an aircraft becomes a Category 2 Hangar Queen, assign a SNCO or officer (or civilian equivalent) to manage the Hangar Queen. The MXG/CC or designated representative must approve any further CANNs, transfer, and diversion actions from the Hangar Queen aircraft. Brief aircraft maintenance and supply status at the daily wing standup meeting. Units will report aircraft tail number(s) to MAJCOM. POC is HQ AFMC/A4MM. **(T-2)** Category II aircraft are managed locally.

11.14.3.2.1. **(Added-AFMC)** The minimum information required to report is **(T-2)**:

11.14.3.2.1.1. **(Added-AFMC)** Hangar Queen Management POC.

11.14.3.2.2. **(Added-AFMC)** Aircraft MDS and Tail #.

11.14.3.2.3. **(Added-AFMC)** Unit of Assignment/Possession.

11.14.3.2.4. **(Added-AFMC)** Date last flown.

11.14.3.2.5. **(Added-AFMC)** Reason for Hangar Queen status.

11.14.3.2.6. **(Added-AFMC)** Document number for parts on order for reason of Hangar Queen status with estimated delivery dates.

11.14.3.2.7. **(Added-AFMC)** Estimated next fly day.

11.14.3.2.8. **(Added-AFMC)** Identify if MAJCOM/weapons system team Lead Command assistance is required.

11.14.3.3. **(Added-AFMC)** When an aircraft becomes a Category 3 Hangar Queen, units will report aircraft tail number(s) to MAJCOM and continue Category 2 Hangar Queen actions. Units will send an updated report to MAJCOM at least monthly. **(T-2)**

11.14.3.3.1. **(Added-AFMC)** MXG/CC will determine the need for an FCF/Operational Check Flight (OCF) (if not otherwise required by the aircraft specific TO). **(T-2)**

11.14.3.3.2. **(Added-AFMC)** QA will perform a final review of all aircraft forms initiated since the last flight prior to the first flight. **(T-2)**

11.14.4. An aircraft is released from Hangar Queen status after the first flight. The following examples are provided to clarify when an aircraft becomes a Hangar Queen:

11.14.4.1. A unit-possessed aircraft has not flown for 20 calendar days, enters depot status for 5 more calendar days, and then returns to unit possession on the 26th non-fly day; the unit has up to 10 calendar days to fly the aircraft to avoid Hangar Queen status. If this

aircraft does not fly on the 10th calendar day (35th non-fly day), the aircraft will become 36-day Category 1 Hangar Queen on the next day.

11.14.4.2. A unit-possessed aircraft has not flown for 2 calendar days, then enters depot status for 1 calendar day and is returned to unit possession, the unit must fly the aircraft in the next 27 calendar days to avoid becoming a Category 1 Hangar Queen.

11.15. Ground Instructional Trainer Aircraft (GITA). GITA are permanently grounded aircraft declared excess to future operations or flying requirements by higher headquarters and will be re-designated by the addition of the prefix "G" to the basic MDS. GITA are not maintained in a flyable condition but maintain system/subsystem operational condition for purposes of maintenance training and will be carried in assignment/PIC outlined in DAFI 21-103 and AFI 16-402. This section does not apply to ABDR training aircraft. ABDR training aircraft are managed by AFSC/LZDB (ABDR Program Office (PO)). This **Chapter** does not apply to training equipment maintained by Contract Logistics Support contracts administered by commands other than AETC.

11.15.1. Only those items requested by the PM are considered for removal. If the item does not affect training and if approved by MXG/CC, the part will be removed and turned in as per the ALC MXG/CC's (or equivalent) instructions. **(T-2)**

11.15.1.1. Units are responsible for storing uninstalled or removed equipment that is not required for training. **(T-3)**

11.15.2. Training Aid Aircraft (TAA). TAA are permanently grounded aircraft that, at a minimum, require an aircraft fuselage that was previously in the AF inventory as an aircraft.

11.15.2.1. Assigned TAA are not maintained in airworthy condition, and only the system/subsystem required for the specific training requirements will be maintained in operational condition for purposes of required maintenance/organizational training. **(T-2)**

11.15.2.2. TAA used for training are not terminated from the AF inventory IAW AFI 16-402. TAA requests for use by non-maintenance AFSCs require coordination through AFMC/LCMC and the PM prior to approval of assignment via an AF Form 913. **(T-3)**

11.15.2.3. Questions about the designation of an aircraft used for training should be directed to the MAJCOM AVDO.

11.15.2.4. Permanently grounded missiles retain their original MDS without a prefix.

11.15.2.5. Upon assignment of a permanently grounded GITA/TAA, the MXG will contact the applicable MAJCOM to coordinate "save list" requirements identified by the applicable PM. **(T-2)**

11.15.2.5.1. "Save list" items removed will be turned into LRS for shipment. **(T-3)**

11.15.2.5.2. If an item on the "save list" is not removed, the reason for not removing it will be annotated and coordinated with the applicable MAJCOM. **(T-2)**

11.15.2.5.3. If items on the "save list" are required for training and an unserviceable item will suffice, units will coordinate with the applicable MAJCOM for receipt of the unserviceable item(s). **(T-2)**

11.15.2.5.4. All unserviceable items furnished by ALC will be marked/identified as "unserviceable" in a conspicuous manner (such as, Red X or Red dot system). **(T-2)**

11.15.3. MAJCOM Responsibilities. MAJCOMs will determine use of MIS for permanently grounded GITA records management.

11.15.3. **(AFMC)** Units will use the applicable MIS to document records management. **(T-2)**

11.15.3.1. MAJCOMs will coordinate "save list" requirements/changes with the applicable PMs.

11.15.4. MXG/CC Responsibilities. MXG/CC or equivalent will:

11.15.4.1. - 11.15.4.2. DELETED.

11.15.4.2.1. GITA maintenance includes on- and off-equipment maintenance of active systems and subsystems and necessary actions to maintain the aircraft in a safe and presentable condition.

11.15.4.2.2. TAA require minimal maintenance on systems/subsystems used to meet training requirements and should be maintained in a safe and presentable condition.

11.15.4.2.3. Determine which system and subsystem are required to support the training. Consider present, future, and cross-utilization of systems when making determinations. These systems will be maintained in the same configuration as operational equipment. **(T-3)**

11.15.4.2.4. - 11.15.4.2.7. DELETED.

11.15.4.2.7.1. Prior-to-use inspections will be conducted by the using organization employing a tailored weapon system pre-/post-dock checklist. **(T-3)**

11.15.4.2.7.2. Conduct periodic maintenance inspections using a tailored work deck. **(T-3)**

11.15.4.2.8. DELETED.

11.15.4.2.8.1. When changes in requirements occur, initiate a new memorandum.

11.15.4.2.8.2. Ensures copies of all GITA/TAA memorandums to the MAJCOM AVDO. **(T-2)**

11.15.4.2.9. DELETED.

11.15.4.2.10. DELETED.

11.15.4.2.10.1. Owning units not having maintenance capability will establish MOAs or MOUs with organizations which can provide maintenance support. **(T-2)**

11.15.4.2.11. - 11.15.4.2.11.2. DELETED.

11.15.4.2.12. DELETED.

11.15.4.2.12.1. When an item is removed or replaced, supervisors will ensure this action is documented in the aircraft forms. **(T-2)** Include the authority for removal (such as, message number, telecon, letters, and dates) and condition of installed/replacement items.

11.15.4.2.12.2. When the limited save list actions have been done, a copy of the completed list will be forwarded to the appropriate PM and the local documentation function which will be added to the TAA historical record. **(T-2)**

11.15.4.2.12.3. W&B handbook requirements will be maintained IAW TO 1-1B-50 and applicable -5 series TOs. **(T-2)**

11.15.4.2.12.4. Operating and maintenance technical data will be readily accessible whenever the GITA/TAA is in use or undergoing inspection. **(T-2)**

11.15.4.2.12.5. DELETED.

11.15.4.2.12.5.1. The GITA/TAA Manager must be qualified to operate GITA/TAA systems and appropriate support equipment to conduct GITA/TAA maintenance. **(T-3)**

11.15.4.2.12.5.2. The GITA/TAA Manager will accomplish and/or coordinate maintenance actions for the GITA/TAA and ensure GITA/TAA documentation is accurate and complete. **(T-3)**

11.15.4.2.13. For equipment designated as trainers, only the systems required for technical training (or those required to ensure safety or system integrity) need to be maintained. **Note:** This does not apply to "temporarily" grounded aircraft or operational equipment or systems on loan from MAJCOMs or ALCs.

11.15.5. Technical Data Applicability.

11.15.5.1. Operational systems on GITA/TAA are maintained IAW applicable technical data. The specific policy governing the use and modification of technical data is contained in TO 00-5-1.

11.15.5.1.1. Some systems may be operated and maintained with original contractor data because formal technical data was never developed and/or the contractor data was never assigned a TO number.

11.15.5.2. Inspection and lubrication requirements may be adjusted to correspond with training requirements and equipment usage and to prevent over or under inspection.

11.15.5.3. When significant savings may be achieved, the commander or contract project manager must request deviations or changes to technical data requirements, including substitution of materiel from the weapon system program manager.

11.15.5.3.1. If deviations are approved, the unit will retain approved deviations/changes in the GITA historical records. **(T-3)** In all cases, safety or design function must not be compromised.

11.15.5.4. TCTOs. The QA function or other designated agency will be responsible for determining applicability of TCTOs for GITAs. **(T-2)** TCTO upgrades are not required on TAA.

11.15.5.4.1. Ensure timely completion of TCTOs on systems designated for configuration management and proper configuration status accounting is maintained.

11.15.5.4.2. Accomplish TCTOs on systems not designated for configuration management as required to ensure safety of operation or as directed by the PM.

11.15.4.2.11.2. DELETED.

11.16. Aircraft Inlet/Intake/Exhaust Certification.

11.16.1. MAJCOMs will determine the requirement to implement an Aircraft Inlet/Intake/Exhaust Certification program and certification frequency requirements.

11.16.1.1. **(Added-AFMC)** This program is applicable to B-1, B-2, F-15, F-16, A-10, F-22, F-35, RPA, and propeller-driven aircraft/helicopters. (**Exception:** reciprocating engines.). **(T-2)**

11.16.1.2. **(Added-AFMC)** Units will have an installed and uninstalled aircraft/engine intake/inlet/exhaust training program. **(T-2)**

11.16.2. DELETED.

11.17. Engine Run Training and Certification Program.

11.17.1. A comprehensive engine run certification program will be developed and strictly enforced to prevent safety mishaps and potential loss of life. **(T-2)**

11.17.1.1. DELETED.

11.17.1.2. All maintenance personnel authorized to start and operate aircraft engines, APUs, and uninstalled engines and APUs will be trained and certified to operate engines at TO determined power settings. **(T-2)**

11.17.1.2. **(AFMC) Note:** F-35 Integrated Power Package (IPP) is considered an APU.

11.17.1.3. Aircraft engine motoring will only be performed by qualified engine run personnel. **(T-2)** Exception: Rotary wing maintenance personnel qualified through OJT may motor engines IAW prescribed TO.

11.17.1.4. The following minimum requirements will be used to certify engine run personnel:

11.17.1.4.1. The MT will serve as the OPR and focal point for the management and development of the engine run certification program, engine run certification test question bank, and written tests for their respective weapon system. **(T-2)**

11.17.1.4.2. Pre-run training will be conducted in the trainee's work center through OJT. **(T-2)** Pre-run training is designed to prepare the trainee for successful completion of initial engine-run training. As a minimum, pre-run training will include:

11.17.1.4.2.1. An evaluation by immediate supervisor or NCOIC/Flight Chief to determine the individual’s level of maturity and experience prior to being selected for engine-run training. **(T-2)**

11.17.1.4.2.2. The trainee will review and become familiar with engine-run operations to include emergency procedures IAW the applicable aircraft general system type TO and engine run checklist. **(T-2)**

11.17.1.4.2.3. MTs may develop a handout to facilitate learning engine-run procedures, engine limitations, and emergency procedures.

11.17.1.4.3. **(Added-AFMC)** The MTs will review approved engine run certification procedures annually for accuracy. **(T-2)**

11.17.1.4.4. **(Added-AFMC)** Units will develop local procedures clearly depicting authorized run locations. **(T-2)**

11.17.2. Certifying Officials. Certifying official certification requirements are listed in **Table 11.1**.

11.17.2.1. Instructor Pilots (IP) can also be used as certifying officials during the practical engine-run demonstration.

11.17.2.2. Certifying officials must maintain proficiency in the same manner as other technicians; certifying officials must recertify each other. **(T-2)**

11.17.3. Instructors. Individuals selected as instructors will possess a 7-skill level or civilian equivalent, a qualified contractor, or AFETS/CETS personnel. **(T-2)**

11.17.3.1. AFMAN 11-218, *Aircraft Operations and Movement on the Ground*, aircraft and engine TOs, commercial aircraft/engine operating procedures, and special test project engineering procedures will be used to develop engine run certification training programs. **(T-2)**

11.17.4. Installed Engine Run Personnel. Prior to entering engine run training, personnel will meet the following requirements:

11.17.4.1. Personnel will be selected IAW criteria established in **Table 11.1**. **(T-2)** MXG/CCs may designate contractors in writing to run aircraft engines.

11.17.4.2. Qualified to operate the aircraft APU as applicable. **(T-2)**

11.17.4.3. Qualified as a brake operator. **(T-2)**

11.17.4.4. Qualified in basic radio and interphone systems operation. **(T-2)**

11.17.4.5. Qualified on marshalling signals. **(T-2)**

11.17.5. The initial engine run certification program will consist of following three phases, each of which will be successfully completed before progressing to the next phase:

11.17.5.1. Phase 1. **(T-2)** Phase 1 is formal classroom training. Classroom instruction will include:

11.17.5.1.1. General aircraft familiarization to include, as a minimum, basic MDS airframe characteristic, aircraft safe-for-maintenance procedures, cockpit configuration and systems, throttles and aircraft controls, egress, normal and emergency braking systems, and aircraft system/subsystems related to safe engine operation. **(T-2)**

11.17.5.1.2. A thorough review of TO procedures with emphasis on and notes, cautions, and warnings. **(T-2)**

11.17.5.1.3. Engine/APU operation, to include normal operational parameters and limitations. **(T-2)**

11.17.5.1.4. Ensuring aircraft, engine, and APU emergency procedures are memorized. **(T-2)**

11.17.5.1.5. UHF/VHF radio operation, Air Traffic Control tower procedures, and emergency radio transmissions. **(T-2)**

11.17.5.1.6. A two-part closed book examination (students will successfully complete **Part I** before taking **Part II**). **(T-2)** The examination will consist of the following:

11.17.5.1.6.1. **Part I** - Students will be given a written/computer-based examination on all boldface emergency procedures or all emergency procedures identified in applicable technical data requiring a passing score of 100 percent. **(T-2)**

11.17.5.1.6.2. **Part II** - Students will be given a written examination covering normal engine run procedures and limitations requiring a minimum passing score of 90 percent, corrected to 100 percent. **(T-2)**

11.17.5.1.7. Personnel failing the written/computer-based examination will receive additional instruction before being re-tested. **(T-2)**

11.17.5.1.8. Students will not be given the same **Part II** test during re-testing efforts. **(T-2)**

11.17.5.1.9. After a second failure of the two-part closed book examination, the SQ/CC (or equivalent) will determine if personnel may retest and continue with the program. **(T-3)**

11.17.5.2. Phase 2. **(T-2)** Phase 2 is simulator training. All maintenance personnel requiring engine run certification will receive simulator training on each specific aircraft MDS and APU. **(T-2)**

11.17.5.2.1. Phase 2 simulator training will be accomplished in an Aircrew Training Device, Cockpit Trainer, flight simulator, Maintenance Training Device, or approved Technology Development Trainer. **(T-2) Note:** If any of the above are not available, a similar MD(S) simulator may be used if the procedures are the same or "dry run" procedures will be accomplished in an aircraft to ensure procedural knowledge.

11.17.5.2.2. As a minimum, students will demonstrate knowledge and proficiency in the following areas:

11.17.5.2.2.1. Proper run clearance procedures. **(T-2)**

11.17.5.2.2.2. UHF/VHF radio operation, Air Traffic Control tower procedures, and emergency radio transmissions. **(T-2)**

11.17.5.2.2.3. Normal APU, engine start, run, and shutdown procedures. **(T-2)**

11.17.5.2.2.4. Augmenter or thrust reverser operation (as applicable). **(T-2)**

11.17.5.2.2.5. Applicable aircraft systems/subsystems normal operating parameters. **(T-2)**

11.17.5.2.2.6. Ensure TO emergency bold face items are memorized. **(T-2)**

11.17.5.2.2.6.1. Instructors will evaluate the student on response time and ability to handle emergency situations to include egress procedures. **(T-2)**

11.17.5.3. Phase 3. **(T-2)** Phase 3 is practical demonstration. Each individual will receive a practical engine run evaluation after successful completion of Phase 1 and Phase 2 training. **(T-2)** For fighter-type aircraft, it is preferable to conduct the evaluation in an NSS,

or on a trim pad. As a minimum, the student will demonstrate successful completion of the following areas without any discrepancies based on a go/no-go standard:

11.17.5.3.1. Run clearance procedures. **(T-2)**

11.17.5.3.2. UHF/VHF radio operation, Air Traffic Control tower procedures, and emergency radio transmissions. **(T-2)**

11.17.5.3.3. Normal APU, engine start, run, and shutdown procedures, including notes, cautions, and warnings. **(T-2)**

11.17.5.3.4. Augmenter or thrust reverser operation as applicable, including notes, cautions, and warnings. **(T-2)**

11.17.5.3.5. Applicable aircraft systems/subsystems normal operating parameters, including notes, cautions, and warnings. **(T-2)**

11.17.5.3.6. Ensure TO emergency bold face items are memorized. **(T-2)** Instructors will evaluate the student on response time and ability to handle emergency situations. **(T-2)**

11.17.5.3.7. Egress procedures. MAJCOM/Lead Command, TO, and checklist procedures for the applicable MDS will be demonstrated without error. **(T-2)**

11.17.6. Annual recertification for certifying officials and engine run certified personnel will be accomplished by successfully completing the written test (**Part I** and **Part II**) administered by the MT and demonstrating proficiency of normal and emergency procedures to a certifying official by operating one of the following: Aircrew Training Device, Cockpit Trainer, authorized Technology Development Trainer (if assigned or available), or aircraft as appropriate. **(T-2)**

11.17.6.1. Personnel failing the written examination will receive additional instruction before being re-tested. **(T-2)**

11.17.6.2. Students will not be given the same **Part II** test during re-testing efforts. **(T-2)**

11.17.6.3. After a second failure of the two-part closed book examination, the individual will be decertified. **(T-2)**

11.17.6.3.1. The SQ/CC (or equivalent) will determine if personnel may retest and continue with the program, and whether they must attend all three phases of initial training prior to being recertified. **(T-2)**

11.17.6.4. Certified individuals who PCS to the same MDS, and engine type and model must be approved by the SQ/CC (or equivalent) and complete an initial evaluation by a certifying official prior to becoming run qualified at the gaining base. **(T-2) Note:** MAJCOMs will determine if additional training is required for the specific engine series.

11.17.6.4. **(AFMC)** Provided certified individuals remain proficient IAW paragraph 11.17.8 while transitioning to gaining unit, no additional training is necessary beyond the initial evaluation. **(T-2)**

11.17.6.4.1. The evaluation will include, as a minimum, familiarization of local procedures and requirements. **(T-2)**

11.17.6.4.2. Carry over the date of original class completion from previous documentation (certificate, training record, MIS printout).

11.17.7. Documentation. Qualifications of installed engine run certifying officials and engine run certified personnel, will be documented in the MIS and entered on the SCR. **(T-2)**

11.17.8. Proficiency. MAJCOMs will determine proficiency requirements for maintenance personnel authorized to operate installed engines.

11.17.8. **(AFMC)** To maintain proficiency, maintenance personnel authorized to operate engine(s) will perform at least one engine run on each aircraft engine type or weapons system they are certified to operate, in order to maintain currency and remain qualified, every 90 days. **(T-2)**

11.17.8.1. Units will track run proficiency requirements in the MIS. **(T-2)**

11.17.8.2. Supervisors will ensure individuals who fail to maintain proficiency are decertified. **(T-2)**

11.17.8.2.1. Decertified individuals will recertify IAW **Paragraph 11.17.6**. **(T-2)**

11.17.9. Engine run certification tests are controlled items and will be handled IAW DAFMAN 36-2664, *Personnel Assessment Program*, and administered only by MT personnel or regional training centers. **(T-2)**

11.17.10. Aircraft APU Installed Operation Training. The following requirements and standards will apply to qualifying maintenance personnel on operating the aircraft APU:

11.17.10.1. When conducting initial operator qualification training for APU, use the applicable video or other training program. **(T-2)**

11.17.10.2. A two-part closed book examination consisting of the following: **Note:** MAJCOMs will determine examination applicability requirements for PMA only APU operations in their supplement/addendum to this DAFI.

11.17.10.2. **(AFMC)** Two-part testing does not apply to F-22 APU and F-35 IPP operators using the Portable Maintenance Aid (PMA) from the ground. **(T-2)**

11.17.10.2.1. **Part I** - Students will be given a written/computer-based examination on all boldface emergency procedures or all emergency procedures identified in applicable technical data requiring a passing score of 100 percent**. (T-2)**

11.17.10.2.2. Students will successfully complete **Part I** before taking **Part II. (T-2)**

11.17.10.2.3. **Part II** - Students will be given a written/computer-based examination covering normal APU run procedures and limitations requiring a minimum passing score of 90 percent, corrected to 100 percent. **(T-2)**

11.17.10.3. Personnel failing the examination will receive additional instruction before being re-tested. **(T-2)**

11.17.10.4. Students will not be given the same **Part II** test during re-testing efforts. **(T-2)**

11.17.10.5. After a second failure of the two-part closed book examination, the individual will be decertified. **(T-2)**

11.17.10.5.1. The SQ/CC (or equivalent) will determine if personnel may retest and continue with the program prior to being recertified. **(T-2)**

11.17.10.5.2. Individuals must attend all three phases of initial training prior to being recertified. **(T-2)**

11.17.10.6. **Part III** Personnel must then accomplish an on-equipment practical evaluation for certification completion. **(T-2)**

11.17.10.7. Personnel will be recertified annually using the initial certification procedures. The practical evaluation portion will be accomplished by operating one of the following: Aircrew Training Device, Cockpit Trainer, authorized Technology Development Trainer (if assigned or available), or aircraft as appropriate (or as determined by the certifying official). **(T-2) Note:** Recertification is not required if the individual is engine run certified and has maintained annual engine-run certification requirements.

11.17.11. Documentation. Qualifications of APU run certifying officials and APU run certified personnel, will be documented in the MIS and entered on the SCR. **(T-2)**

11.17.11.1. If applicable, MAJCOMs will define SCR applicability requirements for PMA only APU operations in their supplement/addendum to this instruction.

11.17.11.1.1. **(Added-AFMC)** Any personnel performing F-22 APU or F-35 IPP operation from the cockpit will be engine run qualified and listed on the SCR. **(T-2)**

11.17.11.1.2. **(Added-AFMC)** When another person is in the cockpit, personnel performing F-22 APU or F-35 IPP operations using the PMA from the ground will be engine run qualified and listed on the SCR. **(T-2)**

11.17.11.1.3. **(Added-AFMC)** Personnel performing F-22 APU or F-35 IPP operations using the PMA from the ground when nobody is in the cockpit do not need to be listed on the SCR. **(T-2)**

11.17.12. Proficiency. MAJCOMs will determine proficiency requirements for maintenance personnel authorized to operate APUs.

11.17.12. **(AFMC)** To maintain proficiency, maintenance personnel authorized to operate APU/GTC/IPP will perform at least one APU/GTC/IPP run every 180 days if applicable to the MDS. (**T-2**)

11.17.12.1. Units will track run proficiency requirements in the MIS. **(T-2)**

11.17.12.2. Supervisors will ensure individuals who fail to maintain proficiency are decertified. **(T-2)**

11.17.13. Certification tests are controlled items and will be handled IAW AFI 36-2605 and administered only by MT personnel. **(T-2)**

11.17.14. Uninstalled Engine Operation on Test Stands and Cells (includes Jet Fuel Starter /APU uninstalled operations). All personnel identified for uninstalled engine run qualification will complete an uninstalled engine run training program prior to certification. **(T-2)** The following minimum requirements will apply:

11.17.14.1. Certification Requirements. Individuals will be certified for each specific engine TMS authorized to run IAW criteria established in **Table 11.1**. **(T-2)**

11.17.14.1. **(AFMC)** Certified contractor personnel may be authorized to run uninstalled engines and/or small gas turbine engines when designated in writing by the respective MXG/CC/CL or equivalent. **(T-2)**

11.17.14.2. Certifying Officials will be designated IAW criteria established in **Table 11.1**. **(T-2)**

11.17.14.3. Instructors. Individuals selected as instructors will be 7-skill level or civilian equivalent, a qualified contractor, or an AFETS/CETS representative, and be run certified on each TMS (if they are to be certifying officials). **(T-2)**

11.17.14.4. Training. Uninstalled engine run training will consist of the following three phases:

11.17.14.4.1. Phase 1. Phase 1 is formal training. Instruction will include, as a minimum, the following areas:

11.17.14.4.1.1. General engine familiarization to include, as a minimum, basic engine description, component location, and functions. **(T-2)**

11.17.14.4.1.2. Thorough familiarization of control cabs, NSSs, ETSs, and T-9 fire suppression control panels (if applicable). **(T-2)**

11.17.14.4.1.3. Thorough review of TO procedures with emphasis on notes, cautions, and warnings. **(T-2)**

11.17.14.4.1.4. Uninstalled engine operation to include normal operating parameters and limitations. **(T-2)**

11.17.14.4.1.5. Ensuring uninstalled engine emergency procedures are memorized. **(T-2)**

11.17.14.4.1.6. Local communication procedures. **(T-2)**

11.17.14.4.1.7. A two-part closed book examination (students will successfully complete **Part I** before taking **Part II**) consisting of the following:

11.17.14.4.1.7.1. **Part I** - Students will be given a written/computer-based examination on all boldface emergency procedures or all emergency procedures identified in applicable technical data requiring a passing score of 100 percent. **(T-2)**

11.17.14.4.1.7.2. **Part II** - Students will be given a written/computer-based examination covering normal engine run procedures and limitations requiring a minimum passing score of 90 percent, corrected to 100 percent. **(T-2)**

11.17.14.4.1.8. Personnel failing the examination will receive additional instruction before being re-tested. **(T-2)**

11.17.14.4.1.9. Students will not be given the same **Part II** test during re-testing efforts. **(T-2)**

11.17.14.4.1.10. After a second failure of the two-part closed book examination, the individual will be decertified. **(T-2)**

11.17.14.4.1.10.1. The SQ/CC (or equivalent) will determine if personnel may

retest and continue with the program prior to being recertified or be removed from program. **(T-2)**

11.17.14.4.1.10.1.1. Individuals must attend all three phases of initial training prior to being recertified. **(T-2)**

11.17.14.4.2. Phase 2. Phase 2 is the control cab evaluation. After successful completion of formal training, students will properly demonstrate the following minimum requirements to a certifying official without discrepancies using the go/no-go standard:

11.17.14.4.2.1. Proper uninstalled engine start, run, and shutdown procedures, including notes, cautions, and warnings (engine not operating). **(T-2)**

11.17.14.4.2.2. Proper uninstalled engine boldface emergency procedures, including notes, cautions, and warnings (engine not operating). **(T-2)**

11.17.14.4.2.3. Knowledge of normal uninstalled engine operating limits, including notes, cautions, and warnings. **(T-2)**

11.17.14.4.2.4. Augmenter or thrust reverser operation (as applicable), including notes, cautions, warnings, and emergency procedures. **(T-2)**

11.17.14.4.3. Phase 3. Phase 3 is the practical evaluation. Each individual will receive a practical uninstalled engine run evaluation after successful completion of classroom training and control cab evaluation from a certifier. **(T-2)** As a minimum, the student will demonstrate successful completion of the following areas without discrepancies based on a go/no-go standard:

11.17.14.4.3.1. Run clearance procedures. **(T-2)**

11.17.14.4.3.2. Emergency communication procedures. **(T-2)**

11.17.14.4.3.3. Normal uninstalled engine start, run, and shutdown procedures, including notes, cautions, and warnings. **(T-2)**

11.17.14.4.3.4. Augmenter or thrust reverser operation (as applicable), including notes, cautions, and warnings. **(T-2)**

11.17.14.4.3.5. Proper emergency procedure corrective actions during all bold face uninstalled engine emergency conditions. **(T-2)**

11.17.14.5. Recertification. Recertification for certifying officials and uninstalled engine run qualified personnel will be accomplished annually. **(T-2)**

11.17.14.5.1. The following three requirements must be met to obtain recertification:

11.17.14.5.1.1. Successfully completing the written test (**Part I** and **Part II**) administered by the MT. **(T-2)**

11.17.14.5.1.2. Passing a control cab evaluation demonstrating knowledge of normal and emergency procedures to a certifying official. **(T-2)**

11.17.14.5.1.3. Completing a practical engine run demonstration. **(T-2)**

11.17.14.5.2. Personnel failing the written examination will receive additional instruction before being re-tested. **(T-2)**

11.17.14.5.3. Students will not be given the same **Part II** test during re-testing efforts. **(T-2)**

11.17.14.5.4. After a second failure of the two-part closed book examination, the individual will be decertified. **(T-2)**

11.17.14.5.4.1. The SQ/CC (or equivalent) will determine if personnel may retest and continue with the program prior to being recertified. **(T-2)**

11.17.14.5.4.2. Individuals must attend all three phases of initial training prior to being recertified. **(T-2)**

11.17.14.6. Proficiency. MAJCOMs will determine proficiency requirements.

11.17.14.6. **(AFMC)** All uninstalled engine/small gas turbine engine run certified individuals and certifiers will run at least one engine every 180 days. When not feasible to run every 180 days, individuals will complete part I and part II tests to maintain proficiency. **(T-2)**

11.17.14.6.1. Supervisors will ensure individuals who fail to maintain proficiency are decertified. **(T-2)**

11.17.14.6.1.1. Decertified individuals will recertify IAW **Paragraph 11.17.14.5**. **(T-2)**

11.17.15. Fire Control Panel Operation in NSS. This section applies to all NSS designed for enclosed aircraft and uninstalled engine operation (such as, T-9, T-10, T-11, T-12, and T-20) with fire suppression systems. Only qualified personnel will be certified to use the NSS Fire Suppression Systems. **(T-2)** The following certification requirements will apply:

11.17.15.1. Meet criteria established in **Table 11.1**. **(T-2)**

11.17.15.2. Training will consist of formal training using TOs and hands on familiarization and will include the following minimum requirements:

11.17.15.2.1. NSS Fire Suppression System familiarization and operation. **(T-2)**

11.17.15.2.2. Emergency procedures, including local notification procedures. **(T-2)**

11.17.15.3. NSS supervisor, contractor, AFETS/CETS personnel or individual designated by the NSS supervisor will serve as certifying official(s). **(T-2)**

11.17.15.4. Annual recertification of NSS Fire Suppression System certified personnel will be accomplished utilizing the same criteria as initial certification. **(T-2)**

11.17.15.4.1. **(Added-AFMC)** Certification will be tracked on SCR. **(T-2)**

11.17.16. **(Added-AFMC)** Special Test Projects. Some AFMC units will participate in special testing of uninstalled engines and/or small gas turbine engines that do not fall within the scope of the normal certification processes outlined in this chapter. **(T-2)** Under these circumstances, the affected unit will apply the following guidelines to ensure test project engines are operated to the safest extent possible:

11.17.16.1. **(Added-AFMC)** Ensure engine operations on test projects that do not have approved Air Force technical procedures have written operating procedures including

emergency procedures approved by the MXG/CC/CL or equivalent and appropriate safety agencies. **(T-2)**

11.17.16.2. **(Added-AFMC)** When engine operating procedures do not exist in Air Force TOs, test project plans will specify use of commercial engine operating manuals (if they exist) and require a contractor or USAF test engineer for the duration of engine testing. **(T-2)**

11.18. Engine Blade Blending Training and Certification Program.

11.18.1. General. All units will have a comprehensive training program to ensure technical standards are met, and proficiency is maintained. **(T-2)** The number of individuals authorized to inspect and repair blades should be sufficient to meet mission requirements and production needs. Personnel will be selected for certification IAW criteria established in **Table 11.1**. **(T-2)** MXG/CCs may designate contractors in writing to complete blade blending certification.

11.18.2. Responsibilities and Management. The MT or TD will be responsible for management and development of the blade blending training program. **(T-2)**

11.18.2. **(AFMC)** [DEV] Within the 412th TW, Responsibility and management for the Engine Blade Blending Training and Certification process will be completed by the Propulsion Flight. This process includes the Propulsion Flight due to the multiple different engine models. All responsibilities contained in DAFI 21-101 will be adhered to by Propulsion Flight leadership.

11.18.2.1. At a minimum, the course will include care and handling of equipment, applicable technical data, fault isolation/damage assessment/defect size determination, techniques required to correctly inspect and repair blades and performance of an engine blade blend. **(T-2)**

11.18.2.2. Prior to placement on the SCR, members must complete formal blade blending training (MT or TD course) and initial engine blade blending certification will be mandatory. **(T-2)**

11.18.2.3. **(Added-AFMC)** Initial engine blade blending certification. **(T-2)**

11.18.3. - 11.18.3.3. DELETED.

11.18.4. Certification Officials. Certifying officials will be selected IAW criteria established in **Table 11.1**.

11.18.4.1. **(Added-AFMC)** Certification Criteria. Certifying officials will be selected IAW criteria established in AFMC Table 11.1 of this instruction. Units will limit the number of certifiers to a minimum to ensure standardized training and certification. Certifying officials will maintain proficiency in the same manner as other technicians; certifying officials will recertify each other. **Note:** If applicable, assigned AFETS/CETS should be used to certify other certifying officials. **(T-2)**

11.18.4.2. **(Added-AFMC)** Certified individuals who PCS to the same MDS and engine Type/Model/Series and Modification (TMSM), and are still current on their proficiency, may by-pass formal training course. These individuals will be re-certified by a certifier prior to being added to the SCR. Carry over date of original class completion from previous documentation (certificate, training record, MIS printout). **(T-2)**

11.18.4.3. **(Added-AFMC)** Upon completion of the formal training, individuals are task evaluated by the certifying official (an individual other than the instructor who administered the course), and placed on the SCR. **(T-2)**

11.18.5. **(Added-AFMC)** Proficiency Requirements. As a minimum, B-1, B-2, F-15, F-16, F-22, F-35 and U-2 personnel must perform one blend repair every 180 days to maintain proficiency. Work centers may use trainer engine blades to perform blade blend proficiency. Work center supervisors ensure personnel who do not meet this requirement are decertified. (**T-2)**

11.18.6. **(Added-AFMC)** Annual Recertification. Recertification is accomplished by having the technician demonstrate they can perform the task(s). A QA PE may be used to satisfy this requirement if the QA evaluator is a certifying official. **(T-2)**

11.18.7. **(Added-AFMC)** Blade blending procedures for installed/uninstalled engines/modules:

11.18.7.1. **(Added-AFMC)** Notify the Wing/Center FOD Monitor prior to blade blending anytime FOD is identified, other than for minor sand nicks or scratches (i.e. blending with emery cloth). **(T-2)**

11.18.7.2. **(Added-AFMC)** Fill out Blade Blending/FOD Damage worksheet or applicable form with the following information; engine serial number, stage number, number of blades blended, depth of damage before and after blend, area of damage and employee number/stamp number of maintenance personnel. **(T-2)**

11.18.7.3. **(Added-AFMC)** Notify EM section and forward Blade Blending/FOD Damage worksheet or applicable form to EM section for filing. The EM section will transcribe information provided in the Blade Blending/FOD Damage worksheet into the applicable engine/module records (i.e., AFTO Form 95, *Significant Historical Data*; if applicable) and CEMS, IAW TO 00-20-1. **(T-2)**

11.18.7.3.1. **(Added-AFMC)** The EM section will also document CEMS IAW 00-25-254-1, *Comprehensive Engine Management System* (CEMS) (D042) *Engine Status, Configuration, and TCTO Reporting Procedures*. **(T-2)**

11.19. Engine Flexible Borescope Inspection Training and Certification Program. The purpose of this program is to ensure individual knowledge and proficiency levels; proper care and use of equipment; and standardization of program requirements.

11.19.1. All units maintaining engines using flexible borescopes will establish a comprehensive training program. **(T-2)** Certification procedures described here are only for engine borescope certification.

11.19.1.1. Training will be annotated in training records. **(T-2)**

11.19.2. MAJCOMs will:

11.19.2.1. Ensure an engine flexible borescope formal training course is developed, tracked, and managed by MT/TD.

11.19.2.1.1. **(Added-AFMC)** MT, Propulsion Flight, or TD will develop and manage training. MXG/CC will select maintenance instructors or TD instructors to provide training, (AFETS/CETS may be used as alternate instructors). As a minimum, courses

will include care and handling of the equipment, all borescope/ port locations to include all inspection requirements and procedures, all applicable technical data, fault isolation/damage assessment/defect size determination, and performance of an actual engine borescope. **(T-2)**

11.19.2.1.2. **(Added-AFMC)** Course material/instruction will include high interest borescope inspections (e.g., blade tip curl HS3 inspection on the F100-series engine). **(T-2)**

11.19.2.2. Ensure engine flexible borescope proficiency and annual recertification (by a certifying official) requirements are established by course code.

11.19.2.2. **(AFMC)** MT, Propulsion Flight, or TD will ensure engine flexible borescope proficiency and annual recertification requirements are established by course code.

11.19.2.3. Ensure time, training and documentation currency requirements are established for engine flexible borescope certified personnel who PCS to the same MDS/engine.

11.19.2.3. **(AFMC)** Personnel who utilize borescope equipment to inspect or perform maintenance on jet engine modules that are not installed on the final engine do not require recurring borescope training. Final borescope inspection of the completed engine must be performed by a certified technician who has completed this recurring requirement. **(T-2)**

11.19.2.3.1. **(Added-AFMC)** Personnel who utilize borescope equipment to inspect or perform maintenance on jet engine modules not installed on aircraft are required to complete initial borescope training for assigned TMS. **(T-2)**

11.19.2.4. Determine training requirements for personnel using borescopes for non-engine type inspections (such as, behind ejection seats, wing boxes) to include, as a minimum, proper use, and care of borescopes.

11.19.2.5. **(Added-AFMC)** Certified individuals who PCS to the same MDS and engine TMSM may by-pass the formal training course. These individuals must be re-certified by a certifier prior to being added to the SCR. Carry over the date of original class completion from previous documentation (certificate, training record, MIS printout). **Note:** If applicable, assigned AFETS/CETS should be used to certify other certifying officials. **(T-2)**

11.20. Flying Crew Chief (FCC) Program. The purpose of the FCC Program is to enhance mission effectiveness by providing qualified maintenance support for aircraft at locations other than home station. The FCC flies in Mission Essential Personnel status. FCC's typically fly with the aircraft for the purpose of accomplishing ground maintenance at the TDY location. The duty period typically starts when the FCC shows at the aircraft prior to departure. FCCs are qualified in their duty AFSC and are required to obtain, maintain, and apply basic knowledge in several other aircraft maintenance AFSCs. They are responsible for launch, recovery, inspection, servicing, generation, and maintenance of aircraft in austere locations and locations where specific MDS maintenance capability may not be available.

11.20.1. MAJCOMs may authorize/develop an FCC Program under the direction of AF/A4LM for maintainers who are required to regularly fly and maintain aircraft.

11.20.1.1. FCCs will be selected per mission requirements as directed by MAJCOMs and qualify for Special Duty Assignment Pay (SDAP) IAW AFI 36-3017, *Assignment Incentive Pay and Special Duty Assignment Pay*.

11.20.2. The FCC program only applies to personnel assigned to positions on the Unit Manning Document with a "C" prefix for the Duty AFSC.

11.20.3. The following situations will not qualify the FCC for SDAP:

11.20.3.1. Occasional flights where the aircraft is used as transportation in lieu of commercial air.

11.20.3.2. Incentive or indoctrination flights.

11.20.3.3. Deployments where additional maintenance personnel are required at the designated location to supplement assigned maintainers.

11.20.4. Qualifying missions. A mission consists of one or more sorties with a mission number as entered on the AFTO Form 781, *Aircrew/Mission Flight Data Document*. The mission must meet at least one of the following criteria to qualify for this program:

11.20.4.1. The FCC is required to accomplish maintenance at locations other than home station to prepare the aircraft for its next departure. **(T-2)**

11.20.4.2. The FCC is required to perform in flight maintenance tasks, if required IAW MDS TO requirements or Special/Alert sorties assigned with a mission number with documented agreement between the Operations and Maintenance Squadron Commanders. **(T-2)**

11.20.5. FCC Program responsibilities.

11.20.5.1. AF/A1PA oversees the overall SDAP and provides guidance in AFI 36-3017.

11.20.5.2. AF/A4LM is the SDAP functional manager for FCCs.

11.20.5.2.1. AF/A4LM sets criteria for FCCs, validates MAJCOM FCC reports, and forecasts FCC SDAP budget needs.

11.20.5.2.2. AF/A4LM approves/disapproves FCC position increases/decreases in coordination with AF/A1PA.

11.20.5.3. MAJCOMs implement the FCC Program and will appoint in writing an FCC Program Manager to enforce standards and prepare the annual report.

11.20.5.3. **(AFMC)** HQ AFMC/A4MM manages the AFMC FCC program.

11.20.5.4. MAJCOM FCC Program Managers will determine which squadrons will participate in the FCC Program and will:

11.20.5.4.1. Validate and forward squadron FCC SDAP requests (**Attachment 5**) to AF/A4LM and AF/A1PA.

11.20.5.4.2. Annually validate FCC SDAP positions.

11.20.5.4.3. Assign FCC SDAP positions with an AFSC prefix of "C" and an appropriate SEI on command manpower documents.

11.20.5.4.4. Establish command unique training requirements and set additional qualification standards for their FCCs as needed.

11.20.5.4.5. Maintain quarterly and annual FCC reports (**Attachment 3** and **Attachment 4**).

11.20.5.4.6. Prepare and submit the command annual FCC report to AF/A4LM at **AF.A4LM.Workflow@us.af.mil** and AF/A1PA by 15 August each year. Submit the biennial FCC report to AF/A1PA upon request.

11.20.5.4.7. Review and approve/disapprove ACR for changes of the "C" prefix to an AFSC on the UMD.

11.20.5.4.8. Review and recommend approval/disapproval of ACRs for additions, deletions of the "C" prefix to an AFSC on the UMD.

11.20.5.5. MAJCOM (A1M) Command Manpower and Organization Responsibilities. A1M will:

11.20.5.5.1. Coordinate and obtain approval/disapproval from MAJCOM for Installation Manpower and Organization Office ACRs pertaining to validation of "C" prefix to AFSCs on the UMD.

11.20.5.5.2. Assign the "C" prefix to AFSCs upon approval from MAJCOM FCC Program Manager. This provides MAJCOM functional managers and unit senior maintenance managers the visibility of squadron FCC SDAP positions. **Note:** FCC SDAP positions do not affect a unit's manpower authorizations.

11.20.5.6. SQ/CC's will:

11.20.5.6.1. Administer the squadron FCC Program IAW AFI 36-3017, AFMAN 36-2100, *Military Utilization and Classification*, and this instruction.

11.20.5.6.2. Ensure FCCs fly only when required for the mission. **(T-2)**

11.20.5.6.3. Appoint and remove personnel from the FCC Program IAW AFMAN 36-2100.

11.20.5.6.3.1. Assign FCCs for a minimum of one year, unless removed for cause. **(T-2)**

11.20.5.6.4. Ensure only qualified FCCs and assistant FCCs who meet minimum requirements IAW AFI 36-3017 receive SDAP and fly a minimum of three qualifying missions per quarter. **(T-2)** An indicator of having too many FCCs may be reflected in a unit whose FCCs routinely do not meet minimum quarterly requirements.

11.20.5.6.5. Assign no more than two FCCs per aircraft (an FCC and assistant FCC) to each qualifying mission unless otherwise approved by MAJCOM. **(T-2) Exception**: SQ/CC may assign the minimum number of additional FCCs when required to maintain proper work-rest cycles or to meet TO requirements.

11.20.5.6.6. Appoint in writing a Unit FCC Program Manager. **(T-2)**

11.20.5.7. Unit FCC Program Managers will:

11.20.5.7.1. Track status and prepare unit reports. **(T-2)**

11.20.5.7.2. Ensure personnel possess the appropriate SEI for their MDS aircraft. **(T-2)**

11.20.5.7.3. Provide a letter to their Installation Manpower and Organization Office and an information copy to the MAJCOM FCC Program Manager to change, add, or delete a "C" prefix to the AFSC on the UMD. **(T-2)**

11.20.5.7.3.1. The letter will contain the unit designation, function account code, AFSC, position number, and a POC. **(T-2)**

11.20.5.7.4. Ensure FCCs and assistant FCCs are aligned in a duty position with a "C" prefix by initiating an DAF Form 2096, *Classification/On-the-Job Training Action*, or special order. **(T-2)**

11.20.5.7.5. Counsel FCCs and assistant FCCs on SDAP termination (AFI 36-3017, **Table 3** lists reasons for termination). **(T-2)**

11.20.5.7.5.1. SDAP stops on the dates listed in this table. As long as a "C" prefix is attached to an AFSC the member will receive SDAP. **(T-2)**

11.20.5.7.6. Review, update, and authenticate the monthly SDAP roster. **(T-2)** The SDAP roster is the only administrative tool used to start, stop, or continue the FCC pay entitlement.

11.20.5.7.6.1. If changes are made on the monthly SDAP roster, an DAF Form 2096 or special order must be submitted to the Military Personnel Section (MPS). **(T-2)**

11.20.5.7.6.2. Authentication of the monthly SDAP roster validates that each FCC is meeting the full intent of the program. **Note:** AFI 36-3017 provides commanders conditions concerning pay entitlements.

11.20.5.7.7. Submit SDAP position increase/decrease requests to MAJCOM FCC Program Manager by message, e-mail, or letter stating the number of positions to be increased/decreased with a brief justification. **(T-2)**

11.20.5.7.7.1. MAJCOMs will forward requests to AF/A4LM for final approval.

11.20.5.7.8. Provide information for processing DD Form 1610, *Request and Authorization for TDY Travel of DoD Personnel*, for FCCs. **(T-3)**

11.20.5.7.9. Ensure TDY orders authorize FCCs to travel in Mission Essential Personnel status. **(T-2) Note:** Aeronautical orders do not apply to this program, as FCCs are not aircrew members.

11.20.5.7.10. Monitor training qualifications and currency to ensure only qualified FCCs are scheduled for missions. **(T-2)**

11.20.5.7.10.1. As a minimum, maintain a folder for each FCC containing training qualifications, immunizations, military passport information, appointment letters, and FCC Mission Reports. **(T-2)** If the unit mobility section already maintains these source documents, either electronic or paper copies may be maintained.

11.20.5.7.11. Coordinate scheduling of FCCs through Flight CC/Chiefs and operations schedulers. **(T-2)**

11.20.5.7.12. Maintain a Unit FCC Program Manager's Continuity Book. **(T-2)** As a minimum the continuity book will include:

11.20.5.7.12.1. Lists of required instructions with web addresses (including AFI 36-3017, AFMAN 36-2108 and this instruction). **(T-2)**

11.20.5.7.12.2. Unit FCC Program Manager appointment letter, DAF Form 2096 or special orders. **(T-2)**

11.20.5.7.12.3. Manpower correspondence assigning "C" prefix AFSC. **(T-2)**

11.20.5.7.12.4. Quarterly and annual FCC status reports, SDAP position requests and miscellaneous FCC and SDAP correspondence. **(T-2)**

11.20.5.7.13. Report program status by Fiscal Year (FY) quarters to MAJCOM FCC Program Manager NLT the 15th day of the month following each FY quarter and report FY annual program status to the MAJCOM NLT 15 July each year. **(T-2)**

11.20.5.7.13.1. Annual report will consist of the previous FY 4th quarter and current FY 1st, 2nd, and 3rd quarters (1 Jul - 30 Jun). **(T-2)**

11.20.5.7.14. Submit funding requests for flight clothing, per diem, and other related expenses for the annual budget (for safety during flight, flight clothing is mandatory for FCCs and Assistant FCCs). **(T-2)**

11.20.5.8. Installation Manpower and Quality Office will:

11.20.5.8.1. Forward ACR to MAJCOM to add, delete, or change "C" prefixes on AFSCs existing on the UMD. **(T-2)**

11.20.5.9. Enroute supervisors will:

11.20.5.9.1. Not assign FCCs to work other enroute aircraft. **(T-2)** However, FCCs left at an enroute location and awaiting transportation may be assigned to work other enroute aircraft.

11.20.5.9.2. Brief FCCs on local safety precautions, maintenance practices, and limitations. **(T-2)**

11.20.5.9.3. Coordinate with the Pilot in Command (PIC) and FCC to obtain transportation to/from quarters. **(T-2)**

11.20.5.10. The FCC will:

11.20.5.10.1. Establish duty shifts and rest periods with the PIC and enroute supervisor based on maintenance and mission requirements. **(T-2) Note:** Consider the duration of the flight, the ability to rest during the flight, and the quality of the rest during the flight. FCCs do not automatically enter crew rest with the aircrew upon arrival at an enroute/transient location unless the duty day was exceeded.

11.20.5.10.1.1. If the FCC's safety is jeopardized by fatigue, the FCC's duty day must end. **(T-2)**

11.20.5.10.2. Upon arrival at enroute locations, determine their ability to perform duties safely and effectively. **(T-2) Note:** The FCC's primary job is preparing the aircraft (inspect, service, aircraft forms maintenance) for the next mission.

11.20.5.10.2.1. Notify Tanker Airlift Control Center/Logistics Cell of planned crew rest periods and or establish an alternate point of contact during scheduled rest periods to minimize rest cycle interpretations. **(T-2)**

11.20.5.10.3. Coordinate with the PIC to ensure crew integrity for quarters is maintained. **(T-2)**

11.20.5.10.4. Coordinate with the PIC to ensure the FCC Performance Feedback Form in **Attachment 2** is completed by the PIC and provided the Unit FCC Program Manager upon return to home station. **(T-2)**

11.20.6. FCC qualifications and responsibilities.

11.20.6.1. FCCs should be a 2AX AFSC 5- or 7-skill level.

11.20.6.2. As a minimum, the FCC must be qualified and certified on the following MDS applicable items:

11.20.6.2.1. Possess a SEI of the aircraft assigned to the FCC. **(T-2)**

11.20.6.2.2. Refuel/defuel member and supervisor; concurrent servicing supervisor (as applicable). **(T-2)**

11.20.6.2.3. Tow member, tow supervisor, and tow brake operator. **(T-2)**

11.20.6.2.4. LOX/GOX servicing, nitrogen, and tire servicing. **(T-2)**

11.20.6.2.5. Tire and brake change; launch; recovery; marshalling; pre-flight, thru-flight and post-flight inspection. **(T-2)**

11.20.6.2.6. APU operation/quick air start system. **(T-2)**

11.20.6.2.7. Engine run. **(T-2)**

11.20.6.2.8. Kneeling operation and cargo door/ramp/visor operation on applicable MDS. **(T-2)**

11.20.6.2.9. All applicable powered/non-powered AGE. **(T-2)**

11.20.6.2.10. Qualified to operate, troubleshoot, service, and perform maintenance on their aircraft's critical systems as required by the MAJCOM. **(T-2)**

11.20.6.3. Assistant FCC qualifications and responsibilities.

11.20.6.3.1. Assistant FCCs must be a 5-level A1C or above with at least a SEI on their assigned aircraft and must accompany a fully qualified FCC. **(T-2)**

11.20.6.3.2. As a minimum, the Assistant FCC will be qualified and certified on the following MDS applicable items:

11.20.6.3.2.1. Refuel/defuel member. **(T-2)**

11.20.6.3.2.2. Tow member and tow brake operator. **(T-2)**

11.20.6.3.2.3. LOX/GOX servicing, nitrogen, and tire servicing. **(T-2)**

11.20.6.3.2.4. Tire and brake change; launch; recovery; marshalling; pre-flight, thru-flight and post-flight inspection. **(T-2)**

11.20.6.3.2.5. APU operation/quick air start system. **(T-2)**

11.20.6.3.2.6. Cargo door/ramp/visor operation on applicable MDS. **(T-2)**

11.20.6.3.2.7. All applicable powered/non-powered AGE. **(T-2)**

11.20.7. Work/rest plan (see **Chapter 1**).

11.20.8. MAJCOM FCC Program reporting.

11.20.8.1. MAJCOMs will forward a yearly report to AF/A4LM by 15 August.

11.20.8.2. Use previous FY 4th quarter, and current FY 1st, 2nd, and 3rd quarters. Late reports may postpone FCC waiver requests. Refer to **Attachment 3** and **Attachment 4** for reporting criteria.

11.20.9. Waivers.

11.20.9.1. Forward unit waiver requests to the MAJCOM FCC Program Manager, who will either disapprove/return to unit, or recommend approval/forward to AF/A4LM for final approval IAW DAFMAN 90-161. **(T-2)**

11.20.9.1.1. All approved waivers are reviewed annually as part of the annual report unless otherwise stipulated by the approval authority.

11.20.9.1.2. Waiver renewals. Submit a brief justification for waivers requiring renewal.

11.21. Maintenance of Flash Blindness Protective Devices.

11.21.1. MAJCOMs will define responsibilities across maintenance for sustainment of flash blindness protective devices for assigned aircraft in a supplement to this instruction. As a minimum, MAJCOM supplements will assign responsibilities that ensure:

11.21.1. **(AFMC)** AFMC has zero Emergency War Order/Operations Plan 8010 responsibilities requiring flash blindness protective devices.

11.21.1.1. Units maintain aircraft thermal protective devices, shields, and associated hardware IAW aircraft TOs.

11.21.1.2. Units will establish a training program to qualify individuals to install, inspect, and when required, seal aircraft thermal protective devices and shields. **(T-2)**

11.22. WRM External Nestable Fuel Tank Build-Up. MAJCOMs will ensure units sustain the capability to support assigned wartime taskings. External Nestable Fuel Tank Build-Up is a wartime capability, supported/tasked through a UTC to provide a critical wartime skill that compensates for the expenditure of aircraft fuel tanks (refer to **Chapter 4**). With exception of the core 2A6X4 personnel, augmentees may come from any group or squadron within the wing. MAJCOMs, as applicable, will:

11.22.1. Ensure units adhere to the direction outlined in their Mission Capability statement and DOC statement IAW DAFI 10-401 governing the quantity, size, and composition of fuel tank build-up teams.

11.22.2. Provide guidance for UDMs to ensure personnel tasked/selected for WRM Nestable Fuel Tank Build-Up team augmentees are not tasked for other wartime UTCs.

11.22.2.1. MAJCOMs must ensure UDMs responsible for deploying 2A6X4 personnel are designated as the focal point for WRM Nestable Fuel Tank Build-Up team assembly and are required to develop/maintain a written plan. The plan must be kept current, reviewed annually, and contain the following:

11.22.2.1.1. Specific manning positions across the wing to be tasked as Nestable Fuel Tank Build-Up team augmentees. **Note:** The applicable independent Nestable Fuel Tank Build-Up UTC Manpower Force Packaging System will be used as a guide to construct the teams.

11.22.2.1.2. Guidelines for activation of the tank build-up teams are established.

11.23. Protective Aircraft Shelters (PAS). MAJCOMs that possess PAS will publish guidance for aircraft maintenance operations in a PAS environment. At a minimum, MAJCOM guidance and procedures will address:

11.23. (AFMC) Protective Aircraft Shelters (PAS). Units that possess PAS will coordinate aircraft maintenance operations guidance in a PAS environment with HQ AFMC/A4M (**AFMC.A4M.workflow@us.af.mil**).

11.23.1. PAS marking and floor plans.

11.23.2. Electrical Requirements.

11.23.3. Refueling/Defueling Operations.

11.23.4. Shelter Door Operations.

11.23.5. Aircraft Engine Operation.

11.23.6. Aircraft Positioning inside the PAS.

11.23.7. Aircraft Winching (Hot/Cold).

11.23.8. Placement and Storage of Munitions in the PAS.

11.23.9. Collocating Nuclear and Conventional Munitions (AF Munitions).

11.23.10. External Fuel Tank storage.

11.23.11. PAS maintenance and Inspection requirements not covered by existing publications (such as, grounding and ventilation, mods).

11.24. Combat Sortie Generation. Combat sortie generation is a process by which mission capable aircraft are generated in a minimum amount of time, during peacetime or wartime, through separate 2AXXX and 2WXXX tasks or by Concurrent Servicing Operations. Combat sortie generation may include fueling, munitions/ammunition loading/unloading, aircraft reconfiguration, -6 TO inspections, and other servicing requirements, IAW applicable MDS TOs, Technical Order Data (TOD), IETM, TO 11A-1-33, *Handling and Maintenance of Explosives-Loaded Aircraft*, TO 00-25-172 and other applicable directives. Procedures can be compressed through pre-positioning resources and concurrent performance of tasks.

11.24.1. Wings will determine when to exercise combat sortie generation procedures. Procedures may be used during actual contingencies, scheduled exercises, and daily flying operations.

11.24.1. **(AFMC)** AFMC does not have a peacetime or wartime requirement for combat sortie generation. For mission necessity, see **paragraph A8.1.10.4**. Rapid Crew Swap (RCS).

11.25. Hot Refueling Procedures. For the purpose of this instruction hot refueling is the transfer of fuel into an aircraft having one or more engines running. The purpose of hot refueling is to reduce aircraft ground time, personnel and equipment support requirements and increase system reliability by eliminating system shut down and subsequent restart. Refer to the following sources for additional guidance: TO 00-25-172, TO 00-25-172 CL-4, *Checklist -- Aircraft Fuel Servicing with R-9, R-11, and Commercial Fuel Servicing Trucks and with Fuels Operational Readiness Capability Equipment (FORCE)*, TO 37A9-3-11-lCL-l, *Checklist, Operational and Organizational Maintenance Hot Refueling and Hot Integrated Combat Turnaround Procedures, Aircraft Fuel Servicing Unit Type GRU 17/E Pantograph PACAF Type IV Hydrant Servicing*, and DAFMAN 91-203. **Exception:** N/A for MAJCOMs/Mx units not tasked to maintain hot pit refueling capabilities.

11.25.1. Hot Refueling Site Types. In addition, the process for certifying a site on Foreign Nation is outlined in section **11.25.8**. Personnel will not perform hot refueling operations until the location, equipment requirements, and personnel qualifications are certified IAW this instruction and TO 00-25-172. **(T-2)**

11.25.1.1. Site Certification. MAJCOMs will develop hot refueling site certification requirements. **(T-2)** Site certification team composition will be at a minimum IAW TO 00-25-172, **paragraph 6.5**. **(T-2)**

11.25.1.1. **(AFMC)** AFMC will follow the guidance in this instruction. Additional MAJCOM requirements are listed in paragraph **11.25.1.1.2.1** and **11.25.1.1.3**. **(T-2)**

11.25.1.1.1. The following questions will be addressed as part of the site certification:

11.25.1.1.1.1. Has the aircraft been approved by System Safety Engineering Analysis (SSEA) for hot pit refueling? **(T-3)**

11.25.1.1.1.2. Is adequate area provided to position the aircraft safely (evaluate ability to reposition due to wind direction)? **(T-3)**

11.25.1.1.1.3. Is the ramp level to prevent drainage that could cause environmental impact? Request the fire department dump water to verify flow, if questionable. **(T-3)**

11.25.1.1.1.4. Is the location adequate for the number of aircraft to be serviced? **(T-3)**

11.25.1.1.1.5. Has a hot brake holding area been established? **(T-3)**

11.25.1.1.1.6. Is there proper clearance between the hot pit area and hot brake holding area to prevent conflict? **(T-3)**

11.25.1.1.1.7. Is there proper clearance between the hot pit and Explosive Clear Zone/Hot Cargo Pad/Airfield Clearance Zones to prevent violations of any area/zone? **(T-3)**

11.25.1.1.1.8. Is the hot pit adequately clear of the aircraft/vehicle traffic area? **(T-3)**

11.25.1.1.1.9. Is the hot pit and cursory check area of the ramp clear of FOD potential? **(T-3)**

11.25.1.1.1.10. Does the location provide for rapid access of emergency equipment and egress of aircraft/equipment? **(T-3)**

11.25.1.1.1.11. Are adequate grounding points available? **(T-3)**

11.25.1.1.2. QA or responsible unit will maintain site certification documentation and a master listing of hot pit refueling sites administered by the MXG. **(T-2)**

11.25.1.1.2.1. QA or responsible unit will coordinate with Fuels Base Support Manager to ensure Base Support and Expeditionary (BaS&E) Planning Tool is updated for hot pit site certification listing any time sites are added, changed, or deleted. **(T-2)**

11.25.1.1.2.1. **(AFMC)** Send copy to AFMC/A4M Workflow (**AFMC.A4M.Workflow@us.af.mil**).

11.25.1.1.3. Each unit hot refueling site will be certified by a unit certification team, and approved by Installation Commander, when one of the following occurs:

11.25.1.1.3. **(AFMC)** Send copy to AFMC/A4M Workflow (**AFMC.A4M.Workflow@us.af.mil**).

11.25.1.1.3.1. Construction of new hot refueling sites. **(T-2)**

11.25.1.1.3.2. Change in the unit MDS, or when an additional MDS is acquired. **(T-2)**

11.25.1.1.3.3. Change in refueling equipment. **(T-2)**

11.25.1.1.3.4. Changes in the certified site areas which affect/change in the previous certification. **(T-2)**

11.25.1.1.4. - 11.25.1.1.10.4. DELETED.

11.25.2. Hot refueling site master listing. This listing must contain the following information for all hot refueling sites established and/or sustained by an AF installation or equivalent:

11.25.2.1. All sites must be identified by coordinates on a map. **(T-3)**

11.25.2.1.1. Each facility within the distance identified in TO 00-25-172, must be identified as to its use/contents and its distance in feet from the refueling site/operation. **(T-3)**

11.25.2.1.2. Other refueling sites, aircraft parking areas, also need to be identified and all distances must be shown even if a violation exists. **(T-3)**

11.25.2.1.3. The request cover letter will state if there are no violations. **(T-3)**

11.25.2.1.4. Procedures such as aircraft taxi routes should also be shown. Use arrows or dotted lines to show taxi directions, both entry and exit. **(T-3)**

11.25.2.1.5. Address any restrictions to normal operations and actions required IAW with TO 00-25-172. **(T-3)**

11.25.2.2. State the type of equipment used for hot refueling at each site, (such as, hose carts, truck). **(T-3)**

11.25.2.2.1. Show the location of any fixed fuel pits and usual location of cart or truck if used. **(T-3)**

11.25.2.2.2. Unit-approved sites will be identified on the aircraft parking plan. **(T-3)**

11.25.2.2.3. OSS, CE, and QA and will maintain copies of hot refueling sites on file. **(T-3)**

11.25.2.3. State whether or not all hot refueling areas comply with the quantity-distance separation requirements of DESR6055.09_AFMAN 91-201 in relation to surrounding exposed sites/potential explosion sites.

11.25.3. Hot refueling requires detailed procedures be published in appropriate TOs and unit-developed Local Checklists. Unit Local Checklists will be developed IAW **Chapter 6** and include detailed procedures, normal and emergency, to meet requirements of the local environment. **(T-3)**

11.25.3.1. Units will forward Local Checklists to their respective QA office for approval. **(T-3)**

11.25.4. Units will publish procedures to supplement this section and outline local requirements and additional precautions as necessary for hot refueling, including hot refueling with ordnance, when authorized, IAW TO 00-25-172. **(T-3)**

11.25.5. Units tasked to perform hot refueling operations will ensure hot refueling crews are available to meet mission requirements. **(T-3)** MXS maintenance personnel may be utilized.

11.25.6. Temporary Sites. Temporary sites allow site certification and use for specific exercises, contingencies, and bare base operations in support of Agile Combat Employment. Temporary sites will be approved by installation commander or appropriate host. **(T-2)**

11.25.6.1. Temporary Site Certification. MAJCOMs will develop site certification team requirements. **(T-2)**

11.25.6.1.1. and 11.25.6.1.2. DELETED.

11.25.6.2. The following questions will be addressed as part of the site certification:

11.25.6.2.1. Has the aircraft been approved by system safety engineering analysis (SSEA) for hot pit refueling?

11.25.6.2.2. Is adequate area provided to position the aircraft safely (evaluate ability to reposition due to wind direction)?

11.25.6.2.3. Is the ramp level to prevent drainage that could cause environmental impact? Request the fire department dump water to verify flow, if questionable.

11.25.6.2.4. Is the location adequate for the number of aircraft to be serviced?

11.25.6.2.5. Has a hot brake holding area been established?

11.25.6.2.6. Is there proper clearance between the hot pit area and hot brake holding area to prevent conflict?

11.25.6.2.7. Is there proper clearance between the hot pit and Explosive Clear Zone/Hot Cargo Pad/Airfield Clearance Zones to prevent violations of any area/zone?

11.25.6.2.8. Is the hot pit adequately clear of the aircraft/vehicle traffic area?

11.25.6.2.9. Is the hot pit and cursory check area of the ramp clear of FOD potential?

11.25.6.2.10. Does the location provide for rapid access of emergency equipment and egress of aircraft/equipment?

11.25.6.2.11. Are adequate grounding points available?

11.25.6.3. - 11.25.6.5. DELETED.

11.25.7. Units will develop a checklist addressing the following at a minimum:

11.25.7.1. Required conditions and materials **(T-2)**

11.25.7.2. Site certification procedures **(T-2)**

11.25.7.3. Preparation for operations. **(T-2)**

11.25.7.4. Map Overlays **(T-2)**

11.25.8. Foreign nation site certification. Provides maintenance units the capability to conduct hot refueling operations. The completion of the site certification assures by MDS that safe hot refueling can be accomplished at locations in support of operations or exercises.

11.25.8.1. Unit Responsibility. At a minimum, requesting unit will accomplish hot refueling site certification IAW this instruction and TO 00-25-172. Owning/Host MAJCOM/A4 should be included in site certification process to help facilitate and advocate for required resources.

11.25.8.2. MXG QA Responsibility. MXG QA review site certification package prior to approval. Develop and maintain completed and approved site certification package. Completed package will be sent to MAJCOM/A4 for their records.

11.25.8.3. MAJCOM/A4 with Unit responsibility (i.e., ACC, AFGSC, etc.). Facilitate and advocate for requesting unit for required resources, as needed.

11.25.8.4. Host MAJCOM/A4 (i.e., PACAF, USAFE, etc.). In coordination with owing MAJCOM, facilitate and advocate for requesting unit for required resources, as needed. Develop and maintain file of approved packages.

11.25.8.4.1. - 11.25.8.4.5. DELETED.

11.25.8.5. Approval Authority. The requesting unit WG/CC is the approving official for the final site certification package. **Exception:** For deployed unit(s)/Task Force, commander with operational command is the approving authority for the final site certification package.

11.25.8.5.1. - 11.25.8.6. DELETED.

11.25.9. Hot Refueling Team Members and Duties. MAJCOMs will determine team member's prerequisites for each position. **(T-2)**

11.25.9.1. Pad Supervisor. Responsible for overall supervision of hot refueling operations when two or more aircraft are simultaneously hot refueled on the same pad (multiple hot refueling). Maintains full view and control of multiple hot refueling operations.

11.25.9.1.1. - 11.25.9.1.2. DELETED.

11.25.9.2. Refuel supervisor "A" member. Individual will be refuel task qualified, capable of supervising hot refuel crew. **(T-2)**

11.25.9.3. Refuel crew "B" member. Individual will be task qualified. **(T-2)**

11.25.9.4. Fuels specialist with 2F0X1 AFSC, or equivalent, "C" member. Individual will be refuel task certified on the specific facility/equipment, and task qualified for aircraft hot refueling. **(T-2)**

11.25.9.5. Additional refuel crew “D” member. Individual will be task qualified. **(T-2)** Use “D” members as required by applicable aircraft technical data.

11.25.10. Hot refueling team members and QA/Unit certifying officials/evaluators may be multi- MDS qualified.

11.25.10.1. After initial certification on each MDS, personnel must update their hot refueling currency by performing hot refueling on each MDS. **(T-2) Exception:** Fuels specialist will maintain certification and proficiency IAW DAFI 23-201, *Fuels Management*.

11.25.10.2. Section NCOICs/Chiefs will ensure personnel maintain proficiency on each MDS. **(T-2)**

11.25.11. Conducting Hot Refueling Training, Certification and Documentation. [For additional information, refer to AFI 11-235, *Specialized Refueling Operations*]. **Note:** Fuels specialist training will be conducted IAW DAFI 23-201 and CFETP. Qualification training of hot refueling personnel will be conducted in three distinct phases. **(T-2)** The three hot refueling qualification training phases are as follows:

11.25.11. **(AFMC)** Training, certification, and recertification requirements for hot refuel team members are outlined in **Table 11.3**. Personnel are decertified if they miss any semiannual proficiency requirements or annual special requirements. Personnel can also be decertified by any hot/aircraft to- aircraft refueling supervisor, squadron/immediate supervisor, or QA any time one of these individuals sees an unsafe act or demonstrated lack of proficiency on the part of any hot and/or aircraft-to-aircraft refueling team member. **(T-2)**

Table 11.3. (Added-AFMC) Hot/Aircraft-to-Aircraft Refueling Training/Certification Requirements. (T-2)

Position	Required Training	Proficiency Requirements	Special Requirements
QA/Squadron Hot Refueling Certifiers	I, II, III	1 Hot Refuel Semi-Annually	One time PE by hot refueling certified QA evaluator; thereafter, annual

			recertification by another hot refueling certifier or hot refueling certified QA evaluator
Pad Supervisor	I, II, III	1 Multiple Hot Refuel Semi-Annually	Annual recertification by hot refueling certifier
Refuel Member	I, II, III	1 Hot Refuel Semi-Annually	Annual recertification by hot refueling certifier
Fuels Specialist (2F0X1)	I, II, III	1 Hot Refuel Annually	Annual recertification by Fuels 7-level Supervisor
Decertified Certifier	Repeat II, III		Recertification must be started within 90 days or Phase I will also be completed. Recertification by another hot refueling certifier or hot refueling certified QA evaluator
Decertified (Other than certifier)	Repeat II, III		Recertification must be started within 90 days or Phase I will also be completed. Recertification by hot refueling certifier.

Decertified (Other than certifier) Repeat II, III Recertification

11.25.11.1. Phase 1. “Familiarization” phase. Designated instructors familiarize trainees with applicable technical data, procedures, and guidance for hot refueling. Place special emphasis on procedures for hot refueling with ordinance loaded, when authorized.

11.25.11.2. Phase 2. “Hands-on” phase. Apply information learned in Phase 1 to develop in-depth knowledge and proficiency in all facets of hot refueling. Training will include proper operation, preventive maintenance, use of hand signals and emergency procedures. **(T-2)** Simulate hot refueling by performing all hot refueling tasks without aircraft engines running (cold pit). Designated instructors will demonstrate tasks then require trainees to perform tasks, practice emergency procedures, critique performance and provide additional training as required. **(T-2)**

11.25.11.3. Phase 3. “Demonstration/Certification” phase. Trainees will demonstrate hot refueling under the supervision of designated certifying officials with aircraft engine(s) running. **(T-2)** The Squadron Certifying Officials will certify individuals upon successful demonstration of hot refueling. **(T-2)** If Phase 3 training has not been completed within 30 days of Phase 2 training, Phase 2 training must be repeated. **(T-2)**

11.25.11.4. Qualification training will:

11.25.11.4.1. Stress safety requirements, emergency procedures and equipment inspection in all three phases of training. **(T-2)**

11.25.11.4.2. Ensure procedures in TO 37A9-3-11-1CL-1, TO 00-25-172, and TO 00-25-172CL-4 are taught to all team supervisors and members. **(T-2)**

11.25.11.4.3. Allow Phase 2 and Phase 3 training to be conducted utilizing joint sessions including 2F0X1 AFSC personnel and all maintenance AFSCs. **(T-2)**

11.25.11.4.4. Utilize both fuels (2F0X1) and maintenance AFSC instructors for joint sessions (T-2)

11.25.11.4.5. Be conducted by MT (QA if MT not available). **(T-2)**

11.25.11.5. QA hot pit certifying officials and QA hot pit certifier augmentees (squadron certifying officials) will train, evaluate, and certify unit personnel. **(T-2)** QA hot pit certifying officials will ensure augmentees conduct evaluations using procedures outlined in this instruction, applicable aircraft TOs and local procedures. **(T-2)**

11.25.11.6. DELETED.

11.25.12. Document training for personnel performing, evaluating, supervising, or instructing hot refuel operations as follows:

11.25.12.1. Document all aircraft maintenance and 2F0X1 AFSC personnel Phases 1, 2, and 3 initial training in MyLearning. **(T-2)**

11.25.12.1.1. For AFSCs where “refuel aircraft with engines operating” is not contained in MyLearning, use DAF Form 797/MIS to document initial hot refuel training. **(T-2)**

11.25.12.1.2. Track recurring hot refueling certification in the MIS (initial and annual) IAW with TO 00-25-172 and this DAFI. **(T-2)**

11.25.12.2. 2F0X1 AFSC personnel will use MyLearning/DAF Form 1098, *Special Tasks Certification and Recurring Training*, to document Phases 1, 2, and 3 initial/recurring hot refuel training. **(T-2)** Fuels (2F0X1) certifying officials will be appointed by the LRS/CC IAW DAFMAN 36-2689. **(T-2)**

11.25.13. Track hot refueling members, by position, on the SCR. **(T-2)**

11.25.14. Unique proficiency, certifying, and decertifying actions for hot refuel team members will be outlined in MAJCOM supplements/addendums to this DAFI. **(T-2)**

11.26. Aircraft Rapid/Hot Defueling.

11.26.1. Rapid defueling presents hazards which are not normally encountered in normal defueling operations. Owning MAJCOMs will develop and sustain a rapid defueling capability to meet routine and contingency mission requirements IAW TO 00-25-172 and MDS-specific TOs.

11.26.1. **(AFMC)** AFMC does not have routine or contingency requirements for aircraft rapid/hot defueling.

11.26.1.1. Rapid defueling operations are considered hot defueling operations whenever the provider/source aircraft has an engine running.

11.27. 406 MHz Emergency Locator Transmitter Systems Program.

11.27.1. Units will ensure procedures are established to update the Emergency Locator Transmitter registration database whenever 406 MHz Emergency Locator Transmitter–equipped aircraft are transferred to other commands/wings, Emergency Locator Transmitter that are taken out of service, removed for maintenance, or destroyed. **(T-0)**

11.27.2. Aircraft maintenance functions must register and track status of fixed-mounted aircraft 406 MHz Emergency Locator Transmitter systems. **(T-0)**

11.27.3. IAW DoDI 3002.02, *Personnel Recovery and 406 MHZ Search and Rescue (SAR) Emergency Beacons in the Department of Defense*, USAF 406 MHz Emergency Locator Transmitter systems must be registered in the DoD Joint Search and Rescue Satellite Aided Tracking Electronic Tracking System database. **(T-0)**

11.27.3.1. The POC for JSETS registration is the Personnel Recovery Mission Software Help Desk at **PRMSMail@jricp.osis.gov**.

11.27.3.2. The governing agencies are the Joint Personnel Recovery Agency and the Electronic Services Command at Hanscom AFB, MA. Refer to AFMAN 10-207, *Command Posts*, for Command Post or C2 function responsibilities regarding 406 MHz Emergency Locator Transmitter and Personal Locator Beacon systems.

11.28. Crashed, Damaged or Disabled Aircraft Recovery (CDDAR) Program.

11.28.1. Installation/WG/CCs responsible for active airfields/runways, and flying missions will implement a CDDAR Program IAW TO 00-80C-1, *Crashed, Damaged, Disabled Aircraft Recovery Manual*. **(T-2)** The program must be designed to provide a response and/or recovery capability of assigned host, tenant, and consider transient aircraft consistent with the following considerations: (1) urgency to open the runway for operational use; (2) prevention of secondary

damage to the aircraft; and (3) preservation of evidence for mishap or accident investigations IAW DAFI 91-202 and DAFI 91-204.

11.28.2. Responsibilities:

11.28.2.1. MAJCOMs will:

11.28.2.1.1. Ensure flying units maintain a CDDAR capability IAW 00-80C-1.

11.28.2.1.2. Designate a MAJCOM CDDAR OPR. As a minimum, the CDDAR OPR will:

11.28.2.1.2. **(AFMC)** HQ AFMC/A4MM is the MAJCOM CDDAR OPR.

11.28.2.1.2.1. Standardize CDDAR equipment inventory accountability and reporting requirements by MDS for all on hand CDDAR equipment prescribed by TO 00-80C-1, allowance standard and applicable weapons system TOs across assigned units with active airfields/runways.

11.28.2.1.2.1.1. Review unit's annual CDDAR equipment inventories to identify and document equipment shortfalls.

11.28.2.1.2.1.2. Coordinate AS change request with the applicable AFMC AS activity IAW AFI 23-101.

11.28.2.1.2.1.3. Ensure excess CDDAR equipment is redistributed to fill internal shortfalls prior to units turning equipment into supply/DLADS as excess.

11.28.2.1.2.2. Participate as member of CDDAR Working Group. **(T-2)**

11.28.2.2. AETC will:

11.28.2.2.1. Develop, sustain, and administer the CDDAR training program.

11.28.2.3. AFMC will:

11.28.2.3.1. Utilize AFSC/LZDB to serve as AF CDDAR Program OPR (**AFSC.LZDB.CDDAROPR@us.af.mil**) with the following responsibilities:

11.28.2.3.1.1. Provide CDDAR Program guidance to MAJCOM CDDAR OPRs. **(T-2)**

11.28.2.3.1.2. Provide routine communication to MAJCOM CDDAR OPRs on relevant matters such as new technology, guidance changes, significant events, equipment recall notifications, and best practices. **(T-2)**

11.28.2.3.1.3. Coordinate and consolidate MAJCOM CDDAR OPR inputs on CDDAR policy, training, and technical data, and present this information to the appropriate agency as required. **(T-2)**

11.28.2.3.1.4. Develop, manage, and chair a CDDAR Working Group to address equipment requirements, priorities, and funding. The CDDAR Working Group members will consist of MAJCOM CDDAR OPRs. The CDDAR Working Group will make efforts to gain economy of scale when requesting new equipment. **(T-2)**

11.28.2.3.2. Provide approved tech-data outlining equipment procedures to safely respond and/or recover aircraft from a CDDAR event.

11.28.2.3.3. Provide timely engineering support to facilitate resolution of unique CDDAR events which cannot be resolved by existing tech-data.

11.28.2.3.4. Develop, manage, and maintain AS needed to sustain a weapon system for peacetime and wartime operations IAW AFI 23-101.

11.28.2.4. - 11.28.2.5.7. DELETED.

11.28.2.6. CDDAR Team Chief and alternate will:

11.28.2.6.1. Be designated as the unit's subject matter expert on aircraft recovery operations and equipment and will be thoroughly familiar with and perform their Team Chief duties IAW TO 00-80C-1. **(T-2)**

11.29. Aircraft Battle Damage Repair (ABDR). ABDR is an effective force multiplier contributing to wartime sortie production by assessing and repairing battle damaged aircraft rapidly to support flying operations. ABDR repairs will be accomplished during contingency or wartime only. However, weapons system program managers may approve ABDR repairs during peacetime on a case-by-case basis using trained ABDR Technicians.

11.29.1. Responsibilities:

11.29.1.1. The Directorate of Logistics (AF/A4L) will provide overall policy and guidance for the USAF ABDR Program.

11.29.1.2. AFMC will:

11.29.1.2. **(AFMC)** AFSC/LZD is the USAF ABDR Program Office, they will: **(T-2)**

11.29.1.2.1. Assume management responsibility for USAF ABDR Programs.

11.29.1.2.2. Publish a MAJCOM instruction to implement the ABDR requirements contained in this instruction.

11.29.1.2.3. Develop and manage ABDR policy for pre-positioning of tools, materiel kits, related SE, and management of ABDR training aircraft.

11.29.1.2.4. Support development and publication of ABDR TOs for new weapon systems.

11.29.1.2.5. Maintain ABDR UTCs for AFMC organizations.

11.29.1.2.6. Plan for and develop capability to repair battle/crash damaged aircraft.

11.29.1.2.6.1. Ensure plans include procedures to add additional repair capabilities into operating locations and provide aircraft evacuation alternatives.

11.29.1.2.7. Plan, program, and submit ABDR funding requests.

11.29.1.2.8. Maintain an ABDR Technical Support Office to advocate and provide day-to-day management of tasks associated with development, implementation, maintenance, and support needed to enhance the USAF ABDR capability.

11.29.1.2.9. Provide support in determining technical requirements, repair techniques, repair materials, assessment aids and Research & Development (R&D) efforts.

11.29.1.2.10. Manage TO 1-1H-39, *Aircraft Battle Damage Repair General Technical Manual*, and the engineering handbook for ABDR engineers and support initiatives to develop, publish, and maintain weapon system-specific –39 TOs.

11.29.1.2.11. Ensure the status of aircraft permanently grounded for ABDR training is reported IAW DAFI 21-103.

11.29.1.2.12. Establish Aircraft Battle Damage Evaluator training program, manage course documentation and provide training to MT Instructors as required. **(T-2)**

11.29.1.3. MAJCOMs will:

11.29.1.3.1. Establish a command focal point to work ABDR issues with AFMC.

11.29.1.3.1. **(AFMC)** AFSC/LZD is the command focal point.

11.29.1.3.2. In conjunction with AFMC, develop a command ABDR Concept of Operations and ensure Concept of Operations covers unit plans for repair of battle/crash damaged aircraft during combat operations.

11.29.1.3.2.1. Address ABDR in mission need statements for new weapon systems that support or engage in combat operations.

11.29.1.3.2.2. Incorporate ABDR in command war planning documents.

11.29.1.3.2.3. Task AFMC ABDR UTCs to support OPLANs.

11.29.1.3.2.4. Develop plans for the reception and employment of AFMC ABDR teams at the onset of hostilities.

11.29.1.3.2.5. Formalize integration and bed down requirements in applicable BSP IAW AFI 10-404.

11.29.1.3.2.6. Provide unit level weapon-system-specific tools (other than common hand tools) and equipment needed to repair battle/crash damaged aircraft.

11.29.1.3.2.7. USAFE and Pacific Air Forces (PACAF) will store and maintain serviceability, accountability, and status reporting to include Financial Improvement and Audit Readiness reporting of AFMC owned and provided WRM ABDR trailers IAW established procedures. **(T-2)**

11.29.1.3.2.8. Provide technical support to the ABDR Technical Support Office for live fire or similar testing.

11.29.1.3.3. – 11.29.1.3.10. DELETED

11.29.1.4. Unit Responsibilities. Units will:

11.29.1.4.1. Utilize trained Aircraft Battle Damage Evaluators to evaluate aircraft battle damage and mishap damage sustained during combat or contingency operations. **(T-3)**

11.29.1.4.2. Ensure shelf-life items listed in TO 1-1H-39 and weapon system-specific –39 TOs are maintained at required levels to support ABDR requirements. **(T-3)**

11.29.1.4.3. Ensure aircraft battle damage is documented on an AFTO Form 97, *Aerospace Vehicle Battle Damage Incident Debrief/Assessment/Repair Record* or

AFTO Form 97B, *Aircraft Battle Damage Evaluator Checklist* as required IAW TO 1-1H-39. Completed forms will be forwarded to the Aircraft Battle Damage Repair Program Office. **(T-2)** CLASSIFIED messages must be sent to SIPR: **usafsaf.wright-patt.afsc-lg.mbx.afsc-lzd-abdr-tso@mail.smil.mil** and UNCLASSIFIED messages must be sent to NIPR: **AFSC.LZD.ABDRTSO@us.af.mil** for filing in the historical archives. **(T-2)**

11.29.1.5. Aircraft Battle Damage Evaluator/Training:

11.29.1.5.1. Aircraft Battle Damage Evaluator training provides MXG/CC's with ABDR knowledgeable forces and prepares units to execute Air Tasking Orders in a denied airspace with potential heavy losses. Aircraft Battle Damage Evaluator roles and responsibilities are outlined in TO 1-1H-39.

11.29.1.5.2. Maintenance Supervision will determine the proper mixture of personnel to attend Aircraft Battle Damage Evaluator training. **(T-3)**

11.29.1.5.3. Aircraft Battle Damage Evaluator formal training will be IAW the approved ABDR course control documents. **(T-2)**

11.29.1.5.4. Aircraft Battle Damage Evaluators will complete unit provided refresher training every 24 months. **(T-2)**

11.29.1.5.4. **(AFMC)** Field level MTFs will maintain instructors by using train the trainer. In the event a MTF has no trained instructors contact the ABDR Program Office. **(T-3)**

11.29.1.5.5. Course documents and instructor training is provided by the ABDR Technical Support Office NIPR, and can be requested at: **AFSC.LZD.ABDRTSO@us.af.mil**

11.30. Egress/Cockpit Familiarization Training.

11.30.1. All non-egress personnel who access aircraft cockpits with egress systems maintained and managed by 2A6X3 Egress Systems personnel must complete initial and refresher familiarization training. **(T-2)**

11.30.1.1. As a minimum, initial and refresher egress/cockpit familiarization training will include location and installation procedures of egress system safety devices, cockpit entry/exit procedures, procedures for determining whether or not an egress component is expended, emergency procedures associated with an expended egress component, and local maintenance concerns identified by the egress work center supervisor. **(T-2)**

11.30.1.2. New personnel to the unit must receive initial familiarization training prior to accessing cockpits unless last duty position involved same mission design aircraft as current duty position. **(T-2)**

11.30.1.3. Personnel not requiring initial training will attend refresher training when they become due. **(T-2)**

11.30.1.4. Initial egress familiarization training will normally be hands-on using an aircraft. **(T-2)**

11.30.1.4.1. Due to the diverse egress systems across all weapon systems, units desiring to use an aircraft maintenance trainer in lieu of an aircraft must submit a request from the MXG/CC to the MAJCOM for approval/disapproval. Approving MAJCOM must coordinate with the MDS lead command to ensure no technical/other mitigating factors exist and to provide transparency. **(T-2)**

11.30.1.5. Refresher familiarization training will be conducted annually using an aircraft, maintenance trainer or media, which is approved and designated by the egress work center supervisor. **(T-2)**

11.30.1.5.1. Non-egress personnel may administer training media (slideshow/video) during refresher familiarization training.

11.30.1.5.2. Direct students to the egress section if technical assistance is required and/or questions are raised concerning course subject matter.

11.30.1.6. Only egress personnel, certified on assigned egress system(s), should instruct initial egress familiarization training. **(T-2)** MXG/CC may authorize MT personnel to instruct this training provided they are currently certified to perform egress maintenance.

11.30.1.7. Training media must meet approval of the 2A6X3 AFSC MAJCOM Functional Manager (MFM) or current media produced by the 367 Training Support Squadron listed on the Defense Imagery at **http://www.defenseimagery.mil. (T-2)**

11.30.1.8. Individuals overdue for annual egress familiarization training will not access aircraft cockpits until they complete familiarization training. **(T-2)**

11.30.1.9. Units with unique, experimental, or test aircraft requirements.

11.30.1.9.1. If training courses are not available through AETC, units must use interagency training before considering non-government training sources. **(T-2)**

11.30.1.9.1.1. If courses in both of these sources are not available, units will establish a documented training program that meets the intent of this instruction. **(T-2)**

11.30.1.9.1.2. Training will be conducted by the most qualified personnel and must be approved by the MFM prior to implementation. **(T-2)**

11.30.2. MAJCOMs in coordination with the applicable lead command will identify emergency aircraft egress/evacuation training and frequency requirements in their supplement to this DAFI for personnel assigned to weapon systems that do not have aircraft egress systems maintained and managed by 2A6X3 personnel.

11.31. Aircraft Defensive Systems Loading Program.

11.31.1. Aircraft Defensive Systems Loading Program provides instruction required to install/remove chaff/flare on unique mission aircraft in units where there are no 2W1 AFSC authorizations assigned.

11.31.2. Authorized units will establish a program to train and qualify personnel to perform these tasks IAW procedures outlined in DAFMAN 21-201 and this **Chapter**.

11.31.2. **(AFMC)** See AFMAN 21-206, *Aircraft Armament Systems Management*.

11.31.3. Units will work with the installation Weapon System Manager (WSM) and Airfield Operations Flight to develop written instructions for handling chaff/flare-loaded aircraft IAW DESR6055.09_AFMAN 91-201 and DAFI 91-202.

11.31.3.1. As a minimum, written instructions will include procedures for launch/recovery/parking of chaff/flare-loaded aircraft; chaff/flare storage and transportation; and partially ejected flares and minimum requirements outlined in DESR6055.09_AFMAN 91-201.

11.31.4. The MXG/CC will appoint 7-skill or 9-skill level individual with maintenance AFSC as the Weapons Task Qualification Manager (WTQM). **(T-2) Note:** Units with 2W1 AFSCs assigned will comply with training/qualification requirements in **Chapter 10**.

11.31.5. WTQM and Weapons Task Qualification Crew (WTQC) responsibilities. The WTQM/WTQC provide oversight of chaff/flare loading operations to ensure they are conducted safely by providing initial and recurring load training, serving as the focal point for all chaff/flare loading issues, and observing loading operations during training. The WTQM and WTQC will not participate in load operations during training. **(T-2)**

11.31.5.1. WTQM. The WTQM typically holds a 2A871X AFSC; however, other flightline personnel with the 2AX7X AFSC may perform this function. The WTQM develops and oversees the chaff/flare loading standardization program, sets standards, and develops local policies and procedures. The WTQM will be tracked on the SCR. **(T-2)** The WTQM will:

11.31.5.1.1. Receive initial and recurring load qualification training from a WTQC and maintain currency on chaff/flare loading tasks. **(T-2)**

11.31.5.1.2. Once trained and qualified, the WTQM will develop and administer the unit's chaff/ flare load training program and train/qualify home station WTQC personnel. **(T-2) Note:** In the event a unit is initially tasked and has no qualified instructors, it will be necessary for the WTQM to become certified at a unit with qualified trainers. The WTQM will:

11.31.5.1.2.1. Ensure sufficient numbers of personnel are qualified to load chaff/flare to support the unit's mission requirements. **(T-2)**

11.31.5.1.2.1.1. A course code will be loaded in the MIS to identify trained personnel and qualification status. **(T-2)**

11.31.5.1.2.2. Establish time standards for initial and recurring loading tasks. **(T-2)**

11.31.5.1.2.2.1. Lead wings will develop time standards for each MDS for qualification purposes. **(T-2)**

11.31.5.1.2.2.2. The senior evaluator has the discretion to add to the time standard if inclement weather or equipment failure is the cause for exceeding the time standard.

11.31.5.1.2.3. As a minimum, the WTQM will identify the number of qualified personnel, names and employee numbers, MDS qualification, Defensive Systems, equipment type, qualification date, and date(s) recurring training is due. **(T-2)**

11.31.5.1.2.4. The WTQM will select, train, evaluate, and qualify a minimum of

two personnel as the WTQC on safe and reliable munitions loading procedures. **(T-2)**

11.31.5.1.2.4.1. The WTQM will evaluate and re-certify WTQC members annually. **(T-2)** WTQC members will be tracked on the SCR. **(T-2)**

11.31.5.1.3. Review and approve/disapprove RCs that pertain to chaff/flare loading technical data. **(T-2)**

11.31.5.1.4. Develop a local Task Assignment List by utilizing lead wing-developed MDS-specific Task Assignment Lists for use during training for all chaff/flare loading operations. **(T-2)** A Task Assignment List is derived from applicable MDS munitions load checklist (TO 33-1-20-series) and identifies the load crew members' responsibilities by step.

11.31.5.1.5. Ensure chaff/flare loading CTKs are standardized to the maximum extent possible. **(T-2)**

11.31.5.1.5.1. Chaff/flare loading CTKs must include all tools and equipment necessary to support applicable MDSs and AME configurations. **(T-2)**

11.31.5.1.6. Coordinate the scheduling of personnel for chaff/flare load training. **(T-2)**

11.31.5.1.6.1. The WTQM may delegate this duty to the WTQC.

11.31.5.1.7. Coordinate with PS&D, or the Regional Training Center, if applicable, to obtain chaff/flare dispensing system-equipped aircraft for training purposes. **(T-2)**

11.31.5.1.8. Ensure training magazines match the characteristics and "feel" of live magazines (such as, weight, dimensions). **(T-2)**

11.31.5.2. WTQC. The WTQC assists the WTQM in managing the chaff/flare loading standardization program. The WTQC's primary purpose is to train and qualify personnel to load chaff/flares but may also perform chaff/flare load duties. The lead WTQC member is typically a 7-skill level technician with the 2AX7X AFSC. Initial training will be conducted using inert munitions. **(T-2)** The number of trained WTQC members should be based on current/anticipated workloads and their ability to maintain proficiency on all applicable MDSs. WTQC members are qualified by the WTQM. The WTQC members will:

11.31.5.2.1. Provide personnel with initial and recurring load qualification training. **(T-2)** One WTQC member will be required to conduct practical training. **(T-2)**

11.31.5.2.2. Monitor personnel qualifications to ensure required academic and practical training is complete. **(T-2)**

11.31.5.2.2.1. Disqualify individuals if recurring requirements are not met. **(T-2)**

11.31.5.2.3. Spot-check personnel to evaluate proficiency. **(T-2)**

11.31.5.2.3.1. The WTQC will disqualify personnel who violate safety, technical data, and reliability procedures, or fail to demonstrate proficiency. **(T-2)**

11.31.5.2.4. Develop/coordinate training schedules and provide to PS&D for inclusion in the appropriate schedule (monthly, weekly). **(T-2) Note:** Enroute WTQMs forward

training requirements to the UTM, who coordinates for ground training aircraft with the Regional Training Center.

11.31.6. Training Requirements. Personnel are considered qualified upon successful completion of training provided by a qualified WTQC.

11.31.6.1. Initial qualification will be conducted using inert munitions. **(T-2)**

11.31.6.2. Live munitions may be used during annual qualification to maintain currency. Load qualification training consists of academic and practical training.

11.31.6.3. Document the initial and recurring load qualification training requirements in MyLearning or other approved training process if MyLearning is unavailable (i.e., 623A, *Individual Training Record*) **(T-2)**

11.31.6.4. Academic and practical training must be provided during initial and recurring load qualification training. **(T-2)**

11.31.6.4.1. Academic training is required before practical training is accomplished. **(T-2)**

11.31.6.4.2. Initial practical training must be completed within 14 days of successfully completing initial academic training. **(T-2)**

11.31.6.4.2.1. Practical training should duplicate operational conditions as closely as possible.

11.31.6.4.3. Recurring practical task qualification is administered at least annually. **(T-2)**

11.31.6.4.3.1. As a minimum, practical training will include chaff/flare module serviceability criteria, actual chaff/flare loading, and operation of support equipment/AGE used during loading operations. **(T-2) Note:** Weapons task qualification academic training may fulfill the requirements for explosive safety training if the requirements of DAFI 91-202 are included.

11.31.6.5. Academic training is administered every 12 months. **(T-2)** As a minimum, academic training will include:

11.31.6.5.1. Familiarization with chaff/flare loading publications, including TO 11A-1-33, MAJCOM, and local procedures. **(T-2)**

11.31.6.5.2. Aircraft and munitions familiarization. **(T-2)**

11.31.6.5.3. Safety, security, and emergency procedures. **(T-2)**

11.31.6.5.4. Support, test, handling equipment, and special tools familiarization. **(T-2)**

11.31.6.5.5. Task Assignment Lists and aircraft specific 33-1-2 series TOs must be available at the load-training site. **(T-2) Note:** Training course control documents will be coordinated annually through the Wing Safety and MT. **(T-2)**

11.31.6.6. Personnel qualified on a specific task on a specific MDS are considered qualified to perform that task on all series of that MDS; however, the member must be familiar with differences within the MDS (such as, cockpit switch locations). **(T-2)**

11.31.6.6.1. The WTQM or WTQC will provide practical, on-aircraft training on these differences and document these qualifications for each dispensing system in the qualification status or equivalent system. **(T-2)**

11.31.7. Disqualifying Chaff/Flare Load Personnel. Disqualification will be documented in A - Changed to MyLearning, or other approved training process if MyLearning is unavailable (i.e., 623A), and the qualification status system. **(T-2)**

11.31.7.1. Although not all-inclusive, the following criteria constitute grounds for disqualifying personnel from chaff/flare loading duties:

11.31.7.1.1. Failing to complete recurring training.

11.31.7.1.2. Committing a safety or reliability error.

11.31.7.1.3. Lack of proficiency.

11.31.8. Transient Aircraft.

11.31.8.1. Apply the following when working transient aircraft:

11.31.8.1.1. Under no circumstances will personnel attempt chaff/flare load operations without current technical data. **(T-2)**

11.31.8.1.2. If current technical data is available, then qualified personnel may perform chaff/flare load operations. **(T-2)**

11.31.8.1.3. If current technical data is available but no one is qualified on the transient aircraft type, then the MXG/CC (or Air Mobility Squadron (AMS)/CC at enroute locations) may authorize the WTQC or WTQM to de-arm and/or unload the aircraft.

11.31.8.1.3.1. The WTQM will submit a written request to the MXG/CC (or AMS/CC at enroute locations) identifying personnel selected to perform the task, aircraft type and (if applicable) number of aircraft to be de-armed and unloaded. **(T-2)**

11.31.8.1.3.1.1. Approved requests will be maintained for 90 days. **(T-2) Note:** This is a temporary, one-time authorization to facilitate required maintenance when qualified personnel are not available.

11.31.9. Identification of Chaff/Flare-Loaded Aircraft. Verify chaff/flare load status of aircraft by checking AFTO Form 781A/C before performing any maintenance.

11.31.9.1. Explosive placards are not required on AMC aircraft.

11.31.9.2. If an aircraft is loaded with chaff/flare, it will be safed IAW applicable technical data prior to performing any maintenance. **(T-2)**

11.31.9.3. Before loading chaff/flares, review the AFTO Form 781C, A*vionics Configuration and Load Status Document*, for defensive systems inspection status. **(T-2)**

11.31.9.3.1. If chaff/flare is loaded on aircraft, ensure/verify applicable MIS documentation requirements are completed. **Note:** Do not load chaff/flares if the aircraft is overdue a scheduled DS inspection.

11.31.10. Tracking and Reconciliation of Chaff/Flare-Loaded Aircraft.

11.31.10.1. Expenditure tracking and processing will be handled by Munitions Personnel (2W0X1) IAW DAFMAN 21-201, **Chapter 7**.

11.31.10.2. Munitions personnel will not use the direct input method to process flightline chaff/flare expenditures; all chaff/flare expenditures will be returned to the Munitions Storage Area (MSA) for verification by munitions personnel before processing expenditures in Theatre Integrated Combat Munitions Systems (TICMS). **(T-2)**

11.31.11. Additional Requirements (as applicable).

11.31.11.1. Document DS software version data and aircraft inspections (such as, 90-, 120-, or 180-day checks) on AFTO Form 781C. **(T-2)**

11.31.11.2. For software version data, enter the following information in the remarks section for each reprogrammable system: type system; installed Operational Flight Program (OFP) version; and/or Mission Data File (MDF) version (such as, ALE-47, OFP XXXX, MDF XXXX).

11.31.11.2.1. If a system contains multiple OFPs, list all applicable versions (such as, ALE-47, Programmer OFP XXXX, Sequencer OFP XXXX, MDF XXXX).

11.31.12. Chaff/Flare Build-up. Chaff/flare magazine build-up will only be accomplished by personnel with 2W0 AFSC or qualified contractors. **(T-2)**

11.31.12.1. Units will only perform chaff/flare build-up in facilities/locations approved by the installation WSM IAW DESR6055.09_AFMAN 91-201. **(T-2)**

11.31.12.2. Units must have an approved explosive site plan or explosives facility license on file with Wing Safety prior to initiating chaff/flare build-up or storage operations. **(T-2)**

11.32. Aircraft and Equipment Decontamination.

11.32.1. Maintenance organizations need to have the Ability to Survive and Operate in a Chemical, Biological, Radiological, Nuclear, and high-yield Explosives (CBRNE) environment and have the capability to decontaminate operational aircraft, vehicle, and SE.

11.32.2. Units will employ AF and locally developed TTPs IAW DAFI 10-2503, *Chemical Biological, Radiological, Nuclear, (CBRN) Defense Program*. **(T-2)**

11.32.2.1. TTPs provide the fundamental counter-chemical warfare (CCW) tools to survive to operate and maximize combat sortie generation capabilities in a CBRNE environment.

11.32.3. The following references in addition to MDS-specific technical data should be utilized when developing unit decontamination programs: DAFMAN 91-203, AFI 10- 2501, AFTTP 3-4, *Airman's Manual*, TO 00-110A-1, *Guidelines for Identification and Handling of Aircraft and Material Contaminated with Radioactive Debris*, TO 00-20-1, TO 11C15-1-3, *Chemical Warfare Decontamination, Detection and Disposal of Decontamination Agents*, TO 11D1-3-8-1, *Decontamination Apparatus, Power Driven, Portable Type A/E32U- 8, (Engineered Air).*

11.33. End-of-Runway (EOR) Inspection.

11.33.1. EOR is MDS specific, PM directed inspection of aircraft systems identified in the Dash 6 TO or equivalent and published in Dash 6 work cards IAW TO 00-20-1. (N/A to aircraft that do not have an EOR -6 TO requirement).

11.33.2. If local requirements dictate, publish additional guidance to TOs for EOR inspections IAW TO 00-20-1 and TO 00-5-1. **Note:** Safing, arming, and de-arming of live munitions will be accomplished by personnel qualified IAW Chapter **4**, **5**, and **10**.

11.33.3. The EOR team chief (identified by a reflective vest) will carry an EOR checklist and ensures each item is inspected as required. **(T-2)** On aircraft with a ground intercom system, units are only required to establish verbal communications with the pilot when communication beyond the standard EOR marshalling hand signals is required unless otherwise directed by MDS specific technical data.

11.34. Wing Avionics Manager (WAM).

11.34.1. WAM Duties and Responsibilities.

11.34.1.1. Will be in the minimum grade of MSGT, 7 Level or equivalent, and 2A Avionics AFSC. **(T-3)**

11.34.1.2. Act As the Wing Avionics Functional Manager. **(T-2)**

11.34.1.3. Serve as the maintenance group focal point for all avionics related interactions between PMs, MAJCOMs, Lead Commands, Wings, Operations and Maintenance or equivalent activities to discern and implement changes in avionics configuration requirements. **(T-2)**

11.34.1.4. Ensure RAMPOD updates are completed IAW DAFI 21-103.

11.34.1.5. Ensure classified pods/components and equipment are stored in authorized areas IAW AFI 16-1404. **(T-2)**

11.34.1.6. Ensure classified aircraft/Support/Test equipment are stored in authorized areas IAW AFI 16-1404. **(T-2)**

11.34.1.7. Meet quarterly with MXG SEL to review avionics manning status and ensure manning resources are strategically distributed to provide the greatest possibility for mission success. **(T-2)**

11.34.1.8. Serve as the MXG Identify Friend or Foe Program Manager IAW **Paragraph 11.11** of this publication. **(T-2)**

11.34.1.9. Serve as the MXG Radar Warning Receiver/Radar Threat Warning program manager IAW **Paragraph 11.12** of this publication. **(T-2)**

11.34.1.10. Serve as the MXG EW program manager and EW Integrated Reprogramming focal point. **(T-2)**

11.34.1.11. Serve as MXG focal point for external organizations 406 MHz Emergency Locator Transmitter Systems Program IAW **Paragraph 11.27**. **(T-2)**

11.34.1.12. Coordinate with the wing EW POC to ensure compliance with DAFMAN 10-703, *Electronic Warfare Integrated Reprogramming*. **(T-2)**

11.34.1.13. Track wing assigned ECM, electronic attack and sensor pods, and associated support equipment. **(T-2)**

11.34.1.14. Coordinate all pod shipments as directed by MAJCOM to/from base or operating location. **(T-2)**

11.34.1.15. Track all incoming and outgoing pod parts and SE until received or arrived at destination. **(T-2)**

11.34.1.16. Coordinate with Electronic Combat Pilot/Electronic Warfare Officer to ensure most current MDF to configure Radar Warning Receiver/Radar Threat Warning to meet mission requirements. **(T-2)**

11.34.1.17. Serve as the MXG focal point for external organizations on all cybersecurity matters pertaining to aircraft interface equipment. **(T-2)**

11.35. Fire Extinguisher Requirements.

11.35.1. Unit coordinate with Fire Emergency Services Flight and Airfield Operations Flight to ensure required number of portable fire extinguishers are available for on and off installation operational requirements. Refer to DAFMAN 91-203, TO 00-25-172, *Ground Servicing of Aircraft and Static Grounding/Bonding*.

11.36. Air Force Repair Enhancement Program (AFREP).

11.36.1. The AFREP optimizes AF resources and repair capability of aerospace parts and equipment by increasing the wing-level (referred also as field-level) participation with the appropriate Program Office. This is accomplished by field-level identification and recommendation of candidates for reparability consideration by the appropriate PO.

11.36.1.1. The AFREP is an optional program.

11.36.1.2. Unit leadership must verify mission benefit outweighs the cost and obtain MAJCOM AFREP Manager Approval prior to program implementation.

11.36.2. AFREP is repairing consumable/expendable items (XF3 and XB3). Repair Network is repairing XD2 (recoverable) and XF3 (field level/condemnable). The concept is to make recommendations to the PM of consumable/expendable items which the field-level has identified as desirable candidates to be considered for conversion to field-level repair, limited repair or even full depot-level repair, contract or organic as directed by the Depot Source of Repair process in AFI 63-101/20-101, in addition, ensure repair cost/benefit analysis takes into consideration the total costs to the AF as outlined in Air Force TO 00-20-3, *Maintenance Processing of Reparable Property and Repair Cycle Asset Control System*.

11.36.2.1. The approval authority to change the SMR/ERRC codes, or select, use, arrange for, contract with, qualify sources of repair, or authorize the initiation of any local or other repair action rests solely with the PM.

11.36.2.2. Repair Approval. Personnel must coordinate any operational change to the system, end item, modified configuration, or maintenance procedure prior to implementation IAW AFI 63-101/20-101. For example, additional base-level repair or contract repair of any item beyond the provisions which already exist in field-level TOs. **(T-2)**

11.36.2.3. Field-level personnel, including AFREP work centers, will not contract out or arrange for repair services without prior written authorization by the PM and approval by the MAJCOM AFREP Manager. **(T-2)** The PM responsible for the system or end item will retain responsibility for making any decision stemming from such a recommendation. Field-level TOs do not give permission for local contract or off-base repair of any item, only on-base repair to the extent specified by the SMR code for a given item.

11.36.3. Operational Safety, Suitability & Effectiveness (OSS&E). OSS&E is an outcome of properly planned and applied systems engineering.

11.36.3.1. Organizations responsible for preserving OSS&E of AF systems or end items must ensure that operational use, configuration changes, maintenance repairs, aging, part substitutions, and similar activities and events do not degrade baselined characteristics of systems or end items over their operational life.

11.36.3.2. OSS&E is an integrated effort to ensure items are not allowed to degrade as a result of maintenance, repairs, parts substitutions, and similar activities.

11.36.3.2.1. The program manager is responsible for the assurance of OSS&E throughout the life cycle of each configuration of each component of each system. Only the PM may convert an item from non-repairable to reparable.

11.36.4. Requirements. Given authorized repair of a consumable/expendable item, the PM will determine the SMR/ERRC code validity. All resulting changes are linked with Supply Chain Manager Consideration for that item.

11.36.4.1. Supply Chain Management issues must be considered as a matter of law, given these issues affect Congressionally Authorized budget authority and funds allocated to accomplish specifically different supply chain functions (spares buys versus repairs).

11.36.4.1.1. Any personnel, organization, or AFREP work center may request SMR/ERRC code changes IAW TO 00-25-195 and AFH 23-123, Volume 2, **Part 2**, *Integrated Logistics System-Supply (ILS-S), Standard Base Supply System Operations*, **Chapter 8**.

11.36.4.2. Non-repairable (XB3) assets submitted to PM for repair approval will contain RC (SMR/ERRC change) IAW TO 00-25-195 and a MAJCOM published form for the AFREP Vendor Repair Approval that will include all supply data, full justification, and test/repair procedures. This document will be reviewed by the MAJCOM AFREP Manager IAW **Paragraph 11.36.8**.. **(T-2)**

11.36.4.3. Repairable (XF3) assets submitted to PM for vendor repair approval will be submitted on MAJCOM published form. The form must contain all current vendor certifications and governing technical procedures utilized for test/repair by vendor. This document will be reviewed by MAJCOM AFREP Manager and signed off.

11.36.5. Asset Turn-In. Items repaired IAW technical data will be turned in to the supply system following guidance in DAFMAN 23-122 and TO 00-20-3. **(T-2)**

11.36.6. AFREP Reporting Instructions: Wing AFREP Managers will provide quarterly updates to the MAJCOM AFREP Manager. **(T-2)**

11.36.6.1. MAJCOM AFREP Managers will report yearly activity updates from October 1 to September 30 and file their yearly report with AF/A4LM by 15 October. **(T-2)**

11.36.6.1.1. The quarterly/yearly updates will include at a minimum the nomenclature and total number of items repaired, total cost avoidance, and total cost savings. **(T-2)**

11.36.7. AF/A4L Roles and Responsibilities.

11.36.7.1. Prepares, publishes, and reviews AF-level policy and guidance for AFREP.

11.36.7.2. Perform annual program analysis to verify mission benefits outweigh costs.

11.36.8. MAJCOM AFREP Manager Responsibilities.

11.36.8.1. Administer/manage the MAJCOM AFREP in coordination with the applicable Program Office.

11.36.8.2. Perform annual analysis on program viability.

11.36.8.3. Mandate use of Program Office/selection authority criteria guidance and recommendations related to maintenance, supply, and repair sources.

11.36.8.4. Designate a focal point within the applicable Program Office for the MAJCOM and end users to contact regarding AFREP recommendations and OSS&E issues.

11.36.8.5. Establish a process to identify candidate items for submission to Program Office for consideration.

11.36.8.5.1. At a minimum, process will ensure base AFREP certifies the candidate item is beyond repair from all applicable on-base organic maintenance work centers/back shops. Certification must be documented (email, logbook). Parts coded direct NRTS are exempt from this requirement. **(T-2)**

11.36.8.6. Establish and maintain an AFREP Program Office submission and status tracking website.

11.36.8.7. Compile annual program cost benefit analysis information and forward to AF/A4L.

11.36.8.8. Provide Program Office repair recommendations for evaluation and track their status (**Part I** identification, initiation date, submittal date, closure date).

11.36.8.9. Maintain, update, and distribute a list of Wing/Unit AFREP POCs to include representatives from other agencies (such as, DLA, ALCs, MAJCOMs).

11.36.8.10. Review/Validate Wing/Unit SMR/ERCC requests IAW TO 00-25-195 and TO 00-20-3 prior to submission to the applicable Program Office for evaluation/approval.

11.36.8.11. Validate items are approved for contract repair for strategic sourcing opportunities.

11.36.8.12. Initiate the opportunity assessment of strategic sourcing candidates when deemed appropriate or when requested by the Program Office.

11.36.8.13. Provide quarterly MAJCOM cross-tell of AFREP repairs to Base/Center AFREP managers. The minimum data will include nomenclature, total cost, and total amount saved for each item across the MAJCOM.

11.36.9. Wing/Unit Responsibilities:

11.36.9.1. Participate in AFREP when approved by the MAJCOM AFREP Manager. **(T-2)**

11.36.9.1.1. DELETED.

11.36.9.1.2. MXG SEL will ensure AFREP Manning Positions are categorized as an additional duty and not listed as positions on the MXG UMD. **(T-2)**

11.36.9.1.2. **(AFMC)** MXG SEL will ensure military AFREP positions are "Out of Hide" and not listed as positions on the MXG UMD. AFREP civilian manning positions will be coded as RBA (Reimbursable Budget Authority). **(T-2)**

11.36.10. Wing/Unit AFREP Manager Responsibilities:

11.36.10.1. Submit SMR/ERRC change recommendations to the MAJCOM AFREP Manager for Program Office coordination. See TO 00-25-195, and this DAFI. **(T-2) Note:** Field-level TOs do not give authorization for local contract repair of any item, only on-base repair to the extent specified by the SMR code for a given item. AFREP or any other field level personnel will not perform repairs not authorized by technical data. **(T-2)**

11.36.10.1.1. Workcenters participating in AFREP repairs will submit Critical Application Items (CAI) and Critical Safety Items (CSI) repair recommendations to the MAJCOM AFREP Manager for screening to determine appropriate PM for submission. **(T-2) Note:** CAIs, CSIs, or elements of a system which the PM has designated as a CAI system, or a CSI system are prohibited from consideration under the AFREP program unless otherwise approved by an authorized engineering authority.

11.36.10.1.1.1. All Electronic Warfare (EW) Systems are CAI-designated systems. Verify CAI and CSI items on the Joint Services Critical Item Data Viewer, located at: **https://remote3.amrdec.army.mil/csiviewer/**. See **Attachment 2**, CAI & CSI for additional information.

11.36.10.1.1.1. **(AFMC)** See **Attachment 7**, CAI & CSI for additional information.

11.36.10.1.2. Contractor repairs of any item are not allowed without prior written authorization from the PM and approval by the MAJCOM AFREP Manager. **(T-2) Note:** AFREP work centers are prohibited from entering into repair source contractual/obligation relationships with contractors. DoD regulations exist to assign such actions (finding and qualifying repair sources) to the authorized engineering authority.

11.36.10.1.2.1. If contract repairs are authorized, contracts will be established IAW local contracting procedures. **(T-2)**

11.36.10.1.3. The Wing/Unit AFREP Manager will ensure the SOW includes, as a minimum, the following items for contracted repairs:

11.36.10.1.3.1. Requirement for the contractor to perform/document receiving inspections. **(T-2)**

11.36.10.1.3.2. Requirement for the contractor to inspect/test the item following

repair. **(T-2)**

11.36.10.1.3.3. Requirement for the contractor to provide a record of the repair and test results to the AFREP office and made available for the Government Inspector (such as, Defense Contract Management Agency (DCMA)). **(T-2)**

11.36.10.1.3.4. Contract will specify expected shipping, transportation, and repair-estimate completion dates. **(T-2)**

11.36.10.1.3.5. Repair items IAW Technical Data. **(T-2) Note:** Restricted use of systems will result from use of components which have been repaired as described immediately above, without prior written authorization from the program manager.

11.36.10.1.3.6. Enter data into the AFREP Recommendation Submittal site: AFREP - Home (dps.mil) The minimum data consists of a roll up AFREP repair total cost and will include:

11.36.10.1.3.6. **(AFMC)** May enter data into a locally used repository or site. The minimum data consists of a roll up of AFREP repair total cost and will include:

11.36.10.1.3.6.1. Time expended coordinating PM approval for AFREP repair authorization. **(T-2)**

11.36.10.1.3.6.2. Man-hours to complete repair "labor hours". **(T-2)**

11.36.10.1.3.6.3. Nomenclature/Stock Number of part being repaired. **(T-2)**

11.36.10.1.3.6.4. Total cost of item if sourced from the supply chain. **(T-2)**

11.36.10.1.3.6.5. Total cost of tangible resources expended to complete AFREP repair (such as, wiring/consumables). **(T-2)**

11.36.10.1.3.7. Assist technicians by interfacing with engineering, Equipment Specialists (ES), QA, Air Force Engineering Technical Services, and other field-level agencies. **(T-2)**

11.36.10.1.3.8. Accumulate and forward data requested by MAJCOMs concerning AFREP. **(T-2)**

11.36.10.1.3.9. Provides repair authorizations to the MXG QA PIM IAW DAFI 21- 101.

11.36.10.1.3.10. Ensure individuals assigned AFREP extra duties are trained, qualified and maintains a minimum five-skill level in their AFSC IAW DAFMAN 36-2689.

11.36.10.1.3.11. Submit and monitors DR for deficiencies discovered during a Wing/Unit AFREP initiative, IAW TO 00-35D-54 or equivalent PM approved DR process.

11.36.10.1.3.12. Review daily Awaiting Parts (AWP) and MICAP lists from maintenance activities to determine if the AFREP office can recommend to the appropriate PO, items from those lists. **(T-2)**

11.36.10.1.3.13. Provided quarterly AFREP repair updates to MXG/CC. The repair updates at a minimum will include nomenclature, total cost, and total amount

saved for each item. **(T-2)**

11.36.10.1.3.14. Establish collection points within unit supply sections for consumable/expendable items coded with ERRC XB3 to determine reparability of assets prior to permanent disposal.

11.37. Calibration Limitation Approval Certification Program.

11.37.1. General. A limited TMDE calibration could seriously impact mission capability of weapon systems. All units will have a comprehensive training program to ensure authorized personnel can interpret TMDE calibration limitations to the specified requirement of the applicable weapon system. **(T-2)** Personnel will be certified IAW criteria established in **Table 11.1**. **(T-2)** MXG/CCs may designate contractors in writing to authorize calibration limitations.

11.37.2. Responsibilities and Management. The MT or TD will be responsible for management and development of the calibration limitation approval training program. **(T-2)**

11.37.2.1. As a minimum, the course will include when to consider a limited calibration, impact of using improperly calibrated equipment, and how to apply calibration specifications to weapon system requirements. **(T-2)**

11.37.2.2. Prior to placement on the SCR, the calibration limitation approval training (MT or TD course) will be mandatory. **(T-2)**

11.37.3. - 11.37.3.2. DELETED.

11.37.4. Certification Criteria. Certifying officials will be selected IAW criteria established in **Table 11.1**. **(T-2)**

11.38. Oil Analysis Program (OAP).

11.38.1. Program. Oil Analysis is the process of analyzing oil and other fluids used to lubricate or operate mechanical equipment, evaluating the condition of the fluid or the equipment from which the fluid originated, and recommending maintenance actions to the equipment operating activity. An OAP ensures timely and accurate oil analysis support through the strategic location of oil analysis laboratories and the standardization of procedures, data elements, analytical instrumentation, and diagnostic techniques. The Joint Oil Analysis Program (JOAP) is a combined effort of the Army, Navy and AF to set-up and maintain a standard program. Oil analysis requires a centrally managed program and the integration of AF OAP and JOAP plans.

11.38.2. Objectives. The overall objective of the AF OAP is to detect oil-wetted air and space equipment failures before serious malfunction or secondary damage occurs. The specific objectives of the AF OAP and JOAP programs are as follows:

11.38.2.1. Improve the operational safety, readiness, and economy of military equipment through the use of on-board and off-board oil analysis, a condition-monitoring concept that relies on the detection and measurement of wear-metals in the fluid.

11.38.2.2. Collect and analyze oil analysis data in order to increase the effectiveness of oil analysis techniques in the diagnosis of potential equipment failures and lubricant condition; to provide wear metal and lubricant physical property data to the various weapons system managers and others, as required.

11.38.2.3. Ensure oil analysis plans and operations are integrated (where practical) to provide:

11.38.2.3.1. Standard laboratory techniques, procedures, data, calibration standards, and analytical instruments.

11.38.2.3.2. Inter-service oil analysis support to all military departments.

11.38.2.3.3. The most cost–effective means of determining the condition of lubricants, fluids, and mechanical system through the use of various analytical techniques.

11.38.3. Guidance. The OAP helps aircraft technicians and supervisors to make informed, condition-based, preventive maintenance decisions, and can reduce equipment costs, increase equipment availability, and reduce in-flight risk. This is primarily achieved by monitoring the concentration of wear metals in fluids used to lubricate or power mechanical systems.

11.38.3.1. To monitor engine health, the OAP uses a variety of testing mechanisms. These include:

11.38.3.1.1. Atomic Emission spectrometric wear metal analysis.

11.38.3.1.2. Magnetic Chip Detectors/Magnetic Chip Detectors with on-board sensors.

11.38.3.1.3. Scanning Electron Microscope/Energy Dispersive X-ray system technology.

11.38.3.1.4. **(Added-AFMC)** X-ray Fluorescence

11.38.3.2. To the extent deemed cost effective, EOT will be used as the standard time interval between oil analysis sampling when in-service engines are equipped with an Engine Monitoring System or other operating time recorders.

11.38.3.2.1. For engines without Engine Monitoring System or other operating time recorders, oil analysis trending intervals will be standardized based upon Engine Flying Hours.

11.38.3.3. Analyze oil samples from transient aircraft IAW applicable technical data and owning activity requirements.

11.38.3.3.1. For transient aircraft with an EMS, at bases without the capability to download EOT data, Engine Flight Hours will be used to continue oil analysis trending.

11.38.3.4. Responsible activities must ensure the resultant data is accurate and given promptly to all customers so they can effectively monitor the condition of their equipment.

11.38.3.4.1. AF OAP laboratories will evaluate response times to optimize support of the customer. **(T-2)** Evaluation frequency is established by MAJCOM supplemental guidance.

11.38.3.4.1.1. **(Added-AFMC)** See **paragraph 11.38.6** and subparagraphs.

11.38.3.5. Laboratory operation.

11.38.3.5.1. Each OAP laboratory must be certified IAW TOs 33-1-37-1/-2.

11.38.3.5.2. Seek to maximize inter-service use of existing laboratories through laboratory consolidation, workload sharing, and use of standardized instrumentation, techniques, and procedures. **(T-2)**

11.38.3.5.3. AF OAP laboratories must provide oil analysis services at no charge for all US Government, North Atlantic Treaty Organization (NATO) and friendly Foreign Military Sales aircraft. These services include analyzing oil samples from transient aircraft IAW applicable technical data within the AF OAP laboratory capabilities. **(T-2)**

11.38.4. Roles and Responsibilities

11.38.4.1. AF/A4L is the chief Air Staff agency with policy responsibility for the maintenance of air and space equipment.

11.38.4.1.1. Guidance. Prepares, publishes, and reviews AF-level policy and guidance for the AF OAP.

11.38.4.1.1.1. Coordinates with MAJCOMs to review and resolve guidance-related issues.

11.38.4.1.2. Management. Designates a POC for AF OAP. The representative will manage this publication while closely coordinating with functional experts.

11.38.4.1.2.1. Works with AF representative to the JOAP-Coordinating Group (CG) on policy issues.

11.38.4.2. Major Command. Each MAJCOM establishes a headquarters POC for complying with MAJCOM OAP responsibilities.

11.38.4.2.1. Establishes or relocates OAP laboratories to support MAJCOM mission. Coordinates establishment or relocation with the AF OAP Manager.

11.38.4.2.2. Ensures quality deficiency reports are submitted to the unit Product Improvement Manager IAW TO 00-35D-54. DRs must be submitted to engine program offices on all equipment requiring tear down or overhaul due to an OAP laboratory maintenance recommendation and on all oil-wetted component failures where no OAP laboratory maintenance recommendation was made.

11.38.4.2.3. Ensures laboratories provide the depot a computer-generated printout/record, for each engine undergoing scheduled maintenance or overhaul.

11.38.4.2.4. Ensures laboratories process and evaluate samples IAW TO 33-1-371/- 2.

11.38.4.2.5. Ensures proper training of AF OAP laboratory technicians. This includes identifying training requirements to the AETC.

11.38.4.2.6. Ensures assigned shop instrumentation and equipment is not modified or used for non-OAP applications without AF OAP Office approval.

11.38.4.2.7. Provides guidance necessary to execute the AF OAP and ensures all subordinate organizations understand and properly execute AF OAP and JOAP responsibilities. MAJCOMs/ANG may provide additional guidance in their supplements or addendums to this DAFI as required.

11.38.4.2.8. Ensures AF OAP and JOAP requirements are included in planning, programming, and budgeting process. This includes providing needed funds, personnel, facilities, and other resources to maintain an effective program.

11.38.4.2.9. Supports equipment evaluations and field surveys for the AF OAP Office.

11.38.4.3. Air Force Materiel Command (AFMC). AFMC is the lead MAJCOM for the AF OAP and AF participation in the JOAP. AFMC is also responsible for oil analysis Research, Development, Test and Evaluation (RDT&E). The Directorate of Logistics, Civil Engineering, Force Protection, and Nuclear Integration (AFMC/A4/10) is the headquarters office of primary responsibility for AF OAP guidance coordination and implementation within AFMC.

11.38.4.3.1. Propulsion Directorate.

11.38.4.3.1.1. Sustainment Chief Engineer represents the AF on the JOAP-Executive Committee.

11.38.4.3.1.1.1. Establishes, funds, staffs, and directs the AF OAP Office.

11.38.4.3.1.2. The Development Program Manager ensures specification guidance for oil analysis sampling provisions or other appropriate measures to detect oil-wetted failures, where applicable, are included during the design phase.

11.38.4.3.1.3. AF OAP Office, AFLCMC/LP. The AF OAP Office manages the AF OAP, and in coordination with the MAJCOMs and propulsion community, leads AF participation in the JOAP.

11.38.4.3.1.3. **(AFMC)** AF OAP Office, AFLCMC Propulsion Directorate (AFLCMC/LPS).

11.38.4.3.1.3.1. Certifies/decertifies AF OAP laboratories for participation in the JOAP.

11.38.4.3.1.3.2. Maintains a list of AF OAP laboratories, equipment, and customers.

11.38.4.3.1.3.3. Assists MAJCOM/Center managers in program execution. This includes recommending the establishment, location, and relocation of AF OAP laboratories.

11.38.4.3.1.3.3.1. Communicates AF OAP objectives, policies and procedures to the Director of Propulsion, Engine OAP Managers, equipment specialists and program managers.

11.38.4.3.1.3.3.2. Evaluates the need for and performs special studies, as requested by the MAJCOMs or depots.

11.38.4.3.1.3.3.3. The AF OAP Office should periodically conduct laboratory assistance/assessments to determine adequacy and effectiveness of the AF OAP. Identifies problems and recommends solutions.

11.38.4.3.1.3.4. Establishes and manages a data system, meeting tri-service requirements, to evaluate AF OAP participation and effectiveness and to provide engine program offices with historical data on oil sample analysis

results.

11.38.4.3.1.3.5. Coordinates and consolidates AF/MAJCOM requirements with Army/Navy to ensure, where practical, the procurement of common OAP equipment.

11.38.4.3.1.3.6. Represents the AF on the JOAP-CG.

11.38.4.3.1.3.7. Develops a comprehensive OAP laboratory certification and quality control program.

11.38.4.3.1.3.8. Maintains and provides AF inputs to TOs 33-1-37-1/-2/-3/-4, *JOAP Manual*.

11.38.4.3.1.3.9. Reviews and evaluates the JOAP school curriculum.

11.38.4.3.1.3.10. Assists the Engine OAP Manager and serves as an advisor for the engine Maintenance Planning Working Group (MPWG) for OAP issues.

11.38.4.3.2. Director of Propulsion (DOP). The DOP AFLCMC/LP in conjunction with Engine Single Managers at AFLCMC/LPS, AFLCMC/LPA and the AF OAP Office assess existing/potential oil analysis technologies. The DOP also ensures wear metal debris and oil analysis is an integral part of the Engine Health Management Program. The DOP appoints Engine OAP Managers for each AF managed engines.

11.38.4.3.2.1. Engine OAP Managers. The Engine OAP Manager is the engineer in charge of a particular engine and is solely responsible for the OAP related issues on that particular engine.

11.38.4.3.2.1.1. Serves as the focal point for the engine MPWG for OAP issues, provides the guidance necessary to accomplish engine-specific oil analysis.

11.38.4.3.2.1.2. Ensures expeditious handling of equipment returned for tear down or overhaul because of an OAP laboratory maintenance recommendation or where oil analysis results indicated a potential problem.

11.38.4.3.2.1.3. Provides guidance necessary to accomplish engine specific oil analysis. Provides updates for TOs 33-1-37-1/-2/-3/-4 to the OAP Office.

11.38.4.3.2.1.4. Provides accurate and timely feedback to the MPWG and field units on OAP-monitored equipment. This includes maintenance findings on equipment in for tear down or overhaul as a result of an OAP laboratory maintenance recommendation. It also includes failure reports and related wear metal and oil analysis data on oil-wetted components where no OAP laboratory maintenance recommendation was made.

11.38.4.3.2.1.5. Works with the MPWG to establish and maintain wear metal limits, diagnostic criteria, and other oil analysis parameters. This is based on a review of data from equipment tear-down and overhaul findings.

11.38.4.3.2.1.6. Maintains metrics on hits, misses, and escapes. Once oil analysis data is included in the Engine Health Management AF Enterprise Center with software capability to perform metric monitoring, the AF OAP will

concurrently monitor metrics with the Engine OAP managers.

11.38.4.3.2.1.7. Works with the MPWG, MAJCOM customers and the AF OAP Office to establish engine-specific technical and performance requirements for all wear metal debris and oil analysis equipment.

11.38.4.3.2.1.8. Utilizes the Component Improvement Program as needed to evaluate the cost effectiveness of existing and potential wear metal debris and oil analysis applications, establish test programs, and implement the most cost-effective method(s).

11.38.4.3.2.1.9. Sends updates of specific oil sampling intervals and wear metal limits (evaluate for new engines during the design phase; reconsider for existing engines when oil-wetted parts undergo any materiel or strength changes) to the MPWG.

11.38.4.3.3. SE and Vehicles Division (AFLCMC/WNZ) procures oil analysis equipment at the request of the AF OAP Office.

11.38.4.3.3.1. Maintains a contract for procurement of JOAP AE spectrometers used by the Army, Navy, and AF laboratories.

11.38.4.3.3.2. Provides technical order provisioning and support.

11.38.4.3.3.3. Establishes inter-service logistics support. This includes spare parts support, instrument repair and overhaul, procurement of common JOAP laboratory equipment and supplies, item management and equipment specialist activities, and funding status of existing contracts.

11.38.4.3.4. Air Force Research Laboratory. The Air Force Research Laboratory accomplishes RDT&E to improve wear metal and oil analysis instruments, materials, and techniques. **(T-2)**

11.38.4.3.4. **(AFMC)** The Air Force Research Laboratory as well as the AF OAP Office, accomplish RDT&E to improve wear metal and oil analysis instruments, materials, and techniques. **(T-2)**

11.38.4.3.4.1. Supports the AF OAP charter to conduct test and evaluation for the oil analysis programs, including the JOAP and the AF OAP, as applicable. **(T-2)**

11.38.4.3.4.2. Coordinates RDT&E activities with the MAJCOMs, DOP, and AF OAP Office as applicable. **(T-2)**

11.38.4.4. AETC. Provides initial oil analysis training for AFSC 2A7X2 through the Nondestructive Inspection course. The AF OAP Management Office and MAJCOMs may request additional training, as required.

11.38.4.4.1. Coordinates course materiel changes with the AF OAP Office and the MAJCOM POCs.

11.38.5. Reporting and Measurement. All MAJCOMs and laboratories must collect and report metrics to the AF OAP Manager IAW TOs 33-1-37-1/-2/-3/-4.

11.38.5.1. All laboratories must collect, and report hits, misses, and escapes as a minimum. **(T-2)**

11.38.6. **(Added-AFMC)** Oil Analysis Program (OAP) Responsibilities and Requirements.

11.38.6.1. **(Added-AFMC)** MAJCOM A4M Responsibilities. **(T-2)**

11.38.6.1.1. **(Added-AFMC)** Appoints a qualified SNCO to manage Command OAP and serve as a point of contact for OAP activities. **(T-2)**

11.38.6.2. **(Added-AFMC)** MXG/CC Responsibilities. **(T-2)**

11.38.6.2.1. **(Added-AFMC)** Appoints a Wing OAP Manager and alternate. **(T-2)**

11.38.6.2.2. **(Added-AFMC)** Ensure Nondestructive Inspection (NDI)/OAP labs coordinate with base Civil Engineer on electrical power system design for sufficient voltage and any alternate power source to meet mission capacity requirements, e.g., AFMAN 32-1062, *Electric Systems, Power Plants and Generators*. **(T-2)**

11.38.6.2.3. **(Added-AFMC)** Ensures the NDI/OAP facility has a Class A telephone and a direct line to MOC to expedite the reporting of abnormal wear-metal trends. **(T-2)**

11.38.6.3. **(Added-AFMC)** Wing OAP Manager Responsibilities. **(T-2)**

11.38.6.3.1. **(Added-AFMC)** Manages the OAP IAW T.O. 33-1-37-1, *Joint Oil Analysis Program Manual*, and other applicable directives. **(T-2)**

11.38.6.3.2. **(Added-AFMC)** Develops procedures to establish policy and requirements for the wing OAP. Include a standardized method to ensure the total oil serviced since last OAP sample can be tracked and accurately entered on the DD Form 2026, *Oil Analysis Request*. **(T-2)**

11.38.6.3.3. **(Added-AFMC)** Ensures all organizations requiring OAP support appoint an OAP Manager and alternate using an appointment letter. **(T-2)**

11.38.6.3.4. **(Added-AFMC)** Provide consolidated quarterly OAP data report to the MXG/CC, the Squadron Operations Officer/Maintenance Superintendent, Propulsion Flight Chief, all organization OAP Managers, and the NDI Section Chief. **(T-2)** As a minimum, the following information will be included in the quarterly report:

11.38.6.3.4.1. **(Added-AFMC)** Number of OAP samples processed. **(T-2)**

11.38.6.3.4.2. **(Added-AFMC)** Number and percentage of DD Form 2026 errors. **Note:** Count each DD Form 2026 that has any errors as one error, then divide the number of discrepant DD Form 2026s by the total number of OAP samples processed for the error percentage. **(T-2)**

11.38.6.3.4.3. **(Added-AFMC)** Number of OAP laboratory maintenance recommendations. **(T-2)**

11.38.6.3.4.4. **(Added-AFMC)** Average OAP sample response time. **Note:** Enter sample response time to the nearest tenth of an hour and do not include the time when the OAP lab is not manned. **(T-2)**

11.38.6.3.4.5. **(Added-AFMC)** Number and percentage of OAP samples that exceeded required sample response times. Obtain the percentage by dividing the number of OAP samples exceeding required sample response time by the total OAP samples processed. **(T-2)**

11.38.6.3.4.6. **(Added-AFMC)** Number and percentage of aircraft engines not sampled as required by applicable technical order. Accomplish this by dividing the number of aircraft engines not sampled as required by the total OAP samples processed. **(T-2)**

11.38.6.3.4.6.1. **(Added-AFMC)** Number of OAP lab maintenance recommendations confirmed by physical finding of abnormal wear or potential for catastrophic failure "hits". **(T-2)**

11.38.6.3.4.6.2. **(Added-AFMC)** For Scanning Electron Microscope/Energy Dispersive X-ray equipped bases: Number of Magnetic Chip Detector (MCD) analyzed.

11.38.6.3.4.6.3. **(Added-AFMC)** For Scanning electron microscopy with energy dispersive X-ray (SEM)/EDX) quipped bases: Number and percentage of Level 1, 2, and 3 warnings by material type. **(T-2)**

11.38.6.3.4.6.4. **(Added-AFMC)** For SEM/EDX equipped bases: Number of unserviceable MCDs found. **(T-2)**

11.38.6.3.4.6.5. **(Added-AFMC)** For SEM/EDX equipped bases: Number and percentage DD Form 2026 errors. Note: Count each DD Form 2026 that has any errors as one error, and then divide the number of discrepant DD Form 2026s by the total number of MCDs analyzed for the error percentage. **(T-2)**

11.38.6.3.4.6.6. **(Added-AFMC)** For SEM/EDX equipped bases: Average SEM/EDX analysis response time. **Note:** This is the time from when the MCD is pulled to the time results are reported to the MOC. Enter the response time to the nearest tenth of an hour and do not include the time when the OAP lab is not manned. **(T-2)**

11.38.6.3.4.6.7. **(Added-AFMC)** For SEM/EDX equipped bases: Number and percentage of MCDs that exceeded required delivery time. Obtain the percentage by dividing the number of MCDs exceeding required delivery time by the total number of MCDs processed. **(T-2)**

11.38.6.4. **(Added-AFMC)** Operations Officer/Maintenance Superintendent Responsibilities. **(T-2)**

11.38.6.4.1. **(Added-AFMC)** Ensures all assigned aircraft are sampled IAW the applicable aircraft T.O. **(T-2)**

11.38.6.4.2. **(Added-AFMC)** Ensures OAP samples are delivered to the OAP lab with a DD Form 2026 filled out IAW T.O. 33-1-37-1. **(T-2)**

11.38.6.4.2.1. **(Added-AFMC)** In the DD Form 2026 "hours/miles since overhaul block, for engines with engine monitoring systems (EMS) use engine operating time unless the specific technical order directs otherwise (e.g., F110 engine use EMS inflight time). Reconcile NDI/OAP lab and aircraft records using downloaded EMS data accordingly. For engines without EMS, use engine flying hours. **(T-2)**

11.38.6.4.3. **(Added-AFMC)** Ensures flightline personnel verify with the OAP lab that the information entered in the Oil Analysis Record matches during scheduled

aircraft records checks. Verify, as a minimum, engine operating hours, time since oil change, oil serviced since last records check/OAP sample, engine serial number(s) and aircraft serial number. **Note:** MXG/CC or equivalent may waive verification of OAP records against aircraft records when aircraft are deployed and the scheduled aircraft records check is due. **(T-2)**

11.38.6.4.4. **(Added-AFMC)** Identifies AMU OAP Managers and alternates by appointment letter. Forward a copy to the Wing OAP Manager and the OAP laboratory. The letter will include grade, name, duty phone, AFSC, organization and office symbol. OAP manager will be a NCO and will serve as the primary liaison between their AMU and the OAP lab for all OAP issues. Ensures assigned OAP Managers or alternates attend all OAP meetings involving their AMU. **(T-2)**

11.38.6.4.5. **(Added-AFMC)** Ensures all aircraft engines under special OAP codes IAW T.O. 33-1-37 are not flown until results of the OAP sample(s) are known. Ensure no aircraft engines are operated until the following DD Form 2026 discrepancies are corrected and verified with the OAP lab: equipment and/or end item serial number error, hours (EOT or EFH) since overhaul error, oil change error, and oil added since last sample error. These items are essential to oil analysis trending and require removal of that engine from service until the discrepancy is corrected. **(T-2)**

11.38.6.4.6. **(Added-AFMC)** Ensures DD Form 2026 with equipment and/or end item serial number error, hours since overhaul error and oil added since last sample error are corrected immediately. **(T-2)**

11.38.6.4.7. **(Added-AFMC)** Ensures all maintenance actions affecting oil-wetted engine components are provided to the OAP lab using the remarks section of the DD Form 2026 or a suitable local form. **(T-2)**

11.38.6.5. **(Added-AFMC)** Propulsion Flight Chief Responsibilities. **Note:** When no Propulsion Flight exists or the Propulsion Flight performs no maintenance on the affected oil-wetted system, the MXG/CC or designated representative or equivalent assumes these responsibilities. **(T-2)**

11.38.6.5.1. **(Added-AFMC)** Ensures accurate and timely deficiency reports are submitted through the unit PIM to the applicable ALC engine program offices on all engines requiring tear down or overhaul due to an OAP laboratory maintenance recommendation and on all oil-wetted component failures. **(T-2)**

11.38.6.5.2. **(Added-AFMC)** Ensures a copy of the Oil Analysis Record is provided to the applicable repair facility for each engine undergoing scheduled maintenance or overhaul. **(T-2)**

11.38.6.5.3. **(Added-AFMC)** Makes the final decision regarding all OAP engine maintenance action recommendations. **(T-2)**

11.38.6.5.4. **(Added-AFMC)** Ensures all Propulsion Flight maintenance actions which affect oil- wetted engine components are provided to the OAP lab. This will be done by using the remarks section of the DD Form 2026 submitted with the OAP sample. **(T-2)**

11.38.6.5.5. **(Added-AFMC)** Appoints OAP Managers and provides the names by appointment letter to the Wing OAP Manager and the OAP lab. Include grade, name, duty phone, organization, AFSC and office symbol. **(T-2)**

11.38.6.5.6. **(Added-AFMC)** Ensures assigned OAP Managers or alternates attend all OAP meetings. **(T-2)**

11.38.6.6. **(Added-AFMC)** MOC Responsibilities. **(T-2)**

11.38.6.6.1. **(Added-AFMC)** Maintains an OAP status on each assigned aircraft showing all lab recommendation codes next to the aircraft serial number. Prior to commencing the flying day, verify status of aircraft or engines on special surveillance. **(T-2)**

11.38.6.6.2. **(Added-AFMC)** Relays to the OAP lab information regarding engine changes on- and off-station as they occur but no later than 0800 the next duty day. **(T-2)**

11.38.6.6.3. **(Added-AFMC)** Notifies the OAP lab when the cross country/deployed aircraft return.

11.38.6.6.4. **(Added-AFMC)** Initiates follow-up action when the oil analysis record from cross country/deployed aircraft is not returned to the OAP lab. **(T-2)**

11.38.6.7. **(Added-AFMC)** NDI/OAP lab NCOIC Responsibilities.

11.38.6.7.1. **(Added-AFMC)** Ensures the scheduled aircraft records check is documented on the affected engine's Oil Analysis Record with the date the check was accomplished and OAP lab person’s initials. **(T-2)**

11.38.6.7.2. **(Added-AFMC)** Ensures a copy of the Oil Analysis Record or a suitable automated form is provided to the propulsion flight for each engine undergoing scheduled maintenance or overhaul at depot, JEIM or CRF. **(T-2)**

11.38.6.7.3. **(Added-AFMC)** Immediately notifies MOC and the propulsion flight chief when an installed engine is restricted from operation or is placed on special sampling. **(T-2)**

11.38.6.7.4. **(Added-AFMC)** Ensures analysis results on all installed engines are provided to MOC after analysis of the OAP sample is complete. **(T-2)**

11.38.6.7.5. **(Added-AFMC)** Immediately notifies test cell and the propulsion flight chief when abnormal OAP results are discovered on test cell engines. **(T-2)**

11.38.6.7.6. **(Added-AFMC)** Ensures DD Form 2026s with equipment and/or end item serial number error, hours since overhaul error and oil added since last sample error are corrected immediately. **(T-2)**

11.38.6.7.7. **(Added-AFMC)** Tracks aircraft OAP sample response times for all assigned aircraft to ensure response time compliance. **(T-2)**

11.38.6.7.8. **(Added-AFMC)** Maintains a current appointment letter of all customer OAP managers. **(T-2)**

11.38.6.8. **(Added-AFMC)** OAP Sample Response Time Requirements for Routine OAP Samples. **Note:** The OAP sample response time begins at the time the OAP sample is taken and ends at the time the oil analysis results are reported to the MOC. **(T-2)**

11.38.6.8.1. **(Added-AFMC)** 2 1/2 hours for one and two engine aircraft. Deliver the sample to the OAP lab within 75 minutes of engine shutdown. MXG/CC may adjust the 2 1/2 hour rule as needed to accommodate flying windows. When the OAP lab is not manned, they shall provide results to the MOC NLT 2 hours after the beginning of the next shift. **Note:** OAP response time does not apply to aircraft that are geographically separated from the supporting OAP lab; however, aircraft will not fly beyond the applicable -6 T.O. sampling interval. **(T-2)**

11.38.6.8.2. **(Added-AFMC)** Five hours for all other aircraft. When the OAP Lab is not manned, they shall provide results to the MOC NLT 2 hours after the beginning of the next shift. **Note:** OAP response time does not apply to aircraft that are geographically separated from the supporting OAP lab; however, aircraft will not fly beyond the applicable -6 T.O. sampling interval. **(T-2)**

11.38.6.8.3. **(Added-AFMC)** Four hours for engine ground/trim and test cell runs. **(T-2)**

11.38.6.8.4. **(Added-AFMC)** Immediately processes/analyzes special "Red Cap" samples after drawing the sample. The OAP lab will assign priority to "Red Caps" over routine samples and expedite results to MOC. **(T-2)**

11.38.6.9. **(Added-AFMC)** OAP Requirements for Cross-Country Flights/Deployments. **(T-2).**

11.38.6.9.1. **(Added-AFMC)** Flightline personnel place an Oil Analysis Record in the aircraft AFTO Form 781-series forms jacket prior to departure. **(T-2)**

11.38.6.9.2. **(Added-AFMC)** The OAP lab personnel ensure the Oil Analysis Record contains at least the last 10 analyses. The flightline expediter or pro super notifies the OAP lab in advance for cross-country documents. **(T-2)**

11.38.6.9.3. **(Added-AFMC)** Flightline personnel sign for the Oil Analysis Record at the OAP lab and return it to the lab the day the aircraft returns to home station. **(T-2)**

11.38.6.9.4. **(Added-AFMC)** The OAP lab notifies MOC if the Oil Analysis Record is not returned. **(T-2)**

11.38.6.9.5. **(Added-AFMC)** The OAP lab reviews the returned Oil Analysis Record for adverse trends and takes necessary action. **(T-2)**

11.38.6.9.6. **(Added-AFMC)** AMUs will follow the OAP procedures in this supplement at the deployment sites. **(T-2)**

11.38.6.9.7. **(Added-AFMC)** Deployed OAP personnel shall have telephone or radio communication with MOC and the AMU to expedite reporting of abnormal OAP trends. **(T-2)**

11.38.6.10. **(Added-AFMC)** OAP Requirements for Transient Aircraft. **(T-2)**

11.38.6.10.1. **(Added-AFMC)** Transient maintenance personnel sample aircraft as required by this section and applicable -6 technical order. **(T-2)**

11.38.6.10.2. **(Added-AFMC)** Transient maintenance personnel draw the OAP sample and make a Red Dash entry on the AFTO Form 781A, *Maintenance Discrepancy and Work Document*, indicating, "Engine oil analysis results due". **(T-2)**

11.38.6.10.3. **(Added-AFMC)** When OAP capability exists at a transient location and an OAP sample is required, the OAP sample results will be known prior to aircraft departure. **(T-2)**

11.38.6.10.4. **(Added-AFMC)** Transient bases without OAP capability shall take required OAP samples and ensure samples are given to aircrew for processing at next base. **(T-2)**

11.38.6.10.5. **(Added-AFMC)** If OAP sample results are not provided before aircraft departure, the results shall be forwarded via FAX, e-mail or most expeditious manner by the local MOC or transient maintenance to the aircraft's next destination (either MOC, TA, or base operations). **(T-2)**

11.39. Air Force Engineering and Technical Services (AFETS).

11.39.1. General. The AF must maintain its weapons systems and equipment to meet worldwide mission requirements and operational needs at a reasonable cost. To accomplish this, units need the capability to quickly resolve complex or unusual technical problems and provide enhanced system-specific technical training to AF technicians, contractors, and operators. Engineering and Technical Services (ETS) provides this expeditionary resource and is prepared to deploy AFETS and CETS as needed.

11.39.1.1. AFETS personnel are the primary source of Engineering and Technical Services support in the AF. AFETS field engineers are Emergency Essential DoD civilian employees, highly experienced and thoroughly trained technical specialists.

11.39.1.2. CETS can be an important element in developing an independent AF capability on new systems; however, units must develop their own organic capability and/or request AFETS support. **(T-2)** Organizations should normally terminate CETS within 12 months after obtaining self-sufficiency.

11.39.2. AFETS are DoD civilians who provide advantages of long-term continuity and decreased retraining costs provided by a civilian work force. AFETS field technicians will be used and retrained as necessary to meet technical needs and changing mission requirements. **(T-2)**

11.39.2.1. AFETS can design special test equipment, develop special maintenance procedures, develop, and conduct technical training for unit maintainers and operators, and recommend changes to maintenance processes.

11.39.2.2. AFETS also serve as the unit technical liaison and work with MAJCOM functional managers, depot technicians, engineers, item managers, and equipment manufacturers to resolve complex equipment problems.

11.39.2.3. AFETS will certify tasks IAW AFI 36-2650 and DAFMAN 36-2689 in USAF personnel training records when training is provided to the go/no-go level. **(T-2)**

11.39.3. AFETS personnel should be functionally aligned under the local ETS OPR (typically the MXG/CC or equivalent commander) and reside within the assigned organizations maintenance complex.

11.39.3.1. AFETS will not be authorized to overcome manning shortfalls or to perform duties considered organic to the unit's manning. **(T-2)**

11.39.3.2. When AFETS and contracted engineering support (example, CETS, Field Service Engineer or FSR) are assigned to a unit every effort will be made to co-locate these resources to maximize effectiveness of technical support within the organization. **(T-2)**

11.39.4. The AF may utilize CETS to provide on-site proficiency training, technical advice, and technical assistance for initial system bed-down or major modifications when AFETS are unavailable. Units desiring services of strategically assigned CETS will direct their requests to their MAJCOM OPR. **(T-2)**

11.39.4.1. When CETS and AFETS are assigned to the same unit, CETS will support and train AFETS personnel as required. **(T-2)**

11.39.4.2. CETS will certify tasks IAW DAFMAN 36-2689, and AFI 36-2650, in USAF personnel training records when training is provided to the go/no-go level. **(T-2)**

11.39.5. Exclusions. ETS covered in this DAFI excludes:

11.39.5.1. Engineering review and resolution of service-revealed deficiencies reported through normal maintenance information systems.

11.39.5.2. Materiel DR covered in TO 00-35D-54.

11.39.5.3. The engineering determination of materiel integrity.

11.39.5.4. The engineering or technical services used in 61-series instructions.

11.39.6. Limits: CETS are restricted to the duties and responsibilities outlined in this DAFI and specific tasks listed in the Task Work Specification. MAJCOM OPRs (with approval by the applicable MAJCOM Directorate) may grant exceptions to these limits only on a case-by-case basis based on mission need. Do not use CETS to avoid manpower ceilings or other personnel rules and regulations.

11.39.6.1. CETS will not:

11.39.6.1.1. Perform non-ETS duties or normal unit duties. **(T-3)**

11.39.6.1.2. Make policy or represent the using activity at meetings or conferences. **(T-3)**

11.39.6.1.3. Supervise or control AF personnel or personnel of other contractors. **(T-3)**

11.39.6.1.4. Hold engineering decision-making positions. **(T-3)**

11.39.6.1.5. Perform direct maintenance except in emergency situations. **(T-3)**

11.39.7. MAJCOM Responsibilities. MAJCOMs will:

11.39.7.1. Supplement this document as necessary to ensure standardization among subordinate units.

11.39.7.2. Designate a Lead ETS Program Office in a single MAJCOM to administer the activities of the member MAJCOMs merged by MOA/MOU.

11.39.7.3. Designate an ETS OPR in the appropriate Headquarters Directorate or Field Operating Unit (FOA) to serve as the MAJCOM ETS OPR for member MAJCOMs merged by MOA/MOU.

11.39.7.4. Ensure all applicable requirements of this DAFI are met by units authorized to decentralize management of their ETS programs to the using activities.

11.39.7.5. Program and defend MAJCOM ETS funds and manpower requirements over the Future Years Defense Program (FYDP) consistent with AF mission requirements.

11.39.7.6. Budget and fund AFETS PCS, TDY, and Developmental Training costs to support their MAJCOM requirements and submit Program Objective Memorandum (POM) for all ETS requirements.

11.39.7.7. Identify and ensure mobility statements are in AFETS Standard Core Personnel Document and designate positions as Emergency-Essential. Refer to DAFI 36-153, *Civilian Mobility*.

11.39.7.8. Ensure general personnel management records are maintained at the unit of assignment on all ETS personnel to assure proper management and administration of ETS resources.

11.39.7.9. Ensure training for AFETS receive proper priority in AF training plans.

11.39.7.10. Redistribute ETS resources between major activities when mission changes dictate such realignment. All redistribution efforts will be coordinated with losing and gaining organizations.

11.39.7.11. Provide government property support to CETS IAW Federal Acquisition Regulation (FAR), Subpart 45.3, Authorizing the Use and Rental of Government Property.

11.39.7.12. Coordinate CETS contract questions with the AFLCMC/Enterprise Acquisition Division (PZIEB) who performs all central acquisition contracting functions for CETS.

11.39.8. Lead ETS Program Office/Decentralized Management Activities Responsibilities (as applicable). Lead ETS Program Office/Decentralized Management Activities will:

11.39.8.1. Develop Task Work Specification and initiate procurement packages for each approved CETS requirement using appropriate Advisory & Assistance Service policies and procedures IAW FAR Subpart 37.2, *Advisory and Assistance Services*.

11.39.8.2. Oversee Contract Officer Representative (COR) activities.

11.39.8.3. Establish procedures to notify subordinate activities of CETS termination.

11.39.8.4. Develop and utilize assessment criteria to effectively manage, administer, and control ETS activities.

11.39.8.5. Conduct annual assessments of ETS Team performance at each field unit.

11.39.8.6. Validate manpower requirements at least every 24 months. Validation should be accomplished with inputs and recommendations from unit leadership and MAJCOM ETS OPR.

11.39.8.7. Realign AFETS manpower as needed when the mission, system, or equipment changes dictate.

11.39.8.8. Coordinate AFETS placement and Standard Core Personnel Documents with manpower and civilian personnel offices.

11.39.8.9. Maintain the knowledge, training abilities and skills of the AFETS workforce.

11.39.8.10. Update/train AFETS on new weapon systems, equipment conversions and major system modifications.

11.39.8.11. Coordinate with weapon system and equipment managers to program AETC Type I Training and other types of training for AFETS on a priority basis for current and new systems.

11.39.8.12. Provide AFETS instructor training.

11.39.8.13. Verify need for CETS personnel security clearances and take action to maintain access at the minimum level required IAW DoDMAN5200.02_AFMAN 16-1405, *Air Force Personnel Security Program*.

11.39.8.14. Include Personnel Reliability Program guidance in the Task Work Specification of contractors whose duties involve nuclear weapons.

11.39.8.15. Consolidate subordinate units' requirements and establish a validation process through the MAJCOM, FOA, or Direct Reporting Unit (DRU) ETS OPR (as applicable).

11.39.8.16. Establish, maintain, and manage MAJCOM ETS TDY and training budget to include planning and execution of funds.

11.39.9. Using Activity will:

11.39.9.1. Employ ETS resources effectively and efficiently to enhance mission capability IAW AF and MAJCOM guidance. **(T-2)**

11.39.9.2. Provide specific direction and guidance on maintenance activities requiring focused AFETS attention and/or technical support. **(T-2)**

11.39.9.3. Ensure AFETS personnel attend and participate in Group, Squadron, and unit maintenance meetings, as required. **(T-2)**

11.39.9.4. Ensure AFETS personnel are providing desired coverage on all shifts, as required, with a focus on shifts where the significant maintenance and repair activities are ongoing. **(T-3)**

11.39.9.5. Provide local access and oversight of Time and Attendance actions for assigned AFETS personnel.

11.39.9.6. AFETS assigned as tenants will be afforded command/base support by the host commensurate with other assigned DoD civilians to include eligibility for local awards, security clearance processing, annual physicals, passport/visa processing. **(T-2)**

11.39.9.7. Fund AFETS to attend conferences, Technical Interchange Meetings (TIM), and deployments as required. **(T-3)** In addition, fund training requirements to ensure AFETS remain current on assigned and emerging systems. **(T-3)**

11.39.9.8. Provide office supplies, special Information Technology (IT) equipment as necessary to support the unit's mission. **(T-3)**

11.39.9.9. Adhere to AF and ETS TDY/Deployment Policies and regulations. **(T-2)** Units are authorized and encouraged to deploy AFETS to support mission requirements worldwide. AFETS employees must be assigned UTC positions in mobility tasked units on the Deployment Manning Document (DMD) as required. **(T-2)** CETS representatives are typically not deployed but may be deployed on specific approval of the MAJCOM OPR if AFETS personnel are not available, subject to contract provisions and funding availability.

11.39.9.10. UDM/deployment functions/processes are the responsibility of assigned unit for AFETS personnel.

11.39.9.11. Adhere to Adverse Action, Appeal and Grievance Procedures. **(T-2)** Refer to DAFI 36-148, *Discipline and Adverse Actions of Civilian Employees*, issue based 36-Series AFIs, and Negotiated Labor Management (Union) Agreement as applicable, before proceeding.

11.39.9.12. Provide support through the local security manager to process AFETS personnel security periodic reviews and updates.

11.39.9.13. Provide AF Certifying Officer Support. **(T-2)** The using activity OPR will serve as or designate AF Certifying Officers for CETS personnel and provide the name, office symbol, signature, and telephone number of the Air Force Certifying Officer to the Administrative Contracting Officer, with a copy to the MAJCOM OPR, no later than 30 days after the CETS assignment or within five workdays of any AF Certifying Officer change. **(T-2)**

11.39.9.13.1. The CETS contract line items assigned to each certifying officer will be clearly identified in this designation.

11.39.9.13.2. The AF Certifying Officer is responsible for certifying the monthly Certificate of Service. It is recommended that the using activity delegate the AF Certifying Officer responsibilities to the Director of Operations of the squadron primarily using the individual CETS employee’s services.

11.39.9.13.2.1. The designated certifying officer will be a commissioned officer. **(T-2)** When it is impractical to designate a commissioned officer or one is not available, requests for approval to appoint a senior NCO or AF civilian (GS-11 or above) as AF Certifying Officer will be submitted by the using activity OPR in writing to the MAJCOM OPR. Each request will be evaluated on an individual basis.

11.39.10. AFETS Responsibilities. AFETS personnel will:

11.39.10.1. Provide field service engineering, technical advice; and assistance to resolve system anomalies and equipment failures. **(T-2)**

11.39.10.2. Develop and teach specific technical training for maintaining and operating unit equipment and assigned weapons systems. **(T-2)**

11.39.10.3. Investigate equipment failures and mishaps and train personnel to help prevent recurrence. **(T-2)**

11.39.10.4. Develop contacts with contractor, depot, and AFLCMC engineers, technicians, and item managers to resolve maintenance problems, design deficiencies, and supply problems. **(T-2)**

11.39.10.5. Develop special test equipment and maintenance procedures to resolve complex system problems. **(T-2)**

11.39.10.6. Perform emergency maintenance (direct assistance) on equipment when temporary skill or manning shortages prevent accomplishment by other assigned personnel. **(T-2)**

11.39.10.7. Advise the ETS OPR on the best utilization and management of CETS. **(T-2)**

11.39.10.8. Document technical activates and provide stakeholders a written account of maintenance activities by the end of each month. **(T-2)**

11.39.10.9. Maintain mobility readiness and accomplish unit required ancillary training. Training must be kept current. **(T-2)**

11.39.11. CETS Contractor Responsibilities. CETS contractor will:

11.39.11.1. Provide ETS through CETS employees to perform the duties described in the Task Work Specification. **(T-2)**

11.39.11.2. Select, supervise, and exercise sole and autonomous control and direction over CETS employees. **(T-2)**

11.39.11.3. Comply with the administrative and security regulations of the using activities. **(T-2)**

11.39.11.4. Provide copies of the Task Work Specification to CETS employees. **(T-2)**

11.39.11.5. Provide CETS security clearance certification to the unit security office. **(T-2)**

11.40. Senior Leader Mission Generation (SLMG) Course.

11.40.1. The Senior Leader Mission Generation (SLMG) Course was developed in 2013 and focused on wing leadership teaming between maintenance, operations, and logistics support to achieve safe and effective mission generation. SLMG objectives are to 1) comprehend the organizational dynamics and responsibilities of operational, maintenance, logistics support, and medical functions for aircraft mission generation; 2) comprehend the necessary integration and teamwork between operational, maintenance, logistics support, and medical functions for aircraft mission generation.

11.40.2. Only Wing CCs/CVs, Operations Group Commander (OG/CCs), MXG/CCs, Mission Support Group (MSG/CCs), MDG/CCs and their equivalent to be stationed at wings with a flying or nuclear mission are required for attendance. **(T-2) Note:** MAJCOM/CV is the waiver authority for attendance. Officers in Space, Cyber, Intel, Air Base Wings or other "non-traditional" wings are not mandated to attend; however, officers desiring to attend from non-

mandated wings can contact their Senior Leader Management office and be added to the course. Registration for SLMG will be accomplished during registration for Pre-Command Training. **Note:** The SLMG course is a unit funded TDY.

11.41. Condition Based Maintenance Plus (CBM+). CBM+ is the application and integration of appropriate processes, technologies, and knowledge-based capabilities to improve the reliability and maintenance effectiveness of USAF systems and components. CBM+ is maintenance performed based on evidence of need provided by RCM analysis and other enabling processes and technologies. The CBM+ Process has four distinct phases, including Component Selection, Enhanced Reliability Centered Maintenance (eRCM), Sensor-Based Algorithms (SBA), and CBM+ Deployment.

11.41.1. PREDICTIVE MAINTENANCE ALERT LOOP (PMAL). A six-step collaborative process that allows stakeholders to receive and act upon the results of eRCM analyses and/or sensor-based algorithms/modules. PMAL transforms eRCM forecasts and sensor-based algorithm alerts into Mx and supply chain actions and is a critical component in the integration of a successful USAF Predictive Mx culture.

11.41.2. GOVERANCE. A CBM+ Team is assigned per weapon system and includes representatives from AFLCMC/RO, the Program Office, the MAJCOM (including direct CBM+ support, WSMs/WSTs and field unit contacts), AFSC/LZRB, and 420 SCMS. For platforms managing many MOBs, a field representative may also serve as an optional member of the CBM+ Team to provide feedback on the execution of Mx. The CBM+ Team oversees the execution of the PMAL. The CBM+ Team meets on an as-needed basis to discuss the most recently published forecast, the locations of affected aircraft, and the availability of supply to make informed, collaborative decisions and recommendations.

11.41.3. PLAN AND SCHEDULE MAINTENANCE. The PS&D Section will be the focal point for CBM+ management, including ordering parts, ensuring that CBM+ parts desired to be changed are scheduled on the weekly operations and maintenance schedule, and that all documentation is complied with IAW **Chapter 14** of this instruction. Refer to the respective MAJCOM guidance on how and when to schedule maintenance for assigned locations. When supply is limited or aircraft are deployed or preparing for mission deployment, the MAJCOM may prioritize Mx requirements or coordination with applicable supporting organizations.

11.41.3.1. **(Added-AFMC)** Access the CBM+ Forecast via the Predictive Analytics and Decision Assistant tool at: **https://cbmplus.cce.af.mil/**. Units must access the CBM+ Forecast on the 15th of each month (or the next duty day if the 15th falls on a weekend or holiday) in order to forecast the maintenance schedule priority. Communicate any inconsistencies to the MAJCOM weapons system team/managers. **(T-2) Note:** Individual CBM+ component threshold hours will change over time.

11.41.3.1.1. **(Added-AFMC)** Units can contact the AFMC/A4M CBM+ manager at: **AFMC.A4M.Workflow@us.af.mil** to obtain information/sponsorship to gain access to the CBM+ Forecast via the Predictive Analytics and Decision Assistant tool.

11.41.3.2. **(Added-AFMC)** Work Unit Codes/Logistics Control Numbers (components) selected for CBM+ are tracked by individual aircraft tail numbers to predict the component's remaining life to failure. The intent of CBM+ is to drive unscheduled

maintenance into scheduled maintenance, thereby optimizing the limited maintenance resources and improving aircraft availability.

11.41.3.3. **(Added-AFMC)** CBM+ Access. If available, maintenance units will utilize the CBM+ Forecast applicable for assigned MDS(s) to view, schedule maintenance, order parts, complete replacement actions, and record the maintenance actions for each aircraft.

11.41.3.4. **(Added-AFMC)** The scheduling priority is based on the CBM+ threshold with the highest priority for aircraft with components over the threshold, followed by the aircraft with components forecasted for removal within the next 30, 60, 90, 120, 150, 180-day intervals.

11.41.3.5. **(Added-AFMC)** Each CBM+ component forecasted for removal should be scheduled in conjunction with the closest scheduled maintenance event where the CBM+ maintenance action can take place. **Example:** Units may determine, through the local scheduling process, that an aircraft with a component forecasted for removal that is within the 30-day to 180-day threshold is already scheduled down and may elect to complete the CBM+ maintenance action ahead of an aircraft that has a component that is over the threshold. This is a local decision and meets the objective of CBM+ by employing opportunistic maintenance to prevent an anticipated part failure.

11.41.3.5.1. **(Added-AFMC)** Consult maintenance tasks SMEs for the typical time duration the maintenance action takes in order to appropriately schedule the task as required.

11.41.3.5.2. **(Added-AFMC)** Parts loaded into the MIS that are CBM+ items will be loaded to fully track component history and life usage. Automated History Events will be enabled for parts loaded with CBM+ Work Unit Codes/Logistics Control Numbers and the Maintenance Type Interval will be established as “H” (for hours) based on the end item’s operating time.

11.41.3.6. **(Added-AFMC)** Units should align the CBM+ forecasted maintenance with other planned/scheduled events and order the part(s) to ensure the DIFM 14-day policy is not violated.

11.41.3.6.1. **(Added-AFMC)** Units are to use project code 063 and urgency justification when ordering the CBM+ WUC/Logistics Control Number part.

11.41.3.6.2. **(Added-AFMC)** Units should not pull parts from the MRSP to fill CBM+ parts orders.

11.41.3.7. **(Added-AFMC)** Units are encouraged to utilize profile job standards for each CBM+ WUC/Logistics Control Number to standardize maintenance entries.

Chapter 12

MAINTAINING COMMERCIAL DERIVATIVE AIRCRAFT (CDA).

12.1. Background Information, Objective and Roles and Responsibilities. The USAF procures CDA for various missions. These aircraft are originally type certificated to Federal Aviation Administration (FAA) regulations/orders and have FAA-approved aircraft maintenance manuals. If the aircraft are civil registered (N number displayed) in lieu of military registration and military tail number, it may carry an FAA standard airworthiness certificate if operated and maintained in full compliance with civil regulations.

12.1.1. Civil registered aircraft owned and operated by the USAF IAW Title 10 U.S. Code are public use aircraft. When these aircraft are engaged in civil aircraft operations, such operations must be conducted IAW FAA and civil. When these aircraft are engaged in public aircraft operations, they are exempt from civil regulations and FAA oversight. Civil registered aircraft owned and operated by the USAF may be declared public use at any time and are then exempt from civil regulations and FAA oversight. If the aircraft are civil registered but do not have a civil airworthiness certificate or are operated by the USAF under a military registration and tail number, the aircraft are for public use and operate entirely under the authority of the USAF military technical airworthiness authority.

12.1.2. The FAA defines a Commercial Derivative Aircraft (CDA) as a commercial aircraft with an FAA type certificate (TC) and produced under an FAA Production Certificate (PC). Air Force Policy Directive (AFPD) 62-6, *USAF Airworthiness*, establishes policies for CDA to ensure that any USAF operated aircraft are airworthy over their entire life cycle and maintain high levels of safety. The directive also instructs the USAF to Meet the Intent (MTI) of applicable FAA regulations to the maximum extent practical.

12.1.3. All USAF-managed aircraft, and associated modifications, must meet the requirements of DAFI 62-601, *Airworthiness*, and AFI 63-101/20-101. **(T-2)** PMs are ultimately responsible for maintaining configuration control and ensuring flight safety of systems within their portfolio. When a military mission is compatible with a certified civil usage, the USAF will utilize FAA-type certified CDA to the maximum extent practicable. **(T-2)** To ensure safety and support, all modifications performed on CDA type certificated components or systems will be FAA certified (example, supplemental type certificate). Modifications to CDA military type certificated components or systems require approval of AF chief engineer, or delegated authority.

12.1.4. For maintenance and operations of CDA the AF will use AF-managed TOs or FAA-approved aircraft and component maintenance manuals and FAA regulations called out in Title 14 Code of Federal Regulation **Parts 43**, **91**, **121,** and **145** as a guide. **(T-0)** FAA Advisory Circulars, Notices to Airmen, and other FAA information sources may also be used to satisfy all requirements of Title 14 Code of Federal Regulations **Parts 43**, **91**, **121,** and **145.**

12.2. AF/A4L will:

12.2.1. Coordinate relevant policies and procedures with SAF/AQ and the FAA.

12.3. The Program Manager (PM) will:

12.3.1. Be responsible for all elements of life cycle management IAW AFI 63-101/20-101. **(T-2)**

12.3.1.1. Evaluate acquisition and sustainment strategies and determine requirements for Contractor Logistics Support (CLS) structure, Organically Managed support structure, and/or hybrid support structure.

12.3.2. When FAA manuals are used, issue technical data for configuration items and inspection requirements that are not approved by the FAA or supplied by the Original Equipment Manufacturer. **(T-2)**

12.3.3. Review evaluations from the Lead Commands concerning Airworthiness Directives (AD), Service Bulletins (SB), Customer Bulletins (CB), All Operator Letters, and Aircraft Service Changes (ASC) and will determine extensions for each, if required. **(T-2)**

12.3.4. For CDA which maintain an FAA Type Certified, ensure that the MAJCOM performs overhauls, rebuilding, major repairs, major alterations, minor repairs, and minor alterations in FAA-authorized repair facilities with appropriate ratings and authorizations or an AF-approved AFSC Military Repair Station depot facility, as directed by the PM. **(T-2)**

12.3.5. Establish a maintenance plan and Service Action Review process with the Lead Command for aircraft originally Type Certified by the FAA. **(T-2)**

12.3.6. Follow AFI 63-101/20-101 for modification requests and approvals. **(T-2)**

12.3.7. Obtain airworthiness approvals IAW DAFI 62-601. **(T-2)**

12.3.8. Coordinate with the FAA Military Certification Office for approval of modifications that affect commercial derivative aircraft configuration IAW USDOT/FAA Order 8110.101, *Type Certification Procedures for Military Commercial Derivative Aircraft*. **(T-0)**

12.3.9. Ensure FAA ADs and SBs are utilized in place of TCTOs, and commercial maintenance manuals are utilized in lieu of AF TOs to the greatest extent possible. **(T-2)**

12.3.10. For units possessing CDA that strictly utilize commercial manuals, may issue original FAA SBs, ADs or other FAA-approved modifications in-place of TCTOs. **(T-2)**

12.3.11. For units possessing CDA that strictly utilize USAF managed TOs, in order to implement ADs, SBs or other FAA-approved modifications, issue TCTOs IAW TO 00-5-1 and TO 00-5-15. Reference one of the following in each TCTO:

12.3.11.1. The AD and/or SB involved.

12.3.11.2. The Supplemental Type Certificate number.

12.3.11.3. Other FAA approval.

12.3.12. Develop a Maintenance Management Program Plan and ensure maintenance planning data is supplied to MAJCOMs, units, and/or contractors in order to operate and maintain airworthiness for the weapon system lifecycle. The program plan meets the intent of the guidance in FAA Advisory Circular 120-16, *Continuous Airworthiness Maintenance Program (CAMP)*.

12.3.12.1. The Maintenance Management Program Plan will outline the requirements to maintain the lifecycle of continued airworthiness of a weapon system in compliance with the intent of Title 14 Code of Federal Regulations (CFR) Parts 43, 65, 121 and 145 throughout its post-production, post-modification DD Form 250, *Material Inspection and Receiving Report* in-service lifecycle. The Maintenance Management Program Plan should include as a minimum:

12.3.12.1.1. Airworthiness Responsibility

12.3.12.1.2. Air Carrier Maintenance Manual

12.3.12.1.3. Maintenance Organization Structure and Function

12.3.12.1.4. Accomplishment and Approval of Maintenance and Alterations

12.3.12.1.5. Maintenance Schedule

12.3.12.1.6. Required Inspection Items

12.3.12.1.7. Maintenance Record Keeping

12.3.12.1.8. Contract Maintenance

12.3.12.1.9. Personnel Training

12.3.12.1.10. Maintenance Program Development

12.3.12.1.11. Configuration Management and Tracking

12.3.12.1.12. Engineering Services

12.3.12.1.13. Continuing Analysis and Surveillance System (CASS)

12.3.12.1.14. Depot Support Plan (if required)

12.4. Lead Commands will:

12.4.1. Ensure any new or modified configurations or maintenance conditions are coordinated with, and approved by, the designated Lead Command IAW AFPD 10-9 and the PM or equivalent responsible for the reliability, maintainability and availability of the systems and end-items prior to implementation. **(T-2)**

12.4.2. Assist ALC in determining additional inspection and component time-change requirements, intervals, documentation and publication update requirements. **(T-2)**

12.4.3. Review evaluations from their field units on ADs, SBs, CBs, All Operator Letters, or ASCs and make recommendations to the aircraft's PM. **(T-2)**

12.4.4. Ensure depot and contractor maintenance providers are furnished with lead command maintenance program and they meet AF approved FAA equivalent requirements or are a FAA approved repair station, as applicable. **(T-2)**

12.5. Units will:

12.5.1. Participate in the Service Action review process established by the PM. **(T-2)**

12.5.2. Assist Lead Commands and the PM to determine additional inspection intervals and requirements. **(T-2)**

12.5.3. Comply with FAA ADs as directed by the Chief Engineer or delegated authority. **(T-2)**

12.6. Maintenance Personnel Requirements. For AF-managed aircraft that maintain an FAA TC and airworthiness, maintenance personnel will meet the Lead Command and PM established requirements and procedures to maintain airworthiness. **(T-2)** AF maintenance technicians performing organic depot maintenance follow Air Force Sustainment Center Instruction (AFSCI) 62-603, Military Repair Station Program, Air Force Sustainment Center Manual (AFSCMAN) 62-602, Military Repair Station Manual.

12.6.1. For CDA that do and do not maintain a civil airworthiness certificate, maintenance is not required to be completed by an FAA-certificated mechanic. However, a Maintenance Management Program Plan detailing the maintenance personnel requirements will be established between the PM and the Lead Command. **(T-2)**

12.6.1.1. The plan will address as a minimum the training requirements, the level of effort allowed (such as, specific maintenance tasks as identified in the Original Equipment Manufacturer maintenance manuals), and tasks that will be performed by FAA-certified mechanics, repair stations, AFSC Military Repair Stations, or the Original Equipment Manufacturer. **(T-2)**

12.6.1.1.1. CDA Training courseware, as applicable, for unique CDA platform requirements, such as but not limited to; Extended Operations (ETOPS and Polar Operations), Materiel Management, Continuing Analysis Surveillance System (CASS), Maintenance Provider Program, Required Inspection Item (RII), Major/Minor Repair) will be addressed in lead command program training development and remain standardized across the entire weapon system. Individual Main Operating Base (MOB) units and depots may augment training as required.

12.6.1.2. The maintenance plan will be approved by the PM with coordination by the FAA Military Certification Office or Military Repair Station/Flight Standards Management Office as appropriate. **(T-2)**

12.7. Deviations/Changes to Inspection Requirements, Time Change Intervals, and Component/Aircraft Overhaul. Commercial derivative aircraft inspection requirements, time change, component and aircraft overhaul intervals are established and controlled by the Original Equipment Manufacturer and approved by the FAA. When deviation from the Original Equipment Manufacturer established maintenance standards/configuration is needed to meet AF mission requirements, units will send proposed changes to the PM through MAJCOM and for evaluation. **(T-2)**

12.8. Air Force Modifications to CDA and Components. AF modifications to CDA and components are developed following procedures outlined in AFI 63-101/20-101. All AF modification requests require coordination with the Chief Engineer or delegated authority who will provide assistance in determining applicable requirements, forms and coordination necessary to correctly disposition aircraft and component modification requests. **(T-2)**

12.9. Certification Basis for CDA. Elements of the certification basis for any CDA which are not met via FAA certification are satisfied by compliance with approved military airworthiness requirements derived from MIL-HDBK-516C, *DoD Handbook, Airworthiness Certification Criteria*. CDA whose primary mission is the transport of passengers are FAA Type Certified; FAA certification of these CDA passenger carrying aircraft is maintained for the life of the air system.

Chapter 13

CENTRALIZED REPAIR FACILITIES (CRF).

13.1. Introduction. CRFs consolidate off-equipment intermediate-level, and in some instances, depot-level tasks for commodities such as aircraft engines, electronic warfare pods, avionics line replaceable units, wheel and tire assemblies, and other aircraft components. CRFs focus on efficiently providing maintenance, repair, and/or overhaul capabilities, support RN efficiencies and will be fully integrated into the AF Supply Chain. CRFs are considered part of the repair network and exist to ensure responsiveness to MGN requirements to sustain operations both at home station and/or when deployed. Management and control procedures are outlined in AFI 20-117.

13.2. Organization. CRFs will be established within existing maintenance organizations (EMS, CMS, MXS), minimizing requirements for overhead and support. **(T-2)** Production oversight and monitoring of repair operations is the responsibility of the owning maintenance organization in which the CRF is established. Commanders with CRFs will manage the personnel, facilities, and processes for the CRF following the policies and procedures in this DAFI and DAFI 20-117. **(T-2)**

13.3. CRF Production Requirements. Maintenance Squadron (EMS, CMS, MXS) Director of Operations/MX SUPT will:

13.3.1. Ensure the Node Manager (NM) executes enterprise production duties as outlined in AFI 20-117. **(T-2)**

13.3.2. Ensure the NM identifies and up-channels repair constraints that affect CRF repair/RN CAP2 to the MFM and RNM as prescribed in DAFI 20-117.

13.3.3. Ensure the NM utilizes information management systems and participates in RNM collaboration calls to provide timely status reports, resolve repair constraints, and receive revised repair requirements/RN changes. **(T-2)**

13.3.4. Identify systemic distribution, transportation, and supply difficulties and coordinate with base LRS leadership and/or up-channel concerns to the RNM and MFM for resolution. **(T-2)**

13.3.5. Follow established procedures to ensure the rapid movement of retrograde and sustainment assets to support enterprise requirements.

13.4. MGN Support. Units supported by CRFs will maintain the level of intermediate-level repair capability necessary to sustain MGN operations. MAJCOMs must identify intermediate-level tasks and resources required to perform MGN maintenance tasks for assigned weapon systems (for example, repair of XF3 assets, hose/tube testing, functional checks, NRTS screening).

13.4.1. Rotable Pools. Customer Wait Time and transportation constraints may drive the establishment of a Centralized Rotable Pool for Class VII end items such as engines and pods to meet established weapons system availability goals. Use of a Centralized Rotable Pool can enhance mission capability by placing serviceable assets closer to the user when the repair capability is off installation. Centralized Rotable Pool size, compared to support unit spare levels, will be determined during deliberate planning between the appropriate RNM and MAJCOMs.

13.4.2. Cannibalization at supported units. When commodity LRU local retail stocks fall below mission requirements, retention of CRF-repaired end items as "CANN assets" may be necessary. However, this will be by exception and must be approved by the appropriate RNM in coordination with the supporting MAJCOM CRF Manager. **(T-2)** CRFs will document their cannibalization process and notify supporting unit of approval to retain CANN assets. **(T-2)**

13.4.3. Provide CRF node performance, CAP2, and commodity status reports and metrics as defined in DAFI 20-117. **(T-2)**

13.4.4. Ensure the NM utilizes information management systems to provide timely status reports and receive workload requirements/ changes for commodity group repairs supported by the CRF IAW with AFI 20-117. **(T-2)**

13.5. CRF Enterprise Information Management. Managers require accurate, timely, and enterprise repair data to make CRF command and control and production decisions. To facilitate this requirement, NMs will utilize systems, processes, and business rules prescribed by AFI 20-117 to provide repair data and ensure enterprise visibility. **(T-2)**

13.6. Documentation. The CRF and supported units will maintain all required status, inventory, and historical record documentation on CRF-repaired assets, IAW TO 00-20-1 and DAFI 21-103.

13.7. Metrics. CRFs will report performance against metrics IAW DAFI 20-117.

Chapter 14

MAINTENANCE PLANS, SCHEDULING AND DOCUMENTATION (PS&D).

14.1. Responsibilities:

14.1.1. and 14.1.1.1. DELETED.

14.1.2. MAJCOMs will:

14.1.2.1. Supplement this instruction to establish minimum requirements for the following:

14.1.2.1.1. TCTO folders and monthly/weekly utilization and maintenance schedules.

14.1.2.1.1. **(AFMC)** Refer to paragraph 14.1.6.5.1. (T-2)

14.1.2.1.2. Publish MAJCOM procedures for verification of configuration items.

14.1.2.1.2. **(AFMC)** Refer to paragraph 14.1.6.5. (T-2)

14.1.2.1.3. Determine whether to ship removed engines to depot or induct into CRF repair.

14.1.2.1.3. **(AFMC)** Refer to paragraph 14.4. (T-2)

14.1.2.1.4. Determine routing and approval for AF Form 2407.

14.1.2.1.4. **(AFMC)** Refer to paragraph 14.5.6.3.8. (T-2)

14.1.3. PS&D will:

14.1.3. **(AFMC)** Maintenance schedulers are centralized (physically) in the MO PS&D section.

14.1.3.1. Maintain historical documents and maintenance data essential for the development of wing plans, schedules, and analysis of historical maintenance events. **(T-2)**

14.1.3.2. Maintain historical maintenance data within the MIS. **(T-2)**

14.1.3.3. Develop wing maintenance plans using MIS aircraft/system historical data input by all maintenance personnel. **(T-2)**

14.1.4. The PS&D Section NCOIC/Chief (or equivalent) will:

14.1.4.1. DELETED.

14.1.4.2. Establish and coordinate plans for rotating 2R1XX personnel through various duty positions to increase field knowledge and experience every 24 months, not to exceed 36 months. **(T-2)**

14.1.4.2.1. This rotation plan applies to TSgts and below as well as 3- or 5-skill level personnel of any rank. **(T-3)**

14.1.4.3. Evaluate quarterly the performance of workcenters performing scheduling functions to include TCTO, SI, and Job Standard Master Listing (JML) management (such as, AGE, Armament, Egress, Fuels, MXO, PS&D). **(T-2)**

14.1.4.3.1. During the visit, ensure historical documents are properly maintained and review and discuss the 2R1X1 training and rotation plan with each section NCOIC that have 2R1s assigned. **(T-2)**

14.1.4.3.2. Provide formal written reports of deficiencies found during the visits to the MXO DO/SUPT and applicable section NCOIC. **(T-2)**

14.1.4.3.2.1. Deficiencies will not be closed until validated by the MXO DO/SUPT. **(T-2)**

14.1.4.4. - 14.1.4.4.2. DELETED.

14.1.4.5. Provide SME on all maintenance scheduling issues and equipment historical document AFTO Form 95 management to QA during inspection/evaluations. **(T-2)**

14.1.4.6. Designate the MSM administrator from within PS&D (for units utilizing IMDS only). **(T-2)**

14.1.5. The Wing AVDO will:

14.1.5.1. Complete AVDO duties IAW DAFI 21-103 and maintain the inventory and utilization portion of the MIS Inventory, status, and utilization subsystem. **(T-2)**

14.1.5.1. **(AFMC)** Coordinate all assignment/possession planning and changes through the AFMC AVDO, (AFMC/A4M) and the AFMC Aerospace Vehicle Inventory and Fleet Manager (AFMC/A3F) **(T-2)**

14.1.5.2. Maintain a PDM schedule by tail/serial number for all assigned aircraft and equipment in support of AFMC and Lead Command plans and requirements. **(T-2)**

14.1.6. Unit Dedicated Scheduler will:

14.1.6.1. Provide dedicated support to AMXS/AMU. **(T-2)**

14.1.6.2. Attend and actively participate in daily, weekly, and monthly scheduling, and quarterly and yearly planning programs and meetings. **(T-2)**

14.1.6.2.1. Inform Unit supervision of maintenance capabilities or limiting factors that could affect maintenance production. **(T-2)**

14.1.6.3. Coordinate with Unit supervision and Operational Squadron (OS) operations schedulers when scheduling Unit aircraft to meet flying requirements. **(T-2)**

14.1.6.3. **(AFMC)** In addition to the coordination with internal agencies including AMXS and OS operations schedulers, coordinate with applicable outside agencies directly involved in the aircraft flying requirements (i.e. engineers/test customers, other government agencies, sympathy organizations), as needed with the execution of flying scheduling processes and procedures.

14.1.6.4. Provide a listing of JCNs for following week's scheduled maintenance. **(T-2)**

14.1.6.4.1. This list will be used to track Maintenance Scheduling Effectiveness (MSE). **(T-2)**

14.1.6.4.2. PS&D will determine causes of missed maintenance for reporting MSE. **(T-2)**

14.1.6.5. Manage TCTOs, TCIs, and Sis (including installed engine inspections) for aircraft assigned to their appointed AMXS/AMU. **(T-2)**

14.1.6.5. **(AFMC)** Individual work centers accomplishing time change items (TCI) are responsible for changing configuration information in MIS. The performing work center supervisor and PS&D must conduct supervisory reviews of configuration change, TCTO, SI and TCI events using MIS on-line capabilities. Unless otherwise specified in local procedures, schedulers must process IMDS screen 128 for all removal, installation, TCI, SI and TCTO compliance updates for any items installed on the aircraft. **(T-2)** EM must process IMDS screen 128 for engines and engine components. In addition:

14.1.6.5.1. **(Added-AFMC)** Attend TCTO planning meetings when aircraft or equipment are affected and initiate/maintain folders for applicable TCTOs. TCTO folders will be standardized IAW TO 00-5-15, *Air Force Time Compliance Technical Order (TCTO)*. Process as a minimum, and may be augmented with additional information provided in MOF PS&D guidance. AMU PS&D will attend the monthly TCTO review meeting hosted by MOF PS&D. **(T-2)**

14.1.6.5.2. **(Added-AFMC)** Manually update MIS products as changes occur IAW procedures established by MOF PS&D. **(T-2)**

14.1.6.5.3. **(Added-AFMC)** Correct TCI and SI program discrepancies identified by the MOF PS&D time change monitor as soon as possible. Provide MOF PS&D with MIS products or other documentation of the corrections. **(T-2)**

14.1.6.6. Generate AFTO Form 103, *Aircraft/Missile Condition Data*, to record certified maintenance needs for aircraft requiring depot maintenance IAW TO 00-25-4, *Depot Maintenance of Aerospace Vehicles and Training Equipment*, coordinate it with PS&D, QA, and AMXS maintenance supervision. **(T-2)**

14.2. Data Documentation.

14.2.1. Maintenance Historical Documentation.

14.2.1.1. Maintenance historical documentation will be accomplished IAW TO 00-20-1 which outlines the requirements to capture and record the significant maintenance actions on aerospace vehicles and equipment. **(T-2)**

14.2.1.2. Historical documentation will be entered and tracked in the authorized MDS MIS. **(T-2)** When the MIS is not available, historical documentation will be documented and tracked on the AFTO Form 95, or equivalent. **(T-2)**

14.2.1.3. MAJCOMs should develop supplements to this instruction to identify aerospace vehicle and support equipment historical file content and retention requirements needed beyond the minimum requirements outlined in this instruction and TO 00-20-1.

14.2.2. Aircraft jacket files. Units will develop and maintain a standardized master aircraft jacket file for use throughout the wing following the requirements listed in this instruction, TO 00-20-1 and AFI 33-322.

14.2.2.1. MAJCOMs will standardize MDS-specific requirements not captured in this instruction in supplements and addendums to this instruction.

14.2.2.1. **(AFMC)** Local procedures are authorized for compliance of any established AF policy unless specified in supplemental guidance for this paragraph.

14.2.2.2. Aircraft jacket files will be maintained in PS&D and standardized IAW the master aircraft historical file developed by the PS&D NCOIC. **(T-2)**

14.2.2.2.1. Off-equipment maintenance documents may be decentralized to sections maintaining installed-on equipment assets (examples include fuel cell records at fuel systems section, landing gear strut records at hydraulics section).

14.2.2.2.1.1. Decentralized records are filed by and are the responsibility of the owning work center.

14.2.2.2.2. PS&D will list all historical records, in their file plan or office of record. **(T-2)**

14.2.2.2.2.1. The DD Form 2861, *Cross-Reference*, will be used to cross-reference documents decentralized from PS&D to other sections and will be filed to cross reference AFTO Form 95 records that are maintained in the MIS. **(T-2)**

14.2.2.2.3. Wing-assigned aircraft jacket files may be maintained electronically; however, they must mirror the standardized master aircraft jacket file in organization and appearance. **(T-2) Note:** Slight variations in composition are allowed between different MDS weapons systems located within the same wing.

14.2.2.3. MXG/CC may identify additional local items for inclusion in aircraft jacket files. Aircraft jacket files as a minimum will include:

14.2.2.3.1. Packages for one complete inspection cycle. **(T-2)** Units may download paperless inspections to automated storage media from MIS for filing in aircraft jacket files.

14.2.2.3.2. Last FCF documentation (such as, FCF certification letter/FCF checklist). **(T-2)**

14.2.2.3.3. Last depot package. **(T-2)**

14.2.2.3.4. Transfer packages. **(T-2)**

14.2.2.3.5. Applicable weapon system -6 TO AFTO Form 95s. **(T-2)**

14.2.2.3.6. W&B records. **(T-2)**

14.2.2.3.7. Engine Records. **(T-2)**

14.2.2.3.8. Document review records/checklists. **(T-2)**

14.2.2.3.9. NDI records. **(T-2)**

14.2.2.3.10. AF Form 2411, Inspection Document (or equivalent). **(T-2)**

14.2.2.3.11. Annual aircraft jacket file review checklist. **(T-2)**

14.2.2.3.12. Authorized TO variances. **(T-2)**

14.2.2.3.13. Requests for assistance meeting the requirements for retention as historical records IAW TO 00-25-107, or equivalent/like MDS specific requirements for retention of documents as historical records. **(T-2) Note:** Contact the Lead

Command as identified in AFPD 10-9 for guidance for meeting retention as historical records requirements outside the scope of TO 00-25-107.

14.2.2.3.14. Pulled AFTO Form 781-series aircraft forms. **(T-2)**

14.2.2.3.14.1. Pulled paper forms retained as part of the jacket file will be destroyed after 3 months if they do not contain historical information IAW AFRIMS. **(T-2)**

14.2.2.3.14.2. Fusing fully automated forms will maintain the last 7 copies of the pulled aircraft forms and destroy the earliest record when the 8th report is received IAW AFRIMS. **(T-2)**

14.2.2.3.14.3. Units not required to use a MIS will use aircraft forms and maintain the current and the last 3 months' worth of pulled aircraft forms. **(T-2)**

14.2.2.3.14.4. Pulled 781 forms will be filed in order by sets identified by the "From and To" date at the top of each 781-series form (see TO 00-20-1). **(T-2)**

14.2.2.3.14.5. Sets of forms may or may not include an AFTO Form 781J, *Aerospace Vehicle – Engine Flight Document* and AFTO Form 781K, *Aerospace Vehicle Inspection, Engine Data, Calendar Inspection and Delayed Discrepancy Document*. AFTO Forms 781J and K will be included in the set of forms they were pulled with and retained for the same period of time. **(T-2)**

14.2.2.3.14.6. When PS&D discovers the AFTO Form 781-series missing during a jacket file inspection, a missing-forms letter will be sent to the appropriate Maintenance Supervision of the maintenance unit responsible for pulling the forms with a 5 duty-day suspense. **(T-2)**

14.2.2.3.14.6.1. If a response is not returned within 5 duty days, notify the applicable maintenance unit supervision. **(T-2)**

14.2.2.3.14.6.2. If the forms cannot be located, file the missing forms letter, endorsed by the Maintenance Supervision in place of the missing forms. **(T-2)** See TO 00-20-1 for missing form procedures and AFI 33-322 for records management and disposition instructions.

14.2.2.3.15. **(Added-AFMC)** Temporary modification packages.

14.2.2.4. Annual jacket file review. Review aircraft jacket files annually using a locally-developed PS&D checklist. **(T-2)**

14.2.2.4.1. The last completed checklist will be kept on file in each aircraft jacket file. **(T-2)**

14.2.3. Aircraft Document Reviews (ADR). ADRs validate and correct any errors on airframe and engine operating times and cycles, TCTO documentation, TCI component operating times, time remaining to the next inspection, backordered supply document numbers and open deferred discrepancies. The aircraft AFTO Form 781-series for possessed aircraft are reviewed by aircraft crew chiefs, flightline maintenance functions, PS&D, Engine Management (EM) and LRS personnel to ensure the accuracy and validity of entries.

14.2.3.1. MAJCOMs will standardize the MIS/on-line products used to perform ADR on like-MDS weapons systems.

14.2.3.1. **(AFMC)** Test units can standardize MIS/on-line products as required for respective test aircraft management.

14.2.3.1.1. Units using MDS-specific laptop forms (for example Integrated Maintenance Information System (IMIS), Autonomic Logistics Information System (ALIS)) must develop procedures to ensure intent of ADRs is implemented. **(T-2)**

14.2.3.2. An ADR will be accomplished at least every 60 days for units using the fully automated AFTO Form 781-series (AFTO Form 781A, AFTO Form 781J, *Aerospace Vehicle – Engine Flight Document*, AFTO Form 781K, *Aerospace Vehicle Inspection, Engine Data, Calendar Inspection, and Delayed Discrepancy Document*. **(T-2)**

14.2.3.2.1. Units without access to a MIS and authorized to use manual AFTO Form 781-series, must accomplish an ADR at least every 30 days. **(T-2)**

14.2.3.2.1. **(AFMC)** Modular engine flying hours and manual cycles are verified with the EM section during document reviews. **(T-2)**

14.2.3.2.1.1. **(Added-AFMC)** If the aircraft has been away from home station and it has been more than 30 days since the last document review, check the documents within 7 days after return. **(T-2)**

14.2.3.2.2. ADRs will also be accomplished when an aircraft is transferred, before and after scheduled inspections (PH or ISO), before and after storage and after fatigue tests. **(T-2)**

14.2.3.2.3. For CANN aircraft, conduct ADRs at least every 30 days. **(T-2)**

14.2.3.2.3. **(AFMC)** PS&D will conduct document reviews and validations at least every 14 days on CANN aircraft. The documents are also reviewed before and upon completion of other major programs such as Programmed Depot Maintenance (PDM), analytical condition inspections, and special tests. The AFTO Forms 781F, *Aerospace Vehicle Flight Report and Maintenance Document*, will also be reviewed for accuracy. **(T-2)**

14.2.3.3. Units will develop and publish an ADR checklist for use by home station and deployed units. **(T-2)**

14.2.3.3.1. This checklist will identify who initiates the ADR, reviewing agencies (to include the OAP lab), AFTO Form 781-series entry requirements, agency responsible for completing the AFTO Form 781-series/MIS entry, and outline any configuration verification requirements. **(T-2)**

14.2.3.4. ADR Procedures.

14.2.3.4. **(AFMC)** The AMXS will develop local procedures to review aircraft forms and MIS automated products as part of document reviews and/or validations. This process will include a review by the aircraft section/flight commander or NCOIC, then the AMU PS&D. This process will allow each level of supervision to review and maintain continuity of the process. **(T-2)**

14.2.3.4.1. PS&D will create a JST for ADRs on a red dash symbol and ensure it is loaded against all assigned aircraft. **(T-3)**

14.2.3.4.2. ADRs will be scheduled and added to the appropriate maintenance plan. **(T-2)** An ADR is a scheduled maintenance action and will be included in MSE computations. **(T-2)**

14.2.3.4.3. PS&D and EM will validate applicable inspection, TCI, TCTO data for correct due dates/time or expiration dates, airframe, and engine operating times (or flight times if applicable) and appropriate symbol entry IAW TO 00-20-1. **(T-2)**

14.2.3.4.4. Units will coordinate with /LRS to run a tail number inquiry to validate backorders and correct any discrepancies discovered. **(T-2)**

14.2.3.4.5. Maintenance personnel will correct all discrepancies discovered during the ADR, prior to signing off the ADR JCN. **(T-2)**

14.2.3.4.5.1. If an ADR discrepancy cannot be corrected immediately, document the ADR discrepancy in the AFTO Form 781A with a JCN and applicable symbol and retain it in the AFTO Form 781-series forms until corrected and signed off. **(T-2)** Once all the uncorrected discrepancies are documented in the AFTO Form 781- series the ADR can be signed off as complete.

14.2.4. Pre-Dock Meetings. PS&D personnel will:

14.2.4.1. Review planned aircraft inspection schedules and initiate an AF Form 2410, *Inspection/TCTO Planning Checklist*, or locally developed product for each aircraft prior to the pre-inspection meeting. **(T-2)**

14.2.4.1.1. MAJCOMs may determine if the pre/post dock requirement for inspections with less than a 200-hourly or 200-calendar day cycle is required. If it is determined that a pre/post dock meeting is not required, initiation of an AF Form 2410 is not necessary.

14.2.4.2. Host meetings and notify the appropriate Maintenance Supervision and flight supervisors of any recurring problems with attendance. **(T-2)** Prior to the pre-dock meeting, PS&D will:

14.2.4.2.1. Determine pre-dock meeting attendees. **(T-2)**

14.2.4.2.1.1. The following personnel will attend the meeting as a minimum: PS&D, Pro Super, Inspection Dock NCOIC, aircraft crew chief, DMS/Supply representative, and EM representative. **(T-2)**

14.2.4.2.1.2. Include other agencies as required for performance of the work package.

14.2.4.2.2. Review and list all known aircraft and equipment TCTOs, TCIs, SIs, and other major requirements to be accomplished during the inspection on the AF Form 2410, or locally developed product. **(T-2)**

14.2.4.2.3. Identify requirements for kits or parts. **(T-2)**

14.2.4.2.4. List all Delayed Discrepancies to be accomplished during the inspection on the AF Form 2410 keeping the original JCN. **(T-2)**

14.2.4.2.5. Incorporate all requirements against the aircraft into a work package. **(T-2)**

14.2.4.2.6. List specialist tasks required in addition to normal inspection needs. **(T-2)**

14.2.4.2.7. Develop a list of items identified as out-of-configuration for verification/correction during the inspection. **(T-2)**

14.2.4.2.7.1. For non-configuration tracked aircraft, compile a list of missing serially controlled items and coordinate/forward them to Inspection Dock NCOIC for verification. **(T-2)**

14.2.4.3. At the pre-dock meeting, PS&D will brief representatives of the inspection schedule and scope, including TCTOs, TCIs, SIs, DDs, and special requirements to be accomplished. **(T-2)**

14.2.4.3.1. Agency representatives will inform PS&D of limiting factors that might affect the schedule. **(T-2)**

14.2.4.3.2. PS&D will discuss aircraft configuration during all aircraft pre-dock meetings. **(T-2)**

14.2.4.3.3. Wings will use the AF Form 2410, or locally developed product to record additional information discussed during the pre-dock meeting. **(T-2)**

14.2.4.3.3.1. Maintain the original AF Form 2410, or locally developed product on file in the aircraft jacket file for use as a guide when conducting the post-dock meeting. **(T-2)**

14.2.4.3.3.2. Provide a copy to the Inspection Dock NCOIC or equivalent for use during the post-dock meeting. **(T-2)**

14.2.4.3.4. PS&D will provide a copy of the applicable "out of configuration" MIS products (such as, IMDS screen 810 and 990; FMxC2, screen 8110; serial number checklists) to Inspection Dock NCOIC in pre-dock package for verification/correction. **(T-2)**

14.2.4.3.4.1. The responsible work center will correct verified erroneous data and "out of configurations" in the MIS prior to post-dock. **(T-2)**

14.2.4.3.5. As a minimum, the following will also be discussed at the pre-dock meeting:

14.2.4.3.5.1. The type and number (if applicable) of the inspection to be performed. **(T-2)**

14.2.4.3.5.2. Validation of current aircraft and engine operating times. **(T-2)**

14.2.4.3.5.3. Parts in the TNB that require aircraft installation. **(T-2)**

14.2.4.3.5.4. Any known post inspection fuel cell work required. **(T-2)**

14.2.4.3.5.5. Date the aircraft is to be ready for the flightline to accept back. **(T-2)**

14.2.4.3.5.6. All known engines requiring replacement. **(T-2)**

14.2.4.3.5.7. Review of the aircraft forms open discrepancies including Delayed Discrepancies and develop a joint plan to work as many discrepancies as

feasible/applicable. **(T-2)**

14.2.4.3.5.8. Any inspections that will require maintenance personnel to stop work (such as, NDI shop requirements) and when the maintenance dock needs to be clear of personnel to perform the inspections. **(T-2)**

14.2.4.3.5.9. All meeting attendees will sign the AF Form 2410. **(T-2)**

14.2.5. Post-Dock Meetings. Units will hold a post-dock meeting as soon as possible after the inspection but no later than before the FCF or first flight. **(T-2)** PS&D will:

14.2.5.1. Lead a post-dock meeting for all inspections that required a pre-dock meeting. **(T-2)** As a minimum, discuss and validate the following information at the post-dock meeting:

14.2.5.1.1. PS&D, Pro Super, Inspection Dock NCOIC, Aircraft Section representative/crew chief and other locally determined attendees will discuss open discrepancies, review any significant inspection events, and identify any problems that may adversely affect future scheduling. **(T-2)**

14.2.5.1.2. The Inspection Dock NCOIC will provide the completed inspection work package to PS&D for filing until it is replaced by the next similar inspection work package. **(T-2)** For example, an HPO1 will be replaced by the next HPO1 and the HPO2 will be replaced with the next HPO2.

14.2.5.1.3. The Inspection Dock NCOIC will return the completed serial number verification sheet to the PS&D representative. **(T-2**

14.2.5.1.4. The Inspection Dock NCOIC or designated representative and the aircraft crew chief or equivalent will perform an aircraft documents review. **(T-2)**

14.2.5.1.5. PS&D personnel will validate TCTOs, TCIs, and SIs scheduled during the inspection were completed and signed off in the MIS prior to the post dock meeting. **(T-2)**

14.2.4.1.5.1. DELETED.

14.2.5.1.5.2. Validate that any TCTO/TCI/SI not complied with will not ground the aircraft before releasing the aircraft back to flightline maintenance personnel. **(T-2)**

14.2.5.1.6. Verify all parts placed on order during the inspection but not received have valid document numbers.

14.2.5.1.7. If maintenance actions previously identified for completion at the pre-dock meeting were not accomplished, establish agreements as to how these inspection requirements will be completed and documented on the AF Form 2410 or locally developed product with the reason why it was not performed. **(T-2)**

14.2.5.1.7.1. The Inspection Dock NCOIC and flightline maintenance supervisor (Pro Super or above) agree that all inspection requirements are completed and the flightline supervisor agrees to accept or “buy back” the aircraft. **(T-2)**

14.2.5.1.8. PS&D will file the completed AF Form 2410, or locally developed product, and completed/verified copies of the output products in the aircraft jacket file (PS&D

retains completed package until the next scheduled PH/ISO inspection for that aircraft). **(T-2)** Electronic versions may be saved to digital media.

14.2.6. MIS (FMxC2/IMDS) extended downtime (more than 48 hours).

14.2.6.1. If the MIS is not available for more than 48 hours, maintenance organizations will use the most current data contained in MSM for IMDS units and "Global Reach" system products for FMxC2 units. The MSM database will be refreshed with new MIS products daily. **(T-2)**

14.2.6.1.1. MSM usage may continue in a digital format as long as updates can be made and retained.

14.2.6.2. Units will develop guidance to establish procedures to accurately manage Time Change Items, Special Inspections, and Time Compliance Technical Orders during Maintenance Information System (MIS) extended downtime, outage, or intermittent operability. These procedures are to ensure health of fleet and aircraft safety of flight. **(T-2)**

14.2.6.2.1. For PS&D and workcenters performing maintenance scheduling functions local guidance will address procedures to validate operating times (i.e., hours, cycles, starts, landings, rounds, due dates...etc.) affecting the accurate management of Time Change Items, Special Inspections, and Time Compliance Technical Orders weekly during extended or intermittent MIS outage or faulty operability. **(T-2)** Document completion on AF Form 2411, *Inspection Documentation*. **(T-2)**

14.2.6.2.2. MSM, Global Reach system products, or local tracking products may be used until the MIS is restored and backlog data has been loaded and verified in the system. If debriefed sorties are unable to be loaded into the MIS, the debrief section will provide a cumulative breakdown of sorties and hours completed daily to PS&D and MMA to ensure operating times reflect accordingly on locally tracked products and use to forecast scheduled maintenance actions. **(T-2)**

14.2.6.2.3. Munitions, Engine Management, AGE, and workcenters performing scheduling functions will provide operating times (i.e., hours, cycles, starts, rounds, due dates...etc.) as required by PS&D. **(T-2)**

14.2.6.3. If data cannot be retained by MSM or Global Reach, the use of AFTO Form 349, *Maintenance Data Collection Record*, or local developed form, will be initiated for use in data collection/completion. **(T-2)**

14.2.6.3.1. The most current paper or electronic version of MIS products will be used once AFTO Form 349 or electronic equivalent usage is initiated. **(T-2)**

14.2.6.3.2. The AFTO Form 349 or electronic equivalent, will be used to update applicable MIS products once brought back online. **(T-2)**

14.2.6.3.3. The AFTO Form 349 or electronic equivalent, will be maintained until the data listed on it has been verified as captured/loaded in the MIS. **(T-2)**

14.2.6.3.4. After all changes have been verified in the MIS, destroy the AFTO Form 349 or electronic equivalent.

14.2.6.4. If an aircraft is temporarily moved to an operating location away from the unit of assignment and connectivity to the MIS is unavailable, units will send only those documents necessary to ensure safety of flight and current aircraft status. **(T-2)**

14.2.7. Aerospace Vehicle and Equipment Mishap Response Procedures:

14.2.7.1. PS&D will coordinate with MMA or equivalent to ensure MIS lock out procedures to prevent further manipulation of data concerning the aerospace vehicle and/or equipment used during maintenance prior to the mishap event are completed IAW **Chapter 5**. **(T-2)**

14.2.7.2. At a minimum, produce, consolidate, and impound the following products: aircraft jacket file, aircraft AFTO Form 95s, TCTO history, debriefing records, pulled AFTO Form 781-series forms, SI/TCI data, maintenance history, automated records check. **(T-2)** Include any additional significant historical data, and other decentralized records. **(T-2)**

14.2.7.3. EM will download and impound engine records from the applicable MIS and CEMS. **(T-2)**

14.2.8. **(Added-AFMC)** Status Reporting. Accomplish status reporting IAW DAFI 21-103. Aircraft transfer Possession Purpose Code (PPC) "BT" may only be used within the following guidelines. These guidelines also apply for permanent aircraft transfers via PDM/modification or contract repair facility. **Note:** If the gaining organization utilizes the aircraft prior to the transfer inspection, use of "BT" status will not be authorized. **(T-2)**

14.2.8.1. **(Added-AFMC)** Intra-command (within AFMC) transferring aircraft may be possessed in this code no longer than 2 workdays for the losing organization (to prepare the aircraft) and no longer than 2 workdays for the gaining organization (to accept the aircraft). This code allows the organization time to prepare and complete pre-transfer inspections and actions in a non-possessed status. HQ AFMC/A4M must approve any extension. **(T-2)**

14.2.8.2. **(Added-AFMC)** Inter-command (into or out of AFMC) transferring aircraft may be possessed in this code no longer than 2 workdays for both accepting and preparing the aircraft. This code allows the organization time to prepare and complete pre-transfer inspections and actions in a non-possessed status. Any extension must be approved by HQ AFMC/A4M. **(T-2)**

14.2.8.3. **(Added-AFMC)** Aircraft going to programmed depot maintenance (PDM)/modification may be possessed in this code for no longer than 2 workdays to prepare the aircraft. Aircraft returning from PDM/modification, contract field team or depot field team maintenance units may use this PPC no longer than 2 workdays after return of the aircraft. PPC "BT" is not authorized for the sole purpose of accomplishing scheduled maintenance such as TCTO, time change items and special inspections before or after PDM/modification input. HQ AFMC/A4M must approve any extension. **(T-2)**

14.3. Configuration, TCTO, SI and TCI Management.

14.3.1. Responsibilities. MAJCOMs will establish PS&D requirements and responsibilities to support work centers whose duties require scheduling functions for the equipment they

maintain (such as, Egress, Armament, Aerospace Ground Equipment, and Fuels) in a supplement to this instruction.

14.3.1. **(AFMC)** Local procedures are authorized for compliance of any established AF policy unless specified in supplemental guidance for this paragraph.

14.3.1.1. PS&D will provide work centers whose duties require scheduling functions (such as, Egress, Armament, and Aerospace Ground Equipment, Fuels) SME training support and oversight of scheduling products necessary to ensure configuration data integrity is maintained. **(T-2)** PS&D will:

14.3.1.1.1. Outline procedures for ordering hazardous materials for TCIs and TCTOs (such as, batteries). **(T-2)**

14.3.1.1.2. Units using a MIS will not delegate suspense validation processing for TCIs installed on aircraft to the performing work center unless the written procedures include the following: a list of work centers and specific technicians authorized to process suspenses; a list of the specific suspenses authorized to be cleared; and the method for notifying PS&D of the work completed (an audit trail) (IMDS units only). **(T-2)**

14.3.1.1.2.1. Ensure EM processes all IMDS suspense validations for engines and engine components. **(T-2)**

14.3.1.1.3. Validate that data errors are corrected with appropriate personnel and updated in the MIS weekly. **(T-2)**

14.3.1.1.4. Submit MSM trouble tickets at **https://midtier.gunter.af.mil/**, call the Field Assistance Branch at DSN 596-5771, or e-mail **team4@gunter.af.mil** to correct program deficiencies. **(T-2)**

14.3.2. Configuration Management. Configuration management provides unit managers the capability to determine the actual versus approved configuration of an aircraft or equipment. The intent of configuration management is to ensure selected serially controlled and/or TCIs are properly loaded to the MIS database with accurate, approved part numbers, Quantity per Assembly and Next Higher Assembly items by WUC/LCN. PS&D has overall responsibility for the Equipment Configuration Management or Aircraft Configuration Management subsystem of the MIS and will provide assistance to maintenance personnel. **(T-2)** The performing work center supervisor and PS&D will validate configuration change, TCTO, SI and TCI events using MIS on-line capabilities. **(T-2)** Individual work centers accomplishing TCIs are responsible for changing configuration information in MIS. Unless otherwise specified in local procedures, schedulers will process all removal, installation, TCI, SI and TCTO compliance updates for aircraft and equipment in the applicable MIS and EM will process engines and engine components in applicable engine information system. **(T-2)**

14.3.2.1. Lead Commands will ensure procedures exist and are executed to provide system configuration tables which are updated, validated, and provided to field maintenance personnel as configurations change. **(T-2)**

14.3.2.1.1. Items not accessed or visible during field-level maintenance will be identified to Lead Command and AFSC managers for disposition. **(T-2)**

14.3.2.2. Maintenance personnel discovering an item with a missing data plate, or one which does not have a serial number, will contact PS&D who will coordinate with the Lead Command system functional manager and/or AFSC item manager for disposition. **(T-2)**

14.3.2.3. For those aircraft that do not currently have an established configuration table, the Lead Command will develop procedures to identify, track and validate installed configuration managed items against the data in the MIS.

14.3.2.4. PS&D will coordinate the daily resolution of IMDS configuration management notices with the appropriate maintenance section utilizing the applicable MIS screen. **(T-2)**

14.3.2.4.1. Uncorrected discrepancies will be briefed weekly at the daily production/scheduling meeting and forwarded to the appropriate maintenance supervision for corrective action. **(T-2)**

14.3.2.5. When out of configuration items or mismatched serially tracked items are discovered, establish a single DD for the "out-of-configuration" condition. **(T-2)**

14.3.2.5.1. Additionally, add a MIS WCE for each WUC/LCN and part/serial number item requiring verification to the single DD. **(T-2)**

14.3.2.6. **(Added-AFMC)** If ACM responsibilities are delegated to other work centers, MOF PS&D will develop a formal agreement between AFMC and the delegated work center to assign responsibilities and guidelines for updating/correcting the IMDS database (e.g., check suspense validation records in IMDS and coordinate to ensure the appropriate work center processes their suspense's daily). **(T-2)**

14.3.2.6.1. **(Added-AFMC)** When assigning these responsibilities, consideration must be given to the agency performing removal and replacement actions for the configured item, (i.e., the egress shop has the responsibility for CAD/PAD items, while flightline avionics has the responsibility for tracked LRUs, etc.). **(T-2)**

14.3.2.6.2. **(Added-AFMC)** Additionally, this guidance establishes procedures to ensure configuration data is maintained during routine maintenance actions. **(T-2)**

14.3.2.6.3. **(Added-AFMC)** All TO 00-25-107, *Maintenance Assistance*, requests must be certified at the appropriate center level and coordinated through HQ AFMC/A4MM. **(T-2)**

14.3.3. TCTO Management. TCTOs are AF, MAJCOM/Lead Command or Numbered Air Force (NAF) directed modifications and inspections that provide units with instructions for doing a one-time change, modification, or inspection of equipment, (includes applicable FAA Airworthiness Directives, original equipment manufacturer service bulletins and service instructions, after concurrence by Lead Command). Lead Command, NAF and local inspections are considered OTIs. Use the MIS to process Lead Command and NAF OTIs or modifications in the same manner as TCTOs with compliance periods, remove from service dates and rescission dates IAW TO 00-5-15. TCTOs, with the exception of immediate and urgent action, are considered scheduled maintenance and integrated into maintenance planning cycles. **(T-2)** Consider concurrent accomplishment of TCTO work with other unscheduled or scheduled maintenance (such as, PH, ISO, HSC, HPO). Manage TCTOs using the MIS, TO 00-5-15 and specific MAJCOM instructions. **Note:** Communication of a grounding, or potential grounding event (TCTO, OTI, PM event) to weapon systems operating in a deployed

environment and/or under the authority of an operational command must follow the grounding procedures outlined in DAFMAN 11-401 and 63-101/20-101. Units must not directly communicate home station requirements to its deployed forces but provide communication through their respective MAJCOM. **(T-2)**

14.3.3. **(AFMC)** Command-directed temporary (T-1 and T-2) modifications are temporary, and there is no corresponding change to approved weapon system technical data. The approved modification instruction is the authority for both modification and maintenance as long as the modification is installed. **(T-2)**

14.3.3.1. PS&D is responsible for managing all assigned weapon system TCTO programs and will monitor/provide oversight of all AF owned aircraft, weapon system, AGE, and commodity TCTOs to ensure all compliance requirements are met. **(T-2)**

14.3.3.1. **(AFMC)** MOF PS&D or equivalent has the overall responsibility on depot level TCTO management. MOF PS&D will coordinate with System Program Managers, Lead Commands, and AFMC/A4MM. **(T-2)**

14.3.3.1.1. Munitions-related TCTOs will be managed by the munition's scheduler (if assigned) and engine-related TCTOs will be managed by EM schedulers. **(T-2)**

14.3.3.1.1.1. **(Added-AFMC)** AGE-related TCTOs will be managed by the AGE scheduler (if assigned). **(T-2)**

14.3.3.1.2. PMEL TCTOs will be managed by the owning agency with PS&D oversight. **(T-2)**

14.3.3.1.3. The parent technical training center manages and schedules all TCTOs for training equipment assigned to a TD or Mobile Training Team.

14.3.3.2. PS&D will review MIS products weekly to ensure proper documentation and management by owning and managing TCTO agencies. **(T-2)**

14.3.3.2.1. When an error is detected, PS&D will inform affected work centers and provide assistance to correct the discrepancy IAW TO 00-20-2.

14.3.3.2.2. Units will complete an annual TCTO status review between MIS and REMIS or equivalent systems. **(T-2)**

14.3.3.2.2.1. Units will reconcile rescinded TCTOs using a REMIS Master TCTO report or equivalent annually (NLT 30 Sep) and before deleting/retiring TCTO records from the appropriate MIS. **(T-2)**

14.3.3.2.2.2. If REMIS or equivalent access is not available, request a REMIS Master TCTO report or equivalent from the MAJCOM MDS Weapon Systems Team/Program Office identified in the subject TCTO. If TCTO status conflicts are identified, units will contact the applicable Lead Command to establish the process for resolving conflicts and facilitating status correction in REMIS or equivalent system. **(T-2)**

14.3.3.2.2.3. Once all status errors are corrected, and reconciliation is complete and verified, IMDS units can delete the TCTO from the MIS. FMxC2 automatically retires TCTOs 60 days after rescission, and all equipment shows as complete.

14.3.3.2.2.3.1. Document completion on AF Form 2411. **(T-2)**

14.3.3.2.3. PS&D will brief the MXG/CC (or equivalent) weekly on unaccomplished TCTOs that are within 60 days of grounding. **(T-2)**

14.3.3.2.3.1. Significant problems or potential delays in TCTO accomplishment will be brought to the immediate attention of the MXO DO/SUPT and MXG/CC (or equivalent). **(T-2)**

14.3.3.2.4. PS&D will chair a TCTO review meeting attended by all TCTO owning and managing agencies after the monthly supply TCTO reconciliation meeting. **(T-2)** These meetings may be combined.

14.3.3.2.4.1. PS&D will discuss the supply reconciliation, supply status, scheduling factors, current TCTO status and anticipated problems for all active TCTOs. **(T-2)**

14.3.3.2.4.2. PS&D will produce meeting minutes on the AF Form 2410 and distribute to all affected agencies. **(T-3)**

14.3.3.2.5. Depot-level TCTOs, excluding commodities, will be loaded, and tracked in the MIS for auditing compliance and applicability. **(T-2)**

14.3.3.2.5.1. Depot-level engine TCTOs will be loaded in CEMS only. **(T-2)**

14.3.3.2.5.2. Units will ensure dual reporting of completed depot-level TCTOs is prevented. **(T-2)**

14.3.3.2.5.3. All field-level companion TCTOs for commodities must be loaded in the MIS. **(T-2)**

14.3.3.2.6. PS&D will monitor, track, and administer all applicable CPINS as commodity TCTOs for configuration management purposes IAW TO 00-5-15 and TO 00-5-16.

14.3.3.2.6.1. PS&D will coordinate reprogramming of all passive/active aircraft internal and external electronic warfare systems and equipment with the wing Electronic Warfare Officer or equivalent before implementing any CPIN changes. **(T-2)**

14.3.3.2.6.2. PS&D will coordinate with EM before issuing NSS/ETS CPINS. **(T-2)**

14.3.3.2.7. When TCTOs are directed for items without serial numbers, assign permanent serial numbers IAW TO 00-20-2 and AFI 23-101. **(T-2)**

14.3.3.2.7.1. For serial numbers that cannot be created IAW TO 00-20-2 or AFI 23-101, use the associated equipment serial number the item is assigned to (for example, an aircraft chock serial number would be 0000AXXXC1).

14.3.3.2.8. Control and Transfer of TCTO Kits. Units will transfer aircraft or equipment, with any TCTOs still pending completion, with their applicable TCTO kits. **(T-2)**

14.3.3.2.8.1. Retain engine TCTO kits for engines installed on aircraft at depot

locations if the aircraft is returning to that unit for TCTO compliance. **(T-2)**

14.3.3.2.8.2. Transfer TCTO kits IAW AFI 23-101, TO 00-5-15 and TO 00-5-1.

14.3.3.2.8.3. **(Added-AFMC)** Until proper disposition is determined, MOF PS&D will record and maintain TCTO kits received for aircraft in waivered status for applicable TCTOs. Consult with TCTO single managers and test program managers for proper disposition of kitted TCTOs IAW TO 00-5-15. If unable to attain disposition of received kitted TCTOs, contact AFMC/A4MM for guidance. **(T-2)**

14.3.3.2.9. **(Added-AFMC)** MOF PS&D will manage the status reporting for command-directed assessments, modifications, and modification rescissions that require de-compliance work, as well as organizational, intermediate, and depot TCTOs. Command-directed modifications will be identified by a title, command data code, and rescission date. **(T-2)**

14.3.3.3. Specific TCTO Responsibilities.

14.3.3.3.1. QA personnel will:

14.3.3.3.1.1. Review all new and revised technical data and TCTO's for completeness, accuracy, and applicability. **(T-2)** Inform applicable work centers of changes and up channel any problems discovered during this review. **(T-2)**

14.3.3.3.1.2. Determine if the TCTO impacts W&B. **(T-2)**

14.3.3.3.1.3. Distribute copies of TCTOs to the managing agency, performing work centers, and LRS. **(T-2)**

14.3.3.3.1.4. Provide a supply cover letter requesting the number of items in supply (including WRM) affected by the TCTO. **(T-2)**

14.3.3.3.1.5. Report all deficiencies in technical instructions and kit-proofing to the appropriate TCTO Manager IAW TOs 00-5-1 and 00-5-15. **(T-2)**

14.3.3.3.1.6. Attend TCTO planning meetings. **(T-2)**

14.3.3.3.1.7. Provide technical support to performing work centers. **(T-3)**

14.3.3.3.2. PS&D personnel will:

14.3.3.3.2.1. Determine the total number of end items applicable to the TCTO. **(T-2)**

14.3.3.3.2.1.1. Items that are assigned with the same Mission Design Series, WUC, Part Number, but are not applicable to the TCTO will be loaded in "22" status. **(T-2)** This ensures accurate accountability that all equipment has been verified as being affected or not applicable to TCTO.

14.3.3.3.2.2. Chair a TCTO planning meeting with attendees from QA, owning and performing work centers and Flight Service Center/LRS IAW AFI 23-101, **Chapter 4**. **(T-2)**

14.3.3.3.2.2.1. Record meeting minutes on AF Form 2410, or locally developed product and provide an overall plan to implement the TCTO. **(T-2)**

14.3.3.3.2.2.2. Minutes will include TCTO applicability by ID number (or applicable part number or serial number for commodity TCTOs), purpose of the inspection/modification and clearly identify and document the performing work centers, training requirements, scheduling parameters, remove from service date, a review of the TCTO procedures, form entries and supply requirements prior to scheduling the TCTO for completion. **(T-2)**

14.3.3.3.2.2.3. All attendees sign the AF Form 2410, or locally developed product, at the conclusion of the planning meeting indicating agreement with the conditions. **(T-2)**

14.3.3.3.2.3. Establish and maintain a TCTO folder for each active TCTO. **(T-2)**

14.3.3.3.2.3.1. TCTO folders will be standardized and include the basic TCTO and any supplements, completed AF Form 2410, or locally developed product, AF Form 2001, *Notification of TCTO Kit Requirements* (if required), messages and the supply cover letter from QA (if required). **(T-2)**

14.3.3.3.2.3.2. Once the TCTO has reached its rescission date, print a MIS product showing the current status of equipment and place it in the TCTO folder. **(T-2)**

14.3.3.3.2.3.2.1. Move the folder to an inactive TCTO file. **(T-2)**

14.3.3.3.2.3.2.2. The TCTO managing agency will maintain the folder until the TCTO is rescinded in the applicable MIS IAW TO 00-5-15. MIS TCTO records will be deleted (scheduled to retire for FMxC2 users) at that time. **(T- 2)**

14.3.3.3.2.3.3. TCTOs will not be deleted from the MIS prior to the rescission date. **(T-2)**

14.3.3.3.2.3.4. Validate in REMIS that no additional requirements have been submitted or extensions applied. **(T-2)**

14.3.3.3.2.4. If an initial TCTO load is not received from REMIS or equivalent, notify the single manager and/or equipment specialist IAW TO 00-5-15. **(T-2)**

14.3.3.3.2.5. Use the ILS-S to order required kits/parts/tools IAW MIS manuals. **(T-2)** Kits, parts, tools, and software will be ordered within 24 hours of the TCTO meeting and document numbers input on the AF Form 2001. **(T-2)** Locally manufactured and obtained parts will be documented on the AF 2001. **(T-2)**

14.3.3.3.2.5.1. When ILS-S is not available, initiate three copies of the AF Form 2001 and forward two copies of the Form with a copy of the TCTO to the supply TCTO monitor. **(T-2)**

14.3.3.3.2.5.2. For locally obtained parts, prepare an AF Form 2001 listing each item by NSN, noun and quantity required. **(T-2)**

14.3.3.3.2.6. Assign ID numbers to kits as they are received. **(T-2)**

14.3.3.3.2.6.1. Use **Part II** of the AF Form 2001 to manage kit/part assignment and track individual end items, date issued, document numbers and the number

of kits remaining. **(T-2)**

14.3.3.3.2.6.2. The LRS/Flight Service Center TCTO monitor will ensure kits and/or parts are assembled prior to release. **(T-2)**

14.3.3.3.2.7. Control and release TCTO kits from LRS. **(T-2)**

14.3.3.3.2.8. Notify appropriate MAJCOM, by message, when local managers anticipate a problem with TCTO compliance within prescribed time limits. **(T-2)**

14.3.3.3.2.8.1. The message should include the TCTO number and narrative, total units affected, total units complete, kits on hand, kits on order, estimated delivery date, requisition number and a narrative of the problem.

14.3.3.3.2.8.2. The message will be endorsed by MXG/CC (or equivalent) prior to submission to MAJCOM. **(T-2)**

14.3.3.3.2.9. Report status of TCTOs that cannot be reported under "HOW MAL" codes 793, 797, 798, 801, 802, or 911 IAW the MIS, and 00-20 series TOs. **(T-2)**

14.3.3.3.2.10. Schedule, track and monitor TCTO accomplishment. **(T-2)**

14.3.3.3.2.10.1. Prepare a work order in the MIS for each affected end-item, including spares. Agencies owning installed on-equipment TCTOs will coordinate with PS&D prior to scheduling on-aircraft TCTOs. **(T-3)**

14.3.3.3.2.11. Review suspense validation or equivalent inputs prior to processing TCTO suspenses and updating the MIS. **(T-2)**

14.3.3.3.2.12. Annotate back-up MIS products as changes occur. **(T-2)**

14.3.3.3.2.13. Ensure TCTOs are scheduled for completion prior to expiration or grounding date whichever comes first. **(T-2)**

14.3.3.3.2.14. Schedule all workable TCTOs for accomplishment prior to permanent equipment transfer or storage input. **(T-2)**

14.3.3.3.2.15. For TCTOs with compliance periods calculated in operating time (hours, cycles, starts, landings, or rounds) create a local JST and load the JST to the equipment, schedule for completion prior to expiration of the compliance period operating time. **(T-2) Note:** Once compliance period operating time has been reached remove equipment from service until the TCTO has been completed. **(T-2)**

14.3.3.3.2.15.1. Document the JST number in the TCTO notes.

14.3.3.3.2.16. **(Added-AFMC)** The owning scheduling agencies will ensure all automated reports are verified and corrected IAW TO 00-20-2, *Maintenance Data Documentation*. **(T-2)**

14.3.3.3.2.17. **(Added-AFMC)** Each month, MOF PS&D, EM section, and TCTO managing agencies receive a copy of the TCTO reconciliation listing and must annotate this listing with TCTO requirements. **(T-2)**

14.3.3.3.2.18. **(Added-AFMC)** MOF PS&D monitors completion of commodity TCTOs and coordinates with performing work centers to ensure compliance within specified time limits. AMU PS&D establishes schedules for completion of installed

commodity TCTOs with companion aircraft TCTOs. AMU PS&D schedules aircraft TCTOs or on-equipment munitions type TCTOs. The EM section schedules non-installed engine-related TCTOs and coordinates with AMU PS&D on installed engine related TCTOs. The munitions flight scheduler manages TCTOs for all air-launched surface attack guided missiles and associated support equipment, munitions, nuclear ordnance controlled material items, guidance kits, air intercept/aerial guided missiles and air intercept/aerial rockets. **(T-2)**

14.3.3.3.2.19. **(Added-AFMC)** TCTO Waiver Procedures. Commanders of AFMC organizations may request to temporarily waive compliance with routine, immediate and urgent action TCTOs (Organizational, Intermediate and Depot Level) only when TCTOs are applicable to components of a system or subsystem that are deactivated or removed from the aircraft because of RDT&E missions or compliance of the TCTO would adversely affect the test mission. Procedures are outlined in DAFI 21-101, Aircraft and Equipment Maintenance Management. **(T-2)**

14.3.3.3.2.19.1. **(Added-AFMC)** Owning organization will initiate the waiver request for Organizational, Intermediate, and Depot Level TCTOs. **(T-2)**

14.3.3.3.2.19.2. **(Added-AFMC)** The request will include the affected TCTO number, T-2 modification number and specifically why the modification interferes with accomplishing the TCTO. **(T-2)**

14.3.3.3.2.19.3. **(Added-AFMC)** The owning organization will contact the TCTO single manager to ensure that noncompliance will not result in a safety of flight condition, and to see if there are any safeguards that should be taken until the TCTO can be complied with. Include the single manager's name and phone number on the waiver request. **(T-2)**

14.3.3.3.2.19.4. **(Added-AFMC)** Owning organization will coordinate

14.3.3.3.2.19.5. **(Added-AFMC)** Once the request has been signed, forward a copy of the signed request to the TCTO monitor, command aircraft manager, and the TCTO single manager. Wing TCTO monitor will retain a record copy. **(T-2)**

14.3.3.3.2.20. **(Added-AFMC)** Noncompliance will be recorded on the AFTO Form 95, and appropriate Maintenance Information System, as applicable IAW TO 00-20-1 and TO 00-5-15. The entry will indicate the reason and estimated date of compliance. **(T-2)**

14.3.3.3.2.20.1. **(Added-AFMC)** Conduct an annual review and validate all waived TCTOs using the original MXG/CC/CL or equivalent approved date or last review date. **(T-2)**

14.3.3.3.2.20.2. **(Added-AFMC)** Send a list of waived TCTOs to HQ AFMC/A4MM no later than the first workday of January and July or when a significant change to a TCTO or waiver has occurred. **(T-2)**

14.3.4. SI and TCI Management:

14.3.4.1. Job Standard Master Listing (JML) Management.

14.3.4.1.1. PS&D will maintain (load, change, and delete) the JML for all inspections and time changes listed in the applicable aircraft/system -6 TO and commodity TOs. **(T-2)**

14.3.4.1.1. **(AFMC)** Schedulers will monitor -6 and associated TOs to ensure time change/inspection frequencies align and support the maintenance concept of the weapon system. Scheduled maintenance requirements that do not align with the established maintenance concept and affect aircraft availability will be identified and forwarded to the Product Improvement Working Group (PIWG) for consideration and/or resolution. **(T-2)**

14.3.4.2. Develop a matrix/chart depicting the total number of SIs and TCIs to be loaded in the MIS for each assigned aircraft/system. **(T-2)**

14.3.4.2.1. Maintain JMLs for off-equipment items in the OWC. PS&D will provide written guidance and training for JML management of off-equipment JSTs when PS&D authorizes OWCs to maintain it. **(T-3)**

14.3.4.2.2. For units using FMxC2, Lead Commands must maintain master inspection and time change requirements.

14.3.4.2.3. Once Master Job Standard Numbers are fielded for a weapon system, local PS&D will review TO 00-20-2 for Master Job Standard Numbers procedures. **(T-2)**

14.3.4.2.4. PS&D will load, change, and delete JSTs in the MIS as soon as possible after receipt of any -6 TOs, or other TO, TCI or inspection change and will promptly notify all affected PS&D sections for action. **(T-2)** PS&D will:

14.3.4.2.4.1. Load separate JSTs for all aircraft/systems -6 TOs special/scheduled inspections with frequencies greater than 30 days or 50 hours in the MIS. **(T-2)**

14.3.4.2.4.1.1. Load PE, PH, engine changes and other event type inspections (such as, hard landing) as a JST in the MIS as they occur. **(T-2)**

14.3.4.2.4.1.2. Provide training for maintaining JSTs as necessary. **(T-2)**

14.3.4.2.4.2. Perform a semi-annual review of the JML and all JSTs for accuracy and currency to include off-equipment and decentralized activities. **(T-2)**

14.3.4.2.4.2.1. Review matrix/chart depicting the total number of SIs and TCI requirements to be loaded in the MIS for each assigned aircraft/system. **(T-2)**

14.3.4.2.4.2.2. Reconcile TCI and SI JSTs with the aircraft/systems -6 TOs and applicable commodity TOs and document the semi-annual review on AF Form 2411. **(T-2)**

14.3.4.2.4.2.3. Units may create JSTs in the MIS to automate required documentation of repetitive or complex tasks (such as engine change, tire change, phase inspection, flight control maintenance).

14.3.4.2.4.3. Monitor the inspection and time change subsystems in the MIS. **(T-2)**

14.3.4.2.4.3.1. Perform a monthly review of all inspections, SIs, and TCI JSTs for each assigned aircraft. **(T-2)**

14.3.4.2.4.3.2. Look for missing and/or excess inspections and TCIs loaded to the aircraft and ensure the accuracy of all due dates/times for TCIs and verify the Date of Manufacture (DOM) and Date of Installation (DOI). **(T-2)**

14.3.4.2.4.3.2.1. TCIs will be loaded in the MIS with a part number and serial number. **(T-2)**

14.3.4.2.4.3.3. Document the review and ensure corrections are made to the MIS. **(T-2)**

14.3.4.2.4.3.4. Maintain the report on file with corrective actions until the next review. **(T-2)** The use of automated verification tools is encouraged provided MIS data is the source for verification.

14.3.4.3. PS&D will manage the assigned weapon systems TCI program. **(T-2)** PS&D Personnel will:

14.3.4.3.1. At least annually, meet with Egress and Aircrew Flight Equipment activities to verify each aircraft's egress data. **(T-2) Note:** MAJCOM/Units may direct more frequent verification of Egress and Aircrew Flight Equipment as required to maintain system integrity.

14.3.4.3.1.1. Document all verification of aircraft's egress data on the AF Form 2411 maintained in the aircraft jacket file. **(T-2)**

14.3.4.3.2. Identify, monitor, forecast and schedule only those selected items specifically identified in TO 00-20-9, *Forecasting Replacement Requirements for Selected Calendar and Hourly Time Change Items*; applicable commodity TOs; the aircraft -6 TO, DAFMAN 21-201 or identified as Federal Supply Group 13 and Materiel Management Code AQ Items. **(T-2)**

14.3.4.3.2.1. TCIs will be loaded in the MIS with a part number and serial number. **(T-2)**

14.3.4.3.3. Establish a JST for both the DOM and DOI for Cartridge-Actuated Devices (CAD), Propellant Actuated Devices (PAD), life sustaining, and other TCI items listed in the aircraft -6 TO and applicable commodity TOs. **(T-2)**

14.3.4.3.3.1. Load only the DOI or DOM JST that comes due first, in the MIS against a specific part or serial number. **(T-2)**

14.3.4.3.3.2. As a minimum, when the DOI and DOM frequencies are identical, maintain the JST for the DOM. **(T-2)** (N/A for FMxC2 units).

14.3.4.3.3.3. Ensure component background information is provided by Egress to include a list of all components having multiple part numbers with a different service life. **(T-2)**

14.3.4.3.3.3.1. Forecasting of CAD/PAD items for long-term CAD/PAD spare requirements will be accomplished by Ogden Air Logistics Complex through use of the Requirements Determination Module to extract installation and due dates from REMIS. **(T-2)**

14.3.4.3.3.3.2. When CAD/PAD items or forecast requirements are not visible

within the maintenance data system (for example, Contract Logistics Support managed components), units will forecast for TCIs IAW TO 00-20-9 and DAFMAN 21-201.

14.3.4.3.3.3.3. Validate and consolidate TCI forecasts for items listed in TO 00-20-9, commodity TOs, and aircraft specific -6 TOs. **(T-2)**

14.3.4.3.3.3.4. Submit consolidated forecasts to the appropriate Lead Command representative with an info copy to munitions operations. **(T-2)**

14.3.4.3.3.3.5. Forward any quarterly updated forecasts to munitions operations. **(T-2)**

14.3.4.3.4. Initiate, validate, and submit TCI extension requests to the Air Force Sustainment Center item manager with an info copy to munitions operations. **(T-2)**

14.3.4.3.4.1. Ensure a copy of approved extension are placed in the affected aircraft's forms and removed when no longer required. **(T-2)**

14.3.4.3.4.2. Maintain and monitor a suspense copy of the extension request and follow up prior to the grounding date of the TCI. **(T-2)**

14.3.4.3.4.3. Refer to TO 00-20-1 and 00-20-9 for additional guidance on TCI extensions and maintain a copy of the Air Force Sustainment Center/System Program Director approved message until the item is replaced. **(T-2)**

14.3.4.3.4.4. EM will generate engine TCI extension requests and coordinate through the Command Engine Manager to the appropriate Engine Program Office in AFLCMC. **(T-2)**

14.3.4.3.5. Perform monthly reconciliation of all TCIs with LRS. **(T-2)**

14.3.4.3.5.1. The reconciliation will consist of 100 percent validation of existing due outs. **(T-2)**

14.3.4.3.5.2. Inform FSC of any "mark for" changes or items no longer required. **(T-2)**

14.3.4.3.6. Monitor and requisition TCI requirements based on projected equipment utilization. **(T-2)**

14.3.4.3.6.1. Order parts using MIS or coordinate with LRS/DMS to order parts using AF Form 2005, unless otherwise specified in -11, -14 and -6 TOs. **(T-2)**

14.3.4.3.6.2. Schedule TCIs for replacement to minimize aircraft downtime at the next applicable inspection IAW TO 00-20-1 (i.e., schedule a TCI at the next inspection interval, HPO, PH, PE, HSC, 60-month, letter check). **(T-2) Note:** Life sustaining, or CAD/PAD TCIs cannot exceed replacement interval in applicable -6 and commodity TOs without an approved extension/waiver from the Program Office/appropriate item manager.

14.3.4.3.6.3. Notify the Munitions Flight of the need to order munitions items IAW TO 00-20-9 and DAFMAN 21-201. **(T-2)**

14.3.4.3.6.3.1. Serviceable CAD/PAD TCIs components will not be turned

into munitions operations until the remaining service life reaches 9-months or less. **(T-2)** Serviceable CAD/PAD TCIs components with less than 9-months service life remaining will not be reissued. **(T-2)**

14.3.4.3.6.3.2. Maintenance plans must reflect replacement dates to coincide within the 9-month parameter. **(T-2)**

14.3.4.3.6.3.3. **(Added-AFMC)** Agencies requesting changes or updates to the need date of any cartridge actuated device/propellant actuated device (CAD/PAD) item must coordinate with AMU PS&D, MOF PS&D, and MASO in sufficient time to preclude emergency issue requests. Agencies requesting a change shall provide justification if need dates are changed by more than 30 days. No justification is necessary if changes are within 30 days, regardless of the quarter, or if the change is driven by Ogden Air Logistics Complex. Issue requests for unforecasted or improperly forecasted items (except for replacement of inadvertently fired items or items found to be unserviceable) will be supported by a letter of justification signed by the MXG/CC/CL or designated representative. **(T-2)**

14.3.4.3.6.4. Order non-CAD/PAD or engine TCIs IAW AFI 23-101.

14.3.4.3.7. Schedule the time change in the MIS and incorporate it in the monthly/weekly/ quarterly maintenance schedule. **(T-2)**

14.3.4.3.8. Review the data (DOM, DOI, LOT number, JST, and Due Date) entered by the performing work center. Ensure the suspense validation is updated in the MIS when the time change is completed (Suspense validation N/A for FMxC2). **(T-2)**

14.3.4.3.9. Coordinate management of respective TCIs with applicable maintenance and operation work centers. **(T-2)**

14.3.4.3.10. Schedule drogue chute TCIs, except chute harnesses, for replacement during the drogue chute repack before the expiration of the component service or shelf life. **(T-2)**

14.3.4.3.10.1. These components will not be over flown without an approved extension from the appropriate item manager. **(T-2)**

14.3.4.3.10.1.1. A copy of approved extensions must be maintained in the affected aircraft's forms and removed when no longer required. **(T-2)**

14.3.4.3.11. Prepare TCI forecasts IAW TO 00-20-9.

14.3.4.3.11. **(AFMC)** Prepare AFTO Form 223, *Time Change Requirements Forecast*, for TO 00-20-9, *Forecasting Replacement Requirements for Selected Calendar and Hourly Time Change Items* and applicable -6 TO requirements. Forward AFTO Form 223 to the MOF PS&D section. MOF PS&D will review all AMU AFTO Form 223 and will forward a combined list of requirements to the appropriate supply agency. AMU PS&D is responsible for forecasting only those selected items specifically identified in TO 00-20-9, and the aircraft -6 TO. **(T-2)**

14.3.4.3.11.1. Provide squadron Maintenance Supervision a forecast for non-munitions items for their supply section. **(T-2)**

14.3.4.3.12. To facilitate quarterly requisitioning, P&S will submit the quarterly validated time-change AFTO Form 223, Spreadsheet, or IMDS/FMxC2 generated forecast to the Munitions Accountable System Officer. **(T-2)** This must be submitted between 45 and 60 calendar days Continental United States (CONUS) or between 90 and 120 calendar days Outside Continental United States (OCONUS) before the next calendar year quarter IAW DAFMAN 21-201 **Table 7.3**, OCONUS and CONUS Time Change Requisitioning Schedule. **(T-2)** If aircraft is scheduled to be in depot when time change will be required, home station P&S will coordinate with depot P&S to facilitate requisition of required assets by the depot munition's function IAW DAFMAN 21-201 **Table 7.3** time frames. **(T-2)**

14.3.4.3.12.1. Validate current requirements against the annual forecast and make corrections based on aircraft utilization. **(T-2)**

14.3.5. Major Maintenance Work Processing. PS&D will:

14.3.5.1. Coordinate on all TO 00-25-107 requests for DAFI 21-103 reporting. **(T-2)**

14.3.5.1.1. The work center discovering the discrepancy is responsible for drafting the TO 00-25-107 request and forwarding the request to QA for coordination and release.

14.3.5.1.2. PS&D will make the appropriate PIC changes in the MIS when the 00-25-107 request has been submitted and/or received. **(T-2)**

14.3.5.1.3. Depot-level assistance provided by contractor support will be accomplished IAW contract specifications. **(T-2)**

14.3.5.2. Develop procedures in conjunction with QA for routing all major maintenance requests to ensure all affected parties are informed. **(T-2)**

14.3.5.3. Conduct an initial meeting upon arrival of a DFT to validate maintenance support requirements are in place. **(T-2)**

14.3.5.3. **(AFMC)** Conduct an initial meeting upon arrival of a Depot Pre-Induction Inspection team (DPII). **(T-2)**

14.3.5.3.1. The meeting will be documented on an AF Form 2410, or locally-developed product. **(T-2)**

14.3.5.3.2. PS&D will initiate/accomplish all PIC changes in the MIS. **(T-2)**

14.3.5.3.3. Once work is completed, PS&D will ensure appropriate PIC are changed and a completed copy of the work package is placed in the aircraft jacket file. **(T-2)**

14.3.5.3.3.1. PS&D will document significant historical data on the appropriate AFTO Form 95 IAW 00-20 series TOs. **(T-2)**

14.3.6. Transfer Inspections.

14.3.6.1. Units will perform gaining/losing transfer inspections IAW TO 00-20-1, MAJCOM guidance and this instruction. **(T-2)**

14.3.6.1.1. In conjunction with QA, develop a local JST for both gaining and losing aircraft and equipment transfer. **(T-2)**

14.3.6.1.1.1. This JST must meet all TO 00-20-1, 2J-1-18, *Preparation for*

Shipment and Storage of Gas Turbine Engines, applicable aircraft -6 and -21 TOs, as well as DAFI 21-103 and MAJCOM specific transfer requirements. **(T-2)**

14.3.6.1.1.2. Include all historical records (example, NDI records, Egress records, W&B records, OAP records, strut records) and other applicable items. **(T-2)**

14.3.6.1.1.3. **(Added-AFMC)** MOF PS&D personnel will:

14.3.6.1.1.3.1. **(Added-AFMC)** Ensure a minimum of 6 months of engine operating time are remaining on TCI such as modules/compressors/turbines/gearboxes and other major components at time of transfer or according to established MOA. **(T-2)**

14.3.6.1.1.3.2. **(Added-AFMC)** Ensure all engines have a comprehensive borescope inspection accomplished unless a borescope was accomplished within the last 10 aircraft flying hours. **(T-2)**

14.3.6.1.1.3.3. **(Added-AFMC)** Verify TCIs have a minimum of 60 days or 50 hours remaining at time of transfer. **(T-2)**

14.3.6.1.1.3.4. **(Added-AFMC)** Verify all SI have at least 25 percent of the usable time remaining at time of transfer. **(T-2)**

14.3.6.1.1.3.5. **(Added-AFMC)** Include two copies of the IMDS pilot reported discrepancy (PRD) report (NFS1SO screen 179) for the previous 90-day period in the aircraft records file. **(T-2)**

14.3.6.1.2. Losing PS&D ensures all actions are completed in the MIS prior to permanently transferring an aircraft to another unit. **(T-2)**

14.3.6.1.3. Losing PS&D conducts a transfer pre-dock meeting one duty day prior to start of the aircraft transfer inspection. **(T-2)**

14.3.6.1.4. All items to be accomplished during the transfer inspection will be documented on an AF Form 2410, or locally developed product, and scheduled in the MIS. **(T-2)**

14.3.6.2. Losing PS&D will complete a total verification of all TCIs installed on the transferring aircraft. **(T-2)**

14.3.6.2.1. Verify the correct computation of all due dates/hour/cycles based on DOM, DOI, installed times, or equivalent factors. **(T-2)**

14.3.6.2.2. For IMDS units only:

14.3.6.2.2.1. Ensure the IMDS-REMIS synchronization programs are processed, and errors are corrected prior to transfer. **(T-2)**

14.3.6.2.2.2. Ensure an up-to-date Transfer of Equipment report and an AFTO Form 95 with current engine trend and performance data are placed in the aircraft jacket file. **(T-2)**

14.3.6.2.2.3. Ensure a backup copy is maintained until receipt is verified by the gaining unit. **(T-2)**

14.3.6.3. Losing PS&D will ensure an ADR is performed and conduct a transfer post dock meeting to ensure all required actions have been completed, all forms are current/accurate, and the MXG/CC (or equivalent) has certified each aircraft ready to transfer aircraft IAW TO 00-20-1, AFI 16-402, and DAFI 21-103.

14.3.6.3.1. **(Added-AFMC)** MOF/AMU PS&D will generate two copies of the following products listed below and forward one copy of each to the gaining unit NLT 30 days before the aircraft is scheduled to be transferred. Annotate changes in red on the remaining copy and include it with the aircraft record jacket file at time of transfer. **(T-2)**

14.3.6.3.1.1. **(Added-AFMC)** Automated records check. **(T-2)**

14.3.6.3.1.2. **(Added-AFMC)** Transfer of equipment for aircraft and engine(s). **(T-2)**

14.3.6.3.1.3. **(Added-AFMC)** Significant history data for aircraft and engine(s). **(T-2)**

14.3.6.3.1.4. **(Added-AFMC)** AFTO Form 427, *Aircraft Integral Fuel Tank Repair Historical Record*, or AFTO Form 428, *B-1B Aircraft Integral Fuel Tank Repair History Record*. **(T-2)**

14.3.6.4. **(Added-AFMC)** During AGE Transfers. In addition to the inspection requirements in TO 00-20-1 and TO 35-1-4, *Processing and Inspection of Support Equipment for Storage and Shipment*, QA may perform an assessment on each AGE item. The inspection includes a complete records check. When the capability exists, ensure all discrepancies are corrected prior to equipment transfer. Annotate in the equipment forms any discrepancy not corrected due to lack of parts. **(T-2)**

14.3.6.5. **(Added-AFMC)** Foreign country, other US Government agency, contractor owned, and/or sister service aircraft that will be flown/maintained by AFMC units will require a transfer inspection (reference T.O. 00-20-1 and 00-35D-54, *USAF-Deficiency Reporting, Investigating, and Resolution*, Chapter 8). **(T-2)**

14.3.6.5.1. **(Added-AFMC)** Foreign country, other U.S. Government agencies, contractor-owned, and/or sister service loaner aircraft require a signed MOA, IAW AFI 16-402, A*erospace Vehicle Programming, Assignment, Distribution, Accounting, and Termination*, and its AFMC Supplement, and DAFI 21-103 and its AFMC Supplement. Also ensure the agreement contains the inspection requirements listed in **paragraph 14.3.6.1.1**. – **paragraph 14.3.6.2.1**. Include any additional requirements the loaning organization affirms needed. **(T-2)**

14.3.6.6. **(Added-AFMC)** Loaner aircraft from another USAF unit require a signed MOA, IAW AFI 16-402 and its AFMC Supplement, and DAFI 21-103 and its AFMC Supplement. Also ensure the agreement contains, the inspection requirements listed in **paragraph 14.3.6.1.1**. – **paragraph 14.3.6.2.1** (T-2)

14.3.6.7. **(Added-AFMC)** In conjunction with MMA DBM, MOF PS&D personnel will process the aircraft transfer file from REMIS IAW AFCSM 21-576V2, Generic Configuration Status Accounting System (GCSAS), Software User Manual. **Note:** DBMs must process NFS6A0, screen 47 and 942, upon receipt of approved file from REMIS. This

must be accomplished prior to processing IMDS-CDB program NFS3W0 (actual configuration). **(T-2)**

14.3.6.8. **(Added-AFMC)** MOF PS&D personnel will request approved configuration tables for B-1, B-2, F-15 and F-16 units and actual configuration tables for all other MDSs using IMDS screen 334 (IMDS-CDB units only). **(T-2)**

14.3.6.9. **(Added-AFMC)** Ensure maintenance performs a complete aircraft Dash 21 series TO equipment inventory IAW DAFI 21-103. **(T-2)**

14.3.6.10. **(Added-AFMC)** Complete an egress system CAD/PAD inspection on newly assigned aircraft and upon those returning from depot/PDM where the egress system has been work on by depot personnel. Inspection will include 100% MIS (IMDS/FMxC2) verification, and all visible and disturbed systems. **(T-2)**

14.3.6.11. **(Added-AFMC)** An ADR must be completed prior to first flight after inspection. AMU PS&D or equivalent will make appropriate entries in the MIS aircraft automated history for all aircraft transfers detailing the date, current operating time, location, unit, and a general synopsis of worked performed or reason for transfer. **(T-2)**

14.3.7. Acceptance Inspections. Units perform acceptance inspections IAW TO 00-20-1, MAJCOM guidance and this instruction.

14.4. ENGINE MANAGEMENT (EM).

14.4. (AFMC) [DEV] Engine Management (EM). The 412 MXS Propulsion Flight Business Unit will perform duties and responsibilities of EM Section listed in DAFI 21-101.

14.4.1. Engine Management (EM). EM manages unit efforts to maintain adequate engine support for mission requirements. EM monitors engine removals and replacements, component tracking, engine TCTOs and TCIs, engine records in the MIS and CEMS and performs engine manager duties.

14.4.1.1. - 14.4.1.1.2. DELETED.

14.4.1.2. EM will:

14.4.1.2.1. Manage the MIS and CEMS by referencing applicable information in this instruction, AFI 23-101, TO 00-25-254-1, TO 00-20-5-1-3, *Instructions for Jet Engine Parts Tracking of OC-ALC/LPA Managed Engines*. Reference AFCSM 21-558, *Comprehensive Engine Management System* and applicable aircraft -6 TOs.

14.4.1.2.2. Coordinate with Propulsion Flight CC/SUPT and organization leadership to support WRE requirements. **(T-2)**

14.4.1.2.3. Ensure plans, schedules, and maintenance actions are documented on assigned engines. **(T-2)**

14.4.1.2.4. Perform quarterly inventory IAW TO 00-25-254-1. Coordinate with the PS&D AVDO or equivalent to ensure accurate aircraft accountability in support of engine quarterly inventory completion.

14.4.1.2.5. Review Engine Managers Data List and reconcile after review IAW 00-25-254-1.

14.4.1.2.6. Ensure IMDS suspense validations/FMxC2 Engine Managers Data List transactions are accomplished no later than the close of business of the next business day following the date/time of occurrence. All transactions must be reported and entered in the order of occurrence. **(T-2)** Note**:** **Paragraph 14.4.1.3.4.2** provides unit level procedures requirements to support EM execution of duties for **Paragraph 14.4.1.2.6**.

14.4.1.2.7. Perform a monthly reconciliation of total assigned assets between the local MIS and CEMS, document the completed reconciliation on AF Form 2411 and develop a file plan for historical purposes. (**T-2)** At a minimum the reconciliation will include SRAN location, total engines assigned (to include serial numbers), engine condition/status, installed engines, uninstalled engines.

14.4.1.2.7. **(AFMC)** Ensure engine SIs are loaded in IMDS or FMxC2 to ensure availability of information to all maintenance activities. Units are authorized to use a separate AFTO Form 781K for each engine to document and monitor SIs and TCTOs. **(T-2)**

14.4.1.2.7.1. Engine managers will take immediate action to correct all reporting errors along with the competed AF Form 2411. **(T-2)** Contact MAJCOM engine manager or CEMS PMO Help Desk, **cems.pmo.helpdesk@us.af.mil** to solve reporting problems requiring clarification for completion of monthly reconciliation

14.4.1.2.7.2. **Command e** ngine managers will develop a standard procedure to extract monthly reconciliation data from engines located at a contractor repair facility whom contractors' sites do not have access to CEMS or the MIS. **(T-2)** At a minimum, reconciliation data must include SRAN location, total engines assigned (to include serial numbers), engine condition/status, installed engines, and uninstalled engines.

14.4.1.2.8. Provide TCI information (cycles remaining, EOT) on serially controlled items to the Propulsion Flight and AMXS/AMU for engine and engine component CANN actions. **(T-2)**

14.4.1.2.9. Ensure all engine SIs are loaded in MIS against the engine, not against the aircraft. **(T-2)**

14.4.1.2.10. Ensure all engine/module inspections/TCIs tracked by EOT, Calculated Cycles (CCY), Total Accumulated Cycles (TAC), are loaded/tracked in the MIS and CEMS databases. **(T-2)**

14.4.1.2.11. Ensure serial numbers erroneously input into CEMS are followed by a Possessor Change (6D) Transaction Condition Code (TCC). **(T-2)**

14.4.1.2.11.1. After the TCC has successfully processed, notify the CEMS Program Management Office help desk stipulating the serial number was erroneously input and will be deleted from CEMS, **cems.pmo.helpdesk@us.af.mil**. **(T-2)**

14.4.1.2.11.2. Create a JCN for engine, module or component data plate changes, modifications, re-identifications and new etchings and document changes in CEMS automated history. **(T-2)**

14.4.1.2.11.3. A matrix by engine type will be developed to depict specific inspection and TCI quantities for each TMS. Inspections tracked by flight hours must be loaded in the MIS. **(T-2)**

14.4.1.2.12. Items that are assigned with the same TMS, WUC, Part Number, but are not applicable to the TCTO will be loaded in "22" status (refer to TO 00-20-2). **(T-2)** This ensures accurate accountability that all equipment has been verified as being affected or not applicable to TCTO.

14.4.1.2.12.1. Comply with TCTO duties and responsibilities for engine items IAW this **Chapter**. **(T-2)**

14.4.1.2.13. Manage TCTOs on all assigned engines and engine components, installed and uninstalled, as well as manage TCTOs for support equipment to include engine trailers. **(T-2)**

14.4.1.2.14. Accomplish quarterly TCTO status reviews and reconciliations IAW TO 00-25-254-1. **(T-2)**

14.4.1.2.15. Maintain records on TCTO kits and status for all engines installed on aircraft sent to depot. **(T-2)**

14.4.1.2.16. Manage time changes on all engines and engine components. **(T-2)**

14.4.1.2.16.1. EM will forecast parts requests and ensure requests are submitted to LRS up to 60 days (but not less than 10 days) prior to the need date of the scheduled time change or JEIM/CRF induction (see sections 14.2. and 14.3.). **(T-2)**

14.4.1.2.17. Reconcile all TCIs during the monthly TCI meeting with PS&D and LRS. **(T-2)**

14.4.1.2.17. **(AFMC)** [DEV] Within the 412 MXG, EM will provide data to PS&D for incorporation of engine-installed items for the monthly TCI meeting chaired by PS&D.

14.4.1.2.17.1. Reconciliation will consist of 100 percent validation of existing due-outs and a complete physical inventory of all issued TCIs. **(T-2)**

14.4.1.2.17.2. Inform FSC of any “mark for” changes or items no longer required. **(T-2)**

14.4.1.2.18. Maintain and update historical documents for all assigned engines, modules, and major assemblies that are not managed by a Performance Based Logistics or contractually by a Contract Logistics Support contract. **(T-2)**

14.4.1.2.18. **(AFMC)** Perform and document annual reviews of both paper (if applicable/maintained) and automated engine records IAW TO 00-20-1. (Paper history can be maintained in conjunction with automated history). **(T-2).** In addition to TO 00-20-1 requirements, the following historical entries in MIS are required for serially tracked engines, modules, components and major assemblies:

14.4.1.2.18.1. **(Added-AFMC)** Engine removal. Conduct a detailed records review to ensure utilization was properly recorded against the engine and sub-components. A removal narrative is required for all removal actions, except for

HOW MAL codes 866, 879, and 800. Use the correct Automated History Event indicator when processing MIS transactions to ensure removal times are posted to automated AFTO IMT 95. Ensure utilization is updated in the MIS before processing the engine removal. **(T-2)**

14.4.1.2.18.2. **(Added-AFMC)** SIs, occurrences, and all borescope inspections. Include total time (EOT, TACs, CCYs, etc.), findings (no defect noted, discrepancies noted, etc.) and other pertinent information. **(T-2)**

14.4.1.2.18.3. **(Added-AFMC)** Engine test cell rejects IAW TO 00-25-254-1. Include total time (EOT, TACs, CCYs, etc.), the reason for reject, any specific test cell data and a summary of work performed at the test cell. **(T-2)**

14.4.1.2.18.4. **(Added-AFMC)** FSE or modifications. Include total time (EOT, TACs, CCYs, etc.), a brief summary of modifications to include part number and serial number of modified items, and other pertinent information. **(T-2)**

14.4.1.2.18.5. **(Added-AFMC)** Shipping engine, module (uninstalled), and major assemblies (uninstalled). Include the reason for shipment, the destination, and other pertinent information.

14.4.1.2.18.6. **(Added-AFMC)** Receiving/acceptance inspections, transfers and shipment of engines to designated repair facilities for engine, module (uninstalled), and major assemblies (uninstalled). Include received or departing from unit, discrepancies, and other pertinent information. **(T-2)**

14.4.1.2.18.7. **(Added-AFMC)** Data corrections. Include corrections to erroneous entries and other pertinent information. **(T-2)**

14.4.1.2.18.8. **(Added-AFMC)** Work completed on engine, module (uninstalled), and major assemblies (uninstalled). Include total time (EOT, TACs, CCYs, etc.) and a brief maintenance summary to include major assemblies replaced, test cell run, engine preservation, annual engine records reviews and other pertinent information (e.g., FOD). **(T-2)**

14.4.1.2.19. Check life-limited components forecast for additional component changes, TCTOs and SIs on all removed engines. **(T-2)**

14.4.1.2.20. Coordinate with the propulsion Flight CC/SUPT to develop a detailed 6-month engine and module TCI removal forecast and publish the forecast in the monthly flying and maintenance schedule. **(T-2)**

14.4.1.2.20.1. This 6-month forecast must be accomplished monthly using CEMS product E373/MIS products and the projected unscheduled removals based on the Unscheduled Removal Rate. **(T-2)**

14.4.1.2.20.2. Removal rate formula (total number of unscheduled removals divided by flying hours, multiplied by 1000). Provide a copy of the forecast to maintenance leaders, PS&D, unit, and the MAJCOM engine manager. **(T-2)**

14.4.1.2.21. Publish scheduled engine changes in the weekly and monthly maintenance schedule. **(T-2)**

14.4.1.2.22. Verify engine total time versus aircraft total time, flying hours and manual cycles with PS&D during aircraft document reviews. **(T-2)**

14.4.1.2.23. Maintain the portion of the JML for engine inspections and time changes. **(T-2)**

14.4.1.2.23.1. Maintain (load, delete, and change) and conduct a semi-annual review of the JML for engine inspections and time changes listed in the aircraft -6 TO. **(T-2)**

14.4.1.2.24. Establish a CEMS and MIS contingency plan for when either or both systems are down for more than 48 hours. **(T-2)**

14.4.1.2.24.1. The plan will include procedures for retaining data in date-time order for input when MIS/CEMS operation resumes. **(T-2)** The plan will also address both home station and deployed procedures. **(T-2)**

14.4.1.2.25. Develop local engine tracking procedures and documentation methods to be used at deployed locations. **(T-2)**

14.4.1.2.25.1. Procedures must include the method of communication (message, e-mail or FAX), documentation and shipping responsibilities with SRAN addresses, and reporting procedures for CANNs and engine removals. **(T-2)**

14.4.1.2.25.2. Procedures will ensure units take immediate action to correct all reporting errors between the base MIS and CEMS using the engine manager's data list. **(T-2)**

14.4.1.2.26. Update CEMS to reflect the four letter GEOLOC for uninstalled engines in a deployed location and the Organization Code is showing "1" per TO 00-25-254-1.

14.4.1.2.27. Accomplish UEM duties IAW, AFI 23-101, AFPAM 63-129, TOs 00-25-254-1, 00-20-5-1-3, AFCSM 21-558, applicable aircraft -6 TOs and this instruction.

14.4.1.2.27.1. Act as liaison to the SRAN engine manager when part of a tenant unit is supported by the host base engine manager. **(T-2)**

14.4.1.2.27.2. Provide the primary SRAN engine manager all quarterly reporting information required for submission to higher headquarters. **(T-2)**

14.4.1.2.27.3. **(Added-AFMC)** Coordinate with the propulsion flight to ensure a copy of the latest work folder accompanies permanent and transferred engines. **(T-2)**

14.4.1.3. The SRAN engine manager will:

14.4.1.3.1. Be selected from AFSC 2A or 2R, minimum 7-skill level (or civilian equivalent). **(T-2)**

14.4.1.3.1.1. The assistant will be a minimum 5-skill level from the same AFSCs or civilian equivalent. **(T-2)**

14.4.1.3.1.2. Both individuals will be aligned under EM. **(T-2)**

14.4.1.3.2. Advise CMS or MXS/CC and MXG/CC (or equivalent), on administration of the base EM Program, engine maintenance concepts, principles, policies, procedures, and techniques. **(T-2)**

14.4.1.3.3. Act as the single point of contact between the unit, MAJCOM, and MMA for EM questions. **(T-2)**

14.4.1.3.4. Establish written procedures to support EM responsibilities IAW TO 00-25-254-1 and this instruction. Unit procedures must:

14.4.1.3.4.1. Specify responsibilities of affected work centers for accurate and timely MIS/CEMS reporting of TCTO, SI, TCI, and other documentation requirements (such as, borescope inspections, blade blending, CANN actions). **(T-2)**

14.4.1.3.4.2. Ensure engine, module, and component operation and maintenance data is reported to EM no later than close of business the first duty day after the event (for example, part removal, installation, time update, TCTO status change, and EMS operational/performance data where applicable). **(T-2)**

14.4.1.3.4.2.1. Ensure Engine Monitoring System (EMS) operational/performance data is transferred to CEMS where applicable. **(T-2)**

14.4.1.3.4.3. Address tenant, transportation, maintenance, aircraft distribution, supply, and support personnel requirements and be coordinated with the MAJCOM EM prior to publication. **(T-2)**

14.4.1.3.5. Request Initialization Decks (I-Deck) for engines and major modules (cores, High Pressure Turbine (HPT), Low Pressure Turbine (LPT), fans), to include embedded parts, part number, serial number, EOT, inspections, active TCTOs and TCIs, from CEMS Central Database (CDB) and ensure data in the MIS matches the CEMS CDB. **(T-2)**

14.4.1.3.6. Ensure deployed engine monitors are identified and trained to perform duties while deployed. **(T-2)**

14.4.1.3.6.1. Designated engine monitors will ensure all deployed spare engines have a copy (paper or electronic) of CEMS product E407, option 1 and 4, included in the deployment package. **(T-2)**

14.4.1.3.6.2. **(Added-AFMC)** Maintain original copies of historical records at home station unless directed to transfer the engine to a gaining Stock Record Account Number. **(T-2)**

14.4.1.3.6.3. **(Added-AFMC)** Perform engine manager duties for shipment and receipt of all assigned modules, related equipment, and shipping devices. **(T-2)**

14.4.1.3.7. Perform engine manager duties for shipment and receipt of all assigned engines. **(T-2)**

14.4.1.3.8. Perform periodic quality audits to monitor accuracy and timeliness of reporting. **(T-2)**

14.4.1.3.9. Perform annual EM training for all affected personnel (back shop, test cell, flightline, aircraft maintenance scheduler) who report engine status or are responsible for engine documentation and scheduling IAW AFCSM 21-558, TOs 00-25-254-1 and 00-20-1/-2.

14.4.1.3.10. Maintain a jacket file of engine shipping documents and receipts. **(T-2)**

14.4.1.3.10.1. Obtain MAJCOM EM approval prior to returning engines to CRF/depot. **(T-2)**

14.4.1.3.11. Perform duties and requirements for engine shipments IAW TOs 00-25-254-1, 00-85-20, 2J-1-18, and 2-1-18.

14.4.1.3.11.1. Engines requiring off-base shipment must be delivered to transportation within 24 hours of notification/decision to ship the engine and/or the engine change is complete. **(T-2)** Notify MAJCOM EM and the owning SRAN EM if this time frame cannot be met.

14.4.1.3.12. The work folder will transfer with the engine. **(T-2)**

14.4.1.3.12.1. A copy will be maintained by the losing organization until verification of receipt by gaining unit. **(T-2)**

14.4.1.3.12.2. Gaining units will maintain the work folders and ship the documents with the engine to depot when appropriate. **(T-2)**

14.4.1.3.12.2.1. Gaining units will retrieve a copy of the previous EAWP from the Data Repository Center or equivalent data in the applicable MIS upon receipt of the engine. **(T-2)**

14.4.1.3.12.2.2. EAWP users are required to send completed EAWP files to the Data Repository Center or MIS equivalent within 3 duty days of EAWP close-out. **(T-2)**

14.4.1.3.13. The SRAN EM will report the following in CEMS:

14.4.1.3.13.1. Receipt transactions for engines as of the date and time engines are delivered from the transportation hold area and accepted at the JEIM facility. **(T-2)**

14.4.1.3.13.2. Shipment transactions with the "as of" date and time the engine(s) physically leave the base. **(T-2)**

14.4.1.3.13.2.1. Once engine is received at gaining unit, ensure trailer and adapter are transferred in MIS. **(T-2)**

14.4.1.3.13.3. All engine and tracked item removals, installations, and engine status changes. **(T-2)**

14.4.1.3.13.4. All engine status transaction removals, installations, gains, Engine-Not-Mission Capable for Supply (ENMCS), work completed, test cell rejects, work stopped, work started, change in level of maintenance, awaiting disposition, intra-AF receipt and intra-AF shipments, transfer, and HOW MAL codes IAW TO 00-25-254-1.

14.4.1.3.13.5. **(Added-AFMC)** Use the MAJCOM's command code the aircraft

is assigned to when processing engine installation in CEMS. Refer to TO 00-25-254-1 for guidance. **(T-2)**

14.4.1.3.13.6. **(Added-AFMC)** Develop a MOA IAW AFI 25-201, *Intra-Service, Intra-Agency, and Inter-Agency Support Agreements Procedures*,, with Stock Record Account Number s at pre-positioned sites to manage engines, and manage tenant spare engines. **(T-2)**

14.4.1.3.14. Verify all update transactions (such as, times, TCTO, part removal and installations) are input before reporting an engine removal or installation. **(T-2)**

14.5. Maintenance and FHP Planning Cycle.

14.5.1. Responsibilities. MAJCOMs will develop procedures to ensure the intent of the maintenance and FHP planning cycle is met. The objective of the planning cycle is to execute the wing FHP consistent with operational requirements and maintenance capabilities. The maintenance and FHP planning cycle begins with the annual allocation of flying hours. Maintenance and operations schedulers propose an annual flying plan that balances both operational requirements and maintenance capabilities. Units should commit the fewest number of aircraft possible to meet programmed Utilization (UTE) rate standards and goals.

14.5.1. **(AFMC)** Refer to Attachment 8.

14.5.1.1. If applicable, MAJCOMs will develop scheduling procedures for units involved in Operational Test and Evaluation, Developmental Test and Evaluation, or Initial Operational Test and Evaluation to ensure the intent of the maintenance and FHP planning cycle is met.

14.5.1.2. AMC units tasked by the 618th Air and Space Operations Center will adhere to Commander, Air Force Forces (COMAFFOR) Apportionment and Allocation Process (CAAP) policies and procedures.

14.5.1.3. The annual plan, detailed by month, will evaluate the capability of maintenance to support the annual FHP.

14.5.1.4. When developing the annual plan, units will utilize the MDS specific sortie production model, or equivalent, if available.

14.5.1.5. Maintenance PS&D. PS&D builds, coordinates, publishes, and distributes an integrated aircraft/system annual and quarterly plan & monthly and weekly schedule to support maintenance and operational requirements.

14.5.1.5. **(AFMC)** See **Attachment 8** for additional information on planning for test unit scheduling.

14.5.1.5.1. Plans will be developed, coordinated, and consolidated jointly by the OSS's Current Operations Flight Scheduling, and PS&D. **(T-2)**

14.5.1.5.2. The printed wing plan will include an assessment of the wing's ability to execute the FHP and will be coordinated with the OG/CC and MXG/CC. **(T-2)**

14.5.1.5.3. Plans and schedules may be published via electronic means (such as, web pages, SharePoint®, or e-mail) provided operations security is not compromised. **Note:** TORQUE will be used as it becomes available.

14.5.1.5.4. Normal daily operations and training schedules are For Official Use Only (FOUO) and will not be restricted to classified systems. **(T-2)**

14.5.1.5.5. **(Added-AFMC)** The principal area of concern is the overall flying/ground test schedule. For mission accomplishment and improved efficiency, the following must be considered: maximize crew training on all flights, plan alternate missions when possible, ensure configurations and fuel loads are accurate, and establish launch and recovery patterns, and utilize historical attrition data. **(T-2)**

14.5.2. First Look Requirements. The First Look report is an internal wing document intended to highlight potential maintenance-capacity and operational-requirement disconnects in the upcoming year. Every year, NLT 15 March, PS&D will coordinate with MMA, flightline and backshop maintenance work centers to provide PS&D with historical attrition and projected manning production. **(T-2)** This assessment will take into account personnel, facilities, and airfield infrastructure for each aircraft maintenance organization.

14.5.2.1. In wings operating aircraft supported by a sortie production model, PS&D will coordinate with the Unit Maintenance Supervision to establish local requirements, responsibilities and procedures for utilizing the model to develop, sustain or reflow FHP/contingency requirements. **(T-2)**

14.5.2.2. The assessment will be provided to PS&D NLT the last workday of March. **(T- 2)** **Note:** AMC units tasked by the 618th Air and Space Operations Center will adhere to the CAAP policies and procedures.

14.5.2.3. PS&D will provide copies of the capability assessment to each OS scheduling section and maintenance supervision. **(T-2)**

14.5.2.3.1. The assessment will provide first look maintenance capability projections in a monthly format IAW MAJCOM guidance. **(T-2)**

14.5.2.3.1. **(AFMC)** Refer to Attachment 8. **(T-2)**

14.5.2.3.2. The assessment will include operational requirements taking into consideration historical data that determines the average number of aircrews not available per month (DNIF, PME attendance, Leave, TDY, deployments), an assessment of maintenance ability to support the monthly requirement and an overall assessment of the unit's maintenance capability to meet the annual FHP (N/A for AMC units). **(T-2)**

14.5.2.4. OS and maintenance responses are sent to PS&D and OSS's Current Operations Flight Scheduling and will be consolidated into a comprehensive package that includes a breakdown of the following items by OS:

14.5.2.4.1. Sortie UTE Rates (N/A to AMC units). **(T-2)** Compute UTE rates by month for the entire FY for contracted (required) sorties and scheduled sorties using the formula: (number of sorties per month) divided by (number of Primary Aerospace Vehicle (Aircraft) Inventory (PAI) aircraft).

14.5.2.4.2. Sorties contracted/scheduled per day (N/A to AMC units). **(T-2)** Compute the number of sorties required per operations and maintenance (O&M) day to meet the operational requirement using the following formula: (Number of Sorties Required)

divided by (Number of O&M days in a Given Month). Sorties per day need to be computed by month for the entire FY.

14.5.2.4.3. Monthly scheduled sorties (N/A to AMC units). **(T-2)** Compute monthly scheduled sortie requirements using the following formula: (Number of Sorties or Hours Required) divided by (1 Minus the Attrition Factor). For example, (1,000 sorties or hours required) divided by (1 minus 0.15) equals 1,177 sorties or hours to schedule. Round any part to the next whole sortie or hour.

14.5.2.4.4. Inspection dock capability. **(T-2)**

14.5.2.4.4.1. Compute the number of major inspections (i.e., PH/ISO) to be accomplished for each maintenance unit, by month, for the entire FY in order to meet operational requirements.

14.5.2.4.4.2. Compute dock capability using the following formula: (Number of O&M Days) divided by (Number of PH/ISO Days) multiplied by (Inspection Cycle) = Inspection Dock Capability. Inspection dock capability is provided at the wing level and provided by the squadron performing inspections.

14.5.2.5. Once compiled, first look packages will be presented to the OG and MXG/CCs before being presented to the WG/CC. **(T-2)**

14.5.2.5. **(AFMC)** Final assessments of maintenance capabilities to support the operations "first look" projections will be sent to AFMC/A4/10, for awareness. If there are unresolved conflicts between operations and maintenance also send the assessment to AFMC Air, Space and Cyberspace Operations Directorate (AFMC/A3/6) and include any actions requested of MAJCOM **(T-2)**

14.5.3. Annual Maintenance Planning Cycle.

14.5.3.1. MAJCOMs will develop procedures to ensure the objectives of the annual maintenance planning cycle are met.

14.5.3.1.1. At a minimum, MAJCOM procedures will produce an annual flying and maintenance plan that allocates sorties and hours into quarters, is approved by the WG/CC, and published prior to the beginning of the FY.

14.5.3.1.1. **(AFMC)** Refer to Attachment 8 for Annual Maintenance Planning Cycle procedures.

14.5.3.1.2. Due to the unpredictable nature of most future AMC mission requirements, units tasked by 618th Air and Space Operations Center will prepare flying and maintenance plans with focus on supporting local operational training requirements based on historical data as well as all known future maintenance and operational requirements.

14.5.3.2. Flying Hour Allocation. Using the MAJCOM Baseline Allocation message, PS&D, the OS, and OSS's Operations Scheduling will provide affected work centers the following planning factors NLT 20 August each year, or within 10 working days after receipt of the flying hour allocations:

14.5.3.2.1. PS&D will provide updated capabilities which are computed by MMA and the PDM schedule. **(T-2)**

14.5.3.2.2. OSS will provide the:

14.5.3.2.2.1. Required flying hours and estimated sorties and missions in monthly increments. **(T-2)**

14.5.3.2.2.2. Flying days in each month. **(T-2)**

14.5.3.2.2.3. Aircraft and aircrew alert requirements. **(T-2)**

14.5.3.2.2.4. Known and projected TDYs and special mission requirements. **(T-2)**

14.5.3.2.2.5. Configuration and munitions requirements. **(T-2)**

14.5.3.3. NLT 1 September, or within 10 working days after receipt of the planning factors, maintenance supervision will provide PS&D, SQ/CCs, and OSS's Operations Scheduling the following planning factors:

14.5.3.3.1. Estimated number of aircraft available by month, taking into consideration aircraft required for training. **(T-2)**

14.5.3.3.2. A projected airframe capability statement. **(T-2)**

14.5.3.3.3. Forecasted personnel capability, taking into consideration required training for maintenance personnel. **(T-2)** (N/A to contract maintenance organizations).

14.5.3.3.4. The number of supportable sorties for each month. **(T-2)**

14.5.3.3.5. An estimated monthly attrition factor provided by MMA. **(T-2)**

14.5.3.3.5.1. The factor combines operations, weather, and materiel (maintenance and supply) factors.

14.5.3.3.5.2. Maintenance is responsible for adding the attrition factor to operational requirements.

14.5.3.3.6. A recommended block scheduling pattern. **(T-2)**

14.5.3.3.7. A statement of limitations. **(T-2)**

14.5.3.3.7. **(AFMC)** Refer to Attachment 8 for test organization limitations and planning factors.

14.5.4. Quarterly Maintenance and FHP Planning. Quarterly planning starts with the operational requirement for flying hours, UTE rate, airframe availability, alert, and other related scheduling data.

14.5.4.1. MAJCOMs will develop procedures to ensure the objectives of the Quarterly Planning cycle are met.

14.5.4.1. **(AFMC)** Refer to Attachment 8 for Test Wing scheduling planning factors. **(T-2)**

14.5.4.2. The OS Director of Operations will provide these requirements to maintenance supervision and PS&D NLT 25 days before the beginning of the quarter. **(T-2)**

14.5.4.3. Maintenance supervision and the OS Director of Operations will discuss these requirements at the scheduling meeting before the quarter being planned. **(T-2)**

14.5.4.4. Schedulers will ensure quarterly plans are as detailed and accurate as possible. **(T-2)**

14.5.4.4.1. Plans should include known special missions, PDM schedules, HHQ commitments and lateral command support requirements.

14.5.4.4.2. All maintenance requirements will be consolidated into a single, quarterly plan using AF Form 2401, Equipment Utilization and Maintenance Schedule, or computer-generated form. **(T-2)**

14.5.4.4.2.1. Specific locally developed codes will be used to identify inspections, SI, TCI, and TCTO on the AF Form 2401. **(T-2)**

14.5.4.4.3. As a minimum, the quarterly plan will show the next 3 months planned sorties and known maintenance requirements. **(T-2)**

14.5.4.4.3.1. Known maintenance requirements include all maintenance events that impact aircraft availability and require management attention to ensure proper Time Distributed Index flow.

14.5.4.4.3.2. Consolidate as many scheduled maintenance events as practical, to reduce individual aircraft downtime, increase Aircraft availability, and minimize the number of times per month an aircraft is removed from the schedule due to scheduled maintenance requirements.

14.5.4.4.3.2.1. The intent is to reduce the number of times per month an aircraft is removed from the schedule for scheduled maintenance, thus increasing aircraft availability.

14.5.4.4.3.2.2. Unit/Wing/MAJCOM requests to change the frequency of -6 TO requirements to increase bundling opportunities will be submitted through the applicable Lead Command for consideration and/or resolution. **(T-2)**

14.5.4.4.3.3. To prevent operational utilization for that day(s) flying schedule, the quarterly plans will include, at a minimum, calendar inspections that hold an aircraft down, calendar TCIs, TCTOs in workable status, PDM schedules, training aircraft, cannibalization aircraft and aircraft ISO/PE/PH inspections. **(T-2)**

14.5.4.4.3.4. Other maintenance requirements, such as engine changes, hourly requirements, acceptance/transfer inspections, training aircraft and cannibalization aircraft will be posted as they become known or planned. **(T-2)**

14.5.4.4.3.5. Add AME inspections to the quarterly plan if the aircraft is scheduled to stay in that configuration to ensure the inspections are included in the monthly and weekly schedules. **(T-2)**

14.5.4.4.4. Revise weekly schedule and monthly plan to meet the quarterly plan objectives while staying within the maintenance capability. **(T-2)**

14.5.4.4.5. Use the following priority to determine which objectives to support if a lack of resources prevents meeting requirements:

14.5.4.4.5.1. Alert commitments. **(T-2)**

14.5.4.4.5.2. HHQ directed missions. **(T-2)**

14.5.4.4.5.3. Training. **(T-2)**

14.5.4.4.5.4. OSS's Current Operations Flight Scheduling will compile, coordinate, and brief the unit's quarterly plan and include operational requirements, support capability and any difficulties expected. **(T-2)**

14.5.4.4.5.5. Once an approved quarterly plan is established, OSS's Current Operations Flight, Scheduling will forward a copy to the OS, AMXS, OG/CC and MXG/CC along with all scheduling agencies. **(T-2)**

14.5.4.4.5.6. The plan will be posted so it may be viewed by both maintenance and operations. **(T-2)**

14.5.4.5. - 14.5.4.5.3. DELETED.

14.5.4.5. **(AFMC)** Refer to Attachment 8 for annual flying plan additional requirements. **(T-2)**

14.5.5. Monthly Maintenance and FHP Planning.

14.5.5.1. Wings will develop procedures to ensure the objectives of the monthly planning cycle are met. **(T-2)**

14.5.5.1. **(AFMC)** Refer to Attachment 8 for test scheduling flexibility. **(T-2)**

14.5.5.1.1. Include predictable maintenance factors based on historical data along with other inputs, such as flow times for maintenance, turnaround times and parts replacement schedules.

14.5.5.1.2. MAJCOMs will develop maintenance scheduling effectiveness guidance in their supplements to this DAFI. **(T-2)**

14.5.5.1.2. **(AFMC)** Refer to Attachment 8 for maintenance scheduling effectiveness guidance. **(T-2)**

14.5.5.2. The monthly flying and maintenance plan schedule refines the quarterly plan by combining all aspects of aircraft utilization and will include:

14.5.5.2.1. A detailed monthly operations utilization calendar that specifies total aircraft flying hours, total sorties and missions, alert requirements, scheduled sortie or mission requirements and daily turn plans for each MDS by squadron, group, or wing. **(T-2)**

14.5.5.2.1.1. Do not assign attrition sorties to a specific aircrew/mission for the monthly planning process. **(T-2)**

14.5.5.2.2. Monthly maintenance requirements (as required). **(T-2)**

14.5.5.2.3. Transient work schedule, if applicable. **(T-2)**

14.5.5.2.4. Scheduled inspections, TCTOs, engine changes, time changes, DDs, contract or depot maintenance, washes, corrosion control, training aircraft and all other known maintenance requirements. **(T-2)**

14.5.5.2.5. SE scheduled inspections, contract, or depot maintenance, TCTOs, time changes, DDs, washes, and corrosion control. **(T-2)**

14.5.5.2.6. Avionics and other off-equipment maintenance scheduled inspections, TCTOs, assembly or repair operations. **(T-2)**

14.5.5.2.7. Engine/module 6-month removal forecast and in-shop inspection requirements. **(T-2)**

14.5.5.2.8. Munitions, ECM, and other mission loading or configuration requirements, including ammunition changes. **(T-2)**

14.5.5.2.9. Total ordnance requirements for aircraft support. **(T-2)**

14.5.5.2.10. Tanks, Racks, Adapters and Pylons and WRM scheduled inspections, TCTOs, assembly or repair operations. **(T-2)**

14.5.5.2.11. Monthly training schedules, if not published separately. **(T-2)**

14.5.5.2.12. Detailed support requirements (such as, Petroleum, Oil, and Lubricants servicing, supply, food service, fire department, security, civil engineer, and airfield operations requirements). **(T-2)**

14.5.5.2.13. All known operational events (such as, exercises, deployments, surges) to determine maintenance capability to meet operational needs. **(T-2)**

14.5.5.2.13. **(AFMC)** Refer to Attachment 8 for test operational needs and maintenance requirements for monthly scheduling.

14.5.5.2.14. **(Added-AFMC)** The monthly maintenance plan may be published separately or as an annex to the last weekly schedule of the preceding month. If published separately, distribute the plan NLT five calendar days before the beginning of the month. **(T-2)**

14.5.5.3. Monthly planning cycle requirements.

14.5.5.3.1. NLT the first weekly scheduling meeting of the month, the OS Director of Operations will provide maintenance supervision and PS&D with the estimated operational needs for the following month in as much detail as possible. **(T-2)**

14.5.5.3.1.1. To optimize aircraft and munitions support, CMS, EMS, MUNS, MXS, AMXS, and OS will ensure the number of aircraft, and/or munitions configurations, are minimized and standardized. **(T-2)**

14.5.5.3.1.2. Include known takeoff times, landing times and flying hour windows. **(T-2) Note:** Landing times are not required if the unit has an established and constant average sortie duration.

14.5.5.3.2. The OS Director of Operations and maintenance supervision will review their applicable portion of the monthly maintenance plan and weekly schedule prior to submission to PS&D. **(T-2)**

14.5.5.3.3. NLT the second weekly scheduling meeting of the month, AMXS maintenance supervision will notify the OS Director of Operations whether requirements can be met or limitations exist and collectively make necessary adjustments to the proposed schedule to satisfy maintenance and operational requirements. **(T-2)**

14.5.5.3.4. MXG/CC and OG/CC will formalize the next month's flying and maintenance plan prior to presenting it to the WG/CC for approval NLT the third scheduling meeting of the preceding month. **(T-2)**

14.5.5.4. WG/CC's monthly scheduling meeting.

14.5.5.4.1. OS scheduling will outline past accomplishments, status of flying goals, problems encountered and detailed needs for the next month. **(T-2)**

14.5.5.4.2. PS&D will outline projected maintenance capability and aircraft/equipment availability. **(T-2)**

14.5.5.4.3. DELETED.

14.5.5.5. When the WG/CC approves/signs the proposed monthly flying plan, PS&D will include it as a portion of the monthly flying and maintenance plan. **(T-2)** Monthly plans may be published electronically provided local security requirements are met.

14.5.6. Weekly Scheduling. The weekly schedule is the final refinement to the monthly plan and results in the weekly flying and maintenance schedule.

14.5.6.1. Wings will develop procedures to ensure the objectives of the weekly scheduling process are met. **(T-2)**

14.5.6.1. **(AFMC)** Refer to Attachment 8 for Test Wing weekly scheduling guidance.

14.5.6.1.1. PS&D will review matrix/chart depicting the total number of SI and TCI requirements to be loaded in the MIS for each assigned aircraft/system and verify against the MIS totals weekly. **(T-2)** Overdue and uncorrected discrepancies will be briefed weekly during a daily production/scheduling meeting chaired by the MXG/CD (or equivalent). **(T-2)**

14.5.6.2. NLT 2 workdays before the weekly scheduling meeting, the OS Director of Operations will provide maintenance supervision the following information (as required for missile units):

14.5.6.2.1. Aircraft takeoff and landing times. **(T-2)**

14.5.6.2.2. Configuration requirements. **(T-2)**

14.5.6.2.3. Munitions requirements. **(T-2)**

14.5.6.2.4. Fuel loads. **(T-2)**

14.5.6.2.5. Special or peculiar mission support requirements. **(T-2)**

14.5.6.2.6. Alert requirements. **(T-2)**

14.5.6.2.7. Exercise vulnerability. **(T-2)**

14.5.6.2.8. Deployments. **(T-2)**

14.5.6.2.9. Off-base sorties. **(T-2)**

14.5.6.2.10. On-equipment training requirements. **(T-2)**

14.5.6.2.11. Other special requirements. **(T-2)**

14.5.6.2.12. All mission unique requirements are annotated by OS Director of Operations on the weekly and daily flying schedule. **(T-2)**

14.5.6.2.12. **(AFMC)** Refer to Attachment 8 for additional maintenance debrief considerations for sorties considered "flown as scheduled".

14.5.6.3. Home and deployed units will publish a weekly schedule. **(T-2)** Include the following in the weekly flying and maintenance schedule:

14.5.6.3.1. Sortie sequence numbers, aircraft tail numbers (primary and spares), scheduled takeoff and landing times, aircraft or equipment scheduled use times, configurations, fuel loads, and special equipment requirements. **(T-2)** Units that fly a published and constant average sortie duration need not publish land times.

14.5.6.3.2. Spare aircraft requirements. **(T-2)** Spare requirements are printed by day for each maintenance unit. Generate only the absolute minimum of spare aircraft.

14.5.6.3.2. **(AFMC)** Refer to Attachment 8 for test mission spare aircraft requirements. **(T-2)**

14.5.6.3.3. Scheduled maintenance actions, by aircraft and equipment serial number, to include inspections, TCTOs, time changes, contract and depot inputs, engine changes, washes or corrosion control, document reviews and DDs. **(T-2)**

14.5.6.3.4. Required pre-inspection and other maintenance/scheduling meetings. **(T-2)**

14.5.6.3.5. Wash rack use. **(T-2)**

14.5.6.3.6. On-equipment training requirements. **(T-2)**

14.5.6.3.7. AGE inspections or maintenance schedule by type and ID number. **(T-2)**

14.5.6.3.8. MAJCOMs will develop standardized procedures to record and coordinate changes to the weekly schedule using an AF Form 2407 or electronic equivalent. Include minimum approval levels for approving changes to the weekly schedule.

14.5.6.3.8. **(AFMC)** See **Attachment 8** for changes to the weekly schedule and Pen-and-Ink processes.

14.5.6.3.9. Any change to the printed schedule will require an AF Form 2407 with the following exceptions: a change to the original printed takeoff or landing time of 15 minutes or less; a change of aircrew names, ranges, or airspace; or a change arising after the first crew ready time for the squadron's current day's scheduled flying window. **(T-2)**

14.5.6.3.9. **(AFMC)** Refer to Attachment 8 for specific AF Form 2407, *Approval Authority*, requirements in test organizations. **(T-2)**

14.5.6.3.9.1. Changes made during the daily scheduling meeting also require an AF Form 2407. **(T-2)**

14.5.6.3.9.2. The agency requesting the change initiates the AF Form 2407 and coordinates it IAW MAJCOM procedures. **(T-2)**

14.5.6.3.9.2. **(AFMC)** Refer to Attachment 8 under Changes to the Weekly Schedule. **(T-2)**

14.5.6.4. The OS Director of Operations and Maintenance Supervision will review and coordinate on the proposed weekly flying and maintenance schedule with OS, AMXS, MXS, CMS, and EMS prior to presenting it to the OG/CC and MXG/CC (or equivalent). **(T-2)**

14.5.6.5. The approved schedule will be submitted to PS&D for compilation and a complete copy provided to the WG/CC. **(T-3)**

14.5.6.6. At the weekly scheduling meeting wings will evaluate the past week's accomplishments (to include flying and MSE) and negotiate/approve refinements to the coming week's schedule. **(T-2)**

14.5.6.6.1. The AF Form 2402, *Weekly Equipment Utilization and Maintenance Schedule,* or locally developed product, will be used to summarize the upcoming week's schedule. **(T-2)**

14.5.6.6.2. The AF Form 2403, *Weekly Aircraft Utilization/Maintenance Schedule*, or locally developed equivalent product that contains all requirements and creates a finite depiction of aircraft utilization and maintenance. **(T-2)**

14.5.6.7. Once the weekly schedule is reviewed and signed by the OG/CC, MXG/CC (or equivalent), and WG/CC it becomes the final planning guide for both operations and maintenance and the basis for deviation reporting. **(T-2)**

14.5.6.7.1. The schedule will be followed as printed or as amended by coordinated changes. **(T-2)**

14.5.6.7.2. Coordinated changes do not negate reporting deviations IAW MAJCOM guidance.

14.5.6.8. PS&D will distribute the schedule to each appropriate activity and work center NLT time determined in MAJCOM supplements to this DAFI. **(T-2)**

14.5.6.8. **(AFMC)** PS&D distributes the schedule to each appropriate activity and work center NLT 1200 Friday preceding the effective week. **(T-2)**

14.5.6.8.1. Weekly schedules may be published electronically provided local security requirements are met.

14.6. Contingency and Expeditionary Responsibilities.

14.6.1. Responsibilities of Contingency/Expeditionary (Cont/Exp) units, for example, maintaining non-assigned aircraft. **Note:** This does not include AMC-established enroute stations.

14.6.1.1. Most planning and scheduling is the responsibility of units with assigned aircraft and is provided through reach back support to home stations. Contingency units have fewer responsibilities as described below. Commanders of expeditionary units will ensure the intent of the guidance is met when the dynamic nature of a Cont/Exp organization make strict adherence impossible. **(T-2)**

14.6.1.2. Cont/Exp PS&D will conduct the following programs as outlined below:

14.6.1.2.1. ADR, pre- and post-dock meetings, acceptance inspections (from DFT/CFTs) and the major work program. **(T-2)**

14.6.1.2.1.1. Cont/Exp PS&D will use the procedures developed by the aircraft-owning organizations. **(T-2)**

14.6.1.2.1.2. If aircraft from multiple bases/units are supported, procedures do not have to be standardized.

14.6.1.3. Ensure discrepancies noted by the aircraft-owning PS&D for the Aircraft Configuration Management, TCI and SI programs are corrected. **(T-2)** Cont/Exp PS&D will not develop programs independent of the aircraft-owning organization. **(T-2)**

14.6.1.4. Develop local coordination procedures for contingency aircraft affected by Immediate and Urgent Action (I/UA) TCTOs. **(T-2)**

14.6.1.5. When notified by the aircraft-owning organization of an I/UA TCTO, the Cont/Exp PS&D will host a TCTO meeting. **(T-2)**

14.6.1.5.1. The purpose of the meeting is to determine if the Cont/Exp unit has the maintenance capability to perform the TCTO.

14.6.1.5.1.1. Invite AMU, QA and affected work centers. Cont/Exp will notify the Commander of the unit's capability to perform the TCTO. **(T-2)**

14.6.1.5.1.2. Develop and implement local tracking methodology to track TCTO completion. **(T-2)**

14.6.1.5.1.3. Update MIS when the aircraft owning Cont/Exp PS&D loads the requisite JST/JCNS. **(T-2)**

14.6.1.5.1.4. If the Cont/Exp unit does not have the maintenance capability to perform the TCTO, Cont/Exp PS&D will notify the owning organization of that inability. **(T-2)**

14.6.1.5.2. Cont/Exp PS&D will only maintain aircraft I/UA TCTO files while active. **(T-2)**

14.6.1.5.2.1. Once TCTOs are completed and loaded in MIS, records will be sent to home station for filing. **(T-2)**

14.6.1.5.2.2. A TCTO meeting is not necessary for Routine Action aircraft TCTOs.

14.6.1.5.3. A full TCTO program IAW this **Chapter** is required for AGE and other special equipment which is assigned to the contingency unit. **(T-2)** This is intended to cover equipment that does not rotate with aviation packages.

14.6.1.5.4. Monthly and weekly maintenance planning. Cont/Exp PS&D will produce maintenance plans detailing all known maintenance requirements for the upcoming month/week. **(T-2)**

14.6.1.5.4.1. This plan will detail by tail number, due date, JST and a description of the scheduled maintenance required for the time period. **(T-2)**

14.6.1.5.4.1.1. Use of the AF Form 2401 is not required.

14.6.1.5.4.1.2. The list will be published 2-days prior to the covered time period, coordinated through maintenance supervision, and approved by the Commander. **(T-3)**

14.6.1.5.4.2. The weekly schedule will additionally identify those actions which will be deferred. **(T-2)**

14.6.1.5.4.2.1. It will specifically identify if the action is deferred for mission requirements or due to a lack of capability. **(T-2)**

14.6.1.5.4.2.2. Actions which are not identified as "pre-deferred" are expected to be accomplished during the upcoming week.

14.6.1.5.4.3. MSE will not be calculated for Cont/Exp units. **(T-2)** It is anticipated that Cont/Exp units require a great deal of flexibility to meet mission requirements.

14.6.1.6. Cont/Exp organizations are not responsible for Aircraft Generation Planning, Transfer Inspections, Flying/Maintenance Planning Cycle, First Look, Annual/Quarterly Mx Planning, AVDO and 2R1 functional management responsibilities.

14.6.1.7. Cont/Exp PS&D will develop procedures with home station AVDOs to communicate and ensure AVDO responsibilities are performed. **(T-2)**

14.7. (Added-AFMC) Depot Pre-Induction Inspection (DPII). DPIIs are designed to determine the aircraft condition lead-time before induction to depot maintenance. Depot teams use the aircraft condition knowledge to establish a supportable Bill of Work process for specific repair actions, parts, tools, and equipment; i.e., generating what needs to be done, with what, and ensuring it will be there when planned. DPIIs are synchronized with the HSC, Phase, or Isochronal (ISO) prior to depot input on MDS directed by program office. DPII can be a physical inspection or a pre-induction analysis of records. The determination on who will accomplish the DPII (Depot Team, Contract Team, or Field Unit) will be negotiated during the Fleet Scheduling Conferences. At that time the Depot schedules are established with arrival dates of the aircraft at the respective Air Logistic Complexes. As part of the conference the DPII will be "scheduled" ~9 months prior to induction to PDM. The MAJCOM/Unit Representative will declare their intent to accomplish "Organizationally" or request Depot Field Team/CFT support for the DPII. If none of these options are feasible at a particular time, any aircraft not receiving a DPII would still be inducted into PDM. **(T-2)**

14.7.1. **(Added-AFMC)** If a depot or contract team is accomplishing the PII, the PS&D Section Chief or designated representative will:

14.7.1.1. **(Added-AFMC)** Be provided a visit agenda by the DPII team lead. **(T-2)**

14.7.1.2. **(Added-AFMC)** Coordinate with PII team lead to present situational awareness in/out-brief to the MXG as required. **(T-2)**

14.7.1.3. **(Added-AFMC)** Coordinate/request tail number data pull queries from applicable MIS. **(T-2)**

14.7.1.4. **(Added-AFMC)** Act as liaison and refer DPII team to maintenance units for equipment support as required. **(T-2)**

14.7.1.5. **(Added-AFMC)** Maintain a file copy of the DPII report from DPII Team Lead. **(T-2)**

Chapter 15

AIRCRAFT SUNSHADE MANAGEMENT

15.1. Purpose: This **Chapter** outlines overall management responsibilities for aircraft sunshades, crew shelters and portable/inflatable shelters. Additionally, it provides guidance and a standardized approach to procure and sustain these equipment items.

15.2. Scope: Applies to all AF units possessing or procuring these equipment items to support logistics operations and flight line or maintenance areas. This includes conventional force logistics units; nuclear and nuclear support units; cyber and space; special operations; wholesale-level procurement, sustainment, and maintenance; aerial port squadrons; logistics readiness squadrons; research, test, and development units. It does not include medical logistics, civil engineering logistics, security forces units and communications units.

15.2.1. Aircraft sunshades and other shelter types as defined below and addressed in this **Chapter** are considered equipment items for accountability purposes and will be tracked and accounted for IAW DAFI 23-101 and DAFMAN 23-122.

15.2.1.1. These assets are not listed on unit authorized equipment tables of allowance because authorizations are ***not required*** support equipment for weapons systems. These assets are the responsibility of the owning organization.

15.2.1.2. Commanders will ensure all other applicable organizations with appropriate subject matter expertise are involved in the procurement, installation, repair, sustainment, and replacement of these assets. **(T-2)**

15.2.1.3. Aircraft sunshades are not considered as temporary or relocatable facilities associated with Military Construction (MILCON) projects. Procedures for temporary or relocatable facilities associated with MILCON projects are contained in DAFI 32-1020, *Planning and Programming Built Infrastructure Projects*. Relocatable buildings are designed to be readily erected, disassembled, stored, moved, and reused to meet a short-term requirement for facilities due to either transitory peak requirements or urgent requirements pending approval, such as construction of facilities via normal military construction programs. According to DAFI 32-1020, **Chapter 8**, Relocatable and Temporary Facilities, the terms temporary and relocatable are interchangeable. Furthermore, DAFI 32-1020 provides guidance on satisfying interim facility requirements using relocatable or temporary facilities. Aircraft sunshades are not incident or interim to a MILCON project, nor are they in place to meet a temporary requirement; therefore, the terms temporary or relocatable as used in DAFI 32-1020 do not apply to aircraft sunshades.

15.2.1.4. Aircraft sunshades are not intended to replace aircraft maintenance hangars and as such there is no requirement associated with aircraft specifications or square footage allowances contained in DAFMAN 32-1084.

15.3. Definitions.

15.3.1. Aircraft Sunshades: Structure with the sole purpose of providing minimal protection for personnel from the elements (sun, wind, rain, snow, excluding lightning) with a roof and a maximum of two sides.

15.3.2. Crew Shelters: Any portable or prefabricated structure placed inside, under or erected to support activities within an aircraft sunshade, hangar, hardened aircraft shelter or protective aircraft shelter. These should not be confused with other shelters that are designed for personnel protection only (such as, End-of-Runway shacks).

15.3.3. Portable/Inflatable shelters: A structure constructed using layers of membrane connected together using pressurized air to produce a structure which covers limited areas (such as, cargo aircraft engine, open fuel tanks) of an aircraft or equipment.

15.3.4. Allied support: Utilities (electrical, water, air, communications) provided up to a stub at each aircraft sunshade. Utilities are real property assets. Installation of new utility services carries a work classification of Construction. See **Paragraph 15.11**.

15.4. Headquarters Air Force.

15.4.1. AF/A4L will:

15.4.1.1. Develop, articulate, and clarify all AF aircraft sunshade policies.

15.4.2. The Directorate of Civil Engineers (AF/A4C) will:

15.4.2.1. Provide input for development and clarification of aircraft sunshade policies.

15.4.2.2. Determine appropriate A4C agencies responsible for procurement consultation to the field.

15.4.2.3. Develop allied support guidance in relation to aircraft sunshades.

15.5. MAJCOM A4s will:

15.5.1. Provide input to AF/A4L on AF aircraft sunshade policies.

15.5.2. Review all plans for procurement, installation, and sustainment of aircraft sunshades with assistance from MAJCOM/A3, A4, Safety, AF Installation and Mission Support Center (AFIMSC) or Primary Subordinate Unit (PSU) Air Force Civil Engineer Center (AFCEC) as outlined in Paragraphs **15.7.1.11** and **15.8**, prior to approving the procurement or installation.

15.5.3. Develop weather event baseline criteria if not previously developed so an organization can procure an acceptable aircraft sunshade suitable to its specific location. Consider annual days of sun, average wet bulb globe temperature, annual rainfall, and annual snowfall, UV index, snow load, snow removal capability, wind sustainment requirements and temperatures that may affect work-rest cycles and overall productivity.

15.5.4. DELETED.

15.5.5. Approve/disapprove submitted request(s) based on information contained under the owning organization responsibilities section of this **Chapter**.

15.5.6. Provide field operating and sustaining organizations direction on the specific types and sizes of aircraft sunshades to use for each application.

15.6. Owning Organization.

15.6.1. The owning organization is the group level or equivalent organization responsible for the planning, funding, procurement, installation, maintenance, and lifetime sustainment of the aircraft sunshade including any installed utilities (lighting, obstructions in front of lighting,

electrical, communications included in the project scope as defined in Paragraphs **15.3.4** and **15.8.8**. **(T-2)**

15.6.1.1. Owning organizations may install aircraft sunshades, upon MAJCOM/A4 concurrence, to provide protection from the sun or other weather events such as: rain, sleet, snow. **(T-2)**

15.6.2. Owning organizations are responsible for the aircraft sunshade grounding system beginning at the single point service ground. Owning organizations will test the aircraft sunshade grounding system IAW AFMAN 32-1065, *Grounding & Electrical Systems*, Chapter 10, *Testing and Inspecting Static and Lightning Protection Systems and Grounding*. **(T-2)** **Note:** These responsibilities can be contracted as outlined in **Paragraph 15.8.8**.

15.6.2.1. Owning organizations will ensure that Base Civil Engineer accepts the Lightning Protection Systems prior to accepting the facility, ensuring the lightning protection system is compliant and that the facility is immediately usable IAW AFMAN 32-1065. **(T-2)**

15.6.2.1. **(AFMC)** Owning organizations will ensure that a certified LPS inspection is conducted by someone other than the designer/installer of the system prior to the acceptance of the project, per AFMAN 32-1065, paragraph 2.6.

15.6.3. The owning organization is responsible for maintaining and storing manufacturer design drawings and specifications and providing ready access to installation agencies as necessary. **(T-2)**

15.6.4. For each procurement, and each aircraft sunshade related contracting effort (such as inspection and sustainment contracts) referenced in **Paragraph 15.10.3** of this publication, the owning organization will provide and appoint a Contracting Officer Representative (COR) who reports to the contracting officer on all aspects of implementation of the contract. **(T-2)**

15.6.4.1. The COR will coordinate agreement with local Civil Engineering organization request a Civil Engineering representative to advise on technical requirements of the procurement and sustainment. **(T-2)**

15.6.4.2. The CE advisor will assist the COR in:

15.6.4.2.1. Enforcing established engineering standards or inspection criteria as defined and incorporated in the procurement/sustainment contract in order to address safety and integrity of the structure. **(T-2)**

15.6.4.2.2. Providing quality oversight authority for any technical/serviceability inspections performed on the structure by the contractor. **(T-2)**

15.7. New Procurements. To minimize diversity of types of aircraft sunshades, the AF utilizes strategic sourcing to establish efficiencies in procurement and long-term sustainment. For additional guidance, contact the MAJCOM A4 on aircraft sunshade standardization.

15.7.1. The owning organization will:

15.7.1.1. Develop plans for purchase, installation, acceptance, and sustainment then submit information to the MAJCOM/A4 for concurrence. **(T-2)** If the owning organization intends to request allied support, the owning organization will include these costs in the forecast. If allied support is required, submit an AF Form 332, *Base Civil Engineer Work*

Request, or service/work request to Civil Engineering Customer Service prior to solicitations, purchases, or procurements. **(T-2)**

15.7.1.1. **(AFMC)** If allied support is required, submit a CE service request using the AF-mandated enterprise IT system (when available) or an AF Form 332, *Base Civil Engineer Work Request*, prior to solicitations, purchases, or procurements.

15.7.1.2. Refer to AFI 65-601, Volume 1, *Budget Guidance and Procedures*, and AFI 65-601, Volume 2, *Budget Management for Operations* to determine correct appropriation for procurement and installation of aircraft sunshades or crew shelters described in this publication. Plan and fund current equipment sustainment IAW **Paragraph 15.10**.

15.7.1.3. Use local contracting office for strategic sourcing, if available, to standardize and leverage buying power from an enterprise perspective. (T-2)

15.7.1.4. Ensure all facets of aircraft sunshade sustainment are considered; all applicable installation functional areas (CE community planner, Airfield Management, Comm, Safety, Medical) are involved in the siting and selection process and is approved by the installation's facility board prior to solicitation and procurement. **(T-2)**

15.7.1.4.1. If expertise is not available in the owning organization, contact the appropriate MAJCOM representative for further guidance. All new sunshades will be compliant with current applicable structural, environmental and safety standards. **(T-2)** Ensure compliance with criteria and design as outlined in **Paragraph 15.8**.

15.7.1.5. Submit planning materials to include drawings, dimensions, cost estimates and statements of work, as well as ramp work striping and support equipment requirements to their applicable MAJCOM/A4. **(T-2) Note:** This must be completed as early as possible in the planning phase to facilitate a smooth execution phase.

15.7.1.6. Ensure use of reflective markings on corner structural supports regardless of lighting used. **(T-2)**

15.7.1.7. Ensure use of retro reflective beads for all apron, taxiway, and taxi lane markings near aircraft sunshades. **(T-2)** Reference Engineering Technical Letter 04-2, *Standard Airfield Pavement Marking Schemes* and Unified Facilities Guide Specifications 32-17-23, *Pavement Markings* and AFMAN 32-1040, *Civil Engineering Airfield Infrastructure Systems.*

15.7.1.8. Perform an evaluation of all proposed sunshade installation plans to ensure requirements in Unified Facilities Criteria (UFC) 3-260-01, *Airfield and Heliport Planning and Design* are met. If waivers are required, those waivers must be approved prior to contract award of the sunshade. **(T-2)**

15.7.1.8.1. Ensure waiver authorities do not undermine contract requirements or federal regulations (example Federal Acquisition Regulation, Department of Defense Federal Acquisition Regulation Supplement). **(T-0)**

15.7.1.9. Ensure aircraft sunshades that penetrate an airfield imaginary surface defined in UFC 3-260-01 have obstruction lights installed IAW UFC 3-535-01, *Visual Air Navigation Facilities*, and conform to requirements contained in AFMAN 32-1040.

15.7.1.10. Ensure compliance with fuel servicing safety requirements IAW TO 00-25-172.

15.7.1.11. Address the following items below and those in **Paragraph 15.8** as a minimum in installation plans submitted to applicable MAJCOM/A4 for approval. **(T-2)**

15.7.1.11.1. Aircraft sunshade protection characteristics such as hail size, snow, ice, and water accumulation (weight and/or inches, consider water runoff, ice mitigation), maximum winds (sustained and gusts), aircraft tie down and lightning/grounding protection determined in conjunction with proper local authorities or AFIMSC/PSU AFCEC. **(T-2)**

15.7.1.11.2. Ensure sunshades are designed IAW UFC 3-301-01, *Structural Engineering*. **Note:** Consider jet blast potential in aircraft sunshade design and siting.

15.7.1.11.3. Number of spots to be covered. Consider ramp space capacity for both peacetime and contingency requirements (reception and bed down). **(T-2)**

15.7.1.11.4. Number of parking spaces lost due to aircraft sunshade placement. Consider ramp requirements, future mission changes, and impact on overall aircraft parking plan. **(T-2)**

15.7.1.11.5. Design life expectancy and warranty information. **(T-2)**

15.7.1.11.6. Expected costs including procurement, installation, and projected annual sustainment costs, to include inspection services. **Note:** Ensure adequate resources are established to maintain aircraft sunshade serviceability and mission accomplishment.

15.7.1.11.7. **Sustainment methodology (owning organization supported, manufacturer supported, supplemental contractor).** Ensure minimum requirements outlined in **Paragraph 15.10.3** are addressed. **(T-2)** Pre-posture sustainment through end-of-year.

15.7.1.11.7.1. DELETED.

15.7.1.11.8. Foreign Object Damage (FOD) mitigation and prevention. **(T-2)**

15.7.1.11.9. Planned storage of Aerospace Ground Equipment (AGE) and other related support equipment in aircraft sunshades within the maintenance area. **(T-2)**

15.7.1.11.10. Provisions for electrical power units or powered AGE placement and protection from equipment exhaust and aircraft jet blast while operating in/near aircraft sunshades. **(T-2)**

15.7.1.11.11. Environmental impacts and mitigation plans. Ensure the host Environmental Planning Function reviews proposals for installation of aircraft sunshades and associated utilities as early in the planning process as possible. The Environmental Planning Function is responsible for determining the level of environmental impact analysis required. Environmental impact analysis must be completed prior to contract award or implementation. **(T-0)**

15.7.1.11.12. Conduct and include a Risk Management (RM) assessment IAW AFPD 90-8, *Environment, Safety, and Occupational Health Management and Risk Management*, and AFI 90-802, to determine potential impact to personnel, environmental, safety, occupational health and airfield operations before procurement and installation activities begin. **(T-2)**

15.7.1.11.13. Mitigate any issues affecting control tower visibility or security to include security lighting and illumination and any issues affecting navigational aids. **(T-2)**

15.7.1.11.14. Mitigate obstructions to airspace and file FAA Form 7460-1, *Notice of Proposed Construction or Alteration*, with the FAA for new construction. **(T-0)**

15.7.1.11.15. Communications requirements. If communication (Local Area Network (LAN), wireless communications) is placed in aircraft sunshades, provide explanation of how communications will be procured, installed, and sustained. Include in procurement, installation, and sustainment cost estimates. **(T-2)**

15.7.1.11.16. Mitigate any limitations on emergency and maintenance vehicle access to aircraft. **(T-2)**

15.7.1.11.17. Fall protection for personnel working above 4’ off the ground IAW DAFMAN 91-203. **(T-2)**

15.8. Design Criteria/Standards of Aircraft Sunshades.

15.8.1. Current aircraft sunshades that don't meet these requirements of current design standardization are authorized for use until time of replacement. If repairs become significant or not cost effective, owning organizations should consider replacement to the new standard or removal of the aircraft sunshade.

15.8.1.1. Coordinate with Air Traffic Control, Base/Unit Terminal Instrument Procedures specialist (SEI 357), to evaluate plans for potential impacts on aircraft approach/departure procedures. **(T-2)**

15.8.1.2. **&**

15.8.1.3. DELETED.

15.8.2. **Type of materiel used.** Select materiel as appropriate for location, type of aircraft sunshade, and the mission being served. The aircraft sunshade and side materiel will be fabric, reinforced aramid fiber, carbon fiber, galvanized steel or shop/factory painted steel. The frame materiel should be galvanized steel or other suitable newly designed building materiel with strength and durability characteristics that replicate or exceed that of galvanized steel.

15.8.3. **Aircraft sunshade height and width.** Aircraft sunshades must meet minimum clearance requirements for interior aircraft movement, as well as entrances and exit points including taxi lanes or taxiways in close proximity to the exterior of the structure. Refer to UFC 3-260-01, **Chapters 6** and **8** for these dimensions. (**T-2)**

15.8.4. Consider AGE transport and maintenance/emergency vehicle height requirements. Also, consider exhaust blast from jet engines and auxiliary power units, and future mission changes that may impact aircraft sunshade height and width requirements.

15.8.5. Wing tip and tail height clearances will meet the requirements outlined in DAFMAN 91-203 and UFC 03-260-01.

15.8.6. Bird nesting/roosting mitigation. Ensure no bird roosting or nesting locations are present, all holes in support structure and beneath roof must be covered or screened. **(T-2)**

15.8.6.1. The metal structure should minimize potential nesting sites for birds.

15.8.6.2. Anti-perching devices will be installed to deter birds from perching on or under the structure. **(T-2)**

15.8.7. Airfield waivers. The owning organization will make every effort to prevent submission of new airfield waivers. **(T-2)**

15.8.8. Lighting requirements. If lighting is placed in or on the aircraft sun shades, provide explanation of how electricity will be provided to and within the aircraft sunshade. Ensure lighting provides sufficient illumination for security but does not interfere with flightline visual references. The procurement of lighting, installation, maintenance, and sustainment will be included in aircraft sunshade cost estimates. **(T-2)**

15.8.8.1. Lighting designs and fixture specifications will be IAW UFC 3-530-01, *Interior and Exterior Lighting Systems*.

15.8.8.2. Light mounts will be designed to handle wind loads on the light fixtures without damaging the lights and without compromising the structural integrity of the aircraft sunshade. **(T-2)**

15.8.9. Electrical requirements. If electrical power is to be placed in aircraft sunshades, provide explanation of how electricity will be provided to and within the aircraft sunshade. Procurement, installation, maintenance, and sustainment of electrical requirements on the load side of the power supply point/stub will be included in aircraft sunshade cost estimates. **(T-2)**

15.8.9.1. Electrical power system design will be IAW AFMAN 32-1062, *Electric Systems, Power Plants and Generators*, AFMAN 32-1065. Reference UFC 3-501-01, *Electrical Engineering*, UFC 3-520-01, *Interior Electrical Systems*, UFC 3-530-01, and UFC 3-550-01, *Exterior Electrical Power Distribution*, UFC 3-600-01, *Fire Protection Engineering for Facilities*, and Engineering Technical Letter 02-15, F*ire Protection Engineering Criteria - New Aircraft Facilities*.

15.8.10. The cost of running power (such as, lighting, and electrical) from prime or temporary power sources to the sunshades will be included in the cost estimate and will be borne by the owning unit. **(T-2)**

15.8.11. In no case will the loss of parking spots resulting from the purchase or installation of aircraft sunshades be a primary factor used to propose or construct additional aircraft parking spots or upgrade existing pavements to support aircraft. Procedures on construction of additional parking spots or pavement upgrades will be IAW AFI 32-1020, *Planning and Programming Built Infrastructure Projects* and DAFI 32-1020.

15.8.11.1. Whenever possible, aircraft sunshades will be installed on existing pavement without driving a requirement to expand or upgrade the underlying features. When the addition of pavement is necessary to make the sunshade complete and usable, the cost of the structure must be included with the cost to installation of the pavement for a total sunshade construction project cost. **(T-2)**

15.8.12. Aircraft sunshades are airfield fixed-structure obstructions and will meet MIL-STD-3007, *Department of Defense Standard Practice for Unified Facilities Criteria and Unified Facilities Guide Specifications* and AF criteria included in referenced UFCs and AFIs, as well as, local and national fire, safety, and electrical standards and codes as applicable. **(T-2)**

15.8.13. Consider location specific design requirements (snow load, maximum wind ratings). In addition to the requirements in UFC 3-301-01, for adjacent sunshades which share bracing frames, each fourth frame will provide twice the required lateral bracing strength. **(T-2)**

15.8.13.1. Particular attention will be focused on the design of lateral bracing for wind and jet blast loads, fatigue strength of steel lateral bracing members, and the design and location of anchor bolts near joints in runway aprons. **(T-2)**

15.8.14. Aircraft sunshades must meet ventilation and exhaust air requirements referenced in the International Mechanical Code, UFC 1-200-01, *DoD Building Code*. **(T-0) Note:** Achieve minimum exhaust and/or ventilation air requirements utilizing either mechanical or natural systems.

15.9. Current Aircraft Sunshades.

15.9.1. To replace currently installed aircraft sunshades, select materiel as appropriate for location, type of aircraft sunshade, and the mission being served. The aircraft sunshade and side materiel will be fabric, reinforced aramid fiber, carbon fiber, galvanized steel or shop/factory painted steel. **(T-2)**

15.9.1.1. The frame materiel will be galvanized steel or other suitable newly designed building materiel with strength and durability characteristics that replicate or exceed that of galvanized steel. **(T-2)**

15.9.1.2. Current aircraft sunshades that don't meet these requirements of current design standardization are authorized for use until time of replacement. **(T-2)**

15.9.1.3. If repairs become significant or not cost effective, owning organizations will consider replacement to the new standard or removal of the aircraft sunshade. **(T-2)**

15.9.1.4. To minimize diversity of types of aircraft sunshades across the enterprise and to establish better efficiencies in procurement and long-term sustainment, use strategic sourcing, if available. **(T-2)**

15.9.1.5. If strategic sourcing is not available, seek guidance from higher headquarters on aircraft sunshade standardization. **(T-2)**

15.9.2. Owning organizations will sustain aircraft sunshades and other shelter types described in this publication IAW AFI 65-601, Volume 2, *Financial Management*, to ensure no degradation of safety to personnel or mission accomplishment occurs. **(T-2)**

15.9.3. Owning organizations will conduct a risk management assessment IAW AFI 90-802 in conjunction with proper local authorities on current aircraft sunshades for FOD mitigation, markings, bird roosting mitigation, sustainability, safety, structural integrity, wind-load capacity, snow-load capacity, and ability to withstand hail. **(T-2)**

15.9.3.1. The owning organization is responsible for repairing any deficient areas identified during the risk management assessment. **(T-2)**

15.9.4. There is no requirement to upgrade current aircraft sunshades to meet the full requirements outlined in Paragraphs **15.7.1** through **15.7.1.11.16** and **15.8.1** through **15.8.13** except where codes apply and are mandatory. However, any replacement or significant repair of current aircraft sunshades will meet requirements as outlined in Paragraphs **15.7.1** through **15.7.1.11.16** and **15.8.1** through **15.8.13** **(T-2)**

15.9.5. Alterations can significantly impact loading on the structure. Conduct a structural analysis certified by a licensed professional engineer or government structural engineer to ensure the alterations do not compromise the structural integrity of the sunshade structure. Sunshades will not be altered to have more than two total walls or bay(s).

15.10. Sustainment and Accountability. Owning organization Group Commander or equivalent will:

15.10.1. Appoint a primary and alternate aircraft sunshade managers.

15.10.2. Develop local procedures establishing management and maintenance of sunshades.

15.10.3. Ensure assets are tracked as equipment items on the appropriate documents IAW DAFI 23-101 and DAFMAN 23-122.

15.10.4. Ensure appropriate authorities (Maintenance Group (MXG), Contracting, Comm, CE, Safety, Airfield Management) at the installation level accept installation of aircraft sunshades and allied support IAW manufacturer specifications, contract specifications, and AFIs. **(T-2)**

15.10.5. Develop an organically supported or contractor supported replacement/sustainment plan that will be included in the 10-year facility plan IAW this DAFI. **(T-2) Note:** This statement does not limit the plan from exceeding the minimum 10-year requirement.

15.10.5.1. Sustainment plan at a minimum will address the following as appropriate:

15.10.5.1.1. Certification and acceptance of installation as needed. **(T-2)**

15.10.5.1.2. Routine maintenance/sustainment and inspection criteria/services which meet manufacturer or qualified engineering authority (such as, personnel officially trained to inspect or provide technical guidance on that specific Sunshade) guidance as needed and appropriate. **(T-2) Note:** Ensure manufacture warranty is not voided.

15.10.5.1.3. Periodic and recurring in-depth inspections with prescribed interval/frequency and items to be inspected to determine structural integrity as appropriately defined for each type of structure by the manufacturer or qualified engineering authority as needed and appropriate. **(T-2)**

15.10.5.1.4. Non-routine maintenance/sustainment and inspection services as needed and appropriate. **(T-2)**

15.10.6. Develop local guidance that establishes periodic inspection criteria as recommended by original manufacturer (interval not to exceed 6 months) approved by a qualified engineering authority for aircraft sunshades and document on AFTO Form 244, or equivalent. **(T-2)**

15.10.6.1. Required inspections will be tracked in MIS. **(T-2)**

15.10.6.2. As a minimum, the owning organization will ensure inspection of:

15.10.6.2.1. Structural Supports for signs of corrosion, excessive movement, cracking, or damage; lateral bracing for signs of corrosion, loosening, cracking, missing hardware, or fatigue damage, particularly at the connections; structural anchorage for signs of loosening and cracking around anchorage points on apron. **(T-2)**

15.10.6.2.2. Foreign Object, cleanliness, and serviceability of installed safety markings. **(T-2)**

15.10.6.2.3. Lighting (if installed) for security and serviceability. **(T-2)**

15.10.6.2.4. Electrical system equipment, including electrical panels, receptacles, lighting systems, solar panels, and conduit (if installed) for security and serviceability. **(T-2)**

15.10.6.2.5. Following extreme/unique weather events (i.e., microburst, flooding).

15.10.7. Any discrepancy identified during routine, non-routine maintenance or other visual inspection will be documented on the AFTO Form 244 or equivalent; appropriate condition symbol will be used IAW TO 00-20-1.

15.10.7.1. Additionally, all discrepancies will be annotated in the MIS. **(T-2)**

15.10.7.2. If a discrepancy is discovered which could affect safety or structural integrity this information will immediately be directed to the owning organization group commander or equivalent. **(T-2)**

15.10.7.3. The owning organization group commander or equivalent with advisement from the qualified engineering authority and safety personnel will decide whether the sunshade is in an acceptable condition to continue performing its intended purpose or whether the sunshade will be removed from service and all assets and personnel vacated from the sunshade until repair is completed. **(T-2)**

15.10.8. Establish an emergency evacuation plan to protect both personnel and equipment from inclement/severe weather that may compromise the integrity of the aircraft sunshade roof or support structure. **(T-2)**

15.10.8.1. During inclement/severe weather situations including but not limited to high winds, thunderstorms, lightning, hail, tornadoes, hurricanes or other adverse weather warning situation, personnel will be evacuated from the aircraft sunshade, not to the aircraft sunshade. **(T-2)**

15.10.8.1.1. The aircraft sunshade is not intended to be used as a protective shelter in these situations.

15.11. Allied Support (Electricity, Water, Communications, Lighting).

15.11.1. If elected, electrical power will be provided to a connection point at the aircraft sunshade IAW UFC 3-550-01, *Exterior Electrical Power Distribution* and will be installed and maintained by Civil Engineering as real property IAW DAFI 32-9005, *Real Property Accountability and Reporting*. **(T-2)**

15.11.1.1. Allied support funding requirements will be prioritized along with other facility and infrastructure requirements through the Facilities Board process or applicable installation facility project prioritization process. **(T-2)**

15.11.2. If elected, water lines can be installed to a stub at the aircraft sunshade site and will be installed and maintained by Civil Engineering as real property IAW DAFI 32-9005. Water stubs and/or connections must be appropriately protected from frost/freeze and other environmental conditions. **(T-2)**

15.11.2.1. These stubs and/or connections must be prominently marked to prevent hazards and accidents. **(T-2)**

15.11.2.2. Allied support funding requirements will be prioritized along with other facility and infrastructure requirements through the facilities board process or applicable installation facility project prioritization process. **(T-2)**

15.11.3. If elected, communications will be provided to the aircraft sunshade IAW UFC 3-580-01, *Telecommunications Interior Infrastructure Planning and Design*, and will be installed and maintained by qualified communications personnel.

15.11.3.1. Allied support funding requirements for the communication real property elements of a project (example, conduit, manholes, duct banks) will be prioritized along with other facility and infrastructure requirements through the facilities board process or applicable installation facility project prioritization process. **(T-2)**

15.11.4. Lighting installation will be considered during procurement and/or sustainment phases and will be purchased and maintained by the owning organization from the light to the power stub. **(T-2)**

15.11.4.1. If lighting is to be included in an aircraft sunshade, the owning organization will include this in the procurement, installation, and sustainment plan. **(T-2)**

15.11.4.2. Owning organizations will use criteria and consultation as outlined in Paragraphs **15.7** through **15.9**. Energy efficient lighting will be considered where allowable. **(T-2)**

15.11.5. Power receptacles will be considered during the procurement and/or sustainment phases and will be purchased and maintained by the owning organization from the power receptacles to the power stub. **(T-2)**

15.11.5.1. If power is to be included, the owning organization will include this in the procurement, installation, and sustainment plan. **(T-2)** Owning organizations will use criteria and consultation as outlined in Paragraphs **15.7** through **15.9**. **(T-2)**

15.12. Crew Shelters and Portable/Inflatable Shelters.

15.12.1. Owning organizations will address the following items as a minimum in installation plans submitted to applicable MAJCOM/A4 for approval:

15.12.1.1. Annual procurement and sustainment costs, including allied support infrastructure costs. See **Paragraph 15.11** for allied support procedures. **(T-2)**

15.12.1.2. Ensure shelter meets all standards and codes as applicable. Crew shelters and inflatable or fabric structures must comply with the minimum Antiterrorism Standard requirements of UFC 4-010-01, *DoD Minimum Antiterrorism Standards for Buildings*. **(T-0)**

15.12.1.3. Ensure a FOD mitigation plan is implemented. **(T-2)**

15.12.1.4. Warranty information. **(T-2)**

15.12.1.5. Wing tip and tail height clearances will meet the requirements outlined in DAFMAN 91-203 and UFC 03-260-01. **(T-0)**

15.12.1.6. Appoint a shelter manager and establish a shelter user and operations plan. Ensure assets are tracked as equipment items on the appropriate documents on the CA/CRL IAW DAFMAN 23-122. **(T-2)**

15.12.2. Crew shelters will be configured to minimum ventilation standards referenced in the International Mechanical Code when occupied. **(T-0)**

Chapter 16 (Added-AFMC)

412TH LOGISTICS TEST SQUADRON

16.1. (Added-AFMC) Purpose: This Chapter outlines general LT&E activities. Provides continued emphasis and awareness of LT&E activities and the necessity for early LT&E involvement in Developmental Testing. Establishes standards and procedures for accomplishment of LT&E. Ensures compliance with statutory directives; Air Force environmental management, safety, security instructions; and TOs. Develops and provides training and tools to improve the LT&E process within the acquisition community. Assists in developing logistics test plans, data analysis plans, and technical reports. Assists local management in test execution during LT&E program involvement. Acts as a focal point with Engineering and Program Management for LT&E activities. Technical and Safety reviews independent of test program are required IAW DAFI 99-103, *Capabilities-Based Test and Evaluation*. Plans, directs, and manages the development and acquisition of technical data in support of System Groups. Manages the Operating Locations' Central Technical Order Control Units. Provides staff assistance in the development and implementation of Technical Order Development policies and procedures.

16.2. (Added-AFMC) Logistics Test and Evaluation. The test methodology, criteria, and tools for evaluating and analyzing the product support elements as they apply to a system under test. The objective is to influence system design as early as possible in the acquisition cycle, and verify that the logistics support being developed is capable of meeting peacetime and wartime employment objectives.

16.3. (Added-AFMC) Test Support Manager. Responsibilities include, but are not limited to:

16.3.1. **(Added-AFMC)** Identify opportunities for LT&E, initiate the test process, and assign management responsibilities.

16.3.2. **(Added-AFMC)** Manage the LT&E plan and report writing process.

16.3.3. **(Added-AFMC)** Manage the creation, modification, and usage of data collection tools.

16.3.4. **(Added-AFMC)** Manage test data software development, installation and training.

16.3.5. **(Added-AFMC)** Identifies test support needed in training, facilities, special equipment, funding, and other logistics areas as required.

16.3.6. **(Added-AFMC)** Provide long range and intermediate logistical planning and support for tests and programs.

16.3.7. **(Added-AFMC)** Work with program engineers, test designers, developmental engineers, and program managers regarding plans for the test and evaluation workload.

16.4. (Added-AFMC) Technical Order Development. Plan, direct and manage the development and acquisition of technical data in support of system group requirements.

16.5. (Added-AFMC) Technical Order Control Unit Manager. Responsibilities include, but are not limited to:

16.5.1. **(Added-AFMC)** Assist Systems Technical Order Manager in development of technical manual contractual requirements.

16.5.2. **(Added-AFMC)** Assist in establishing and supporting technical order control units.

16.5.3. **(Added-AFMC)** Schedule/witness technical order certification/verification events.

16.5.4. **(Added-AFMC)** Participates in contractor quality assurance (QA) processes.

16.5.5. **(Added-AFMC)** Schedules/attends guidance conferences.

16.5.6. **(Added-AFMC)** Chairs technical interchange meetings.

16.5.7. **(Added-AFMC)** Supervises/performs in-process reviews.

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Attachment 1

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AF Form 2401, *Equipment Utilization and Maintenance Schedule*

AF Form 2402, *Weekly Equipment Utilization and Maintenance Schedule*

AF Form 2403, *Weekly Aircraft Utilization/Maintenance Schedule*

AF Form 2407, *Weekly/Daily Flying Schedule Coordination*

AF Form 2410, *Inspection/TCTO Planning Checklist*

AF Form 2411, *Inspection Document*

AF Form 2426, *Training Request and Completion*

AF Form 2408, Generation Maintenance Plan

AF Form 2409, Generation Sequence Action Schedule

(**Added-AFMC**) AFMC Form 61, *Missing/Removed Tools and Equipment*

(**Added-AFMC**) AFMC Form 62, *CTK Inventory and Control Log*

(**Added-AFMC**) AFMC Form 64, *Request for Special Certification*

(**Added-AFMC**) AFMC Form 310, *Lost/Found Item Report*

Adopted Forms

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AF Form 332, *Base Civil Engineer Work Request*

AF Form 623, *Individual Training Record*

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AF Form 3580, *USAF Museum Aerospace Vehicle Static Display Egress and Safety Certificate*

AFTO Form 20, *Caution and Inspection Record*

AFTO Form 66, *TMDE Bar Codes (Polyester Film)*

AFTO Form 82, *TCTO Verification Certificate*

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AFTO Form 97, *Aerospace Vehicle Battle Damage Incident Debrief/Assessment/Repair Record*

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(**Added-AFMC**) AFTO Forms 428, B-1B *Aircraft Integral Fuel Tank Repair History Record*

AFTO Form 492, *Maintenance Warning Tag*

AFTO Form 726, *Transient Aircraft Service Record*

AFTO Form 781, *Arms Aircrew/Mission Flight Data Document*

AFTO Form 781A, *Maintenance Discrepancy and Work Document*

AFTO Form 781C, *Avionics Configuration and Load Status Document*

(**Added-AFMC**) AFTO Forms 781F, *Aerospace Vehicle Flight Report and Maintenance Document*

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DD Form 1348-6, *DoD Single Line Item Requisition System Document*

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(**Added-AFMC**) DD Form 2026, *Oil Analysis Request*

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FAA Form 7460-1, *Notice of Proposed Construction or Alteration*

Abbreviations and Acronyms

(Added-AFMC) AA—Air Abort

ABDR—Aircraft Battle Damage Repair

AC—Aircraft Commander

ACC—Air Combat Command

ACR—Authorization Change Requests

AD—Airworthiness Directives

ADR—Aircraft Document Review

(Added-AFMC) AEDC—Arnold Engineering and Development Complex

AETC—Air Education and Training Command

AFCSM—Air Force Computer Systems Manual

AFETS—Air Force Engineering and Technical Service

AFI—Air Force Instruction

AFLCMC—Air Force Life Cycle Management Center

AFMAN—Air Force Manual

AFMC—Air Force Materiel Command

AFPAM—Air Force Pamphlet

AFPD—Air Force Policy Directive

AFR—Air Force Reserve Command

AFREP—Air Force Repair and Enhancement Program

AFRIMS—Air Force Records Information Management System

AFSC—Air Force Specialty Code

AFSCI—Air Force Sustainment Center Instruction

(Added-AFMC) AFTC—Air Force Test Center

AFTO—Air Force Technical Order

AGE—Aerospace Ground Equipment

(Added-AFMC) AI—Aircrew Inflight

AIRCAT—Automated Inspection, Repair, Corrosion, and Aircraft Tracking

ALC—Air Logistics Complex

ALIS—Autonomic Logistics Information System

AMC—Air Mobility Command

AME—Alternate Mission Equipment

AMU—Aircraft Maintenance Unit

AMS—Air Mobility Squadron

AMXS—Aircraft Maintenance Squadron

ANG—Air National Guard

(Added-AFMC) APSR—Accountable Property System of Record

APU—Auxiliary Power Unit

AQL—Acceptable Quality Level

ARC—Air Reserve Component

AS—Allowance Standard

ASC—Aircraft Service Changes

ASIP—Aircraft Structural Integrity Program

ASM—Aircraft Structural Maintenance

AVDO—Aerospace Vehicle Distribution Office

(Added-AFMC) AvSE—Aviation Support Equipment

AWM—Awaiting Maintenance

AWP—Awaiting Parts

(Added-AFMC) BB—Black Belt

BCS—Bench Check Serviceable

(Added-AFMC) BPA—Blanket Purchasing Agreement

BSL—Basic Systems Listing

BSP—Base Support Plan

CA—Cannibalization Authority

CAAP—COMAFFOR Apportionment and Allocation Process

CA/CRL—Custodian Authorization/Custody Receipt Listing

CAD/PAD—Cartridge Actuated Device/Propellant Actuated Device

CAI—Critical Application Items

CANN—Cannibalization

(Added-AFMC) CAP—Corrective Action Plan

CAT I—Category I

CAT II—Category II

(Added-AFMC) CAvSE—Common Aviation Support Equipment

CB—Customer Bulletins

CBM+—Condition-Based Maintenance Plus

CBRNE—Chemical, Biological, Radiological, Nuclear, and high-yield Explosive

CBU—Cluster Bomb Unit

CCY—Calculated Cycles

CD—Deputy Commander (MXG/CD)

CDA—Commercial Derivative Aircraft

CDB—Central Database

CDDAR—Crashed, Damaged or Disabled Aircraft Recovery

CE—Civil Engineer

CEMS—Comprehensive Engine Management System

CETS—Contractor Engineering and Technical Services

CFETP—Career Field Education and Training Plan

CFT—Contract Field Team

CM—Configuration Management

CMD—Commercial Mobile Device

CMS—Component Maintenance Squadron

CND—Can Not Duplicate

(Added-AFMC) COA—Course of actions

COMAFFOR—Commander, Air Force Forces

Cont/Exp—Contingency/Expeditionary

COMSEC—Communications Security

CONUS—Continental United States

(Added-AFMC) COR—Contracting Officer Representative

(Added-AFMC) CPI—Continuous Process Improvement

CPINS—Computer Program Identification Numbering System

CRF—Centralized Repair Facility

CSI—Critical Safety Items

CTK—Composite Tool Kit

CCW—Counter Chemical Warfare

(Added-AFMC) CX—Cancellation

DAFI—Department of Air Force Instruction

DBM—Database Manager

DCA—Design Control Activity

DCC—Dedicated Crew Chief

DCMA—Defense Contract Management Agency

DDR—Daily Demand Rate

(Added-AFMC) DEMIL—Demilitarization

DEV—Deviation

DFT—Depot Field Team

DIAMONDS—Defense Integration and Management of Nuclear Data Services

DIFM—Due-in From Maintenance

DISA—Defense Information System Agency

DIT—Data Integrity Team

DLA—Defense Logistics Agency

(Added-AFMC) DLADS—Defense Logistics Agency Disposition Services

DMS—Decentralized Materiel Support

D23—Repair Cycle Asset Management Listing

DOC—Designed Operational Capability

DoD—Department of Defense

DoDI—Department of Defense Instruction

DoDD—Department of Defense Directive

DOI—Date of Installation

DOM—Date of Manufacture / Director of Maintenance

DOP—Dropped Object Prevention / Director of Propulsion

(Added-AFMC) DPII—Depot Pre-Induction Inspection

DR—Deficiency Report

DLADS—Defense Logistics Agency Disposition Service

DRU—Direct Report Unit

DSV—Detected Safety Violations

(Added-AFMC) EL—Early Landing

(Added-AFMC) ET—Early Takeoff

eTools—Electronic Tools

E&E—Electrical & Environmental

EAWP—Engine Automated Work Package

ECM—Electronic Countermeasures

ED—Incapacitated

EHM+—Engine Health Management

EHR—Event History Recorder

E&I—Evaluation and Inspection

EID—Event Identification Description / Equipment Identification Designator

EM—Engine Management/Emergency Management

EMS—Equipment Maintenance Squadron

EMXG/CC—Expeditionary Maintenance Group Commander

ENMCS—Engine Not Mission Capable for Supply

EOD—Explosive Ordnance Disposal

EOR—End of Runway

EOT—Engine Operating Time

EPE—Evaluator Proficiency Evaluation

ER—Exceptional Release

ERRC—Expendability, Recoverability, Reparability Code

ES—Equipment Specialist

ESOH—Environment Safety and Occupational Health

ESOHMS—Environment, Safety, and Occupational Health Management System

ESP—Expeditionary Site Plan

ESS—Environmental Stress Screening

ETIC—Estimated Time in Commission

ETIMS—Enhanced Technical Information Management System

ET&D—Engine Trending and Diagnostics

ETS—Engineering and Technical Services/Engine Test Stand

EW—Electronic Warfare

EWS—Electronic Warfare System

EX—Exercises/Contingencies

(Added-AFMC) EXH—Exercise, Higher Headquarters

(Added-AFMC) EXL—Exercise, Local

FAA—Federal Aviation Administration

FAR—Federal Acquisition Regulation

FCC—Flying Crew Chief

FCF—Functional Check Flight

(Added-AFMC) FE—Flight Emergency

FHP—Flying Hour Program

FMxC2—Field Maintenance Command and Control

FO—Foreign Object

FOA—Field Operating Unit

FOD—Foreign Object Damage

FOM—Facilitate Other Maintenance

FOUO—For Official Use Only

FSL—Full Systems Listing

FSR—Field Service Representatives

FY—Fiscal Year

(Added-AFMC) GB—Green Belt

GBU—Guided Bomb Unit

GEOLOC—Geographical Location

GITA—Ground Instructional Trainer Aircraft

GOX—Gaseous Oxygen

GP—Group

GSU—Geographically Separated Units

HAF—Headquarters, US Air Force

HAZMAT—Hazardous Material

HF—High Frequency

HHQ—Higher Headquarters

HPO—Hourly Post-flight

HPT—High Pressure Turbine

HQ—Headquarters

HSC—Home Station Check

(Added-AFMC) IAT—Individual Aircraft Tracking

IAW—In Accordance With

ID—Identification / Integrated Defense

I-Deck—Initialization Deck

IETM—Interactive Electronic Technical Manuals

IFCS—Instrument and Flight Control Systems

IFE—In-Flight Emergency

IFR—In Flight Refueling

ILS-S—Integrated Logistics Systems-Supply

IMDS—Integrated Maintenance Data System

IMIS—Integrated Maintenance Information System

(Added-AFMC) IOT&E—Initial Operational Testing and Evaluation

IP—Instructor Pilot

IPCOT—In-Place Consecutive Overseas Tour

IPI—In-Process Inspection

(Added-AFMC) IPP—Integrated Power Package

(Added-AFMC) IPS—Integrated Product Support

ISO—Isochronal Inspection

I/UA—Immediate and Urgent Action

ISU/DOR—Issue/Due-Out Release

JCN—Job Control Number

JDD—Job Data Documentation

JDRS—Joint Deficiency Reporting System

JEDMICS—Joint Engineering Data Management Information and Control System

JEIM—Jet Engine Intermediate Maintenance

JML—Job Standard Master Listing

JOAP—Joint Oil Analysis Program

JST—Job Standard

KTL—Key Task List

LAN—Local Area Network

(Added-AFMC) LCL—Local Checklists

(Added-AFMC) LCLMS—Life Cycle Logistics Management Specialist

LCN—Logistics Control Number

LEAP—Logistics Evaluation Assurance Program

(Added-AFMC) LL—Late Landing

LM—Limited-use Munition

LME—Locally Manufactured Equipment

LMR—Land Mobile Radio

LO—Low Observable

LOX—Liquid Oxygen

LPT—Low Pressure Turbine

LRS—Logistics Readiness Squadron

LRU—Line Replaceable Unit

LSC—Load Standardization Crew

(Added-AFMC) LT—Late Takeoff

(Added-AFMC) LT&E—Logistics Test and Evaluation

LV—Emergency Leave

MAJCOM—Major Command

MC—Mission Capable

(Added-AFMC) MCD—Magnetic Chip Detector

MDF—Mission Data File

MDS—Mission Design Series

MEL—Minimum Equipment Level

MESL—Minimum Essential Subsystems List

MFG—Munitions Family Group

MFM—MAJCOM Functional Manager

MGN—Mission Generation Networks

MI—Management Inspection

MICAP—Mission Impaired Capability Awaiting Parts

MIL—Master Inventory List

MILSPEC—Military Specification

MIL-STD—Military Standard

MIS—Maintenance Information Systems

MMA—Maintenance Management Analysis

MMHE—Munitions Materiel Handling Equipment

MOA—Memorandum of Agreement

MOC—Maintenance Operations Center

(Added-AFMC) MOF—Maintenance Operations Flight

MOU—Memorandum of Understanding

MPS—Military Personnel Section

MRPL—Minimum Required Proficiency Load

MRSP—Mobility Readiness Spares Package

MSA—Munitions Storage Area

MSE—Maintenance Scheduling Effectiveness

MSEP—Maintenance Standardization & Evaluation Program

MSG—Mission Support Group

MT—Maintenance Training

MSM—DS Maintenance Scheduling Module

MUNS—Munitions Squadron

MX—Maintenance

MXG—Maintenance Group

MXG/CC—Maintenance Group Commander

MXG/CD—Maintenance Group Deputy Commander

MXO—Maintenance Operations

MXS—Maintenance Squadron

MX SUPT—Maintenance Superintendent

NAF—Numbered Air Force

NATO—North Atlantic Treaty Organization

NCE—Nuclear Certified Equipment

NCOIC—Non-Commissioned Officer in Charge

NDI—Nondestructive Inspection

NIE—Normally Installed Equipment

NLT—Not Later Than

NMC—Non Mission Capable

(Added-AFMC) NMCS—Not Mission Capable Supply

NPA—Non-Powered AGE

NORAD—North American Aerospace Defense Command

NRTS—Not Repairable This Station

NSN—National Stock Number

NSS—Noise Suppression System

NWRM—Nuclear Weapons-Related Materiel

O&M—Operations and Maintenance

OAP—Oil Analysis Program

OBIGGS—On-Board Inert Gas Generating Systems

OBOGS—On-Board Oxygen Generating Systems

OCF—Operational Check Flight

OCONUS—Outside Continental U.S.

OFP—Operations Flight Program

OG—Operations Group

G/CC—Operations Group Commander

OI—Operating Instruction

OIC—Operations-in-Charge

OJT—On-the-Job Training

OPLAN—Operational Plan

OPR—Office of Primary Responsibility

ORE—Operational Readiness Exercises

OS—Operational Squadron

OSS—Operations Support Squadron

OSS&E—Operational Safety Suitability and Effectiveness

(Added-AFMC) OT&E—Operational Test and Evaluation

OTI—One Time Inspection

OWC—Owning Work Center

P&R—Programs and Resources

PAA—Primary Aerospace Vehicle (Aircraft) Authorized

PACAF—Pacific Air Forces

PAFSC—Primary AFSC

PAI—Primary Aerospace Vehicle (Aircraft) Inventory

PAS—Protective Aircraft Shelter / Personnel Assignment Symbol (Code)

PBR—Percent of Base Repair

PCS—Permanent Change of Station

PDM—Programmed Depot Maintenance

PE—Personnel Evaluation/Periodic Inspection

PED—Portable Electronic Device

PH—Phase

PIC—Purpose Identifier Code / Pilot in Command

PIM—Product Improvement Manager

PIP—Product Improvement Program

PIT—Platform Information Technology

PM—Primary Munition/Program Manager

PMA—Portable Maintenance Aids

PMC—Partially Mission Capable

PME—Precision Measurement Equipment

PMEL—Precision Measurement Equipment Laboratory

PMO—Program Management Office

PO—Program Office

POC—Point of Contact

(Added-AFMC) POM—Program Objective Memorandum

(Added-AFMC) PPC—Possession Purpose Code

PPE—Personal Protective Equipment

(Added-AFMC) PPSM—Practical Problem Solving Method

(Added-AFMC) PRA—Planning requirements

(Added-AFMC) PRD—Pilot reported discrepancy

PRP—Personnel Reliability Program

PRS—Performance Requirements Statement

PS&D—Plans, Scheduling, and Documentation

PWS—Performance Work Statement

QA—Quality Assurance

QASP—Quality Assurance Surveillance Plan

(Added-AFMC) QAWG—Quality Assurance Working Group

QE—Quarterly Evaluation

QEC—Quick Engine Change

QRL—Quick Reference List

QVI—Quality Verification Inspections

RAMPOD—Reliability, Availability, Maintainability for Pods

RC—Recommended Change

(Added-AFMC) RCA—Root Cause Analysis

(Added-AFMC) RCS—Rapid Crew Swap

RegAF—Regular Air Force

REMIS—Reliability and Maintainability Information System

RIL—Routine Inspection List

RN—Repair Network

(Added-AFMC) RNEP—Repair Network Enhancement Program

RNM—Repair Network Manager

(Added-AFMC) ROM—Rough-Order-Magnitude

RPA—Remotely Piloted Aircraft

(Added-AFMC) SAC—Self Assessment Checklist

SB—Service Bulletins

SCR—Special Certification Roster

SDAP—Special Duty Assignment Pay

SE—Support Equipment

SEI—Special Experience Identifier

(Added-AFMC) SEM/EDX—Scanning electron microscopy with energy dispersive X-ray

SI—Special Inspection

SIPRNET—Secret Internet Protocol Router Network

SM—Support Munitions

SME—Subject Matter Expert

SMR—Source of Maintenance and Recoverability

SNCO—Senior Non-Commissioned Officer

(Added-AFMC) SOC—Statement of capabilities

(Added-AFMC) SOIs—Sustainment Operating Instructions

SOW—Statement of Work

(Added-AFMC) SP—Spare

SPRAM—Special Purpose Recoverable Authorized Maintenance

SQ—Squadron

SQ/CC—Squadron Commander

(Added-AFMC) SR—Service Report

SRAN—Stock Record Account Number

SRU—Shop Replaceable Unit

SEL—Senior Enlisted Leader (Enlisted Duties)

TA—Transient Alert

TAA—Training Aid Aircraft

TAC—Total Accumulated Cycles

TBA—Training Business Area

TCC—Transaction Condition Code

TCI—Time Change Item

TCTO—Time Compliance Technical Order

TD—Training Detachment

(Added-AFMC) TDI—Time distribution index

TDV—Technical Data Violation

TDY—Temporary Duty

TFI—Total Force Integration

(Added-AFMC) TG—Test Group

(Added-AFMC) TK—Tool Kit

(Added-AFMC) TM/SM—Type/Model/Series and Modification

TMDE—Test Measurement and Diagnostic Equipment

TMS—Type Model Series

TNB—Tail Number Bin

(Added-AFMC) TNS—Tail Number Swap

TO—Technical Order

TODA—Technical Order Distribution Account

TODO—Technical Order Distribution Office

(Added-AFMC) TSE—Total Scheduling Effectiveness

TTML—Test/Training Munitions List

TTP—Tactics, Techniques & Procedures

(Added-AFMC) TW—Test Wing

UCML—Unit Committed Munitions List

UCR—Unsatisfactory Condition Report

UEM—Unit Engine Manager

UFC—Unified Facilities Criteria

UHF—Ultra High Frequency

UII—Unique Item Identifier

UMD—Unit Manpower Document

UPMR—Unit Personnel Management Roster

USAF—United States Air Force

USAFE—United States Air Forces in Europe

UTC—Unit Type Code

UTE—Utilization (rate)

UTM—Unit Training Manager

VHF—Very High Frequency

W&B—Weight and Balance

WASP—Web Applications Software Product

WCE—Work Center Event

WG—Wing

WG/CC—Wing Commander

WG/CD—Vice Wing Commander

WJQS—Work Center Job Qualification Standard

WLCMT—Weapons Load Crew Management Tool

WLCTP—Weapons Load Crew Training Program

WLT—Weapons Load Training

WRE—War Readiness Engine

WRM—War Reserve Materiel

WS—Weapons Standardization

WSM—Weapon System Manager

TQC—Weapons Task Qualification Crew

WTQM—Weapons Task Qualification Training Manager

(Added-AFMC) WTR—Workable TCTO report

WWID—Worldwide Identification (code for TCMax®)

WWM—Wing Weapons Manager

WX—Weather

WUC—Work Unit Code

Office Symbols

(Added-AFMC) AF/A4LM—Air Force Deputy Chief of Staff, Directorate of Logistics, Maintenance Division

(Added-AFMC) AFMC/IG—AFMC Inspector General

(Added-AFMC) AFSC/LZD—Air Force Sustainment Center USAF Aircraft Battle Damage Repair Office

(Added-AFMC) AFTC_SMO—Air Force Test Center Specialized Management Office

(Added-AFMC) AFLCMC/EBHMA—AFLCMC Global Ammunitions Control Point, Afloat Preposition Fleet (APF) Flight/Cell

(Added-AFMC) AFLCMC/EB—AFLCMC Armament Sustainment Division

(Added-AFMC) AFLCMC/EBRM—AFLCMC GACP Missiles and Weapons

(Added-AFMC) AFLCMC/EBWWE—AFLCMC Weapons Engineering Laboratories

(Added-AFMC) AFLCMC/LPS—AFLCMC Propulsion Directorate

(Added-AFMC) AFLCMC/WN—AFLCMC Technology Modernization Program

(Added-AFMC) AFLCMC/WNA—AFLCMC Automatic Test Systems

(Added-AFMC) AFLCMC/WNZ—AFLCMC Support Equipment

(Added-AFMC) AFLCMC/USAF Gunsmith Shop—USAF Gunsmith Shop (Small Arms/Light Weapons)

(Added-AFMC) AFMC/A1MR—Air Force Materiel Commend (AFMC) Manpower Requirements Branch

(Added-AFMC) AFMC/A3/6—AFMC Air, Space and Cyberspace Operations Directorate

(Added-AFMC) AFMC/A3F—AFMC Aerospace Vehicle Fleet Management

(Added-AFMC) AFMC/A4/10—Directorate of Logistics, Civil Engineering, Force Protection, and Nuclear Integration

(Added-AFMC) AFMC/A4M—AFMC Maintenance Division

(Added-AFMC) AFMC/A4MM—AFMC Maintenance Management Branch

(Added-AFMC) AFMC/A4/10/A4—EN—AFMC/A4/10 Engineering Divion

(Added-AFMC) 412 TW—412th Test Wing

(Added-AFMC) 412 EWG—412th Electronic Warfare Group

(Added-AFMC) 412 MXS/MXMX—412 MXS, Maintenance Flight

(Added-AFMC) 412 MXG/MXOPL—412 MXG Logistics Programs

(Added-AFMC) 412 MXG/MXOPO—412 MXG, Organizational Design and Improvement

(Added-AFMC) 412 MXG/MXOPM—412 MXG, Manpower

(Added-AFMC) 412 MXG/MXOL—412 MXG, Logistics Site Support

(Added-AFMC) 412 MXS—412th Maintenance Squadron

(Added-AFMC) 412 MXS/MXMA—412th Maintenance Squadron, Component Repair Flight

(Added-AFMC) 412 MXS/MXMC—Accessories Flight

(Added-AFMC) 412 MXS/MXMR—Armament Flight

(Added-AFMC) 412 MXS/MXMT—412 MXS Support Flight

(Added-AFMC) 412 MXS/MXMW—412th Maintenance Squadron, Munitions Flight

(Added-AFMC) 72 LRS/LGRM—72nd Logistics Readiness Squadron Munitions Section

(Added-AFMC) 763 ESS/PKB—763d Enterprise Sourcing Squadron, Contracting

(Added-AFMC) 78 LRS/LGRW—78th Logistics Readiness Squadron Ammo

(Added-AFMC) 96 TW—96th Test Wing

Terms

Aircraft and Equipment Impoundment—Isolation of an aircraft or equipment due to an unknown malfunction or condition making it unsafe for use or flight.

Aircraft Purpose Identifier Codes (PIC)—specified in DAFI 21-103, PIC are applied to assigned aerospace vehicles to facilitate standardization of reporting. Examples of PIC are: CC=Combat, BQ=major maintenance awaiting AFMC decision or action; DJ=awaiting depot level maintenance work. Refer to DAFI 21-103 for a listing of all specific PIC.

Air Reserve Component—The Air National Guard and Air Force Reserve together form the ARC.

Allowance Standard (AS)—Authorized document that identifies the amount and type of equipment for an organization.

Alternate Mission Equipment (AME)—Equipment identified to a higher end-item, not listed in the table of allowance. Normally, -21 equipment.

Automated Inspection, Repair, Corrosion, and Aircraft Tracking (AIRCAT)—is the Individual Aircraft Tracking Program (IATP) of record for the C-130 as mandated by the USAF Aircraft Structural Integrity Program (ASIP). This effort includes development and maintenance

of an extensive Oracle database and a wide variety of both client, server, and web-based applications to provide data entry, reporting, and analysis.

Awaiting Maintenance (AWM)—Designation for a deferred discrepancy on an aircraft awaiting maintenance.

Awaiting Parts (AWP)—Designation for a deferred discrepancy on an aircraft awaiting parts.

Bench Stocks—Stores of expendability, recoverability, reparability coded (ERRC) XB3 items kept on-hand in a work center to enhance maintenance productivity.

Cannibalization—Authorized removals of a specific assembly, subassembly, or part from one weapons system, system, support system, or equipment end-item for installation on another end-item to meet priority mission requirements with an obligation to replace the removed item.

Centralized Repair Facility—A facility that performs repairs for a specified region or bases.

Certified Load Crew Member—A load crew member trained and certified by position according to **Chapter 10**.

Classified Processing Area (CPA)—Areas identified by the unit which have had an Emission Security assessment and have been approved by the by the wing Information Assurance office to be utilized to discuss or process classified information IAW AFI 16-1404.

Code 1, Code 2, Code 3, Code 4, Code 5—Landing status codes used by aircrew to inform maintenance of their inbound aircraft's condition. A Code 1 aircraft has no additional discrepancies other than those it had when it last departed; a code 2 aircraft has minor discrepancies, but is capable of further mission assignments; a code 3 aircraft has major discrepancies in mission-essential equipment that may require repair or replacement prior to further mission tasking; a code 4 indicates suspected or known nuclear, biological, or chemical contamination; and a code 5 indicates battle damage. Codes 4 and 5 are entered into the MIS as code 8.

Commercial Derivative Aircraft (CDA)—Any fixed or rotary-wing aircraft procured as a commercial Type Certified off-the-shelf aircraft, and whose serial number is listed on an FAA-approved Type Certified Data Sheet.

Commodity Time Compliance Technical Order—TCTO concerning a designated item, subsystem, or system that is not identified as a weapon or military system.

(Added-AFMC) Common Aviation Support Equipment—This supplement further defines and expands the definition in T.O. 00-20-1: Aviation maintenance (AF Specialty Code 2A/2W) support equipment currently in use within the DoD inventory and are applicable to multiple weapon systems, centrally managed and asset controlled by PEO Agile Combat Support (ACS), AFLCMC/WN, for Active, Guard, and Reserve. Specific examples include functional area/type equipment: Accessories, Aerospace Ground Equipment, Aircraft Maintenance Testers, Armament, Automatic Test Systems, Fabrication, Munitions Materiel Handling Equipment, and Propulsion.

Composite Tool Kit (CTK)—A controlled area or container used to store tools or equipment and maintain order, positive control, and ease of inventory. CTKs are assembled as a kit and designed to provide quick, easy visual inventory and accountability of all tools and equipment. CTKs may be in the form of a toolbox, a shadow board, shelves, system of drawers (Stanley Vidmar®,

Lista®), cabinets, or other similar areas or containers. The CTK contains tools and equipment necessary to accomplish maintenance tasks, troubleshooting, and repair.

Condition—**Based Maintenance Plus**—A set of maintenance processes and capabilities derived from real-time assessment of weapon system condition obtained from embedded sensors, external tests and measurements using portable equipment. The goal of CBM+ is to perform maintenance only when internal and/or external sensors indicate the need instead of performing maintenance on a periodic basis. **Consumable Items**—Also known as "Consumption" or "Expendable" Items designated XB3. Items which are consumed in use, or which lose their original identity during periods of use by incorporation into or attachments upon another assembly. Issued on an as required basis and consist of such supplies as maintenance parts or office supplies.

Contracting Officer Representative (COR)—A COR is an individual designated IAW Department of Defense Federal Acquisition Regulation Supplement subsection 201.602-2 and authorized in writing by the contracting officer to perform specific technical or administrative functions.

Crashed, Damaged or Disabled Aircraft Recovery (CDDAR)—The ability to move damaged or disabled aircraft using specialized equipment.

Critical Application Item (CAI)—An item that is essential to weapon system performance or operation, or the preservation of life or safety of operating personnel, as determined by the military services. The subset of CAI whose failure could have catastrophic or critical safety consequences is called CSIs. Refer to **Attachment 7**.

Critical Safety Item (CSI)—A part, assembly, installation equipment, launch equipment, recovery equipment, or support equipment for an aircraft or aviation weapons system that contains a characteristic any failure, malfunction, or absence of which could cause a catastrophic or critical failure resulting in the loss or serious damage to the aircraft or weapons system, an unacceptable risk of personal injury or loss of life, or an uncommanded engine shutdown that jeopardizes safety. Damage is considered serious or substantial when sufficient to cause a 'Class A' mishap. The determining factor in CSIs is the consequence of failure, not the probability that the failure or consequence may occur. For the purpose of this instruction "Critical Safety Item", "Flight Safety Critical Aircraft Part", "Flight Safety Part", "Safety of Flight Item", and similar terms are synonymous.

Cross-tell—Cross-tells are used to highlight trends, benchmarks or safety conditions relating to maintenance equipment, personnel, training, or processes. A cross-tell is initiated to assist other maintenance or logistics personnel with similar equipment to do their jobs more safely and/or efficiently. Typically, a cross-tell will be initiated when a condition or trend is discovered regarding but not limited to, a weapon system or common components that should be shared with other users or potential users. This information should be transmitted using signed and encrypted e-mail to ensure widest dissemination and ensure it is brought to the attention of unit commanders in order to prevent or mitigate mishaps, injury or damage to AF personnel, equipment or property. Typically, cross-tells will provide relevant background information and history and can include such information as NSNs, part numbers, specific location of problem areas.

Customer Wait Time—Customer Wait Time for LRUs is the total elapsed time between the issuance of a customer order and satisfaction of that order, regardless of source (immediate issues or backorders), and can include issues from wholesale and/or retail stocks as well as various other

arrangements. Customer Wait Time for end items (engines and pods) includes time for the retrograde and serviceable transportation legs.

Debriefing—Program designed to ensure malfunctions identified by aircrews are properly reported and documented.

Decertification—The removal of certification status from a person for a specific task

Dedicated Crew Chief—DCCs are first-level supervisors in the flightline management structure who manage and supervise all maintenance on their aircraft, and are selected on the basis of initiative, management and leadership ability, and technical knowledge.

Delayed or Deferred Discrepancies—Malfunctions or discrepancies not creating NMC or PMC status that are not immediately corrected.

Delayed Release—Munition or store that fails to eject from an aircraft upon firing of impulse cartridge, but releases sometime afterwards. Release times qualifying "delayed" bombs are outlined in MDS-specific technical orders.

Demand Response Team—Two-member team where one person reads technical order steps and the other person performs the task and responds when each step is completed.

Depot Level Maintenance—Provides the capability to maintain materiel coded for organizational, intermediate and depot levels of maintenance. Includes maintenance requiring the overhaul, upgrading, or rebuilding of parts, assemblies, or subassemblies, and the testing and reclamation of equipment as necessary IAW AFPD 21-1

Dispatchable CTK—CTK issued out and is designed to be used outside the work center.

Equipment Custodian—Individual responsible for all in-use equipment at the organizational level whose duties include requisitioning, receiving, and controlling of all equipment assets.

Equipment Identification Designator (EID)—A number assigned to a piece of shop equipment, used to track status and accountability.

Equipment Items—Item authorized in the allowance standard within an organization.

Evaluated Load—A loading task that is assessed according to **Chapter 10**.

Expendability, Recoverability, Reparability, Category (ERRC)—Used to categorize Air Force inventory into various management groupings. The grouping determines the type of management used throughout the logistics cycle, designates the process to be used in computing requirements and are used in the correction and reporting of asset and usage data. (such as, XB3, XF3, XD2, NF2, NF4).

Flight Chief—NCO responsible to the maintenance officer or superintendent for management, supervision, and training of assigned personnel.

FK or FV—Prefix used to identify the munitions supply account. FV denotes units utilizing the TICMS and FK denotes units utilizing ILS-S or manual records supply point within a munitions'' operations unit for conventional munitions.

Functional Checklist—locally developed checklists used to identify the steps required to react to specific events. Functional checklists are required for use by functional area(s) during actions such as aircraft crash, mass loads, severe weather warning or evacuation, self-inspections.

Hung Ordnance—Any item attached to the aircraft for the purpose of dropping or firing which has malfunctioned or failed to release. In addition, hung ordnance includes the following items: (1) External fuel tanks after unsuccessful jettison attempt; (2) Remaining ordnance after an inadvertent release; (3) 20/30 mm ammunition after a gun malfunction (no fire, unplanned cease fire, runaway gun, or gun unsafe indication); (4) Any stores determined to be in an unsafe condition.

Inadvertent Release—Uncommanded launch or release of a store or ordnance or launch or release of a store or ordnance other than those selected when a launch or release command was generated (system malfunction); does not include an unintentional release. If commanding a single release, do not consider a double bomb release as an inadvertent release if the releases occur from a practice bomb dispenser.

Individual Tools and Equipment—Tools and equipment that are available for individual sign-out but stored in the tool room in storage bins, cabinets, shelves with every item having an assigned location (example, flashlights, ladders). **Integrated Logistics System-Supply (ILS-S)**—is the overarching term used to describe the system(s) used by base retail materiel management operations. ILS-S is comprised of the Enterprise Solution–Supply. In many cases the term ILS-S is used to identify system related functions and/or references.

Intermediate-Level Maintenance—Maintenance consisting of those off-equipment tasks normally performed using the resources of the operating command at an operating location or at a centralized intermediate repair facility.

In Process Inspection (IPI)—Inspection performed during the assembly or reassembly of systems, subsystems, or components with applicable technical orders. An IPI is accomplished and documented by an authorized IPI inspector other than the technician performing the specific step of a task that requires the IPI.

"Knock It Off"—"Knock it Off" empowers all Airman regardless of rank to terminate an operation or situation which they perceive is unsafe or too dangerous. "Knock it Off" includes using a recognizable "audible" (capable of being heard) from anyone in an effort to prevent a potential mishap.

Lead Crews—A load crew certified by the load standardization crew (LSC), which is assigned to WS to assist in conducting the weapons standardization program. **Levels**—Computed and authorized requirements for a quantity of assets.

Loading Standardization Crew (LSC)—A load crew designated by the WWM and the WS superintendent to administer the weapons standardization program. LSC members have certification and decertification authority.

Loading Task—The actions required by one crew member, in a designated position, to accomplish a munitions load.

Local Commander—The group commander with responsibility for maintenance (as applicable to loading technical data).

Maintenance Capability—Unit's ability to generate and sustain weapon systems to support the mission. It is composed of personnel, capacity (facilities, support equipment, and parts), and weapons systems and is affected by policies and business practices.

Maintenance Cyber Discipline—A focus on daily cyber hygiene activities which requires continuous attention in order to mitigate daily threats by creating a culture of cyber awareness, discipline, and strict compliance.

Maintenance Training—Any proficiency, qualification, or certification tasking required by a technician to perform duties in their primary AFSC.

Master Inventory List (MIL)—Primary source document for inventory of CTKs. The MIL indicates the total number of items in each drawer or section of the tool kit. MIL may be automated.

Mission Design Series (MDS)—Alpha and numeric characters denoting primary mission and model of a military weapons system.

Mission Generation Network—The MGN supports all Organizational-level, on-equipment and off-equipment maintenance and is optimized at the Wing-level across the USAF. MGN consists of the cumulative effort required to generate and sustain sortie and mission production to meet assigned mission requirements.

Minimum Required Proficiency Load (MRPL)—Recurring loading of munitions for which a person is certified.

Munitions Decertification—Removal of the certification status of a person that precludes them from loading a specific type of munitions or MFG.

Normally Installed Equipment (NIE)—launchers, and pylons normally installed on an aircraft.

No-Lone Zone—Area where the two-person concept must be enforced because it contains nuclear weapons, nuclear weapons systems, or certified critical components.

Non-Consumable Item—Also referred to as a "non-expendable" or "equipment" item. Durable items that are capable of continuing or repetitive use by an individual or organization.

Non-Release—System malfunction in which a weapon does not release from the delivery system.

Off-Equipment Maintenance—Maintenance tasks that are not or cannot be effectively accomplished on or at the weapon system or end-item of equipment but require the removal of the component to a shop or facility for repair.

On-Equipment Maintenance—Maintenance tasks that are or can be effectively performed on or at the weapon system or end-item of equipment.

Operating Stock—The bits and pieces needed to support a maintenance work center that does not meet the criteria of bench stock. It includes reusable items such as dust covers, hydraulic line covers, caps, items leftover from work orders, TCTOs. Items deleted from bench stock that are less than a full Unit of Issue (UI) are not considered operating stock but may be retained as work order residue.

Operational Safety, Suitability & Effectiveness (OSS&E)—OSS&E is an outcome of properly applied systems engineering principles, processes, and practices. Well-integrated configuration management and control, deficiency reporting and response, reliability, maintainability, integrity, and other engineering practices ensure that base-lined engineering characteristics of systems and end items are not allowed to degrade as a result of maintenance, repairs, parts substitutions, and similar activities. The PM is responsible for the assurance OSS&E throughout the life cycle of each configuration of each component of each system.

Organizational Level Maintenance—Maintenance consisting of those on-equipment tasks normally performed using the resources of an operating command at an operating location.

PACER WARE—is the unclassified term for an actual change or notification of a deficiency to an Electronic Warfare system.

Personnel Protective Equipment (PPE)—Equipment required to do a job or task in a safe manner.

Plan—A forecasted scheme of sequenced and timed events for accomplishing broad objectives. The plan is the product of annual, quarterly, and monthly planning of scalable operations and maintenance activities necessary to achieve long term mission requirements.

Preload—A complete munition and suspension equipment package ready for loading.

Primary Aerospace Vehicle Authorization (PAA)—The number of aircraft authorized to a unit for performance of its operational mission. The primary authorization forms the basis for the allocation of operating resources to include manpower, support equipment, and flying-hour funds.

Primary Aerospace Vehicle Inventory (PAI)—The aircraft assigned to meet the primary aircraft authorization. Includes PMAI, PTAI, PDAI and POAI.

Production Superintendent (Pro Super)—Senior NCO responsible for squadron maintenance production. Directs the maintenance repair effort.

Program Manager (PM)—The designated individual with responsibility for and authority to accomplish program objectives for development, production, and sustainment to meet the user's operational needs. The PM will be accountable for credible cost, schedule, and performance reporting to the Milestone Decision Authority.

Programmed Depot Maintenance (PDM)—Maintenance activities requiring skills, equipment, or facilities not normally possessed by operating locations.

Project Funds Management Records—a record maintained in the material accounting system to provide for control over that portion of each responsibility center manager operating budget programmed for purchase of expense materiel from the Defense Business Operations Funds stock activity fund. It is used to record available expense authority, current month and fiscal year-to-date sales, sales returns, and due outs for both supplies and expense equipment.

Quality Assurance (QA)—Office or individual who monitors maintenance (organic or contractor) on a daily basis.

Quarterly Evaluation (QE)—Recurring calendar task evaluations required by munitions and weapons personnel.

Quick Reference List (QRL)—Listing of fast moving, high use items required for primary mission aircraft. The basic purpose of the QRL is to provide maintenance personnel with a speedy way to place a demand on the supply system.

Rag—A remnant of cloth purchased in bulk or a standardized, commercial quality, vendor-supplied shop cloth (uniform size and color) or similar material used in general industrial, shop, and flightline operations.

(Added-AFMC) Ready Line—Designated parking area for AGE that is ready for use. Prior to parking AGE on the ready line a service inspection shall be performed and annotated in Part II of

the AFTO Form 244. Ready lines are established by the AGE Flight Chief (or equivalent) as needed, in coordination with OSS's Airfield Operations Flight. The terms READY LINE and SUB-POOL are not synonymous. See SUB-POOL for definition.

Reclama—A request to a duly constituted authority to re-consider its decision or its proposed action (see JP 1-02).

Recoverability Code—A one position code assigned to end items and support items to indicate the recoverability intention and the level of maintenance authorized disposition action on unserviceable support items; and for reparable items, it is used to indicate the lowest maintenance level responsible for repair, disposition, or condemnation of the item.

Recurring Discrepancy—A recurring discrepancy is one that occurs on the second through fourth sortie or attempted sortie after corrective action has been taken and the system or sub- system indicates the same malfunction when operated.

Reliability-Centered Maintenance—A logical discipline for developing a scheduled-maintenance program that will realize the inherent reliability levels of complex equipment at minimum cost.

Remote Split Operations—Occurs when the ground control stations, the Unmanned Aerial Vehicle (UAV) launch and recovery functions, and the satellite uplink are geographically separated.

Repair Cycle Asset—Any recoverable item with an expendability, recoverability, reparability code (ERRC) category of XD or XF.

Repair Recommendation—An idea or proposal to repair an item that is not currently repaired or is beyond the capability of the work center. An AFREP initiative is generated when an asset has a demand level of "greater than three" per calendar year. All new AFREP initiatives will be staffed through the applicable organizations.

Repairable—Unserviceable items that can be economically repaired and restored to a serviceable condition.

Repeat Discrepancy—A repeat discrepancy is a pilot reported discrepancy (PRD) occurring on the same system or subsystem on the first sortie or sortie attempt after that PRD has been signed off.

Retrograde—Returning assets (reparable assets) from the field to their source of repair.

Schedule—Planned events that result in final review and agreement of how to execute a proposed plan of sequenced and timed events. Results in a binding commitment captured in writing and approved by signature between operations and maintenance to complete activities required to accomplish agreed upon objectives. Refers to the execution phase of weekly and daily operations and maintenance activities.

SEEK EAGLE—The Air Force certification program for determining safe carriage, employment and jettison limits, safe escape, and ballistics accuracy, when applicable, for all stores in specified loading configurations on USAF aircraft.

SERENE BYTE—is the unclassified term for an exercise change or deficiency notification to an Electronic Warfare system.

Shop CTK—Tool kits (not dispatched) used by work center personnel during a shift, provided a single person is responsible for the tool kit.

Shop Stock—Includes items such as sheet metal, electrical wire, fabric, and metal stock, used and stored within a maintenance work center to facilitate maintenance.

Source Code—Codes assigned to end items and support items to indicate the manner of acquiring items for the maintenance, repair, or overhaul of end items.

Source, Maintenance, Recoverability (SMR) Code—A code assigned to parts and assemblies that provides maintenance activities with repair level responsibilities, support method and disposition instructions. The SMR codes are also input into the supply and maintenance automated data system. The uniform SMR code is composed of three parts, consisting of a two-position source code, a two-position maintenance code, and one position recoverability code.

Spares—Serviceable assets that are available for future use, and in the logistics pipeline. The term spare carries the assumption that there are already enough assets in the AF inventory to satisfy end item or quantity per aircraft requirements.

Special Certification Roster (SCR)—Management tool that provides supervisors a listing of personnel authorized to perform, evaluate, and inspect critical work.

Special Purpose CTK—Small individually issued tool kits that because of the nature of contents or type of container could preclude shadowing or silhouetting (example, launch kits, recovery kits, cartridge cleaning kits, oxygen servicing kits).

Subcrew—Two or more certified and/or qualified personnel who may perform specific tasks.

Supply Point—Forward warehouse located within or near the maintenance work center.

Supply Reports—There are many examples of "Supply Reports" used to record supply transactions. The Daily Document Register (D04) provides a means for organizations to review all document numbers processed during the day by the SBSS. The Project Funds Management Records and Organization Cost Center Record Update and Reconciliation (D11) show the current status and internal balance of the Project Funds Management Record by supplies and equipment. The Repair Cycle Asset Management Listing (D23) is used to monitor repair cycle assets and as a management product to monitor the stock position and repair cycle status of repairable (DIFM) assets. It may be produced in several sequences and is provided to the customer daily.

Sub-Pool—A parking area designated by the Airfield Operations Flight that provides authorized pooling of serviceable AGE to enhance close proximity support to using organizations.

Tactical/Theater Airborne Reconnaissance System (TARS)—is a sensor package offers improved timeliness, reduced support costs, and improved operational capability over film systems. Once fielded, this system will provide the tactical commander with an organic system capable of responding in Near Real-Time (NRT) (in time) to battlefield requirements.

Tail Number Bins (TNB)—Locations established and controlled to store issued parts awaiting installation and parts removed to FOM. Holding bins are set up by tail number, serial number, or identification number.

Task Assignment List—Functional grouping of procedural steps from applicable -33 series TOs, by crew position, to be accomplished in sequence by each crew member during an operation.

Technical Administrative Function—Function responsible for ordering and posting instructions, processing all orders, enlisted performance ratings, and general administrative tasks for the section.

Technical Data—Information (regardless of the form or method of the recording) of a scientific or technical nature, including computer software documentation. As applied in this publication, it includes information required for the design, development, production, manufacture, assembly, operation, training, testing, repair, maintenance, or modification of defense articles.

Technical Order Distribution Office (TODO)—Function required to maintain records on TOs received and distributed.

(Added-AFMC) Temporary Modifications—Temporary modifications change the configuration of an item to enable short-term operational mission accomplishment, or to conduct T&E of new and modified equipment. Temporary modification proposals are validated, reviewed, approved as described in AFI 10-601, *Operational Capability Requirements Documentation and Validation*, and DAFI 63-101/20-101. Refer to DAFMAN 65-605, Vol. 1, Budget Guidance and Technical Procedures, for AF policy on funding. There are two kinds of temporary modifications: Temporary-1 (T-1) and Temporary-2 (T-2).

Time Compliance Technical Order (TCTO)—Authorized method of directing and providing instructions for modifying equipment and performing or initially establishing one-time inspections.

Tool Storage Facility/Tool Room—A controlled area within a work center designated for storage and issue of tools and equipment.

Total Asset Visibility—The capability to provide users with timely and accurate information on the location, movement, status, and identity of units, personnel, equipment, materiel, and supplies. It also includes the capability to act upon that information to improve overall performance of the Department of Defense's logistic practices.

Unintentional Release—Store or ordnance launched or released through pilot error.

Unique Item Identifier (UII)—The set of data elements marked on items that are globally unique, unambiguous, and robust enough to ensure data information quality throughout life, and to support multi-faceted business applications and users.

Unit Committed Munitions List (UCML)/Test/Training Munitions List (TTML)—The UCML/TTML is a list of primary munitions (PM), support munitions (SM), and limited-use munitions (LM) necessary to meet unit operational and training requirements.

Unmanned Aerial Vehicle (UAV)—An unmanned aircraft that is either remotely piloted (such as, Predator) or programmed (such as, Global Hawk).

Urgency Justification Code—Two-digit code used to reflect the impact and type of need. The Urgency of Need Designator fills the first position of the Urgency Justification Code. Use of Urgency of Need Designator 1, A and J is restricted and is verified by designated personnel.

Utilization Rate (UTE Rate)—Average number of sorties or hours flown per primary assigned aircraft per period. Usually, time period is based on a monthly rate.

War Readiness Engine (WRE) Levels—The quantity of net serviceable engines required to support the Air Force war tasking and to sustain operational units' war efforts until pipelines are filled and repair capabilities are available. These engines are to be available to support a weapon

system from the start of the war until re-supply (via base, intermediate, CRF or depot repair) is established.

War Reserve Materiel (WRM)—Consists of enterprise managed, dynamically positioned equipment and consumables that contribute to initial operations and provide initial support cross the full range of military operations. It enhances Agile Combat Support capability to reduce the time required to achieve an operational capability and/or produce an operational effect.

Weapons Certification—The act of verifying and documenting a person's ability to load a particular type of aircraft, and munition or MFG within established standards.

Weapons Locally-Manufactured Equipment (LME)—All equipment that measures, tests, or verifies system, subsystem, component, or item integrity. It also includes equipment such as handling dollies, storage racks (except storage shelves), maintenance stands, or transport adapters. It does not include simple adapter cables and plugs constructed as troubleshooting aids to replace pin-to-pin jumper wires specified in TOs.

Weapons Standardization (WS)—Organization comprised of the WWM, a Superintendent, the Load Standardization Crew, an academic instructor, and lead crews.

Weapons Task Qualification—A munitions related task not requiring certification.

Weight and Balance (W&B) Program—Program used in calculating, verifying, updating, and computing weight and balance on a weapon system.

Attachment 2

AIRCRAFT COMMANDER FEEDBACK ON FCC

Figure A2.1. Aircraft Commander Feedback on FCC.

MEMORANDUM FOR <Unit Designation/Office Symbol> *Date*
<Street>
<Base, State, and Zip Code>
FROM: <Aircraft Commander>
<Street>
<Base, State, and Zip Code>
SUBJECT: Aircraft Commander Feedback of the Flying Crew Chief
(FCC) Was the FCC knowledgeable of the aircraft and the systems?
a - Extremely knowledgeable c - Lacks knowledge
b - Sufficient knowledge d - Not observed
Did the FCC know the status of PMC and NMC discrepancies?
a - Always c - Rarely
b - Most of the time d - Never
Did the FCC perform duties willingly and enthusiastically?
a - Always c - Never b
- Sometimes d - Not
Observed
What type of working relationship did the FCC have with the aircrew?
a - Outstanding c - Fair
b - Good d - Poor
Rate the overall maintenance support provided by the FCC:
a - Outstanding c - Fair
b - Good d - Poor
This FCC was:
a - An asset to the FCC program c - Just getting by b - A hard
worker, but needs more experience d - Detriment to the FCC program
Remarks:
POC is <FCC Program Manager's Name, office symbol, duty phone number>.<signed>
Aircraft Commander

Note*:* Please fold and return to the squadron FCC Program Manager upon return to home station.

Attachment 3

QUARTERLY FCC REPORT FORMAT

Figure A3.1. Quarterly FCC Report Format.

MEMORANDUM FOR MAJCOM/A4L *Date*
FROM: <Unit Designation/Office Symbol>
<Street>
<Base and Zip Code>
SUBJECT: <State fiscal quarter (FY20/1)> Quarterly Flying Crew Chief Report (RCS: AF/A4L(Q&A)0011)

IAW AFI 21-101 <unit designations> report is submitted.
Number of C-coded FCC positions on the Unit Manpower Document entitled to be filled. Include approved changes (losses/increases):
Number of people filling C-coded positions:
Number of qualifying missions flown per quarter by C-coded crew chiefs. Include the number of TO directed missions:
Number of qualifying missions flown by personnel without C-coded prefix. Include TO directed missions flown by non-C-coded prefix personnel:
Number of all missions away from home station that required FCCs:
Total number of days TDY for all C-coded crew chiefs on qualifying missions:
Total number of days TDY for all non-C-coded crew chiefs on qualifying missions:
Unit and MAJCOM remarks and overall program assessment. Include remarks to justify vacant positions: FCC Program Manager is <rank, name>, office symbol, DSN number.
<Sign>
Commander, <Unit Designation>

Attachment 4

ANNUAL FCC REPORT

Figure A4.1. Annual FCC Report.

MEMORANDUM FOR MAJCOM/A4L or DOM *Date*

FROM: <Unit designation/Office Symbol>
<Street>
<Base and Zip Code>
SUBJECT:<state fiscal year (FY20)> Annual Flying Crew Chief Report RCS: AF/A4L(Q&A)0011) IAW AFI 21-101<unit designations> report is submitted.
Number of C-coded FCC positions on the Unit Manpower Document entitled to be filled. Include approved changes (losses/increases):
Number of people filling C-coded positions:
Number of qualifying missions flown per quarter by C-coded crew chiefs. Include the number of TO directed missions:
Number of qualifying missions flown by personnel without C-coded prefix. Include TO directed missions flown by non-C-coded prefix personnel:
Number of all missions away from home station that required FCCs:
Total number of days TDY for all C-coded crew chiefs on qualifying missions:
Total number of days TDY for all non-C-coded crew chiefs on qualifying missions:
Unit and MAJCOM remarks and overall program assessment. Include remarks to justify vacant positions: FCC Program Manager is <rank, name>, office symbol, DSN number.
<Sign>
Commander, <Unit Designation>

Attachment 5

FCC SDAP REQUEST

Figure A5.1. FCC SDAP Request.

MEMORANDUM FOR MAJCOM/A4L or DOM *Date*

FROM: <Unit Designation/Office Symbol>
<Street>
<Base and Zip Code>

SUBJECT: Flying Crew Chief (FCC) SDAP Positions <Increase/Decrease> Request

IAW <unit designations> requests <increase or decrease> of <state quantity of positions>. Provide brief justification; include comments about force structure changes, additional mission requirements. FCC Program Manager is <rank, name>, office symbol, DSN number.
<Sign>
Commander, <Unit Designation>

Attachment 6

FOREIGN OBJECT DAMAGE (FOD) REPORT

Figure A6.1. Foreign Object Damage (FOD) Report.

MEMORANDUM FOR *Date*

FROM:<Unit Designation/Office Symbol> <Street> <Base and Zip Code>
SUBJECT: <Foreign Object Report>. FOD program report number (unit, year, and month, followed by sequence number -- example, 301FW-060501).

Type of report: Initial/Formal Update/Final FOD Report
Date and Time of Incident:
Unit and Base of Incident:
Origin of Sortie:
When discovered (Preflight, Postflight, In-Coming, ETS)
Owning Unit, Base and MAJCOM
MDS and Tail Number (N/A for ETS incidents)
Engine Type, Model, Series (TMS):
Engine S/N:
Engine Position (If Applicable):
Time Since Overhaul:
Description of Incident:
Material Failure: (Yes or No)
Tech Data Deficiency: (Yes/No) Preventable/Non-Preventable:
Investigation Findings:
Action Taken to Prevent Recurrence:
Parts Cost: Labor Cost: Total Cost:
Additional Comments (if necessary):

<Sign>
FOD Monitor, <Unit Designation>

Attachment 7

CRITICAL APPLICATION ITEMS & CRITICAL SAFETY ITEMS

A7.1. CRITICAL APPLICATION ITEMS (CAIs). For the purpose of this instruction, it is an item that is essential to weapon system performance or operation, or the operating personnel as determined by AFI 20-106IP, *Management of Aviation Critical Safety Items*.

A7.1.1. Includes flight safety items, life support, critical safety items (CSI), and nuclear hardened items. For systems including radar, avionics, munitions, contact the PO for the system to obtain the CAI designation.

A7.1.2. The management of CAIs (contains unique repair and manufacturing qualifications; material and manufacturing process requirements; and extensive testing requirements after repair) is a complex process. These specified procedures rest with the program manager.

A7.1.3. Other than TO and PO approved repairs, Electronic Warfare (EW) Systems are Critical Application Items and prohibited from consideration under the AFREP program. Performing repairs on EW system components may render the entire EW system degraded. Environmental Stress Screening (ESS) of repaired EW components is mandatory. Many EW components cannot be repaired due to electrical characteristics that are not visible or evident without special test facilities and procedures, none of which are available to field personnel or unqualified contractors.

A7.2. CRITICAL SAFETY ITEMS (CSIs). For the purpose of this instruction, CSIs are items whose failure may cause loss of life, permanent physical disability or major injury, loss of a system, or significant damage to equipment.

A7.2.1. Special attention is placed on CSIs due to potential catastrophic or critical consequences of failure; Public Law 108-136, section 802, *Quality Control in Procurement of Aviation Critical Safety Items and Related Services*, was enacted to address aviation CSIs. The public law addresses three concerns:

A7.2.1.1. The Design Control Activity (DCA) is responsible for processes related to identification and management of CSIs used in procurement, modification, repair, and overhaul of aviation systems. The DCA is defined as the systems command of a military Service responsible for the airworthiness certification of the system in which a CSI is used.

A7.2.1.2. For contracts involving CSIs, DoD is restricted to DCA approved sources.

A7.2.1.3. The law requires that CSI deliveries and services meet the technical and quality requirements established by the DCA.

A7.2.2. ODM 4140.01, *DoD Supply Chain Materiel Management Procedures*, establishes procedures for the management of aviation CSIs. AFI 20-106IP, *Management of Aviation Critical Safety Items*, addresses requirements for identification, acquisition, quality assurance, management, repair, and disposition of aviation CSIs.

Attachment 8 (Added-AFMC)

TEST WING MAINTENANCE AND FLYING SCHEDULING PROCEDURES

A8.1. (Added-AFMC) General Responsibility and Policy.

A8.1.1. **(Added-AFMC)** Goals and Objectives. The goal of this attachment is to provide the most effective guidance for unit planning and execution of operations flying and maintenance scheduling activities in support of the AFMC flying hour program. The principal area of concern is the overall flying/ground test schedule execution balanced against the health of the fleet. For mission accomplishment and improved efficiency, the following must be considered: maximize crew training on all flights, plan alternate missions when possible, ensure configurations and fuel loads are accurate, and establish launch and recovery patterns, and utilize historical attrition data. The objectives of this guidance are to:

A8.1.1.1. **(Added-AFMC)** Provide methods for units to develop and improve their flying and maintenance scheduling programs.

A8.1.1.2. **(Added-AFMC)** Provide audit methods so that units may analyze their effectiveness.

A8.1.1.3. **(Added-AFMC)** Provide methods for local commanders, and the MAJCOM, to assess effectiveness of unit performance in schedule planning and execution.

A8.1.2. **(Added-AFMC)** Guidance. These instructions provide procedures and audit methods for units to develop their flying and maintenance scheduling program and analyze their effectiveness. It is intended to be a local tool for operations and maintenance activities to use in support of their programs. Reviewing flying and maintenance schedule deviations, and the reasons for those deviations, allows wing commanders (WG/CC) and their staffs/subordinates to evaluate unit flying program and scheduling procedures for efficiencies and effectiveness. Higher Headquarters (HHQ) management attention is directed to those areas beyond a unit's control.

A8.1.3. **(Added-AFMC)** Applicability. This publication is applicable to all AFMC Centers and activities listed in the PURPOSE section, possessing, flying or supporting aircraft, to include contractor maintained operations.

A8.1.4. **(Added-AFMC)** Standardization. Wings will develop a maintenance and operations coordinated supplement to this instruction standardizing attrition rate, spare factors, maximum aircraft commitment rate, and scheduling practices for the wing and each assigned mission design series (MDS). This local supplement will be reviewed and updated annually. Minimum topics should include quiet hour policies, cross country takeoffs and returns, minimum turn times, crew ready times, etc. Supplements will include local schedule input and publishing deadlines along with any wing unique requirements. **(T-2)**

A8.1.5. **(Added-AFMC)** Reporting Requirements. Flying scheduling deviation recording and reporting is required for all AFTC and AFLCMC possessed aircraft while status, inventory, and utilization reporting is required for all AFMC units and activities possessing or supporting AFMC possessed aircraft to include those possessed for the AFMC mission by contractors.

A8.1.5.1. **(Added-AFMC)** Reporting procedures are contained in **Chapter 2**. Operations, maintenance, test agencies, and others share in the responsibility for

monitoring and controlling deviations. Schedule deviations altering the planned use of resources generally have a domino effect that upsets future and near-term events. Every effort must be made to reduce or eliminate deviations.

A8.1.5.2. **(Added-AFMC)** Units will use approved MIS reporting procedures. This includes contractor-maintained, AFMC assigned and possessed aircraft. In the event that the contractor is not obligated to utilize the MIS, the MXG commander or equivalent is responsible to ensure all reporting procedures are complied with. **(T-2)**

A8.1.6. **(Added-AFMC)** First Look. The maintenance planning cycle ensures proper and effective use of maintenance resources. Maintenance schedulers use both maintenance and operational long-range planning to assess maintenance's ability to support the wing flying program, Programmed Depot Maintenance (PDM) schedules, Time Compliance Technical Order programs, scheduled inspections and exercises. PS&D, or equivalent, will coordinate with AMUs in performing the long range planning function and use local products and MIS products such as time distribution index (TDI), planning requirements (PRA), and workable TCTO report (WTR), and modification schedules to determine long range maintenance requirements. PS&D will forecast and monitor requirements for the planned fiscal year. Maintenance requirements are included in long-range plans to project aircraft availability.

A8.1.6.1. **(Added-AFMC)** Every year, no later than 5 March, MOF PS&D will request Maintenance Management Analysis (MMA) accomplish airframe, personnel, and facility capabilities for each AMU based on historical data to start the First Look requirements. The capabilities are due back to MOF PS&D NLT the last workday of March.

A8.1.6.2. **(Added-AFMC)** MOF PS&D will provide copies of the capability study to each Operations Squadron (OS) operations scheduling and maintenance supervision. AMU PS&D will refine the initial MMA airframe capability by applying projected maintenance requirements to the historical data. AMU PS&D will provide first look maintenance capability projections in a monthly format to OS operations officer, AMU OIC/ NCOIC, AMXS/CC/Maintenance Officer and MOF PS&D by 15 May. Projections include known operational and test requirements, an assessment of maintenance's ability to support the monthly contract requirements, and an overall assessment of the unit's maintenance capability to meet the annual flying hour program. **(T-2)**

A8.1.6.3. **(Added-AFMC)** MOF PS&D will gather the AMXS and OS coordinated operational capability and forward them to OSS Operations Flight scheduling section for consolidation into a comprehensive package that includes a breakdown of the following items by OS/AMU: **(T-2)**

A8.1.6.3.1. **(Added-AFMC)** Sorties contracted/scheduled per day. Compute the number of sorties (hours) required per operations and maintenance (O&M) day to meet the operational and test requirement using the formula: Number of sorties (hours) required divided by number of O&M days in a given month. Sorties (hours) per day will be computed by month for each operational squadron/AMU.

A8.1.6.3.2. **(Added-AFMC)** Monthly scheduled sorties. Compute monthly scheduled sortie requirements using the formula: (Number of sorties or hours required) divided by (1 minus the attrition factor). Example: 1,000 sorties or hours required

divided by (1 minus 0.15) equals 1,177 sorties or hours to schedule. Remember to round any part up to the next whole sortie or hour.

A8.1.6.3.3. **(Added-AFMC)** Inspection dock capabilities. Compute the number of Phases/ISO inspections to be accomplished in order to meet operational and test requirements for each AMU, by month, for the entire fiscal year. Compute dock capability using the formula (number of O&M days) divided by (number of phase days) times (inspection cycle). Inspection dock capabilities is provided at the wing level and provided by the squadron performing inspections.

A8.1.6.3.4. **(Added-AFMC)** Sortie UTE Rates. Compute UTE rates by month for the entire fiscal year for contracted (required) sorties and scheduled sorties using formula: (number of sorties per month) divided by (number of PAI aircraft).

A8.1.6.4. **(Added-AFMC)** Once compiled, the annual maintenance plan will be presented to the OG and MXG CCs (or equivalents) before being presented to the WG/CC for final approval. MOF PS&D will review the comprehensive package submitted to OSS Operations Flight scheduling section and provide feedback to AMU PS&D and MXG/CC if required. Final assessments of maintenance capabilities to support the operations ―first look‖ projections are sent to AFMC/A4/10. **(T-2)**

A8.1.6.4.1. **(Added-AFMC)** Final assessments of maintenance capabilities to support the operations "first look" projections will be sent to AFMC A4 NLT 30 June.

A8.1.7. **(Added-AFMC)** Flying Hour Allocation/Annual Maintenance Planning Cycle.

A8.1.7.1. **(Added-AFMC)** Flying Hour Allocation. Using funding guidance from the Financial Management organization in the chain of command, PS&D and OS/OSS scheduling provide affected work centers the following planning factors NLT 20 August each year, or within 10 working days after receipt of the flying hour allocations:

A8.1.7.1.1. **(Added-AFMC)** Updated capabilities computed by MMA, provided by PS&D.

A8.1.7.1.2. **(Added-AFMC)** Required flying hours and estimated sorties and missions, provided by OSS, in monthly increments.

A8.1.7.1.3. **(Added-AFMC)** Flying days in each month, provided by OSS.

A8.1.7.1.4. **(Added-AFMC)** Aircraft and aircrew alert requirements, provided by OSS.

A8.1.7.1.5. **(Added-AFMC)** Known and projected TDY and special mission requirements, provided by OSS.

A8.1.7.1.6. **(Added-AFMC)** PDM schedule, provided by PS&D.

A8.1.7.1.7. **(Added-AFMC)** Configuration and munitions requirements, provided by OSS.

A8.1.7.1.8. **(Added-AFMC)** Special mission and test requirements.

A8.1.7.2. **(Added-AFMC)** NLT 1 September, or within 10 working days after receipt of the planning factors, AMU and AMXS maintenance supervision provides PS&D, SQ/CCs, and OSS operations scheduling the following planning factors:

A8.1.7.2.1. **(Added-AFMC)** Estimated number of aircraft available by month, taking into consideration aircraft required for training.

A8.1.7.2.2. **(Added-AFMC)** A projected airframe capability statement.

A8.1.7.2.3. **(Added-AFMC)** Forecasted personnel capability, taking into consideration required annual training for maintenance personnel. (Not applicable to contract maintenance organizations).

A8.1.7.2.4. **(Added-AFMC)** The number of supportable sorties for each month.

A8.1.7.2.5. **(Added-AFMC)** An estimated monthly attrition factor provided by MMA. This factor combines operations weather and materiel (maintenance and supply) factors. Maintenance is responsible for adding the attrition factor to operational requirements. AMUs supporting multiple flying squadrons will not utilize a joint, or combined, attrition rate.

A8.1.7.2.6. **(Added-AFMC)** A recommended block scheduling pattern.

A8.1.7.2.7. **(Added-AFMC)** A statement of limitations.

A8.1.7.3. **(Added-AFMC)** The OSS Operations Flight commander consolidates unit inputs into a final package and coordinates a combined OG and MXG CC concurrence meeting before final approval from the WG/CC. MOF PS&D will review the package and provide feedback to AMU PS&D and MXG/CC, if required. The final approved package constitutes the wing annual FHP. A copy of the approved annual plan and changes thereafter will be forwarded to MOF PS&D and the Analysis section. Wings will submit the annual FHP (contract) to AFMC/A3F (AFMC Flying Hour Program) and AFMC/A4M around the 15th of September for awareness. Advise HQ AFMC for any specific MAJCOM-level awareness needs or support required. **(T-2)**

A8.1.7.4. **(Added-AFMC)** The annual plan will show the number of flying hours projected to fly by aircraft MDS. AFTC flying hours will be subdivided into planned test/test support (Program Element Code (PEC) 65807F), test pilot school (PEC 65807F), and an all other category (i.e. proficiency, maintenance, ferry, etc. (by applicable PEC)). **(T-2)**

A8.1.7.5. **(Added-AFMC)** Maintenance requirements are included in long-range plans to project aircraft availability. Develop long-range maintenance plans in as much detail as possible. All maintenance requirements will be consolidated into a single long-range plan using AF Form 2401, Equipment Utilization and Maintenance Schedule, or computer generated form. As a minimum, the long-range plan shows the current month and the next 2 months known maintenance requirements. As a minimum, include the next due date for all calendar inspections, calendar time change items, TCTOs in workable status, PDM schedules, and at least the next 3 aircraft ISO/Periodic/Phase inspections. Other maintenance requirements such as engine changes, hourly requirements, acceptance/transfer inspections, training and cannibalization aircraft will be posted as they become known or planned. Add Alternate Mission Equipment (AME) inspections to the long-range plan if the aircraft is scheduled to stay in that configuration to ensure the inspections are included in the monthly and weekly schedules. **(T-2)**

A8.1.7.6. **(Added-AFMC)** Maintenance planning includes predictable maintenance factors based on historical data and other staff inputs (e.g., flow times for maintenance, turnaround times, and parts replacement schedules). **(T-2)**

A8.1.7.7. **(Added-AFMC)** Include all known operational events (e.g., exercises, deployments, and development tests) during maintenance planning to determine the maintenance capability to meet operational needs. **(T-2)**

A8.1.7.8. **(Added-AFMC)** The Test Wing scheduling office will use the following priority as a guide to determine which objectives to support if a lack of resources prevents meeting requirements such as: alert commitments, HHQ directed missions, training, and test and test support missions.

A8.1.8. **(Added-AFMC)** Quarterly Scheduling. The quarterly plan applies to Test Wing training missions and all known test missions.

A8.1.8.1. **(Added-AFMC)** Quarterly planning starts with the operational requirements for the flying hours, UTE rate, and airframe availability, alert and other related scheduling data. The OS operations officer provides these requirements to maintenance supervision and PS&D NLT 25 days before the beginning of the quarter. Maintenance supervision and the OS operations officer discuss these requirements at the scheduling meeting before the quarter being planned.

A8.1.8.1.1. **(Added-AFMC)** The Test Wing scheduling office will use the following priority as a guide to determine which objectives to support if a lack of resources prevents meeting requirements such as: alert commitments, HHQ directed missions, training, and test and test support missions.

A8.1.8.2. **(Added-AFMC)** The unit's rolling 3-month maintenance plan briefed each month meets the intent of the quarterly scheduling process. The meeting may be held in conjunction with the weekly or as a separate scheduling meeting. The plan will be posted so it may be viewed by both maintenance and operations.

A8.1.9. **(Added-AFMC)** Monthly Scheduling. Due to the nature of the test mission, publishing the monthly flying schedule according to DAFI 21-101 may not be conducive to mission requirements; however, a tentative monthly plan will be created for planning purposes. Every effort will be made to create a realistic monthly plan to accurately plan resource utilization. **(T-2)**

A8.1.9.1. **(Added-AFMC)** The monthly scheduling meeting may be held in conjunction with the weekly scheduling meeting or as a separate scheduling meeting, NLT the third week of the month prior.

A8.1.9.2. **(Added-AFMC)** Included in the monthly flying and maintenance plan are:

A8.1.9.2.1. **(Added-AFMC)** Follow guidelines in **Chapter 14** with the following exceptions:

A8.1.9.2.1.1. **(Added-AFMC)** Only known operational and test requirements are reported (not detailed utilization).

A8.1.9.2.1.2. **(Added-AFMC)** Known ground test/mount requirements are included.

A8.1.9.3. **(Added-AFMC)** Post the monthly flying and maintenance plan NLT 1200 Friday of the third week. All agencies will submit their monthly schedule inputs to PS&D for consolidation and coordination.

A8.1.9.4. **(Added-AFMC)** QA scheduled inspections listed by type and quantities are included in the monthly flying and maintenance plan unless published separately by QA. **(T-2)**

A8.1.10. **(Added-AFMC)** Weekly Scheduling. Refer to details within this attachment along with **paragraph 14.5.5** for any additional guidance.

A8.1.10.1. **(Added-AFMC)** Also included in the weekly flying and maintenance schedule will be:

A8.1.10.1.1. **(Added-AFMC)** Sortie sequence numbers, aircraft tail numbers (primary and spares), scheduled take-off and landing times, aircraft or equipment scheduled use times, configurations, fuel loads and special equipment requirements. **(T-2)**

A8.1.10.1.2. **(Added-AFMC)** Scheduled maintenance actions by aircraft and equipment serial number to include inspections, TCTOs, time changes, contract and depot inputs, engine changes, washes or corrosion control, documents review, deferred discrepancies, modifications and hours remaining to phase. **(T-2)**

A8.1.10.1.3. **(Added-AFMC)** Spare aircraft requirements are based on projected test mission needs. The following factors are used to determine spares and are not hard limits but guidelines, and as such, it is up to the AMU to actually schedule what it can support:

A8.1.10.1.3.1. **(Added-AFMC)** The spare requirement may be adjusted to compensate for multiple configurations and test or school syllabus constraints. **(T-2)**

A8.1.10.1.3.2. **(Added-AFMC)** Additional spares are authorized to support HHQ taskings and special missions (if required by the tasking). **(T-2)**

A8.1.10.2. **(Added-AFMC)** AF Form 2407, *Approval Authority*. All AF Form 2407 changes that add aircraft, sorties, or increase the flying window require both OG and MXG commander (or designated group level representative) approval. MXG/CD or equivalent will approve changes to the final schedule and coordinated through affected AMXS Operations Officer/MX SUPT, AMU OIC/SUPT, MXS Operations Officer/MX SUPT, operations group representative, maintenance group, and wing staff agencies. (e.g., MOC, PS&D, Analysis, etc.) using an AF Form 2407. MOC will coordinate higher headquarters directed taskings that require immediate execution. **(T-2)**

A8.1.10.2.1. **(Added-AFMC)** Changes to the Weekly Schedule. Pen-and-Ink. The pen-and-ink AF Form 2407, Weekly/Daily Flying Schedule Coordination is not intended to be used as a tool to extend the scheduling process by another day. Pen-and-ink changes made to next week's schedule are authorized, provided they are submitted to the PS&D by 1600 hours Friday or at the daily maintenance scheduling/production meeting, whichever occurs first (**Exception:** NLT 2 hours after the last landing during printed wing night flying weeks). They are non-reportable and become part of the

printed weekly flying schedule. An AF Form 2407 is required stating the changes are pen-and-ink. **Note:** The intent of the pen-and-ink AF Form 2407 is not to create complete revisions of the previously Wing/CC approved schedule. **(T-2)**

A8.1.10.2.2. **(Added-AFMC)** AFMC units are authorized to use the Center Scheduling Enterprise in recording and coordinating changes to the weekly schedule in place of AF Form 2407. Electronic coordination is acceptable provided receipt is acknowledged and the sender enters the name of the person notified, along with date/time of notification, and logged-in accordance with requirements on the AF Form 2407. **(T-2)**

A8.1.10.3. **(Added-AFMC)** The MXG/CC/CL, OG/CC, and/or director(s) will approve the proposed weekly flying schedule and forward to the WG/CC for final approval. Once the weekly schedule is reviewed and signed by the OG/CC, MXG/CC and WG/CC it becomes the final planning guide for both operations and maintenance and the basis for deviation reporting. The schedule will be followed as printed or as amended by coordinated changes. MOF PS&D will publish the final weekly flying schedule. **(T-2)**

A8.1.10.4. **(Added-AFMC)** Rapid Crew Swap (RCS). To meet mission requirements, MXG/CC may authorize the use of RCSs for fighter/attack aircraft. Local Checklists (LCL) will be developed in coordination with applicable MDS specific -6 TO or maintenance manual, that standardizes specific procedures when performing RCS tasks. RCSs involves turnaround of an aircraft incorporating partial or full crew change between two separate sorties after engine shutdown. The returning aircrew must brief the departing aircrew at the aircraft. The difference between RCS and the policy for continuation sorties is RCSs allow minor servicing (i.e. fuel, hydraulics, oil/sampling) to be performed between sorties. If at any time the aircraft is returned to maintenance for anything other than minor servicing, the RCS is terminated. RCS sorties will be published in the weekly flying schedule; deviations will apply to each sortie.

A8.1.10.5. **(Added-AFMC)** MOF PS&D will record maintenance scheduling deviations and forward the computations to MOF maintenance analysis weekly for publication in the monthly maintenance summary. **(T-2)**

A8.1.11. **(Added-AFMC)** Daily Scheduling Process. Daily Scheduling Process. Daily scheduling must be approved by the MXG/CC. If utilized, units must effectively plan and execute operations and maintenance schedules. Each day, no later than 1600 local time, the OG and MXG CCs or their designated group representative will review and publish the daily flying and maintenance schedule, agree on revisions, and coordinate with other agencies to confirm the next day's activities. Use AF IMT 2407 or authorized automated system in coordinating the final schedule through the affected production superintendent, AMU OIC/SUPT, AMXS Maintenance Supervision (MS), MXS MS, operations group representative, maintenance group, and wing staff agencies. (i.e. MOC/ PS&D, Analysis, etc.). Deviations will be recorded based on this final daily flying schedule. Additional scheduling reporting procedures are located in paragraph **2.3** and **5.1** of this Attachment. **(T-2)**

A8.1.11.1. **(Added-AFMC)** The finalized daily flying schedule will include, at a minimum, the following details: **(T-2)**

A8.1.11.1.1. **(Added-AFMC)** Aircraft tail numbers.

A8.1.11.1.2. **(Added-AFMC)** Aircraft takeoff and landing times.

A8.1.11.1.3. **(Added-AFMC)** Configuration requirements.

A8.1.11.1.4. **(Added-AFMC)** Munitions requirements.

A8.1.11.1.5. **(Added-AFMC)** Fuel loads.

A8.1.11.1.6. **(Added-AFMC)** Special or peculiar test support requirements.

A8.1.11.1.7. **(Added-AFMC)** Special or peculiar mission support requirements.

A8.1.11.1.8. **(Added-AFMC)** Off station sorties.

A8.1.11.1.9. **(Added-AFMC)** On equipment training requirements.

A8.1.11.1.10. **(Added-AFMC)** Other special requirements (i.e. ground test/mount).

A8.1.11.2. **(Added-AFMC)** Maintenance Debrief will load all Functional Check Flights (FCF)/OCF, Ferry and cross country returns as "flown as scheduled" in the MIS. (**T-2)**

A8.2. (Added-AFMC) Total Scheduling Effectiveness (TSE).

A8.2.1. **(Added-AFMC)** Purpose. This section defines flying schedule deviations and provides formulas for computing TSE. A cornerstone of successful flying scheduling and execution is an understanding of how the schedule is executed versus how it was planned to be executed. These differences in scheduled versus actual events, are only recorded in the execution phase of the scheduling process and are called deviations. Deviation data must be recorded so that follow-up analysis can identify appropriate corrective actions, if any. Deviation data recording and analysis is the beginning of the process that will, in the end, improve unit's flying operations. The unit is responsible for documenting deviations to the flying schedule and determining the cause for each deviation. Deviations must be coordinated with the appropriate agency before being assigned to a specific category. Schedule deviations that result from a sequence of events will be assigned a primary cause. A determination of the primary cause will be made by the parties involved to arrive at a unit position. The Operations representative and the AMU/AMXS MS will monitor deviations to ensure they meet the criteria in this publication. When conflicts arise, leadership of involved units will resolve them at the lowest level. All deviations will be recorded as described in this publication. Although every attempt will be made to properly schedule and reduce deviations, this metric must be viewed in conjunction with additional mission execution metrics to allow a more comprehensive assessment of maintenance effectiveness. Add narrative to read: Although every attempt will be made to properly schedule and reduce deviations, this metric must be viewed in conjunction with additional mission execution metrics to allow a more comprehensive assessment of maintenance effectiveness. The previous week's deviations will be briefed by the responsible organizations at the Weekly Wing Scheduling Meeting. **(T-2)**

A8.2.2. **(Added-AFMC)** Requirements. Total scheduling effectiveness computation and deviation recording are required for all AFMC assigned and possession-only aircraft. Reporting procedures are contained in **paragraph 2.5**.

A8.2.3. **(Added-AFMC)** Deviations.

A8.2.3.1. **(Added-AFMC)** Deviations apply to the respective Test Wing's final/approved weekly or daily (if utilized) flying schedule. Any change to the approved daily flying

schedule after 1600 the day prior will be recorded as a deviation. Multiple chargeable deviations will not be recorded against a single line entry except for (a) additions that air abort, (b) additions that cancel or ground abort, (c) added sorties that take-off late, and (d) late take-offs that air abort; however, all deviations will be recorded in the MIS. The AFTO Form 781, ARMS Aircrew/Mission Flight Data Document is the official source document for takeoff and landing data. For all deviations, the person recording the deviations in the MIS will provide a detailed explanation in the remarks section and a Job Control Number/Event ID in IMDS screen 350, if applicable. Schedule deviations fall into one of the following categories:

A8.2.3.2. **(Added-AFMC)** Ground Deviations. Ground deviations are events occurring before aircraft takeoff or scheduled ground test. All ground deviations are recorded and used in total scheduling effectiveness calculations unless otherwise noted. FCF and OCF whose primary purpose is to perform maintenance checks and sorties originating off-station without home-unit support will be considered "flown as scheduled" without recording deviations with the exception of aborts. Specific ground deviations are:

A8.2.3.2.1. **(Added-AFMC)** Addition. An aircraft or event (including both sortie and ground test) added to the approved flying schedule. Will be recorded against the agency (OP, MX, HQ, Test Agency, etc.) requesting the additional sortie or aircraft. (**T-2)**

A8.2.3.2.2. **(Added-AFMC)** Cancellation (CX). Cancellation (CX). A sortie or ground-test event that is removed from the printed schedule for any reason before crew show. Cancellations occur when it is determined the originally scheduled mission cannot be met. For training sorties, if the sortie can launch and recover during the squadron's flying window and perform its original mission with the original crew, a cancellation is not recorded. If any sortie does not launch within the late takeoff criteria, a late takeoff is recorded. Any cancellations will be recorded against the agency (OP, MX, HQ, Test Agency, etc.) that is responsible for the cancellation. **(T-2)**

A8.2.3.2.3. **(Added-AFMC)** Early Takeoff (ET). An early takeoff is a scheduled sortie launching more than 30 minutes prior to the published takeoff time.

A8.2.3.2.4. **(Added-AFMC)** Ground Abort (GA). A ground abort is an event, after crew show, which prevents a ―crew ready aircraft against the responsible agency or condition that caused the abort. Ground aborts are categorized as maintenance (GAA/GAB/GAC), operations, higher headquarters, weather, sympathy, project, other, etc.

A8.2.3.2.5. **(Added-AFMC)** Late Takeoff (LT). A late takeoff occurs when a scheduled sortie becomes airborne more than 30 minutes after the scheduled takeoff time. If a LT is driven by a late land, the deviation for the LT will be charged against the responsible agency for the late land.

A8.2.3.2.6. **(Added-AFMC)** Spare (SP). A spare is a designated aircraft on the printed schedule to be used in case a scheduled aircraft cannot fly its scheduled sortie. Spares can include aircraft that have canceled, aborted, flown earlier or an aircraft released after FCF/OCF. The aircraft production supervisor will manage scheduling and use of spare aircraft. Do not count printed spares flown in scheduled lines as deviations when computing TSE. Spare aircraft requirements are based on projected

test mission needs. The following factors are used to determine spares and are not hard limits but guidelines, and as such, it is up to the AMU to actually schedule what it can support.

A8.2.3.2.6.1. **(Added-AFMC)** The spare requirements may be canceled (when deviations are not used for computing TSE) if test requirements are accomplished/met.

A8.2.3.2.6.2. **(Added-AFMC)** Additional spares are authorized to support HHQ taskings and special missions (if required by tasking).

A8.2.3.2.7. **(Added-AFMC)** Interchange/Tail Number Swap (TS). Interchanges are changes to the printed flying schedule involving aircraft tail numbers printed on the flying schedule. Interchanges may be used to prevent reconfigurations and unnecessary expenditures of man-hours, or to replace a printed tail number with another tail number on the signed scheduled. Interchanges between aircraft on the flying schedule, including FCF/OCF and cross country returns that have flown, will not be recorded as deviations. Interchanges include the following:

A8.2.3.2.7.1. **(Added-AFMC)** Changing aircraft in printed line numbers with printed spare aircraft.

A8.2.3.2.7.2. **(Added-AFMC)** Changing aircraft in printed line numbers to different printed line numbers (tail number swap).

A8.2.3.2.7.3. **(Added-AFMC)** Interchanges may be made up to any time prior to aircraft takeoff. The MOC must be notified of all interchanges for proper coordination with affected work centers. Record all interchanges in the MIS. Do not count Interchanges/Tail Number Swaps as deviations when computing TSE.

A8.2.3.3. **(Added-AFMC)** Air Deviations. Air deviations are events occurring after takeoff. They are recorded in the MIS, but are not included in total scheduling effectiveness calculations. Air deviations fall into the following categories:

A8.2.3.3.1. **(Added-AFMC)** Air Abort (AA). An aircraft/sortie that cannot complete its primary or alternate mission for any reason. Air aborts are considered a sortie flown when reporting total sorties flown. Air aborts will be recorded against the agency or condition that caused the aborted mission. An air abort will not be recorded when malfunctions occur during the "Before Takeoff Checklist" portion of helicopter sorties. Air aborts caused by specific test mission will not be counted in abort rate computations. **Note:** Effective sortie decisions will be made by operations.

A8.2.3.3.2. **(Added-AFMC)** IFE Aircrew Inflight (AI). An air aborted aircraft/sortie with a situation resulting in an in-flight emergency declared by the aircrew.

A8.2.3.3.3. **(Added-AFMC)** IFE Flight Emergency (FE). An aircraft/sortie with a situation resulting in an in-flight emergency declared by the aircrew after the mission is accomplished.

A8.2.3.3.4. **(Added-AFMC)** Late Landing (LL). An aircraft/sortie landing more than 15 minutes after the scheduled landing time. If the sortie originated on time, record any subsequent late takeoff or cancellation against the agency that caused the late landing. Late landings are included in TSE calculations. **Note:** LTs do not justify LLs.

A8.2.3.3.5. **(Added-AFMC)** Early Landing (EL). An early landing is an aircraft/sortie landing more than 30 minutes before the scheduled landing time. Early Landing deviations are not used when computing TSE.

Table A8.1. (Added-AFMC) Common Total Scheduling Effectiveness (TSE) Deviation Determination Matrix

Event	Is the deviation		Remarks
	Recorded in IMDS/MIS?	Calculated in TSE?	
Pen-and-Ink changes to the schedule are made on an AF Form 2407 (IAW Chapter 14)	No	No	Pen-and-Ink changes are not deviations and are considered part of the printed schedule. See Chapter 14
Take-off or landing time is changed after approved Pen-and-Ink submissions via AF Form 2407	Yes	Yes	See paragraphs in **Chapter 14**. Calculation in TSE is determined by late and early criteria
Aircraft configuration is changed after approved Pen-and-Ink submissions via AF Form 2407	Yes	No	Changes will be tracked to prevent reoccurrence and get a true picture of the total scheduling turmoil.
A sortie is added to the flying schedule (excluding OCFs/FCFs, XC return)	Yes	Yes	
A sortie is added for an OCF/FCF/OCF/FCF “Chase”	No	No	These are considered flown as scheduled.
A sortie is canceled	Yes	Yes	Once the decision is made to cancel the sortie, it is a cancel. If a decision is made after the cancel to go ahead and fly the sortie, it becomes an added line.
A sortie is determined to be non-effective	No	No	Not a deviation. The determination is made by operations and has no bearing on TSE.
A take-off is determined to be late	Yes	Yes	

A take-off is determined to be early	Yes	Yes	
A landing is determined to be early	Yes	No	
A landing is determined to be late	Yes	Yes	A late landing may result in a late take-off on a subsequent sortie.
During a surge, more sorties are flown than were printed and the statement “Sortie Surge” is NOT printed in the remarks section of the affected day’s flying schedule	Yes	Yes	Sorties printed in the weekly schedule will be flown as printed. Additional sorties not printed will be considered added lines.
During a surge, more sorties are flown than were printed and the weekly schedule contains “Sortie Surge” in the remarks section of the affected day	No	No	During planned and printed surges and combat sortie generations, additional lines are considered flown as scheduled.
During a surge, an aircraft turn sortie takes off early or late	No	No	Units should track late take-offs of turn sorties locally during surges to prevent reoccurrence. Late take-offs are recorded for surge first go sorties.
Maintenance is performed during a stop in a continuation sortie and the mission continues	Yes	Yes	An “add” is recorded for the subsequent sortie. Be sure the added line is designated as a continuation sortie to prevent further deviations for other scheduled stops.
A sortie is added to the schedule for weather attrition	Yes	No	
A sortie is added to the schedule for weather attrition when planned weather attrition has been meet	Yes	No	See A8.2.6.1

A sortie is canceled at any time due to weather	Yes	Yes	Prior to crew show it is a cancel, after crew show, it is a weather abort.
A spare aircraft printed on the flying schedule is used in a printed line.	Yes	No	
An aircraft in the printed schedule is swapped with an aircraft in another printed line.	Yes	No	
An aircraft not printed in the flying schedule is used in a printed line. (excluding aircraft already flown that day such as OCF/FCF, X-Country returns)	Yes	Yes	One deviation is recorded for the added aircraft. The result is the same as adding an aircraft as a spare, then tail swapping it into a printed line.
An aircraft not on the printed flying schedule is added as a spare.	No	Yes	Counts as a TSE deviation even if the aircraft does not fly. This has to be manually done by MMA because there is no required IMDS/MIS transaction that captures this.
An aircraft not printed in the flying schedule that has flown that day is flown/used in a printed line	Yes	No	Examples include previously flown FCF/OCF aircraft as well as cross country returns.
A ground abort is replaced with a another aircraft/spare on the printed schedule	Yes	No	Both the ground abort and spare action will be recorded in IMDS/MIS. If the replacement aircraft takes-off on time, no deviation is recorded.

A printed aircraft ground aborts and is replaced with an aircraft **NOT** on the printed schedule and the second aircraft also ground aborts and the original aircraft is fixed, takes off late, and flies the sortie.	Yes	Yes	The original aircraft is recorded as a ground abort and late take-off. The second aircraft is recorded as an "add" and a ground abort. Ground aborts in themselves are not deviations calculated in the TSE rate, but are calculated in the ground abort rate.

A8.2.4. **(Added-AFMC)** Deviation Causes. Deviations will be assigned a primary cause. The WG/CC will resolve questions concerning assigning deviations between maintenance and operations. The OS/CC, AMXS/CC, and LRS/CC will resolve questions concerning deviations involving supply and POL. Deviations will be assigned one of the following causes: (**T-2)**

A8.2.4.1. **(Added-AFMC)** Maintenance (MT). Deviations resulting from aircraft discrepancies, unscheduled maintenance, or for actions taken for maintenance consideration. **Note:** Maintenance-related ground aborts will use one of the ―GAx cause codes (See **paragraph 5.2.4**.).

A8.2.4.2. **(Added-AFMC)** Operations (OP). Deviations resulting from operations/aircrew actions, mission changes causing an early/late takeoff, or cancellation including substitution/aircrew illness, and over- stressing the aircraft. OP are also deviations resulting from unit controlled operations factors including those caused by mission/load planning, life support, intelligence, base operations, and passengers.

A8.2.4.3. **(Added-AFMC)** Project/Test Agency. Deviations to flight that result from rescheduling, reconfiguration, test item malfunctions, failures, or necessary adjustments to equipment undergoing tests or evaluations associated with Operational Testing and Evaluation (OT&E). Development Testing and Evaluation (DT&E), or Initial Operational Testing and Evaluation (IOT&E).

A8.2.4.4. **(Added-AFMC)** Supply (SU). Deviations resulting from a Partially Mission Capable Supply (PMCS) or Not Mission Capable Supply (NMCS) condition or for late Supply or Petroleum Oil Lubricant (POL) delivery. Contract Logistics Support (CLS) items are included. In accordance with local MOAs/operating instructions. **Note:** The actual time required for installation and ops check will be considered. Maintenance has up to the end of the original parts plus estimated time in commission (ETIC), deviations after the ETIC has expired are charged against maintenance.

A8.2.4.5. **(Added-AFMC)** Higher Headquarters (HHQ). Deviations resulting from a higher headquarters tasking including closing of low level routes/ranges or external customer driven mission change. When an aircraft that was scheduled for a higher headquarters directed off-base mission is replaced by a spare refer to paragraphs **A8.2.3.2.6** and **A8.2.3.2.7** for unit options.

A8.2.4.6. **(Added-AFMC)** Weather. Deviations for aircraft which takeoff early, late, abort, or are added or canceled due to weather conditions. For example, if an aircraft taxied to the end of runway and the wing commander cancels all flying due to weather, the aircraft is a weather abort. Sorties/Aircraft cancelled prior to crew show are weather cancels.

A8.2.4.7. **(Added-AFMC)** Sympathy Deviations occurring when an event of two or more aircraft, under the command of a flight leader or instructor pilot are canceled, aborted, or late due to a cancellation, abort, or delay of one of the aircraft in the flight or a supporting flight. Events, which are to replace sympathy aborts or cancellations on the same day, will be recorded as sympathy additions. Events lost caused by the aircraft's scheduled mated tanker/receiver/mission event will be recorded as sympathy. Examples of mission events are: loss of release times, Airborne Warning and Control System support, Minimum Interval Take Off causing takeoff delay or cancellation, or for another unit's or command's support should be coded as sympathy deviations. **Note:** Deviations caused by aircraft/missions earlier scheduled lines will be assigned to the cause of the earlier deviation, not sympathy.

A8.2.4.8. **(Added-AFMC)** Air Traffic Control. Deviations resulting from air traffic control problems (for example, flight clearance delays, tower communication failure, conflicting air traffic, runway change, or runway closure).

A8.2.4.9. **(Added-AFMC)** Contractor Support. A deviation resulting from a test or system contractor not being able to support the mission in accordance with the requirements of the contract.

A8.2.4.10. **(Added-AFMC)** Range Availability. A deviation resulting from the Range not being available.

A8.2.4.11. **(Added-AFMC)** Other. Deviations resulting from unusual circumstances not covered by other causes listed. OT may include:

A8.2.4.11.1. **(Added-AFMC)** Unusual circumstances (e.g.., bird strikes, damage during air refueling, unscheduled exercises) may use this code.

A8.2.4.11.2. **(Added-AFMC)** Equipment, non-AFMC. Deviations caused by National Airborne Operations Center or Air Intelligence Agency, and other non-AFMC support and equipment.

A8.2.4.11.3. **(Added-AFMC)** When an aircraft is off station and cannot return for its scheduled sortie/mission, a deviation will be recorded for the reasons the aircraft was unable to return. The reasons will be specific (maintenance, operations, weather, etc.). A printed spare aircraft may be used for any sorties scheduled for the aircraft that did not return.

A8.2.4.11.4. **(Added-AFMC)** Ferry Sorties. Ferry sortie requirement too late for inclusion in the flying schedule will be documented as a new line for the day and "flown as scheduled".

A8.2.4.11.5. **(Added-AFMC)** Hot Spare Aircraft. Properly configured aerospace vehicle that is listed on the flying schedule as a spare with a specific takeoff time and not utilized. If not used will be recorded as a cancel other.

A8.2.4.11.6. **(Added-AFMC)** Airborne Instrumentation.

A8.2.4.11.7. **(Added-AFMC)** Laboratories.

A8.2.4.12. **(Added-AFMC)** Exercise, Higher Headquarters (EXH). Deviations resulting from higher headquarters directed exercises, including alarm/force protection conditions. Deviations in this category will not be included in TSE calculations.

A8.2.4.13. **(Added-AFMC)** Exercise, Local (EXL). Deviations resulting from wing/unit directed exercises, including alarm/force protection conditions. Deviations in this category will not be included in TSE calculations.

A8.2.4.14. **(Added-AFMC)** Special Instrumentation. Deviations that directly result from issues with SI support.

A8.2.5. **(Added-AFMC)** Rules for Reporting Schedule Deviations. The unit is responsible for documenting deviations to the weekly flying schedule and determining the cause for each deviation. Deviations must be coordinated with the appropriate agency before being assigned to a specific category. The operations representative and AMU/AMXS MS will monitor deviations to make sure they meet the criteria in this instruction. When disparities exist, they will be referred to the OG and MXG CCs or designated representative for resolution. Schedule deviations that result from a sequence of events will be assigned a primary cause. A determination of the primary cause will be decided upon by all parties involved to arrive at a unit position.

A8.2.6. **(Added-AFMC)** Scheduling Exceptions.

A8.2.6.1. **(Added-AFMC)** Adverse Weather. Units/flying squadron(s) may add sorties to the flying schedule to make up for weather losses. Sorties will only be added to the schedule when the planned weather attrition for the month has been exceeded for that month. The number of sorties added will not exceed the difference between the planned weather attrition and actual weather losses. Weather attrition will be computed by the MMA Section. (Example: Planned weather attrition for the month equals 30 sorties. On the 20th of the month 40 sorties are lost. Ten non-chargeable sorties may be added. The maintenance schedule and the ability of maintenance to support the additional requirements must be carefully considered before adding sorties.

A8.2.6.2. **(Added-AFMC)** Unscheduled Tasking. When a unit is tasked with an unscheduled higher headquarters tasking or self-initiated tasking (mobility exercises or weather evacuations), or other services tasking which significantly impacts the flying schedule, the printed schedule may be revised or deleted from the MIS by AMU PS&D and replaced with a new schedule without recording deviations. For weather evacuations, the schedule will be cancelled in the MIS, not deleted, so the data is available for historical attrition. (**T-2) Note: Local unscheduled tasking will not be used solely to recoup sortie losses.**

A8.2.6.2.1. **(Added-AFMC)** If the schedule is revised or canceled and reprinted, the following procedures will be used: **(T-2)**

A8.2.6.2.1.1. **(Added-AFMC)** Follow normal deviation reporting procedures once the revised/reprinted schedule is finalized. The revised schedule will be finalized a minimum of 2 hours before the first scheduled launch.

A8.2.6.2.1.2. **(Added-AFMC)** Once the tasking terminates, the original schedule

may be used or it may be revised or reprinted for the tasking period, as required. Normal deviation reporting is to be used once the revised or reprinted schedule is finalized.

A8.2.6.2.1.3. **(Added-AFMC)** Normal deviation reporting procedures will be followed after a takeoff time is established to a tasking by higher headquarters or other services.

A8.2.7. **(Added-AFMC)** Utilization Reporting. All aircraft flying hours are required to be reported in REMIS in accordance with DAFI 21-103. This requirement is not waived for contractor-maintained aircraft.

A8.2.8. **(Added-AFMC)** Scheduling Effectiveness Computation

A8.2.8.1. **(Added-AFMC)** TSE Rate. Similar to the Flying Schedule Effectiveness for the CAF, TSE is a leading indicator and is a measure of how well the unit planned and executed the weekly flying and ground test event schedule. TSE is derived by comparing each day's deviations to the flying and ground test schedule. Deviations that decrease the TSE from 100 percent include: scheduled sorties/ground test events not accomplished because of maintenance, supply, operations, weather, HHQ, air traffic control, sympathy, or other reasons; scheduled events that actually start more than 30 minutes prior to scheduled start time; scheduled events that begin more than 30 minutes after their scheduled start time and events that are added to the schedule. Disruptions to the flying or ground test schedule, or constant configuration changes to test assets can cause turmoil on the flightline, send a ripple effect throughout other agencies, and adversely impact scheduled maintenance actions.

Table A8.2. (Added-AFMC) Test Scheduling Effectiveness Computation

TSE (%) = Total Ground and Sortie Events Scheduled – Total Deviations x 100 / Total Ground and Sortie Events Scheduled

A8.2.9. **(Added-AFMC)** Maintenance Deviation Rate (MX Dev). Although all deviations to the test schedule are counted in the TSE calculation, the Maintenance Deviation Rate represents deviations to the flying/ground test schedule that are, for the most part, are within the maintenance community's control. All maintenance late take offs, ground aborts, cancellations, and maintenance additions are counted towards the rate. Although all deviations from the published schedule can cause turmoil on the flightline, maintenance supervision should focus on eliminating or minimizing schedule deviations due to maintenance.

Table A8.3. (Added-AFMC) Maintenance Deviation Rate Computation

Maint Dev Rate (%) = Total Maintenance Deviations x 100 / Total Ground and Sortie Events Scheduled

A8.3. (Added-AFMC) Maintenance Scheduling Effectiveness (MSE).

A8.3.1. **(Added-AFMC)** Purpose. This chapter provides a means to measure maintenance management effectiveness, reflected by how well the maintenance schedule is carried out. Maintenance efforts should be directed toward the timely accomplishment of all scheduled maintenance actions. The AFMC Maintenance Scheduling Effectiveness (MSE) goal is 95%.

A8.3.2. **(Added-AFMC)** Computations.

A8.3.2.1. **(Added-AFMC)** Scheduled maintenance events and respective weighted factor points in **Table 3.1** are used to compute the MSE rate. Credit is received for actions completed on, or prior to, the scheduled date as printed in the weekly schedule. The MIS database and the published schedule will be used to determine whether or not the maintenance actions were completed on-time. **Exception:** Phase/PE and ISO inspection, and all maintenance requiring more than 24 work hours to complete will be measured against the scheduled start time/date.

A8.3.2.2. **(Added-AFMC)** The MXG/CC may select additional areas (such as AGE, AIS, AME, etc.) for local scheduling effectiveness tracking. The unit will establish standards for these programs and develop a computation table in the format of **Table 3.1**. This data will not be included in aircraft maintenance scheduling effectiveness when reported to higher headquarters.

A8.3.2.3. **(Added-AFMC)** MOF PS&D will record maintenance scheduling deviations and forward the computations to MOF maintenance analysis weekly for publications in the monthly maintenance summary. **(T-2)**

A8.3.2.4. **(Added-AFMC)** When a unit is tasked with a test acceleration, quick reaction test, unscheduled tasking, unannounced exercise/real world contingency, or higher headquarters exercise that significantly impacts the printed weekly maintenance schedule, the plan may be revised or reprinted without incurring deviations. Normal deviation reporting procedures will be followed once the revised or reprinted plan is finalized. The unaccomplished portion of the maintenance schedule that was revised will not be included in the scheduling effectiveness formula. **(T-2)**

A8.3.2.5. **(Added-AFMC)** Establish local written procedures to ensure coordination of changed/cancelled/added scheduled maintenance.

A8.3.2.6. **(Added-AFMC)** Maintenance Scheduling Deviation Categories.

A8.3.2.6.1. **(Added-AFMC)** Maintenance (MT). Actions canceled or not completed as a result of adding aircraft to the flying schedule, mismanagement, or a lack of manpower or equipment.

A8.3.2.6.2. **(Added-AFMC)** Higher headquarters (HHQ). Actions canceled or not completed as a result of higher headquarters tasking from outside of the wing.

A8.3.2.6.3. **(Added-AFMC)** Weather (WX). Actions canceled or not completed as a result of weather conditions.

A8.3.2.6.4. **(Added-AFMC)** Supply (SU). Actions canceled or not completed as a result of verified parts back order condition.

A8.3.2.6.5. **(Added-AFMC)** Other (OT). Actions canceled or not completed as a result of impounded aircraft (after the weekly schedule is published), major maintenance where technical data restrictions do not allow the scheduled maintenance, or aircraft is off-base and unable to return.

Table A8.4. (Added-AFMC) Maintenance Scheduling Effectiveness Computation

SCHEDULED EVENT	A WEIGHTED POINTS	B NUMBER OF EVENTS	C POSSIBLE POINTS (A x B)	D COMPLETED SCHEDULED	E POINTS EARNED (A x D)
Periodic/Isochronal/ Phase Inspections	5				
Home Station Checks/ Hourly Post Flights	5				
Engine Changes	5				
Time Changes	4				
TCTOs	4				
Corrosion Control/ Paint	4				
Special Inspections	3				
Document Reviews	2				
Delayed Discrepancies	3				
Total Points Possible:_____ Total Points Earned: Total O&M Points Possible: _____ Total O&M Points Earned:					

A8.4. (Added-AFMC) TDY/Deployed Sorties.

A8.4.1. **(Added-AFMC)** Purpose. This chapter establishes rules and procedures used in planning, executing, evaluating, and reporting of unit flying and maintenance schedules at deployed/TDY locations where unit maintenance is provided. Sorties flown at deployed/TDY locations where no parent unit maintenance is provided are considered off-station sorties. If parent unit support is deployed, this is considered the same as home station support and normal deviation reporting applies. Limited launch support is not considered parent unit maintenance.

A8.4.2. **(Added-AFMC)** General. Normal deviation reporting applies to deployed operations except as noted in this chapter. Data from deployed operations will be transmitted or forwarded back to home station.

A8.4.3. **(Added-AFMC)** Scheduling. In addition to the procedures established at home station for scheduling and reporting, deployed units will use the following procedures when developing a flying schedule and reporting deviations: Separate block(s) of sortie sequence numbers will be assigned for deployment/TDY location(s). **(T-2)**

A8.4.3.1. **(Added-AFMC)** Additions and cancellations at deployed/TDY locations, which are required to accomplish specific test and aircrew training requirements and make optimum use of available range time, are considered flown as scheduled. This does not relieve operations and maintenance from developing a viable and realistic flying schedule at the deployed location. The primary purpose of this flexibility is to allow the unit to make up non-effective sorties to ensure accomplishment of the training plan. Procedures for changing the flying schedule in **Chapter 2** apply to deployment/TDY location flying and maintenance schedules. Additions and cancellations caused by ineffective planning are recorded.

A8.4.3.2. **(Added-AFMC)** Sorties scheduled at other than home station are not required to be input into the home station's operations scheduling system to be counted as scheduled sorties. Once tail numbers are plugged in at the daily meeting; this becomes the firm schedule‖ and deviations are reported from that point on.

A8.4.4. **(Added-AFMC)** Deployed/TDY Daily Activity Report. Required information for deployed/TDY AFMC units will be transmitted to home station IAW applicable unit deployment/TDY plans. **(T-2)**

A8.4.5. **(Added-AFMC)** Off-Station Sorties. Off-station sorties are sorties flown from other than home station where parent unit maintenance is not provided (e.g., cross-country sorties). **Note:** If the off-station aircraft can fly its scheduled mission from its location, no deviation is recorded.

A8.5. (Added-AFMC) Scheduling Reporting Procedures.

A8.5.1. **(Added-AFMC)** Purpose. This chapter provides instructions on flying scheduling reporting procedures. The flying schedule must be loaded into the MIS to track scheduling and deviation data. Once loaded, the mission schedule or proposed maintenance plan background programs will provide base-level retrieval of flying and maintenance schedule information, and will complete higher headquarters reporting of aircraft utilization.

A8.5.2. **(Added-AFMC)** Responsibilities.

A8.5.2.1. **(Added-AFMC)** The MXG/CC will ensure procedures are established to verify the accuracy of all scheduling and deviation data. **(T-2)**

A8.5.2.2. **(Added-AFMC)** The OG and MXG CCs or their designated group representative will publish the flying schedule IAW **Chapter 2** or computer generated forms. Units will load changes to the flying schedule into the MIS by 1600L one workday prior to date of execution. **(T-2)**

A8.5.2.3. **(Added-AFMC)** The MOC will review the previous day's activity reports to ensure accurate deviation accounting and reporting. The MOC will record additions, cancellations before crew show, interchanges, and the use of spare aircraft as deviations occur. The debrief section will record, late and early takeoffs and landing aborts, and in-flight emergency incidents during the automated debriefing process. The MOC will coordinate with the squadron/AMU on all changes and deviations to the flying schedule to assist in determining correct debriefing status codes. The MOC will provide sortie sequence numbers and sortie numbers to the squadron/AMU for all additions and cross-country sorties. Sortie numbers assigned to a specific tail number must be in sequential

order (for example sortie number 101 must be used on a specific tail number before sortie number 102). Unique sortie sequence numbers will be developed for deployed sorties. **(T-2)**

A8.5.2.4. **(Added-AFMC)** Cause Code. Enter one of the following codes to indicate the reason for a deviation or the agency, which caused a deviation.

Table A8.5. (Added-AFMC) Code Function

ATx . . . . . Air Traffic
EXH . . . . . .Exercise, HHQ
EXL . . . . . .Exercise, Local
GAA . . . . . Ground Abort, before engine start, maintenance
GAB . . . . . .Ground Abort, after engine start, before taxi, maintenance
GAC . . . . . .Ground Abort, after taxi, maintenance
GAT. . . . . Ground Abort, test induced
HQT . . . . . .Higher Headquarters, MAJCOM (non-exercise)
HQN . . . . . Higher Headquarters, NAF (non-exercise)
HQP . . . . . .Higher Headquarters, other (non-exercise)
MTx . . . . . .Maintenance
OPx . . . . . Operations
SIx . . . . . .Special Instrumentation
SUx . . . . . .Supply
SYx . . . . . .Sympathy
RNx. . . . . Range
WXx . . . . . Weather
OTx . . . . . .Other
PJx……...Project/Test Agency
CSx. . . . . Contractor Support

A8.6. (Added-AFMC) Attrition and Spare Factors.

A8.6.1. **(Added-AFMC)** Attrition. Attrition factors represent historical percentage of scheduled sorties lost to causes outside unit control. Maintenance and operations schedulers add attrition sorties to monthly contracts to ensure mission goals are met. Operations and maintenance personnel must ensure the contract includes flexibility to accept attrition requirements during acquisition planning. Units may make a conscious decision, with MXG/CC approval, to use different attrition factors from statistical attrition rates calculated by MMA.

A8.6.1.1. **(Added-AFMC)** Attrition sorties are not substitutes for capability shortfalls, they are additive to the contract to ensure mission goals are met. The monthly flying and maintenance plan will clearly identify attrition sorties for management purposes. If attrition is less or more than planned, adjustments to the weekly flying and maintenance schedule will be made to prevent over-extending maintenance or exceeding the unit's contract. A sortie lost will normally be flown in the same month the loss occurred. If at the end of a quarter combined losses exceed attrition figures, the OG and MXG CCs will negotiate a resolution to the shortfall.

A8.6.1.2. **(Added-AFMC)** The factors used to compute attrition will be MXx, OPx, SUx, WXx, ATx, SYx, OTx, EXH, EXL, and HQx sortie losses. Attrition factors are used to compensate for non-unit controlled factors. Spares are used to compensate for unit controlled factors. Attrition and spare factors will be computed for and applied to each flying squadron/AMU. Monthly statistical attrition anomalies should be identified, documented and factored out of attrition calculations if necessary. MMA will compute attrition factors monthly for each OS/AMU and provide the results to MOF PS&D for dissemination to the OS and AMU PS&D schedulers. During the annual first look, MMA will provide attrition factors by month for the entire next fiscal year. **(T-2) Note:** Attrition and spare factors need not be developed for test and evaluation Combat Tactics Development and Equipment Evaluation possession identifier coded aircraft.

A8.6.2. **(Added-AFMC)** Attrition Factor Application.

A8.6.2.1. **(Added-AFMC)** Attrition computation is based on at least 5 years of unit historical data from previous flying months. For example, when computing attrition for Jan 12, use historical data for Jan 11, Jan 10, Jan 09, Jan 08, Jan 07, etc. Use as much historical data as required ensuring seasonal variations are considered to determine a basis for attrition. When computing attrition, use the total sorties lost in a particular category. Do not use the difference between the sorties lost and those sorties added to make up for the losses. The formula for computing the attrition factor is Historical Sorties Lost divided by Historical Sorties Scheduled.

Table A8.6. (Added-AFMC) Attrition Computation Example

Sortie Losses:
MX Sortie Losses .02
OP Sortie Losses .01 SU Sortie Losses .01
OT Sortie Losses .01
AT Sortie Losses .01
EXH Sortie Losses .01
EXL Sortie Losses .01
HQ Sortie Losses .01
Sortie loss attrition factor: 0.09
WX Sortie Losses: .03
Total Attrition factor: .12
Overall attrition factor is: .12 or 12 %

Table A8.7. (Added-AFMC) Sample Application of Total Attrition Factor

Sorties Required 1,000
Subtract attrition factor from 1 (1-.12)=0.88
Divide 1,000 by 0.88
Required sorties to schedule 1,136.36 , round up to 1,137.
Based on historical attrition of 12%, the unit can expect to lose 137 sorties to meet the required 1,000 sorties.

A8.6.3. **(Added-AFMC)** Spares. The spare requirements will not exceed 20 percent (30 percent for training units owning TF coded aircraft) of aircraft committed to the flying schedule, rounded up to the next whole aircraft. **Note:** During Planned Sortie Surges the MXG/CC determines the amount of spares that will be committed. However, leadership must consider health of the fleet when authorizing scheduled spares above 40 percent. **Note:** Units should be cognizant of their historical break rates and spare constraints when scheduling surges. Spares can be quickly used during surges and once spares are exhausted the capability to meet surge goals is severely limited.

A8.6.3.1. **(Added-AFMC)** MMA computes annual spare aircraft requirements by month, using historical aircraft first sortie logistics losses and provides this information to the MO PS&D for use in computing spare aircraft requirements. Spare computation is based on 5 years of unit historical data from previous flying months. For example, when computing spares for Jan 06, use historical data for Jan 05, Jan 04, Jan 03, Jan 02, Jan 01, etc. Use a minimum of 5 years historical data to ensure seasonal variations are included to determine a basis. The formula for computing spare factors is Historical First Sortie Deletions/Cancellation divided by historical first sorties scheduled.

A8.6.3.2. **(Added-AFMC)** A first sortie is defined as a sortie flown by an aircraft that has not previously flown for the day (0001-2400 flying period). For example, if 8 aircraft are committed to the schedule and there are 14 total sorties scheduled, the first 8 sortie line numbers (i.e., 101-108) should reflect all 8 committed aircraft tail numbers before they are re-scheduled (turned) against the last 6 (i.e., 109-114). This would be reflected as an 8 x 6 and will not be reflected as a 6 x 8 because of scheduled take-off times. Additionally lines are listed in sequential order with take-off times that reflect this order. Operations may define first sorties or turns by mission profile, take-off times, but for the purposes of this instruction, the sortie turn pattern is defined against initial aircraft flown and scheduled turns of the same or a portion of the same aircraft.

Table A8.8. (Added-AFMC) Sample Application of Spare Factors

1st Sortie Maintenance Cancellations	.10
1st Sortie Supply Cancellations	.03
1st Sortie Ground Aborts	.05
Spare factor	.18 or 18%
A sample figure of 12 first sorties is used in the following computation: Spare aircraft required equals 1st sorties scheduled times the spare factor and rounded up to the next whole number. 12 x .18 = 2.16 Spares Required is 3	

A8.6.3.3. **(Added-AFMC)** The computed spare requirement may be adjusted to compensate for multiple configurations and syllabus constraints. When additional spares are added for multiple configurations, units will not exceed one spare per configuration.

A8.6.3.4. **(Added-AFMC)** Additional spares are authorized to support higher headquarters taskings and special missions (if required by the tasking).

A8.6.3.5. **(Added-AFMC)** At least one spare aircraft is authorized per MDS for each flying day.

A8.6.3.6. **(AFMC) (Added** Unmanned Aerial Systems training missions are authorized an additional spare to support increased aircrew training requirements due to crew size ratio.

Attachment 9 (Added-AFMC)

SAMPLE AIRCRAFT AND EQUIPMENT IMPOUNDMENT CHECKLIST

Table A9.1. (Added-AFMC) Sample Aircraft and Equipment Impoundment Checklist

	Impoundment Actions	Status
1	Impoundment authority directs impoundment and assigns an impoundment official.	
2	Enter a Red X in the applicable forms and MIS indicating the reason for impoundment and the name of individual assigned as impoundment official.	
3	Notify the MOC of impoundment decision.	
4	Select a team of highly qualified technicians to determine cause of problem that led to the impoundment.	
5	Control access to impounded aircraft or equipment and determine if an entry control point (ECP) is required. If ECP is established, use an access control log.	
6	Determine necessary controls required for aircraft and equipment records.	
7	Review aircraft or equipment forms and MIS for historical data related to the malfunction causing impoundment.	
8	Ensure maintenance is limited until the cause of impoundment is determined.	
9	Ensure parts removed are carefully controlled.	
10	Once the cause of the malfunction has been determined and corrected, determine if an OCF or FCF is necessary or required.	
11	Ensure QA is intimately involved in the impoundment process and has reviewed all actions taken to correct the malfunction.	
12	QA determines the need for cross-tell based on the potential effect to other aircraft and equipment.	
13	Brief release authority on findings and corrective actions, and request release from impoundment.	
14	Impoundment release authority clears or directs the impoundment be cleared in the applicable aircraft or equipment forms.	